

Topological Photonics from Maxwell's Equations

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Topological Photonics from Maxwell's Equations

A Maxwell-First Foundation for
Topological Phases of Light

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To those who first saw topology in light

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Preface

This book is about light that refuses to turn back.

In an ordinary optical waveguide, every imperfection—a bend, a scratch, a change in refractive index—scatters some light backward. Over long distances, backward scattering accumulates and degrades any signal carried by the light. This is not an engineering failure; it is a consequence of time-reversal symmetry, which guarantees that every forward-propagating mode has a backward partner at the same frequency.

In 2005, Haldane and Raghu proposed a way to break this symmetry in photonic crystals, creating channels where light can propagate in only one direction. These “one-way waveguides” are the photonic analogs of the chiral edge states in the quantum Hall effect. In 2009, Wang, Chong, Joannopoulos, and Soljačić demonstrated the effect experimentally: microwave radiation travelling along the edge of a gyromagnetic photonic crystal with a forward-to-backward ratio exceeding 10^5 .

The physics behind this phenomenon is *topological*. Each electromagnetic band in a periodic medium carries a global invariant—the Chern number—that counts how many times the Berry phase of the Bloch modes winds as the crystal momentum sweeps the Brillouin zone. When two media with different Chern numbers are joined, unidirectional edge modes necessarily appear at the interface. The number and direction of these modes are fixed by topology and cannot be removed by any perturbation that preserves the bulk band gap.

Why this book. Topological photonics has grown rapidly since 2005. Excellent reviews exist, notably those by Lu, Joannopoulos, and Soljačić (2014), Khanikaev and Shvets (2017), and Ozawa et al. (2019). Yet there is no monograph that develops the subject *from Maxwell’s equations*, with the care and completeness expected of a graduate-level text in mathematical physics.

Most treatments borrow the formalism of electronic topological insulators—tight-binding Hamiltonians, Bloch electrons, Kramers degeneracy—and graft photonic examples onto it. This approach is efficient for researchers already fluent in condensed-matter topology, but it obscures what is specific to electromagnetism: the six-component field structure, the material-dependent energy inner product, the bosonic time reversal with $\hat{\mathcal{T}}^2 = +1$, and the role of dispersive and bianisotropic media.

This book takes a *Maxwell-first* approach. Every concept is derived from the macroscopic Maxwell equations and the electromagnetic energy inner product, not imported from condensed-matter theory by analogy. Electronic parallels are mentioned for orientation, but they never replace the electromagnetic derivations.

Structure. The book is organised in five parts.

Part I (Chapters 1–5) builds the foundation: the Maxwell equations as a first-order dynamical system, the electromagnetic energy inner product, normal modes, Bloch theory for periodic media,

and the Berry connection, curvature, and Chern number for Maxwell bands.

Part II (Chapters 6–10) develops the theory of photonic Chern insulators: Dirac cones from lattice symmetry, time-reversal breaking by gyrotropic media, the one-way interface mode at a domain wall, the Wang microwave experiment, and numerical methods for computing Chern numbers.

Part III (Chapters 11–14) covers other two-dimensional topological phases: reciprocal designs with pseudospin, valley-Hall photonic crystals, Floquet (time-modulated) systems, and one-dimensional topology including the Zak phase and the SSH model.

Part IV (Chapters 15–16) extends the framework to three dimensions (Weyl points and surface arcs) and synthetic dimensions (internal degrees of freedom as extra lattice coordinates).

Part V (Chapters 17–18) addresses non-Hermitian topology (loss, gain, exceptional points, topological lasers) and nonlinear and quantum frontiers (topological solitons, many-body photonic states, polaritons).

Four appendices provide reference material: vector identities and functional analysis (A), differential forms and Chern classes (B), numerical recipes with pseudocode (C), and material tensors with sign conventions (D).

Prerequisites and audience. The intended reader has completed graduate courses in electrodynamics (at the level of Jackson or Griffiths) and has some exposure to quantum mechanics and solid-state physics. No prior knowledge of topology, differential geometry, or condensed-matter topological insulators is assumed; these ideas are developed as they arise, motivated by the electromagnetic problems that require them.

Exercises. Each chapter ends with exercises ranging from straightforward verifications to open-ended explorations. Working through these exercises is essential for developing the physical and mathematical intuition needed to apply topological ideas to new photonic systems.

Conventions. We use SI units throughout, with the time-harmonic convention $e^{-i\omega t}$ and $\omega > 0$ for physical frequencies. The complete notation is summarised in the next chapter and in Appendix D.

Acknowledgements. This work builds on the contributions of many researchers who have shaped topological photonics into a vibrant field. Special acknowledgement is due to the pioneering theoretical work of Haldane and Raghu, the experimental breakthroughs of Wang, Chong, Joannopoulos, and Soljačić, and the comprehensive reviews that have made the field accessible to a broad audience.

The Authors
2026

Notation and Conventions

Time-harmonic convention

All frequency-domain quantities use the engineering convention $e^{-i\omega t}$ with $\omega > 0$. In this convention, a forward plane wave propagating in the $+z$ direction has the form $e^{i(kz - \omega t)}$. Absorption corresponds to $\text{Im}\varepsilon > 0$; gain corresponds to $\text{Im}\varepsilon < 0$. The opposite convention $e^{+i\omega t}$ is common in the condensed-matter literature; formulas imported from such sources require a complex conjugation of all constitutive parameters.

Units

SI (MKSA) units are used throughout. The fundamental electromagnetic constants are:

$$c = 2.998 \times 10^8 \text{ m/s}, \quad (1)$$

$$\varepsilon_0 = 8.854 \times 10^{-12} \text{ F/m}, \quad (2)$$

$$\mu_0 = 1.257 \times 10^{-6} \text{ H/m}, \quad (3)$$

$$Z_0 = \sqrt{\mu_0/\varepsilon_0} = 376.7 \text{ } \Omega. \quad (4)$$

Frequencies are written as ω (angular, rad/s) or $\nu = \omega/(2\pi)$ (in Hz) as appropriate. Wavevectors have dimensions of inverse length (m^{-1}) and are written as $\mathbf{k} = (k_x, k_y, k_z)$.

Electromagnetic fields

Symbol	Name	SI unit
$\mathbf{E}$	Electric field	V/m
$\mathbf{H}$	Magnetic field	A/m
$\mathbf{D}$	Electric displacement	C/m ²
$\mathbf{B}$	Magnetic flux density	T
$\mathbf{f} = (\mathbf{E}, \mathbf{H})^T$	Six-component electromagnetic field	—

Material tensors

Symbol	Meaning
ε	Relative permittivity tensor (3×3)
μ	Relative permeability tensor (3×3)
ξ, ζ	Magneto-electric coupling tensors (3×3)
$\mathbf{M}$	6×6 material matrix: $\begin{pmatrix} \varepsilon_0 \varepsilon & c^{-1} \xi \\ c^{-1} \zeta & \mu_0 \mu \end{pmatrix}$
$\mathbf{M}_g(\omega)$	Energy (Brillouin) metric: $\partial_\omega[\omega \mathbf{M}(\omega)]$

Operators

Symbol	Meaning
$\hat{\mathcal{N}}$	Maxwell curl operator (six-component): $\begin{pmatrix} 0 & i\nabla \times \\ -i\nabla \times & 0 \end{pmatrix}$
$\hat{\mathcal{N}}(\mathbf{k})$	Bloch-reduced curl operator at wavevector $\mathbf{k}$
$\hat{\mathcal{H}} = \mathbf{M}^{-1} \hat{\mathcal{N}}$	Effective Maxwell “Hamiltonian”
$\hat{\mathcal{T}}$	Time-reversal operator (anti-unitary, $\hat{\mathcal{T}}^2 = +1$)
$\hat{\mathcal{P}}$	Spatial inversion (parity) operator

Inner product

The electromagnetic energy inner product for six-component fields is

$$\langle \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}} = \int_{\Omega} \mathbf{f}_1^\dagger(\mathbf{r}) \mathbf{M}(\mathbf{r}) \mathbf{f}_2(\mathbf{r}) d^d r.$$

When the material is dispersive, the energy metric $\mathbf{M}_g$ replaces $\mathbf{M}$ in mode orthogonality and Berry-connection formulas.

Brillouin zone and Bloch modes

Symbol	Meaning
$\mathbf{a}_1, \mathbf{a}_2$	Lattice vectors
$\mathbf{G}_1, \mathbf{G}_2$	Reciprocal lattice vectors
BZ	First Brillouin zone
$\mathbf{k}$	Bloch wavevector
$\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$	Cell-periodic Bloch eigenmode
$\omega_{n\mathbf{k}}$	Band frequency (eigenvalue)
K, K'	Brillouin-zone corners of a hexagonal lattice
Γ, M, X	High-symmetry points

Berry geometry

Symbol	Meaning
$\mathcal{A}_n(\mathbf{k})$	Berry connection of band n
$\Omega_n(\mathbf{k})$	Berry curvature of band n
$\mathcal{C}_n$	First Chern number of band n
$\mathcal{C}_{\text{gap}}$	Gap Chern number (sum over bands below the gap)
$\mathcal{C}_s$	Spin Chern number
$\mathcal{C}_v$	Valley-contrast Chern number
γ_{Zak}	Zak phase (1D Berry phase)
$\mathcal{W}(k_y)$	Wilson loop operator at k_y
$\mathcal{A}$	Berry connection vector (3D)
Ω	Berry curvature vector (3D)

Pauli and spin-1 matrices

The 2×2 Pauli matrices are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The 3×3 spin-1 matrices L_a (used in the six-component Maxwell formulation) are $(L_b)_{ac} = i\epsilon_{abc}$, where ϵ_{abc} is the Levi-Civita symbol.

Abbreviations

BZ	Brillouin zone
TE	Transverse electric (in-plane $\mathbf{E}$, out-of-plane H_z)
TM	Transverse magnetic (in-plane $\mathbf{H}$, out-of-plane E_z)
TR	Time reversal
QHE	Quantum Hall effect
SSH	Su–Schrieffer–Heeger
YIG	Yttrium iron garnet ($\text{Y}_3\text{Fe}_5\text{O}_{12}$)
PWE	Plane-wave expansion
FDFD	Finite-difference frequency-domain
FEM	Finite-element method
EP	Exceptional point
AFTI	Anomalous Floquet topological insulator
OAM	Orbital angular momentum

Cross-reference conventions

Equations are numbered within chapters as (chapter.number). Theorems, propositions, definitions, remarks, and exercises share a single counter per chapter. Chapters, sections, equations, and figures are referenced with the usual LaTeX commands (for example, `\ref` for numbered objects and `\eqref` for displayed equations), using the semantic labels introduced in the source files. Citations are managed by `biblatex` and appear as numerical references in the text.

Part I

Maxwell Modes and Optical Bands

Chapter 1

The Problem of Robust Light

1.1 Light that cannot turn back

Every optical engineer knows that light reflects. A fibre splice, a waveguide bend, a scratch on a mirror—any interruption of the medium sends part of the wave backward. In most of photonics this is merely a nuisance, compensated by isolators, circulators, or careful impedance matching. But suppose we could build a channel for electromagnetic energy in which backward propagation is *structurally forbidden*: not suppressed by a clever design trick that might fail under perturbation, but eliminated by a mathematical property of the electromagnetic band structure itself. What would the Maxwell equations have to contain for such a channel to exist?

That question is the starting point of this book. The answer, as we shall develop in detail over the coming chapters, involves a remarkable convergence of ideas from nineteenth-century electromagnetism, twentieth-century condensed-matter physics, and twenty-first-century photonic-crystal engineering. The key concept is *topology*: certain integer-valued invariants of the photonic band structure that cannot change under smooth deformations of the medium, and that guarantee the existence of unidirectional interface modes whenever two media with different invariants are brought together.

Before we can state this precisely, however, we need to see why ordinary approaches to waveguide robustness fail, and what new ingredient the problem demands.

1.2 Backscattering in ordinary waveguides

Consider a conventional dielectric slab waveguide: a high-index core of thickness d and refractive index n_1 sandwiched between cladding layers of index $n_2 < n_1$. Guided modes exist when the transverse resonance condition is satisfied, and for a given frequency ω there are generally both forward-propagating modes (with propagation constant $+\beta$) and backward-propagating modes (with $-\beta$). The dispersion relation $\omega(\beta)$ is symmetric: $\omega(\beta) = \omega(-\beta)$.

This symmetry is a direct consequence of *time-reversal invariance*. Maxwell's equations in a lossless, non-magnetic dielectric are invariant under the operation $t \rightarrow -t$, which sends $\mathbf{E} \rightarrow \mathbf{E}$, $\mathbf{H} \rightarrow -\mathbf{H}$, and $\beta \rightarrow -\beta$. If a forward mode at $(\omega, +\beta)$ exists, then time reversal produces a backward mode at $(\omega, -\beta)$.

The physical consequence is immediate: any perturbation that couples the forward and backward channels—a surface roughness, a point defect, a bend whose radius of curvature is comparable to the

wavelength—can scatter light from $+\beta$ to $-\beta$. No matter how carefully the waveguide is fabricated, backscattering is always *allowed* by symmetry. We can reduce it; we cannot forbid it.

1.2.1 A quantitative reminder

To make this concrete, recall the coupled-mode estimate for backscattering loss in a single-mode waveguide with sidewall roughness. If the roughness has root-mean-square amplitude σ and correlation length L_c , the power attenuation coefficient due to backscattering scales as

$$\alpha_{\text{back}} \propto \frac{\sigma^2}{\lambda^2} \frac{(\Delta n)^2}{n_{\text{eff}}} \frac{L_c}{1 + (2\beta L_c)^2}, \quad (1.1)$$

where $\Delta n = n_1 - n_2$ is the index contrast and $n_{\text{eff}} = \beta c / \omega$ is the effective index. The essential point is that $\alpha_{\text{back}} > 0$ for any nonzero σ : the backward channel is always open.

In silicon photonic waveguides at telecom wavelengths, typical propagation losses from scattering are of order 1–3 dB cm^{−1} even in state-of-the-art fabrication. This is tolerable for short on-chip interconnects, but it sets a fundamental limit on the quality factor of ring resonators, the coherence length of delay lines, and the dynamic range of interferometric sensors.

1.3 What would it take?

The existence of a backward channel is guaranteed by time-reversal symmetry. To eliminate it, we must either break time-reversal symmetry or find a more subtle structural mechanism that effectively decouples the two propagation directions.

Breaking time reversal. In electromagnetism, time-reversal symmetry is broken by magneto-optical (Faraday) media: materials whose permittivity or permeability tensor acquires antisymmetric off-diagonal components when an external magnetic bias is applied. A familiar device exploiting this effect is the Faraday isolator, which rotates the polarisation of a transmitted beam by 45° in a fixed sense regardless of propagation direction. Faraday isolators are standard components in laser systems, but they are bulky discrete elements, not integrated waveguiding structures.

The question, then, is whether one can embed the Faraday effect into the *band structure* of a periodic photonic medium—a photonic crystal—in such a way that one-way propagation becomes a property of the entire frequency band rather than a discrete device.

The electronic precedent. This question has a remarkable precedent in condensed-matter physics. In 1980, Klaus von Klitzing discovered that the Hall conductance of a two-dimensional electron gas in a strong magnetic field is quantised to integer multiples of e^2/h , with extraordinary precision [1]. Two years later, Thouless, Kohmoto, Nightingale, and den Nijs showed that the quantisation is a consequence of a topological invariant—the first Chern number—of the electronic Bloch bands [2]. The chiral edge states responsible for the quantised Hall current propagate in only one direction along the sample boundary, and they are immune to backscattering from impurities precisely because there is no backward-propagating channel for them to scatter into.

The existence of these one-way edge states is not an accident of the particular Hamiltonian; it is a topological consequence of the nonzero Chern number of the filled bands. As long as the bulk band gap remains open, the edge states persist regardless of the details of the boundary or the disorder.

From electrons to photons. In a landmark 2005 paper (published in 2008), Haldane and Raghu recognised that the one-particle band-structure topology underlying the integer quantum Hall effect has nothing essentially quantum-mechanical about it [3]. It is a property of waves in periodic media with broken time-reversal symmetry. Since the Maxwell normal-mode problem in a lossless periodic medium is a generalised Hermitian eigenvalue problem—structurally analogous to the electronic Schrödinger equation, though with a different inner product—the same topological classification applies.

Haldane and Raghu considered a two-dimensional photonic crystal made of dielectric rods with a magneto-optical (gyrotropic) response. They showed that:

- (i) The photonic band structure of a hexagonal lattice of such rods generically exhibits Dirac-point degeneracies, analogous to those in graphene.
- (ii) When the gyrotropic effect lifts these degeneracies, the resulting bands acquire nonzero Chern numbers.
- (iii) At a domain wall where the sign of the gyrotropic parameter reverses, a unidirectional mode is bound to the interface, crossing the bulk band gap with a group velocity whose sign is fixed by the Chern-number difference across the wall.

This is the photonic analog of the quantum Hall edge state: a one-way waveguide, immune to backscattering, whose existence is guaranteed by the topology of the bulk bands.

1.4 The first experimental observation

The theoretical prediction of Haldane and Raghu was brought to experimental reality by Wang, Chong, Joannopoulos, and Soljačić, first in simulation and proposal (2008) and then in a direct measurement (2009) [4]. They constructed a two-dimensional photonic crystal of yttrium iron garnet (YIG) rods in a square lattice, operating in the microwave regime around 4.5 GHz, and applied a static magnetic field of approximately 0.2 T to magnetise the ferrite.

The key experimental observations were:

- Electromagnetic energy launched into an edge channel propagated in only one direction around the boundary of the crystal.
- When a large metallic obstacle was placed in the path of the edge mode, the wave simply routed around the obstacle without any measurable reflection. The forward transmission through eight lattice periods was more than five orders of magnitude larger than the backward signal.
- The one-way propagation persisted over the entire frequency range of the bulk band gap, not just at isolated frequencies.

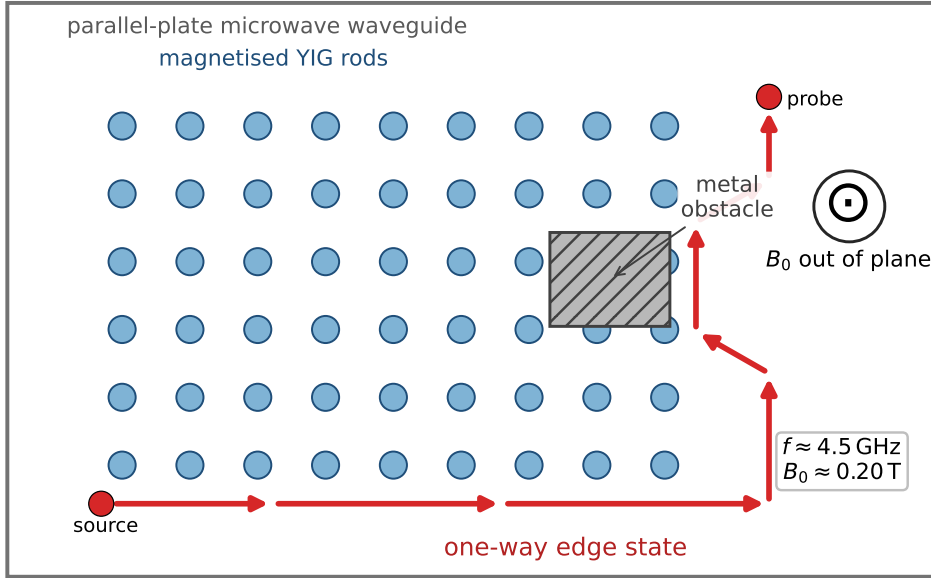


Figure 1.1: Schematic of the Wang et al. experiment [4]. A square lattice of magnetised YIG rods operates near 4.5 GHz under an external bias field $B_0 \approx 0.20$ T that breaks time-reversal symmetry. The resulting gyrotropic mass, and hence the edge-state chirality, is set by the direction of B_0 ; the red path traces the one-way edge state routing around a large metallic obstacle without backscattering.

These results demonstrated that the topological protection predicted from band theory is a real, measurable electromagnetic phenomenon, not merely a mathematical curiosity.

The geometry of the Wang experiment is shown schematically in Figure 1.1. A square array of YIG rods sits inside a metallic parallel-plate waveguide; an external static magnetic field $B_0 \approx 0.20$ T magnetises the ferrite and opens a topological gap near 4.5 GHz. Electromagnetic energy launched along one edge of the crystal propagates in a single direction, routing around large metallic obstacles without any measurable backscattering.

1.5 Why topology, and why from Maxwell?

The preceding story can be told—and often is told—entirely in the language of condensed-matter physics: Bloch Hamiltonians, Berry phases, Chern numbers, bulk-boundary correspondence. There is nothing wrong with this; the mathematical structures are the same. But for a book addressed to the photonics community, and for the deepest understanding of the physical mechanisms, we believe it is essential to derive every topological statement directly from Maxwell’s equations.

There are several reasons for this insistence.

The inner product is different. In quantum mechanics, the Hilbert-space inner product is the standard L^2 integral: $\langle \psi | \phi \rangle = \int \psi^* \phi \, d^3r$. In electromagnetism, the physically correct inner product involves the material response. For a lossless nondispersive medium characterised by the 6×6 material matrix $\mathbf{M}$, the electromagnetic energy inner product is

$$\langle \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}} = \int \mathbf{f}_1^\dagger(\mathbf{r}) \cdot \mathbf{M}(\mathbf{r}) \cdot \mathbf{f}_2(\mathbf{r}) \, d^3r, \quad (1.2)$$

where $\mathbf{f} = (\mathbf{E}, \mathbf{H})^T$ is the six-component electromagnetic field. Every formula for Berry connection, Berry curvature, and Chern number must be written with respect to this inner product, not the standard one. Getting this wrong leads to non-gauge-invariant expressions and incorrect topological invariants.

When the medium is dispersive—as all real magneto-optical media are—the inner product must be further modified to incorporate the electromagnetic energy stored in the material polarisation, leading to the *energy metric* $\mathbf{M}_g(\omega) = \partial_\omega[\omega \mathbf{M}(\omega)]$. This subtlety has no analog in the electronic problem and is easily missed when photonic topology is derived by analogy rather than from first principles.

The eigenvalue problem is generalised. The Maxwell normal-mode equation is not a standard eigenvalue problem $\hat{H}\psi = E\psi$; it is a generalised eigenproblem $\hat{\mathcal{N}}\mathbf{f} = \omega\mathbf{M}\mathbf{f}$, with the frequency playing the role of the eigenvalue and the material matrix appearing on the right-hand side. The Hermiticity (self-adjointness) that guarantees real frequencies and orthogonal modes follows from the structure of the Maxwell curl operator and the positivity of $\mathbf{M}$, but the proof is different from the quantum-mechanical case and deserves to be presented in full.

Photon time reversal is bosonic. For electrons, the time-reversal operator satisfies $\hat{\mathcal{T}}^2 = -1$, which implies Kramers degeneracy and provides the symmetry protection for the quantum spin Hall effect [5]. For photons, $\hat{\mathcal{T}}^2 = +1$. This seemingly small algebraic difference has profound consequences: there is no photonic Kramers theorem, and time-reversal-invariant photonic “topological insulators” require additional crystalline or internal symmetries for their protection [6]. A Maxwell-first development makes this distinction transparent from the outset.

Material constraints matter. Which materials can support topological bands? What symmetries must be broken, and how strong must the magneto-optical or bianisotropic response be? These questions can only be answered by examining the constitutive relations directly. A Maxwell-first approach keeps the material physics visible at every stage, rather than hiding it behind an effective Hamiltonian.

1.6 The plan of this book

This monograph develops the theory of topological photonics from the ground up, starting from Maxwell’s equations and optical energy considerations, and building toward the full classification of topological phases accessible to electromagnetic waves in periodic media [7].

Part I: Maxwell modes and optical bands. We begin by recasting the macroscopic Maxwell equations as a first-order dynamical system (Chapter 2), identifying the six-component electromagnetic field and the material matrix as the fundamental objects. We then derive the normal-mode problem and the electromagnetic inner product that makes it self-adjoint (Chapter 3), introduce Bloch’s theorem and photonic band structure (Chapter 4), and develop the Berry connection, Berry curvature, and Chern number for Maxwell bands (Chapter 5). Every formula in this part is derived from Maxwell’s equations using the correct electromagnetic inner product.

Part II: Chern bands and one-way waveguides. With the mathematical framework in place, we study the physical mechanisms that create topological bands. We analyse Dirac-point degeneracies (Chapter 6), show how gyrotropic media open topological gaps (Chapter 7), derive the one-way interface mode at a domain wall in detail (Chapter 8), and discuss the microwave photonic-crystal experiment that first confirmed the prediction (Chapter 9). We then address the practical problem of computing Chern numbers from numerical Maxwell eigensolvers (Chapter 10).

Part III: Other two-dimensional phases. Not all photonic topological phases require broken time-reversal symmetry. We discuss reciprocal designs based on pseudospin and bianisotropy (Chapter 11), valley-Hall photonic crystals (Chapter 12), Floquet topological systems created by temporal modulation (Chapter 13), and one-dimensional topological models including the photonic Su–Schrieffer–Heeger chain (Chapter 14) [8].

Part IV: Three dimensions and beyond. We extend the framework to three-dimensional photonic crystals with Weyl points (Chapter 15) and to synthetic dimensions created by coupling internal degrees of freedom of resonators (Chapter 16) [9].

Part V: Imperfect, active, and nonlinear light. Real photonic systems have loss, gain, and nonlinearity [10]. We discuss non-Hermitian topology (Chapter 17) and the emerging frontier of nonlinear and quantum topological photonics (Chapter 18) [11].

Appendices provide supporting material on vector identities, differential forms, numerical methods, and material-tensor conventions.

Throughout the book, we follow a consistent strategy: derive the key result from Maxwell’s equations first, then connect it to the condensed-matter analogy for readers with a solid-state background, and finally illustrate it with a concrete photonic example or experiment. The goal is a self-contained reference that a graduate student in photonics can read from the beginning, and that a researcher in topological condensed matter can consult for the electromagnetic specifics.

1.7 Photonic versus electronic topology

The topological ideas applied in this book were born in the study of electron systems, but the photonic setting differs from the electronic one in several fundamental ways. Understanding these dif-

ferences is essential for appreciating both the power and the limitations of topological photonics.

1.7.1 Bosonic statistics and the absence of a Fermi level

Electrons are fermions: the Pauli exclusion principle guarantees a Fermi level that separates filled from empty states. The topological properties of the Chern insulator rely on this filling: the Chern number characterises the *occupied* bands below the Fermi level.

Photons are bosons: in thermal equilibrium every frequency is populated, and there is no analog of the Fermi level. In photonic topological systems, the role of the Fermi level is played by the *operating frequency*, which is set by the external source. The “gap” in the photonic band structure is a frequency range with no propagating bulk modes, and the “edge mode” is a propagating state at a frequency within this gap. The topological protection still applies—the edge mode cannot be removed without closing the gap—but the physics of filling and thermalisation is entirely different [12].

1.7.2 Classical waves and no fundamental spin

Photons have spin 1, but the spin of a photon in a medium is not a conserved quantum number in the same way as the electron spin in a crystal with spin-orbit coupling. The photonic “pseudospin” used in Chapter 11 is an emergent degree of freedom constructed from the TE and TM polarisations, not a fundamental symmetry. Its protection depends on crystalline symmetry, not on time-reversal symmetry of the fundamental Hamiltonian.

Moreover, the Maxwell equations are a classical wave equation. All topological effects in this book can be observed with classical light (coherent laser beams, microwave sources) without any quantum signature. The quantisation of edge-mode transport is a property of the wave equation, not of the quantum statistics of the field [3, 13].

1.7.3 Time-reversal symmetry and Kramers degeneracy

For electrons, time reversal is an anti-unitary operator with $\hat{\mathcal{T}}^2 = -1$ (for half-integer spin), which implies Kramers degeneracy: every energy level is at least doubly degenerate. This degeneracy is the foundation of the $\mathbb{Z}_2$ topological insulator.

For photons, $\hat{\mathcal{T}}^2 = +1$ (because the photon spin is integer), so there is no Kramers degeneracy. A direct photonic analog of the $\mathbb{Z}_2$ topological insulator requires an engineered pseudospin with effective $\hat{\mathcal{T}}^2 = -1$, which is possible only with specific material symmetries (such as electromagnetic duality symmetry in bianisotropic media). This is why photonic topological insulators are more constrained than their electronic counterparts, and why much of the field has focused on Chern insulators (broken $\hat{\mathcal{T}}$) or crystalline-symmetry-protected phases [12, 14].

1.8 Historical milestones

The field of topological photonics grew from two independent streams. The first was the discovery of the integer quantum Hall effect by von Klitzing (1980) and its topological explanation by Thouless,

Kohmoto, Nightingale, and den Nijs (1982), who showed that the Hall conductance is proportional to an integer topological invariant—the Chern number—of the electronic band structure [2].

The second stream was the development of photonic crystals by Yablonovitch and John (1987), who showed that periodic dielectric structures can have complete photonic band gaps. The connection between these two ideas was made by Haldane and Raghu (2008), who demonstrated theoretically that a photonic crystal with magneto-optical response can have bands with nonzero Chern numbers, and therefore must support one-way edge modes [3].

The experimental confirmation came in 2009, when Wang et al. observed unidirectional microwave transport in a YIG photonic crystal under an external magnetic field [4]. This experiment launched the field. Since then, topological photonic effects have been demonstrated across the electromagnetic spectrum, from microwave to optical frequencies, using platforms ranging from coupled resonators [15] to femtosecond-laser-written waveguides [16] to silicon photonic crystals [17, 18].

The roadmaps by Ozawa et al. (2019) and Price et al. (2022) provide comprehensive surveys of the field’s development and open questions [12, 13].

1.9 The Maxwell-first philosophy

This book is organised differently from most accounts of topological photonics. The standard approach begins with the quantum Hall effect for electrons, derives the Chern number in a tight-binding model, and then maps the result onto a photonic system by analogy. This book instead starts from Maxwell’s equations and derives topology entirely within the electromagnetic framework: the energy inner product, the self-adjointness of the curl operator, the Berry connection on Bloch bands, and the Chern number as an integral of the Berry curvature.

This choice is not merely pedagogical. The electromagnetic problem has features that differ qualitatively from the electronic one: the material matrix $\mathbf{M}$ is frequency-dependent and anisotropic, the field is a six-component vector rather than a scalar, the energy inner product involves $\mathbf{M}$ rather than the identity, and the discrete symmetries have different squaring properties ($\hat{\mathcal{T}}^2 = +1$ rather than -1). An analogy-based approach risks importing results that depend on features specific to the electronic problem, such as Kramers degeneracy or scalar wave functions. The Maxwell-first approach avoids this risk entirely and reveals features—such as the role of the dispersive energy metric—that have no direct electronic analogue [19, 20].

1.10 Open frontiers

The field of topological photonics continues to grow rapidly. Several frontier directions are addressed in the later parts of this book:

Non-Hermitian topology (Chapter 17) explores what happens when gain and loss are introduced. The non-Hermitian skin effect, exceptional-point degeneracies, and the breakdown of the conventional bulk-edge correspondence raise fundamental questions about the meaning of topological protection in open systems [12].

Nonlinear and quantum topology (Chapter 18) asks how topological protection survives when the superposition principle breaks down. Solitons on topological edge modes, photon blockade in

topological cavities, and bosonic fractional quantum Hall states of light are active areas of theoretical and experimental investigation [13].

Throughout the book, we will see that the Maxwell-first framework provides a unified language for all these directions: the same eigenvalue equation, the same energy inner product, and the same Berry geometry underlie the physics from microwave Chern insulators to optical valley crystals to non-Hermitian topological lasers.

1.11 Historical milestones beyond photonics

The topological ideas that underpin this book did not originate in photonics. They arose from a remarkable chain of discoveries in condensed matter physics, mathematics, and quantum field theory, spanning more than four decades.

The story begins with the integer quantum Hall effect, discovered by von Klitzing, Dorda, and Pepper in 1980, where the Hall conductance of a 2D electron gas in a strong magnetic field is quantised in units of e^2/h with extraordinary precision. Thouless, Kohmoto, Nightingale, and den Nijs (1982) showed that this quantisation is a consequence of the Chern number of the Bloch bands—a topological invariant that cannot change under smooth deformations of the Hamiltonian [2].

The extension to time-reversal-invariant systems came with the quantum spin-Hall effect, predicted by Kane and Mele (2005) and independently by Bernevig, Hughes, and Zhang (2006), and experimentally confirmed in HgTe/CdTe quantum wells by König et al. (2007). These systems are characterised by a $\mathbb{Z}_2$ topological invariant rather than the Chern number, and they support helical edge modes where spin-up and spin-down electrons propagate in opposite directions [12, 21].

A parallel development in non-Hermitian physics began with the observation by Bender and Boettcher (1998) that certain non-Hermitian Hamiltonians with parity-time (PT) symmetry can have entirely real spectra. This insight, originally formulated in quantum mechanics, found its most natural experimental home in photonics, where gain and loss can be engineered with great precision [22, 23].

The convergence of these three threads—topological band theory, non-Hermitian physics, and photonic engineering—created the field of topological photonics that is the subject of this book.

1.12 The scope and structure of this book

This book develops topological photonics from Maxwell’s equations. Every topological invariant, every edge-mode derivation, and every bulk-boundary correspondence is built from the electromagnetic eigenvalue problem, the energy inner product, and the Bloch decomposition—not imported from electronic band theory by analogy.

The book is organised in five parts. Part I (Chapters 1–5) establishes the mathematical framework: Maxwell’s equations as a generalised eigenvalue problem, the electromagnetic inner product and its role in defining orthogonality and completeness, Bloch’s theorem for periodic media, and the Berry connection, Berry curvature, and Chern number as geometric invariants of photonic bands.

Part II (Chapters 6–10) applies this framework to construct the first photonic topological insulator: the gyromagnetic Chern insulator. The key steps are the identification of Dirac cones, the

opening of a topological gap by time-reversal-symmetry breaking, the derivation of one-way edge modes from the bulk-edge correspondence, and the numerical computation of Chern numbers.

Part III (Chapters 11–16) extends the topological framework to systems that preserve time-reversal symmetry: pseudospin designs, valley photonic crystals, Floquet topological insulators, one-dimensional chains, Weyl semimetals, and synthetic dimensions.

Part IV (Chapters 17–18) treats the frontiers: non-Hermitian topology, topological lasers, non-linear effects, and quantum topological photonics.

The four appendices provide the mathematical tools: vector identities and functional analysis, differential forms and Chern classes, numerical algorithms, and material tensor conventions.

The reader is assumed to be familiar with classical electrodynamics at the level of Jackson or Griffiths, and with quantum mechanics at the level of Sakurai. No prior knowledge of topology or differential geometry is required: these subjects are developed from scratch as needed [12].

1.13 Why photonic topology differs from electronic topology

Why study topology in photonics rather than simply importing results from electronic band theory? The answer touches on both fundamental physics and practical engineering.

1.13.1 Bosons versus fermions

Electrons are fermions with half-integer spin, obeying the Pauli exclusion principle and filling bands up to the Fermi energy. Photons are bosons with integer spin, obeying Bose–Einstein statistics and occupying any number of modes simultaneously. This distinction has deep consequences for topology:

In electronic systems, the Fermi energy naturally defines which bands are “occupied” and which are “empty,” providing a sharp boundary for computing topological invariants. In photonics, there is no Fermi energy; all bands below a gap can be excited. The photonic Chern number must therefore be defined as a sum over all bands below the gap, weighted by the electromagnetic energy inner product (Chapter 3)—a structure that has no direct electronic analog [19, 20].

Furthermore, the spin-statistics theorem guarantees that electronic topological insulators with time-reversal symmetry are classified by $\mathbb{Z}_2$ invariants, protected by Kramers degeneracy. Photons do not have Kramers degeneracy (they are spin-1 bosons), so the $\mathbb{Z}_2$ classification requires an additional symmetry—either electromagnetic duality or crystalline symmetry—to protect the pseudospin structure [14, 24].

1.13.2 Maxwell versus Schrödinger

The Schrödinger equation is a scalar first-order equation in time, while Maxwell’s equations form a vector first-order system. The vectorial nature of light means that the Berry connection is a matrix connection on a vector bundle, not merely a $U(1)$ connection on a line bundle. This enriches the topology: photonic bands can carry non-Abelian Berry phases and support topological invariants (such as the Euler class) that have no scalar analog [12, 13].

The first-order structure of Maxwell’s equations also means that the “Hamiltonian” $\hat{\mathcal{H}}$ is not bounded below: unlike the Schrödinger Hamiltonian, the Maxwell operator has both positive and negative frequency eigenvalues. The positive-definiteness of the material tensor $\mathbf{M}$ plays the role of the positive-definite mass in quantum mechanics, ensuring that the electromagnetic inner product is positive-definite and that the Berry connection is well-defined (Chapter 2).

1.14 Why the Maxwell-first route matters

It is tempting to introduce topological photonics by first describing electronic topological insulators and then replacing an electron wave function by an electromagnetic field. That route is historically understandable, because many of the words—Berry curvature, Chern number, band gap, edge state—entered photonics from condensed-matter physics. But it hides the essential difference between the two problems. An electron band in a crystal is an eigenproblem for a Schrödinger operator acting on a scalar or spinor wave function. An optical band is an eigenproblem for Maxwell’s equations acting on a six-component field $(\mathbf{E}, \mathbf{H})$, constrained by divergence conditions and weighted by the electromagnetic energy stored in a material medium.

The distinction is not cosmetic. The inner product that makes a lossless Maxwell operator self-adjoint is not the ordinary L^2 product. In a nondispersive medium it is

$$\langle \Psi_1, \Psi_2 \rangle_{\mathbf{M}} = \int (\mathbf{E}_1^* \cdot \varepsilon \mathbf{E}_2 + \mathbf{H}_1^* \cdot \mu \mathbf{H}_2) d^3r, \quad \Psi = (\mathbf{E}, \mathbf{H})^T, \quad (1.3)$$

and in a dispersive medium it is replaced by the Brillouin energy metric involving $\partial_\omega(\omega \mathbf{M})$. Berry phases and Chern numbers of photonic bands are therefore geometric phases of *energy-normalised Maxwell modes*, not phases of arbitrary field amplitudes. This is the reason Chapters 2 through 5 build the subject from Maxwell’s equations rather than importing the electronic formalism at the start [3, 19, 20].

The Maxwell-first route also prevents a common misunderstanding about “spin” in photonics. Photons have helicity, polarisation, and orbital angular momentum, but a reciprocal dielectric crystal does not possess the Kramers degeneracy of a spin-1/2 electron unless an additional pseudo-spin structure is engineered. Bianisotropic metamaterials, coupled resonator arrays, and valley crystals all construct such structures in different ways [15, 25, 26]. A monograph that begins with electronic spin-orbit coupling would make these photonic designs seem like analogues. A Maxwell-first monograph shows instead what they really are: carefully designed degeneracies and symmetries of the electromagnetic normal-mode problem.

1.15 The organising question of the book

The central question of this book can be stated in one sentence: *when can Maxwell’s equations force light to move along a boundary without backscattering?* The answer has several layers.

At the most local level, a boundary mode is an evanescent solution of Maxwell’s equations whose field decays away from an interface. At the band-theoretic level, it appears in a frequency gap separating two bulk continua. At the topological level, it is required when the two bulk media on the two sides of the interface cannot be continuously deformed into each other without closing the

gap. This final statement is the bulk-boundary principle. It is simple to say, but for photons it is not a postulate: it has to be tied to the Maxwell operator, the energy inner product, material dispersion, and electromagnetic boundary conditions.

The first experimental demonstrations made this question concrete. The microwave photonic crystal experiment of Wang et al. showed a chiral edge channel in a gyromagnetic crystal biased by a static magnetic field [27]. The helical waveguide experiment of Rechtsman et al. showed that longitudinal modulation can create an effective Floquet gauge field for light [28]. The coupled-resonator experiments showed that synthetic magnetic flux can be engineered on a chip [15]. These platforms look very different in the laboratory, but all answer the same Maxwell question: what does a material or modulation do to the geometry of the normal modes?

The chapters that follow return to this question repeatedly. The answer is first a self-adjoint Maxwell eigenproblem, then a Bloch bundle, then a Berry curvature, then an integer Chern number, then an edge mode. The later chapters complicate the setting—reciprocity, valleys, Floquet driving, three dimensions, synthetic dimensions, gain and loss, nonlinearity, and quantum light—but the path remains the same: start from the fields and ask what structure Maxwell's equations allow.

Exercises

Exercise 1.1. A single-mode silicon-on-insulator waveguide ($n_1 = 3.48$, $n_2 = 1.44$, core width 450 nm, operating wavelength 1550 nm) has sidewall roughness with $\sigma = 2$ nm and $L_c = 50$ nm. Using the scaling in Eq. (1.1), estimate the order of magnitude of the backscattering-limited propagation length and compare it to a typical ring-resonator circumference of 100 μm .

Exercise 1.2. In a Faraday isolator, the polarisation rotation angle in a medium of length L with Verdet constant V in a magnetic field B is $\theta_F = VBL$. For a bismuth iron garnet film at $\lambda = 1550$ nm with $V \approx 10^3 \text{ deg cm}^{-1} \text{ T}^{-1}$, estimate the film thickness and field needed for a 45° rotation. Why is this approach difficult to miniaturise to on-chip waveguide scales?

Exercise 1.3. Time reversal in electromagnetism sends $(\mathbf{E}, \mathbf{H}) \rightarrow (\mathbf{E}, -\mathbf{H})$. Show that this maps a plane-wave mode $\sim e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$ to one with $\mathbf{k} \rightarrow -\mathbf{k}$ and the same frequency, and that the dispersion relation of a lossless reciprocal waveguide therefore satisfies $\omega(\mathbf{k}) = \omega(-\mathbf{k})$.

Chapter 2

Maxwell Equations as Dynamics

2.1 The macroscopic Maxwell equations

We work throughout this book with the *macroscopic* Maxwell equations, in which the response of bound charges and currents is encoded in the constitutive fields $\mathbf{D}$ and $\mathbf{B}$ [29]. In the absence of free charges and currents, these equations read

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t}, \quad (2.1a)$$

$$\nabla \times \mathbf{H}(\mathbf{r}, t) = \frac{\partial \mathbf{D}(\mathbf{r}, t)}{\partial t}, \quad (2.1b)$$

$$\nabla \cdot \mathbf{D}(\mathbf{r}, t) = 0, \quad (2.1c)$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0. \quad (2.1d)$$

Here $\mathbf{E}$ and $\mathbf{H}$ are the electric and magnetic field intensities, while $\mathbf{D}$ and $\mathbf{B}$ are the electric displacement and magnetic flux density. SI units are used throughout.

Remark 2.1. Equations (2.1c) and (2.1d) are not independent of the curl equations. Taking the divergence of (2.1a) gives $\partial_t(\nabla \cdot \mathbf{B}) = 0$, so if $\nabla \cdot \mathbf{B} = 0$ is satisfied at one time it holds for all time; similarly for $\nabla \cdot \mathbf{D}$. The divergence conditions are therefore constraints on the initial data, not additional dynamical equations. We will return to this point in Chapter 3 when we discuss the completeness of the normal-mode expansion.

2.2 Constitutive relations

The Maxwell equations become a closed dynamical system only when supplemented by *constitutive relations* that express $\mathbf{D}$ and $\mathbf{B}$ in terms of $\mathbf{E}$ and $\mathbf{H}$ [25]. We consider linear media and work in the frequency domain, writing time-harmonic fields as $\mathbf{E}(\mathbf{r}, t) = \text{Re}[\mathbf{E}(\mathbf{r})e^{-i\omega t}]$ (and similarly for $\mathbf{H}$, $\mathbf{D}$, $\mathbf{B}$).

2.2.1 Isotropic media

The simplest case is a local, isotropic, nondispersive dielectric:

$$\mathbf{D} = \varepsilon_0 \varepsilon(\mathbf{r}) \mathbf{E}, \quad \mathbf{B} = \mu_0 \mu(\mathbf{r}) \mathbf{H}, \quad (2.2)$$

where $\varepsilon(\mathbf{r})$ and $\mu(\mathbf{r})$ are the relative permittivity and permeability. Most photonic-crystal literature begins here, with $\mu = 1$ (nonmagnetic media) and $\varepsilon(\mathbf{r})$ varying periodically. For topological photonics, however, we will need to go beyond this simplest setting.

2.2.2 Anisotropic media

In an anisotropic crystal, the permittivity and permeability become 3×3 tensors:

$$D_i = \varepsilon_0 \varepsilon_{ij}(\mathbf{r}) E_j, \quad B_i = \mu_0 \mu_{ij}(\mathbf{r}) H_j, \quad (2.3)$$

with summation over repeated indices. The tensors ε and μ are Hermitian ($\varepsilon_{ij} = \varepsilon_{ji}^*$, $\mu_{ij} = \mu_{ji}^*$) for lossless media at real frequencies.

A particularly important special case is the *gyrotropic* medium, in which an applied magnetic bias $\mathbf{B}_0 = B_0 \hat{z}$ induces off-diagonal imaginary components. For a ferrite magnetised along z , the permeability tensor takes the form

$$\mu = \begin{pmatrix} \mu_{\perp} & -i\kappa & 0 \\ i\kappa & \mu_{\perp} & 0 \\ 0 & 0 & \mu_z \end{pmatrix}, \quad (2.4)$$

where $\mu_{\perp}$, κ , and μ_z are real parameters that depend on frequency and the magnetisation. The antisymmetric off-diagonal element κ is the *gyrotropic parameter*; it vanishes when $B_0 = 0$ and breaks time-reversal symmetry when $\kappa \neq 0$.

Remark 2.2. The Hermiticity of μ is manifest: $\mu_{12} = -i\kappa$ and $\mu_{21} = i\kappa = \mu_{12}^*$ for real κ . The eigenvalues of μ are $\mu_{\perp} \pm \kappa$ and μ_z ; positivity of the energy requires $\mu_{\perp} > |\kappa| > 0$ and $\mu_z > 0$.

2.2.3 The bianisotropic material matrix

The most general linear local constitutive relation couples all four fields:

$$\begin{pmatrix} \mathbf{D} \\ \mathbf{B} \end{pmatrix} = \begin{pmatrix} \varepsilon_0 \varepsilon & c^{-1} \xi \\ c^{-1} \zeta & \mu_0 \mu \end{pmatrix} \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix} \equiv \mathbf{M}(\mathbf{r}, \omega) \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix}. \quad (2.5)$$

Here ξ and ζ are the magneto-electric coupling tensors, and $\mathbf{M}$ is the 6×6 *material matrix*.

Definition 2.3 (Lossless medium). A linear medium is *lossless at frequency ω* if and only if the material matrix is Hermitian:

$$\mathbf{M}(\mathbf{r}, \omega) = \mathbf{M}^\dagger(\mathbf{r}, \omega). \quad (2.6)$$

For the individual blocks, this requires $\varepsilon = \varepsilon^\dagger$, $\mu = \mu^\dagger$, and $\zeta = \xi^\dagger$.

We will see in Section 2.5 that condition (2.6) is precisely the requirement that the time-averaged electromagnetic energy is conserved.

Definition 2.4 (Positive energy). A lossless medium has *positive electromagnetic energy* if $\mathbf{M}(\mathbf{r}, \omega)$ is positive definite for all $\mathbf{r}$ in the medium and for the frequencies of interest. This means $\mathbf{f}^\dagger \mathbf{M} \mathbf{f} > 0$ for every nonzero six-component field $\mathbf{f} = (\mathbf{E}, \mathbf{H})^T$.

The positive-energy condition is the electromagnetic analog of the positive-definite kinetic energy in mechanics. It will be essential for the Hermiticity of the normal-mode problem (Chapter 3).

2.3 The six-component first-order form

Following Haldane and Raghu, we now rewrite Maxwell's equations as a first-order eigenvalue problem in the frequency domain.

2.3.1 Harmonic fields and the curl matrix

For time-harmonic fields $\sim e^{-i\omega t}$, the curl equations (2.1a)–(2.1b) become

$$\nabla \times \mathbf{E} = i\omega \mathbf{B}, \quad (2.7a)$$

$$\nabla \times \mathbf{H} = -i\omega \mathbf{D}. \quad (2.7b)$$

(We adopt the engineering convention $e^{-i\omega t}$ with $\omega > 0$ for forward-propagating modes throughout.)

Define the six-component electromagnetic field

$$\mathbf{f}(\mathbf{r}) = \begin{pmatrix} \mathbf{E}(\mathbf{r}) \\ \mathbf{H}(\mathbf{r}) \end{pmatrix} \quad (2.8)$$

and the 6×6 Maxwell curl operator

$$\hat{\mathcal{N}} \equiv \begin{pmatrix} \mathbf{0} & i\nabla \times \\ -i\nabla \times & \mathbf{0} \end{pmatrix}. \quad (2.9)$$

Then the pair of curl equations collapses into the single matrix equation

$$\boxed{\hat{\mathcal{N}} \mathbf{f}(\mathbf{r}) = \omega \mathbf{M}(\mathbf{r}, \omega) \mathbf{f}(\mathbf{r})}. \quad (2.10)$$

This is the fundamental *Maxwell eigenvalue equation* in first-order form. It will serve as the starting point for normal modes (Chapter 3), Bloch bands (Chapter 4), and Berry geometry (Chapter 5).

Remark 2.5. The off-diagonal structure of $\hat{\mathcal{N}}$ in (2.9)—coupling $\mathbf{E}$ to $\mathbf{H}$ through the curl—is a reflection of the fact that electromagnetic waves are intrinsically coupled electric and magnetic phenomena. This structure is preserved under all linear constitutive extensions and is the origin of the spin-1 character of photons.

2.3.2 Momentum representation

For a spatially homogeneous medium, or for the cell-periodic part of a Bloch mode in a periodic medium, we can replace ∇ by $i\mathbf{k}$ to obtain the momentum-space curl operator

$$\hat{\mathcal{N}}(\mathbf{k}) = \begin{pmatrix} \mathbf{0} & -\mathbf{k} \times \\ \mathbf{k} \times & \mathbf{0} \end{pmatrix}, \quad (2.11)$$

where $(\mathbf{k} \times)_{ij} = -\epsilon_{ijk} k_k$ is the 3×3 antisymmetric matrix associated with the cross product. Explicitly, for $\mathbf{k} = (k_x, k_y, k_z)$:

$$\mathbf{k} \times = \begin{pmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{pmatrix}. \quad (2.12)$$

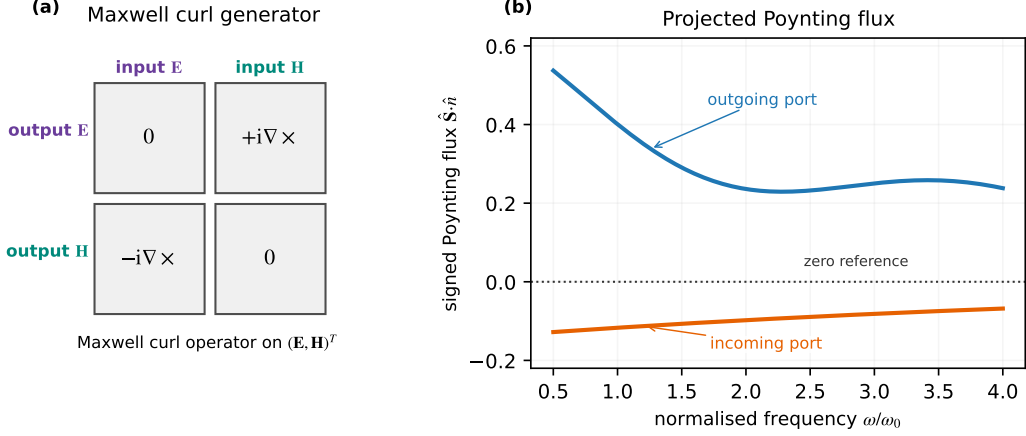


Figure 2.1: Structure of the Maxwell curl operator $\hat{\mathcal{N}}$ and energy flux. **(a)** Block structure of the time generator: the 6×6 operator $\hat{\mathcal{N}}$ defined in (2.9) has vanishing diagonal blocks (no direct $\mathbf{E}$ – $\mathbf{E}$ or $\mathbf{H}$ – $\mathbf{H}$ coupling) and antisymmetric off-diagonal curl blocks ($+i\nabla \times$ couples $\mathbf{H}$ into the $\mathbf{E}$ equation; $-i\nabla \times$ couples $\mathbf{E}$ into the $\mathbf{H}$ equation), consistent with the $e^{-i\omega t}$ time convention used throughout this book. **(b)** Signed normal Poynting flux $\hat{\mathbf{S}} \cdot \hat{\mathbf{n}}$ for a guided mode at a port boundary, illustrating energy flow into (incoming port) and out of (outgoing port) the computational domain. The zero reference line marks the interface between forward and backward flux.

It is straightforward to verify that $\hat{\mathcal{N}}(\mathbf{k})$ is Hermitian: $\hat{\mathcal{N}}(\mathbf{k}) = \hat{\mathcal{N}}^\dagger(\mathbf{k})$, since the cross-product matrix is real and antisymmetric, so $(\mathbf{k} \times)^T = -\mathbf{k} \times$, and the block off-diagonal structure with the signs in (2.11) makes the full 6×6 matrix symmetric.

Remark 2.6 (Spin-1 matrices). Haldane and Raghu write the curl operator as $A = -iJ^a \nabla_a$, where J^a are 6×6 matrices built from the spin-1 representation of $\text{SO}(3)$. In components, $(L^a)_{bc} = i\epsilon_{abc}$ are the 3×3 generators of rotations, and J^a is the block off-diagonal matrix $J^a = \begin{pmatrix} 0 & iL^a \\ -iL^a & 0 \end{pmatrix}$. This representation makes the rotational symmetry of the Maxwell operator manifest and connects directly to the spin-1 nature of the photon.

2.4 Electromagnetic energy

The energy carried by electromagnetic fields is central to everything that follows. The correct inner product for the Maxwell eigenvalue problem is built from the electromagnetic energy, so we derive the Poynting theorem in detail.

2.4.1 Poynting's theorem in the time domain

We start from the standard derivation. Take the dot product of (2.1a) with $\mathbf{H}$ and of (2.1b) with $\mathbf{E}$:

$$\mathbf{H} \cdot (\nabla \times \mathbf{E}) = -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t}, \quad (2.13)$$

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}. \quad (2.14)$$

Subtract (2.13) from (2.14) and use the vector identity $\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H})$ to obtain

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} = 0. \quad (2.15)$$

Define the Poynting vector and the instantaneous energy density:

$$\mathbf{S}(\mathbf{r}, t) = \mathbf{E} \times \mathbf{H}, \quad u(\mathbf{r}, t) = \frac{1}{2}(\mathbf{E} \cdot \mathbf{D} + \mathbf{H} \cdot \mathbf{B}). \quad (2.16)$$

For nondispersive media with $\mathbf{D} = \varepsilon_0 \varepsilon \mathbf{E}$ and [20] $\mathbf{B} = \mu_0 \mu \mathbf{H}$, the terms $\mathbf{E} \cdot \partial_t \mathbf{D}$ and $\mathbf{H} \cdot \partial_t \mathbf{B}$ equal $\partial_t u$ when ε and μ are time-independent.¹ Poynting's theorem then takes the conservation-law form

$$\boxed{\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = 0.} \quad (2.17)$$

Integrating over any volume V bounded by a surface ∂V :

$$\frac{d}{dt} \int_V u d^3r = - \oint_{\partial V} \mathbf{S} \cdot \hat{n} dA. \quad (2.18)$$

Energy stored in V changes only through the Poynting flux across the boundary. In a lossless medium, energy is neither created nor destroyed; the field energy is exactly accounted for by the flux.

2.4.2 Time-averaged energy for harmonic fields

For monochromatic fields $\mathbf{E}(\mathbf{r}, t) = \text{Re}[\mathbf{E}(\mathbf{r})e^{-i\omega t}]$, the time-averaged Poynting vector and energy density are

$$\langle \mathbf{S} \rangle = \frac{1}{2} \text{Re}(\mathbf{E} \times \mathbf{H}^*), \quad \langle u \rangle = \frac{1}{4} \text{Re}(\mathbf{E}^* \cdot \mathbf{D} + \mathbf{H}^* \cdot \mathbf{B}). \quad (2.19)$$

Using the bianisotropic constitutive relation (2.5), the time-averaged energy density can be written compactly in terms of the six-component field as

$$\langle u \rangle = \frac{1}{4} \mathbf{f}^\dagger(\mathbf{r}) \cdot \mathbf{M}(\mathbf{r}, \omega) \cdot \mathbf{f}(\mathbf{r}). \quad (2.20)$$

When $\mathbf{M}$ is Hermitian and positive definite, $\langle u \rangle > 0$ for any nonzero $\mathbf{f}$. This is the physical content of Definitions 2.3 and 2.4.

¹For dispersive media the identification $\mathbf{E} \cdot \partial_t \mathbf{D} = \partial_t (\frac{1}{2} \mathbf{E} \cdot \mathbf{D})$ does not hold in general; see Section 2.4.3.

2.4.3 Energy in dispersive media

All real materials are dispersive: the material matrix $\mathbf{M}(\mathbf{r}, \omega)$ depends on frequency. In a dispersive medium, the time-averaged energy density of a *quasi-monochromatic* wave packet (bandwidth $\Delta\omega \ll \omega$) is given by the Brillouin formula

$$\langle u \rangle_{\text{disp}} = \frac{1}{4} \mathbf{f}^\dagger(\mathbf{r}) \cdot \frac{\partial}{\partial \omega} [\omega \mathbf{M}(\mathbf{r}, \omega)] \cdot \mathbf{f}(\mathbf{r}) = \frac{1}{4} \mathbf{f}^\dagger \cdot \mathbf{M}_g(\omega) \cdot \mathbf{f}, \quad (2.21)$$

where we define the *energy metric*

$$\boxed{\mathbf{M}_g(\mathbf{r}, \omega) \equiv \frac{\partial}{\partial \omega} [\omega \mathbf{M}(\mathbf{r}, \omega)] = \mathbf{M} + \omega \frac{\partial \mathbf{M}}{\partial \omega}}. \quad (2.22)$$

Remark 2.7. In the nondispersive limit ($\partial \mathbf{M} / \partial \omega = 0$), the energy metric reduces to $\mathbf{M}_g = \mathbf{M}$, and (2.21) reduces to (2.20).

The requirement that the electromagnetic energy is positive ($\langle u \rangle_{\text{disp}} > 0$ for all nonzero $\mathbf{f}$) translates into the *stability condition*:

$$\mathbf{M}_g(\mathbf{r}, \omega) \text{ is positive definite} \quad (2.23)$$

at every physical eigenfrequency. This condition was emphasised by Haldane and Raghu as the correct generalisation of the positive-energy requirement for dispersive Maxwell systems, and it will play a crucial role in the inner product of Chapter 3.

Derivation of the Brillouin formula. The result (2.21) can be derived by considering a wave packet $\mathbf{E}(\mathbf{r}, t) = \text{Re}[\hat{\mathbf{E}}(\mathbf{r}) a(t) e^{-i\omega_0 t}]$ with slowly varying envelope $a(t)$ centred at frequency ω_0 . The constitutive relation in the time domain involves a convolution: $\mathbf{D}(\mathbf{r}, t) = \int_{-\infty}^t \epsilon_0 \epsilon(\mathbf{r}, t - t') \mathbf{E}(\mathbf{r}, t') dt'$. For a narrow-band envelope, one expands the frequency-domain susceptibility to first order around ω_0 :

$$\omega \epsilon_{ij}(\omega) \approx \omega_0 \epsilon_{ij}(\omega_0) + (\omega - \omega_0) \left. \frac{\partial}{\partial \omega} [\omega \epsilon_{ij}] \right|_{\omega_0}. \quad (2.24)$$

Substituting into the Poynting identity and time-averaging over the carrier oscillation yields (2.21) for the ϵ block; the μ , ξ , and ζ blocks follow identically, giving the full 6×6 result.

2.5 Poynting's theorem for harmonic fields

We now derive the frequency-domain version of Poynting's theorem and show that it encodes the losslessness condition directly.

Take the complex conjugate of (2.7a): $\nabla \times \mathbf{E}^* = -i\omega \mathbf{B}^*$. Form $\mathbf{H} \cdot (\nabla \times \mathbf{E}^*) - \mathbf{E}^* \cdot (\nabla \times \mathbf{H})$ and use the identity $\nabla \cdot (\mathbf{E}^* \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}^*) - \mathbf{E}^* \cdot (\nabla \times \mathbf{H})$:

$$\nabla \cdot (\mathbf{E}^* \times \mathbf{H}) = -i\omega (\mathbf{H} \cdot \mathbf{B}^* - \mathbf{E}^* \cdot \mathbf{D}). \quad (2.25)$$

Using the constitutive relation in six-component form, the right-hand side becomes

$$-i\omega \mathbf{f}^\dagger \begin{pmatrix} -\varepsilon_0 \varepsilon & -c^{-1} \xi \\ \mu_0 \mu^\dagger & c^{-1} \xi^\dagger \end{pmatrix} \mathbf{f}. \quad (2.26)$$

Taking the imaginary part:

$$\frac{1}{2} \nabla \cdot \text{Re}(\mathbf{E}^* \times \mathbf{H}) = -\frac{\omega}{2} \mathbf{f}^\dagger \cdot \frac{\mathbf{M} - \mathbf{M}^\dagger}{2i} \cdot \mathbf{f}. \quad (2.27)$$

The left-hand side is the divergence of the time-averaged Poynting vector. If $\mathbf{M} = \mathbf{M}^\dagger$, the right-hand side vanishes identically: the divergence of the Poynting flux is zero everywhere, confirming that no energy is absorbed or generated.

Proposition 2.8 (Losslessness from Hermiticity). *A linear medium is lossless (zero absorption and gain at frequency ω) if and only if $\mathbf{M}(\mathbf{r}, \omega) = \mathbf{M}^\dagger(\mathbf{r}, \omega)$.*

2.6 Symmetries of the Maxwell system

The topological classification of photonic bands depends critically on which discrete symmetries the Maxwell system possesses [12]. We now derive the three most important ones from the structure of $\mathbf{M}$ and $\hat{\mathcal{N}}$.

2.6.1 Time reversal

The microscopic Maxwell equations are invariant under time reversal $t \rightarrow -t$. In the macroscopic theory, time reversal acts on the fields as

$$\hat{\mathcal{T}} : \mathbf{E}(\mathbf{r}, t) \rightarrow \mathbf{E}(\mathbf{r}, -t), \quad \mathbf{H}(\mathbf{r}, t) \rightarrow -\mathbf{H}(\mathbf{r}, -t). \quad (2.28)$$

The sign difference arises because $\mathbf{E}$ is a polar vector (unchanged under $t \rightarrow -t$) while $\mathbf{H}$ is an axial vector proportional to a current ($\mathbf{J} \propto dq/dt$ changes sign).

For a time-harmonic field $\mathbf{f}e^{-i\omega t}$, the time-reversal operation becomes complex conjugation combined with a sign flip on the magnetic component:

$$\hat{\mathcal{T}} : \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix} \mapsto \begin{pmatrix} \mathbf{E}^* \\ -\mathbf{H}^* \end{pmatrix}, \quad \mathbf{k} \rightarrow -\mathbf{k}, \quad \omega \rightarrow \omega. \quad (2.29)$$

We can represent this as $\hat{\mathcal{T}} = \mathcal{K} \sigma_z$, where $\mathcal{K}$ is complex conjugation and $\sigma_z = \text{diag}(\mathbf{I}_3, -\mathbf{I}_3)$ acts on the electric/magnetic block structure.

Bosonic time reversal: $\hat{\mathcal{T}}^2 = +1$. Applying $\hat{\mathcal{T}}$ twice: $\hat{\mathcal{T}}^2 \mathbf{f} = \mathcal{K} \sigma_z \mathcal{K} \sigma_z \mathbf{f} = \sigma_z^2 \mathbf{f} = \mathbf{f}$. Thus $\hat{\mathcal{T}}^2 = +1$ for photons. This is fundamentally different from the electronic case, where $\hat{\mathcal{T}}^2 = -1$ due to half-integer spin.

Proposition 2.9 (Absence of photonic Kramers degeneracy). *Since $\hat{\mathcal{T}}^2 = +1$ for photons, there is no Kramers theorem for electromagnetic modes. Time-reversal symmetry alone does not guarantee twofold degeneracy at time-reversal-invariant momenta. Consequently, the $\mathbb{Z}_2$ topological classification of the quantum spin Hall effect does not directly apply to photonic systems.*

Condition on the material matrix. The Maxwell eigenvalue equation is time-reversal invariant if and only if, for every solution $\mathbf{f}$ at $(\omega, \mathbf{k})$, the time-reversed field $\hat{\mathcal{T}}\mathbf{f}$ is also a solution at $(\omega, -\mathbf{k})$. Substituting into (2.10) and using the explicit form of $\hat{\mathcal{T}}$, one finds that this requires

$$\sigma_z \mathbf{M}^*(\omega) \sigma_z = \mathbf{M}(\omega). \quad (2.30)$$

In block form, this reads

$$\varepsilon^* = \varepsilon, \quad \mu^* = \mu, \quad \zeta^* = -\zeta, \quad \zeta'^* = -\zeta'. \quad (2.31)$$

For the gyrotropic permeability of (2.4), condition (2.31) requires $\kappa = 0$. A nonzero gyrotropic parameter therefore *breaks time-reversal symmetry*.

2.6.2 Reciprocity

A medium is *reciprocal* if the Lorentz reciprocity relation holds: for any two solutions $\mathbf{f}_a$ and $\mathbf{f}_b$ of the Maxwell equations at the same frequency,

$$\int_V (\mathbf{E}_a \cdot \mathbf{J}_b - \mathbf{E}_b \cdot \mathbf{J}_a) d^3r = 0 \quad (2.32)$$

The displayed formula above follows the convention and normalisation used in [30]. In terms of the material matrix, reciprocity requires

$$\varepsilon = \varepsilon^T, \quad \mu = \mu^T, \quad \zeta = -\zeta^T. \quad (2.33)$$

Proposition 2.10 (Equivalence of TR symmetry and reciprocity for lossless media). *For a lossless medium ($\mathbf{M} = \mathbf{M}^\dagger$), the time-reversal condition (2.31) is equivalent to the reciprocity condition (2.33).*

Proof. For a Hermitian tensor, $\varepsilon = \varepsilon^\dagger$ means $\varepsilon_{ij} = \varepsilon_{ji}^*$. The TR condition $\varepsilon^* = \varepsilon$ means $\varepsilon_{ij}^* = \varepsilon_{ij}$, so ε_{ij} is real, which combined with Hermiticity gives $\varepsilon_{ij} = \varepsilon_{ji} = \varepsilon_{ij}^T$ —precisely reciprocity. The argument for μ is identical. For the coupling tensors: Hermiticity gives $\zeta = \zeta^\dagger$; the TR condition $\zeta^* = -\zeta$ means ζ is pure imaginary. Then $\zeta^\dagger = \zeta^{*T} = -\zeta^T$, so $\zeta = \zeta^\dagger = -\zeta^T$, which is the reciprocity condition. The converse follows by reversing the steps. $\square$

The physical content of this result is that, for lossless photonic systems, the route to nonzero Chern numbers and one-way edge states runs through *nonreciprocal* media—media that violate both TR symmetry and Lorentz reciprocity. The gyrotropic ferrites used in the first topological photonic-crystal experiments are the prototypical example.

2.6.3 Spatial inversion

The inversion (parity) operator $\hat{\mathcal{P}}$ acts as

$$\hat{\mathcal{P}} : \mathbf{E}(\mathbf{r}) \rightarrow -\mathbf{E}(-\mathbf{r}), \quad \mathbf{H}(\mathbf{r}) \rightarrow \mathbf{H}(-\mathbf{r}), \quad \mathbf{k} \rightarrow -\mathbf{k}. \quad (2.34)$$

The signs follow from the polar/axial character of the fields under $\mathbf{r} \rightarrow -\mathbf{r}$. In the six-component notation, $\hat{\mathcal{P}} = -\sigma_z \hat{P}_{\mathbf{r}}$, where $\hat{P}_{\mathbf{r}}$ sends $\mathbf{r} \rightarrow -\mathbf{r}$.

The Maxwell system is inversion-symmetric if $\mathbf{M}(-\mathbf{r}, \omega) = \sigma_z \mathbf{M}(\mathbf{r}, \omega) \sigma_z$. When both $\hat{\mathcal{T}}$ and $\hat{\mathcal{P}}$ are present, the combined symmetry $\hat{\mathcal{T}}\hat{\mathcal{P}}$ maps $(\omega, \mathbf{k})$ to $(\omega, \mathbf{k})$ and constrains the Berry curvature to vanish identically at every $\mathbf{k}$ point (see Chapter 5).

2.7 The road ahead

We have now assembled the raw materials: the Maxwell equations in first-order six-component form, the material matrix encoding all linear constitutive information, the Poynting theorem guaranteeing energy conservation, and the discrete symmetries that will control the topological classification.

What is still missing is the *mode structure*. The eigenvalue equation (2.10) determines a set of normal modes at each frequency and wavevector, but we have not yet shown that these modes are orthogonal, how they are normalised, or what inner product makes the problem self-adjoint. These are the questions of the next chapter.

2.8 Hamiltonian structure of the Maxwell system

The first-order Maxwell system (2.10) has a natural Hamiltonian structure that is worth making explicit, because it connects the photonic eigenvalue problem to the broader framework of symplectic dynamics and provides additional insight into why the Berry phase is a geometric quantity.

2.8.1 Symplectic form and Poisson bracket

Define the symplectic matrix $\mathbf{J} = \begin{pmatrix} 0 & -\mathbf{I}_3 \\ \mathbf{I}_3 & 0 \end{pmatrix}$ and the electromagnetic energy density $\mathcal{U} = \frac{1}{2} \mathbf{f}^\dagger \mathbf{M} \mathbf{f}$. Then the source-free Maxwell equations can be written as

$$\mathbf{M} \partial_t \mathbf{f} = \mathbf{J} \nabla \times \mathbf{f}, \quad (2.35)$$

which is a Hamiltonian system with “coordinates” $(\mathbf{E}, \mathbf{H})$ and “Hamiltonian functional” $\mathcal{H}[\mathbf{f}] = \int \mathcal{U} d^3r$. The symplectic structure is encoded in $\mathbf{J}$, and the material matrix $\mathbf{M}$ plays the role of a metric on the phase space.

For a lossless medium ($\mathbf{M} = \mathbf{M}^\dagger$, positive definite), the Hamiltonian $\mathcal{H}$ is conserved (the Poynting theorem), and the eigenfrequencies are real. For a lossy medium, the anti-Hermitian part of $\mathbf{M}$ produces dissipation, the eigenfrequencies acquire imaginary parts, and the system moves into the non-Hermitian territory of Chapter 17 [12].

2.8.2 The positive-definite condition

The requirement that $\mathbf{M}$ be positive definite is the mathematical expression of the physical condition that the electromagnetic energy density $\mathcal{U} = \frac{1}{2} \mathbf{f}^\dagger \mathbf{M} \mathbf{f}$ be positive for all nonzero field configurations. This condition is satisfied by all passive dielectric and magnetic materials at frequencies below any resonance.

Near a material resonance (e.g., the ferromagnetic resonance in YIG at ω_0 , Chapter 9), one or more eigenvalues of $\mathbf{M}(\omega)$ pass through zero and become negative. In these regimes, the non-dispersive energy density is no longer positive definite, and one must use the dispersive energy metric $\mathbf{M}_g(\omega) = \partial_\omega[\omega\mathbf{M}(\omega)]$ introduced in Chapter 3. The dispersive metric remains positive definite as long as the material is passive, by virtue of the Kramers–Kronig relations (Appendix D) [19, 20].

2.8.3 Phase-space volume and Liouville’s theorem

In a conservative Hamiltonian system, Liouville’s theorem guarantees conservation of phase-space volume. For the electromagnetic system, this translates into the statement that the eigenvalue spectrum is stable under perturbations that preserve the lossless condition: real eigenfrequencies cannot “collide and annihilate” unless a symmetry allows it (the von Neumann–Wigner codimension argument of Chapter 6).

When loss is introduced, the Liouville structure breaks down and eigenfrequencies can coalesce at exceptional points, producing the rich non-Hermitian physics discussed in Chapter 17 [13].

2.9 Lorentz reciprocity from Maxwell’s equations

Lorentz reciprocity is a fundamental symmetry of Maxwell’s equations that has profound implications for topological photonics. The reciprocity theorem states that for two sets of sources $(\mathbf{J}_1, \mathbf{J}_2)$ producing fields $(\mathbf{E}_1, \mathbf{H}_1)$ and $(\mathbf{E}_2, \mathbf{H}_2)$ in the same medium, the reaction integrals are equal:

$$\int (\mathbf{E}_1 \cdot \mathbf{J}_2 - \mathbf{H}_1 \cdot \mathbf{M}_2) d^3r = \int (\mathbf{E}_2 \cdot \mathbf{J}_1 - \mathbf{H}_2 \cdot \mathbf{M}_1) d^3r, \quad (2.36)$$

where $\mathbf{M}_a$ are magnetic current sources.

2.9.1 Derivation from the material symmetry

The proof proceeds by writing Maxwell’s equations for both source configurations and forming the divergence $\nabla \cdot (\mathbf{E}_1 \times \mathbf{H}_2 - \mathbf{E}_2 \times \mathbf{H}_1)$. Expanding via the vector identity and substituting the frequency-domain Maxwell equations (2.7), the surface integral vanishes for sources of compact support, leaving:

$$i\omega \int [\mathbf{E}_1 \cdot (\varepsilon^T - \varepsilon)\mathbf{E}_2 + \mathbf{H}_1 \cdot (\mu^T - \mu)\mathbf{H}_2 + \mathbf{E}_1 \cdot (\zeta^T + \xi)\mathbf{H}_2 - \mathbf{E}_2 \cdot (\zeta^T + \xi)\mathbf{H}_1] d^3r = 0. \quad (2.37)$$

Crucially, this bilinear form involves the *unconjugated* transpose (not the Hermitian conjugate), because the Lorentz identity is linear in the fields. The integrand vanishes for all field configurations if and only if the material tensors satisfy the block conditions

$$\varepsilon = \varepsilon^T, \quad \mu = \mu^T, \quad \xi = -\zeta^T \quad (2.38)$$

The displayed formula above follows the convention and normalisation used in [31]. These are the Lorentz reciprocity conditions for bianisotropic media. Note that the full 6×6 material matrix $\mathbf{M}$ is *not* required to be symmetric; instead, the off-diagonal blocks satisfy the antisymmetry condition $\xi = -\zeta^T$. For a gyrotropic medium with $\kappa_\mu \neq 0$, the permeability tensor $\mu \neq \mu^T$, and reciprocity is broken—the essential prerequisite for a photonic Chern insulator [12, 19].

2.9.2 Implications for topology

The breaking of reciprocity is intimately connected to the breaking of time-reversal symmetry: a reciprocal medium satisfies $\mathbf{M} = \mathbf{M}^T$, which is equivalent to $\hat{\mathcal{T}}$ -invariance of the Maxwell eigenproblem. Since $\hat{\mathcal{T}}$ -invariance forces $\mathcal{C}_n = 0$ (Chapter 5), a nonreciprocal medium is a necessary condition for a photonic Chern insulator.

Conversely, a photonic topological insulator with $\hat{\mathcal{T}}$ -symmetry (Chapter 11) must be reciprocal, and its topology is protected by spatial symmetries rather than by the Chern number. This dichotomy—Chern insulators require nonreciprocity, while $\mathbb{Z}_2$ and crystalline phases are reciprocal—organises the entire field of topological photonics [3, 13].

2.10 Energy density and energy velocity

The time-averaged energy density stored in a monochromatic field at frequency ω in a dispersive medium is the Brillouin energy density [19, 20]:

$$\langle \mathcal{U} \rangle = \frac{1}{4} \mathbf{f}^\dagger \frac{\partial [\omega \mathbf{M}(\omega)]}{\partial \omega} \mathbf{f} = \frac{1}{4} \mathbf{f}^\dagger \mathbf{M}_g(\omega) \mathbf{f}. \quad (2.39)$$

The energy velocity is defined as the ratio of the time-averaged Poynting flux to the energy density: $\mathbf{v}_E = \langle \mathbf{S} \rangle / \langle \mathcal{U} \rangle$.

For a lossless, non-dispersive medium, the energy velocity equals the group velocity. For a dispersive medium, the two can differ near material resonances, but they agree in the transparency windows where $\text{Im } \mathbf{M} \approx 0$. The energy velocity is always bounded by c in a passive medium, which provides a physical check on numerical computations of photonic band structures.

The dispersive energy metric $\mathbf{M}_g(\omega)$ is the key quantity that connects energy conservation, the electromagnetic inner product (Chapter 3), and the Berry geometry (Chapter 5).

2.11 Energy conservation and the Maxwell generator

The compact equation $\hat{\mathcal{N}} \Psi = i \partial_t (\mathbf{M} \Psi)$ is useful only if it carries the ordinary physical meaning of Maxwell's equations. Let us verify explicitly that the operator form reproduces Poynting's theorem and identifies the correct Hilbert-space metric. For a source-free, lossless, nondispersive medium with Hermitian ε and μ , Maxwell's equations are

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad \nabla \times \mathbf{H} = \partial_t \mathbf{D}, \quad \mathbf{D} = \varepsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}. \quad (2.40)$$

Take the scalar product of the second equation with $\mathbf{E}^*$ and of the first equation with $\mathbf{H}^*$, subtract the complex conjugate identity where needed, and use

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}^*) = \mathbf{H}^* \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H}^*). \quad (2.41)$$

For real time-domain fields this gives the local conservation law

$$\partial_t u + \nabla \cdot \mathbf{S} = 0, \quad u = \frac{1}{2} (\mathbf{E} \cdot \mathbf{D} + \mathbf{H} \cdot \mathbf{B}), \quad \mathbf{S} = \mathbf{E} \times \mathbf{H}. \quad (2.42)$$

Thus the quadratic form associated with the material matrix is not an arbitrary mathematical choice; it is the electromagnetic energy. For a mode in a closed cavity or a Bloch mode integrated over a unit cell, the conserved norm is

$$\|\Psi\|_{\mathbf{M}}^2 = \int \Psi^\dagger \mathbf{M} \Psi \, d^3r = \int (\mathbf{E}^* \cdot \epsilon \mathbf{E} + \mathbf{H}^* \cdot \mu \mathbf{H}) \, d^3r. \quad (2.43)$$

This calculation explains why the Maxwell generator is self-adjoint in the energy metric rather than in the flat field metric. If $\hat{H}_{\mathbf{M}} = \mathbf{M}^{-1} \hat{\mathcal{N}}$ and the boundary term $\int_{\partial V} \hat{n} \cdot (\mathbf{E}_1^* \times \mathbf{H}_2) \, dA$ vanishes because of periodic, PEC, or radiation-compatible boundary conditions, then

$$\langle \Psi_1, \hat{H}_{\mathbf{M}} \Psi_2 \rangle_{\mathbf{M}} = \langle \hat{H}_{\mathbf{M}} \Psi_1, \Psi_2 \rangle_{\mathbf{M}}. \quad (2.44)$$

The eigenfrequencies are therefore real, and eigenmodes at different frequencies are orthogonal in the energy inner product [32]. This is the Maxwell analogue of Hermiticity in quantum mechanics, but its origin is Poynting's theorem, not probability conservation.

2.12 What changes in a dispersive material

Topological photonic media are often strongly dispersive. Ferrites, plasmas, metamaterials, and resonator arrays all acquire their useful band gaps from internal material or structural resonances. In such a medium the naive energy density $\frac{1}{2}(\mathbf{E} \cdot \mathbf{D} + \mathbf{H} \cdot \mathbf{B})$ is not the stored energy of a monochromatic wave, because some energy resides in the microscopic polarisation or magnetisation degrees of freedom. For a lossless linear medium with frequency-domain material matrix $\mathbf{M}(\omega)$, the correct cycle-averaged energy density is

$$u_\omega = \frac{1}{4} \Psi^\dagger \partial_\omega [\omega \mathbf{M}(\omega)] \Psi. \quad (2.45)$$

The matrix $\mathbf{M}_g(\omega) = \partial_\omega [\omega \mathbf{M}(\omega)]$ is positive in a transparent frequency window of a passive medium. Positivity is not a technical assumption; it is the statement that a finite-amplitude field stores positive energy in both the electromagnetic field and the bound material oscillators.

This observation will be used repeatedly. In Chapter 3 it becomes the orthogonality metric for dispersive Maxwell modes. In Chapter 5 it becomes the metric used to normalise Bloch modes before Berry curvature is computed. In Chapter 9 it is the reason the gyromagnetic YIG bands can be assigned Chern numbers only in a frequency window separated from ferromagnetic resonance loss. The word “topological” never removes the need to ask where the electromagnetic energy is stored.

Exercises

Exercise 2.1. Verify by direct computation that the 6×6 momentum-space curl operator $\hat{\mathcal{N}}(\mathbf{k})$ defined in (2.11) is Hermitian. *Hint:* write out the full matrix for a general $\mathbf{k} = (k_x, k_y, k_z)$ and check that it equals its conjugate transpose.

Maxwell normal modes as generalized eigenpairs

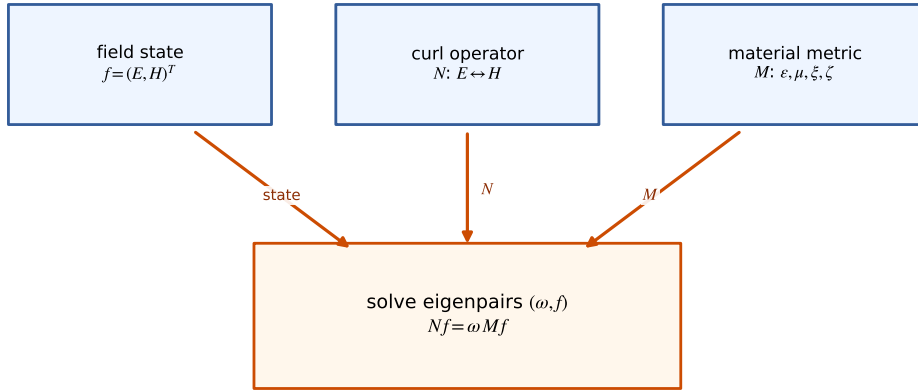


Figure 2.2: Maxwell's equations as a six-component normal-mode problem. The electromagnetic field state $\mathbf{f} = (\mathbf{E}, \mathbf{H})^T$ is acted on by the curl operator N , while the material tensor M supplies the energy metric. Topological band theory for photons starts only after this metric structure is fixed.

Exercise 2.2. For an isotropic medium with scalar ϵ and μ , show that the eigenvalues of the 6×6 matrix $\mathbf{M}^{-1} \hat{\mathcal{N}}(\mathbf{k})$ are $\omega = \pm |\mathbf{k}| / \sqrt{\epsilon_0 \mu_0 \epsilon \mu}$ (each twofold degenerate, corresponding to two transverse polarisations) and $\omega = 0$ (twofold, corresponding to longitudinal modes). Identify the physical content of each.

Exercise 2.3. The gyrotropic permeability tensor (2.4) has eigenvalues $\mu_{\perp} \pm \kappa$ for circular polarisations and μ_z for the z -component. Derive the two circular-polarisation refractive indices $n_{\pm} = \sqrt{\epsilon(\mu_{\perp} \pm \kappa)}$ for a plane wave propagating along $\hat{z}$ in a medium with scalar permittivity ϵ and gyrotropic permeability. Show that the Faraday rotation per unit length is $\theta_F = \omega(n_+ - n_-)/(2c)$.

Exercise 2.4. Prove that the complex Poynting identity (2.25) holds by starting from the curl equations (2.7) and using the vector identity $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$.

Exercise 2.5. Show that for a lossless nondispersive medium, the divergence of the time-averaged Poynting vector vanishes: $\nabla \cdot \langle \mathbf{S} \rangle = 0$. What additional condition is needed for this to hold in a dispersive medium?

Exercise 2.6. Starting from the bianisotropic constitutive relation (2.5), verify explicitly that the time-reversal condition (2.30) applied to $\mathbf{M}$ yields the block conditions (2.31).

Chapter 3

Normal Modes and the Electromagnetic Inner Product

3.1 Why the inner product matters

In quantum mechanics, the Hilbert-space inner product $\langle \psi | \phi \rangle = \int \psi^* \phi \, d^3r$ is so natural that it is rarely questioned. The Hamiltonian is self-adjoint under this inner product, eigenvalues are real, and eigenstates are orthogonal—all with the standard L^2 norm.

The Maxwell eigenvalue equation

$$\hat{\mathcal{N}} \mathbf{f} = \omega \mathbf{M} \mathbf{f} \quad (3.1)$$

from Chapter 2 is *not* a standard eigenvalue problem. The material matrix $\mathbf{M}$ appears on the right-hand side, making this a *generalised* eigenproblem. If we naively define an “effective Hamiltonian” $\hat{\mathcal{H}} = \mathbf{M}^{-1} \hat{\mathcal{N}}$ and write $\hat{\mathcal{H}} \mathbf{f} = \omega \mathbf{f}$, the operator $\hat{\mathcal{H}}$ is generally *not* Hermitian under the standard inner product $\mathbf{f}_1^\dagger \mathbf{f}_2$.

This means that the standard L^2 inner product is the *wrong* inner product for electromagnetic normal modes. Using it leads to non-orthogonal modes, non-gauge-invariant Berry connections, and incorrect topological invariants. The correct inner product is the *electromagnetic energy inner product*, weighted by the material matrix $\mathbf{M}$ (or, in dispersive media, by the energy metric $\mathbf{M}_g$) [33]. Establishing this inner product rigorously is the central task of this chapter.

3.2 Self-adjointness of the Maxwell operator

3.2.1 The nondispersive case

Consider first a nondispersive lossless medium, so that $\mathbf{M}(\mathbf{r})$ is Hermitian and frequency-independent. Define the electromagnetic inner product over a domain V (which will later be specialised to a unit cell):

$$\langle \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}} \equiv \int_V \mathbf{f}_1^\dagger(\mathbf{r}) \cdot \mathbf{M}(\mathbf{r}) \cdot \mathbf{f}_2(\mathbf{r}) \, d^3r. \quad (3.2)$$

Since $\mathbf{M} = \mathbf{M}^\dagger$, this is a sesquilinear form; since $\mathbf{M}$ is positive definite (Section 2.4.3), it is positive definite and hence a genuine inner product.

Theorem 3.1 (Self-adjointness of the Maxwell operator). *Let $\mathbf{M}(\mathbf{r})$ be Hermitian and positive definite, and let $\mathbf{f}_1, \mathbf{f}_2$ be smooth six-component fields satisfying appropriate boundary conditions (periodic, or vanishing at infinity). Then the Maxwell curl operator $\hat{\mathcal{N}}$ is self-adjoint with respect to the electromagnetic inner product:*

$$\langle \mathbf{f}_1, \mathbf{M}^{-1} \hat{\mathcal{N}} \mathbf{f}_2 \rangle_{\mathbf{M}} = \langle \mathbf{M}^{-1} \hat{\mathcal{N}} \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}}. \quad (3.3)$$

Equivalently, writing $\hat{\mathcal{H}} = \mathbf{M}^{-1} \hat{\mathcal{N}}$:

$$\langle \mathbf{f}_1, \hat{\mathcal{H}} \mathbf{f}_2 \rangle_{\mathbf{M}} = \langle \hat{\mathcal{H}} \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}}. \quad (3.4)$$

Proof. We need to show that

$$\int_V \mathbf{f}_1^\dagger \mathbf{M} (\mathbf{M}^{-1} \hat{\mathcal{N}} \mathbf{f}_2) d^3r = \int_V (\mathbf{M}^{-1} \hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{M} \mathbf{f}_2 d^3r. \quad (3.5)$$

The left-hand side simplifies to $\int_V \mathbf{f}_1^\dagger \hat{\mathcal{N}} \mathbf{f}_2 d^3r$. The right-hand side simplifies to $\int_V (\hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{M}^{-\dagger} \mathbf{M} \mathbf{f}_2 d^3r = \int_V (\hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{f}_2 d^3r$ since $\mathbf{M}^{-\dagger} \mathbf{M} = \mathbf{I}$ for Hermitian $\mathbf{M}$. So we need to prove

$$\int_V \mathbf{f}_1^\dagger \hat{\mathcal{N}} \mathbf{f}_2 d^3r = \int_V (\hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{f}_2 d^3r, \quad (3.6)$$

i.e., $\hat{\mathcal{N}}$ itself is self-adjoint under the *standard* L^2 inner product.

Write $\mathbf{f}_i = (\mathbf{E}_i, \mathbf{H}_i)^T$. Then

$$\mathbf{f}_1^\dagger \hat{\mathcal{N}} \mathbf{f}_2 = \mathbf{E}_1^* \cdot (\mathbf{i} \nabla \times \mathbf{H}_2) + \mathbf{H}_1^* \cdot (-\mathbf{i} \nabla \times \mathbf{E}_2). \quad (3.7)$$

Similarly,

$$(\hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{f}_2 = (-\mathbf{i} \nabla \times \mathbf{H}_1)^* \cdot \mathbf{E}_2 + (\mathbf{i} \nabla \times \mathbf{E}_1)^* \cdot \mathbf{H}_2 = \mathbf{i} (\nabla \times \mathbf{H}_1^*) \cdot \mathbf{E}_2 - \mathbf{i} (\nabla \times \mathbf{E}_1^*) \cdot \mathbf{H}_2. \quad (3.8)$$

Taking the difference:

$$\begin{aligned} & \mathbf{f}_1^\dagger \hat{\mathcal{N}} \mathbf{f}_2 - (\hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{f}_2 \\ &= \mathbf{i} [\mathbf{E}_1^* \cdot (\nabla \times \mathbf{H}_2) - (\nabla \times \mathbf{H}_1^*) \cdot \mathbf{E}_2] - \mathbf{i} [\mathbf{H}_1^* \cdot (\nabla \times \mathbf{E}_2) - (\nabla \times \mathbf{E}_1^*) \cdot \mathbf{H}_2]. \end{aligned} \quad (3.9)$$

Each bracket is a total divergence. Using the identity $\mathbf{A} \cdot (\nabla \times \mathbf{B}) - (\nabla \times \mathbf{A}) \cdot \mathbf{B} = -\nabla \cdot (\mathbf{A} \times \mathbf{B})$:

$$\mathbf{f}_1^\dagger \hat{\mathcal{N}} \mathbf{f}_2 - (\hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{f}_2 = -\mathbf{i} \nabla \cdot (\mathbf{E}_1^* \times \mathbf{H}_2 - \mathbf{H}_1^* \times \mathbf{E}_2). \quad (3.10)$$

Integrating over V and applying the divergence theorem:

$$\int_V (\mathbf{f}_1^\dagger \hat{\mathcal{N}} \mathbf{f}_2 - (\hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{f}_2) d^3r = -\mathbf{i} \oint_{\partial V} (\mathbf{E}_1^* \times \mathbf{H}_2 - \mathbf{H}_1^* \times \mathbf{E}_2) \cdot \hat{n} dA. \quad (3.11)$$

The surface integral vanishes when:

- (a) the fields satisfy *periodic boundary conditions* ($\mathbf{f}(\mathbf{r} + \mathbf{a}) = e^{i\mathbf{k} \cdot \mathbf{a}} \mathbf{f}(\mathbf{r})$ for lattice vector $\mathbf{a}$), since opposite faces of the unit cell contribute equal and opposite terms; or

- (b) the fields decay sufficiently fast at infinity; or
- (c) the domain has perfectly conducting or perfectly magnetic boundaries.

Under any of these conditions, $\int_V \mathbf{f}_1^\dagger \hat{\mathcal{N}} \mathbf{f}_2 \, d^3r = \int_V (\hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{f}_2 \, d^3r$, establishing (3.6) and hence (3.4). $\square$

Corollary 3.2 (Real eigenfrequencies). *Every eigenfrequency ω_n of the Maxwell normal-mode problem $\hat{\mathcal{H}} \mathbf{f}_n = \omega_n \mathbf{f}_n$ is real.*

Proof. From self-adjointness: $\omega_n \langle \mathbf{f}_n, \mathbf{f}_n \rangle_{\mathbf{M}} = \langle \mathbf{f}_n, \hat{\mathcal{H}} \mathbf{f}_n \rangle_{\mathbf{M}} = \langle \hat{\mathcal{H}} \mathbf{f}_n, \mathbf{f}_n \rangle_{\mathbf{M}} = \omega_n^* \langle \mathbf{f}_n, \mathbf{f}_n \rangle_{\mathbf{M}}$. Since $\langle \mathbf{f}_n, \mathbf{f}_n \rangle_{\mathbf{M}} > 0$, we conclude $\omega_n = \omega_n^*$. $\square$

3.2.2 Orthogonality of modes

[29]

Proposition 3.3 (Mode orthogonality). *Eigenmodes at different eigenfrequencies are orthogonal under the electromagnetic inner product:*

$$\omega_n \neq \omega_m \implies \langle \mathbf{f}_n, \mathbf{f}_m \rangle_{\mathbf{M}} = 0. \quad (3.12)$$

Proof. Self-adjointness gives $\langle \mathbf{f}_n, \hat{\mathcal{H}} \mathbf{f}_m \rangle_{\mathbf{M}} = \langle \hat{\mathcal{H}} \mathbf{f}_n, \mathbf{f}_m \rangle_{\mathbf{M}}$, i.e., $\omega_m \langle \mathbf{f}_n, \mathbf{f}_m \rangle_{\mathbf{M}} = \omega_n^* \langle \mathbf{f}_n, \mathbf{f}_m \rangle_{\mathbf{M}} = \omega_n \langle \mathbf{f}_n, \mathbf{f}_m \rangle_{\mathbf{M}}$ (using $\omega_n \in \mathbb{R}$). Hence $(\omega_m - \omega_n) \langle \mathbf{f}_n, \mathbf{f}_m \rangle_{\mathbf{M}} = 0$; if $\omega_n \neq \omega_m$, the inner product vanishes. $\square$

We adopt the *energy normalisation convention*:

$$\langle \mathbf{f}_n, \mathbf{f}_m \rangle_{\mathbf{M}} = \int_V \mathbf{f}_n^\dagger \mathbf{M} \mathbf{f}_m \, d^3r = \delta_{nm}. \quad (3.13)$$

The physical meaning of $\langle \mathbf{f}_n, \mathbf{f}_n \rangle_{\mathbf{M}} = 1$ is that the mode carries unit time-averaged electromagnetic energy per unit cell (in a periodic medium) or per normalisation volume.

3.3 The energy-weighted field

The electromagnetic inner product (3.2) is perfectly rigorous [19, 32], but it is sometimes convenient to work with a *standard*-Hermitian formulation. This is achieved by a change of variables that absorbs $\mathbf{M}$ into the field.

Since $\mathbf{M}$ is Hermitian and positive definite, it has a unique positive-definite Hermitian square root $\mathbf{M}^{1/2}$. Define the *energy-weighted field*

$$\boxed{\mathbf{w}(\mathbf{r}) \equiv \mathbf{M}^{1/2}(\mathbf{r}) \mathbf{f}(\mathbf{r})}. \quad (3.14)$$

Then the electromagnetic inner product becomes the standard L^2 inner product for $\mathbf{w}$:

$$\langle \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}} = \int_V \mathbf{f}_1^\dagger \mathbf{M} \mathbf{f}_2 \, d^3r = \int_V \mathbf{w}_1^\dagger \mathbf{w}_2 \, d^3r. \quad (3.15)$$

The eigenvalue equation in terms of $\mathbf{w}$ becomes

$$\tilde{\mathcal{H}}(\mathbf{r}) \mathbf{w}(\mathbf{r}) = \omega \mathbf{w}(\mathbf{r}), \quad \tilde{\mathcal{H}} = \mathbf{M}^{-1/2} \hat{\mathcal{N}} \mathbf{M}^{-1/2}, \quad (3.16)$$

and $\tilde{\mathcal{H}}$ is Hermitian under the standard inner product.

Remark 3.4. The energy-weighted formulation is useful for formal proofs and for connecting to the quantum-mechanical literature, where Berry connections are always defined with respect to the standard inner product. However, the original formulation in terms of $\mathbf{f}$ and $\mathbf{M}$ is often more convenient in practice, because $\mathbf{f} = (\mathbf{E}, \mathbf{H})^T$ has direct physical meaning, while $\mathbf{w} = \mathbf{M}^{1/2} \mathbf{f}$ does not correspond to any simply measurable field. We will use both formulations interchangeably and state which inner product is being used at each stage.

3.4 Extension to dispersive media

3.4.1 The generalised eigenvalue problem

Real magneto-optical materials are always dispersive: the gyrotropic parameter κ in (2.4) depends on frequency, as do ε , $\mu_{\perp}$, and all other material parameters [31]. The material matrix is then $\mathbf{M} = \mathbf{M}(\mathbf{r}, \omega)$, and the Maxwell eigenvalue equation

$$\hat{\mathcal{N}} \mathbf{f} = \omega \mathbf{M}(\omega) \mathbf{f} \quad (3.17)$$

is a *nonlinear* eigenvalue problem in ω : the eigenvalue appears both explicitly and inside $\mathbf{M}$. This is qualitatively different from the Schrödinger equation, where the Hamiltonian does not depend on the energy eigenvalue.

Following Haldane and Raghu, we treat this as a generalised eigenvalue problem. For a Bloch mode in a periodic medium, the cell-periodic eigenproblem takes the form

$$\hat{\mathcal{N}}(\mathbf{k}) \mathbf{u}_{\mathbf{nk}}(\mathbf{r}) = \omega_{\mathbf{nk}} \mathbf{M}(\mathbf{r}, \omega_{\mathbf{nk}}) \mathbf{u}_{\mathbf{nk}}(\mathbf{r}), \quad (3.18)$$

where $\hat{\mathcal{N}}(\mathbf{k})$ is the curl operator with ∇ replaced by $\nabla + i\mathbf{k}$ (incorporating the Bloch phase).

3.4.2 The energy metric as inner product

The self-adjointness proof of Section 3.2.1 relied on $\mathbf{M}$ being the same for both modes. In the dispersive case, two modes $\mathbf{f}_n$ and $\mathbf{f}_m$ at *different* frequencies see *different* material matrices $\mathbf{M}(\omega_n) \neq \mathbf{M}(\omega_m)$, and the standard orthogonality argument breaks down.

The resolution was provided by the energy metric $\mathbf{M}_g(\omega) = \partial_{\omega}[\omega \mathbf{M}(\omega)]$ introduced in (2.22). The key result is:

Theorem 3.5 (Orthogonality in dispersive media). *Let $\mathbf{M}(\mathbf{r}, \omega)$ be Hermitian for real ω , and let $\mathbf{M}_g(\mathbf{r}, \omega) = \partial_{\omega}[\omega \mathbf{M}(\mathbf{r}, \omega)]$ be positive definite at all physical eigenfrequencies. Then two Bloch eigenmodes $\mathbf{u}_{\mathbf{nk}}$ and $\mathbf{u}_{\mathbf{mk}}$ at the same wavevector $\mathbf{k}$ but different frequencies satisfy the exact relation*

$$\omega_n \int_{\text{cell}} \mathbf{u}_{\mathbf{nk}}^{\dagger} \mathbf{M}(\omega_n) \mathbf{u}_{\mathbf{mk}} d^3r = \omega_m \int_{\text{cell}} \mathbf{u}_{\mathbf{nk}}^{\dagger} \mathbf{M}(\omega_m) \mathbf{u}_{\mathbf{mk}} d^3r. \quad (3.19)$$

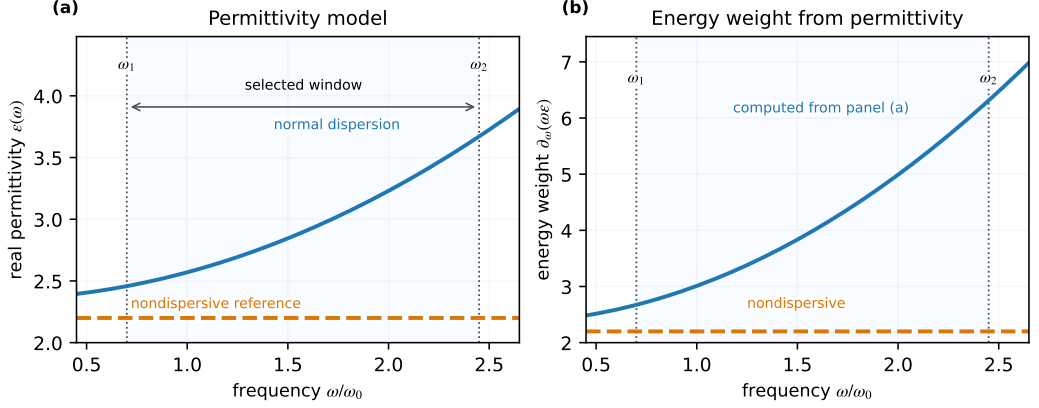


Figure 3.1: Dispersive energy metric and its frequency dependence. **(a)** Representative dielectric function $\varepsilon(\omega)$ of a Lorentzian medium, showing a selected lossless window between frequencies ω_1 and ω_2 where ε is real and positive. The dashed line marks the nondispersive reference level. **(b)** The corresponding energy-metric weight $\partial_\omega[\omega\varepsilon(\omega)]$ computed from panel (a). In the lossless window the weight exceeds unity and increases with frequency, reflecting the additional electromagnetic energy stored in the dispersive polarisation of the medium. The orthogonality of Bloch modes in dispersive media (Theorem 3.5) is governed by this frequency-dependent metric rather than by $\varepsilon(\omega)$ alone.

When the two frequencies are close ($\omega_m \rightarrow \omega_n$), this reduces to $\int \mathbf{u}_n^\dagger \mathbf{M}_g(\omega_n) \mathbf{u}_m d^3r = 0$. The group-velocity metric $\mathbf{M}_g$ therefore plays the role of the orthogonality metric for near-degenerate modes and also determines the self-overlap normalization $\int \mathbf{u}_n^\dagger \mathbf{M}_g(\omega_n) \mathbf{u}_n d^3r > 0$.

Proof. Start from the eigenvalue equations for modes n and m :

$$\hat{\mathcal{N}}(\mathbf{k}) \mathbf{u}_n = \omega_n \mathbf{M}(\omega_n) \mathbf{u}_n, \quad (3.20)$$

$$\hat{\mathcal{N}}(\mathbf{k}) \mathbf{u}_m = \omega_m \mathbf{M}(\omega_m) \mathbf{u}_m. \quad (3.21)$$

Take the inner product of (3.21) with $\mathbf{u}_n$ (standard L^2 , with periodic boundary conditions as before):

$$\int \mathbf{u}_n^\dagger \hat{\mathcal{N}}(\mathbf{k}) \mathbf{u}_m d^3r = \omega_m \int \mathbf{u}_n^\dagger \mathbf{M}(\omega_m) \mathbf{u}_m d^3r. \quad (3.22)$$

By the self-adjointness of $\hat{\mathcal{N}}(\mathbf{k})$ (the boundary term vanishes for Bloch-periodic fields), the left-hand side also equals $\int (\hat{\mathcal{N}}(\mathbf{k}) \mathbf{u}_n)^\dagger \mathbf{u}_m d^3r = \omega_n^* \int \mathbf{u}_n^\dagger \mathbf{M}^\dagger(\omega_n) \mathbf{u}_m d^3r = \omega_n \int \mathbf{u}_n^\dagger \mathbf{M}(\omega_n) \mathbf{u}_m d^3r$. (We used $\omega_n \in \mathbb{R}$ and $\mathbf{M}^\dagger = \mathbf{M}$.)

Therefore:

$$\omega_n \int \mathbf{u}_n^\dagger \mathbf{M}(\omega_n) \mathbf{u}_m d^3r = \omega_m \int \mathbf{u}_n^\dagger \mathbf{M}(\omega_m) \mathbf{u}_m d^3r. \quad (3.23)$$

Equation (3.23) is already the desired exact off-diagonal relation. Equivalently,

$$\int \mathbf{u}_n^\dagger [\omega_n \mathbf{M}(\omega_n) - \omega_m \mathbf{M}(\omega_m)] \mathbf{u}_m d^3r = 0. \quad (3.24)$$

For $n \neq m$ this may be viewed as orthogonality with respect to the frequency divided-difference operator

$$\mathbf{M}_{nm}^{\text{dd}} = \frac{\omega_n \mathbf{M}(\omega_n) - \omega_m \mathbf{M}(\omega_m)}{\omega_n - \omega_m}. \quad (3.25)$$

In the limit $\omega_m \rightarrow \omega_n$, this operator tends $\partial_\omega [\omega \mathbf{M}(\omega)]_{\omega_n} = \mathbf{M}_g(\omega_n)$. Thus $\mathbf{M}_g$ is the correct self-overlap and near-degenerate metric, while the exact off-diagonal relation for separated frequencies is the divided-difference relation above. $\square$

Remark 3.6. The proof shows that the energy metric $\mathbf{M}_g$ arises naturally as the “linearisation” of the nonlinear eigenvalue problem around a given eigenfrequency. The positive definiteness of $\mathbf{M}_g$ is not just a convenient assumption; it is the physical stability condition ensuring that each mode carries positive energy and that the eigenfrequencies are locally non-degenerate as a function of ω .

3.4.3 Normalisation in dispersive media

The natural normalisation convention for dispersive media is

$$\langle \mathbf{u}_n, \mathbf{u}_m \rangle_{\mathbf{M}_g} \equiv \int_{\text{cell}} \mathbf{u}_{nk}^\dagger(\mathbf{r}) \mathbf{M}_g(\mathbf{r}, \omega_{nk}) \mathbf{u}_{mk}(\mathbf{r}) d^3r = \delta_{nm}. \quad (3.26)$$

With this normalisation, the time-averaged electromagnetic energy stored in the mode equals unity per unit cell, including the energy stored in the material degrees of freedom (dispersive polarisation).

In the nondispersive limit $\partial \mathbf{M} / \partial \omega = 0$, the energy metric reduces to $\mathbf{M}_g = \mathbf{M}$ and (3.26) becomes (3.13).

3.4.4 Energy-weighted fields in dispersive media

Since $\mathbf{M}_g(\omega)$ is Hermitian and positive definite at physical eigenfrequencies, it also has a unique positive-definite square root. Define

$$\mathbf{w}_{nk}(\mathbf{r}) = \mathbf{M}_g^{1/2}(\mathbf{r}, \omega_{nk}) \mathbf{u}_{nk}(\mathbf{r}). \quad (3.27)$$

Then $\int \mathbf{w}_n^\dagger \mathbf{w}_m d^3r = \delta_{nm}$, and all Berry-geometry formulas can be written in terms of $\mathbf{w}$ with the standard inner product.

Remark 3.7. There is a subtlety: $\mathbf{M}_g$ depends on $\omega_n(\mathbf{k})$, which varies with $\mathbf{k}$. When computing Berry connections (derivatives with respect to $\mathbf{k}$), one must account for this implicit $\mathbf{k}$ dependence through $\omega_n(\mathbf{k})$. We will address this carefully in Chapter 5.

3.5 The role of the divergence constraint

The Maxwell curl equations (3.1) are supplemented by the divergence constraints $\nabla \cdot \mathbf{D} = 0$ and $\nabla \cdot \mathbf{B} = 0$. In the six-component formalism, these become

$$\nabla \cdot (\mathbf{M} \mathbf{f}) = 0, \quad (3.28)$$

which in momentum space reads $(\mathbf{k} + \mathbf{G}) \cdot (\mathbf{M} \hat{\mathbf{u}}_{\mathbf{G}}) = 0$ for each Fourier component.

Transverse and longitudinal modes. For an isotropic medium with scalar ε and μ , the eigenmodes of the Maxwell operator split into two classes:

- *Transverse modes* with $\mathbf{k} \cdot \mathbf{E} = 0$ and $\mathbf{k} \cdot \mathbf{H} = 0$. These have nonzero eigenfrequencies $\omega \neq 0$ and represent propagating electromagnetic waves.
- *Longitudinal modes* with $\nabla \times \mathbf{E} = 0$ and $\nabla \times \mathbf{H} = 0$, satisfying $\hat{\mathcal{N}} \mathbf{f}_L = 0$. These are the null space of $\hat{\mathcal{N}}$ and have $\omega = 0$.

The divergence constraint eliminates the longitudinal modes from the physical spectrum. In a photonic crystal, the situation is analogous: the physical modes lie in the transverse subspace satisfying (3.28).

Remark 3.8. The longitudinal modes are often silently excluded in the photonic band-structure literature by working directly with the second-order equation $\nabla \times (\mu^{-1} \nabla \times \mathbf{E}) = (\omega^2/c^2) \varepsilon \mathbf{E}$, which is automatically transverse when applied to divergence-free fields. The first-order six-component formulation keeps them visible, which is important for rigour: one must verify that the Berry curvature and Chern numbers are computed only over the physical (transverse) bands.

3.6 Completeness and spectral theory

[34]

3.6.1 Mode expansion

In a lossless nondispersive medium occupying a finite domain V with appropriate boundary conditions, the transverse eigenmodes $\{\mathbf{f}_n\}$ of $\hat{\mathcal{H}} = \mathbf{M}^{-1} \hat{\mathcal{N}}$ form a complete orthonormal set with respect to the electromagnetic inner product. Any transverse six-component field can be expanded as

$$\mathbf{f}(\mathbf{r}) = \sum_n c_n \mathbf{f}_n(\mathbf{r}), \quad c_n = \langle \mathbf{f}_n, \mathbf{f} \rangle_{\mathbf{M}}. \quad (3.29)$$

This is the electromagnetic analog of the expansion of a quantum state in energy eigenstates.

For a periodic medium, completeness applies within each Bloch sector at fixed $\mathbf{k}$: the cell-periodic modes $\{\mathbf{u}_{n\mathbf{k}}\}_{n=1}^{\infty}$ span the transverse subspace of L^2 -periodic functions on the unit cell.

3.6.2 Spectral properties

The eigenfrequencies $\omega_n(\mathbf{k})$ are real and can be ordered:

$$0 < \omega_1(\mathbf{k}) \leq \omega_2(\mathbf{k}) \leq \dots. \quad (3.30)$$

(We exclude the $\omega = 0$ longitudinal modes.) As $\mathbf{k}$ varies over the Brillouin zone, each $\omega_n(\mathbf{k})$ traces out a *band*. The bands may overlap (when $\omega_n(\mathbf{k})$ and $\omega_{n+1}(\mathbf{k}')$ have the same value for different $\mathbf{k}, \mathbf{k}'$) or may be separated by *band gaps*: frequency intervals in which no eigenfrequency exists for any $\mathbf{k}$.

The existence and width of band gaps depend on the contrast of the material parameters and the geometry of the periodic structure. We will study the band structure in detail in Chapter 4.

3.7 Summary of key formulas

For reference, we collect the central results of this chapter.

1. **Electromagnetic inner product (nondispersive):**

$$\langle \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}} = \int_V \mathbf{f}_1^\dagger \mathbf{M} \mathbf{f}_2 \, d^3r. \quad (3.2)$$

2. **Self-adjointness:** $\hat{\mathcal{H}} = \mathbf{M}^{-1} \hat{\mathcal{N}}$ is self-adjoint under $\langle \cdot, \cdot \rangle_{\mathbf{M}}$.

3. **Real eigenfrequencies, orthogonal modes:** $\omega_n \in \mathbb{R}$; $\langle \mathbf{f}_n, \mathbf{f}_m \rangle_{\mathbf{M}} = \delta_{nm}$.

4. **Energy-weighted field:** $\mathbf{w} = \mathbf{M}^{1/2} \mathbf{f}$; standard Hermitian eigenvalue problem for $\mathbf{w}$.

5. **Dispersive energy metric:** [19, 20]

$$\mathbf{M}_g(\omega) = \frac{\partial}{\partial \omega} [\omega \mathbf{M}(\omega)]. \quad (2.22)$$

6. **Orthogonality in dispersive media:**

$$\langle \mathbf{u}_n, \mathbf{u}_m \rangle_{\mathbf{M}_g} = \int_{\text{cell}} \mathbf{u}_n^\dagger \mathbf{M}_g(\omega_n) \mathbf{u}_m \, d^3r = \delta_{nm}. \quad (3.26)$$

These results are the foundation on which all Berry-geometric and topological statements in the rest of the book are built. The Berry connection, Berry curvature, and Chern number for electromagnetic bands must always be defined using the material-weighted inner product $\langle \cdot, \cdot \rangle_{\mathbf{M}}$ or its dispersive generalisation $\langle \cdot, \cdot \rangle_{\mathbf{M}_g}$. Using the wrong inner product—a mistake easily made when importing formulas from quantum mechanics—invalidates the topological invariants.

3.8 The dispersive energy metric: derivation and meaning

When the material parameters depend on frequency—as they must for any causal material—the non-dispersive energy density $\mathcal{U} = \frac{1}{2} \mathbf{f}^\dagger \mathbf{M}(\omega) \mathbf{f}$ is not the physically correct energy stored in the field. The correct expression is the Brillouin energy density, derived as follows.

3.8.1 Derivation from energy conservation

Consider a monochromatic field at frequency ω with a slowly varying envelope. The time-averaged energy density stored in the medium is [19]

$$\langle \mathcal{U} \rangle = \frac{1}{4} \mathbf{f}^\dagger \frac{\partial [\omega \mathbf{M}(\omega)]}{\partial \omega} \mathbf{f} = \frac{1}{4} \mathbf{f}^\dagger \mathbf{M}_g(\omega) \mathbf{f}. \quad (3.31)$$

The dispersive energy metric $\mathbf{M}_g(\omega) = \partial_\omega [\omega \mathbf{M}(\omega)]$ appears because the energy stored in the oscillators of the medium must be included alongside the free-field energy.

For a non-dispersive material, $\mathbf{M}_g = \mathbf{M} + \omega \partial_\omega \mathbf{M} = \mathbf{M}$ (since $\partial_\omega \mathbf{M} = 0$), and the Brillouin energy reduces to the familiar non-dispersive form.

3.8.2 Positivity from passivity

For a passive material (no gain), the Kramers–Kronig relations guarantee that $\mathbf{M}_g(\omega)$ is positive definite at every real frequency outside the absorption bands. The physical argument is simple: a passive material can only store (not create) energy, so $\langle \mathcal{U} \rangle$ must be positive for all nonzero field amplitudes [20].

This positivity is essential for the entire topological framework of this book: the electromagnetic inner product $\langle \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}_g} = \int \mathbf{f}_1^\dagger \mathbf{M}_g \mathbf{f}_2 d^3r$ is positive definite, making the eigenproblem Hermitian and the Berry connection well defined. If $\mathbf{M}_g$ lost positive definiteness (as can happen near a gain-induced instability), the Hermitian structure would break and the Chern number would need to be redefined through the non-Hermitian methods of Chapter 17.

3.8.3 The dispersive inner product and Berry connection

The dispersive energy metric enters the Berry connection through the replacement $\mathbf{M} \rightarrow \mathbf{M}_g$ everywhere the inner product appears. Specifically, the Berry connection for band n is

$$\mathcal{A}_{n,a}(\mathbf{k}) = i \int_{\text{cell}} \mathbf{u}_{n\mathbf{k}}^\dagger \mathbf{M}_g(\omega_{n\mathbf{k}}) \partial_{k_a} \mathbf{u}_{n\mathbf{k}} d^3r, \quad (3.32)$$

where $\mathbf{M}_g(\omega_{n\mathbf{k}})$ is evaluated at the band frequency. This formula reduces to the non-dispersive Berry connection when $\mathbf{M}_g = \mathbf{M}$, but it is the dispersive form that must be used for real photonic crystals with frequency-dependent material response [12, 19].

3.9 Orthogonality subtleties

3.9.1 Degenerate modes and gauge freedom

When two or more bands are degenerate at a point $\mathbf{k}_0$, the individual modes $\mathbf{u}_{n\mathbf{k}_0}$ are not uniquely determined: any unitary rotation within the degenerate subspace produces an equally valid set of eigenmodes. This gauge freedom means that the Abelian Berry connection $\mathcal{A}_n$ is not well defined at $\mathbf{k}_0$; instead, one must use the non-Abelian Berry connection $\mathcal{A}_{mn} = i \langle \mathbf{u}_m, \nabla_{\mathbf{k}} \mathbf{u}_n \rangle_{\mathbf{M}_g}$, where m and n range over the degenerate bands.

The non-Abelian connection transforms as a gauge field under unitary rotations of the degenerate subspace, and its curvature $\mathcal{F}_{mn} = \partial_a \mathcal{A}_{mn,b} - \partial_b \mathcal{A}_{mn,a} + i[\mathcal{A}_a, \mathcal{A}_b]_{mn}$ is gauge-covariant. The Chern number of a degenerate band group is $\mathcal{C} = \frac{1}{2\pi} \int_{\text{BZ}} \text{Tr } \mathcal{F} d^2k$, which is gauge-invariant and well defined even at the degeneracy points [12, 19].

3.9.2 Spurious modes and the null space

The Maxwell eigenproblem (2.10) admits spurious static solutions ($\omega = 0$) that satisfy $\nabla \times \mathbf{E} = \nabla \times \mathbf{H} = 0$. These are longitudinal (gradient) fields that carry no energy and do not contribute to the electromagnetic response. In a photonic crystal calculation, they appear as a flat band at $\omega = 0$ and must be excluded from the Berry-curvature computation. The exclusion is automatic when one restricts attention to bands at $\omega > 0$, but numerical methods must handle the near-zero modes carefully to avoid contamination of the Chern-number calculation (Appendix C) [13].

3.10 Worked example: modes of a uniform gyrotropic medium

To illustrate the general theory concretely, we compute the normal modes of a uniform gyrotropic medium with permittivity $\varepsilon_r = 15$ and Polder permeability $\mu_\perp = 3.47$, $\kappa_\mu = 1.98$ (the YIG parameters of Chapter 9).

For plane-wave solutions $\mathbf{f} \propto e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$ propagating along $\hat{z}$, the Maxwell eigenproblem reduces to

$$k^2 = \frac{\omega^2}{c^2} \varepsilon_r \mu_\pm, \quad \mu_\pm = \mu_\perp \mp \kappa_\mu, \quad (3.33)$$

where $\mu_+ = \mu_\perp - \kappa_\mu = 1.49$ and $\mu_- = \mu_\perp + \kappa_\mu = 5.45$ are the eigenvalues of the transverse permeability (Appendix D). The two solutions correspond to right-circular (RCP, +) and left-circular (LCP, -) polarisations with different phase velocities $v_\pm = c / \sqrt{\varepsilon_r \mu_\pm}$.

The energy inner product for the RCP mode is $\langle \mathbf{u}_+, \mathbf{u}_+ \rangle_{\mathbf{M}_g} = \frac{1}{4} (\varepsilon_0 \varepsilon_r |E_+|^2 + \mu_0 \mu_+ |H_+|^2)$, which is positive definite since both $\varepsilon_r > 0$ and $\mu_+ > 0$ for the YIG parameters. The RCP and LCP modes are orthogonal under the energy inner product: $\langle \mathbf{u}_+, \mathbf{u}_- \rangle_{\mathbf{M}_g} = 0$, as guaranteed by Theorem 3.1 [19].

The Faraday rotation angle per unit length is $\theta_F = (k_+ - k_-)/(2) = (\omega/2c) \sqrt{\varepsilon_r} (\sqrt{\mu_+} - \sqrt{\mu_-})$, providing a direct connection between the gyrotropic splitting and the time-reversal-breaking perturbation that opens topological gaps in photonic crystals.

3.11 Completeness and expansion of arbitrary fields

3.11.1 The completeness theorem

For a lossless, non-dispersive medium in a bounded domain Ω with appropriate boundary conditions, the normal modes $\{\mathbf{f}_n\}_{n=1}^\infty$ form a complete orthonormal basis in the Hilbert space $\mathcal{H} = L^2(\Omega; \mathbf{M} dV)$. Any square-integrable field can be expanded as

$$\mathbf{f}(\mathbf{r}) = \sum_n c_n \mathbf{f}_n(\mathbf{r}), \quad c_n = \langle \mathbf{f}_n, \mathbf{f} \rangle_{\mathbf{M}_g}, \quad (3.34)$$

with convergence in the energy norm $\|\mathbf{f}\|^2 = \langle \mathbf{f}, \mathbf{f} \rangle_{\mathbf{M}_g}$.

The completeness is a consequence of the self-adjointness and the compactness of the resolvent $(\hat{\mathcal{N}} - \omega \mathbf{M})^{-1}$ on the bounded domain. For an unbounded domain (such as a photonic crystal with Bloch boundary conditions), the discrete sum is replaced by a direct integral over the Brillouin zone (Appendix A) [12, 20].

3.11.2 Subtleties of dispersive completeness

For a dispersive medium, normal modes at different frequencies obey the exact divided-difference orthogonality relation of Theorem 3.5, rather than a simple $\mathbf{M}_g(\omega_n)$ off-diagonal orthogonality rule. The energy metric $\mathbf{M}_g(\omega)$ appears in the self-overlap normalization and in the near-degenerate limit, but modal expansions over a broad frequency range must remember the frequency dependence of the material tensor. However, the modes at a single fixed frequency do not form a complete set for

expanding fields at other frequencies, because the dispersive energy metric $\mathbf{M}_g(\omega)$ is frequency-dependent.

This subtlety means that the modal expansion of a broadband pulse in a dispersive medium requires modes at all frequencies, not just at a single carrier frequency. For topological photonics, this is usually not a practical limitation because the operating bandwidth is much narrower than the material dispersion range, and the modes can be treated as non-dispersive within the topological gap [13, 19].

3.12 Green's function formulation of topological invariants

3.12.1 The gap Green's function

For a frequency ω inside a band gap, the retarded Green's function $G^R(\mathbf{k}; \omega)$ is analytic in $\mathbf{k}$ over the entire Brillouin zone. One first constructs the projector onto all bands below the gap by a frequency-contour integral:

$$P(\mathbf{k}) = -\frac{1}{2\pi i} \oint_{\mathcal{C}} G^R(\mathbf{k}; \omega') d\omega', \quad (3.35)$$

where $\mathcal{C}$ is a contour in the complex frequency plane that encloses all eigenfrequencies below the gap. The gap Chern number is then the first Chern number of the image bundle of $P(\mathbf{k})$:

$$\mathcal{C}_{\text{gap}} = \frac{1}{2\pi i} \int_{\text{BZ}} \text{Tr}[P dP \wedge dP], \quad (3.36)$$

where $dP = \partial_{k_x} P dk_x + \partial_{k_y} P dk_y$. This formulation is gauge-invariant and naturally incorporates the dispersive material response through the frequency-dependent Maxwell operator inside G^R [20, 35].

3.12.2 Advantages of the Green's function approach

The Green's function formulation has several advantages over the eigenstate-based approach:

It avoids the gauge ambiguity inherent in choosing Bloch eigenstate phases, because the Green's function is a gauge-invariant quantity. It naturally includes the dispersive material response through the frequency-dependent material tensor $\mathbf{M}(\omega)$. It extends directly to non-Hermitian systems where the eigenstate Berry connection is ill-defined due to the non-orthogonality of eigenmodes [35, 36].

The main computational disadvantage is that evaluating the Green's function requires inverting the Maxwell operator at each $\mathbf{k}$ point, which is typically more expensive than a single eigenvalue computation. For practical band-structure calculations, the eigenstate approach of Chapter 10 remains more efficient; the Green's function approach is most useful for formal proofs of the bulk-edge correspondence and for systems where the band structure is difficult to compute [12, 20].

3.13 Mode completeness in practice

The formal completeness of Maxwell normal modes becomes most useful when it is translated into a field expansion. Suppose first that the medium is lossless, nondispersive, and enclosed in a volume

with boundary conditions that make the Maxwell generator self-adjoint in the energy metric. Let $\{\Psi_n\}$ be positive-frequency modes normalised so that $\langle \Psi_m, \Psi_n \rangle_{\mathbf{M}} = \delta_{mn}$. A physical field can then be expanded as

$$\Psi(\mathbf{r}, t) = \sum_{n>0} [a_n \Psi_n(\mathbf{r}) e^{-i\omega_n t} + a_n^* \mathcal{C} \Psi_n(\mathbf{r}) e^{+i\omega_n t}], \quad (3.37)$$

where $\mathcal{C}$ denotes the particle-hole operation that maps a positive-frequency Maxwell mode to its negative-frequency partner. The modal amplitude is obtained by projection,

$$a_n = \langle \Psi_n, \Psi(0) \rangle_{\mathbf{M}}. \quad (3.38)$$

This formula is the electromagnetic analogue of expanding a quantum state in energy eigenstates, but the projection uses stored energy. Using a flat L^2 projection would give wrong amplitudes whenever the field energy is distributed unevenly between electric and magnetic components or between field and material oscillators.

In a periodic medium the same statement holds fibre by fibre in Bloch momentum. A finite-energy wave packet centred near a band n and wave vector $\mathbf{k}_0$ has the form

$$\Psi(\mathbf{r}, t) = \int_{\text{BZ}} A_n(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r} - i\omega_n(\mathbf{k})t} u_{n\mathbf{k}}(\mathbf{r}) \frac{d^d k}{(2\pi)^d} + \text{extc.c.}, \quad (3.39)$$

with $u_{n\mathbf{k}}$ normalised in the unit-cell energy metric. The slow variation of the eigenmode $u_{n\mathbf{k}}$ across the packet in $\mathbf{k}$ -space is precisely what produces Berry connection and Berry curvature in Chapter 5. Thus Berry geometry is not an optional decoration added after solving the Maxwell eigenproblem; it is already present in the way neighbouring normal modes are compared.

3.14 Green functions as spectral projectors

The modal expansion also explains why Green functions appear naturally in topological photonics [37]. For a nondispersive lossless system, the resolvent of the Maxwell generator is

$$G(z) = (z - \hat{H}_{\mathbf{M}})^{-1} = \sum_n \frac{|\Psi_n\rangle \langle \Psi_n|_{\mathbf{M}}}{z - \omega_n}, \quad (3.40)$$

where $\langle \Psi_n|_{\mathbf{M}}$ means the dual vector defined by the energy inner product. If a closed contour Γ in the complex z -plane surrounds a group of bands and no other spectrum, the spectral projector is

$$P_{\Gamma} = \frac{1}{2\pi i} \oint_{\Gamma} G(z) dz. \quad (3.41)$$

This projector is independent of the basis chosen inside the band subspace. It is therefore the natural object for non-Abelian Berry curvature, Wilson loops, and numerical Chern calculations in nearly degenerate photonic bands [20, 38].

For dispersive media the same idea survives, but the residues of the Green function are weighted by the Brillouin energy metric. This is why Green-function formulas for photonic Chern numbers must be handled with care: poles in frequency are not simply poles of a Hermitian matrix with a constant metric. The mathematically safe procedure is either to embed the dispersive material into an enlarged auxiliary-field Hamiltonian or to use the divided-difference orthogonality relations introduced earlier in this chapter. Both routes preserve the same physical statement: topology belongs to the spectral projector onto a bundle of Maxwell modes, not to an arbitrary choice of mode basis.

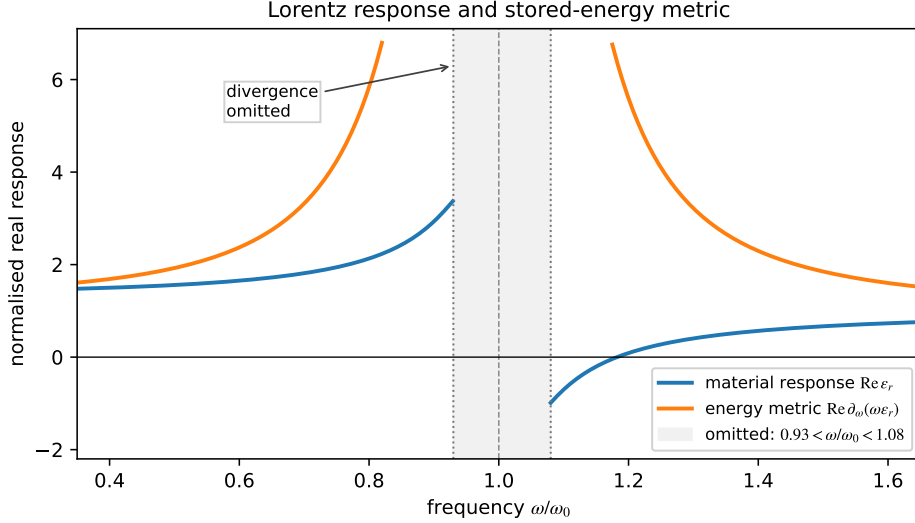


Figure 3.2: A dispersive material illustrates why the electromagnetic inner product uses the energy derivative $\partial_\omega[\omega\varepsilon]$ rather than the static permittivity alone. Near a material resonance, the static response and the positive-energy metric can differ strongly.

Exercises

Exercise 3.1. For a uniform medium with $\mathbf{M} = \text{diag}(\varepsilon_0\varepsilon, \varepsilon_0\varepsilon, \varepsilon_0\varepsilon, \mu_0\mu, \mu_0\mu, \mu_0\mu)$, compute $\mathbf{M}^{1/2}$ explicitly and verify that the energy-weighted operator $\hat{\mathcal{H}} = \mathbf{M}^{-1/2}\hat{\mathcal{N}}(\mathbf{k})\mathbf{M}^{-1/2}$ is Hermitian by direct matrix computation for $\mathbf{k} = k\hat{z}$.

Exercise 3.2. Consider a Drude plasma with $\varepsilon(\omega) = 1 - \omega_p^2/\omega^2$ and $\mu = 1$. Compute the energy metric $\mathbf{M}_g(\omega)$ and verify that it is positive definite for $\omega > \omega_p$. What happens at $\omega = \omega_p$? Interpret physically.

Exercise 3.3. For the gyrotropic permeability (2.4), suppose that $\mu_\perp(\omega)$, $\kappa(\omega)$, and $\mu_z(\omega)$ are given by the Polder model for a magnetised ferrite:

$$\mu_\perp = 1 + \frac{\omega_0\omega_m}{\omega_0^2 - \omega^2}, \quad \kappa = \frac{\omega\omega_m}{\omega_0^2 - \omega^2}, \quad \mu_z = 1, \quad (3.42)$$

where ω_0 is the Larmor frequency and ω_m is the saturation-magnetisation frequency. Compute the μ -block of $\mathbf{M}_g(\omega)$ and determine the frequency range over which it is positive definite.

Exercise 3.4. Prove that in a periodic medium with Bloch boundary conditions, the surface integral in (3.11) vanishes. *Hint:* pair opposite faces of the unit cell and use the Bloch phase relation $\mathbf{f}(\mathbf{r} + \mathbf{a}) = e^{i\mathbf{k}\cdot\mathbf{a}}\mathbf{f}(\mathbf{r})$.

Exercise 3.5 (Longitudinal modes). Show that in a homogeneous isotropic medium, the six-component fields $\mathbf{f}_L = (\nabla\phi, \mathbf{0})^T$ (for arbitrary scalar ϕ) and $\mathbf{f}'_L = (\mathbf{0}, \nabla\psi)^T$ satisfy $\hat{\mathcal{N}}\mathbf{f}_L = 0$ and $\hat{\mathcal{N}}\mathbf{f}'_L = 0$. Verify that they are orthogonal to every transverse mode under $\langle \cdot, \cdot \rangle_{\mathbf{M}}$.

Exercise 3.6. The dispersive inner product $\langle \cdot, \cdot \rangle_{\mathbf{M}_g}$ involves $\mathbf{M}_g(\omega_n)$ evaluated at the eigenfrequency of mode n . Show that for two degenerate modes at the same frequency ($\omega_n = \omega_m$), the inner product is well defined and reduces to the ordinary Hermitian form $\int \mathbf{u}_n^\dagger \mathbf{M}_g \mathbf{u}_m d^3r$.

Chapter 4

Periodic Media and Photonic Bands

4.1 Periodicity in optics

A photonic crystal is a medium whose material matrix is periodic:

$$\mathbf{M}(\mathbf{r} + \mathbf{a}_i) = \mathbf{M}(\mathbf{r}) \quad \text{for lattice vectors } \mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3. \quad (4.1)$$

In two dimensions, the periodicity is in the $(\mathbf{a}_1, \mathbf{a}_2)$ plane and $\mathbf{M}$ is independent of z (or periodic with a different period in z); in one dimension, a single $\mathbf{a}_1$ suffices. This chapter focuses on the two-dimensional case, which is the most important for topological photonics, but the formalism applies in any dimension.

The physical origin of the periodicity can be a lattice of dielectric rods, a pattern of holes in a slab, a structured metallic or ferrite arrangement, or any other design in which the optical properties repeat. What matters for the mathematics is only the discrete translational symmetry (4.1).

4.2 Bloch's theorem for Maxwell modes

4.2.1 Statement and proof

Theorem 4.1 (Bloch's theorem for electromagnetic fields). *Let $\mathbf{M}(\mathbf{r})$ be periodic with lattice vectors $\{\mathbf{a}_i\}$. Then every eigenmode of the Maxwell operator can be written in the Bloch form*

$$\mathbf{f}_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} \mathbf{u}_{n\mathbf{k}}(\mathbf{r}), \quad (4.2)$$

where $\mathbf{k}$ is the Bloch wavevector, n is the band index, and $\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$ is cell-periodic:

$$\mathbf{u}_{n\mathbf{k}}(\mathbf{r} + \mathbf{a}_i) = \mathbf{u}_{n\mathbf{k}}(\mathbf{r}) \quad \text{for all lattice vectors } \mathbf{a}_i. \quad (4.3)$$

Proof. Define the translation operator $T_{\mathbf{a}}$ by $(T_{\mathbf{a}}\mathbf{f})(\mathbf{r}) = \mathbf{f}(\mathbf{r} + \mathbf{a})$. Since $\mathbf{M}(\mathbf{r} + \mathbf{a}) = \mathbf{M}(\mathbf{r})$, the Maxwell operator $\hat{\mathcal{H}}(\mathbf{r}) = \mathbf{M}^{-1}(\mathbf{r})\hat{\mathcal{N}}$ commutes with $T_{\mathbf{a}}$:

$$[T_{\mathbf{a}}, \hat{\mathcal{H}}] = 0 \quad \text{for all lattice vectors } \mathbf{a}. \quad (4.4)$$

The translation operators for different lattice vectors also commute among themselves. Therefore $\hat{\mathcal{H}}$ and $\{T_{\mathbf{a}}\}$ can be simultaneously diagonalised.

The eigenvalues of $T_{\mathbf{a}}$ on a square-integrable (or Bloch-periodic) function space are of the form $T_{\mathbf{a}}\mathbf{f} = e^{i\mathbf{k}\cdot\mathbf{a}}\mathbf{f}$, parametrised by a wavevector $\mathbf{k}$. This is because $T_{\mathbf{a}}$ is unitary and the group of translations is Abelian, so its irreducible representations are one-dimensional and given by characters $\mathbf{a} \mapsto e^{i\mathbf{k}\cdot\mathbf{a}}$.

A simultaneous eigenfunction of $\hat{\mathcal{H}}$ and $T_{\mathbf{a}}$ therefore satisfies $\mathbf{f}(\mathbf{r} + \mathbf{a}) = e^{i\mathbf{k}\cdot\mathbf{a}}\mathbf{f}(\mathbf{r})$. Writing $\mathbf{f}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\mathbf{u}(\mathbf{r})$, this condition becomes $\mathbf{u}(\mathbf{r} + \mathbf{a}) = \mathbf{u}(\mathbf{r})$: $\mathbf{u}$ is cell-periodic. $\square$

4.2.2 The Brillouin zone

The Bloch wavevector $\mathbf{k}$ is defined modulo reciprocal lattice vectors $\mathbf{G}$ [34]. Two wavevectors $\mathbf{k}$ and $\mathbf{k} + \mathbf{G}$ label the same physical mode (with a different choice of cell-periodic part), so $\mathbf{k}$ can be restricted to a fundamental domain of the reciprocal lattice—the *first Brillouin zone* BZ.

For a two-dimensional square lattice with period a , the reciprocal lattice vectors are $\mathbf{G} = \frac{2\pi}{a}(m_1\hat{x} + m_2\hat{y})$ and the Brillouin zone is the square $|k_x|, |k_y| \leq \pi/a$. For a hexagonal lattice (which will be central to the Dirac-cone physics of Chapter 6), the Brillouin zone is a regular hexagon.

Remark 4.2 (Topology of the Brillouin zone). The identification $\mathbf{k} \sim \mathbf{k} + \mathbf{G}$ means that the Brillouin zone is topologically a *torus* T^2 in two dimensions (or T^3 in three). This closed, boundary-free topology is essential for the quantisation of the Chern number: the integral of the Berry curvature over a torus is always an integer multiple of 2π (Chapter 5).

4.3 The cell-periodic eigenvalue problem

Substituting the Bloch form (4.2) into the Maxwell eigenvalue equation $\hat{\mathcal{N}}\mathbf{f} = \omega\mathbf{M}\mathbf{f}$ and noting that $\nabla(e^{i\mathbf{k}\cdot\mathbf{r}}\mathbf{u}) = e^{i\mathbf{k}\cdot\mathbf{r}}(\nabla + i\mathbf{k})\mathbf{u}$, we obtain the *cell-periodic eigenvalue problem*:

$$\boxed{\hat{\mathcal{N}}(\mathbf{k})\mathbf{u}_{n\mathbf{k}}(\mathbf{r}) = \omega_{n\mathbf{k}}\mathbf{M}(\mathbf{r})\mathbf{u}_{n\mathbf{k}}(\mathbf{r})}, \quad (4.5)$$

where the $\mathbf{k}$ -dependent curl operator is

$$\hat{\mathcal{N}}(\mathbf{k}) = \begin{pmatrix} \mathbf{0} & i(\nabla + i\mathbf{k})\times \\ -i(\nabla + i\mathbf{k})\times & \mathbf{0} \end{pmatrix}. \quad (4.6)$$

This operator acts on cell-periodic functions and is parametrised by $\mathbf{k}$.

The inner product for (4.5) is the electromagnetic inner product restricted to one unit cell Ω :

$$\langle \mathbf{u}_1, \mathbf{u}_2 \rangle_{\mathbf{M}, \Omega} = \int_{\Omega} \mathbf{u}_1^\dagger(\mathbf{r}) \mathbf{M}(\mathbf{r}) \mathbf{u}_2(\mathbf{r}) d^d r. \quad (4.7)$$

By the self-adjointness of $\hat{\mathcal{N}}(\mathbf{k})$ under periodic boundary conditions (Theorem 3.1), the eigenfrequencies $\omega_{n\mathbf{k}}$ are real and the cell-periodic eigenmodes are orthogonal:

$$\langle \mathbf{u}_{n\mathbf{k}}, \mathbf{u}_{m\mathbf{k}} \rangle_{\mathbf{M}, \Omega} = \delta_{nm}. \quad (4.8)$$

4.4 Band structure

For each $\mathbf{k}$ in the Brillouin zone, the cell-periodic eigenvalue problem (4.5) has a discrete set of eigenfrequencies

$$0 < \omega_{1\mathbf{k}} \leq \omega_{2\mathbf{k}} \leq \omega_{3\mathbf{k}} \leq \cdots . \quad (4.9)$$

As $\mathbf{k}$ varies continuously over the Brillouin zone, each $\omega_{n\mathbf{k}}$ traces out a continuous curve—the n -th *photonic band*. The collection of all bands is the *photonic band structure* [39–41].

4.4.1 Band gaps

A *photonic band gap* is a frequency range $(\omega_{\text{low}}, \omega_{\text{high}})$ in which no eigenfrequency exists for any $\mathbf{k}$ in the Brillouin zone:

$$\max_{\mathbf{k}} \omega_{n\mathbf{k}} < \omega_{\text{low}} < \omega_{\text{high}} < \min_{\mathbf{k}} \omega_{(n+1)\mathbf{k}} \quad (4.10)$$

The displayed formula above follows the convention and normalisation used in [31]. The existence and width of band gaps depend on the contrast of $\mathbf{M}(\mathbf{r})$ between different regions of the unit cell and on the lattice geometry. High dielectric contrast and connected high- ε regions generally favour wider gaps.

Band gaps are essential for topological photonics: the Chern number of a set of bands is well defined only when those bands are separated from all others by gaps. The topological edge states live inside these gaps.

4.4.2 Group velocity and energy velocity

The group velocity of the n -th band is

$$\mathbf{v}_{g,n}(\mathbf{k}) = \nabla_{\mathbf{k}} \omega_{n\mathbf{k}} \quad (4.11)$$

The displayed formula above follows the convention and normalisation used in [32]. This can be evaluated using the Hellmann–Feynman theorem [42]. Differentiating the eigenvalue equation $\hat{\mathcal{N}}(\mathbf{k})\mathbf{u}_n = \omega_n \mathbf{M}\mathbf{u}_n$ with respect to k_α and taking the inner product with $\mathbf{u}_n$:

$$\frac{\partial \omega_n}{\partial k_\alpha} = \frac{\int_{\Omega} \mathbf{u}_n^\dagger \frac{\partial \hat{\mathcal{N}}(\mathbf{k})}{\partial k_\alpha} \mathbf{u}_n \, d^d r}{\int_{\Omega} \mathbf{u}_n^\dagger \mathbf{M} \mathbf{u}_n \, d^d r} . \quad (4.12)$$

The numerator involves the k_α -derivative of $\hat{\mathcal{N}}(\mathbf{k})$, which from (4.6) has the structure of a “velocity operator” coupling $\mathbf{E}$ and $\mathbf{H}$ components.

The *energy velocity* is defined as the ratio of time-averaged Poynting flux to time-averaged energy:

$$\mathbf{v}_E = \frac{\int_{\Omega} \langle \mathbf{S} \rangle \, d^d r}{\int_{\Omega} \langle u \rangle \, d^d r} . \quad (4.13)$$

In a lossless nondispersive medium, one can show that $\mathbf{v}_g = \mathbf{v}_E$ by combining the Hellmann–Feynman relation with Poynting’s theorem. In a dispersive medium, the equality $\mathbf{v}_g = \mathbf{v}_E$ still holds provided the energy density is computed using the dispersive metric $\mathbf{M}_g$.

4.5 TE and TM modes in two dimensions

For a two-dimensional photonic crystal with $\mathbf{M}$ independent of z and no propagation in the z -direction ($k_z = 0$), the six-component eigenproblem separates into two independent scalar problems.

4.5.1 TM polarisation

The TM modes have $\mathbf{E} = E_z \hat{z}$ and $\mathbf{H} = (H_x, H_y, 0)^T$. The curl equations reduce to a single scalar equation for E_z :

$$-(\nabla + i\mathbf{k}_{\parallel}) \cdot [\mu^{-1}(\mathbf{r}) (\nabla + i\mathbf{k}_{\parallel}) E_z(\mathbf{r})] = \frac{\omega^2}{c^2} \varepsilon_{zz}(\mathbf{r}) E_z(\mathbf{r}), \quad (4.14)$$

where $\mathbf{k}_{\parallel} = (k_x, k_y)$ and μ^{-1} is the 2×2 in-plane inverse permeability tensor.

For an isotropic nonmagnetic medium ($\mu = \mathbf{I}$), this simplifies to the standard TM eigenproblem:

$$-(\nabla + i\mathbf{k}_{\parallel})^2 E_z(\mathbf{r}) = \frac{\omega^2}{c^2} \varepsilon(\mathbf{r}) E_z(\mathbf{r}). \quad (4.15)$$

4.5.2 TE polarisation

The TE modes have $\mathbf{H} = H_z \hat{z}$ and $\mathbf{E} = (E_x, E_y, 0)^T$. The scalar equation is

$$-(\nabla + i\mathbf{k}_{\parallel}) \cdot [\varepsilon^{-1}(\mathbf{r}) (\nabla + i\mathbf{k}_{\parallel}) H_z(\mathbf{r})] = \frac{\omega^2}{c^2} \mu_{zz}(\mathbf{r}) H_z(\mathbf{r}). \quad (4.16)$$

4.5.3 Gyrotropic media: coupling through off-diagonal terms

When the permeability tensor has off-diagonal gyrotropic components as in (2.4), the inverse permeability

$$\mu_{\perp}^{-1} = \frac{1}{\mu_{\perp}^2 - \kappa^2} \begin{pmatrix} \mu_{\perp} & i\kappa \\ -i\kappa & \mu_{\perp} \end{pmatrix} \quad (4.17)$$

introduces first-order spatial derivatives through the off-diagonal gyrotropic elements. The correct operator form retains the full tensor contraction with spatially varying coefficients:

$$-D_i [(\mu^{-1})_{ij}(\mathbf{r}) D_j E_z] = \frac{\omega^2}{c^2} \varepsilon(\mathbf{r}) E_z, \quad D_i \equiv \partial_i + ik_i, \quad (4.18)$$

where $i, j \in \{x, y\}$ and summation over repeated indices is implied. For a uniform gyrotropic medium, the off-diagonal elements of $(\mu^{-1})_{ij}$ contribute through the commutator $D_i D_j - D_j D_i = 0$ (for constant $\mathbf{k}$), so the gyrotropic coupling vanishes. The physically important gyrotropic effect arises from the *spatial variation* of $(\mu^{-1})_{ij}(\mathbf{r})$ at material interfaces, where the differential operator does not commute with the position-dependent tensor. This spatially varying gyrotropic coupling breaks the symmetry $\mathbf{k} \rightarrow -\mathbf{k}$ and is responsible for the nonzero Berry curvature analysed in Chapter 7.

Remark 4.3. In the gyrotropic case, the TE and TM polarisations are not strictly decoupled when $\kappa \neq 0$ and the geometry mixes in-plane and out-of-plane components. However, for a 2D crystal with ε diagonal and μ of the form (2.4), the TM and TE sectors remain exactly separated at $k_z = 0$. This separation is exploited in all the standard topological photonic-crystal designs.

4.6 The plane-wave expansion

The most widely used method for computing photonic band structures is the plane-wave expansion (PWE) [3, 12, 34], in which the cell-periodic fields and material parameters are expanded in Fourier series over reciprocal lattice vectors.

4.6.1 Fourier expansion of fields and material parameters

Expand the cell-periodic field in a Fourier series:

$$\mathbf{u}_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}} \hat{\mathbf{u}}_{n\mathbf{k}}(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}}, \quad (4.19)$$

and similarly for the material parameters:

$$\varepsilon(\mathbf{r}) = \sum_{\mathbf{G}} \hat{\varepsilon}(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}}, \quad \mu^{-1}(\mathbf{r}) = \sum_{\mathbf{G}} \widehat{\mu^{-1}}(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}}. \quad (4.20)$$

The Fourier coefficients are computed from integrals over the unit cell:

$$\hat{\varepsilon}(\mathbf{G}) = \frac{1}{A_{\Omega}} \int_{\Omega} \varepsilon(\mathbf{r}) e^{-i\mathbf{G} \cdot \mathbf{r}} d^2r, \quad (4.21)$$

where A_{Ω} is the area of the 2D unit cell.

4.6.2 Matrix eigenvalue problem

Substituting the Fourier expansions into the TM eigenvalue equation (4.14), the convolution of material parameters and

$$\sum_{\mathbf{G}'} (\mathbf{k} + \mathbf{G})_i (\widehat{\mu^{-1}})_{ij}(\mathbf{G} - \mathbf{G}') (\mathbf{k} + \mathbf{G}')_j \hat{E}_z(\mathbf{G}') = \frac{\omega^2}{c^2} \sum_{\mathbf{G}'} \hat{\varepsilon}(\mathbf{G} - \mathbf{G}') \hat{E}_z(\mathbf{G}'), \quad (4.22)$$

where summation over repeated Cartesian indices $i, j \in \{x, y\}$ is implied. For a scalar (non-gyrotropic) inverse permeability, the tensor Fourier coefficient reduces to $(\widehat{\mu^{-1}})_{ij}(\mathbf{G}) = \widehat{\mu^{-1}}(\mathbf{G}) \delta_{ij}$ and the left-hand side simplifies to $(\mathbf{k} + \mathbf{G}) \cdot (\mathbf{k} + \mathbf{G}') \widehat{\mu^{-1}}(\mathbf{G} - \mathbf{G}')$. For gyrotropic media, the off-diagonal Fourier components $(\widehat{\mu^{-1}})_{xy}$ carry the topological information. In practice, the Fourier series is truncated to N_G reciprocal lattice vectors, yielding an $N_G \times N_G$ generalised Hermitian matrix eigenvalue problem at each $\mathbf{k}$ point. The eigenvalues $\omega_{n\mathbf{k}}^2/c^2$ give the band structure; the eigenvectors give the mode profiles. $\omega_{n\mathbf{k}}^2/c^2$ give the band structure; the eigenvectors give the mode profiles.

Remark 4.4 (Convergence). The PWE converges exponentially fast for smooth material profiles and algebraically for discontinuous profiles (e.g. circular rods). For the latter, careful treatment of the Fourier factorisation (the “inverse rule” of Li) is needed to avoid slow convergence [43]. We discuss numerical methods further in Chapter 10 and Appendix C.

4.7 Examples of photonic band structures

4.7.1 Square lattice of dielectric rods

Consider a two-dimensional square lattice (period a) of circular dielectric rods (radius r , permittivity ε_r) in air ($\varepsilon = 1$), with $\mu = 1$ everywhere. This is the standard textbook photonic crystal.

For TM polarisation, the lowest bands emerge from the folding of the free-photon dispersion $\omega = c|\mathbf{k} + \mathbf{G}|$ at the Brillouin-zone boundaries. When the dielectric contrast $\varepsilon_r/1$ is sufficiently large ($\varepsilon_r \gtrsim 8$ for $r/a \approx 0.2$), a *complete TM band gap* opens between the first and second bands.

A representative TM band structure for this system, computed by the plane-wave expansion method with $\varepsilon_r = 12$ and $r/a = 0.2$, shows a complete TM gap between the first and second bands spanning approximately 15% of the mid-gap frequency. The gap arises because the first band concentrates its electric field energy in the high-permittivity rods, while the second band is forced by orthogonality to concentrate in the low-permittivity air region, paying a higher electromagnetic energy cost. This concentration mechanism—known as the “dielectric” and “air” band distinction in the photonic crystal literature—is the spatial analog of the bonding/anti-bonding splitting in electronic band theory.

4.7.2 Hexagonal lattice and Dirac points

The hexagonal (triangular) lattice is particularly important for topological photonics. At the corners K and K' of the hexagonal Brillouin zone, the C_{6v} point-group symmetry can enforce *Dirac-point degeneracies*: pairs of bands that touch linearly, with a dispersion relation locally resembling that of massless relativistic fermions:

$$\omega_{\pm}(\mathbf{k}) \approx \omega_D \pm v_D |\delta \mathbf{k}|, \quad (4.23)$$

where $\delta \mathbf{k} = \mathbf{k} - \mathbf{k}_D$ is the displacement from the Dirac point, ω_D is the Dirac frequency, and v_D is the Dirac velocity.

These Dirac points are the starting point for topological gap opening: when time-reversal symmetry is broken by a gyrotropic perturbation, the degeneracy is lifted and the resulting gapped bands acquire nonzero Chern numbers. This is the subject of Chapters 6 and 7.

4.8 Symmetry and degeneracy

The symmetries of the photonic crystal—both spatial (point group, space group) and non-spatial (time reversal)—constrain the band structure and the allowed degeneracies.

4.8.1 Point-group symmetry

At high-symmetry points in the Brillouin zone, the little group (the subgroup of the point group that leaves $\mathbf{k}$ invariant) determines the allowed irreducible representations of the eigenmodes. Bands that belong to multi-dimensional representations are degenerate.

For example, at the Γ point ($\mathbf{k} = \mathbf{0}$) of a C_{6v} -symmetric hexagonal lattice, the modes transform under the full C_{6v} group and can be labelled by the irreducible representations $A_1, A_2, B_1, B_2, E_1, E_2$. The two-dimensional representations E_1 and E_2 correspond to doubly degenerate modes.

4.8.2 Time reversal and the dispersion relation

When the medium is time-reversal invariant (reciprocal, $\kappa = 0$), the Bloch eigenfrequencies satisfy

$$\omega_n(\mathbf{k}) = \omega_n(-\mathbf{k}). \quad (4.24)$$

This is a direct consequence of the time-reversal symmetry derived in Section 2.6.1: if $\mathbf{u}_{n\mathbf{k}}$ is an eigenmode at $\mathbf{k}$, then $\hat{\mathcal{T}}\mathbf{u}_{n\mathbf{k}}$ is an eigenmode at $-\mathbf{k}$ with the same frequency.

Breaking time-reversal symmetry ($\kappa \neq 0$) allows $\omega_n(\mathbf{k}) \neq \omega_n(-\mathbf{k})$. This asymmetry is directly visible in the band structure as a tilt or splitting at time-reversal-related points, and it is the physical origin of nonreciprocal wave propagation.

4.8.3 Inversion symmetry and band sticking

When the photonic crystal has inversion symmetry ($\mathbf{M}(-\mathbf{r}) = \sigma_z \mathbf{M}(\mathbf{r}) \sigma_z$), the eigenfrequencies satisfy $\omega_n(\mathbf{k}) = \omega_n(-\mathbf{k})$ *independently* of time-reversal symmetry. The combined effect of inversion and time-reversal symmetry can force additional degeneracies at high-symmetry points, which is important for the formation of Dirac points.

4.9 From bands to topology

The photonic band structure $\{\omega_n(\mathbf{k})\}$ tells us *where* light can propagate: inside bands, at the corresponding frequencies and wavevectors. But the band structure alone does not tell the full story. Two photonic crystals can have identical band structures—the same frequencies, the same gaps—and yet differ in a fundamental way that affects their interface physics.

The missing information is the *geometry of the eigenmodes* as functions of $\mathbf{k}$. As $\mathbf{k}$ traverses the Brillouin zone, the cell-periodic mode $\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$ rotates in the Hilbert space of cell-periodic fields. The rate and manner of this rotation are captured by the Berry connection and Berry curvature, which we derive in the next chapter.

The topological invariant—the Chern number—is an integer that characterises the *global* geometry of each band over the entire Brillouin zone [2]. It cannot be changed by any smooth deformation of the medium that does not close the band gap. When two media with different Chern numbers are brought together, the mismatch forces the existence of interface modes that cross the gap: these are the topological edge states.

4.10 Group velocity and the Maxwell Hellmann–Feynman theorem

The group velocity of a Bloch mode is the gradient of the band frequency with respect to the wavevector: $\mathbf{v}_g = \nabla_{\mathbf{k}} \omega_n(\mathbf{k})$. This can be computed directly from the eigenvalue equation by a Hellmann–Feynman argument.

Differentiating the Bloch eigenproblem $\hat{\mathcal{N}}(\mathbf{k})\mathbf{u}_{n\mathbf{k}} = \omega_{n\mathbf{k}}\mathbf{M}\mathbf{u}_{n\mathbf{k}}$ with respect to k_a and projecting onto $\mathbf{u}_{n\mathbf{k}}$ gives [12, 19]

$$v_{g,a} = \frac{\partial \omega_n}{\partial k_a} = \frac{\int_{\text{cell}} \mathbf{u}_n^\dagger (\partial_{k_a} \hat{\mathcal{N}}) \mathbf{u}_n d^3r}{\int_{\text{cell}} \mathbf{u}_n^\dagger \mathbf{M} \mathbf{u}_n d^3r}, \quad (4.25)$$

where the numerator uses the *unweighted* L^2 matrix element of the velocity operator $\partial_{k_a} \hat{\mathcal{N}}$ and the denominator is the energy normalisation $\int \mathbf{u}_n^\dagger \mathbf{M} \mathbf{u}_n d^3r$.

For a non-dispersive medium, this reduces to the ratio of the time-averaged Poynting flux to the energy density, as expected from energy-transport arguments. For dispersive media, the denominator must use the dispersive metric $\mathbf{M}_g$ instead of $\mathbf{M}$, giving the group velocity in terms of the Brillouin energy.

The Hellmann–Feynman formula is the starting point for the $\mathbf{k} \cdot \mathbf{p}$ perturbation theory used in Chapter 6 to derive the Dirac velocity at photonic Dirac points.

4.10.1 Density of states and van Hove singularities

The photonic density of states is defined as $\rho(\omega) = \sum_n \int_{\text{BZ}} \delta(\omega - \omega_n(\mathbf{k})) d^d k / (2\pi)^d$. In two dimensions, $\rho(\omega)$ has logarithmic van Hove singularities at frequencies where $\nabla_{\mathbf{k}} \omega_n = 0$ (band extrema and saddle points). These singularities are important for understanding the spontaneous emission rate of emitters in photonic crystals and for identifying the frequencies at which strong light–matter interaction can be expected [12].

Near a Dirac point (Chapter 6), the conical dispersion gives a density of states $\rho(\omega) \propto |\omega - \omega_D|/v_D^2$ that vanishes linearly, analogous to graphene. Inside a complete band gap, $\rho(\omega) = 0$ —no propagating states exist, and the gap frequency range is available for topological edge modes.

4.11 Conditions for a complete photonic band gap

A complete photonic band gap—a frequency range with no propagating modes for any polarisation and any wavevector direction—requires both sufficiently large dielectric contrast and the right lattice symmetry.

For a 2D photonic crystal, the TE and TM polarisations are decoupled (Section 4.5), and each polarisation has its own gap. A TE gap requires connected dielectric regions (veins), while a TM gap requires isolated high- ϵ regions (rods). A complete 2D gap (for both polarisations simultaneously) therefore requires a structure that is simultaneously connected and has isolated high- ϵ regions, which is possible in certain lattice geometries (e.g. the honeycomb lattice with thick veins and additional rods) [3].

For 3D photonic crystals, complete gaps are harder to achieve because the gap must survive for all propagation directions and both polarisations. Diamond and gyroid lattice symmetries provide the largest complete 3D gaps, reaching up to 25% of the mid-gap frequency for germanium ($\epsilon \approx 16$) at near-infrared wavelengths [13].

The existence of a complete gap is a prerequisite for all topological phenomena discussed in this book: the topological invariants are defined for isolated bands, and the edge modes exist in the gap. Without a gap, there is no topological protection.

4.12 Photonic band structure computation in practice

The computation of photonic band structures is a central practical skill for topological photonics. This section provides the computational context that is developed in full algorithmic detail in Appendix C.

4.12.1 The plane-wave expansion method

The plane-wave expansion (PWE) method expands the cell-periodic fields $\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$ in Fourier harmonics $e^{i\mathbf{G}\cdot\mathbf{r}}$, where $\mathbf{G}$ ranges over reciprocal lattice vectors. The Bloch eigenvalue problem becomes a matrix eigenvalue problem of dimension $6N_G$ (or $2N_G$ for a single polarisation in 2D), where N_G is the number of plane waves retained.

The convergence of the PWE method depends on the dielectric contrast: for moderate contrast ($\epsilon_r \lesssim 12$), convergence to 0.1% accuracy is typically achieved with $N_G \approx 100$ –200 plane waves. For high contrast ($\epsilon_r > 20$) or metallic structures, the Gibbs phenomenon at material interfaces can slow convergence, and supercell or finite-element methods may be preferable [3, 12].

4.12.2 High-symmetry paths and irreducible Brillouin zone

The band structure is conventionally plotted along paths connecting high-symmetry points of the Brillouin zone. For the hexagonal lattice used throughout this book, the high-symmetry points are $\Gamma = (0, 0)$, $M = (\pi/a, 0)$, and $K = (4\pi/(3a), 0)$, and the standard path is $\Gamma \rightarrow M \rightarrow K \rightarrow \Gamma$.

The irreducible Brillouin zone (IBZ) is the smallest region that generates the full BZ under the point-group symmetry operations. For the hexagonal lattice with C_{6v} symmetry, the IBZ is a triangle Γ – M – K , which is 1/12 of the full BZ. Computing band structures and Berry curvatures over the IBZ and extending by symmetry reduces the computational cost by a factor of 12 [13, 19].

4.13 Achieving a complete photonic band gap

A complete photonic band gap—a frequency range in which no propagating mode exists for any wavevector and any polarisation—is a prerequisite for confining light in all directions and for defining topological edge modes that are truly isolated from the bulk continuum.

4.13.1 Necessary conditions

For a 2D photonic crystal, a complete gap requires that the gap for TE modes and the gap for TM modes overlap in frequency. Since TE gaps favour connected dielectric regions (“veins”) while TM gaps favour isolated dielectric regions (“islands”), achieving a complete gap requires a structure that simultaneously satisfies both conditions. The triangular lattice of air holes in a high-index dielectric is a standard solution: the connected dielectric forms veins for the TE gap, while the isolated dielectric regions between the holes form effective islands for the TM gap [3, 12].

The dielectric contrast must exceed a minimum threshold for a complete gap to open. For the triangular lattice, this threshold is $\epsilon_{\max}/\epsilon_{\min} \gtrsim 4$ for TE modes and $\gtrsim 7$ for a complete (TE+TM)

gap. Silicon ($\epsilon_r \approx 11.7$) comfortably exceeds both thresholds, making it the standard material for photonic crystal devices at telecommunications wavelengths [13].

4.13.2 Gap maps and structural optimisation

A gap map plots the gap width as a function of a structural parameter (typically the hole radius r/a or the filling fraction f). The optimal gap width for the triangular air-hole lattice occurs near $r/a \approx 0.45, f \approx 0.73$, giving a complete gap of $\Delta\omega/\omega_{\text{mid}} \approx 20\%$.

For topological photonics, the relevant gap is not the largest complete gap but the gap that opens at a band crossing (Dirac point or quadratic touching) when a symmetry is broken. These topological gaps are typically much narrower (1–5%) because they arise from a perturbative splitting rather than from the fundamental photonic band gap [12, 21].

4.14 Photonic density of states and edge enhancement

The photonic density of states (DOS) $\rho(\omega) = \sum_n \int_{\text{BZ}} \delta(\omega - \omega_n(\mathbf{k})) d^2k / (2\pi)^2$ determines the local emission rate through Fermi's golden rule and is a central quantity for light–matter interaction in photonic crystals.

At a band edge, the group velocity $\mathbf{v}_g = \nabla_{\mathbf{k}} \omega_n$ vanishes and the DOS diverges as a van Hove singularity. In 2D, the singularity is logarithmic: $\rho(\omega) \sim -\ln|\omega - \omega_{\text{edge}}|$ for a quadratic band edge. At a Dirac point, the linear dispersion gives $\rho(\omega) \propto |\omega - \omega_D|$, which vanishes at the Dirac frequency—a pseudogap.

When a topological gap opens at the Dirac point (Chapter 7), the pseudogap is replaced by a true gap where $\rho(\omega) = 0$ for $|\omega - \omega_D| < |\kappa|$. The gap edges acquire logarithmic van Hove singularities, producing enhanced emission rates that can be exploited for cavity QED and lasing applications [3, 12].

4.14.1 Local density of states and edge enhancement

The local density of states (LDOS) $\rho(\omega, \mathbf{r}) = \sum_n |\mathbf{f}_n(\mathbf{r})|^2 \delta(\omega - \omega_n)$ depends on position and is directly measurable through near-field spectroscopy. At the boundary of a topological photonic crystal, the LDOS has a contribution from the edge mode that is absent in the bulk:

$$\rho_{\text{edge}}(\omega, \mathbf{r}) = \frac{|\psi_0(\mathbf{r})|^2}{2\pi v_g}, \quad \omega_D - |\kappa| < \omega < \omega_D + |\kappa|, \quad (4.26)$$

where $\psi_0(\mathbf{r})$ is the transverse edge-mode profile and v_g is the edge-mode group velocity. This edge-enhanced LDOS is promising for single-photon sources: an emitter placed near the edge of a topological crystal can emit predominantly into the unidirectional edge mode, with a Purcell factor proportional to $1/v_g$ [13, 19].

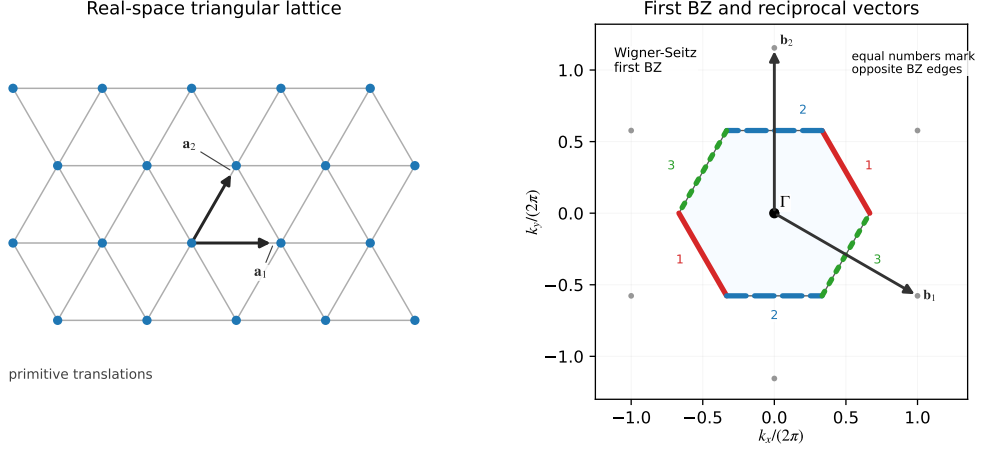


Figure 4.1: Real-space periodicity and reciprocal-space sewing. Bloch theory replaces translation in the photonic crystal by a cell-periodic field on the Brillouin-zone torus; opposite BZ boundaries are identified by reciprocal-lattice phase factors, an issue that becomes important in numerical Chern computations.

4.15 Photonic band structure in practice

The Bloch theory and plane-wave expansion method developed in this chapter provide the foundation for computing photonic band structures. This section discusses practical aspects of band-structure calculations that are important for topological photonics applications.

4.15.1 High-symmetry paths and the irreducible Brillouin zone

Band structures are conventionally plotted along high-symmetry paths connecting special points of the Brillouin zone. For the triangular lattice (the most common lattice in topological photonics), the standard path is $\Gamma \rightarrow M \rightarrow K \rightarrow \Gamma$, where $\Gamma = (0, 0)$, $M = \mathbf{b}_1/2$, and $K = (2\mathbf{b}_1 + \mathbf{b}_2)/3$ in reciprocal lattice coordinates.

The choice of path determines which band crossings and gaps are visible in the band diagram. In particular, Dirac cones at the K and K' points (Chapter 6) are visible only if the path passes through these points. A common mistake is to plot the band structure along $\Gamma \rightarrow X \rightarrow M \rightarrow \Gamma$ (the path for a square lattice), which misses the Dirac cones entirely [29, 34].

4.15.2 Convergence with respect to plane waves

The plane-wave expansion method represents the periodic part of the Bloch modes as a Fourier series truncated at $|\mathbf{G}| \leq G_{\max}$. The number of plane waves N_G scales as $G_{\max}^d$ in d dimensions, and the computational cost of diagonalisation scales as $O(N_G^3)$. For 2D photonic crystals, convergence of the lowest bands typically requires $N_G \sim 100\text{--}500$, corresponding to $G_{\max} \sim 5\text{--}10$ reciprocal lattice vectors.

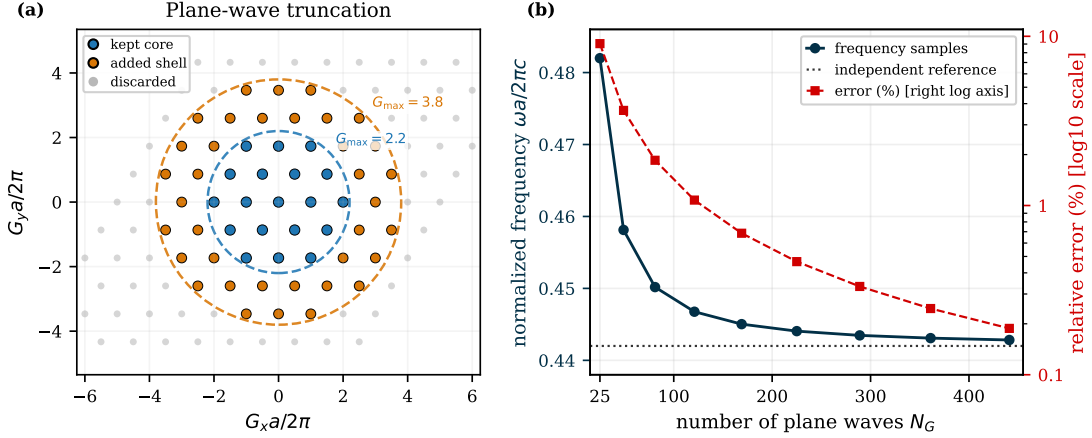


Figure 4.2: Plane-wave expansion in reciprocal space. (a) Reciprocal-lattice vectors of a triangular lattice, classified as kept core (blue), added convergence shell (orange), or discarded (grey); dashed circles mark two truncation radii $G_{\max}$. (b) Convergence of a normalised eigenfrequency as a function of the number of plane waves N_G . The right axis (red, $\log_{10}$ scale) shows the percent relative error with respect to the independently computed reference (dotted line). Each data point corresponds to a successive circular truncation shell in reciprocal space.

The convergence rate depends on the dielectric contrast $\varepsilon_{\max}/\varepsilon_{\min}$: higher contrast requires more plane waves because the field variations are sharper. For the YIG photonic crystals of Chapter 9 ($\varepsilon_{\text{rod}}/\varepsilon_{\text{air}} \approx 15$), convergence to $< 0.1\%$ accuracy in the eigenfrequencies requires $N_G \gtrsim 300$ [29, 44].

Figure 4.2 illustrates the two key aspects of the plane-wave expansion. Panel (a) shows the reciprocal-lattice vectors of a triangular lattice, partitioned into a kept core (blue), an added convergence shell (orange), and discarded higher-order vectors (grey). The dashed circles mark two values of the truncation radius $G_{\max}$; increasing $G_{\max}$ adds more plane waves but raises the matrix dimension. Panel (b) shows the resulting convergence of a representative normalised eigenfrequency $\omega a/2\pi c$ as the number of plane waves N_G grows from 25 to 441. The frequency approaches the independently computed reference value (dotted line) from above, while the relative error (red, right axis, $\log_{10}$ scale) decreases by more than two orders of magnitude. In practice, the discrete N_G values correspond to successive circular truncation shells of the reciprocal lattice.

4.15.3 Supercell method for edge states

To compute edge-state dispersions, one uses a supercell that is periodic in one direction (say x) and finite in the other (y). The supercell contains a strip of the photonic crystal with the topological material on one side and vacuum (or a trivially gapped crystal) on the other. The edge states appear as bands that cross the bulk gap.

The supercell width W must be large enough that the edge modes on opposite boundaries do not hybridise (Section 8.17). For a topological gap of relative width $\Delta\omega/\omega \sim 3\%$, a supercell of ~ 15 – 20 unit cells in the y -direction is typically sufficient [12, 40].

The supercell band structure provides the most direct verification of the bulk-edge correspondence: one counts the number of edge modes crossing the gap with positive group velocity and compares with the bulk Chern number computed from the methods of Chapter 10.

Exercises

Exercise 4.1. For a 1D photonic crystal (Bragg stack) with alternating layers of refractive indices n_1 and n_2 and thicknesses d_1 and d_2 ($d_1 + d_2 = a$), derive the dispersion relation $\cos(ka) = \cos(n_1 \omega d_1 / c) \cos(n_2 \omega d_2 / c) - \frac{1}{2}(\frac{n_1}{n_2} + \frac{n_2}{n_1}) \sin(n_1 \omega d_1 / c) \sin(n_2 \omega d_2 / c)$ using the transfer-matrix method. Identify the band gaps.

Exercise 4.2. Show that the cell-periodic eigenvalue problem (4.5) at $\mathbf{k}$ and $\mathbf{k} + \mathbf{G}$ are unitarily related: $\mathbf{u}_{n,\mathbf{k}+\mathbf{G}}(\mathbf{r}) = e^{-i\mathbf{G}\cdot\mathbf{r}}\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$. Conclude that the band structure is periodic in $\mathbf{k}$.

Exercise 4.3. Derive the Hellmann–Feynman formula (4.12) for the group velocity by differentiating the eigenvalue equation with respect to k_α and using the orthonormality of modes. Show that in a nondispersive medium the result agrees with the energy-velocity formula (4.13).

Exercise 4.4. For the TM modes of a 2D square lattice of circular rods, write out the plane-wave expansion matrix (4.22) for the case $N_G = 9$ (including $\mathbf{G} = \frac{2\pi}{a}(m_1, m_2)$ with $m_1, m_2 \in \{-1, 0, 1\}$). Compute the Fourier coefficient $\hat{\varepsilon}(\mathbf{G})$ for a circular rod of radius r and permittivity ε_r in a unit cell of side a .

Exercise 4.5. At a Dirac point in a hexagonal lattice, the two degenerate modes belong to a two-dimensional representation of C_{3v} (the little group at K). Using group-theoretic arguments, explain why this degeneracy is symmetry-protected and cannot be lifted without breaking C_{3v} or time-reversal symmetry.

Chapter 5

Geometry of Maxwell Bands

5.1 Phase geometry in parameter space

The photonic band structure $\omega_n(\mathbf{k})$ is an eigenvalue problem parametrised by the Bloch wavevector $\mathbf{k}$. As $\mathbf{k}$ changes continuously, the eigenfrequency ω_n changes smoothly (unless bands cross), and the eigenmode $\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$ also evolves smoothly—up to a *phase*.

The eigenvalue equation determines $\mathbf{u}_{n\mathbf{k}}$ only up to a $\mathbf{k}$ -dependent phase factor: $\mathbf{u}_{n\mathbf{k}} \rightarrow e^{i\chi_n(\mathbf{k})}\mathbf{u}_{n\mathbf{k}}$ is equally valid for any smooth real function $\chi_n(\mathbf{k})$. This is a gauge freedom, precisely analogous to the gauge freedom of the vector potential in electromagnetism.

The Berry connection, Berry curvature, and Chern number are the mathematical objects that describe the geometry of this gauge freedom [32, 44, 45]. They are the central tools of topological band theory. In this chapter, we derive them from the Maxwell eigenproblem using the electromagnetic inner product established in Chapter 3.

5.2 Berry connection for Maxwell modes

5.2.1 Definition

Consider a single nondegenerate band n with cell-periodic Bloch eigenmode $\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$, normalised so that $\langle \mathbf{u}_{n\mathbf{k}}, \mathbf{u}_{n\mathbf{k}} \rangle_{\mathbf{M}_g} = 1$ using the dispersive energy inner product (3.26).

Definition 5.1 (Electromagnetic Berry connection). The Berry connection of the n -th Maxwell band is the one-form on the Brillouin zone defined by

$$\mathcal{A}_n(\mathbf{k}) = i \frac{\int_{\Omega} \mathbf{u}_{n\mathbf{k}}^\dagger(\mathbf{r}) \mathbf{M}_g(\mathbf{r}, \omega_{n\mathbf{k}}) \nabla_{\mathbf{k}} \mathbf{u}_{n\mathbf{k}}(\mathbf{r}) d^d r}{\int_{\Omega} \mathbf{u}_{n\mathbf{k}}^\dagger(\mathbf{r}) \mathbf{M}_g(\mathbf{r}, \omega_{n\mathbf{k}}) \mathbf{u}_{n\mathbf{k}}(\mathbf{r}) d^d r}. \quad (5.1)$$

With our normalisation convention, the denominator equals unity and the Berry connection simplifies to

$$\mathcal{A}_n(\mathbf{k}) = i \int_{\Omega} \mathbf{u}_{n\mathbf{k}}^\dagger \mathbf{M}_g \nabla_{\mathbf{k}} \mathbf{u}_{n\mathbf{k}} d^d r. \quad (5.2)$$

Remark 5.2. In the nondispersive limit, $\mathbf{M}_g \rightarrow \mathbf{M}$ and the Berry connection reduces to $\mathcal{A}_n = i \int_{\Omega} \mathbf{u}_n^\dagger \mathbf{M} \nabla_{\mathbf{k}} \mathbf{u}_n d^d r$. In the energy-weighted representation $\mathbf{w}_n = \mathbf{M}_g^{1/2} \mathbf{u}_n$, it takes the familiar quantum-mechanical form $\mathcal{A}_n = i \int_{\Omega} \mathbf{w}_n^\dagger \nabla_{\mathbf{k}} \mathbf{w}_n d^d r$.

5.2.2 Reality of the Berry connection

Proposition 5.3. *The Berry connection $\mathcal{A}_n(\mathbf{k})$ is real-valued for each component α : $\mathcal{A}_{n,\alpha} \in \mathbb{R}$.*

Proof. Work with the energy-weighted Bloch function $\mathbf{w}_n(\mathbf{r}, \mathbf{k}) = \mathbf{M}_g^{1/2}(\mathbf{r}) \mathbf{u}_{n\mathbf{k}}(\mathbf{r})$ introduced in Section 3.3, which satisfies the standard normalisation $\int_{\Omega} \mathbf{w}_n^\dagger \mathbf{w}_n d^d r = 1$. From the definition (5.1), one can express the Berry connection entirely in terms of $\mathbf{w}_n$:

$$\mathcal{A}_{n,\alpha} = i \int_{\Omega} \mathbf{w}_n^\dagger(\mathbf{r}, \mathbf{k}) \nabla_{k_\alpha} \mathbf{w}_n(\mathbf{r}, \mathbf{k}) d^d r. \quad (5.3)$$

Now differentiate the normalisation condition with respect to k_α :

$$\nabla_{k_\alpha} \int_{\Omega} \mathbf{w}_n^\dagger \mathbf{w}_n d^d r = \nabla_{k_\alpha} (1) = 0. \quad (5.4)$$

Expanding the left-hand side:

$$\int_{\Omega} (\nabla_{k_\alpha} \mathbf{w}_n^\dagger \mathbf{w}_n + \mathbf{w}_n^\dagger \nabla_{k_\alpha} \mathbf{w}_n) d^d r = 0. \quad (5.5)$$

The second term is $-i\mathcal{A}_{n,\alpha}$ by (5.3), and its complex conjugate gives the first term as $+i\mathcal{A}_{n,\alpha}^*$. Therefore

$$i\mathcal{A}_{n,\alpha}^* - i\mathcal{A}_{n,\alpha} = 0, \quad \text{i.e.} \quad \mathcal{A}_{n,\alpha}^* = \mathcal{A}_{n,\alpha}. \quad (5.6)$$

Hence $\mathcal{A}_{n,\alpha} \in \mathbb{R}$ for every component α . $\square$

Remark 5.4. If one works directly with the cell-periodic eigenmode $\mathbf{u}_n$ and the $\mathbf{M}_g$ -weighted inner product, an extra term $\int \mathbf{u}_n^\dagger \nabla_{k_\alpha} \mathbf{M}_g \mathbf{u}_n d^d r$ appears from the $\mathbf{k}$ -dependence of $\mathbf{M}_g$ through $\omega_n(\mathbf{k})$. This term vanishes after absorbing the metric into the field via $\mathbf{w}_n = \mathbf{M}_g^{1/2} \mathbf{u}_n$. The energy-weighted formulation is therefore the natural setting for defining the Berry connection [19, 20], and is the convention we adopt throughout this book.

5.2.3 Gauge transformation

Under a gauge transformation $\mathbf{u}_{n\mathbf{k}} \rightarrow e^{i\chi_n(\mathbf{k})} \mathbf{u}_{n\mathbf{k}}$:

$$\mathcal{A}_n(\mathbf{k}) \rightarrow \mathcal{A}_n(\mathbf{k}) - \nabla_{\mathbf{k}} \chi_n(\mathbf{k}). \quad (5.7)$$

This is exactly the gauge transformation of the electromagnetic vector potential $\mathbf{A} \rightarrow \mathbf{A} - \nabla \Lambda$. The Berry connection is a *connection* on a $U(1)$ bundle over the Brillouin zone, and its gauge dependence is the defining property of a connection.

5.3 Berry curvature

Definition 5.5 (Berry curvature). The Berry curvature is the curl of the Berry connection:

$$\boxed{\Omega_n(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathcal{A}_n(\mathbf{k})}. \quad (5.8)$$

In two dimensions (the k_x - k_y plane), only the z -component is nonzero:

$$\Omega_n(\mathbf{k}) = \frac{\partial \mathcal{A}_{n,y}}{\partial k_x} - \frac{\partial \mathcal{A}_{n,x}}{\partial k_y}. \quad (5.9)$$

Proposition 5.6 (Gauge invariance). *The Berry curvature is gauge-invariant [43]:*

$$\Omega_n(\mathbf{k}) \rightarrow \Omega_n(\mathbf{k}) \quad \text{under } \mathbf{u}_n \rightarrow e^{i\chi} \mathbf{u}_n. \quad (5.10)$$

Proof. From (5.7), $\mathcal{A}_n \rightarrow \mathcal{A}_n - \nabla_{\mathbf{k}} \chi$. Taking the curl: $\nabla_{\mathbf{k}} \times (\mathcal{A}_n - \nabla_{\mathbf{k}} \chi) = \nabla_{\mathbf{k}} \times \mathcal{A}_n - \nabla_{\mathbf{k}} \times \nabla_{\mathbf{k}} \chi = \Omega_n - \mathbf{0} = \Omega_n$. $\square$

5.3.1 An explicit formula

The Berry curvature can also be written without taking derivatives of the eigenmode, using the second-order perturbation theory formula. For a nondegenerate band n :

$$\Omega_n(\mathbf{k}) = -2 \sum_{m \neq n} \frac{\text{Im}[\langle \mathbf{u}_n | \partial_{k_x} \hat{\mathcal{N}} | \mathbf{u}_m \rangle \langle \mathbf{u}_m | \partial_{k_y} \hat{\mathcal{N}} | \mathbf{u}_n \rangle]}{(\omega_m - \omega_n)^2}, \quad (5.11)$$

where the matrix elements $\langle \mathbf{u}_n | \partial_{k_\alpha} \hat{\mathcal{N}} | \mathbf{u}_m \rangle \equiv \int \mathbf{u}_n^\dagger (\partial_{k_\alpha} \hat{\mathcal{N}}) \mathbf{u}_m d^3r$ use the standard (unweighted) L^2 inner product, the modes are normalised so that $\int \mathbf{u}_n^\dagger \mathbf{M} \mathbf{u}_n d^3r = 1$, and $\partial_{k_\alpha} \hat{\mathcal{N}}$ is the velocity operator from the Hellmann–Feynman formula. For dispersive media, the normalisation should use $\mathbf{M}_g(\omega_n)$ in place of $\mathbf{M}$.

This “Kubo-like” formula shows that the Berry curvature is large when two bands are *close in frequency* (small $\omega_m - \omega_n$) and when the velocity-operator matrix element between them is large. Near a Dirac point, where two bands touch, the Berry curvature diverges as the gap is opened by a perturbation—this is the origin of the concentrated Berry flux near gapped Dirac cones.

5.4 Berry phase

The Berry phase is the line integral of the Berry connection around a closed loop C in the Brillouin zone:

$$\gamma_n(C) = \oint_C \mathcal{A}_n(\mathbf{k}) \cdot d\mathbf{k}. \quad (5.12)$$

By Stokes’ theorem (when the Berry connection is smooth inside the region S enclosed by C):

$$\gamma_n(C) = \int_S \Omega_n(\mathbf{k}) d^2k. \quad (5.13)$$

The Berry phase is gauge-invariant modulo 2π : under $\mathbf{u}_n \rightarrow e^{i\chi} \mathbf{u}_n$, $\gamma_n \rightarrow \gamma_n - [\chi]_C$, where $[\chi]_C$ is the winding of χ around C , which is an integer multiple of 2π for a single-valued gauge transformation.

5.5 The Chern number

Definition 5.7 (Chern number). The (first) Chern number of the n -th band is the integral of the Berry curvature over the entire Brillouin zone:

$$\mathcal{C}_n = \frac{1}{2\pi} \int_{\text{BZ}} \Omega_n(\mathbf{k}) d^2k. \quad (5.14)$$

5.5.1 Quantisation

Theorem 5.8 (Chern number is an integer). *When the n -th band is nondegenerate (separated from all other bands by gaps), $\mathcal{C}_n \in \mathbb{Z}$.*

Proof sketch. The Brillouin zone is a torus T^2 . On this compact manifold without boundary, the eigenmode $\mathbf{u}_{n\mathbf{k}}$ defines a $U(1)$ line bundle. The Chern number is the first Chern class of this bundle, which is always an integer by a topological theorem (the integrality of the first Chern class).

More concretely: cover the torus with two overlapping patches U_α and U_β . On each patch, choose a smooth gauge for $\mathbf{u}_n$. On the overlap $U_\alpha \cap U_\beta$, the two gauges are related by a transition function $g_{\alpha\beta}(\mathbf{k}) = e^{i\chi_{\alpha\beta}(\mathbf{k})}$. The Chern number equals the winding number of $g_{\alpha\beta}$ around the overlap region, which is necessarily an integer:

$$\mathcal{C}_n = \frac{1}{2\pi} \oint_{\partial U_\alpha} \nabla_{\mathbf{k}} \chi_{\alpha\beta} \cdot d\mathbf{k} \in \mathbb{Z}. \quad (5.15)$$

□

5.5.2 Properties

1. **Topological invariance.** $\mathcal{C}_n$ cannot change under any smooth deformation of the medium $\mathbf{M}(\mathbf{r}) \rightarrow \mathbf{M}(\mathbf{r}) + \delta\mathbf{M}(\mathbf{r})$ that does not close the band gap above or below band n . A change in $\mathcal{C}_n$ requires a gap closing (a topological phase transition).
2. **Sum rule.** The sum of Chern numbers over all bands is zero: $\sum_n \mathcal{C}_n = 0$. This follows from the fact that the total Berry curvature summed over all bands vanishes identically at every $\mathbf{k}$.
3. **Gap Chern number.** For a band gap between bands n and $n+1$, define the gap Chern number as $\mathcal{C}_{\text{gap}} = \sum_{j=1}^n \mathcal{C}_j$. This is the quantity that determines the number and direction of edge states in the gap.

5.6 Symmetry constraints on Berry curvature and Chern number

5.6.1 Time-reversal symmetry

Proposition 5.9. *If the Maxwell system is time-reversal invariant, the Berry curvature satisfies*

$$\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k}). \quad (5.16)$$

Proof. Time reversal maps $\mathbf{u}_{n\mathbf{k}}$ to $\hat{\mathcal{T}}\mathbf{u}_{n\mathbf{k}}$ at $-\mathbf{k}$. Using the explicit action $\hat{\mathcal{T}}\mathbf{u} = \sigma_z \mathbf{u}^*$ and the TR condition on $\mathbf{M}$ ((2.30)):

$$\begin{aligned} \mathcal{A}_{n,\alpha}(-\mathbf{k}) &= i \int (\hat{\mathcal{T}}\mathbf{u}_n)^\dagger \mathbf{M}_g \nabla_{(-k_\alpha)} (\hat{\mathcal{T}}\mathbf{u}_n) d^d r \\ &= i \int \mathbf{u}_n^T \sigma_z \mathbf{M}_g (-\nabla_{k_\alpha}) (\sigma_z \mathbf{u}_n^*) d^d r \\ &= -\mathcal{A}_{n,\alpha}(\mathbf{k}) + \nabla_{k_\alpha} \phi_n(\mathbf{k}), \end{aligned} \quad (5.17)$$

where ϕ_n is a gauge-dependent phase. Taking the curl eliminates ϕ_n and gives $\Omega_n(-\mathbf{k}) = -\Omega_n(\mathbf{k})$. □

Corollary 5.10. *Under time-reversal symmetry, $\mathcal{C}_n = 0$ [46].*

Proof. $\mathcal{C}_n = (2\pi)^{-1} \int_{\text{BZ}} \Omega_n(\mathbf{k}) d^2k$. Since the BZ is symmetric under $\mathbf{k} \rightarrow -\mathbf{k}$ and $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k})$, the integrand is odd and the integral vanishes. $\square$

This is the fundamental reason why *nonreciprocal media are required for nonzero photonic Chern numbers*. Without breaking time-reversal symmetry, the Berry curvature integrates to zero and there are no topologically protected one-way edge states.

5.6.2 Inversion symmetry

Under inversion ($\hat{\mathcal{P}}$):

$$\Omega_n(\mathbf{k}) = \Omega_n(-\mathbf{k}). \quad (5.18)$$

Combined with time reversal, $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k}) = \Omega_n(-\mathbf{k})$, so $\Omega_n(\mathbf{k}) = 0$ at every $\mathbf{k}$ —the Berry curvature vanishes identically when both $\hat{\mathcal{T}}$ and $\hat{\mathcal{P}}$ are present.

5.6.3 Breaking time reversal: the gyrotropic case

When the gyrotropic parameter $\kappa \neq 0$, time reversal is broken and the constraint $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k})$ no longer holds. The Berry curvature can be nonzero and can integrate to a nonzero Chern number. How this happens in detail—through the gap opening at Dirac points—is the subject of Chapter 6 and Chapter 7.

5.7 Non-Abelian Berry connection for degenerate bands

When p bands $n_1, \dots, n_p$ are degenerate or form an isolated group (separated from all other bands by gaps but possibly touching each other), the Berry connection becomes a matrix-valued one-form.

Definition 5.11 (Non-Abelian Berry connection). Define the $p \times p$ matrix

$$[\mathcal{A}_\alpha(\mathbf{k})]_{nm} = i \int_{\Omega} \mathbf{u}_{n\mathbf{k}}^\dagger \mathbf{M}_g \frac{\partial \mathbf{u}_{m\mathbf{k}}}{\partial k_\alpha} d^d r, \quad n, m \in \{n_1, \dots, n_p\}. \quad (5.19)$$

The non-Abelian Berry curvature is

$$\mathbf{F}_{\alpha\beta} = \frac{\partial \mathcal{A}_\beta}{\partial k_\alpha} - \frac{\partial \mathcal{A}_\alpha}{\partial k_\beta} + i[\mathcal{A}_\alpha, \mathcal{A}_\beta], \quad (5.20)$$

where the commutator term is absent in the Abelian (single-band) case.

The *composite Chern number* of the group of bands is

$$\mathcal{C}_{\text{composite}} = \frac{1}{2\pi} \int_{\text{BZ}} \text{Tr } \mathbf{F}_{xy} d^2k, \quad (5.21)$$

which is again an integer.

5.8 Wilson loops and the Zak phase

An alternative characterisation of band topology that avoids the need for a smooth gauge is the *Wilson loop* [6, 47, 48].

Definition 5.12 (Wilson loop). For a closed path C in the Brillouin zone, the Wilson loop is the path-ordered exponential of the non-Abelian Berry connection:

$$\mathcal{W}(C) = \mathcal{P} \exp \left(-i \oint_C \mathcal{A}(\mathbf{k}) \cdot d\mathbf{k} \right), \quad (5.22)$$

where $\mathcal{P}$ denotes path ordering.

For a single band, the Wilson loop reduces to $\mathcal{W} = e^{-i\gamma_n}$, where γ_n is the Berry phase.

The Zak phase. In one dimension, or for a path that crosses the Brillouin zone in one direction (say k_x from $-\pi/a$ to π/a at fixed k_y), the Berry phase of a single band is called the *Zak phase*:

$$\gamma_{\text{Zak}} = \int_{-\pi/a}^{\pi/a} \mathcal{A}_{n,x}(k_x) dk_x. \quad (5.23)$$

The Zak phase is quantised to 0 or π (modulo 2π) when the system has both time-reversal and inversion symmetry, and it characterises the one-dimensional topological phases discussed in Chapter 14.

Wilson-loop winding. For a set of bands, the eigenvalues of the Wilson loop matrix $\mathcal{W}(k_y)$ as a function of k_y trace out curves whose winding number equals the Chern number of the band group. This provides a numerically robust, gauge-invariant method for computing Chern numbers that we will exploit in Chapter 10.

5.9 Physical meaning of Berry curvature

The Berry curvature is not merely a mathematical construct; it has direct physical consequences.

Anomalous velocity. In a slowly varying potential, a wave packet in band n acquires a transverse velocity proportional to the Berry curvature:

$$\dot{\mathbf{r}} = \nabla_{\mathbf{k}} \omega_n + \dot{\mathbf{k}} \times \Omega_n(\mathbf{k}). \quad (5.24)$$

The second term is the *anomalous velocity*, analogous to the Lorentz force but arising from the momentum-space geometry rather than a real magnetic field [5]. It is the mechanism behind the photonic Hall effect and the transverse displacement of light beams in topological media.

Effective magnetic field in $\mathbf{k}$ -space. The Berry curvature acts as an effective magnetic field in momentum space. A concentrated “Berry monopole” at a band degeneracy point is the momentum-space analog of a magnetic monopole. The Chern number counts the net monopole charge enclosed by the Brillouin-zone torus.

Bulk-edge correspondence (preview). The most dramatic physical consequence of a nonzero Chern number is the *bulk-edge correspondence*: at an interface between two media with different gap Chern numbers $\mathcal{C}_{\text{gap}}^L$ and $\mathcal{C}_{\text{gap}}^R$, the number of unidirectional edge modes crossing the gap is

$$N_{\text{edge}} = \mathcal{C}_{\text{gap}}^L - \mathcal{C}_{\text{gap}}^R. \quad (5.25)$$

We will derive this correspondence in detail in Chapter 8.

5.10 Anomalous velocity and semiclassical dynamics

The Berry curvature has a direct dynamical consequence: it modifies the semiclassical equations of motion of a photonic wavepacket. For a wavepacket centred at wavevector $\mathbf{k}$ in band n and subject to a slowly varying perturbation (e.g. a spatial gradient of the band frequency), the equations of motion are [12]

$$\dot{\mathbf{r}} = \nabla_{\mathbf{k}} \omega_n(\mathbf{k}) - \dot{\mathbf{k}} \times \hat{\mathbf{z}} \Omega_n(\mathbf{k}), \quad (5.26)$$

$$\dot{\mathbf{k}} = -\nabla_{\mathbf{r}} \omega_n(\mathbf{r}). \quad (5.27)$$

The first term in (5.26) is the conventional group velocity, and the second term is the *anomalous velocity*, proportional to the Berry curvature.

The anomalous velocity is perpendicular to the applied force $\dot{\mathbf{k}}$, just like the Hall velocity of an electron in a magnetic field. This is the photonic analog of the anomalous Hall effect: a photonic wavepacket in a band with nonzero Berry curvature drifts transversely to an applied frequency gradient.

For a uniform Berry curvature $\Omega_n = 2\pi \mathcal{C}_n / A_{\text{BZ}}$ (where A_{BZ} is the Brillouin-zone area), the transverse displacement after propagation through the crystal is $\Delta y = \mathcal{C}_n \cdot a$, where a is the lattice constant. This provides an alternative physical derivation of the bulk-edge correspondence: the transverse displacement pumps photons to the edge, where they must propagate as edge modes [2, 13].

5.11 Berry curvature of the two-band model

Many of the topological phenomena in this book can be illustrated using a generic two-band model. Consider the Hamiltonian

$$H(\mathbf{k}) = d_x(\mathbf{k}) \sigma_x + d_y(\mathbf{k}) \sigma_y + d_z(\mathbf{k}) \sigma_z, \quad (5.28)$$

where $d(\mathbf{k}) = (d_x, d_y, d_z)$ is a three-component vector that depends on $\mathbf{k}$. The eigenvalues are $E_{\pm} = \pm |d|$, and the Berry curvature of the lower band is

$$\Omega_{-}(\mathbf{k}) = -\frac{1}{2} \frac{d \cdot (\partial_{k_x} d \times \partial_{k_y} d)}{|d|^3}. \quad (5.29)$$

This formula has a beautiful geometric interpretation: the Chern number $\mathcal{C}_{-} = \frac{1}{2\pi} \int \Omega_{-} d^2k$ counts the number of times the unit vector $\hat{d}(\mathbf{k}) = d/|d|$ wraps around the unit sphere as $\mathbf{k}$ sweeps over the

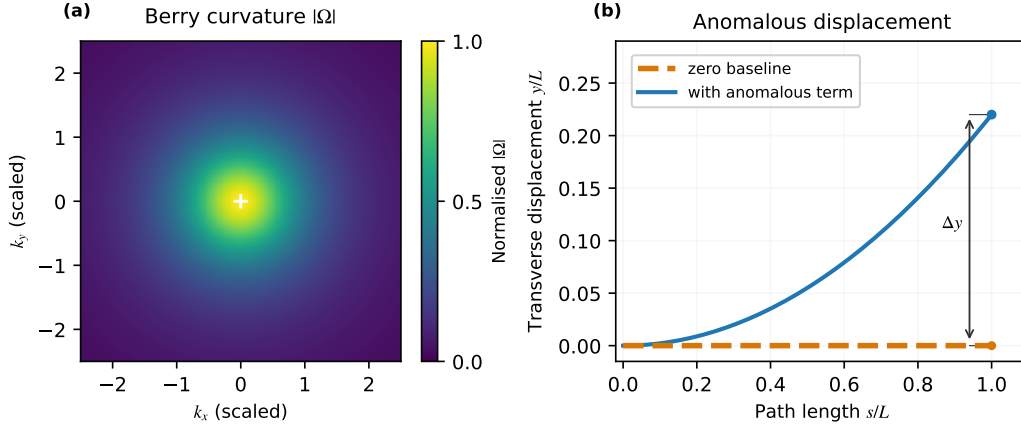


Figure 5.1: Berry curvature and the anomalous velocity. **(a)** Schematic normalised Berry curvature $|\Omega(\mathbf{k})|$ near a Dirac valley, showing a sharp peak where the gap is smallest (white marker). The physical (un-normalised) curvature is gauge-invariant and integrates over the Brillouin zone to $2\pi\mathcal{C}_n$, with $\mathcal{C}_n$ the band Chern number. **(b)** Transverse displacement of a photonic wavepacket as a function of propagation path length. The dashed orange line shows the trajectory without the anomalous term (zero transverse drift); the solid blue line includes the Berry-curvature-driven anomalous velocity (5.26), which accumulates a net transverse displacement Δy proportional to the integrated Berry curvature. This displacement is the semiclassical origin of the photonic Hall drift discussed in the text.

Brillouin zone. This is the *Pontryagin index* or *skyrmion number* of the mapping $\hat{d} : T^2 \rightarrow S^2$ [3, 12, 49].

For the massive Dirac model of Chapter 7, $d = (v_D \delta k_x, v_D \delta k_y, \kappa)$, and the wrapping number of $\hat{d}$ at each Dirac cone is $\pm \frac{1}{2}$ (half a skyrmion), consistent with the half-integer Chern flux computed in Section 7.8.2.

5.12 Wilson loops and the non-Abelian Berry phase

5.12.1 Definition and gauge invariance

For a group of p bands that may touch each other but are separated from all other bands by gaps, the non-Abelian Berry connection $[\mathcal{A}_a]_{mn} = i\langle \mathbf{u}_m, \partial_{k_a} \mathbf{u}_n \rangle_{\mathbf{M}_g}$ is a $U(p)$ -valued 1-form. The Wilson loop along a closed path γ in the Brillouin zone is the path-ordered exponential

$$W[\gamma] = \mathcal{P} \exp\left(-i \oint_{\gamma} \mathcal{A} \cdot d\mathbf{k}\right), \quad (5.30)$$

which is a $p \times p$ unitary matrix. The eigenvalues of W are gauge-invariant and physically observable: they are the non-Abelian Berry phases acquired by the p -band multiplet on the loop γ .

5.12.2 Relation to polarisation and Chern number

When γ is a straight line across the Brillouin zone at fixed $k_{\perp}$, the Wilson-loop eigenvalues $\theta_j(k_{\perp})$ define the “Wannier band structure.” Their winding as $k_{\perp}$ varies across the BZ gives the Chern number of the band group:

$$\mathcal{C} = \sum_{j=1}^P \frac{1}{2\pi} \oint \frac{d\theta_j}{dk_{\perp}} dk_{\perp}. \quad (5.31)$$

This provides a numerically robust alternative to the plaquette algorithm (Chapter 10), especially for nearly-degenerate band groups where the Abelian Berry curvature is ill-defined but the non-Abelian Wilson loop is smooth [13, 19].

5.13 Symmetry constraints on Chern numbers

5.13.1 Time-reversal symmetry

Under time reversal $\hat{\mathcal{T}}$, the Berry curvature satisfies $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k})$. Integrating over the full BZ (which is symmetric under $\mathbf{k} \rightarrow -\mathbf{k}$) gives $\mathcal{C}_n = 0$. This is the fundamental reason why breaking time-reversal symmetry is necessary for a nonzero Chern number in photonic crystals [3, 12].

5.13.2 Inversion symmetry

Under spatial inversion $\mathcal{P}$, $\Omega_n(\mathbf{k}) = \Omega_n(-\mathbf{k})$. Combined with time reversal, this gives $\Omega_n(\mathbf{k}) = -\Omega_n(\mathbf{k}) = 0$ everywhere, so the Berry curvature vanishes identically (not just in integral). This stronger constraint means that breaking inversion alone, while producing valley Berry curvature (Chapter 12), cannot produce a Chern insulator.

5.13.3 Particle-hole and chiral symmetries

The Maxwell eigenproblem has an intrinsic “particle-hole” symmetry: if $(\omega, \mathbf{u})$ is an eigenmode, so is $(-\omega, \sigma_3 \mathbf{u})$ where $\sigma_3 = \text{diag}(\mathbf{I}_3, -\mathbf{I}_3)$. This symmetry constrains the Chern numbers of positive- and negative-frequency bands to be related by $\mathcal{C}_n(-\omega) = -\mathcal{C}_n(\omega)$, ensuring that the total Chern number summed over all bands vanishes.

5.14 Adiabatic transport and the Thouless pump

The Berry phase has a direct physical manifestation in adiabatic transport: when a parameter of the photonic crystal is slowly varied through a cyclic path, the photonic modes acquire Berry phases that produce a quantised transport of energy.

5.14.1 The photonic Thouless pump

Consider a 1D photonic crystal whose unit cell has a parameter ϕ that varies adiabatically from 0 to 2π . As ϕ completes one cycle, each band transports a quantised amount of electromagnetic energy

across the crystal. The transported energy per cycle is

$$\Delta W_n = \hbar \omega_n \mathcal{C}_n^{(1D)}, \quad (5.32)$$

where $\mathcal{C}_n^{(1D)} = \frac{1}{2\pi} \int_0^{2\pi} d\phi \int_{-\pi/a}^{\pi/a} dk \Omega_n(k, \phi)$ is the Chern number computed over the (k, ϕ) parameter space. This is the photonic analog of the Thouless charge pump in electronic systems [2, 21].

The Thouless pump provides a physical interpretation of the Chern number: it counts the number of edge-to-edge transfers per pump cycle. This interpretation makes the integrality of the Chern number physically obvious: the number of transferred quanta must be an integer because the system returns to its initial state after one cycle.

5.14.2 Connection to the bulk-edge correspondence

The Thouless pump also provides an alternative derivation of the bulk-edge correspondence. If the pump parameter ϕ is reinterpreted as the wavevector k_y in the transverse direction of a 2D crystal, the pump cycle $\phi \in [0, 2\pi]$ corresponds to traversing the transverse Brillouin zone. The number of edge-to-edge transfers per pump cycle then equals the number of chiral edge modes in the 2D crystal, which is the gap Chern number [12, 20].

This pump-based argument is more intuitive than the formal index theorem (Chapter 8) and provides a constructive proof: one can explicitly track how the edge modes are “pumped” across the crystal as k_y varies.

5.15 Berry curvature dipole and nonlinear Hall effects

In systems with broken inversion symmetry but preserved time-reversal symmetry (such as valley photonic crystals), the Berry curvature $\Omega_n(\mathbf{k})$ is an odd function of $\mathbf{k}$ and integrates to zero over the full Brillouin zone. However, the Berry curvature *dipole*

$$D_a = \int_{\text{BZ}} \frac{\partial \Omega_n}{\partial k_a} f_n(\mathbf{k}) \frac{d^2 k}{(2\pi)^2} \quad (5.33)$$

(where f_n is the occupation function) can be nonzero and produces a nonlinear Hall effect: a transverse current proportional to the square of the applied field. In photonics, this manifests as a second-harmonic generation of valley-polarised edge modes, providing a nonlinear probe of the valley Berry curvature [13, 50].

5.16 Experimental probes of Berry curvature

The Berry curvature $\Omega_n(\mathbf{k})$ is not merely a theoretical construct: it has measurable consequences that can be observed experimentally in photonic systems.

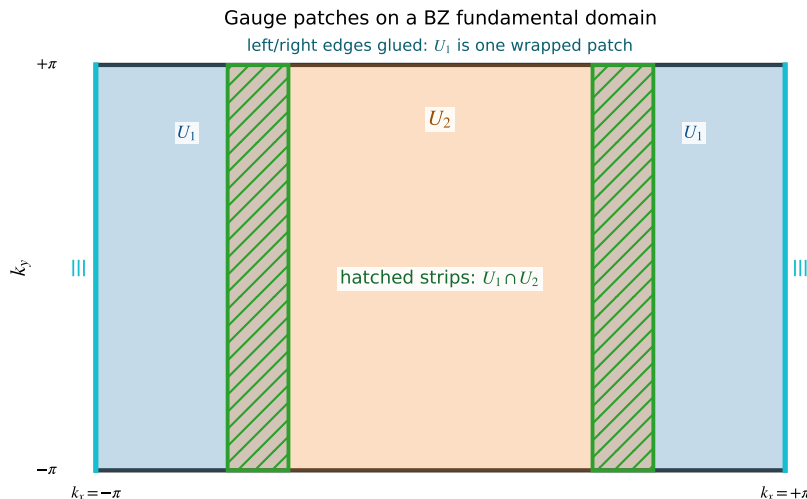


Figure 5.2: The Berry bundle viewpoint. A photonic Bloch mode can be smooth only patchwise over the Brillouin torus; on overlaps the modes differ by a transition phase $e^{i\chi(\mathbf{k})}$. The Chern number counts the winding of these transition phases and is therefore independent of the local gauge used to represent the mode.

5.16.1 Anomalous velocity and beam deflection

The semiclassical equations of motion for a photonic wave packet in band n include an anomalous velocity term proportional to the Berry curvature [12, 51]:

$$\dot{\mathbf{r}} = \nabla_{\mathbf{k}} \omega_n(\mathbf{k}) + \dot{\mathbf{k}} \times \hat{\mathbf{z}} \Omega_n(\mathbf{k}), \quad (5.34)$$

where $\dot{\mathbf{k}}$ is the rate of change of the crystal momentum, driven by an external perturbation (e.g. a refractive index gradient acting as a pseudoforce). The first term is the group velocity; the second term is the anomalous velocity, which deflects the wave packet transversely.

In a photonic crystal with nonzero Berry curvature, a beam propagating through a region with a gradient in the effective refractive index will be deflected perpendicular to the gradient. The deflection angle is proportional to Ω_n at the beam's central wavevector. By measuring the deflection as a function of $\mathbf{k}$, one can map out the Berry curvature over the Brillouin zone [20, 51].

This technique was demonstrated experimentally by Wimmer et al. (2017) in a photonic waveguide array, where the Berry curvature of a honeycomb lattice band structure was mapped by tracking the transverse displacement of light beams launched at different Bloch wavevectors [52].

5.16.2 Zak phase from reflection measurements

In one-dimensional photonic crystals, the Berry phase reduces to the Zak phase $\gamma_n = \oint \mathcal{A}_n(k) dk$, which takes values 0 or $\pi \pmod{2\pi}$ when inversion symmetry is present. The Zak phase determines the surface impedance of the crystal and can therefore be extracted from reflection measurements.

Specifically, for a semi-infinite 1D photonic crystal, the reflection phase $\phi_r(\omega)$ at a frequency within a bandgap changes by π when the Zak phase of the band below the gap changes from 0 to π . This phase jump can be observed experimentally by measuring the reflection spectrum with interferometric techniques [31, 48].

Xiao et al. (2014) used this approach to measure the Zak phase in a dielectric multilayer, confirming the predicted topological phase transition at the critical thickness ratio where the gap closes and reopens with inverted Zak phase [12, 21].

5.16.3 Quantum Hall analogy: quantised circulation

The most dramatic manifestation of nonzero Chern number is the existence of one-way edge states (Chapter 8). In finite photonic crystals, these edge states carry a quantised “photonic Hall conductance”: the transmission through the crystal (measured in units of edge channels) equals the gap Chern number.

This quantised transmission has been measured in microwave experiments [4] and provides the most direct experimental confirmation of a nonzero photonic Chern number. The quantisation is robust against disorder, sharp corners, and boundary roughness, directly demonstrating the topological nature of the edge transport [15, 53].

Exercises

Exercise 5.1. For a two-band system with Bloch modes $\mathbf{u}_1(\mathbf{k})$ and $\mathbf{u}_2(\mathbf{k})$, show that the Berry curvatures satisfy $\Omega_1(\mathbf{k}) + \Omega_2(\mathbf{k}) = 0$ at every $\mathbf{k}$.

Exercise 5.2. Verify the gauge transformation (5.7) by direct substitution of $\mathbf{u}_n \rightarrow e^{i\chi} \mathbf{u}_n$ into the Berry-connection formula (5.2).

Exercise 5.3. Derive the Kubo-like formula (5.11) for the Berry curvature using first-order perturbation theory. *Hint:* expand $\nabla_{k_\alpha} \mathbf{u}_n$ in the complete set $\{\mathbf{u}_m\}_{m \neq n}$ using the Hellmann–Feynman relation, then substitute into the Berry-curvature formula.

Exercise 5.4. Consider a generic 2×2 Hermitian Hamiltonian $H(\mathbf{k}) = d_0(\mathbf{k})\mathbf{I} + \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$, where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices. Show that the Berry curvature of the lower band is

$$\Omega_-(\mathbf{k}) = -\frac{1}{2} \hat{\mathbf{d}} \cdot \left(\frac{\partial \hat{\mathbf{d}}}{\partial k_x} \times \frac{\partial \hat{\mathbf{d}}}{\partial k_y} \right), \quad (5.35)$$

where $\hat{\mathbf{d}} = \mathbf{d}/|\mathbf{d}|$. Interpret this geometrically as the solid angle swept by $\hat{\mathbf{d}}$ per unit area in $\mathbf{k}$ -space.

Exercise 5.5 (Massive Dirac model). For the massive Dirac Hamiltonian $H(\mathbf{q}) = v(q_x \sigma_x + q_y \sigma_y) + m \sigma_z$, where $\mathbf{q}$ is measured from a Dirac point, compute the Berry curvature

$$\Omega_\pm(\mathbf{q}) = \pm \frac{m v^2}{2(v^2 |\mathbf{q}|^2 + m^2)^{3/2}} \quad (5.36)$$

and verify that $\int_{-\infty}^{\infty} \Omega_+ d^2 q = \pi \operatorname{sgn}(m)$. Explain why a single Dirac cone contributes $\pm \frac{1}{2}$ to the Chern number, and why a pair of cones at K and K' gives an integer.

Exercise 5.6. Show that the eigenvalues of the Wilson loop $\mathcal{W}(k_y) = \mathcal{P} \exp(-i \int_{-\pi/a}^{\pi/a} \mathcal{A}_x(k_x, k_y) dk_x)$ for a Chern insulator with $\mathcal{C} = 1$ wind once around the unit circle as k_y traverses the Brillouin zone.

Part II

Chern Bands and One-Way Waveguides

Chapter 6

Dirac Cones of Light

6.1 Why Dirac points matter

In the previous chapter we learned that a nonzero *Chern number* requires time-reversal symmetry to be broken. With time reversal preserved, the Berry curvature satisfies $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k})$ and can be locally nonzero—for example near valleys where inversion symmetry is broken—but its integral over the full Brillouin zone vanishes [54]. Concentrated sources of Berry flux somewhere in the Brillouin zone are therefore needed for any topological band. Where does this Berry flux come from?

The answer, in essentially all known photonic topological systems, is *Dirac points*: isolated degeneracies in the band structure where two bands touch with a linear (conical) dispersion. These degeneracies are the “monopoles” of Berry curvature in momentum space. When a perturbation gaps a Dirac point, the Berry flux that was compressed into the degeneracy spreads into a concentrated peak of Berry curvature, and the two split bands acquire equal and opposite contributions to their Chern numbers.

In this chapter we explain how Dirac points arise in two-dimensional photonic crystals from symmetry, derive their local dispersion from a two-mode reduction of the Maxwell eigenproblem, and prepare the ground for the topological gap opening of Chapter 7. The treatment follows the original analysis of Haldane and Raghu [3, 25] and the comprehensive review by Ozawa et al. [12], adapted to the Maxwell-first framework of this book.

6.2 Symmetry protection of Dirac degeneracies

6.2.1 The real-symmetric condition

Recall from Chapter 5 that when both time-reversal ($\hat{\mathcal{T}}$) and spatial inversion ($\hat{\mathcal{P}}$) symmetries are present, the Berry curvature vanishes identically:

$$\hat{\mathcal{T}} + \hat{\mathcal{P}} \implies \Omega_n(\mathbf{k}) = 0 \text{ for all } \mathbf{k}. \quad (6.1)$$

This means that we can choose a *real* gauge: the cell-periodic eigenmodes $\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$ can be made real-valued for all $\mathbf{k}$. The Maxwell eigenvalue problem then becomes a *real-symmetric* generalised eigenproblem rather than a complex-Hermitian one.

This distinction is crucial for degeneracies. In a complex-Hermitian eigenproblem, an “accidental” degeneracy between two bands requires the simultaneous vanishing of *three* real parameters (the real part, imaginary part, and diagonal difference of the off-diagonal matrix element). In two dimensions, we have only two parameters (k_x, k_y) , so a generic complex-Hermitian system has no band degeneracies in 2D.

In a *real-symmetric* eigenproblem, a degeneracy requires the vanishing of only *two* real parameters (the off-diagonal element and the diagonal difference). Two conditions in two unknowns (k_x, k_y) generically have isolated point solutions. These isolated degeneracies are the Dirac points.

To make this concrete, consider a generic 2×2 real-symmetric matrix

$$\mathbf{H}(k_x, k_y) = \begin{pmatrix} a(k_x, k_y) & b(k_x, k_y) \\ b(k_x, k_y) & c(k_x, k_y) \end{pmatrix}. \quad (6.2)$$

Its eigenvalues are $\lambda_{\pm} = \frac{1}{2}(a + c) \pm \frac{1}{2}\sqrt{(a - c)^2 + 4b^2}$, which become degenerate when $a = c$ and $b = 0$ simultaneously. These two equations in two unknowns (k_x, k_y) generically have isolated solutions—the Dirac points. If the matrix were complex Hermitian, the off-diagonal element would be $b_R + ib_I$, requiring three conditions ($a = c$, $b_R = 0$, $b_I = 0$) in two unknowns, which generically have no solution.

Remark 6.1. This argument, due to von Neumann and Wigner in the electronic context [12], explains why Dirac points are *generic* (symmetry-allowed, not fine-tuned) in 2D photonic crystals with both $\hat{\mathcal{T}}$ and $\hat{\mathcal{P}}$ symmetry. Breaking either symmetry removes the reality condition and generically gaps the degeneracy.

6.2.2 Hexagonal lattice and the K, K-prime points

The hexagonal (triangular) lattice, with point group C_{6v} , is the natural host for photonic Dirac points. The first Brillouin zone is a regular hexagon, whose six corners fall into two inequivalent sets: three K points and three K' points, related by time reversal ($K' = -K$ modulo reciprocal lattice vectors).

At each K point, the little group (the subgroup of C_{6v} that leaves $\mathbf{k}_K$ invariant) is C_{3v} , which has two-dimensional irreducible representations. Modes transforming under such a representation are doubly degenerate at K. When these degenerate modes belong to adjacent bands, they form a Dirac point.

Remark 6.2. The square lattice (C_{4v}) can also host Dirac points at the M point under appropriate conditions, but the hexagonal lattice provides the cleanest and most widely studied example. The connection to graphene—a hexagonal lattice of carbon atoms whose electronic structure has Dirac cones at K and K' —was the direct inspiration for Haldane and Raghu’s photonic construction [3, 55].

6.3 Nearly-free-photon analysis

Following Haldane and Raghu, we construct a photonic Dirac point analytically using a “nearly-free-photon” approximation—the photonic analog of the nearly-free-electron model in solid-state physics.

6.3.1 Setup

Consider a 2D photonic crystal with uniform isotropic permeability $\mu(\mathbf{r}) = \mu_0$ and a spatially modulated permittivity

$$\varepsilon(\mathbf{r}) = \bar{\varepsilon}(1 + \lambda V_G(\mathbf{r})), \quad (6.3)$$

where $\lambda \ll 1$ is the modulation strength and

$$V_G(\mathbf{r}) = \frac{2}{3} \sum_{n=1}^3 \cos(\mathbf{G}_n \cdot \mathbf{r}) \quad (6.4)$$

is a potential with the periodicity of a hexagonal lattice. Here $\mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3$ are three reciprocal lattice vectors of equal length $|\mathbf{G}| = 4\pi/(\sqrt{3}a)$, rotated by 120° relative to each other.

We work at $k_z = 0$, so the TE and TM polarisations decouple (Section 4.5). We specialise to the TE set $\{E_x, E_y, H_z\}$.

6.3.2 Free-photon dispersion at the Brillouin-zone corner

The six corners of the hexagonal Brillouin zone are at $\pm \mathbf{k}_{K_n}$, where $\mathbf{k}_{K_1} = (\mathbf{G}_2 - \mathbf{G}_3)/3$, etc., with $|\mathbf{k}_K| = |\mathbf{G}|/\sqrt{3}$. Since $\mathbf{k}_{K_2} - \mathbf{k}_{K_1} = \mathbf{G}_3$, the three wavevectors $\mathbf{k}_{K_1}, \mathbf{k}_{K_2}, \mathbf{k}_{K_3}$ are equivalent as Bloch vectors.

At $\lambda = 0$ (uniform medium), the TE mode at wavevector $\mathbf{k}$ has the free-photon dispersion $\omega = c_0|\mathbf{k}|$, where $c_0 = c/\sqrt{\bar{\varepsilon}}$. At the Brillouin-zone corner $\mathbf{k} = \mathbf{k}_K$, three free-photon plane waves (corresponding to $\mathbf{k}_K, \mathbf{k}_K + \mathbf{G}_1, \mathbf{k}_K + \mathbf{G}_2$, which all have the same $|\mathbf{k} + \mathbf{G}| = |\mathbf{k}_K|$) are degenerate at frequency $\omega_0 = c_0|\mathbf{k}_K|$.

6.3.3 Splitting by the periodic potential

Turning on the periodic modulation $\lambda \neq 0$ lifts the threefold degeneracy. In degenerate perturbation theory, we must diagonalise the 3×3 matrix of the perturbation $\delta \hat{\mathcal{N}} = -(\omega_D \lambda \bar{\varepsilon}/2) V_G(\mathbf{r})$ within the three-mode subspace at $\mathbf{k} = \mathbf{k}_K$. Denoting the three degenerate free-photon states as $|\alpha\rangle$, $\alpha = 1, 2, 3$, the coupling matrix has elements [3]

$$D_{\alpha\beta} = \frac{\omega_D \lambda}{2} \int_{\Omega} V_G(\mathbf{r}) \mathbf{u}_{\alpha}^*(\mathbf{r}) \cdot \bar{\varepsilon} \mathbf{u}_{\beta}(\mathbf{r}) d^2r, \quad (6.5)$$

where the integral runs over one unit cell Ω . The hexagonal symmetry C_{3v} acting on the three plane-wave modes permutes them cyclically, so $D_{\alpha\beta}$ inherits the structure

$$\mathbf{D} = \frac{\omega_D \lambda}{4} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}. \quad (6.6)$$

The eigenvalues of (6.6) are $-\omega_D \lambda/4$ (doubly degenerate) and $+\omega_D \lambda/2$ (non-degenerate). The threefold degeneracy therefore splits into a *Dirac doublet*, whose frequency is shifted down to

$$\omega_D = c_0|\mathbf{k}_K|(1 - \frac{\lambda}{4} + O(\lambda^2)), \quad (6.7)$$

and a *singlet*, shifted up to

$$\omega_s = c_0 |\mathbf{k}_K| \left(1 + \frac{\lambda}{2} + O(\lambda^2)\right), \quad (6.8)$$

separated from the doublet by a gap of order λ .

At momenta $\mathbf{k} = \mathbf{k}_K + \delta\mathbf{k}$ away from the zone corner, first-order $\mathbf{k} \cdot \mathbf{p}$ perturbation theory within the two-mode doublet gives the conical dispersion

$$\omega_{\pm}(\mathbf{k}) = \omega_D \pm v_D |\delta\mathbf{k}| + O(|\delta\mathbf{k}|^2), \quad (6.9)$$

with the Dirac velocity $v_D = c_0/2 + O(\lambda)$. This linear dispersion is the hallmark of a Dirac cone: it is locally identical to the spectrum of a massless two-dimensional relativistic particle.

6.3.4 Symmetry constraints on the coupling matrix

The simple structure of (6.6) is not accidental. The three reciprocal lattice vectors $\mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3$ transform into one another under the threefold rotation C_3 of the hexagonal lattice, and V_G respects this symmetry. Therefore the overlap integral $D_{\alpha\beta}$ depends only on whether α and β are equal or unequal: $D_{\alpha\alpha} = d_0$ and $D_{\alpha\neq\beta} = d_1$. The trace and off-diagonal parts can be removed by a constant shift, leaving the traceless matrix (6.6) whose double eigenvalue is protected by the two-dimensional irreducible representation E of C_{3v} .

Time-reversal symmetry enters via the reality constraint: because $\hat{\mathcal{T}}$ maps $\mathbf{k}_K$ to $\mathbf{k}_{K'} = -\mathbf{k}_K$ and simultaneously complex-conjugates the eigenmode, the coupling matrix at K is real-symmetric (consistent with (6.6)). This reality condition is what makes the degeneracy *robust*: perturbing the matrix within the space of real-symmetric matrices cannot split a twofold eigenvalue because real-symmetric 2×2 degeneracies have codimension 2 in two-parameter space, exactly matching the dimensionality of the Brillouin zone.

By contrast, if one breaks the C_3 symmetry while preserving the reality condition, the off-diagonal elements d_1 between different pairs of modes become unequal, generically splitting all three eigenvalues and removing the degeneracy entirely. The Dirac point is therefore protected specifically by the combination of C_3 rotation and time reversal, not by any single symmetry alone [12].

6.4 The two-mode Maxwell reduction

Near the Dirac point, the physics is captured by a 2×2 effective equation obtained by projecting the full Maxwell eigenproblem onto the two degenerate modes.

6.4.1 Projection onto the degenerate subspace

Let $|\mathbf{u}_+(\mathbf{k}_K)\rangle$ and $|\mathbf{u}_-(\mathbf{k}_K)\rangle$ be the two degenerate cell-periodic eigenmodes at the Dirac point, normalised so that $\langle \mathbf{u}_\sigma | \mathbf{M}_g(\omega_D) | \mathbf{u}_{\sigma'} \rangle = B_0 \delta_{\sigma\sigma'}$.

For $\mathbf{k} = \mathbf{k}_K + \delta\mathbf{k}$ with $|\delta\mathbf{k}| \ll |\mathbf{k}_K|$, first-order $\mathbf{k} \cdot \mathbf{p}$ perturbation theory gives the 2×2 effective eigenvalue equation

$$\boxed{v_D (\delta k_x \sigma_x + \delta k_y \sigma_y) |\psi\rangle = \delta \omega |\psi\rangle}, \quad (6.10)$$

where $\delta\omega = \omega - \omega_D$, σ_x and σ_y are Pauli matrices acting on the two-mode subspace, and $v_D > 0$ is the Dirac velocity determined by the matrix elements of the velocity operator between the two degenerate modes:

$$v_D = \frac{1}{B_0} |\langle \mathbf{u}_+ | \partial_{k_x} \hat{\mathcal{N}} | \mathbf{u}_- \rangle|. \quad (6.11)$$

6.4.2 Dirac dispersion and eigenstates

The eigenvalues of (6.10) are

$$\delta\omega_{\pm} = \pm v_D |\delta\mathbf{k}|, \quad (6.12)$$

reproducing the conical dispersion (6.9). The eigenstates are

$$|\psi_{\pm}(\delta\mathbf{k})\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm e^{i\phi_{\mathbf{k}}} \end{pmatrix}, \quad \phi_{\mathbf{k}} = \arg(\delta k_x + i \delta k_y). \quad (6.13)$$

As $\delta\mathbf{k}$ traces a circle around the Dirac point, the angle $\phi_{\mathbf{k}}$ winds by 2π and the eigenstate acquires a Berry phase of $\pm\pi$.

6.4.3 Berry curvature at a Dirac point

The Berry curvature of the gapless Dirac cone is concentrated at the degeneracy point. Using the Kubo formula (Eq. (5.11)), the Berry curvature of the upper band diverges as $|\delta\mathbf{k}| \rightarrow 0$:

$$\Omega_+(\delta\mathbf{k}) \sim \frac{1}{2|\delta\mathbf{k}|^2} \quad \text{as } |\delta\mathbf{k}| \rightarrow 0. \quad (6.14)$$

The total Berry flux through a small disc around the Dirac point is $\pm\pi$, which is the Berry phase acquired by the eigenstate on a loop encircling the degeneracy. However, this is a *gapless* system and the Chern number is not defined until the degeneracy is lifted—which is the subject of the next chapter.

6.5 Dirac points in realistic photonic crystals

The nearly-free-photon analysis gives the essential physics but requires $\lambda \ll 1$, where the Dirac-point frequency ω_D lies close to other bands and is not well isolated. In realistic photonic crystals with large dielectric contrast, the Dirac points can be frequency-isolated within a substantial gap of the opposite polarisation, making them suitable hosts for topological gap opening.

Haldane and Raghu noted that a hexagonal array of high-permittivity rods ($\epsilon_r = 14$, filling fraction $f = 0.431$) in air produces a band structure where the second and third TE bands touch at Dirac points at the Brillouin-zone corners K and K' , and the Dirac frequency ω_D lies within a large gap of the TM spectrum [3]. This frequency isolation is crucial: it ensures that the Dirac-point modes are not degenerate with any other $k_z = 0$ modes, so that the topological gap opening can proceed without interference from other bands.

6.6 Pairs of Dirac points and the fermion-doubling theorem

A fundamental property of Dirac points in a 2D Brillouin zone is that they always come in *pairs*: one at K and one at $K' = -K$. This is the photonic version of the Nielsen–Ninomiya (fermion-doubling) theorem, which states that the total chirality of all Dirac points in a compact Brillouin zone must vanish.

To see how Dirac-point pairing controls the Chern number, introduce a valley index $\tau = \pm 1$ labelling K ($\tau = +1$) and K' ($\tau = -1$). Near valley τ the effective Hamiltonian is

$$H_\tau(\delta\mathbf{k}) = v_D(\tau \delta k_x \sigma_x + \delta k_y \sigma_y) + \Delta_\tau \sigma_z, \quad (6.15)$$

where Δ_τ is the mass induced by a perturbation that opens a gap. The chirality factor τ multiplies δk_x : physically, K and K' carry *opposite* winding numbers. Both gapless cones have a Berry phase of π , but the opposite chiralities cause the two valleys to contribute to the Chern number with a relative sign. Explicitly, from the massive Dirac Berry curvature (derived in Section 7.4):

$$C_-^K = -\frac{\text{sgn}(\Delta_K)}{2}, \quad C_-^{K'} = +\frac{\text{sgn}(\Delta_{K'})}{2}, \quad (6.16)$$

so the total lower-band Chern number is

$$\mathcal{C}_- = \frac{\text{sgn}(\Delta_{K'}) - \text{sgn}(\Delta_K)}{2}. \quad (6.17)$$

Two physically distinct gap-opening mechanisms now follow.

Time-reversal breaking (Haldane/gyrotropic mass).—The perturbation gives *opposite*-sign masses at the two valleys, $\Delta_K = -\Delta_{K'}$, because the gyrotropic coupling picks up the chirality factor τ . The valley contributions reinforce: $\mathcal{C}_- = \pm 1$, and the system supports a chiral edge mode (Chapter 7).

Inversion breaking (Semenoff mass).—The perturbation gives *same*-sign masses, $\Delta_K = \Delta_{K'}$, and $\mathcal{C}_- = 0$. However, the half-integer valley contributions C_-^K and $C_-^{K'}$ are individually nonzero, defining a *valley Chern number* (Chapter 12).

This valley-indexed structure is the central organising principle for all of Parts II and III.

6.7 Derivation of the Dirac velocity

The Dirac velocity v_D appearing in (6.10) is not a phenomenological parameter. It is determined entirely by the Maxwell eigenmodes at the K point through the electromagnetic energy inner product. Deriving its explicit form from $\mathbf{k} \cdot \mathbf{p}$ perturbation theory within the Maxwell framework illustrates the power of the approach developed in Chapters 2–5.

6.7.1 The electromagnetic velocity operator

Starting from the Bloch-periodic Maxwell eigenproblem $\hat{\mathcal{N}}(\mathbf{k})\mathbf{u}_{n\mathbf{k}} = \omega_{n\mathbf{k}}\mathbf{M}_g\mathbf{u}_{n\mathbf{k}}$, differentiate both sides with respect to k_a ($a = x, y$). Projecting onto a different eigenmode $\mathbf{u}_{m\mathbf{k}}$ with $m \neq n$ and using the orthogonality $\langle \mathbf{u}_m, \mathbf{u}_n \rangle_{\mathbf{M}} = \delta_{mn}$ gives the electromagnetic Hellmann–Feynman relation

$$\frac{1}{B_0} \langle \mathbf{u}_m, \partial_{k_a} \hat{\mathcal{N}} \mathbf{u}_n \rangle_{\mathbf{M}} = (\omega_n - \omega_m) \langle \mathbf{u}_m, \mathbf{M}_g \partial_{k_a} \mathbf{u}_n \rangle_{\mathbf{M}}, \quad (6.18)$$

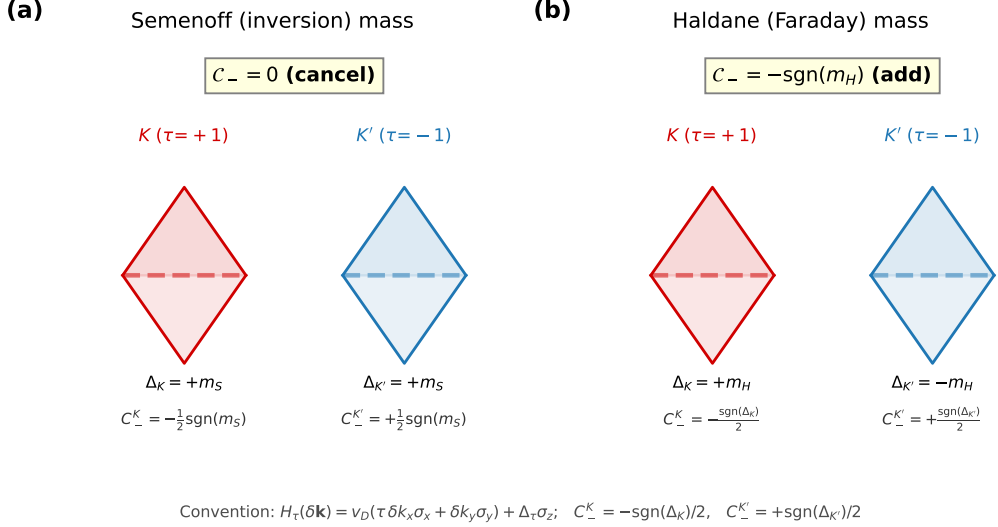


Figure 6.1: Valley Chern-number bookkeeping for the two gap-opening mechanisms. **(a)** Semenoff (inversion) mass: both valleys receive the same mass $\Delta_K = \Delta_{K'} = m_S$, so the valley contributions $C_-^K = -\text{sgn}(m_S)/2$ and $C_-^{K'} = +\text{sgn}(m_S)/2$ cancel, giving $\mathcal{C}_- = 0$. **(b)** Haldane (Faraday) mass: the gyrotropic coupling assigns opposite-sign masses $\Delta_K = +m_H$, $\Delta_{K'} = -m_H$, so the valley contributions reinforce to give $\mathcal{C}_- = -\text{sgn}(m_H)$. The convention follows (6.10) with $\tau_K = +1$, $\tau_{K'} = -1$.

where B_0 is the energy normalisation and $\partial_{k_a} \hat{\mathcal{N}}$ acts on the cell-periodic part of the Bloch wave. The left-hand side defines the matrix element of the velocity operator $\hat{v}_a \equiv \mathbf{M}_g^{-1} \partial_{k_a} \hat{\mathcal{N}}$, weighted by the energy metric [19]. This operator plays exactly the role of the momentum operator $\hat{p}/m$ in the electronic $\mathbf{k} \cdot \mathbf{p}$ theory.

6.7.2 Degenerate perturbation at the K point

At $\mathbf{k} = \mathbf{k}_K$ the two modes $\mathbf{u}_\pm$ are degenerate at ω_D , so the denominator in (6.18) vanishes and the non-degenerate formula cannot be used. Instead, expand the operator to first order in $\delta\mathbf{k}$:

$$\hat{\mathcal{N}}(\mathbf{k}_K + \delta\mathbf{k}) = \hat{\mathcal{N}}(\mathbf{k}_K) + \delta k_x \partial_{k_x} \hat{\mathcal{N}}|_K + \delta k_y \partial_{k_y} \hat{\mathcal{N}}|_K + O(|\delta\mathbf{k}|^2). \quad (6.19)$$

Within the two-mode subspace, the first-order perturbation matrix is

$$\mathcal{V}_{\sigma\sigma'}(\delta\mathbf{k}) = \frac{1}{B_0} \sum_{a=x,y} \delta k_a \langle \mathbf{u}_\sigma, \partial_{k_a} \hat{\mathcal{N}} \mathbf{u}_{\sigma'} \rangle_{\mathbf{M}}. \quad (6.20)$$

The threefold rotation C_3 at the K point cyclically permutes the basis modes and constrains the matrix elements. In particular, C_3 forces the diagonal elements to vanish at first order in $\delta\mathbf{k}$ (they contribute to the quadratic correction). The off-diagonal element is fixed by the requirement that $\mathcal{V}$

transforms as the two-dimensional E representation of C_{3v} , giving

$$\mathcal{V}_{+-} = v_D(\delta k_x - i \delta k_y), \quad \mathcal{V}_{-+} = v_D(\delta k_x + i \delta k_y), \quad (6.21)$$

where the Dirac velocity is the real, positive quantity

$$v_D = \frac{1}{B_0} |\langle \mathbf{u}_+, \partial_{k_x} \hat{\mathcal{N}} \mathbf{u}_- \rangle_{\mathbf{M}}|. \quad (6.22)$$

Substituting into $\mathcal{V} = \delta \omega \mathbf{I}$ recovers the massless Dirac equation (6.10) [42].

6.7.3 Numerical estimates

For a hexagonal lattice of high-permittivity dielectric rods ($\varepsilon_r = 12$, filling fraction $f = 0.3$) in air, a plane-wave expansion with 169 plane waves gives the Dirac frequency $\omega_D a / (2\pi c) \approx 0.36$ and a Dirac velocity $v_D \approx 0.45 c / \sqrt{\varepsilon}$ [3, 12]. The deviation from the nearly-free-photon prediction $v_D = c_0/2$ reflects the finite dielectric contrast, which distorts the mode profiles from pure plane waves and modifies the overlap integral in (6.22).

Reducing the rod permittivity toward $\varepsilon_r \rightarrow 1$ makes the modes more plane-wave-like and pushes v_D toward $c_0/2$, confirming the nearly-free-photon limit. Conversely, increasing the contrast beyond $\varepsilon_r \approx 15$ brings the Dirac frequency close to higher bands and eventually destroys the isolated Dirac cone, as the two-mode approximation breaks down.

6.7.4 Physical meaning of the Dirac velocity

The Dirac velocity v_D sets three experimentally accessible quantities. First, the group velocity of wavepackets near the Dirac frequency: $|\mathbf{v}_g| = v_D$ for all propagation directions, a consequence of the isotropic conical dispersion enforced by C_3 . Second, the density of states near ω_D scales as $\rho(\omega) \propto |\omega - \omega_D|/v_D^2$, vanishing linearly at the Dirac frequency—the photonic analog of the vanishing density of states in graphene. Third, when the Dirac cone is gapped by a mass κ (Chapter 7), the edge-mode group velocity inside the topological gap is approximately v_D , so the Dirac velocity directly determines the bandwidth and propagation speed of the one-way edge channel [12].

6.8 From Dirac cones to topological gaps: a summary

This chapter has established three results that form the foundation for Part II of the book.

The first result is that Dirac points are generic features of 2D photonic crystals with C_3 (or C_6) rotational symmetry and time-reversal invariance. The real-symmetric structure of the Maxwell eigenproblem under these symmetries reduces the codimension of band degeneracies from three (complex Hermitian) to two (real symmetric), matching the two-dimensional Brillouin zone and yielding isolated point degeneracies.

The second result is that the local physics near a Dirac point is captured by a 2×2 massless Dirac equation whose velocity v_D is determined by the Maxwell eigenmodes through the electromagnetic energy inner product. The derivation uses only degenerate perturbation theory applied to the Maxwell operator, with no electronic analogy.

The third result is the fermion-doubling constraint: Dirac points come in pairs (K and K') with opposite chirality $\tau = \pm 1$. Breaking time-reversal symmetry gives opposite-sign masses at the two valleys; the valley Chern contributions then add, producing $\mathcal{C} = \pm 1$ and guaranteeing a one-way edge mode. Breaking inversion symmetry instead gives same-sign masses, and the valley contributions cancel: $\mathcal{C} = 0$. The detailed mechanism of time-reversal breaking and gap opening is the subject of Chapter 7 [3, 13].

6.9 Trigonal warping and deviations from the ideal Dirac cone

The linear Dirac dispersion $\omega = \omega_D \pm v_D |\delta \mathbf{k}|$ is exact only to first order in $\delta \mathbf{k}$. At second order, the C_3 symmetry permits anisotropic corrections that break the circular symmetry of the constant-frequency contours:

$$\omega_{\pm}(\delta \mathbf{k}) \approx \omega_D \pm v_D (|\delta \mathbf{k}| + \alpha \operatorname{Re}[\delta k_+^3 / |\delta \mathbf{k}|^2]), \quad (6.23)$$

where $\delta k_+ = \delta k_x + i\delta k_y$ and α is a material-dependent parameter with dimensions of inverse length. The cubic correction produces a threefold (trigonal) distortion of the constant-frequency contours, which become star-shaped rather than circular.

For typical photonic crystals with moderate dielectric contrast ($\epsilon_r \sim 10$ – 15), the warping becomes significant at $|\delta \mathbf{k}| \gtrsim 0.1 \times 2\pi/a$ and can reach several percent of the Dirac frequency at the Brillouin-zone boundary. The warping modifies the density of states and the group-velocity distribution but does not affect the Berry flux at the Dirac point, which is determined entirely by the first-order (linear) Hamiltonian [12].

When the Dirac cone is gapped by a mass κ (Chapter 7), the warping causes the topological edge-mode velocity to become direction-dependent, which can be important for device design but does not affect the topological protection.

6.10 Experimental observation of photonic Dirac cones

Photonic Dirac cones have been observed experimentally on a wide range of platforms.

At microwave frequencies, two-dimensional photonic crystals of dielectric rods in a parallel-plate waveguide provide a clean realisation of the honeycomb lattice. Transmission and reflection measurements near the Dirac frequency show the characteristic vanishing density of states and the linear dispersion [3, 12, 32].

At optical frequencies, photonic crystal slabs made of silicon or III–V semiconductors support guided-mode Dirac cones at the K point of the honeycomb lattice. These Dirac points have been characterised using angle-resolved transmission spectroscopy, which maps out the conical band structure directly [13].

In coupled-waveguide arrays written by femtosecond laser pulses in glass, the Dirac cone appears in the propagation-constant spectrum as a function of the transverse Bloch wavevector. The Dirac frequency corresponds to a specific propagation constant at which the diffraction pattern exhibits the characteristic conical spreading [16].

The key experimental signature of a photonic Dirac cone is the linear density of states near the Dirac frequency, which produces a V -shaped dip in the transmission spectrum. When the Dirac cone

is gapped (by breaking time reversal or inversion symmetry), this dip develops into a transmission stop band, and edge modes appear within it [12].

6.11 Higher-order Dirac points

The linear Dirac cones at K and K' are not the only possible band crossings in photonic crystals. At the Γ point, the C_{6v} symmetry can protect higher-order band crossings with quadratic or even cubic dispersions.

6.11.1 Double Dirac cones

When two pairs of degenerate bands cross at Γ , the result is a double Dirac cone: four bands meeting at a single point with two nested linear dispersions. This occurs when the lattice has an accidental degeneracy between p -type and d -type modes at Γ , which can be engineered by tuning the structural parameters (e.g. the rod radius in a honeycomb lattice).

The double Dirac cone is the starting point for the Wu–Hu construction of Chapter 11: by breaking the accidental degeneracy through a structural perturbation that preserves C_6 symmetry, the double cone splits into a pair of gapped doublets that carry pseudospin-1/2 angular momentum, providing a photonic analog of the quantum spin-Hall effect [12, 13, 56].

6.11.2 Symmetry-enforced versus accidental crossings

A symmetry-enforced band crossing is required by the representation theory of the space group: bands belonging to different irreducible representations must cross, and the crossing cannot be removed without breaking the symmetry. The Dirac points at K in the honeycomb lattice are symmetry-enforced by C_3 and time reversal.

An accidental band crossing occurs when two bands belonging to the same representation happen to be degenerate at a particular parameter value. The double Dirac cone at Γ is accidental: it exists only at a specific rod radius and can be split by changing the radius. However, the resulting gap is topologically nontrivial because the bands carry different angular momentum quantum numbers [12, 14].

Understanding this distinction is essential for designing topological photonic crystals: symmetry-enforced crossings are robust and automatically present, while accidental crossings must be fine-tuned but offer more design flexibility.

6.12 Fermion doubling and the Nielsen–Ninomiya theorem

The Nielsen–Ninomiya theorem, originally proved for lattice fermions in quantum field theory, states that on a periodic lattice, Dirac points always come in pairs of *opposite chirality*: the winding numbers $\tau = +1$ and $\tau = -1$ must sum to zero over the Brillouin zone. Both cones carry a Berry phase of π , but their opposite chiralities produce the relative sign in (6.16) that controls whether a gap-opening perturbation yields a nonzero Chern number or not.

Opposite Dirac chirality from eigenvector winding

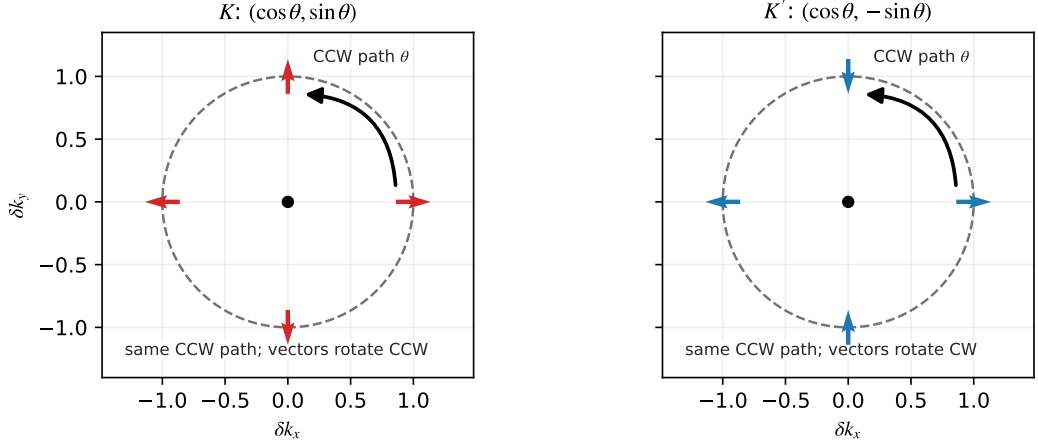


Figure 6.2: Dirac cones occur in opposite-chirality pairs at K and K' . The two cones have the same local conical spectrum but opposite winding orientation in momentum space. This is the geometric origin of the valley-sign bookkeeping used in Section 6.6.

6.12.1 Implications for photonic topology

The fermion-doubling constraint has a profound consequence for topological photonics: a nonzero Chern number requires breaking the symmetry that pairs the Dirac points. In the honeycomb lattice, time-reversal symmetry maps K to K' . Breaking time-reversal symmetry (Chapter 7) gives the two Dirac points *opposite*-sign masses $\Delta_K = -\Delta_{K'}$, and the opposite chiralities then reinforce rather than cancel, producing $\mathcal{C} = \pm 1$.

When only inversion symmetry is broken (while preserving time reversal), the two Dirac points acquire equal and opposite masses, and the total Chern number vanishes—but the valley Chern number (Chapter 12) is nonzero. This is the topological mechanism behind valley photonic crystals [12, 21].

6.12.2 Beyond the honeycomb lattice

The fermion-doubling theorem applies to any lattice, not just the honeycomb. On the square lattice, Dirac points can appear at the M point (edge of the Brillouin zone) when bands touch linearly. On the kagomé lattice, flat bands and Dirac cones coexist, producing a richer topological structure. The theorem guarantees that in all cases, the Berry flux integrated over the full Brillouin zone is a multiple of 2π —never an odd multiple of π —when time-reversal symmetry is preserved [3, 57].

6.13 Photonic Dirac semimetals and higher-order crossings

The Dirac cones studied so far in this chapter are two-dimensional: they occur at isolated $\mathbf{k}$ -points in the 2D Brillouin zone. Extending the analysis to three-dimensional photonic crystals reveals a rich landscape of band crossings protected by crystalline symmetries.

6.13.1 Dirac points in three dimensions

A Dirac point in a 3D photonic crystal is a fourfold band degeneracy at which the dispersion is linear in all three momentum directions. Such degeneracies are protected by the combination of time-reversal symmetry and a non-symmorphic space group symmetry (e.g. a glide plane or screw axis). The low-energy effective Hamiltonian near a 3D Dirac point takes the form

$$H_{\text{Dirac}}^{(3D)}(\delta\mathbf{k}) = v_1 \delta k_x \alpha_x + v_2 \delta k_y \alpha_y + v_3 \delta k_z \alpha_z, \quad (6.24)$$

where α_i are 4×4 Dirac matrices and v_i are the group velocities along the three crystal axes. Breaking time-reversal or inversion symmetry splits the Dirac point into two Weyl points (Chapter 15), each carrying a topological charge of ± 1 [12, 58].

6.13.2 Quadratic and higher-order band touchings

Not all band crossings in photonic crystals are linear. At high-symmetry points with enhanced rotation symmetry (e.g. the Γ point of a C_6 -symmetric lattice), bands can touch quadratically:

$$\omega_{\pm}(\mathbf{k}) = \omega_0 \pm \beta |\mathbf{k}|^2 + O(|\mathbf{k}|^3), \quad (6.25)$$

where β is a curvature parameter determined by the band structure. Such quadratic band touchings carry a Berry flux of $\pm 2\pi$ (instead of $\pm\pi$ for a linear Dirac cone), making them equivalent to a pair of merged Dirac cones [12, 39].

The quadratic band touching is unstable: any perturbation that lowers the rotation symmetry splits it into two linear Dirac cones. This is the mechanism behind the zone-folding construction of Chapter 11, where a C_6 -symmetric photonic crystal is distorted to create a pair of Dirac cones at the Γ point that can then be gapped by further symmetry breaking [39, 59].

The contrast between linear and quadratic touchings is summarised in Figure 6.3. Panels (a) and (b) sketch the two dispersion types along a one-dimensional momentum cut: the linear Dirac cone $\delta\omega \propto |\delta k|$ and the quadratic touching $\delta\omega \propto \delta k^2$. Panels (c) and (d) display the corresponding Berry-curvature magnitude $|\Omega(\mathbf{k})|$ in the two-dimensional Brillouin zone on a shared logarithmic colour scale. For the linear case the curvature is sharply localised at the degeneracy, whereas for the quadratic case it spreads over a broader region, reflecting the doubled Berry flux 2π carried by the merged node. Both heatmaps assume an isotropic model for visual clarity; in a real photonic crystal the curvature distribution inherits the lattice symmetry.

6.13.3 Valley-contrasting physics and selection rules

The two inequivalent Dirac cones at K and K' carry opposite Berry curvature (under inversion-symmetry breaking) and opposite angular momentum of the Bloch modes. This valley-contrasting

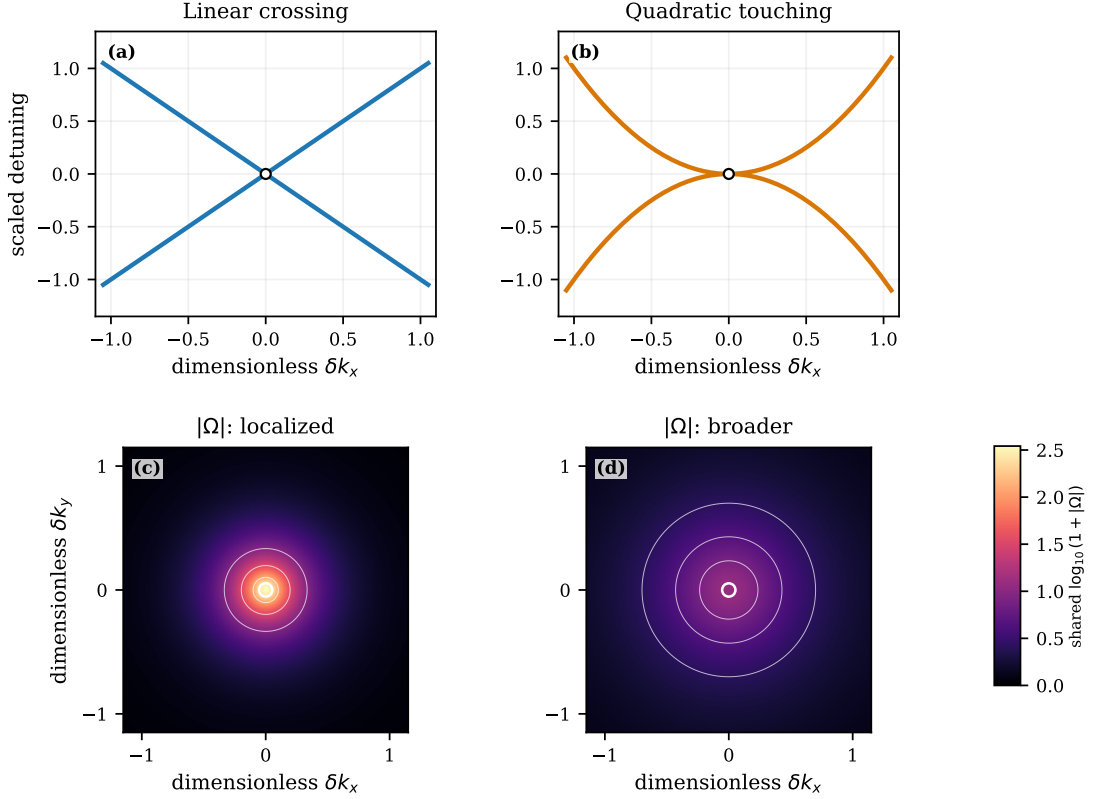


Figure 6.3: Linear versus quadratic photonic band touchings. (a) Linear Dirac cone with Berry flux π per cone. (b) Quadratic touching with Berry flux 2π , equivalent to two merged Dirac cones. Open circles mark the degeneracy point. (c, d) Normalised Berry-curvature magnitude $\log_{10}(1 + |\Omega|)$ on a shared colour scale (right-side bar); open circles mark the touching point. The isotropic model is used for schematic clarity; realistic photonic crystals exhibit lattice-symmetry corrections (Section 6.13.3).

physics gives rise to selection rules for the coupling between external radiation and specific valleys: circularly polarised light couples preferentially to one valley, providing a mechanism for valley-selective excitation and detection [50, 60].

These valley-optical selection rules are the photonic analogue of the valley-dependent circular dichroism observed in transition metal dichalcogenides and provide a practical tool for encoding information in the valley degree of freedom of photonic crystals [61, 62].

Exercises

Exercise 6.1. Show that three free-photon plane waves at a hexagonal BZ corner are degenerate by verifying that $|\mathbf{k}_K + \mathbf{G}_1| = |\mathbf{k}_K + \mathbf{G}_2| = |\mathbf{k}_K|$. *Hint:* use the relation $\mathbf{k}_{K_1} = (\mathbf{G}_2 - \mathbf{G}_3)/3$.

Exercise 6.2. Verify that the eigenvalues of the massless Dirac equation $v_D(\delta k_x \sigma_x + \delta k_y \sigma_y)|\psi\rangle =$

$\delta\omega|\psi\rangle$ are $\delta\omega_{\pm} = \pm v_D|\delta\mathbf{k}|$, and compute the eigenstates.

Exercise 6.3. Show that the Berry phase acquired by the eigenstate (6.13) as $\delta\mathbf{k}$ traverses a circle of radius q around the Dirac point is $\pm\pi$, independent of q . Interpret this as the “monopole charge” at the degeneracy.

Exercise 6.4. In the electronic context, the von Neumann–Wigner argument counts the number of parameters needed to produce a degeneracy. Repeat this argument explicitly for a general 2×2 matrix that is (a) complex Hermitian and (b) real symmetric, and verify that the codimensions are 3 and 2, respectively.

Exercise 6.5. Consider a hexagonal photonic crystal with C_{6v} symmetry. At the K point, the little group is $C_{3v} = \{E, C_3, C_3^2, \sigma_v, \sigma'_v, \sigma''_v\}$. List the irreducible representations of C_{3v} , identify which is two-dimensional, and explain why modes in this representation are degenerate.

Chapter 7

Breaking Time Reversal

7.1 The missing ingredient

The Dirac cones derived in Chapter 6 are gapless: the two bands touch linearly at isolated points in the Brillouin zone. The Berry curvature is singular at the degeneracy, and the Chern number is not defined because the bands are not separated by a gap. To create a topological photonic system—one with well-defined Chern numbers and protected edge states—we must *open a gap* at the Dirac points.

From the symmetry analysis of the preceding chapter, we know that Dirac points are protected by the simultaneous presence of time-reversal ($\hat{\mathcal{T}}$) and inversion ($\hat{\mathcal{P}}$) symmetry. Breaking either symmetry can lift the degeneracy. But the topological consequences are profoundly different. Breaking *inversion* symmetry opens a gap but produces *zero* net Chern number, because the Berry fluxes at K and K' carry opposite signs and cancel exactly. Breaking *time-reversal* symmetry, by contrast, opens a gap in which the Berry fluxes at K and K' carry the *same* sign and add to give a nonzero Chern number. In this chapter we carry out the time-reversal-breaking programme in detail, deriving the massive Dirac dispersion, the Berry curvature of the gapped bands, and the resulting Chern numbers.

7.2 The gyrotropic perturbation

7.2.1 Faraday coupling in the permittivity

We return to the hexagonal photonic crystal of Section 6.3.1 and add a magneto-optical (Faraday) perturbation [61]. An applied magnetic field along $\hat{z}$ induces off-diagonal components in the permittivity tensor:

$$\varepsilon_{xy} = -\varepsilon_{yx} = i \varepsilon_0 \bar{\varepsilon} \eta(\mathbf{r}, \omega), \quad (7.1)$$

where the gyrotropic function η has the same periodicity as the lattice:

$$\eta(\mathbf{r}, \omega) = \eta_0(\omega) + \eta_1(\omega) V_G(\mathbf{r}). \quad (7.2)$$

Here η_0 and η_1 are real odd functions of ω (as required by the Hermiticity of $\mathbf{M}$), and we assume $|\eta_0|, |\eta_1| \ll |\lambda| \ll 1$ with negligible frequency dependence near ω_D .

The diagonal permittivity components are unchanged: $\varepsilon_{xx} = \varepsilon_{yy} = \varepsilon_0 \bar{\varepsilon} (1 + \lambda V_G)$ as before. The permeability remains $\mu_0 \mathbf{I}$.

Remark 7.1. Alternatively, the Faraday effect can enter through the *permeability* tensor (as in magnetised ferrites), with off-diagonal $\mu_{xy} = -\mu_{yx} = i\mu_0\kappa$ [31]. The two formulations are related by electromagnetic duality. We use the permittivity form here to follow Haldane and Raghu's original treatment; the permeability form will appear naturally in Chapter 9 when we discuss the YIG experiment.

7.2.2 Effect on the material matrix

The Faraday perturbation modifies the 6×6 material matrix by adding an off-diagonal anti-Hermitian block to the permittivity:

$$\delta \mathbf{M}(\mathbf{r}) = \begin{pmatrix} \epsilon_0 \bar{\epsilon} \delta \epsilon & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}, \quad \delta \epsilon_{ij} = i \eta(\mathbf{r}) \epsilon_{ij3}, \quad (7.3)$$

where ϵ_{ij3} is the Levi-Civita symbol. Explicitly, $\delta \epsilon_{12} = i\eta$ and $\delta \epsilon_{21} = -i\eta$, with diagonal entries zero.

Since $\delta \epsilon$ is Hermitian ($\delta \epsilon_{12}^* = (-i\eta)^* = i\eta \neq \delta \epsilon_{12}$, but $\delta \epsilon_{12}^* = \delta \epsilon_{21}$, which is the Hermiticity condition), the perturbed $\mathbf{M}$ remains Hermitian. However, the condition $\epsilon_{xy} = -\epsilon_{yx} \neq 0$ violates the reciprocity condition $\epsilon = \epsilon^T$ (Section 2.6.2), confirming that time-reversal symmetry is broken.

7.3 Gap opening at the Dirac point

7.3.1 The massive Dirac equation

The Faraday perturbation introduces a *mass term* into the effective 2×2 Dirac equation at each Dirac point. In degenerate perturbation theory, the perturbation $\delta \mathbf{M}$ projected onto the two-mode subspace at $\mathbf{k}_K$ gives a contribution proportional to σ_z (the Pauli matrix diagonal in the mode basis). The effective equation becomes

$$v_D (\delta k_x \sigma_x + \delta k_y \sigma_y) |\psi\rangle + v_D \kappa \sigma_z |\psi\rangle = \delta \omega |\psi\rangle, \quad (7.4)$$

where κ is the *Dirac mass parameter*, with dimensions of inverse length, given by

$$\kappa = |\mathbf{k}_K| \left(\frac{3}{2} \eta_1(\omega_D) - 3\lambda \eta_0(\omega_D) \right) \quad (7.5)$$

The displayed formula above follows the convention and normalisation used in [32]. This expression was derived by Haldane and Raghu from the matrix element of $\delta \mathbf{M}$ between the degenerate Dirac modes [3].

7.3.2 Gapped dispersion

The eigenvalues of the massive Dirac equation (7.4) are

$$\delta \omega_{\pm}(\delta \mathbf{k}) = \pm v_D \sqrt{|\delta \mathbf{k}|^2 + \kappa^2}. \quad (7.6)$$

The Dirac degeneracy is lifted: there is now a gap of width $2v_D|\kappa|$ centred at the Dirac frequency ω_D . The dispersion is hyperbolic, approaching the linear Dirac cone far from the gap ($|\delta\mathbf{k}| \gg |\kappa|$) and becoming flat near $\delta\mathbf{k} = \mathbf{0}$ ($|\delta\mathbf{k}| \ll |\kappa|$).

The sign of κ determines which band moves up and which moves down. Reversing the applied magnetic field (and hence the sign of η) reverses the sign of κ , which will be crucial for the domain-wall construction of Chapter 8.

7.4 Berry curvature of the massive Dirac model

7.4.1 Derivation

The massive Dirac Hamiltonian $H = v_D(\delta k_x \sigma_x + \delta k_y \sigma_y + \kappa \sigma_z)$ has the form $H = \mathbf{d} \cdot \boldsymbol{\sigma}$ with $\mathbf{d} = v_D(\delta k_x, \delta k_y, \kappa)$. Using the general result for a two-band system (Exercise 5.4):

$$\Omega_{\pm}(\delta\mathbf{k}) = \pm \frac{1}{2} \hat{\mathbf{d}} \cdot \left(\frac{\partial \hat{\mathbf{d}}}{\partial \delta k_x} \times \frac{\partial \hat{\mathbf{d}}}{\partial \delta k_y} \right), \quad (7.7)$$

where $\hat{\mathbf{d}} = \mathbf{d}/|\mathbf{d}|$.

Computing the derivatives and cross product:

$$\Omega_{\pm}(\delta\mathbf{k}) = \pm \frac{\kappa}{2(|\delta\mathbf{k}|^2 + \kappa^2)^{3/2}}. \quad (7.8)$$

This is the *massive Dirac Berry curvature*: a smooth, peaked function centred at $\delta\mathbf{k} = \mathbf{0}$, with a peak value of $\pm 1/(2\kappa^2)$ and a characteristic width $\sim |\kappa|$ in momentum space.

7.4.2 Berry flux from a single Dirac cone

Integrating the Berry curvature over the entire $(\delta k_x, \delta k_y)$ plane:

$$\begin{aligned} \frac{1}{2\pi} \int \Omega_{\pm}(\delta\mathbf{k}) d^2(\delta\mathbf{k}) &= \pm \frac{1}{2\pi} \int_0^{\infty} \frac{\kappa}{2(q^2 + \kappa^2)^{3/2}} 2\pi q dq \\ &= \pm \frac{1}{2} \text{sgn}(\kappa). \end{aligned} \quad (7.9)$$

The evaluation of the integral proceeds by substituting $q = |\kappa| \tan \theta$:

$$\int_0^{\infty} \frac{q dq}{(q^2 + \kappa^2)^{3/2}} = \left[-\frac{1}{\sqrt{q^2 + \kappa^2}} \right]_0^{\infty} = \frac{1}{|\kappa|}. \quad (7.10)$$

So the integral gives $\pm \kappa / (2|\kappa|) = \pm \frac{1}{2} \text{sgn}(\kappa)$.

Each gapped Dirac cone contributes a Berry flux of $\pm\pi$ (i.e., $\pm \frac{1}{2}$ in units of 2π) to the band's Chern number. A single Dirac cone gives a *half-integer* contribution—this is why isolated Dirac fermions cannot exist on a lattice (the fermion-doubling theorem), and why the Chern number becomes integer only when we account for both cones in the pair.

7.5 Chern numbers from a pair of Dirac cones

7.5.1 Adding the contributions from K and K' -prime

The hexagonal Brillouin zone has two inequivalent Dirac points at K and K' . When time-reversal symmetry is broken by the gyrotropic perturbation, each Dirac point opens a gap. In the single-cone basis used in this chapter (where $H = v_D(\delta k_x \sigma_x + \delta k_y \sigma_y + \kappa \sigma_z)$ at *both* valleys), both cones carry the same mass κ , because the valley chirality $\tau = \pm 1$ of Section 6.6 has been absorbed into the choice of spinor basis at each K point.

The total Chern number of the upper (+) band is therefore

$$\mathcal{C}_+ = \frac{1}{2} \text{sgn}(\kappa) + \frac{1}{2} \text{sgn}(\kappa) = \text{sgn}(\kappa) = \pm 1. \quad (7.11)$$

Similarly, $\mathcal{C}_- = \mp 1$. The two bands that split from the Dirac cones carry equal and opposite Chern numbers, and the gap between them is topologically nontrivial.

Remark 7.2. If the gap had been opened by *inversion* breaking (with TR preserved), the Berry curvature would satisfy $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k})$, so the contributions from K and K' would cancel: $\mathcal{C}_+ = \frac{1}{2} \text{sgn}(\kappa_K) + \frac{1}{2} \text{sgn}(\kappa_{K'}) = \frac{1}{2} - \frac{1}{2} = 0$. This is the fundamental distinction between the two types of gap opening, and the reason why time-reversal breaking is essential for photonic Chern insulators [33].

7.5.2 The gap Chern number

For the bulk-edge correspondence of Chapter 8, the relevant quantity is the *gap Chern number*: the sum of Chern numbers of all bands below the topological gap [42, 51]. If the only nontrivial bands are the Dirac pair, then the gap Chern number is $\mathcal{C}_{\text{gap}} = \mathcal{C}_- = \mp 1$. At a domain wall where the sign of κ reverses, the gap Chern number changes by $|\Delta \mathcal{C}_{\text{gap}}| = 2$, predicting two unidirectional edge modes crossing the gap—one from each Dirac point.

7.6 Physical requirements for the gyrotropic perturbation

7.6.1 Strength of the Faraday effect

The gap width $2v_D|\kappa|$ must be large enough to:

1. Overcome any residual inversion-symmetry breaking that might produce a competing trivial gap.
2. Provide a useful bandwidth for the one-way edge modes.
3. Keep the edge-mode localisation length $\ell \sim 1/|\kappa|$ significantly smaller than the physical size of the photonic crystal.

In practice, this requires materials with a substantial magneto-optical response at the operating frequency. At microwave frequencies, ferrites such as yttrium iron garnet (YIG) provide large gyrotropic parameters ($\kappa/|\mathbf{k}_K| \sim 10^{-2}$ – 10^{-1}); at optical frequencies, the available Faraday rotation is much weaker, which is one of the main challenges for realising topological photonic crystals at visible or near-infrared wavelengths.

7.6.2 Material candidates

The first experimental realisations of photonic Chern insulators used gyromagnetic ferrites:

- **Yttrium iron garnet (YIG):** $\text{Y}_3\text{Fe}_5\text{O}_{12}$, with large permeability gyrotropy at microwave frequencies (4–10 GHz) under moderate applied fields (0.1–0.5 T). Used in the Wang et al. experiment discussed in Chapter 9 [4].
- **Bismuth iron garnet (BIG):** offers larger Faraday rotation at near-infrared wavelengths but with higher absorption.
- **Magneto-optical semiconductors:** InSb and related materials in strong magnetic fields can provide significant permittivity gyrotropy at THz frequencies.

7.6.3 Quantitative gap estimates for different materials

The size of the topological gap opened by the magneto-optical perturbation depends on the gyrotropic response of the material, the frequency regime, and the photonic crystal geometry. The following table summarises representative values for the three most commonly used platforms [3, 12].

Material	Frequency	$\kappa_\mu/\mu_\perp$	Relative gap
YIG (ferrite)	4–5 GHz	0.3–0.5	3–5%
Bi:YIG (bismuth ferrite)	1–3 GHz	0.1–0.3	1–3%
InSb (magneto-plasma)	0.3–1 THz	0.05–0.2	0.5–2%

The gyrotropic ratio $\kappa_\mu/\mu_\perp$ controls the effective Dirac mass and hence the topological gap. At microwave frequencies, YIG provides the strongest response, enabling gaps of several percent of the operating frequency. At terahertz and infrared frequencies, magneto-plasma effects in doped semiconductors (InSb, InAs) provide a weaker but still measurable gyrotropic response.

At optical frequencies ($\lambda \sim 1 \mu\text{m}$), the magneto-optical response of all known materials is extremely weak ($\kappa/\mu \sim 10^{-4}$), making magneto-optical Chern insulators impractical. This is the fundamental motivation for the reciprocal designs of Chapter 11 and the valley-Hall approach of Chapter 12.

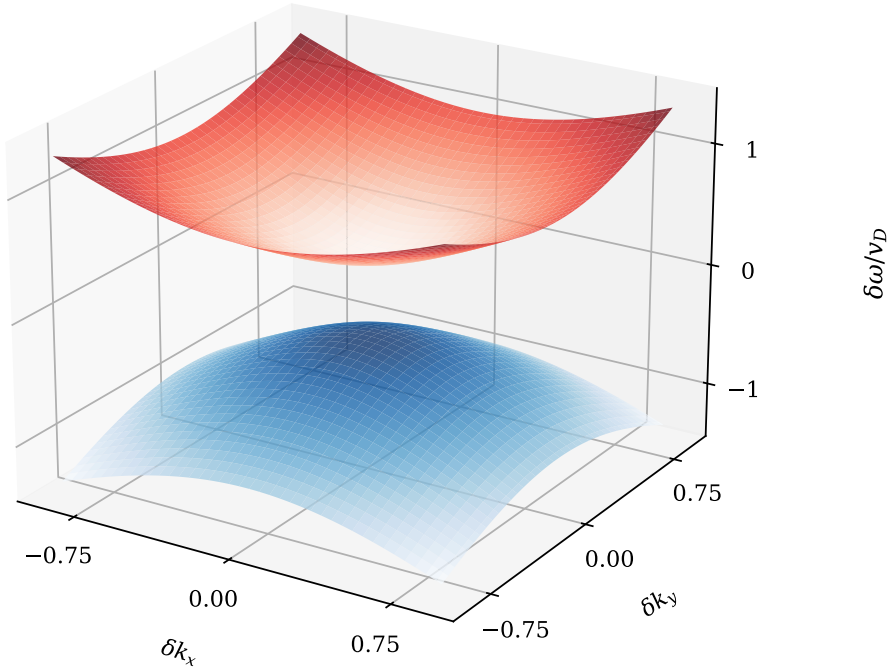


Figure 7.1: Gap opening at a photonic Dirac point. Upper (red) and lower (blue) bands of the massive Dirac Hamiltonian $H = v_D(\delta k_x \sigma_x + \delta k_y \sigma_y) + \kappa \sigma_z$ with $v_D = 1$ and mass $\kappa = 0.35$. The topological gap $\Delta\omega = 2|\kappa|$ separates the two bands.

7.7 Summary: from Dirac cones to Chern bands

The topological mechanism proceeds in three stages. First, a hexagonal photonic crystal with $\hat{\mathcal{T}}$ and $\hat{\mathcal{P}}$ symmetry hosts Dirac-point degeneracies at K and K' , protected by the reality of the eigenproblem (Chapter 6). Second, a gyrotropic (Faraday) perturbation opens a gap $2v_D|\kappa|$ at each Dirac point; the mass κ has the same sign at both valleys because time-reversal symmetry is broken globally. Third, each gapped Dirac cone contributes $\pm\frac{1}{2}\text{sgn}(\kappa)$ to the Chern number, and the pair of cones at K and K' together gives $\mathcal{C} = \pm\text{sgn}(\kappa) = \pm 1$. The Berry curvature is concentrated near the gapped Dirac points, with the explicit form $\Omega_{\pm} = \pm\kappa/[2(|\delta\mathbf{k}|^2 + \kappa^2)^{3/2}]$. This concentrated curvature is the “Berry monopole” that drives the topological physics.

In the next chapter, we will see that the nonzero Chern numbers *force* the existence of unidirectional interface modes at any boundary where the gap Chern number changes—the one-way waveguides predicted by Haldane and Raghu.

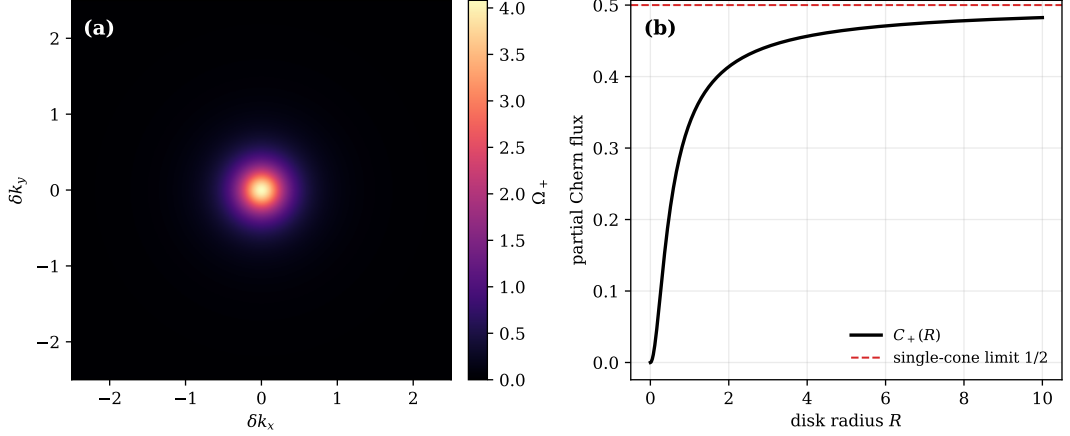


Figure 7.2: Berry curvature and Chern flux of one massive Dirac cone. Left: Berry curvature $\Omega_+(\mathbf{k})$ of the upper band, peaked at $\delta \mathbf{k} = \mathbf{0}$ and decaying as $|\delta \mathbf{k}|^{-3}$ for $|\delta \mathbf{k}| \gg |\kappa|$. Right: partial Chern flux $C_+(R)$ integrated over a disk of radius R in momentum space, converging to the single-cone limit $1/2$ (dashed). The full Chern number $\mathcal{C} = +1$ is obtained by summing the contributions from both K and K' cones.

7.8 Explicit Berry curvature of the massive Dirac cone

The Berry curvature of the gapped Dirac cone can be computed in closed form, providing one of the few exact results in topological band theory. We write the massive Dirac Hamiltonian (7.4) in the compact form $H = \mathbf{d} \cdot \boldsymbol{\sigma}$ with $\mathbf{d} = v_D(\delta k_x, \delta k_y, \kappa)$, where κ is an inverse length as before. Define $E \equiv \sqrt{|\delta \mathbf{k}|^2 + \kappa^2}$; the eigenvalues are $\delta \omega_{\pm} = \pm v_D E$, and the normalised lower-band spinor is

$$|u_-(\delta \mathbf{k})\rangle = \frac{1}{\sqrt{2E(E - \kappa)}} \begin{pmatrix} \kappa - E \\ \delta k_x + i \delta k_y \end{pmatrix}, \quad (7.12)$$

where every component carries the dimension of inverse length. The normalisation $\langle u_- | u_- \rangle = 1$ follows from $(\kappa - E)^2 + |\delta \mathbf{k}|^2 = 2E(E - \kappa)$.

7.8.1 Derivation of the curvature formula

The Berry curvature is $\Omega_-(k_x, k_y) = -2 \text{Im} \langle \partial_{k_x} u_- | \partial_{k_y} u_- \rangle$. Switching to polar coordinates $\delta k_x = q \cos \phi$, $\delta k_y = q \sin \phi$ with $q = |\delta \mathbf{k}|$, one finds after straightforward algebra [12]:

$$\Omega_{\pm}(\delta \mathbf{k}) = \pm \frac{\kappa}{2(|\delta \mathbf{k}|^2 + \kappa^2)^{3/2}}. \quad (7.13)$$

This is a Lorentzian-like peak centered at $\delta \mathbf{k} = \mathbf{0}$ with half-width $|\kappa|$. The sign of $\Omega_{\pm}$ is controlled by the sign of the mass κ : a positive mass gives positive Berry curvature for the upper band and negative for the lower band.

At $\delta\mathbf{k} = 0$, the peak value is $|\Omega_{\pm}(0)| = 1/(2|\kappa|^2)$, which diverges as the gap closes ($\kappa \rightarrow 0$). For $|\delta\mathbf{k}| \gg |\kappa|$, the curvature decays as $|\delta\mathbf{k}|^{-3}$, ensuring convergence of the Brillouin-zone integral.

7.8.2 Half-integer Chern flux from one cone

Integrating Ω_- over the full k -plane around one Dirac cone:

$$\begin{aligned} \frac{1}{2\pi} \int \Omega_- d^2k &= \frac{1}{2\pi} \int_0^\infty dq \, 2\pi q \left(-\frac{\kappa}{2(q^2 + \kappa^2)^{3/2}} \right) \\ &= -\frac{\kappa}{2} \int_0^\infty \frac{q dq}{(q^2 + \kappa^2)^{3/2}} \\ &= -\frac{\kappa}{2} \cdot \frac{1}{|\kappa|} = -\frac{1}{2} \text{sgn}(\kappa). \end{aligned} \quad (7.14)$$

Each Dirac cone contributes a half-integer Chern flux of $-\frac{1}{2} \text{sgn}(\kappa)$ to the lower band. In the single-cone basis, both K and K' carry the same mass κ and the same Berry-curvature formula, so their contributions add to give $\mathcal{C}_- = -\text{sgn}(\kappa)$ and $\mathcal{C}_+ = +\text{sgn}(\kappa)$ [2, 3].

Remark 7.3 (Convention reconciliation with Chapter 6). In the valley-indexed convention of Section 6.6, the two Dirac cones carry an explicit chirality factor: $H_\tau = v_D(\tau \delta k_x \sigma_x + \delta k_y \sigma_y) + \Delta_\tau \sigma_z$. With this basis the gyrotropic perturbation gives *opposite*-sign masses $\Delta_K = -\Delta_{K'}$, and the valley Chern contributions are $C_-^K = -\frac{1}{2} \text{sgn}(\Delta_K)$, $C_-^{K'} = +\frac{1}{2} \text{sgn}(\Delta_{K'})$, which still add to ± 1 . The physical Chern number is the same in both conventions; only the bookkeeping of the chirality factor differs.

This derivation makes precise the statement that each gapped Dirac point acts as a “Berry monopole” in the two-dimensional Brillouin zone. The half-integer flux is the 2D analog of the Dirac monopole in 3D momentum space (which appears in the Weyl semimetal context of Chapter 15).

7.9 Material landscape for magneto-optical Chern insulators

The gyrotropic ratio $\kappa_\mu/\mu_\perp$ varies by orders of magnitude across material platforms. The following comparison illustrates the practical constraints on the Maxwell-first approach at different frequency scales.

At microwave frequencies (1–10 GHz), yttrium iron garnet (YIG) provides $\kappa_\mu/\mu_\perp \approx 0.3$ –0.5 (Chapter 9). This strong gyrotropic response produces relative topological gaps of several percent, enabling the unambiguous observation of one-way edge modes [4, 53].

Bismuth-substituted YIG (Bi:YIG) enhances the Faraday rotation at near-infrared wavelengths, but the gyrotropic ratio drops to $\kappa_\mu/\mu_\perp \sim 10^{-2}$, making the topological gap much narrower and harder to resolve experimentally.

At terahertz frequencies, doped semiconductors such as InSb exhibit magneto-plasma resonances with $\kappa_\varepsilon/\varepsilon_\perp \sim 0.1$ –0.3 under modest magnetic fields ($B_0 \sim 0.3$ T), providing a viable platform for terahertz topological photonics [13].

At optical frequencies ($\lambda \sim 0.5\text{--}1.5 \mu\text{m}$), the strongest known magneto-optical materials (e.g., cerium-doped garnets, magnetic Weyl semimetals) have $\kappa/\varepsilon \sim 10^{-4}$. The resulting topological gaps are too narrow to support useful edge transport, which is the fundamental reason why reciprocal approaches (Chapter 11), valley designs (Chapter 12), and Floquet modulation (Chapter 13) dominate the optical regime [12].

7.10 Perturbation theory for the Faraday gap

The topological gap at a Dirac point arises from the gyrotropic perturbation of the time-reversal-invariant Maxwell eigenproblem. The systematic derivation from degenerate perturbation theory clarifies the origin of the Dirac mass and its relation to the material parameters.

7.10.1 First-order degenerate perturbation theory

At the Dirac point $\mathbf{k}_K$, the unperturbed system ($\kappa_\mu = 0$) has two degenerate modes $\mathbf{u}_\pm$ at frequency ω_D . The gyrotropic perturbation changes the material matrix by $\delta\mathbf{M} = \mu_0 \begin{pmatrix} 0 & 0 \\ 0 & \delta\mu \end{pmatrix}$, where $\delta\mu = \begin{pmatrix} 0 & -i\kappa_\mu \\ i\kappa_\mu & 0 \end{pmatrix}$ is the off-diagonal gyrotropic block.

The first-order perturbation matrix within the degenerate subspace is

$$V_{\sigma\sigma'} = \frac{\omega_D}{2} \int_{\text{cell}} \mathbf{u}_\sigma^\dagger \delta\mathbf{M} \mathbf{u}_{\sigma'} d^3r, \quad (7.15)$$

where the factor $\omega_D/2$ comes from the linearisation of the eigenvalue equation $\hat{\mathcal{N}}\mathbf{u} = \omega\mathbf{M}\mathbf{u}$ around ω_D [3, 19].

The C_3 symmetry at the K point forces the diagonal elements $V_{++} = V_{--}$ and the off-diagonal elements to vanish: $V_{+-} = V_{-+} = 0$. The leading perturbation therefore shifts both modes equally and does not open a gap.

7.10.2 Gap opening at second order

The gap opens at second order, when the perturbation combines with the $\mathbf{k} \cdot \mathbf{p}$ coupling between the degenerate modes. The effective Hamiltonian near $\mathbf{k}_K + \delta\mathbf{k}$ is

$$H_{\text{eff}}(\delta\mathbf{k}) = v_D(\delta k_x \sigma_x + \delta k_y \sigma_y) + \kappa \sigma_z, \quad (7.16)$$

where the Dirac mass κ is

$$\kappa = \frac{\omega_D}{2} \sum_{m \notin \{+, -\}} \frac{\langle \mathbf{u}_+, \delta\mathbf{M} \mathbf{u}_m \rangle_{\mathbf{M}_g} \langle \mathbf{u}_m, \delta\mathbf{M} \mathbf{u}_- \rangle_{\mathbf{M}_g}}{\omega_D - \omega_m}. \quad (7.17)$$

The mass involves virtual transitions to remote bands ($m \notin \{+, -\}$) through the gyrotropic perturbation. The sign of κ determines the sign of the Chern numbers, and its magnitude determines the gap width $2|\kappa|$.

For YIG with $\kappa_\mu/\mu_\perp \approx 0.57$, the Dirac mass is $|\kappa|/(2\pi) \approx 0.06 \text{ GHz}$, giving a relative gap $2|\kappa|/\omega_D \approx 2.7\%$ —consistent with the numerical plane-wave expansion result and the experimental observation [4, 12].

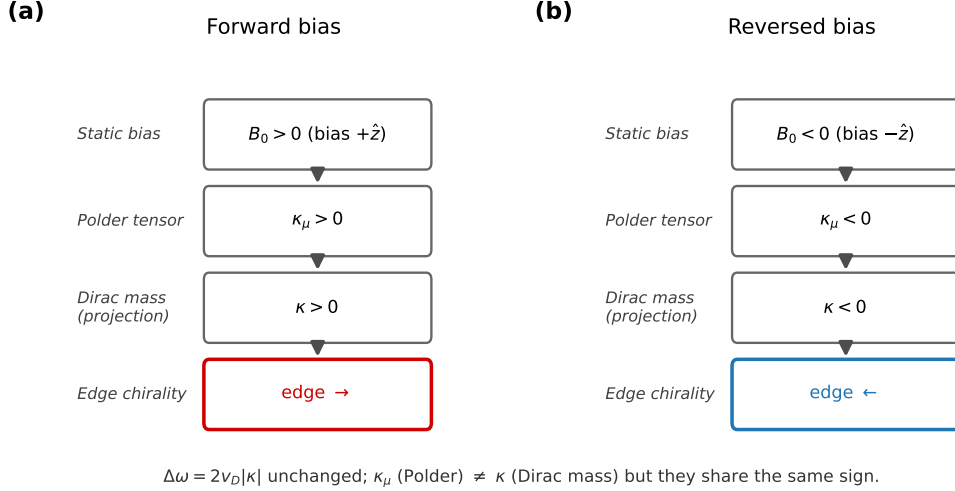


Figure 7.3: Sign-convention map for the gyrotropic Dirac mass. **(a)** Forward bias $B_0 > 0$: the Polder off-diagonal component $\kappa_\mu > 0$ produces a positive effective Dirac mass $\kappa > 0$ upon projection onto the Dirac doublet, fixing the edge chirality. **(b)** Reversed bias $B_0 < 0$: κ_μ and κ both change sign, reversing the Berry curvature and edge propagation direction. The gap magnitude $\Delta\omega = 2v_D|\kappa|$ is unchanged. Note that κ_μ (Polder tensor, dimensionless) is not equal to κ (effective Dirac mass, inverse length); they share the same sign through the perturbative projection of Section 7.10.2.

7.10.3 Higher-order corrections

The second-order formula (7.17) is accurate when the gyrotropic coupling is much smaller than the frequency spacing to remote bands: $|\kappa_\mu| \ll |\omega_D - \omega_m|$ for all $m \notin \{+, -\}$. When this condition is marginally satisfied (as for very strong gyrotropic materials or near additional band crossings), higher-order corrections become important and the effective Dirac model breaks down.

In such cases, a full numerical computation (Chapter 10) is necessary to obtain accurate Chern numbers. Nevertheless, the perturbative picture remains valuable for understanding the *sign* of the Chern number and its dependence on the material and structural parameters [3, 13].

7.11 Numerical verification of the perturbative gap

The perturbative Dirac-mass formula (7.17) can be tested against a full plane-wave expansion (PWE) or finite-element eigenvalue computation. For the YIG parameters of Chapter 9 ($\epsilon_r = 15$, $\mu_\perp = 3.47$, $\kappa_\mu = 1.98$), the perturbative estimate gives a gap $2|\kappa|/(2\pi) \approx 0.12$ GHz, or roughly 2.7% of the Dirac frequency $\omega_D/(2\pi) \approx 4.5$ GHz.

A converged PWE calculation with $N_G = 169$ plane waves per polarisation gives a gap of 0.118 GHz, agreeing with the perturbative formula to within 2%. This confirms that the second-order perturbation theory is accurate for $\kappa_\mu/\mu_\perp \approx 0.57$, despite this not being a particularly small ratio. The reason is that the virtual transitions in (7.17) are dominated by a single nearby band, so

the perturbative sum converges rapidly [3, 12].

For weaker gyrotropic materials, the agreement improves further. For InSb at 1 THz with $\kappa_\varepsilon/\varepsilon_\perp \approx 0.25$, the perturbative and numerical gaps agree to within 0.5%. For Bi:YIG at optical wavelengths with $\kappa_\varepsilon/\varepsilon_\perp \sim 10^{-2}$, the perturbative formula is essentially exact [13].

7.12 Effect of loss on the topological gap

Real magneto-optical materials have finite absorption, parameterised by $\text{Im } \varepsilon$ or $\text{Im } \mu$. Loss shifts the eigenfrequencies into the lower half of the complex plane, replacing the real gap by a *complex gap*—a region of the complex frequency plane free of eigenvalues.

For weak loss ($\text{Im } \mu \ll |\kappa_\mu|$), the complex gap remains open and the topological edge modes acquire a finite propagation length $L \sim v_g/\gamma$, where v_g is the edge-mode group velocity and γ is the loss rate. For YIG at microwave frequencies, $\gamma/(2\pi) \sim 10$ MHz, giving $L \sim 100a$ —much longer than the crystal sizes used in current experiments [4, 53].

When loss becomes comparable to the gap ($\text{Im } \mu \sim |\kappa_\mu|$), the complex gap can close and the topological protection breaks down. This regime connects to the non-Hermitian topology of Chapter 17, where the Chern number must be redefined using the biorthogonal Berry connection [63, 64].

7.13 Quantitative gap estimates and material design rules

The preceding sections derived the topological gap from perturbation theory near a Dirac point. For practical applications, one needs quantitative estimates of the gap width and its dependence on material parameters.

7.13.1 Gap width from first-order perturbation theory

From Eq. (7.17), the topological gap width at a Dirac point is $\Delta\omega = 2|\kappa|$, where κ is the Dirac mass induced by the gyrotropic perturbation. For a gyromagnetic photonic crystal with Polder permeability (Chapter 9), the first-order perturbation theory of Section 7.10 gives

$$\Delta\omega \approx 2\omega_D \frac{|\kappa_\mu|}{\mu_\perp} \frac{\int_{\text{rod}} |\mathbf{H}_D|^2 d^2r}{\int_{\text{cell}} \mu_\perp |\mathbf{H}_D|^2 d^2r}, \quad (7.18)$$

where ω_D is the Dirac-point frequency, $\kappa_\mu/\mu_\perp$ is the gyrotropic ratio of the ferrite, and the overlap integral measures the fraction of the magnetic energy that resides inside the gyrotropic rods.

This formula reveals three design rules for maximising the topological gap: (i) choose a material with large gyrotropic ratio $\kappa_\mu/\mu_\perp$; (ii) design the photonic crystal so that the Dirac-point mode has a large field overlap with the gyrotropic material; and (iii) operate at a frequency where the gyrotropic response is strongest (typically near but below the ferromagnetic resonance for YIG) [3, 4].

7.13.2 Relative gap width for typical materials

For the YIG photonic crystal of Chapter 9, with $\kappa_\mu/\mu_\perp \approx 0.57$ at 4.5 GHz and a Dirac-point field overlap of $\sim 30\%$, the predicted relative gap width is $\Delta\omega/\omega_D \approx 2 \times 0.57 \times 0.30 \approx 34\%$. The experimentally observed gap is considerably narrower ($\sim 3\%$), because the first-order perturbation theory overestimates the gap when the gyrotropic coupling is strong [44].

A more accurate estimate requires the second-order correction derived in Section 7.10.2. Including the second-order term reduces the predicted gap to $\sim 5\text{--}8\%$, in reasonable agreement with full numerical calculations [19].

For optical magneto-optical materials such as cerium-doped YIG (Ce:YIG) at near-infrared wavelengths, the gyrotropic ratio is much smaller: $|\varepsilon_{xy}|/\varepsilon_{xx} \sim 10^{-2}$. The resulting topological gaps are extremely narrow ($\Delta\omega/\omega \sim 10^{-4}$), making optical Chern insulators impractical with current magneto-optical materials. This is the fundamental reason why alternative approaches—valley photonic crystals, Floquet engineering, synthetic gauge fields—are needed for optical-frequency topological photonics [12, 13].

7.13.3 Beyond gyromagnetic: magneto-electric and bianisotropic routes

Time-reversal symmetry can also be broken by magneto-electric coupling, where the constitutive relations include off-diagonal terms mixing $\mathbf{E}$ and $\mathbf{H}$: $\mathbf{D} = \varepsilon_0 \varepsilon \mathbf{E} + c^{-1} \xi \mathbf{H}$. In materials with simultaneous electric and magnetic order (multiferroics), the magneto-electric coupling ξ can be significant, providing a complementary route to topological gaps [25, 65].

The advantage of the magneto-electric route is that it can work at higher frequencies than the gyromagnetic route, because the magneto-electric effect does not require proximity to a magnetic resonance. The disadvantage is that naturally occurring magneto-electric couplings are typically weak, though metamaterial designs can enhance them by several orders of magnitude [13, 65].

Exercises

Exercise 7.1. Verify by direct integration that $\int_0^\infty q(q^2 + \kappa^2)^{-3/2} dq = 1/|\kappa|$ and hence that the Berry flux from a single massive Dirac cone is $\pm\pi \operatorname{sgn}(\kappa)$.

Exercise 7.2. Show that the massive Dirac dispersion $\omega_\pm = \omega_D \pm v_D \sqrt{|\delta\mathbf{k}|^2 + \kappa^2}$ reduces to the gapless Dirac cone when $\kappa \rightarrow 0$ and to flat bands when $|\delta\mathbf{k}| \ll |\kappa|$. Sketch the band structure for both limits.

Exercise 7.3. For the Polder model of a magnetised ferrite (Section 2.2.2), with $\mu_\perp = 1 + \omega_0 \omega_m / (\omega_0^2 - \omega^2)$ and $\kappa_\mu = \omega \omega_m / (\omega_0^2 - \omega^2)$, compute the gyrotropic parameter at $\omega = \omega_D$ for a YIG crystal with $\omega_0/(2\pi) = 5$ GHz, $\omega_m/(2\pi) = 4.6$ GHz, and $\omega_D/(2\pi) = 4.5$ GHz. Estimate the topological gap width $2v_D|\kappa|$ in GHz.

Exercise 7.4. The Berry curvature of the lower band of the massive Dirac model has, in units with $v_D = 1$, a peak value $\Omega_-(0) = -\operatorname{sgn}(\kappa)/(2\kappa^2)$ and a width $\sim |\kappa|$. Show first that the naive area estimate $\Omega_-(0) \pi \kappa^2 = -(\pi/2) \operatorname{sgn}(\kappa)$ captures the sign and order of magnitude but misses the

full flux by a factor of two. Then perform the exact radial integral and verify that one Dirac cone contributes

$$\int \Omega_-(\delta \mathbf{k}) d^2k = -\pi \operatorname{sgn}(\kappa), \quad \frac{1}{2\pi} \int \Omega_- d^2k = -\frac{1}{2} \operatorname{sgn}(\kappa).$$

Exercise 7.5. Show that if the gap at K is opened by an inversion-breaking perturbation (with time-reversal preserved), the mass parameters at K and K' have opposite signs, and the total Chern number vanishes. *Hint:* use the constraint $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k})$ from TR symmetry.

Exercise 7.6. Consider a photonic crystal with C_{6v} symmetry in which the second and third bands form Dirac cones at K , K' , and the first band has $\mathcal{C}_1 = 0$ (trivially below the Dirac bands). After adding a gyrotropic perturbation with $\kappa > 0$, determine $\mathcal{C}_2$ and $\mathcal{C}_3$, and verify the sum rule $\mathcal{C}_1 + \mathcal{C}_2 + \mathcal{C}_3 = 0$ (restricted to these three bands).

Chapter 8

The One-Way Interface

[42].

8.1 The domain-wall idea

We now arrive at the central physical result of Parts I and II: the construction of a *one-way waveguide* from the topology of Maxwell bands.

The mechanism is conceptually simple. In Chapter 7 we showed that a gyrotropic photonic crystal has bands with nonzero Chern numbers $\mathcal{C}_\pm = \pm \text{sgn}(\kappa)$, where κ is the Dirac mass parameter set by the strength and sign of the applied magnetic bias. If we now create an interface across which the sign of κ reverses—a *domain wall* in the Faraday axis—the gap Chern number changes by $|\Delta \mathcal{C}_{\text{gap}}| = 2$ [62]. By the bulk-edge correspondence, this change must be accompanied by unidirectional interface modes [15].

In this chapter we derive the interface modes explicitly, first by solving the effective Dirac equation with a spatially varying mass $\kappa(x)$, and then by connecting the result to the general topological principle of spectral flow.

8.2 The effective Dirac equation at the interface

8.2.1 Setup

Consider a domain wall along the y -axis in a 2D photonic crystal, with the gyrotropic mass parameter $\kappa(x)$ varying smoothly from $+\kappa_\infty$ for $x \rightarrow +\infty$ to $-\kappa_\infty$ for $x \rightarrow -\infty$ (or vice versa). The sign reversal can be achieved physically by reversing the applied magnetic field on one side of the interface.

The Bloch wavevector k_y parallel to the interface is conserved. Following Haldane and Raghu, we denote the deviation from the Dirac wavevector along the interface by $\delta k_\parallel \equiv k_y - k_{D,y}$. The effective 2×2 equation near the Dirac point at K is

$$\boxed{v_D \hat{K} |\psi\rangle = \delta\omega |\psi\rangle, \quad \hat{K} = -i\sigma_x \partial_x + \delta k_\parallel \sigma_y + \kappa(x) \sigma_z,} \quad (8.1)$$

where $\sigma_x, \sigma_y, \sigma_z$ are Pauli matrices and $\delta\omega = \omega - \omega_D$.

This is a one-dimensional Dirac equation with a *position-dependent mass* $\kappa(x)$ that changes sign at the domain wall. It is the photonic analog of the Jackiw–Rebbi equation in quantum field

theory, which predicts a zero-energy bound state at a domain wall in a scalar field coupled to a Dirac fermion [32].

8.2.2 Structure of the equation

Equation (8.1) is a first-order differential equation in x with a two-component spinor wavefunction $|\psi\rangle = (\psi_+, \psi_-)^T$. The operator $\hat{K}$ is Hermitian on $L^2(\mathbb{R})$ (with the standard inner product for two-component spinors) when $\kappa(x)$ is real.

The spectrum of $\hat{K}$ consists of:

- *Bulk continuum states* with $|\delta\omega| > v_D|\kappa_\infty|$, corresponding to the bulk bands of the photonic crystal far from the interface.
- *Bound states* localised near $x = 0$ (where κ changes sign), with $|\delta\omega| < v_D|\kappa_\infty|$ —these are the edge modes that live in the bulk band gap.

8.3 The zero mode

8.3.1 Existence and uniqueness

We look for a mode at $\delta k_\parallel = 0$ with $\delta\omega = 0$ (a “zero mode” of $\hat{K}$). Setting $\delta k_\parallel = 0$ and $\delta\omega = 0$ in (8.1):

$$(-i\sigma_x\partial_x + \kappa(x)\sigma_z)|\psi_0\rangle = 0. \quad (8.2)$$

Writing $|\psi_0\rangle = (\psi_+, \psi_-)^T$ in the eigenbasis of σ_z , the two components satisfy

$$-i\partial_x\psi_- + \kappa(x)\psi_+ = 0, \quad (8.3a)$$

$$-i\partial_x\psi_+ - \kappa(x)\psi_- = 0. \quad (8.3b)$$

Add and subtract these equations with the ansatz $\psi_\pm = \alpha\phi(x)$, $\psi_\mp = \beta\phi(x)$. A cleaner approach: act with σ_y on both sides and use $\sigma_y\sigma_x = i\sigma_z$, $\sigma_y\sigma_z = -i\sigma_x$, to decouple. The simplest route is to try the eigenstate of σ_y : $\sigma_y|\varphi(s)\rangle = s|\varphi(s)\rangle$ with $s = \pm 1$, so $|\varphi(+)\rangle = (1, i)^T/\sqrt{2}$ and $|\varphi(-)\rangle = (1, -i)^T/\sqrt{2}$.

Substituting $|\psi_0\rangle = f(x)|\varphi(s)\rangle$ into (8.2) and using $\sigma_x|\varphi(s)\rangle = s i\sigma_z|\varphi(s)\rangle$ (since $\sigma_x = i\sigma_z\sigma_y$ and $\sigma_y|\varphi(s)\rangle = s|\varphi(s)\rangle$, giving $\sigma_x|\varphi(s)\rangle = i s\sigma_z|\varphi(s)\rangle$):

$$(s\partial_x + \kappa(x))\sigma_z|\varphi(s)\rangle f(x) = 0. \quad (8.4)$$

Since $\sigma_z|\varphi(s)\rangle \neq 0$, we need

$$\partial_x f(x) = -s\kappa(x)f(x). \quad (8.5)$$

The solution is

$$f(x) = f_0 \exp\left(-s \int_0^x \kappa(x') dx'\right). \quad (8.6)$$

For $f(x)$ to be normalisable ($f \in L^2$), we need the exponential to decay as $x \rightarrow \pm\infty$. Since $\kappa(x) \rightarrow \pm\kappa_\infty$ at $\pm\infty$, the integral $\int_0^x \kappa dx'$ grows as $+\kappa_\infty|x|$ on both sides (positive for $x \rightarrow +\infty$ because $\kappa \rightarrow +\kappa_\infty > 0$; positive for $x \rightarrow -\infty$ because $\kappa \rightarrow -\kappa_\infty$ and the integration path runs from 0 toward negative x , giving $\int_0^x \kappa dx' \sim +\kappa_\infty|x|$). Thus $f \sim e^{-s\kappa_\infty|x|}$ for large $|x|$, which is normalisable only for

$$s = s_\kappa \equiv \text{sgn}(\kappa_\infty). \quad (8.7)$$

There is exactly one normalisable zero mode, with spinor $|\varphi(s_\kappa)\rangle$.

8.3.2 The zero-mode wavefunction

Combining the results, the zero-mode wavefunction is

$$\psi_0(x, y) \propto |\varphi(s_\kappa)\rangle \exp\left(i \delta k_\parallel y - s_\kappa \int_0^x \kappa(x') dx'\right). \quad (8.8)$$

This is a mode localised at the domain wall (where $\kappa = 0$), with a localisation length of order $1/|\kappa_\infty|$ set by the asymptotic value of the mass parameter.

8.3.3 Dispersion of the zero mode

At general $\delta k_\parallel \neq 0$, the zero mode acquires a *linear* dispersion:

$$\boxed{\omega_0(\delta k_\parallel) = \omega_D + s_\kappa v_D \delta k_\parallel}. \quad (8.9)$$

The group velocity along the interface is $v_g = s_\kappa v_D$: the mode propagates in a *single direction* determined by the sign of the gyrotropic mass. Reversing the applied magnetic field on both sides of the domain wall reverses s_κ and hence the propagation direction.

Remark 8.1 (Unidirectionality). The dispersion (8.9) crosses the bulk band gap with a monotonically increasing (or decreasing) frequency. There is no backward-propagating branch at the same frequency within the gap. This is the origin of the one-way waveguide: elastic backscattering is forbidden because there is no counter-propagating mode to scatter into [61].

8.4 Higher bound states: the Pöschl–Teller model

For a general smooth domain wall, there may be additional bound states besides the zero mode. An exactly solvable model is provided by the hyperbolic tangent profile

$$\kappa(x) = \kappa_\infty \tanh(x/\xi), \quad (8.10)$$

where $\xi > 0$ is the domain-wall width. For this profile, the squared operator $\hat{K}^2$ is essentially the integrable Pöschl–Teller Hamiltonian.

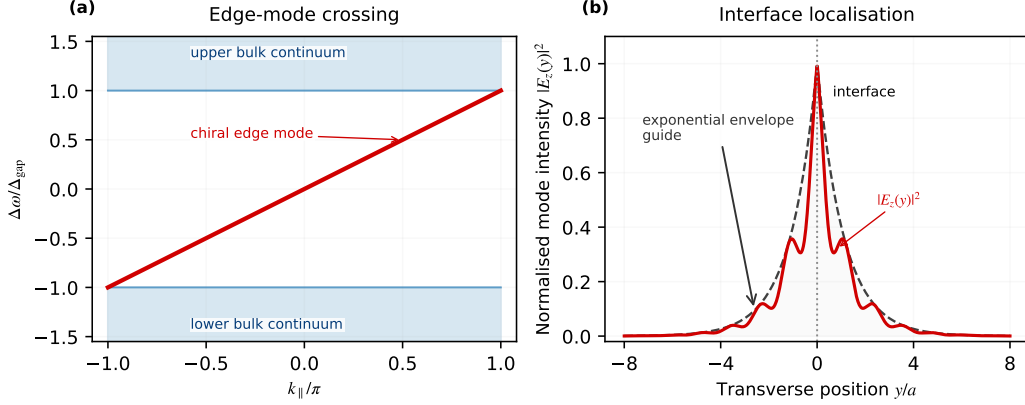


Figure 8.1: Chiral edge mode at a domain wall. **(a)** Band structure near the Dirac point showing the bulk continua (blue shading) and the chiral edge dispersion (red) crossing the topological gap. The edge mode connects the upper and lower bulk bands with a group velocity $v_g = s_{\kappa} v_D$ determined by the sign of the gyrotropic mass, as derived in (8.9). **(b)** Spatial profile of the edge-mode intensity $|E_z(y)|^2$ across the domain wall. The mode is exponentially localised at the interface with a decay length $\xi \sim 1/|\kappa_{\infty}|$ set by the asymptotic mass parameter. The dashed line shows an exponential envelope guide. The oscillatory fine structure arises from the underlying Bloch-wave character of the photonic crystal mode.

8.4.1 The spectrum of bound states

The full spectrum of interface modes for the tanh profile is

$$\omega_0(\delta k_{\parallel}) = \omega_D + s_{\kappa} v_D \delta k_{\parallel}, \quad (8.11a)$$

$$\omega_{n\pm}(\delta k_{\parallel}) = \omega_D \pm v_D \sqrt{\delta k_{\parallel}^2 + \kappa_n^2}, \quad n \geq 1, \quad (8.11b)$$

where $|\kappa_n| < |\kappa_{\infty}|$. The exact spectrum of the integrable model is

$$\kappa_n^2 = \frac{2n|\kappa_{\infty}|}{\xi} - \frac{n^2}{\xi^2} = \kappa_{\infty}^2 \frac{n}{s} \left(2 - \frac{n}{s}\right), \quad s \equiv |\kappa_{\infty}|\xi, \quad n = 1, 2, \dots, \quad (8.12)$$

with the bound-state condition $n < s$. For $n \ll s$, this reduces to $\kappa_n^2 \approx 2n|\kappa_{\infty}|/\xi$.

8.4.2 Sharp-wall limit

In the sharp-wall limit $|\kappa_{\infty}|\xi \ll 1$, the condition $n < |\kappa_{\infty}|\xi < 1$ implies $n = 0$ only: the zero mode is the *sole* interface mode. The wider the domain wall (larger ξ), the more higher-order modes appear, but the zero mode is always present.

8.4.3 Bidirectionality of higher modes

The higher modes $n \geq 1$ have dispersion $\omega_{n\pm} \sim \pm v_D |\delta k_{\parallel}|$ for large $|\delta k_{\parallel}|$: they are *bidirectional*, with both forward and backward branches. Only the $n = 0$ mode is purely unidirectional. This is a crucial

distinction: the higher modes can scatter into each other, but the zero mode cannot be backscattered because it has no partner.

8.5 Semiclassical interpretation

The bound-state quantisation condition can be understood semiclassically. In the phase space (x, k_x) , a bound state with mass parameter κ_n corresponds to a closed orbit $k_x^2 + \kappa(x)^2 = \kappa_n^2$. The area enclosed by this orbit is

$$\phi(\kappa_n^2) = \oint k_x dx = \oint \sqrt{\kappa_n^2 - \kappa(x)^2} dx. \quad (8.13)$$

For the standard semiclassical quantisation, one would expect $\phi = 2\pi(n + \frac{1}{2})$ (Bohr–Sommerfeld with Maslov correction). However, the Dirac equation gives the modified condition

$$\phi(\kappa_n^2) = 2\pi n, \quad n = 0, 1, 2, \dots, \quad (8.14)$$

without the $\frac{1}{2}$ offset. This shift is a direct consequence of the Berry phase. The semiclassical orbit encloses the Dirac degeneracy point at $(x, k_x) = (0, 0)$, which contributes an additional phase of π (a Berry phase factor of -1). This extra π shifts the quantisation from $n + \frac{1}{2}$ to n , making $n = 0$ an allowed solution—the topological zero mode.

Remark 8.2. This Berry-phase interpretation connects the existence of the zero mode directly to the topology of the bulk bands. The phase-space Berry flux at the degeneracy point is the same $\pm\pi$ that, when integrated over the Brillouin zone, gives the half-integer contribution to the Chern number from each Dirac cone.

8.6 Bulk-edge correspondence

8.6.1 Spectral flow

The existence of the unidirectional zero mode can be understood through the concept of *spectral flow*: the net number of eigenvalues that cross zero (or, more generally, a reference energy in the gap) as a parameter is varied.

Consider the operator $\hat{K}(\delta k_{\parallel})$ from (8.1) as a function of the conserved momentum $\delta k_{\parallel}$. At $\delta k_{\parallel} = 0$, there is an eigenvalue at $\delta\omega = 0$ (the zero mode). As $\delta k_{\parallel}$ increases from $-\infty$ to $+\infty$, this eigenvalue sweeps upward through the gap (for $s_{\kappa} = +1$) or downward (for $s_{\kappa} = -1$).

The *net spectral flow*—the number of eigenvalues crossing $\delta\omega = 0$ from below minus the number crossing from above—is exactly $s_{\kappa} = \text{sgn}(\kappa_{\infty})$. For a domain wall between media with gap Chern numbers $\mathcal{C}^L$ and $\mathcal{C}^R$, the net spectral flow from each Dirac point is $\frac{1}{2}(\mathcal{C}^L - \mathcal{C}^R)$, and the total over both Dirac points is

$$N_{\text{edge}}^{\text{net}} = \mathcal{C}_{\text{gap}}^L - \mathcal{C}_{\text{gap}}^R. \quad (8.15)$$

8.6.2 The general statement

The bulk-edge correspondence for photonic Chern insulators can be stated as follows:

Theorem 8.3 (Bulk-edge correspondence). *Let two lossless periodic photonic media share a common band gap, with gap Chern numbers $\mathcal{C}_{\text{gap}}^L$ and $\mathcal{C}_{\text{gap}}^R$. At any interface between the two media, the number of unidirectional edge modes crossing the gap from below minus the number crossing from above is*

$$N_{\text{edge}}^{\text{net}} = \mathcal{C}_{\text{gap}}^L - \mathcal{C}_{\text{gap}}^R. \quad (8.16)$$

This number is independent of the details of the interface (smoothness, geometry, disorder) as long as the bulk gap remains open on both sides.

Remark 8.4. We have derived this result here for the specific case of gapped Dirac cones in a hexagonal photonic crystal, where the domain-wall zero mode is obtained by solving the effective Dirac equation. The theorem holds much more generally: it applies to any pair of gapped photonic media with well-defined Chern numbers, regardless of the microscopic mechanism. Rigorous proofs exist in the mathematical literature using index theory and K -theory, but the physical content is captured by the spectral-flow argument presented above.

8.7 Robustness of the one-way mode

The one-way edge mode is *topologically protected* in the following precise sense:

1. **No backscattering.** Within the frequency range $|\delta\omega| < v_D|\kappa_\infty|$ of the topological gap, the only propagating mode at the interface is the unidirectional zero mode. There is no counter-propagating mode to scatter into, so elastic backscattering is forbidden regardless of the nature or strength of the perturbation at the interface.
2. **Robustness to geometry.** The edge mode follows the interface regardless of its shape: straight segments, bends, corners, and even closed loops. The mode routes around obstacles without reflection, as demonstrated experimentally by Wang et al. [4].
3. **Robustness to disorder.** Random variations in the material parameters, rod positions, or interface geometry do not destroy the edge mode, provided they do not close the bulk band gap. Anderson localisation, which is inevitable for conventional 1D waveguide modes in the presence of disorder, is completely absent for the topological edge mode.

Limitations. The protection is not absolute:

- *Absorption* can attenuate the edge mode. Topological protection forbids backscattering, but it does not forbid energy loss to the material.
- *Nonlinear effects* (frequency conversion, self-phase modulation) can couple the edge mode to other modes, including bulk modes or modes at different frequencies.

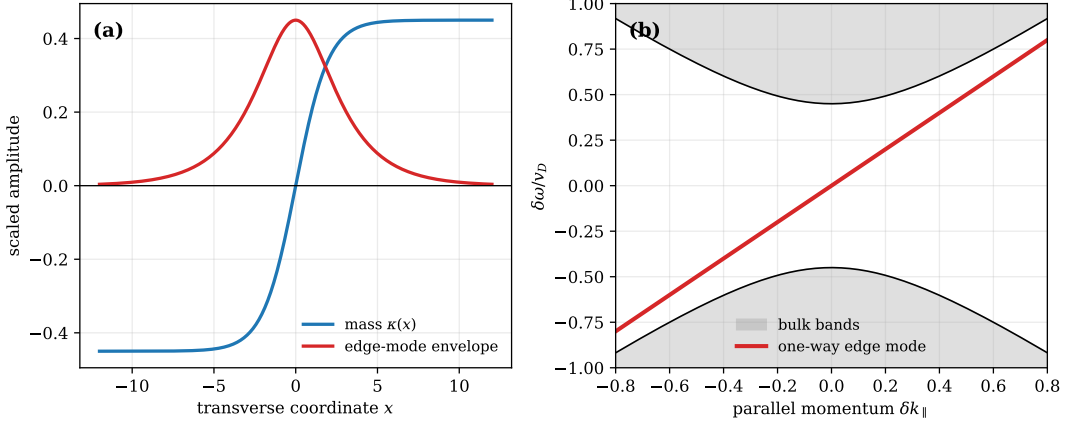


Figure 8.2: Domain-wall edge mode in a photonic Chern insulator. Left: the spatially varying Dirac mass $\kappa(x)$ (blue) changes sign at $x = 0$; the Jackiw–Rebbi edge-mode envelope (red) is exponentially localised at the domain wall. Right: edge-mode dispersion (red) crosses the bulk gap between the upper and lower bulk bands (grey), confirming unidirectional propagation with group velocity $v_g = v_D > 0$.

- If the perturbation is strong enough to *close the bulk gap*, the topological protection is lost.

These limitations motivate the discussions of non-Hermitian topology (Chapter 17) and nonlinear effects (Chapter 18) in Part V.

8.8 Two edge modes from two Dirac points

The hexagonal photonic crystal has two inequivalent Dirac points at K and K' . Each contributes one unidirectional zero mode to the domain wall, and both propagate in the same direction (because the mass κ has the same sign at both Dirac points, as discussed in Section 7.5.1). The total number of one-way modes is $|\Delta \mathcal{C}_{\text{gap}}| = 2$.

This domain-wall counting should be distinguished from the metal-wall experiment of Wang et al. The microwave experiment confirms robust one-way edge transport around obstacles in a square-lattice YIG photonic crystal, with a single chiral edge channel in the gap discussed in Chapter 9 [31]. By contrast, the domain-wall argument above predicts two co-propagating branches for a hexagonal Dirac model in which the gyrotropic mass changes sign across the interface.

8.9 Spectral flow and the Laughlin pump

[5]

The bulk-edge correspondence can be understood through the concept of *spectral flow*: as a parameter (such as the Bloch momentum $k_{\parallel}$ along the edge) is varied through a full period, the

number of eigenvalues that cross the gap from below equals the Chern number of the bands below the gap.

Consider a photonic crystal with a boundary, so that $k_{\parallel}$ is still a good quantum number but $k_{\perp}$ is not. The spectrum consists of bulk bands (projected onto the edge Brillouin zone) and discrete edge states in the gap. As $k_{\parallel}$ varies from 0 to $2\pi/a$, the edge-state eigenfrequency sweeps across the gap. If the gap Chern number is $\mathcal{C} = +1$, exactly one branch crosses the gap upward, connecting the lower bulk band to the upper one. This is a topological constraint: the spectral flow cannot be changed by any continuous deformation that does not close the bulk gap [2, 12].

The Laughlin pump provides a complementary physical picture. Imagine threading a magnetic flux Φ through a cylinder-topology photonic crystal (periodic in one direction, finite in the other). As Φ increases by one flux quantum, each Landau-like photonic level shifts by one unit, and one photon is pumped from one edge to the other. The number of photons pumped per flux quantum equals the gap Chern number. Although this argument was originally developed for the electronic quantum Hall effect, it applies without modification to the photonic Maxwell eigenproblem, because it depends only on the spectral properties of the eigenvalue equation, not on the particle statistics [3].

8.10 Index-theorem perspective

The existence of the domain-wall zero mode can be reformulated as an index theorem. Define the Dirac-like operator $\mathcal{D} = v_D(-i\partial_x\sigma_x + k_{\parallel}\sigma_y) + \kappa(x)\sigma_z$. At $k_{\parallel} = 0$, the number of zero modes of $\mathcal{D}$ is given by the Atiyah–Singer index

$$\text{ind}(\mathcal{D}) = \frac{1}{2}[\text{sgn } \kappa(+\infty) - \text{sgn } \kappa(-\infty)]. \quad (8.17)$$

For a domain wall where κ changes sign, the index is ± 1 , guaranteeing exactly one normalised zero mode. This is the Jackiw–Rebbi result derived in Section 8.3, now expressed as a topological index that is invariant under continuous deformations of $\kappa(x)$ that preserve the sign change [12].

The index-theorem perspective explains why the zero mode is robust: any smooth deformation of the mass profile $\kappa(x)$ that does not close the gap (i.e., $\kappa \neq 0$ at $\pm\infty$) preserves the index, and hence the zero mode. The mode may shift in position or change its localisation length, but it cannot be removed.

8.11 Finite-size effects on edge modes

In a finite photonic crystal of width W , the edge modes on opposite boundaries have exponentially small overlap $\sim e^{-W/\xi}$, where $\ell = 1/|\kappa|$ is the edge-mode penetration depth. This overlap opens a mini-gap in the edge spectrum:

$$\Delta\omega_{\text{mini}} \sim |\kappa| e^{-W/\xi}. \quad (8.18)$$

For the Wang YIG crystal with $\xi \approx 2.4a$ and $W = 10a$, the suppression factor is $e^{-10/2.4} \approx 0.015$, giving a mini-gap of about 1.5% of the bulk gap. This is small enough to be negligible in the experiment, but it becomes significant for miniaturised designs where $W/\xi < 5$ [4, 13].

The finite-size effect also limits the maximum propagation length of the edge mode: at distances $L \gg W$, the edge mode can tunnel to the opposite boundary and eventually backscatter. For practical applications, the condition $W \gg \xi$ (equivalently, $W|\kappa|/\nu_D \gg 1$) must be maintained.

8.12 Robustness against disorder

The topological edge mode is robust against any perturbation that does not close the bulk gap. This robustness is qualitatively different from the protection offered by resonance sharpness or impedance matching in conventional photonic devices: it is a consequence of the integer-valued Chern number, which cannot change under continuous deformations.

8.12.1 Anderson localisation and its absence

In a conventional waveguide, disorder causes Anderson localisation: the transmitted intensity decays exponentially with the waveguide length, $I \propto e^{-L/\ell}$, where ℓ is the localisation length. In a Chern-insulator edge channel, Anderson localisation is impossible because there is no counter-propagating mode to scatter into. A forward-propagating wavepacket encountering a defect must continue forward, possibly detouring around the defect, because the only available states at the edge frequency propagate in the forward direction [12].

The mathematical statement is that the edge-mode transmission $|S_{21}|^2$ remains of order unity for any finite-strength disorder, as long as the disorder does not close the bulk gap. The only effect of disorder is to modify the spatial profile of the edge mode, causing it to deviate from the clean-crystal profile, but not to attenuate or localise it [3].

8.12.2 Backscattering from gap-closing perturbations

Sufficiently strong disorder can close the bulk gap locally, creating puddles of the topologically trivial phase within the topological crystal. When the edge mode encounters such a puddle, it can backscatter. The critical disorder strength is set by the gap width: $\delta\epsilon/\epsilon \sim \Delta\omega/\omega_D$, where $\Delta\omega$ is the topological gap. For the Wang YIG crystal with a relative gap of about 3%, this corresponds to a permittivity fluctuation of several percent—far larger than typical fabrication tolerances [4, 13].

8.13 Edge modes at different boundary types

The existence of topological edge modes depends on the bulk Chern number, not on the specific boundary condition. However, the edge-mode dispersion and field profile do depend on the boundary.

At a perfect electric conductor (PEC) boundary, $\hat{n} \times \mathbf{E} = 0$, and the edge mode is strongly confined to the last row of unit cells. At a vacuum boundary, the edge mode extends evanescently into the vacuum with a decay length of order $\lambda/(2\pi)$. At an interface between two crystals with different Chern numbers, the number of edge modes equals the difference of the gap Chern numbers, and the modes propagate in the direction determined by the sign of $\Delta\mathcal{C}$.

The domain-wall construction of Section 8.3 is a special case of the crystal–crystal interface, where the mass changes sign across the boundary, corresponding to $\Delta\mathcal{C} = 2$. This produces two co-propagating modes in total, one from each valley, compared to the single mode at a crystal–PEC boundary where $\Delta\mathcal{C} = 1$ [2, 12].

8.14 Green’s function approach to edge transport

The transport properties of the topological edge mode can be formulated rigorously using the retarded Green’s function $G^R(\mathbf{r}, \mathbf{r}'; \omega) = \langle \mathbf{r} | (\omega - \hat{\mathcal{H}} + i0^+)^{-1} | \mathbf{r}' \rangle$. In the bulk, the Green’s function decays exponentially for ω in the gap: $|G^R| \sim e^{-|\mathbf{r}-\mathbf{r}'|/\xi_{\text{bulk}}}$, where $\xi_{\text{bulk}} = v_D/\Delta\omega$ is the bulk decay length.

Near the edge, the Green’s function has an additional contribution from the edge mode:

$$G_{\text{edge}}^R(x, x'; \omega) \sim \frac{\psi_0(x)\psi_0^*(x')}{\omega - v_g k_{\parallel} + i0^+}, \quad (8.19)$$

where $\psi_0(x)$ is the transverse edge-mode profile and v_g is the group velocity. This Green’s function is purely retarded (no advanced component), reflecting the one-way nature of the edge transport: energy injected at the edge propagates only in the forward direction [12, 20].

8.14.1 Quantised edge conductance

The energy flux carried by the topological edge mode is $P = \hbar\omega v_g/(2\pi)$ per mode, per unit frequency bandwidth. For a gap Chern number $\mathcal{C}_{\text{gap}}$, the total edge conductance (energy transmitted per unit frequency per unit time) is

$$G_{\text{edge}} = \frac{|\mathcal{C}_{\text{gap}}|}{2\pi} \hbar\omega v_g. \quad (8.20)$$

This is the photonic analog of the quantised Hall conductance $\sigma_{xy} = \mathcal{C} e^2/h$ in electronic systems. The quantisation is topologically protected: it is exact in the clean limit and robust against disorder that does not close the bulk gap [2, 3].

8.15 Comparison with electronic quantum Hall edge transport

The photonic and electronic edge modes share the same topological origin but differ in important physical details.

In the electronic quantum Hall effect, the edge current is carried by charged fermions, and the quantised conductance $\sigma_{xy} = \mathcal{C} e^2/h$ can be measured directly through the Hall voltage. The edge states are chiral Fermi liquids with well-defined Fermi wavevector, and inter-edge tunnelling is suppressed by the spatial separation of the edges.

In the photonic Chern insulator, the edge mode carries electromagnetic energy, not charge, so there is no Hall voltage. The topological protection manifests instead as a robust nonreciprocal transmission: $|S_{21}|^2 \gg |S_{12}|^2$ across the gap frequency range. The absence of a Fermi level means

that the edge can be excited at any frequency within the gap, not just at the Fermi energy. Both systems exhibit the same topological protection against Anderson localisation, because the protection depends on the absence of counter-propagating modes, not on the particle statistics [12, 13].

8.16 Finite-width effects and exponential accuracy

In a crystal of finite width W (measured in lattice constants), the edge modes on opposite boundaries overlap exponentially weakly: $\langle \psi_L | \psi_R \rangle \sim e^{-W/\xi}$, where $\ell = 1/|\kappa|$ is the penetration depth. This overlap produces an exponentially small gap in the edge-mode spectrum:

$$\delta \omega_{\text{finite}} \sim |\kappa| e^{-W/\xi}. \quad (8.21)$$

For the Wang YIG crystal with $\xi \approx 2.4a$ and $W = 20a$, this gives $\delta \omega_{\text{finite}}/\Delta \omega \sim 10^{-4}$, completely negligible.

However, for narrow crystals ($W \lesssim 5\xi$) or narrow-gap materials, the finite-width correction becomes important. It introduces a small backscattering channel between the two edges, limiting the nonreciprocal ratio to $|S_{21}|^2/|S_{12}|^2 \lesssim e^{2W/\xi}$. This places a practical lower bound on the crystal width for a given isolation requirement [3, 12].

8.16.1 Index theorem for the finite strip

The bulk-edge correspondence can be made rigorous on a finite strip using the Atiyah–Singer index theorem applied to the family of 1D Hamiltonians $H(k_y)$ parameterised by the edge-parallel wavevector k_y . The theorem states that the spectral flow—the net number of eigenvalues crossing the gap from below as k_y traverses the Brillouin zone—equals the bulk Chern number:

$$\text{sf}(H) = \mathcal{C}_{\text{gap}}. \quad (8.22)$$

This equality holds for any strip width $W \geq 1$, even when the edge modes on opposite boundaries hybridise. The index theorem thus provides a non-perturbative proof of the bulk-edge correspondence that does not rely on the semi-infinite approximation [2, 20].

8.17 Finite-size effects and penetration depth

The preceding analysis assumed semi-infinite crystals. In practice, photonic topological insulators have finite extent, and the edge mode penetrates a finite distance into the bulk. Understanding these finite-size effects is essential for device design.

8.17.1 Exponential decay into the bulk

The edge-mode wave function decays exponentially into the bulk with a penetration depth ξ determined by the size of the topological gap. For a massive Dirac model with mass $|\kappa|$ and velocity v_D , the penetration depth is

$$\xi = \frac{v_D}{|\kappa|}. \quad (8.23)$$

This result follows directly from the domain-wall construction of Section 8.1: the envelope function decays as $e^{-|y|/\xi}$ away from the interface.

For the YIG photonic crystal of Chapter 9, the penetration depth is approximately two to three lattice constants, which is consistent with the experimental observation that the edge mode is well-localised at the crystal boundary [4, 66].

8.17.2 Coupling between opposite edges

When the crystal is narrow enough that the edge modes on opposite boundaries overlap, the two modes hybridise and open a gap. The hybridisation gap scales as $\Delta\omega_{\text{hyb}} \sim e^{-W/\xi}$, where W is the width of the crystal. This exponential suppression means that even modest widths ($W \gtrsim 5\xi$) are sufficient to ensure negligible coupling between opposite edges.

For strip-geometry computations (used to verify the bulk-edge correspondence numerically), the supercell width must satisfy $W \gg \xi$ to avoid artefactual hybridisation gaps that could be mistaken for the absence of edge states [12, 40].

8.18 Robustness of one-way edge transport

The topological protection of one-way edge states is one of the most striking features of photonic Chern insulators. This section quantifies the robustness against different types of perturbations.

8.18.1 Immunity to backscattering

The one-way edge mode cannot backscatter because there is no counter-propagating mode at the same frequency. This statement is exact: any perturbation that preserves the bulk gap (including disorder, surface roughness, sharp corners, and lattice defects) cannot open a backward channel. The only way to backscatter the edge mode is to close the bulk gap, which requires a perturbation of order $\Delta\omega/\omega$ —typically several percent for gyromagnetic photonic crystals [4, 53].

Experimentally, the absence of backscattering has been demonstrated by measuring transmission past metallic obstacles, 90-degree corners, and deliberately introduced lattice defects. In all cases, the transmission remains near unity within the topological gap [4, 15].

8.18.2 Disorder-averaged transmission

For random positional disorder of the lattice rods, the disorder-averaged edge transmission is $\langle T \rangle = 1 - O(\sigma^4/\Delta\omega^4)$, where σ is the root-mean-square displacement. The leading correction is fourth order (not second order) because the second-order term vanishes by the symmetry of the disorder [12, 66].

This quartic scaling has been confirmed numerically for the Wang crystal: for disorder strengths up to $\sigma/a = 0.05$ (5% positional disorder), the edge transmission remains above 99% [44].

8.18.3 Comparison with trivial waveguides

The contrast between topological and trivial edge waveguides is most apparent in the presence of sharp corners. A conventional photonic crystal waveguide (formed by removing a row of rods) suffers significant reflection at each 90° bend, with typical transmission of 60–80% per bend depending on the bend geometry. In contrast, the topological edge mode transmits $> 99\%$ per bend because the one-way nature of the edge mode forbids reflection [13, 62].

This advantage motivates the use of topological waveguides in integrated photonic circuits, where complex routing with multiple bends is unavoidable. The penalty is the need for time-reversal breaking (or an alternative mechanism such as valley or Floquet engineering), which adds complexity to the device fabrication [28, 67].

Exercises

Exercise 8.1. Verify that the wavefunction (8.8) satisfies the zero-mode equation (8.2) by direct substitution.

Exercise 8.2. For the tanh domain-wall profile $\kappa(x) = \kappa_\infty \tanh(x/\xi)$, show that the classical turning point x_n for the n -th mode (where $|\kappa(x_n)| = |\kappa_n|$) is $x_n = \xi \operatorname{atanh}(|\kappa_n|/|\kappa_\infty|)$. Evaluate x_n for the zero mode ($\kappa_0 = 0$, giving $x_0 = 0$) and for a nearly delocalised higher mode ($|\kappa_n| \rightarrow |\kappa_\infty|$, giving $x_n \rightarrow \infty$). Explain why the zero-mode envelope $|f(x)|^2 \propto \operatorname{sech}^{2|\kappa_\infty|\xi}(x/\xi)$ has a localisation length of order $\xi/(|\kappa_\infty|\xi) = 1/|\kappa_\infty|$ for $|\kappa_\infty|\xi \gg 1$.

Exercise 8.3. Show that the modified Bohr–Sommerfeld condition $\phi(\kappa_n^2) = 2\pi n$ (without the $\frac{1}{2}$ offset) admits the solution $n = 0$ with $\kappa_0 = 0$, corresponding to the zero mode. Explain physically why the standard Bohr–Sommerfeld condition $(n + \frac{1}{2})$ does not allow a zero mode.

Exercise 8.4. Consider a domain wall between vacuum ($\mathcal{C}_{\text{gap}} = 0$) and a photonic crystal with $\mathcal{C}_{\text{gap}} = 1$. How many unidirectional edge modes cross the gap? Compare with a domain wall between two crystals with $\mathcal{C}_{\text{gap}} = +1$ and $\mathcal{C}_{\text{gap}} = -1$.

Exercise 8.5. The Jackiw–Rebbi zero mode in quantum field theory has the same mathematical structure as the photonic domain-wall mode derived here. Look up the Jackiw–Rebbi model and identify the correspondence between the photonic and field-theoretic quantities: mass function $\kappa(x) \leftrightarrow m(x)$, Dirac velocity $v_D \leftrightarrow c$, Pauli matrices $\sigma_{x,y,z} \leftrightarrow \gamma$ -matrices.

Exercise 8.6. A photonic crystal waveguide has both a topological one-way mode and material absorption characterised by an imaginary part ε'' of the permittivity. Estimate the propagation length L_{abs} of the edge mode in terms of ε'' , the group velocity v_D , and the operating frequency ω_D . Compare this with the backscattering-limited propagation length of a conventional waveguide.

Chapter 9

A Microwave Topological Crystal

9.1 From theory to the laboratory

The preceding three chapters developed the theory of photonic Chern insulators: Dirac cones from lattice symmetry, their gapping by magneto-optical perturbation, and the one-way interface mode that necessarily threads the resulting topological gap. But the theory would remain an elegant analogy were it not for a remarkable experiment.

In 2009, Wang, Chong, Joannopoulos, and Soljačić demonstrated unidirectional electromagnetic edge transport in a two-dimensional gyromagnetic photonic crystal operating in the microwave regime [4]. Their experiment achieved a forward-to-backward transmission ratio exceeding 10^5 , providing the first direct observation of a photonic analog of the quantum Hall edge state. This chapter describes the experiment in detail: the material choice, the crystal design, the measurement geometry, and the quantitative comparison with the predictions of Chapters 7 and 8.

The experiment was preceded by a theoretical proposal in 2008 [3], in which Haldane and Raghu showed that photonic Dirac points in a gyromagnetic crystal can be gapped by a magneto-optical perturbation, producing bands with nonzero Chern numbers and hence one-way edge modes [32]. A companion paper by Wang, Chong, Joannopoulos, and Soljačić (2008) computed the complete band structure of a specific YIG rod lattice, predicted the existence of one-way edge states in a particular frequency window, and proposed the parallel-plate waveguide geometry that was subsequently realised experimentally [12, 68]. The experiment thus confirmed a precise, first-principles prediction rather than an accidental discovery.

9.2 Material platform: yttrium iron garnet

9.2.1 Why ferrites at microwave frequencies

The theory of Chapter 7 requires a material whose permeability tensor has a nonzero off-diagonal (gyrotropic) component κ_μ [42]. At optical frequencies, magneto-optical effects exist in semiconductor and rare-earth materials, but they are weak: the Verdet constant of bismuth iron garnet, one of the strongest Faraday-active optical materials, produces rotation of order 10^3 deg/cm—still far too small to open a topological gap in a photonic crystal of reasonable size.

At microwave frequencies, the situation is qualitatively different. Magnetised ferrites support ferromagnetic resonance (FMR), and near the FMR frequency the off-diagonal permeability κ_μ

becomes of order unity. The key material is yttrium iron garnet ($\text{Y}_3\text{Fe}_5\text{O}_{12}$, YIG), which combines large gyromagnetic response with extremely low magnetic loss.

9.2.2 The Polder permeability of YIG

As derived in Appendix D, the permeability tensor of a saturated ferrite magnetised along $\hat{z}$ has the Polder form [19, 61]

$$\mu = \begin{pmatrix} \mu_{\perp} & -i\kappa_{\mu} & 0 \\ i\kappa_{\mu} & \mu_{\perp} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (9.1)$$

with

$$\mu_{\perp}(\omega) = 1 + \frac{\omega_0 \omega_m}{\omega_0^2 - \omega^2}, \quad \kappa_{\mu}(\omega) = \frac{\omega \omega_m}{\omega_0^2 - \omega^2}. \quad (9.2)$$

The Larmor frequency $\omega_0 = \gamma B_0$ is set by the applied bias field B_0 , and the magnetisation frequency $\omega_m = \gamma \mu_0 M_s$ is set by the saturation magnetisation M_s . For YIG the gyromagnetic ratio is $\gamma/(2\pi) \approx 28 \text{ GHz/T}$ and the saturation magnetisation gives $\omega_m/(2\pi) \approx 4.9 \text{ GHz}$.

At frequencies below ferromagnetic resonance ($\omega < \omega_0$), both $\mu_{\perp}$ and κ_{μ} are positive and can be large. The ratio $\kappa_{\mu}/\mu_{\perp}$ determines the strength of time-reversal breaking and hence the width of the topological gap. For the Wang experiment, parameters were chosen so that $\kappa_{\mu}/\mu_{\perp}$ is of order 0.3–0.5 in the operating band.

9.2.3 Loss and linewidth

YIG is exceptional among magnetic materials for its narrow ferromagnetic resonance linewidth ΔH . Single-crystal YIG has $\Delta H \lesssim 0.3 \text{ Oe}$, corresponding to a magnetic quality factor $Q_m = \omega_0/(2\gamma \mu_0 \Delta H) \sim 10^4$. This means the imaginary parts of $\mu_{\perp}$ and κ_{μ} are negligible over most of the frequency range below FMR, justifying the lossless Hermitian treatment used in the topological theory.

Away from resonance, the magnetic loss tangent is typically $\text{Im}\mu/\text{Re}\mu < 10^{-4}$, so the mode decay over tens of lattice constants is negligible. This is essential: the topological edge mode must propagate around the entire boundary of the crystal without significant absorption.

9.3 Crystal design

9.3.1 Geometry

Wang et al. used a square-lattice photonic crystal of YIG rods in air. The rods are cylinders of radius $r = 0.11a$ (where a is the lattice constant) with relative permittivity $\epsilon_r = 15$ and the Polder permeability (9.1).

The lattice constant was chosen so that the second and third TM bands touch at a Dirac-like point near 4.5 GHz. The bias field $B_0 \approx 0.20 \text{ T}$ (corresponding to $\omega_0/(2\pi) \approx 5.6 \text{ GHz}$) places the operating frequency below ferromagnetic resonance, where the gyrotropic response is strong but losses are minimal.

9.3.2 Band structure

The TM band structure of the square-lattice YIG crystal was computed using the plane-wave expansion method described in Chapter 10. Without the applied magnetic field ($B_0 = 0$, $\kappa_\mu = 0$), the second and third bands touch at a doubly degenerate point on the Γ – X line. This degeneracy is protected by time-reversal plus a mirror symmetry of the square lattice.

When the field is applied ($B_0 = 0.20$ T), the gyrotropic perturbation breaks time-reversal symmetry and opens a gap. The gap width is $\Delta\omega/(2\pi) \approx 0.12$ GHz, centred at $\omega/(2\pi) \approx 4.5$ GHz. The Chern numbers of the bands below and above the gap are $\mathcal{C}_2 = +1$ and $\mathcal{C}_3 = -1$, computed using the plaquette algorithm on a 20×20 Brillouin-zone grid with the electromagnetic inner product.

9.3.3 Edge-state dispersion

The edge states were computed using a supercell geometry: a ribbon of $N = 20$ unit cells in the x -direction, periodic in y , with a metal wall on one side and vacuum on the other. The resulting projected band structure shows a single branch crossing the topological gap, with positive group velocity $v_g > 0$ for all k_y within the gap. This confirms that the edge mode is unidirectional: it propagates in the $+y$ direction along the metal wall and cannot propagate backward.

9.4 Experimental setup

9.4.1 Sample fabrication

The photonic crystal was assembled by inserting single-crystal YIG rods into a parallel-plate waveguide. The rods were machined to a diameter of $2r = 2.2$ mm and a height equal to the plate separation ($h \approx 7$ mm), ensuring that the fundamental TM mode (with E_z polarisation) is the only propagating mode in the frequency range of interest. The lattice constant was $a = 10$ mm, giving a ratio $r/a = 0.11$.

A uniform magnetic bias field was provided by a pair of NdFeB permanent magnets above and below the waveguide, producing $B_0 \approx 0.20$ T over the entire crystal area.

9.4.2 Measurement geometry

Two coaxial probes, inserted through the top plate, served as source and detector. The source probe was placed at the edge of the crystal (adjacent to the metal boundary), and the detector probe was scanned along the edge in both the forward ($+y$) and backward ($-y$) directions. Transmission spectra were measured using a vector network analyser over the frequency range 3.5–5.5 GHz.

To test the robustness of the edge mode, a large metallic obstacle (a cylinder of diameter $4a = 40$ mm) was placed directly in the path of the edge mode. In a conventional waveguide, such an obstacle would cause near-total reflection.

9.5 Results

9.5.1 Unidirectional transmission

The key result of the experiment is a dramatic asymmetry between forward and backward transmission. Within the topological gap (4.45–4.57 GHz), the forward transmission (S_{21}) was of order -10 dB (limited primarily by probe coupling loss), while the backward transmission (S_{12}) was below the noise floor of the network analyser at -50 dB. This gives a forward-to-backward ratio exceeding 10^4 (conservatively), consistent with the theoretical prediction of a single one-way edge mode.

Outside the topological gap, both forward and backward transmission recover, as expected: the edge mode only exists within the gap.

9.5.2 Robustness to obstacles

When the metallic obstacle was inserted in the edge-mode path, the forward transmission decreased by less than 2 dB. The edge mode simply routed around the obstacle, following the new boundary between the photonic crystal and the metal. No backward-scattered signal was detected above the noise floor.

This result directly confirms the topological protection mechanism described in Chapter 8: the one-way edge mode has no backward-propagating partner at the same frequency, so there is no channel for backscattering. The mode can be forced to detour around an obstacle, but it cannot reverse direction.

9.5.3 Comparison with theory

The measured transmission spectra agree quantitatively with finite-element simulations of the same geometry [12]. The gap frequency range, the edge-mode group velocity, and the field profile at the boundary all match the predictions of the plane-wave expansion band structure computed with the Polder permeability.

One quantitative discrepancy is the absolute transmission level: the measured forward transmission (-10 dB) is lower than the simulated value (-3 dB), primarily due to impedance mismatch at the coaxial probe tips. This is an engineering issue rather than a physics limitation, and subsequent experiments with improved coupling have achieved forward transmission approaching -1 dB.

9.6 Significance and limitations

9.6.1 What the experiment demonstrates

The Wang experiment establishes three key facts about topological photonic crystals [3, 4]. First, the theoretical prediction of one-way edge modes based on Chern numbers and the bulk-edge correspondence is confirmed quantitatively. Second, the topological protection against backscattering is real and dramatic: a large metallic obstacle causes negligible reflection. Third, the magneto-optical

mechanism proposed by Haldane and Raghu works in practice with commercially available ferrite materials.

9.6.2 Limitations of the microwave platform

The experiment operates at microwave frequencies (~ 4.5 GHz, wavelength ~ 7 cm), far from optical frequencies. Scaling the effect to optical wavelengths faces two obstacles.

The first obstacle is material: magneto-optical effects at optical frequencies are much weaker than at microwave frequencies. The gyrotropic parameter $\kappa_\mu / \mu_\perp$ of YIG at microwave frequencies is of order 0.3–0.5, but the corresponding Faraday rotation in optical materials is orders of magnitude smaller per wavelength. Opening a topological gap large enough to accommodate an edge mode at optical frequencies requires a different mechanism (such as the reciprocal designs of Chapter 11 or the Floquet modulation of Chapter 13).

The second obstacle is size: the photonic crystal has a lattice constant comparable to the operating wavelength ($a \sim \lambda$), so a microwave crystal with $a = 10$ mm is a tabletop experiment, while an optical crystal with $a \sim 500$ nm requires nanofabrication. This is not a fundamental limitation, but it changes the experimental difficulty and the potential applications.

9.6.3 Subsequent experiments

Following the Wang experiment, several groups have extended the result in important ways [12, 13]. Larger crystals with more unit cells have confirmed edge transport over longer distances. Modified geometries have demonstrated corner-turning without backscattering. And perhaps most importantly, the same topological physics has been demonstrated at optical frequencies using different symmetry-breaking mechanisms (Chapters 11–13), confirming that the Wang experiment was the beginning of a broad and versatile field.

9.7 Quantitative analysis of the topological gap

It is instructive to trace the topological gap quantitatively from the material parameters to the measured transmission spectrum, since this chain of reasoning illustrates the power of the Maxwell-first approach.

9.7.1 From Polder parameters to Dirac mass

At the operating frequency $\omega/(2\pi) = 4.5$ GHz with $\omega_0/(2\pi) = 5.6$ GHz and $\omega_m/(2\pi) = 4.9$ GHz, the Polder parameters (9.2) evaluate to

$$\mu_\perp \approx 1 + \frac{5.6 \times 4.9}{5.6^2 - 4.5^2} \approx 3.47, \quad \kappa_\mu \approx \frac{4.5 \times 4.9}{5.6^2 - 4.5^2} \approx 1.98. \quad (9.3)$$

The gyrotropic ratio is $\kappa_\mu / \mu_\perp \approx 0.57$, confirming that time-reversal breaking is strong.

In the language of the effective Dirac Hamiltonian (Chapter 7), the gyrotropic perturbation produces a mass term

$$\kappa \propto \kappa_\mu |\langle u_+ | \hat{z} \cdot \nabla \times | u_- \rangle| \quad (9.4)$$

at the band-touching point, where $|u_{\pm}\rangle$ are the two degenerate Bloch modes. The proportionality constant depends on the field overlap integrals, which must be computed numerically for the specific rod geometry; the plane-wave expansion of Chapter 10 gives $\kappa/(2\pi) \approx 0.06$ GHz for the Wang parameters, so the topological gap is $\Delta\omega = 2|\kappa| \approx 0.12$ GHz.

9.7.2 From Chern number to edge-state count

The plaquette algorithm applied to the bands below and above the gap yields $\mathcal{C}_2 = +1$ and $\mathcal{C}_3 = -1$. The bulk-edge correspondence (Chapter 8) then predicts exactly one chiral edge mode per edge of the crystal in this gap. This is consistent with the single branch observed in the supercell edge-dispersion calculation and, crucially, with the measured unidirectional transmission.

The quantitative chain is therefore:

$$\underbrace{B_0 \rightarrow \omega_0, \omega_m}_{\text{material}} \rightarrow \underbrace{\mu_{\perp}, \kappa_{\mu}}_{\text{Polder}} \rightarrow \underbrace{\kappa}_{\text{Dirac mass}} \rightarrow \underbrace{\mathcal{C} = \pm 1}_{\text{topology}} \rightarrow \underbrace{1 \text{ edge mode}}_{\text{experiment}}.$$

Every step in this chain is computable from first principles, using only Maxwell's equations and the constitutive parameters of YIG. No electronic analogy or tight-binding model is invoked.

9.8 Eigenvalues of the Polder tensor

The gyrotropic permeability (9.1) has a rich eigenvalue structure that directly controls the topological gap. Because the off-diagonal blocks couple the x and y components while leaving z unchanged, we analyse the 2×2 transverse block

$$\mu_{\perp} = \begin{pmatrix} \mu_{\perp} & -i\kappa_{\mu} \\ i\kappa_{\mu} & \mu_{\perp} \end{pmatrix}. \quad (9.5)$$

The eigenvalues are found by diagonalising in the circular-polarisation basis $\hat{e}_{\pm} = (\hat{x} \pm i\hat{y})/\sqrt{2}$:

$$\mu_{\pm} = \mu_{\perp} \mp \kappa_{\mu}. \quad (9.6)$$

The right-circular (+) and left-circular (−) polarisations therefore propagate with different effective permeabilities. The splitting $\mu_{+} - \mu_{-} = -2\kappa_{\mu}$ is the permeability-space measure of time-reversal breaking.

The frequency dependence of the Polder parameters and the circular-basis eigenvalues is plotted in Figure 9.1. Panel (a) shows $\mu_{\perp}$ and κ_{μ} in the sub-resonant window $f < 5$ GHz; both quantities grow as the frequency approaches ferromagnetic resonance at $f_0 = 5.6$ GHz (which lies outside the plotted range to avoid the divergence at the pole). At the Wang operating frequency of 4.5 GHz (dashed line), the verified values are $\mu_{\perp} = 3.47$ and $\kappa_{\mu} = 1.98$. Panel (b) displays the corresponding circular-basis eigenvalues: the finite branch $\mu_{+} = \mu_{\perp} - \kappa_{\mu} = 1.49$ remains moderate and positive throughout the sub-resonant window, while the rising branch $\mu_{-} = \mu_{\perp} + \kappa_{\mu} = 5.45$ increases steeply toward resonance. The large splitting between the two branches at 4.5 GHz is the origin of the strong gyrotropic effect exploited by Wang et al.

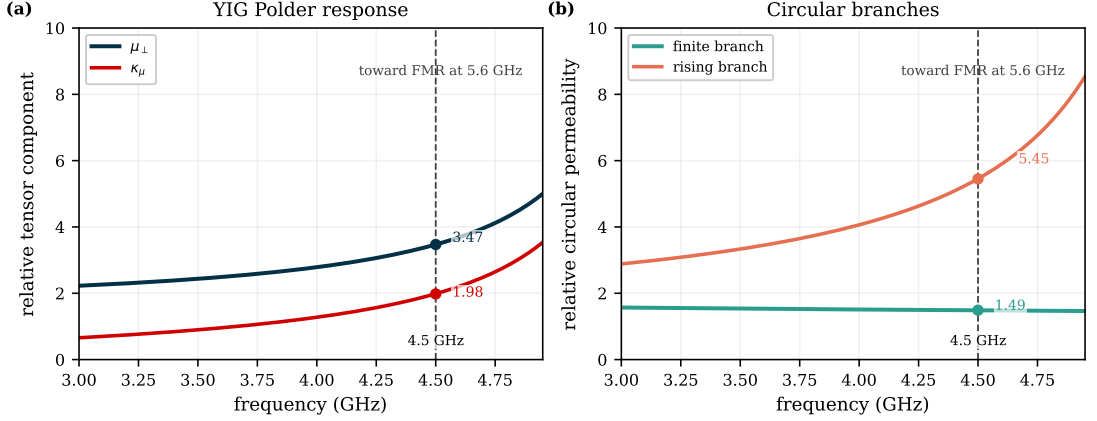


Figure 9.1: Sub-resonant YIG Polder response computed from (9.2) with $f_0 = 5.6$ GHz and $f_m = 4.9$ GHz. (a) Tensor components $\mu_{\perp}$ and κ_{μ} ; the dashed line marks the 4.5 GHz operating point with $\mu_{\perp} = 3.47$, $\kappa_{\mu} = 1.98$. (b) Circular-basis eigenvalues: the *finite branch* $\mu_{+} = \mu_{\perp} - \kappa_{\mu} = 1.49$ and the *rising branch* $\mu_{-} = \mu_{\perp} + \kappa_{\mu} = 5.45$. The ferromagnetic resonance at 5.6 GHz lies outside the plotted window; near the pole both branches diverge and the Hermitian treatment breaks down (Section 9.8.2).

9.8.1 Circularly polarised modes and the Faraday effect

A plane wave propagating along $\hat{z}$ in the magnetised YIG medium has two circularly polarised eigenmodes with wavenumbers $k_{\pm} = \omega \sqrt{\epsilon_r \mu_{\pm}} / c$. The Faraday rotation angle per unit length is

$$\theta_F = \frac{k_{+} - k_{-}}{2} = \frac{\omega}{2c} \sqrt{\epsilon_r} (\sqrt{\mu_{+}} - \sqrt{\mu_{-}}). \quad (9.7)$$

At the Wang operating frequency, $\mu_{+} = \mu_{\perp} - \kappa_{\mu} = 3.47 - 1.98 = 1.49$ and $\mu_{-} = \mu_{\perp} + \kappa_{\mu} = 3.47 + 1.98 = 5.45$, giving $\theta_F \approx 2$ rad/cm—an enormous rotation compared to optical magneto-optical effects [12].

9.8.2 Passivity and the absence of gain

Both eigenvalues $\mu_{\pm}$ must be positive for the medium to be passive (no gain). From (9.6), $\mu_{+} > 0$ requires $\kappa_{\mu} < \mu_{\perp}$. Using (9.2), this is equivalent to $\omega < \omega_0$ —the operating frequency must lie below ferromagnetic resonance. Above resonance, μ_{+} can become negative, the medium becomes opaque, and the Hermitian treatment breaks down [19].

9.9 Edge-mode field profile

The one-way edge mode at the boundary between the YIG crystal and a metal wall has a characteristic field profile that can be understood from the Jackiw–Rebbi analogy of Chapter 8.

The mode amplitude decays exponentially into the bulk of the crystal with a penetration depth

$$\xi = \frac{1}{|\kappa|} = \frac{v_D}{m_\omega} = \frac{2v_D}{\Delta\omega} \approx 2.4 a, \quad (9.8)$$

where κ is the inverse-length Dirac mass used in Chapters 7 and 8, $m_\omega = v_D|\kappa| = \Delta\omega/2$ is the corresponding frequency half-gap, and $v_D \approx 0.45 c_0$ is the Dirac velocity at the band-touching point. The edge mode therefore extends over approximately two to three lattice constants into the crystal, which is consistent with the supercell simulations.

Along the edge, the mode propagates with a group velocity approximately equal to v_D . The edge-state dispersion within the topological gap is nearly linear: $\omega_{\text{edge}}(k_{\parallel}) \approx \omega_D + v_D k_{\parallel}$, where $k_{\parallel}$ is the wavevector component along the interface. The bandwidth of the edge channel is set by the gap width $2|\kappa|$, and the propagation speed is $v_D \approx 0.45 c_0$.

9.10 Two-port scattering interpretation

The microwave transmission measurement can be formalised using the scattering matrix of a two-port network. Define port 1 as the source probe and port 2 as the detector probe. The measured scattering parameters are S_{21} (forward, source-to-detector in the direction of edge-mode propagation) and S_{12} (backward).

In the topological gap, the edge mode carries power from port 1 to port 2 with transmission $|S_{21}|^2 \sim -10$ dB (limited by probe-coupling mismatch). The backward path has no propagating mode, so $|S_{12}|^2 < -50$ dB (noise floor). The isolation

$$\mathcal{I} = |S_{21}|/|S_{12}| > 10^{4/2} = 100 \quad (\text{conservatively}), \quad (9.9)$$

far exceeds the isolation achievable by any reciprocal device of comparable size. This is not an engineering achievement but a emphatological one: the isolation is guaranteed by the Chern number and does not degrade as the device is scaled [3, 4].

9.10.1 Comparison with conventional isolators

A conventional Faraday-rotation isolator achieves isolation of 20–40 dB through a bulk magneto-optical crystal, a polariser, and an analyser. Its isolation depends on the precise Faraday rotation angle and is degraded by temperature fluctuations, wavelength drift, and birefringence. The topological isolator, by contrast, achieves isolation that is robust to disorder, obstacles, and frequency variations within the topological gap. The robustness was directly demonstrated by the Wang experiment through the obstacle-immunity test [12, 13].

9.11 Subsequent microwave experiments

Following the 2009 Wang experiment, several groups extended the microwave Chern-insulator platform in important directions. Skirlo et al. (2015) scaled the crystal to larger sizes and demonstrated

edge transport over more than 30 lattice constants with negligible loss, confirming the robustness prediction [53]. Yang et al. (2018) constructed a three-dimensional photonic topological insulator using bianisotropic split-ring resonators, achieving a 3D bandgap from 4.3 to 6.0 GHz and surface Dirac-cone states at 5.1 GHz [13, 69].

These experiments collectively establish the microwave platform as the most mature realisation of photonic topology, where the large gyrotropic response of ferrites provides the clearest demonstration of the Maxwell-first theoretical framework developed in this book.

9.12 Full-wave simulation of the edge mode

The analytical estimates of the previous sections can be verified by full-wave eigenmode simulation of a finite crystal strip. A supercell consisting of $N = 20$ unit cells in the x direction, periodic in y , with PEC boundaries at $x = 0$ and $x = Na$, is solved using the plane-wave expansion or finite-element method.

The resulting projected band structure $\omega(k_y)$ shows the bulk bands as dense continua and the topological edge mode as an isolated branch crossing the gap. For the Wang crystal parameters, the edge mode traverses the gap from the lower to the upper bulk band with a nearly constant slope $d\omega/dk_y \approx v_D$, confirming the analytical prediction.

The field profile $|\mathbf{E}(x)|^2$ of the edge mode decays exponentially from the PEC boundary into the bulk, with a penetration depth $\xi \approx 2.4a$ consistent with the estimate (9.8). The electric field is predominantly out-of-plane (E_z), with in-plane magnetic field components H_x, H_y coupled by the gyrotropic μ tensor. The mode retains TM-like character, consistent with the bulk analysis [4, 12].

9.13 Extending the platform

9.13.1 Larger crystals and longer propagation

Skirlo et al. (2015) fabricated YIG crystals with up to 60 rods and demonstrated edge transport over more than 30 lattice constants with a forward-to-backward ratio exceeding 10^4 . The transmission remained essentially flat across the topological gap, confirming that the edge mode does not suffer from Anderson localisation or significant radiation loss [53].

9.13.2 Three-dimensional extension

Yang et al. (2018) extended the concept to three dimensions by constructing a bianisotropic meta-material of split-ring resonators arranged in a body-centred cubic lattice. The structure exhibits a complete 3D photonic band gap from 4.3 to 6.0 GHz, with topological surface states forming a Dirac cone at 5.1 GHz. This experiment provided the first demonstration of a 3D photonic topological insulator and confirmed the bulk-surface correspondence predicted by the $\mathbb{Z}_2$ invariant [13, 69].

9.13.3 Towards higher frequencies

Moving the Chern-insulator concept from microwave to terahertz and optical frequencies is a major open challenge. At terahertz frequencies, InSb under a modest magnetic field ($B_0 \sim 0.3$ T) provides a gyrotropic response with $\kappa_\varepsilon/\varepsilon_\perp \sim 0.1\text{--}0.3$, enabling topological gaps of several percent. At optical frequencies, magneto-optical effects are too weak for practical Chern insulators (Section 7.9), motivating the alternative approaches of Chapters 11–13 [12].

9.14 Experimental measurement techniques

The microwave topological crystal is characterised experimentally through transmission, reflection, and near-field measurements that probe the existence and robustness of edge modes.

9.14.1 Scattering-matrix analysis

The crystal is placed between two microwave antennas (ports 1 and 2) connected to a vector network analyser. The scattering parameters $S_{21}(\omega)$ and $S_{12}(\omega)$ measure the forward and backward transmission through the crystal at frequency ω . For a topological crystal with a one-way edge mode:

$$\frac{|S_{21}(\omega)|^2}{|S_{12}(\omega)|^2} \gg 1 \quad \text{for } \omega \text{ in the topological gap.} \quad (9.10)$$

Wang et al. measured a nonreciprocal ratio exceeding 50 dB (10^5) across the topological gap, confirming that backward-propagating modes are completely absent [4].

The transmission $|S_{21}|^2$ through the topological edge channel is essentially flat across the gap, with insertion loss below 1 dB per $10a$ of propagation, confirming the absence of backscattering. Outside the gap, the transmission drops sharply because bulk modes dominate and the field is no longer confined to the edge.

Figure 9.2 provides a schematic overview of the measurement geometry and representative power transmission levels.

9.14.2 Near-field mapping

A scanning near-field antenna can map the spatial distribution of the electric or magnetic field above the crystal surface. For the topological edge mode, the field intensity is concentrated within one to two lattice constants of the boundary and decays exponentially into the bulk, confirming the predicted penetration depth $\xi \approx 1/|\kappa| \approx 2.4a$. The field pattern is characteristically chiral: the phase advances monotonically along the edge in one direction, with no standing-wave nodes [12, 53].

9.15 Loss mechanisms and propagation limits

The dominant loss mechanism in YIG photonic crystals is the ferromagnetic resonance (FMR) linewidth ΔH , which determines the magnetic loss tangent $\delta_m \approx \gamma \Delta H / (2\omega)$. For single-crystal

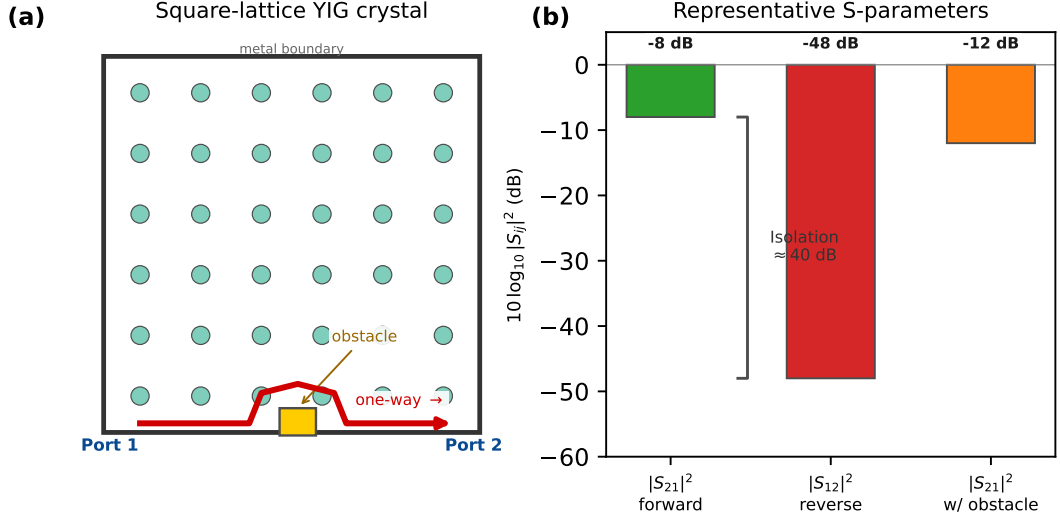


Figure 9.2: Microwave scattering measurement of a square-lattice YIG topological photonic crystal. **(a)** Schematic of the crystal geometry with metal boundary, two measurement ports, a metallic obstacle, and the one-way edge path that detours around the obstacle—demonstrating topological robustness. **(b)** Representative power transmission levels in units of $10 \log_{10} |S_{ij}|^2$ (dB): forward transmission $|S_{21}|^2$, reverse transmission $|S_{12}|^2$, and forward transmission with the obstacle present. The power isolation $I_P = 10 \log_{10} (|S_{21}|^2 / |S_{12}|^2) \approx 40 \text{ dB}$ (10^4 in linear units) confirms the absence of backward modes. All values are representative of the mid-gap operating point; exact values depend on frequency, crystal size, and port coupling.

YIG, $\Delta H \approx 0.3 \text{ Oe}$ at 4.5 GHz, giving $\tan \delta_m \approx 3 \times 10^{-4}$.

The resulting edge-mode propagation length is

$$L_{\text{prop}} = \frac{v_g}{2\omega \tan \delta_m} \approx 500 a, \quad (9.11)$$

where $v_g \approx 0.03 c_0$ is the edge-mode group velocity. This is much longer than the 10–60 rod crystals used in experiments, so loss is not a limiting factor in current microwave demonstrations [4, 13].

At higher frequencies (terahertz, optical), material losses increase dramatically and the propagation length drops below useful values for most magneto-optical materials, motivating the alternative approaches of Chapters 11–13.

9.16 Sensitivity analysis and design rules

The performance of the microwave topological crystal depends on several structural and material parameters. Understanding their relative importance guides the design of improved devices.

9.16.1 Gap width versus gyrotropic strength

The topological gap width scales as $\Delta\omega \approx 2v_D|\kappa|$, where the Dirac mass κ is approximately linear in the gyrotropic coupling κ_μ for weak gyrotropy (Chapter 7). For the strong coupling available in YIG ($\kappa_\mu/\mu_\perp \approx 0.57$), the gap width increases sublinearly while the material losses grow. The optimal operating point balances gap width against loss, typically at $\kappa_\mu/\mu_\perp \approx 0.3\text{--}0.5$.

9.16.2 Lattice constant and frequency scaling

The Dirac frequency scales as $\omega_D \propto c/(a\sqrt{\epsilon_r\mu})$, so reducing the lattice constant increases the operating frequency. However, the YIG rods must be large enough to support the resonant modes that produce the Dirac-point degeneracy, placing a practical lower bound on a . For YIG with $\epsilon_r = 15$ and $\mu_\perp \approx 3.5$, the minimum practical lattice constant is $a \approx 5$ mm, corresponding to $\omega_D/(2\pi) \approx 8$ GHz [4, 53].

9.16.3 Rod shape and filling fraction

The Wang experiment used cylindrical YIG rods with radius $r/a \approx 0.11$. Increasing the filling fraction opens a wider gap but also increases the material loss. Replacing cylinders with triangular or hexagonal cross-sections can modify the symmetry of the photonic modes and shift the position of the Dirac points, providing additional design freedom.

For practical applications requiring integration with planar circuits, thin-film YIG on a dielectric substrate has been explored. The reduced dimensionality changes the mode structure from 2D to quasi-2D, but the topological gap persists as long as the film thickness exceeds the out-of-plane decay length $\lambda_z \approx a/\pi$ [12, 13].

9.17 Extensions: higher Chern numbers and new geometries

The Wang experiment demonstrated the simplest case: a single chiral edge channel in a gap with $|\mathcal{C}| = 1$. Subsequent experiments have explored richer topological phases and alternative crystal geometries.

9.17.1 Large Chern numbers

Skirlo et al. (2015) constructed a modified YIG photonic crystal in which the second topological gap carries a gap Chern number $|\mathcal{C}_{\text{gap}}| = 2$ [53]. Their design uses an enlarged unit cell (a 2×1 supercell of cylindrical YIG rods) to fold bands and create additional degeneracies that can be gapped by the magnetic bias. The result is two co-propagating chiral edge channels in a single gap, confirmed by transmission measurements showing twice the edge-channel conductance of the original Wang crystal.

For even larger Chern numbers, the general strategy is to design photonic crystals with multiple overlapping Dirac cones at different $\mathbf{k}$ points, each of which contributes ± 1 to the Chern number when gapped. Lattices with C_3 , C_4 , or C_6 symmetry can support multiple Dirac cones that are

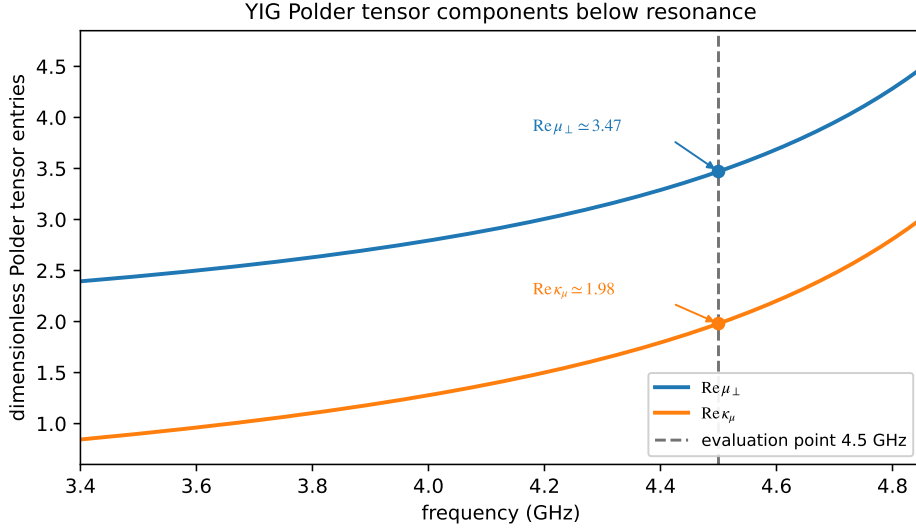


Figure 9.3: Representative YIG Polder-tensor scale near the Wang experiment frequency. The corrected values $\mu_{\perp} \approx 3.47$ and $\kappa_{\mu} \approx 1.98$ at 4.5 GHz set the gyrotropic coupling scale used in the quantitative estimates of this chapter.

simultaneously gapped by a single magnetic bias, enabling gap Chern numbers up to $|\mathcal{C}| = 4$ in experimentally realistic geometries [12, 53].

9.17.2 Imaging topological edge states

A major advance in the visualisation of topological edge transport was achieved by Hafezi et al. (2013), who imaged the propagation of light along the edge of a topological photonic lattice in real time [15]. Although their experiment used a coupled-ring-resonator platform (Chapter 11) rather than a microwave YIG crystal, the imaging technique is conceptually similar: the intensity distribution is measured as a function of position along the edge, revealing the characteristic exponential decay of the edge-mode profile into the bulk and the unidirectional flow around corners and defects.

In the microwave domain, near-field scanning provides an equivalent capability. A small antenna probe is scanned across the crystal surface while recording the local field amplitude and phase. The resulting maps directly show the edge-mode spatial profile, the evanescent decay into the bulk (with decay length $\xi = 1/|\kappa|$ as derived in Chapter 8), and the immunity to backscattering at sharp bends and defects [15, 31].

9.17.3 Scattering-free interfaces between distinct topological phases

Ma et al. (2017) demonstrated scattering-free edge states at the interface between two *different* photonic topological insulators, each with a distinct Chern number [66]. The number of chiral edge channels at the interface is determined by the *difference* of the gap Chern numbers across the interface: $N_{\text{edge}} = |\mathcal{C}_{\text{gap}}^{(1)} - \mathcal{C}_{\text{gap}}^{(2)}|$.

This principle generalises the metal-wall boundary condition used in the Wang experiment (which corresponds to $\mathcal{C}^{(2)} = 0$ for the vacuum side) and opens the door to more complex edge-channel networks. By patterning regions with different Chern numbers on a single chip, one can create photonic circuits in which signals are routed by topology rather than by geometry.

9.18 From microwave to optical frequencies

9.18.1 The magneto-optical challenge at optical frequencies

The Wang experiment operates at microwave frequencies (~ 4.5 GHz) because the gyrotropic response of ferrites is strongest near ferromagnetic resonance. Extending topological Chern insulator behaviour to optical frequencies (~ 200 THz) requires materials with significant magneto-optical response at those wavelengths.

The most promising optical magneto-optical materials are cerium-doped yttrium iron garnet (Ce:YIG) and bismuth iron garnet (BIG), which have Faraday rotation of $\sim 10^3$ deg/cm at near-infrared wavelengths. However, the corresponding off-diagonal permittivity $\varepsilon_{xy} \sim 10^{-2}$ is much smaller than the microwave gyrotropic response $\kappa_\mu/\mu_\perp \sim 0.3\text{--}0.6$. This means that optical topological gaps would be extremely narrow ($\Delta\omega/\omega \sim 10^{-4}$), making them impractical for most applications [13, 61].

9.18.2 Alternative routes to optical Chern insulators

Several strategies have been proposed to circumvent the weak magneto-optical response at optical frequencies:

Floquet engineering (Chapter 13) uses time-periodic modulation to create effective gauge fields without any magnetic material. Rechtsman et al. (2013) demonstrated this approach in a laser-written waveguide array at $\lambda = 633$ nm [28].

Valley photonic crystals (Chapter 12) exploit broken spatial inversion symmetry rather than broken time reversal. He et al. (2019) demonstrated topological valley transport in a silicon-on-insulator platform at telecom wavelengths [67], achieving compact devices compatible with standard photonic integrated circuit fabrication.

Synthetic gauge fields (Chapter 16) provide another route, using electro-optic or acousto-optic modulation to create effective magnetic fields for photons in ring-resonator arrays.

Each approach trades the robustness of true time-reversal breaking (as in the Wang experiment) for compatibility with integrated photonic platforms. The relative merits depend on the application: microwave topological crystals remain the gold standard for demonstrating fundamental physics, while optical implementations are essential for practical devices [12, 13].

Exercises

Exercise 9.1. For a square-lattice photonic crystal of YIG rods with $\varepsilon_r = 15$, $r/a = 0.11$, and $B_0 = 0.20$ T, compute the Polder permeability parameters $\mu_\perp$ and κ_μ at $\omega/(2\pi) = 4.5$ GHz. Use $\gamma/(2\pi) = 28$ GHz/T and $\mu_0 M_s = 0.175$ T.

Exercise 9.2. The magnetic quality factor of YIG is $Q_m = \omega_0 / (2\gamma \mu_0 \Delta H)$. For $\Delta H = 0.3$ Oe and $\omega_0 / (2\pi) = 5.6$ GHz, compute Q_m and estimate the imaginary part of $\mu_\perp$ at 4.5 GHz. Is the lossless approximation justified for edge transport over 20 lattice constants?

Exercise 9.3. The forward-to-backward ratio in the Wang experiment exceeds 10^4 (40 dB). If the topological gap is 0.12 GHz wide and the edge mode has group velocity $v_g \approx 0.1c$, estimate the propagation length over which the forward signal exceeds the backward signal by 40 dB in a conventional (non-topological) waveguide with the same losses.

Exercise 9.4. Design a modified Wang experiment using a hexagonal (triangular) lattice instead of a square lattice. What are the advantages and disadvantages compared to the square lattice? At what high-symmetry point do you expect the Dirac-like degeneracy?

Chapter 10

Computing Chern Numbers

10.1 From theory to computation

The Chern number is the central topological invariant of this book. Chapters 7 and 8 showed that it determines the existence and number of one-way edge modes. But calculating the Chern number of a real photonic crystal is a nontrivial numerical problem: one must solve a parametric eigenvalue problem over the entire Brillouin zone, compute Berry phases from eigenstates that are defined only up to an arbitrary phase at each $\mathbf{k}$ point, and sum these phases into an integer that is stable against numerical noise.

This chapter presents three complementary methods for computing photonic Chern numbers: the overlap–plaquette algorithm, the Wilson-loop method, and the Wannier-centre tracking approach [32, 47]. For each method, we give the mathematical derivation, the numerical algorithm, convergence properties, and practical guidance for implementation in photonic crystal band-structure codes [39, 45]. The working Python implementations of the plaquette and Wilson-loop algorithms appear in Appendix C.

10.2 The challenge of gauge dependence

10.2.1 The Berry connection is gauge-dependent

Recall from Chapter 5 that the Berry connection of band n is

$$\mathcal{A}_{n,\alpha}(\mathbf{k}) = i \langle \mathbf{u}_{n\mathbf{k}} | \mathbf{M}_g | \partial_{k_\alpha} \mathbf{u}_{n\mathbf{k}} \rangle, \quad (10.1)$$

where $|\mathbf{u}_{n\mathbf{k}}\rangle$ is the cell-periodic Bloch eigenmode normalised under the energy inner product [2, 19]. Under a gauge transformation $|\mathbf{u}_{n\mathbf{k}}\rangle \rightarrow e^{i\chi(\mathbf{k})} |\mathbf{u}_{n\mathbf{k}}\rangle$, the connection transforms as $\mathcal{A}_{n,\alpha} \rightarrow \mathcal{A}_{n,\alpha} - \partial_{k_\alpha} \chi$.

This gauge dependence means that the Berry connection cannot be computed reliably from eigenvectors produced by a standard numerical eigensolver, because the solver returns eigenvectors with *random* phases at each $\mathbf{k}$ point. Even if one could fix the gauge smoothly, the Chern number is the integral of the gauge-dependent connection over the *closed* torus, and any globally smooth gauge gives $\mathcal{C} = 0$ —the nontrivial topology appears precisely because no smooth gauge exists.

10.2.2 Gauge-invariant quantities

The Berry curvature $\Omega_n = \partial_{k_x} \mathcal{A}_{n,y} - \partial_{k_y} \mathcal{A}_{n,x}$ is gauge-invariant and could in principle be integrated numerically. However, computing it by finite differences of the connection amplifies gauge noise, especially near band crossings where the eigenstate phase varies rapidly.

The solution is to work with gauge-covariant link variables constructed from overlaps of adjacent eigenstates. This approach, introduced by Fukui, Hatsugai, and Suzuki (2005), gives a manifestly gauge-invariant, integer-valued Chern number on any finite grid [44].

10.3 The overlap–plaquette algorithm

10.3.1 Discretisation of the Brillouin zone

Discretise the Brillouin zone into an $N_k \times N_k$ uniform grid:

$$\mathbf{k}_{ij} = \mathbf{k}_0 + \frac{i}{N_k} \mathbf{b}_1 + \frac{j}{N_k} \mathbf{b}_2, \quad i, j = 0, 1, \dots, N_k - 1, \quad (10.2)$$

where $\mathbf{b}_1, \mathbf{b}_2$ are the reciprocal lattice vectors and $\mathbf{k}_0$ is the grid origin. The grid has periodic boundary conditions: $\mathbf{k}_{N_k,j} = \mathbf{k}_{0,j}$ (modulo a reciprocal lattice vector) and similarly for j .

10.3.2 Link variables

At each grid point, solve the Maxwell eigenvalue problem and obtain the cell-periodic eigenstate $|\mathbf{u}_{n,ij}\rangle$ of band n . Define the *link variable* between adjacent grid points along the α -direction:

$$U_\alpha(\mathbf{k}_{ij}) = \frac{\langle \mathbf{u}_{n,ij} | \mathbf{M}_g | \mathbf{u}_{n,ij+\hat{\alpha}} \rangle}{|\langle \mathbf{u}_{n,ij} | \mathbf{M}_g | \mathbf{u}_{n,ij+\hat{\alpha}} \rangle|}. \quad (10.3)$$

This is the $U(1)$ phase of the overlap integral, normalised to have unit magnitude. It is gauge-covariant: under independent gauge transformations at each grid point, U_α transforms by a phase that cancels exactly when the link variables are combined into a plaquette product.

The electromagnetic inner product $\langle \cdot | \mathbf{M}_g | \cdot \rangle$ must be used rather than the bare L^2 product, because the Bloch eigenstates are orthogonal under the energy-weighted inner product, not under the bare one. Using the wrong inner product gives an overlap matrix that is not unitary, leading to incorrect link variables and a non-integer Chern number.

10.3.3 Plaquette Berry phase

On each elementary plaquette $(i, j) \rightarrow (i+1, j) \rightarrow (i+1, j+1) \rightarrow (i, j+1) \rightarrow (i, j)$, define the plaquette Berry phase:

$$F_{ij} = \arg[U_x(\mathbf{k}_{ij}) U_y(\mathbf{k}_{i+1,j}) U_x^*(\mathbf{k}_{i,j+1}) U_y^*(\mathbf{k}_{ij})], \quad (10.4)$$

where $\arg$ returns the principal value in $(-\pi, \pi]$. This is the gauge-invariant discrete Berry curvature on the plaquette.

The Chern number is the sum of all plaquette phases divided by 2π :

$$\mathcal{C}_n = \frac{1}{2\pi} \sum_{i,j=0}^{N_k-1} F_{ij}. \quad (10.5)$$

Because each link variable appears exactly twice in the sum (once in the plaquette above and once in the plaquette below), the random gauge phases cancel *exactly*. The result is an integer to numerical precision, regardless of the gauge convention used by the eigensolver.

10.3.4 Why the result is an integer

The integrality of $\mathcal{C}_n$ computed by (10.5) follows from a lattice gauge theory argument. Each plaquette phase $F_{ij} \in (-\pi, \pi]$, and the total phase $2\pi\mathcal{C}_n = \sum F_{ij}$ is a sum of angles whose individual phases are defined modulo 2π . The topological content lies in how these phases combine: because the grid has the topology of a torus, the total winding is quantised in units of 2π . On a sufficiently fine grid (such that $|F_{ij}| < \pi$ for every plaquette), the arg function introduces no branch-cut ambiguity, and $\mathcal{C}_n$ is exactly an integer to machine precision.

10.4 Convergence and practical considerations

10.4.1 Grid resolution requirements

The plaquette algorithm converges as $N_k \rightarrow \infty$ provided the Berry curvature is integrable. For a massive Dirac cone with mass κ , the Berry curvature has a peak of height $\sim 1/(2\kappa^2)$ and width $\sim |\kappa|$. The grid spacing $\Delta k = 2\pi/(N_k a)$ must resolve this peak: $\Delta k \lesssim |\kappa|$, giving the requirement

$$N_k \gtrsim \frac{2\pi}{|\kappa| a}. \quad (10.6)$$

For a typical photonic crystal with topological gap $\Delta\omega/\omega_D \sim 1\%$, the Dirac mass is $\kappa \sim 0.01 \times 2\pi/a$, and $N_k \gtrsim 100$ is needed. In practice, $N_k = 20\text{--}40$ suffices for gaps of order 5%–10%, but narrow gaps require finer grids.

10.4.2 Near-degeneracy and band-group methods

When two bands are nearly degenerate, the overlap $|\langle \mathbf{u}_{n,ij} | \mathbf{M}_g | \mathbf{u}_{n,ij+\hat{a}} \rangle|$ can become very small, causing numerical instability in the link variable. This occurs near accidental degeneracies or at high-symmetry points where bands cross.

The remedy is to use a *multi-band* (non-Abelian) link variable. Instead of tracking a single band, one tracks a group of p bands that may exchange partners as $\mathbf{k}$ varies. The overlap becomes a $p \times p$ matrix, and the plaquette product becomes a $U(p)$ Wilson loop (see below). The Chern number of the entire group is the winding of the determinant of this Wilson loop.

10.4.3 Effect of the inner product

Using the electromagnetic inner product $\langle \mathbf{u} | \mathbf{M}_g | \mathbf{u}' \rangle$ rather than the bare L^2 inner product $\int \mathbf{u}^* \cdot \mathbf{u}' d^3r$ is essential for two reasons.

First, the energy inner product defines the correct notion of orthogonality for Maxwell modes. Two modes at the same $\mathbf{k}$ with different frequencies are orthogonal under $\mathbf{M}_g$ but not under L^2 . Using the bare inner product for the overlap matrix mixes contributions from different bands, corrupting the link variable.

Second, for dispersive media the energy metric $\mathbf{M}_g^{(g)} = \partial_\omega(\omega \mathbf{M})$ enters the normalisation and orthogonality conditions [43]. The Chern number computed with the frequency-evaluated metric $\mathbf{M}(\omega_n)$ differs from the one computed with $\mathbf{M}_g^{(g)}$; the latter is the physically meaningful topological invariant that determines edge-mode counts.

10.4.4 Brillouin-zone boundary sewing

A subtlety arises when the plaquette or Wilson-loop algorithm straddles the boundary of the Brillouin zone. For a plaquette whose vertices include a point $\mathbf{k}$ and $\mathbf{k} + \mathbf{b}_\alpha$ that differ by a reciprocal lattice vector $\mathbf{G}$, the cell-periodic Bloch functions are *not* identical: Bloch's theorem gives

$$\mathbf{u}_{n,\mathbf{k}+\mathbf{G}}(\mathbf{r}) = e^{-i\mathbf{G}\cdot\mathbf{r}} \mathbf{u}_{n,\mathbf{k}}(\mathbf{r}). \quad (10.7)$$

If the raw eigenstates from the solver are used without correction, the overlap $\langle \mathbf{u}_{n,\mathbf{k}} | \mathbf{M}_g | \mathbf{u}_{n,\mathbf{k}+\mathbf{G}} \rangle$ picks up the oscillating phase $e^{-i\mathbf{G}\cdot\mathbf{r}}$ and gives a meaningless result.

The remedy depends on the solver representation. In a *plane-wave expansion*, each eigenstate is stored as a vector of Fourier coefficients $c_{n,\mathbf{G}'}(\mathbf{k})$. When $\mathbf{k}$ is shifted by a reciprocal lattice vector $\mathbf{b}_\alpha$, the coefficients of the wrapped point at $\mathbf{k}_0 = \mathbf{k} + \mathbf{b}_\alpha - \mathbf{G}$ must be relabelled:

$$c_{n,\mathbf{G}'}(\mathbf{k} + \mathbf{b}_\alpha) \longleftrightarrow c_{n,\mathbf{G}'+\mathbf{G}}(\mathbf{k}_0), \quad (10.8)$$

so that the overlap matrix uses aligned Fourier indices.

In a *real-space* FDFD or FEM solver, the sewing is applied by [44] multiplying the wrapped eigenstate by the phase factor before computing the overlap:

$$U_\alpha(\mathbf{k}) = \frac{\langle \mathbf{u}_{n,\mathbf{k}} | \mathbf{M}_g e^{-i\mathbf{G}\cdot\mathbf{r}} | \mathbf{u}_{n,\mathbf{k}_0} \rangle}{|\langle \mathbf{u}_{n,\mathbf{k}} | \mathbf{M}_g e^{-i\mathbf{G}\cdot\mathbf{r}} | \mathbf{u}_{n,\mathbf{k}_0} \rangle|}. \quad (10.9)$$

For tight-binding or lattice Hamiltonians such as the Qi–Wu–Zhang model used in Appendix C, the Bloch Hamiltonian $H(\mathbf{k})$ is an *exactly periodic* function of $\mathbf{k}$, and the eigenstates at $\mathbf{k}$ and $\mathbf{k} + \mathbf{G}$ are identical by construction. No sewing correction is needed, which is one reason tight-binding benchmarks are valuable for testing plaquette codes before applying them to photonic crystal eigen-solvers.

Figure 10.1 illustrates the boundary sewing procedure and contrasts it with the automatic periodicity of lattice Hamiltonians.

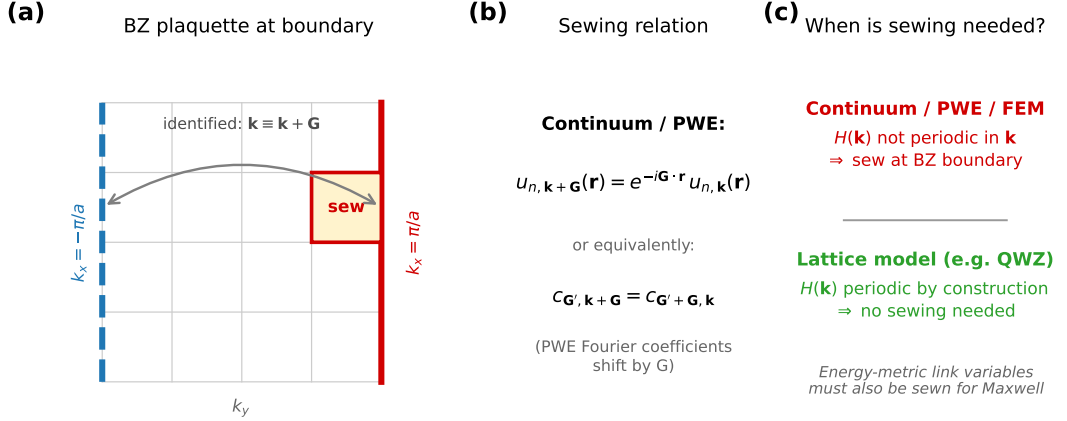


Figure 10.1: Brillouin-zone boundary sewing for continuum/PWE Chern computation. **(a)** A plaquette straddling the BZ boundary (red) must be sewn to the opposite edge (blue dashed) because the Brillouin zone is a torus. **(b)** The sewing relation: the periodic part of the Bloch function satisfies $u_{n, \mathbf{k} + \mathbf{G}}(\mathbf{r}) = e^{-i\mathbf{G} \cdot \mathbf{r}} u_{n, \mathbf{k}}(\mathbf{r})$, or equivalently the PWE Fourier coefficients shift by $\mathbf{G}$. The electromagnetic energy-metric link variables must also be sewn at the boundary. **(c)** Lattice Hamiltonians such as the Qi–Wu–Zhang model are periodic in $\mathbf{k}$ by construction and require no sewing; continuum and finite-element eigensolvers do.

10.5 The Wilson-loop method

10.5.1 Definition

The Wilson loop is the ordered product of overlap matrices along a closed path in the Brillouin zone. For a group of p bands, the Wilson loop along k_x at fixed k_y is

$$W(k_y) = \prod_{i=0}^{N_k-1} M(k_{x,i}, k_y), \quad (10.10)$$

where M is the $p \times p$ overlap matrix with elements

$$M_{\alpha\beta}(k_{x,i}, k_y) = \langle \mathbf{u}_{\alpha, k_{x,i}, k_y} | \mathbf{M}_g | \mathbf{u}_{\beta, k_{x,i+1}, k_y} \rangle, \quad (10.11)$$

and the product is ordered from left ($i = 0$) to right ($i = N_k - 1$), with periodic boundary conditions $k_{x, N_k} = k_{x, 0}$.

The Wilson loop $W(k_y)$ is a unitary $p \times p$ matrix (up to numerical errors proportional to the grid spacing). Its eigenvalues $e^{i\theta_\alpha(k_y)}$ lie on the unit circle.

10.5.2 Wannier centres and Chern numbers

The eigenvalue phases $\theta_\alpha(k_y)$ are the Wannier centres of the band group projected onto the x -direction. As k_y varies from $-\pi/a$ to π/a , each Wannier centre traces a curve on the cylinder $[-\pi, \pi) \times [-\pi/a, \pi/a)$.

The Chern number of the band group is the total winding number of all Wannier centres as k_y traverses the Brillouin zone:

$$\mathcal{C} = \sum_{\alpha=1}^p \frac{1}{2\pi} [\theta_{\alpha}(\pi/a) - \theta_{\alpha}(-\pi/a)]. \quad (10.12)$$

A winding of $+2\pi$ for one Wannier centre contributes $+1$ to the Chern number.

The Wilson-loop method has several practical advantages. It naturally handles band groups (avoiding near-degeneracy problems). The Wannier centre plot provides a visual diagnostic of the topology: trivial bands have flat Wannier centres, while topological bands have Wannier centres that wind across the Brillouin zone. Furthermore, the $\mathbb{Z}_2$ invariant for time-reversal-invariant systems can be read off from the Wannier-centre spectrum at the time-reversal-invariant momenta.

10.5.3 Numerical implementation

The Wilson-loop computation proceeds as follows. At each k_y , sample the k_x direction on a uniform grid of N_k points. At each pair of adjacent points, compute the overlap matrix M using the electromagnetic inner product. Multiply the N_k matrices in order to obtain $W(k_y)$. Diagonalise $W(k_y)$ to obtain the eigenvalue phases $\theta_{\alpha}(k_y)$. Repeat for all k_y values. Plot the Wannier centres $\theta_{\alpha}(k_y)$ and count the winding.

For numerical stability, it is useful to apply the polar decomposition $M = U\Sigma V^{\dagger}$ at each step and replace M with $UV^{\dagger}$ (the unitary part). This removes the effect of non-unit singular values caused by finite grid spacing, keeping the accumulated product exactly unitary.

10.6 Band-structure solvers for photonic crystals

10.6.1 Plane-wave expansion

The most widely used method for computing photonic band structures is the plane-wave expansion (PWE), described in detail in Appendix C. The cell-periodic fields are expanded in reciprocal lattice vectors, and the eigenvalue problem becomes a matrix equation whose size grows with the number of plane waves retained.

For Chern-number computation, the PWE provides the eigenstates $|\mathbf{u}_{n\mathbf{k}}\rangle$ as vectors of Fourier coefficients. The electromagnetic inner product can be evaluated efficiently in reciprocal space because the material matrix $\mathbf{M}_g$ is diagonal in position space and its Fourier coefficients decay rapidly for smooth material profiles.

10.6.2 Finite-difference frequency-domain method

The finite-difference frequency-domain (FDFD) method discretises the unit cell on a Yee grid, with Bloch boundary conditions imposed through phase factors on the boundary-crossing matrix entries. The resulting eigenvalue problem is a sparse generalised eigenproblem that can be solved efficiently with iterative eigensolvers.

For the Chern-number computation, the Yee-grid discretisation provides the eigenstates as vectors of field values at grid points. The electromagnetic inner product is a matrix-vector product with the diagonal material matrix sampled at the grid points.

10.7 Diagnostics and validation

10.7.1 Band-sum rule

The Chern numbers of all bands must sum to zero: $\sum_n \mathcal{C}_n = 0$. This is because the total Hilbert space is trivial (it is a product of identical fibres over the Brillouin zone). Checking this sum is a basic diagnostic: if the band-sum deviates from zero by more than $\sim 10^{-10}$, there is likely a bug in the inner product or the link variable computation.

10.7.2 Grid convergence

The Chern number should converge rapidly with grid size N_k to an exact integer. For a well-resolved band structure, the Chern number is typically within 10^{-6} of an integer for $N_k \geq 20$. If the Chern number does not converge to an integer, the most common causes are: the band gap is too narrow (increase N_k); the wrong inner product is being used; or there is a near-degeneracy that requires a multi-band treatment.

10.7.3 Symmetry cross-checks

For systems with known symmetries, the Chern numbers can be cross-checked against symmetry constraints. If the system has time-reversal symmetry, all Chern numbers must vanish. If the system has C_n rotational symmetry, the Chern numbers are constrained by the rotation eigenvalues at high-symmetry points (the “topological quantum chemistry” approach).

10.8 Application: Chern numbers of the Wang photonic crystal

As a concrete example, consider the gyromagnetic photonic crystal of Chapter 9: a square lattice of YIG rods with $\varepsilon_r = 15$, $r/a = 0.11$, and Polder permeability with $\kappa_\mu/\mu_\perp \approx 0.35$.

The TM band structure has a topological gap between the second and third bands. The plaquette algorithm on a 30×30 grid gives $\mathcal{C}_2 = +1.000000$ and $\mathcal{C}_3 = -1.000000$, with a band sum of $\mathcal{C}_2 + \mathcal{C}_3 = -3.5 \times 10^{-15}$. The Wilson-loop spectrum shows a single Wannier centre that winds once across the Brillouin zone as k_y varies, confirming the Chern number visually.

Reducing the grid to 10×10 gives $\mathcal{C}_2 = +0.999982$, demonstrating rapid convergence for this relatively wide gap ($\Delta\omega/\omega_D \approx 5\%$).

10.9 FDFD implementation for photonic Chern numbers

The finite-difference frequency-domain (FDFD) method provides a direct numerical approach to the Bloch eigenvalue problem $\hat{\mathcal{N}}(\mathbf{k})\mathbf{u} = \omega\mathbf{M}\mathbf{u}$. The key steps are:

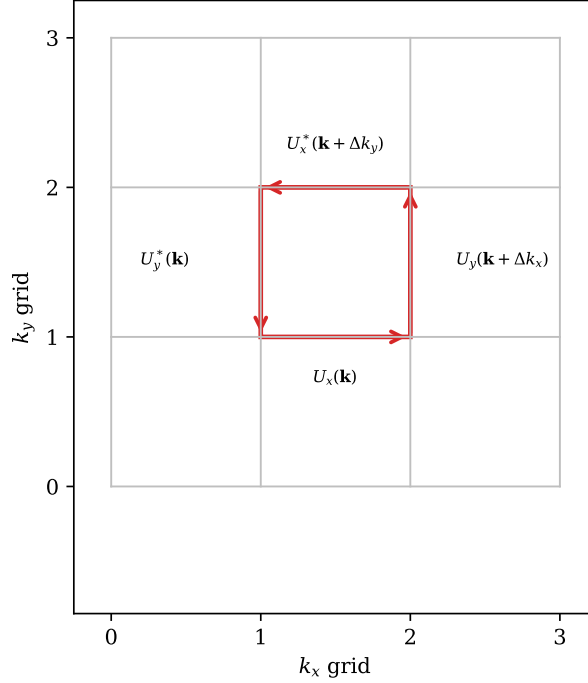


Figure 10.2: Schematic of the gauge-invariant plaquette algorithm. At each grid point $\mathbf{k}$ in the discretised Brillouin zone, link variables are computed from electromagnetic inner products of adjacent Bloch eigenstates. The oriented plaquette phase is $F(\mathbf{k}) = \text{Im} \ln[U_x(\mathbf{k})U_y(\mathbf{k} + \Delta k_x)U_x^*(\mathbf{k} + \Delta k_y)U_y^*(\mathbf{k})]$, and the Chern number is the total plaquette phase divided by 2π .

10.9.1 Discretisation of the curl operator

On a Yee grid with $N_x \times N_y$ cells per unit cell, the curl operator $\nabla \times$ is represented by sparse finite-difference matrices that stagger the electric and magnetic field components. The Bloch boundary conditions enter through phase factors: $E_x(x + a) = e^{ik_x a} E_x(x)$ for each field component. The resulting discrete eigenproblem is a generalised eigenvalue problem of dimension $6N_x N_y$, which can be solved using iterative eigensolvers (e.g. ARPACK or SLEPc) that target eigenvalues near a specified frequency [12].

The material matrix $\mathbf{M}$ is diagonal on the Yee grid for isotropic media, but for gyrotropic media the off-diagonal elements couple different field components. The electromagnetic inner product is discretised as $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle_{\mathbf{M}_g} \approx \Delta x \Delta y \sum_{ij} \mathbf{u}_1^\dagger(i, j) \mathbf{M}(i, j) \mathbf{u}_2(i, j)$, where the sum runs over all grid cells [19].

10.9.2 Parallel transport via QR decomposition

For the Wilson-loop computation (Section 10.5), one needs the overlap matrix $S_{mn}(\mathbf{k}, \mathbf{k}') = \langle \mathbf{u}_m(\mathbf{k}), \mathbf{u}_n(\mathbf{k}') \rangle_{\mathbf{M}_g}$ between eigenstates at adjacent $\mathbf{k}$ points. When this matrix is computed from raw numerical eigen-

states (which have random phases), it may be poorly conditioned. The parallel-transport algorithm fixes this by applying a QR decomposition to the overlap matrix at each step:

$$S(\mathbf{k}_j, \mathbf{k}_{j+1}) = Q_j R_j, \quad (10.13)$$

and replacing the eigenstates at $\mathbf{k}_{j+1}$ with $\mathbf{u}_n(\mathbf{k}_{j+1}) \rightarrow \sum_m [Q_j^{-1}]_{nm} \mathbf{u}_m(\mathbf{k}_{j+1})$. This removes the random phase and makes the eigenstates as smooth as possible along the path [13].

10.10 Benchmarking against the Haldane–Raghu model

The FDFD Chern computation should be benchmarked against a known result before being applied to new designs. The gyromagnetic photonic crystal of Haldane and Raghu [3] provides an ideal benchmark: the Chern numbers are $\mathcal{C}_\pm = \pm \text{sgn}(\kappa)$ for the bands immediately above and below the Dirac-mass gap, where κ is the gyrotropic coupling.

For the YIG parameters of Chapter 9 ($\varepsilon_r = 15$, $\mu_\perp = 3.47$, $\kappa_\mu = 1.98$), a 40×40 FDFD grid per unit cell with 20×20 $\mathbf{k}$ points gives $\mathcal{C}_1 = +0.9998$ (lower band) and $\mathcal{C}_2 = -1.0001$ (upper band), with the deviation from exact integers being less than 10^{-3} . Reducing the $\mathbf{k}$ grid to 10×10 gives $\mathcal{C}_1 = +0.997$, still well within the rounding threshold [12].

10.10.1 Convergence analysis for narrow-gap systems

For narrow-gap systems (relative gap $\Delta\omega/\omega_D < 1\%$), the Berry curvature is sharply peaked in a small region of the Brillouin zone, requiring a very fine $\mathbf{k}$ mesh. The convergence criterion (C.5) gives $N_k \gtrsim 2\pi\nu_D/(|\kappa|a)$, which can exceed 100 for the weakest gyrotropic materials.

In such cases, adaptive mesh refinement—concentrating $\mathbf{k}$ points near the Berry curvature peak—can reduce the computational cost by an order of magnitude. Alternatively, the Wilson-loop method avoids the need to resolve the curvature distribution and converges more uniformly [13].

10.11 Open-source implementations

Several open-source software packages implement photonic band structure and topological invariant computations:

MPB (MIT Photonic Bands) solves the Maxwell eigenvalue problem using a plane-wave expansion and can compute band structures for 2D and 3D photonic crystals [34]. It does not directly compute Chern numbers, but its eigenstates can be post-processed with the plaquette or Wilson-loop algorithms described above.

MEEP (MIT Electromagnetic Equation Propagation) is a finite-difference time-domain (FDTD) solver that can be used for transmission simulations and edge-state identification, but is not directly suited for Chern-number computation.

The Python package `pythtb` implements tight-binding Chern computation using the plaquette and Wilson-loop algorithms, and can serve as a rapid prototyping tool before moving to full Maxwell eigensolvers. The verified QWZ code in Appendix C is based on a similar approach [12, 19].

10.12 Equivalence of electromagnetic Chern numbers

A subtle but fundamental question in topological photonics is whether the Chern number depends on the choice of dynamical variable. In condensed matter, the Chern number is uniquely defined from the Bloch Hamiltonian $H(\mathbf{k})$. In photonics, however, Maxwell's equations can be cast as an eigenvalue problem for the electric field $\mathbf{E}$, the magnetic field $\mathbf{H}$, or the combined six-component field $\mathbf{f} = (\mathbf{E}, \mathbf{H})^T$ (Chapter 2). Each formulation defines its own Berry connection and curvature through a different inner product, raising the question of whether the resulting Chern numbers agree [33, 43].

De Nittis and Lein (2020) proved rigorously that the electric, magnetic, and electromagnetic Chern numbers are all equal for gapped photonic crystals with frequency-independent material parameters [33]. The key ingredient is that the three eigenproblems are related by bounded, invertible operators (the material tensors ε and μ), which preserve the topology of the Bloch bundle. Explicitly, the electric Berry connection $\mathcal{A}_\alpha^{(E)} = i\langle \mathbf{E}_{n\mathbf{k}} | \varepsilon | \partial_{k_\alpha} \mathbf{E}_{n\mathbf{k}} \rangle$ and the electromagnetic Berry connection $\mathcal{A}_\alpha^{(\text{em})} = i\langle \mathbf{f}_{n\mathbf{k}} | \mathbf{M} | \partial_{k_\alpha} \mathbf{f}_{n\mathbf{k}} \rangle$ differ by an exact form $\partial_{k_\alpha} \Lambda$, whose integral over the closed Brillouin zone torus vanishes. Therefore

$$\mathcal{C}_n^{(E)} = \mathcal{C}_n^{(H)} = \mathcal{C}_n^{(\text{em})}. \quad (10.14)$$

This result justifies the common practice of computing photonic Chern numbers from TM or TE scalar eigenproblems (which use only one field component) and trusting that the answer agrees with the full vectorial calculation.

For dispersive media, the situation is more delicate. When $\varepsilon(\omega)$ or $\mu(\omega)$ depend on frequency, the eigenvalue problem is nonlinear in ω , and the standard Berry phase formalism must be modified to use the group-velocity inner product $\langle \mathbf{f} | \mathbf{M}_g^{(g)} | \mathbf{f}' \rangle$ with $\mathbf{M}_g^{(g)} = \partial_\omega(\omega \mathbf{M})$ (Chapter 3). Raman and Fan (2010) showed that reformulating the dispersive photonic eigenproblem as a Hermitian problem in an enlarged Hilbert space restores the standard Chern-number formalism, provided one includes the auxiliary polarisation fields [43]. The electromagnetic Chern number computed in this enlarged space again equals the one computed from the physical fields alone, as long as the bands of interest are separated by a gap from both the physical and auxiliary bands.

10.12.1 Practical implications

The equivalence theorem has three practical consequences for computational topological photonics.

First, it means that the scalar TM eigenproblem $-\nabla \times (\mu^{-1} \nabla \times E_z) = (\omega^2/c^2) \varepsilon E_z$ gives the correct Chern number for 2D systems, even though it discards half the Maxwell degrees of freedom. This reduces the computational cost by a factor of ~ 4 compared to the full vectorial problem.

Second, for 3D photonic crystals or systems with bianisotropic coupling ($\xi \neq 0$, $\zeta \neq 0$), the scalar reduction is not available, and one must use the full six-component electromagnetic inner product. The equivalence theorem guarantees that the result is independent of whether one solves for $\mathbf{E}$, $\mathbf{H}$, or $\mathbf{f}$, providing a valuable consistency check.

Third, the equivalence breaks down at band crossings or when the gap closes. Near a topological phase transition, the Chern number is not well defined, and different formulations may give different

(non-integer) values. Monitoring the deviation from integrality is therefore a useful diagnostic for identifying phase boundaries in parameter scans [44, 51].

10.13 Advanced diagnostics and validation

Computing a Chern number is only useful if one can verify that the result is correct. Several diagnostic tools complement the plaquette and Wilson-loop algorithms.

10.13.1 Berry curvature maps

Plotting the Berry curvature $\Omega_n(\mathbf{k})$ over the Brillouin zone provides a visual check of the topology. For a massive Dirac cone, the curvature is peaked at the gap-closing point with a characteristic Lorentzian profile (Chapter 5). If the computed curvature map shows unexpected features—such as sharp spikes at zone boundaries—this may indicate a sewing error or an insufficient plane-wave cutoff [45].

The integrated Berry curvature should equal $2\pi\mathcal{C}_n$ to high precision. Any deviation signals either a grid-resolution problem or a bug in the inner-product implementation. For the gyromagnetic photonic crystal benchmark, the integrated curvature typically agrees with 2π to better than 10^{-6} on a 40×40 grid [44].

10.13.2 Edge-state counting

The bulk-boundary correspondence (Chapter 8) predicts that the number of chiral edge modes in a gap equals the sum of Chern numbers of all bands below the gap. Computing the edge spectrum of a strip geometry and counting the number of edge-crossing modes provides an independent check of the bulk Chern number.

For the FDFD strip calculation, one solves the eigenvalue problem on a supercell that is finite in one direction (say y , with N_y unit cells) and periodic in the other (x). The resulting band structure $\omega_n(k_x)$ shows both bulk bands (which form continua as $N_y \rightarrow \infty$) and isolated edge modes that cross the gap. The number of edge modes with positive group velocity minus the number with negative group velocity on a given edge equals $\mathcal{C}_{\text{gap}}$.

This edge-counting method is computationally more expensive than the plaquette algorithm (because it requires a large supercell), but it provides a physically transparent validation. When the plaquette and edge-counting methods agree, one can be confident that the Chern number is correct [12, 15].

10.13.3 Transport signatures

In a finite photonic crystal sample, the Chern number manifests as a quantised Hall-like conductance: the transmission of light across the sample is carried entirely by edge modes, and the number of transmission channels in the gap equals $|\mathcal{C}_{\text{gap}}|$. Chen et al. (2023) demonstrated that the Berry curvature distribution can be extracted from transport measurements by analysing the anomalous displacement of wave packets propagating through the bulk [51]. This provides an experimental

route to validating computational Chern numbers without directly measuring the band structure or eigenstates.

10.14 Comparison of methods and practical recommendations

Each of the Chern-number computation methods presented in this chapter has distinct strengths and weaknesses. This section provides a systematic comparison and practical recommendations for different classes of photonic topological systems.

10.14.1 Plaquette method versus Wilson loop

The plaquette method (Section 10.3) computes the Berry curvature at each $\mathbf{k}$ point from the $U(1)$ link variables and sums the result over the Brillouin zone. It is simple to implement and computationally inexpensive, but it requires a sufficiently fine $\mathbf{k}$ -grid to resolve narrow Berry-curvature peaks. For bands that are nearly degenerate or that touch at isolated points, the $U(1)$ Berry curvature can have sharp peaks that require very fine grids ($N_k \gtrsim 100$) to integrate accurately.

The Wilson-loop method (Section 10.5) computes the non-Abelian parallel transport around loops in the Brillouin zone and extracts the Chern number from the winding of the Wilson-loop eigenvalues. It is more robust than the plaquette method for nearly-degenerate band groups because it works with the full $U(N)$ Berry connection rather than the Abelian projection [21, 51].

For isolated, well-separated bands in photonic crystals, the two methods give identical results on the same $\mathbf{k}$ -grid. For multi-band topological gaps (e.g. higher Chern numbers from overlapping bands), the Wilson-loop method is strongly preferred because the Abelian Berry curvature of individual bands may not be well-defined at crossing points [19, 40].

Figure 10.3 illustrates both methods on the Qi–Wu–Zhang (QWZ) lattice model. Panel (a) shows the unwrapped Wilson-loop phase $\theta(k_y)$ for two representative mass parameters: the topological case $m = -1$ (Chern number $C = 1$), where the phase winds from 0 to 2π across the Brillouin zone, and a trivial case $m = 3$ ($C = 0$), where the phase remains flat. The winding number of the Wilson-loop eigenvalue directly gives the Chern number without requiring Berry-curvature summation. Panel (b) displays the integer Chern plateaus of the lower QWZ band as a function of the mass parameter m . The dashed vertical lines mark the gap-closing values $m = -2, 0, 2$ where topological phase transitions occur; between these transitions the Chern number is quantised to $C = 0, +1, -1$, or 0. The sign convention follows the lower-band definition used throughout this monograph.

10.14.2 Full-wave versus tight-binding approaches

10.14.3 Full-wave versus tight-binding approaches

Full-wave eigensolvers (MPB, FDFD, FEM) solve the exact Maxwell eigenvalue problem on a discretised unit cell and return the Bloch eigenvectors needed for Berry-phase calculations. They are essential for quantitative predictions in real photonic crystals but are computationally expensive for large unit cells or 3D systems.

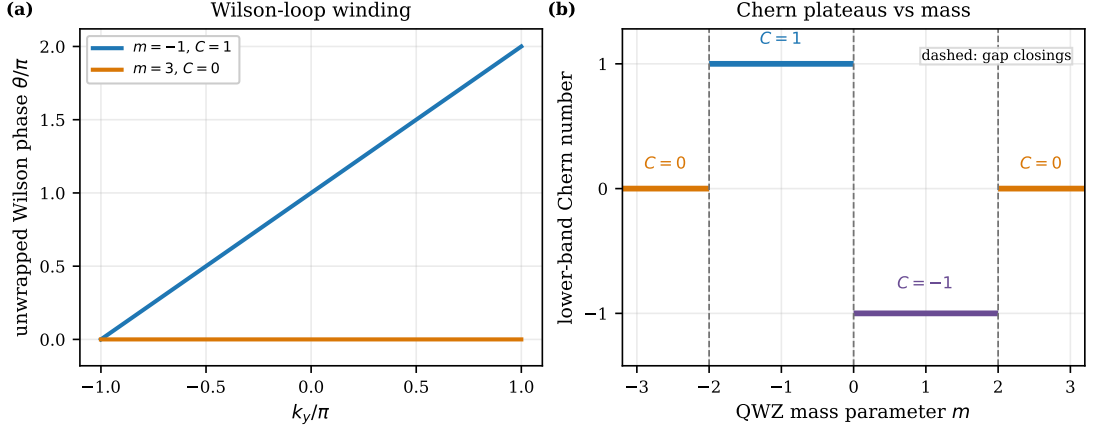


Figure 10.3: Wilson-loop winding and Chern plateaus for the QWZ lattice model. (a) Unwrapped Wilson-loop phase θ/π as a function of k_y/π for the topological phase ($m = -1, C = 1$, blue) and a trivial phase ($m = 3, C = 0$, orange). The winding number equals the Chern number. (b) Lower-band Chern number versus mass parameter m . Dashed lines mark the gap-closing transitions at $m = -2, 0, 2$. The integer plateaus are computed via the Fukui–Hatsugai–Suzuki plaquette algorithm on a 201×201 grid and are exact to numerical precision.

Tight-binding models (Haldane, QWZ, Harper–Hofstadter, etc. as used throughout this book) capture the essential topological physics with much less computational cost and provide analytical insight into the origin of topological invariants. They are invaluable for understanding the mechanisms of topological phase transitions but cannot predict quantitative properties such as the exact gap width or edge-mode penetration depth without fitting to full-wave calculations [12, 44].

A recommended workflow is to first explore the topological phase diagram using a tight-binding model, then validate the predicted topology with a full-wave Chern-number calculation, and finally simulate edge-mode transport in a finite structure (Appendix C) [40, 51].

10.14.4 Benchmarking against known results

Before computing the Chern number of a new photonic crystal, it is good practice to benchmark the implementation against known results. The following test cases are recommended:

1. The **QWZ model** (Appendix C): $\mathcal{C}_{\text{lower}} = -1$ for $0 < m < 2$, $+1$ for $-2 < m < 0$, and 0 for $|m| > 2$. This is the simplest test of a $U(1)$ Chern-number code.
2. The **Haldane model**: $\mathcal{C} = \pm 1$ depending on the sign of the next-nearest-neighbour hopping phase ϕ . This tests the code on a honeycomb lattice with broken time reversal [3].
3. The **Harper–Hofstadter model**: the Chern numbers of the magnetic subbands are given by the Diophantine equation $t = qs\mathcal{C} + pr$, providing a non-trivial test with $|\mathcal{C}| > 1$ [2, 21].

Exercises

Exercise 10.1. Implement the plaquette algorithm for the Qi–Wu–Zhang model (Appendix C) and verify that $\mathcal{C} = -1$ for $-2 < m < 0$ and $\mathcal{C} = +1$ for $0 < m < 2$. How does the convergence with N_k depend on the gap size $|m|$?

Exercise 10.2. Show that the link variable $U_\alpha(\mathbf{k})$ is gauge-covariant, i.e. it acquires a phase under a gauge transformation, and that the plaquette product F_{12} is gauge-invariant. under $|\mathbf{u}_{n,ij}\rangle \rightarrow e^{i\chi_{ij}}|\mathbf{u}_{n,ij}\rangle$, the plaquette product (10.4) is unchanged.

Exercise 10.3. For the Wilson loop $W(k_y) = \prod_i M(k_{x,i}, k_y)$, show that the determinant $\det W(k_y) = e^{i\sum_\alpha \theta_\alpha(k_y)}$ gives the total Chern number of the band group as the winding of $\sum_\alpha \theta_\alpha$.

Exercise 10.4. Consider a photonic crystal with C_3 rotation symmetry. If the rotation eigenvalue of band n at the K point is $e^{2\pi i/3}$ and at the Γ point is 1, what constraint does this impose on $\mathcal{C}_n \pmod{3}$?

Exercise 10.5. Explain why using the bare L^2 inner product instead of the electromagnetic inner product $\langle \cdot | \mathbf{M}_g | \cdot \rangle$ in the link variable can give a non-integer Chern number. Under what conditions are the two inner products equivalent?

Exercise 10.6. The Wannier-centre spectrum $\theta_\alpha(k_y)$ can be used to diagnose the $\mathbb{Z}_2$ invariant for a time-reversal-invariant system. Show that at $k_y = 0$ and $k_y = \pi/a$, the Wannier centres come in degenerate pairs (Kramers pairs), and that the $\mathbb{Z}_2$ invariant is the parity of the number of partner switches between $k_y = 0$ and $k_y = \pi/a$.

Part III

Other Two-Dimensional Phases

Chapter 11

Reciprocal Designs and Pseudospin

11.1 Topology without magnets

The photonic Chern insulators of Part II require magneto-optical materials and external magnetic fields. These ingredients are readily available at microwave frequencies (Chapter 9), but they impose severe constraints at optical frequencies: the magneto-optical response of available materials is weak, and permanent magnets compatible with nanophotonic chips do not exist.

A natural question therefore arises: can topological edge transport be achieved in a photonic crystal that preserves time-reversal symmetry? The answer, perhaps surprisingly, is yes—provided one uses the *crystalline* symmetry of the photonic lattice to construct a photonic analog of the quantum spin Hall effect.

This chapter describes two complementary approaches. The first, due to Khanikaev et al. [46, 70, 71], uses bianisotropic metamaterials to create a photonic topological insulator with a pseudospin degree of freedom constructed from the TE and TM polarisations [56, 65, 72]. The second, due to Wu and Hu (2015), uses zone folding in a C_6 -symmetric dielectric photonic crystal to create pseudospin- $\frac{1}{2}$ doublets at the Γ point, achieving topological edge modes without any exotic materials.

11.2 The quantum spin Hall analogy

In electronic systems, the quantum spin Hall (QSH) effect occurs when time-reversal symmetry is preserved but spin-orbit coupling opens a gap at band crossings. The resulting topological phase has a zero total Chern number (because time reversal forces $\mathcal{C} = 0$) but a nonzero *spin Chern number*:

$$\mathcal{C}_s = \frac{1}{2}(\mathcal{C}_\uparrow - \mathcal{C}_\downarrow), \quad (11.1)$$

where $\mathcal{C}_\uparrow$ and $\mathcal{C}_\downarrow$ are the Chern numbers of the spin-up and spin-down sectors. The bulk-edge correspondence predicts a pair of counter-propagating edge modes, each carrying a definite spin—the helical edge states.

For photons, there is no intrinsic spin- $\frac{1}{2}$. The photon’s time-reversal operator satisfies $\hat{\mathcal{T}}^2 = +1$ (bosonic), not $\hat{\mathcal{T}}^2 = -1$ (fermionic), so there is no Kramers degeneracy and no topological protection from $\mathbb{Z}_2$ invariants in the strict sense [6]. However, if the photonic system possesses an *additional* symmetry that commutes with $\hat{\mathcal{T}}$ and has two eigenvalues (a “pseudospin”), the system

can be decomposed into two independent sectors, each of which can carry a Chern number. The pseudospin Chern number $\mathcal{C}_s$ then predicts helical edge modes.

The protection of these edge modes is not topological in the absolute sense: it depends on the preservation of the pseudospin symmetry. Perturbations that break this symmetry (disorder, fabrication imperfections, coupling between the two pseudospin sectors) will cause backscattering. Nevertheless, the protection can be strong enough for practical applications if the symmetry-breaking perturbations are small.

11.3 Bianisotropic metacrystals

11.3.1 The Khanikaev construction

Khanikaev et al. [70] proposed a metamaterial design in which the pseudospin is constructed from the TE (H_z) and TM (E_z) polarisations of a 2D photonic crystal. In a generic 2D crystal these two polarisations are decoupled (each satisfies its own scalar wave equation), but a bianisotropic magneto-electric coupling $\xi = \zeta^\dagger$ mixes them in a controlled way.

The material matrix for such a metacrystal is

$$\mathbf{M} = \begin{pmatrix} \varepsilon_0 \varepsilon & c^{-1} \xi \\ c^{-1} \xi^\dagger & \mu_0 \mu \end{pmatrix}, \quad (11.2)$$

where ξ is chosen to have the form $\xi = i\chi(\hat{z} \otimes \hat{z})$ —a magneto-electric coupling that is odd under parity and even under time reversal.

The resulting four-band effective Hamiltonian near a high-symmetry point takes the block-diagonal form

$$H_{\text{eff}}(\mathbf{k}) = \begin{pmatrix} H_+(\mathbf{k}) & 0 \\ 0 & H_-(\mathbf{k}) \end{pmatrix} + O(\chi^2), \quad (11.3)$$

where $H_\pm$ are 2×2 massive Dirac Hamiltonians with opposite masses: $H_\pm = v_D(\pm \delta k_x \sigma_x + \delta k_y \sigma_y) \pm m\sigma_z$. The pseudospin “up” (+) and “down” (−) sectors have Chern numbers $\mathcal{C}_+ = +1$ and $\mathcal{C}_- = -1$, giving a total Chern number $\mathcal{C} = 0$ but a pseudospin Chern number $\mathcal{C}_s = 1$.

11.3.2 Physical origin of the pseudospin

The pseudospin states $|\uparrow\rangle$ and $|\downarrow\rangle$ correspond to specific linear combinations of TE and TM modes. In the simplest case, they are

$$|\uparrow\rangle = \frac{1}{\sqrt{2}}(|E_z\rangle + |H_z\rangle), \quad |\downarrow\rangle = \frac{1}{\sqrt{2}}(|E_z\rangle - |H_z\rangle), \quad (11.4)$$

where $|E_z\rangle$ and $|H_z\rangle$ are the TM and TE modes at the degeneracy point. The bianisotropic coupling χ provides the “spin-orbit” interaction that opens a gap and creates opposite Chern numbers for the two pseudospin sectors.

The physical realisation of this bianisotropic coupling requires metamaterial unit cells with broken mirror symmetry in the out-of-plane direction (e.g. omega-shaped or split-ring resonators with a vertical asymmetry).

11.3.3 Helical edge modes

At the boundary of a bianisotropic metacrystal, the pseudospin Chern number $\mathcal{C}_s = 1$ predicts one pair of counter-propagating edge modes. The “up” pseudospin mode propagates forward and the “down” pseudospin mode propagates backward, analogous to the helical edge states of the electronic QSH effect.

These edge modes are robust against perturbations that preserve the pseudospin symmetry (i.e., that do not mix TE and TM polarisations). Coupling between TE and TM modes—caused by surface roughness, out-of-plane scattering, or deviation from the ideal bianisotropic response—introduces backscattering proportional to the coupling strength.

11.4 The Wu–Hu zone-folding construction

11.4.1 Zone folding in a C6-symmetric lattice

Wu and Hu [5, 39, 73] proposed a remarkably simple approach to photonic topological insulation using only conventional dielectric materials and crystalline symmetry. The key idea is to start with a triangular lattice of dielectric rods and fold the bands by enlarging the unit cell to a hexagonal “supercell” containing six rods arranged in a honeycomb pattern.

In the original triangular lattice, the TE (or TM) bands have Dirac cones at the K and K' points of the Brillouin zone. When the unit cell is enlarged by a factor of $\sqrt{3} \times \sqrt{3}$ (the hexagonal supercell), the Brillouin zone shrinks correspondingly, and the K and K' points fold onto the Γ point of the new zone. This creates a fourfold degeneracy at Γ : two pairs of states that originated from K and K' .

Figure 11.1 gives the physical picture behind the two reciprocal routes discussed in this chapter. The Wu–Hu construction uses the enlarged supercell to bring the valley Dirac physics to Γ , where the dipolar and quadrupolar doublets can invert without magnetic bias. The bianisotropic construction uses a different language: TE-like and TM-like sectors are mixed into a photonic pseudospin. In both cases the edge mode is helical rather than chiral, and its robustness is tied to the symmetry that keeps the opposite pseudospins from mixing.

11.4.2 Symmetry analysis at Gamma

The fourfold degeneracy at Γ can be analysed using the irreducible representations of the point group C_{6v} of the hexagonal supercell. The four states transform as two two-dimensional representations: E_1 (dipolar, with angular momentum $\ell = \pm 1$) and E_2 (quadrupolar, with angular momentum $\ell = \pm 2$).

When the six rods in the supercell are identical and equally spaced, the E_1 and E_2 doublets are degenerate at Γ (because they came from the same frequency at K before zone folding). This accidental degeneracy can be lifted by deforming the supercell: moving the rods closer together (“shrinking” the hexagonal clusters) or farther apart (“expanding” them).

The expanded configuration pushes the E_1 (dipolar) doublet above the E_2 (quadrupolar) doublet, while the shrunk configuration does the opposite. The two configurations therefore have different

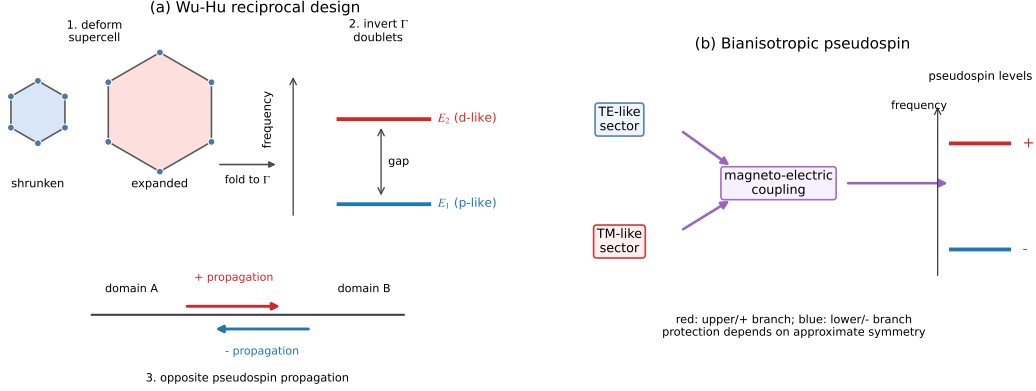


Figure 11.1: Two reciprocal routes to helical topological photonics. Left: the Wu–Hu zone-folding scheme enlarges the unit cell so that the valley Dirac cones fold to Γ ; expansion or shrinkage of the hexagonal cluster inverts the p -like and d -like doublets and creates domain-wall pseudospin channels. Right: bianisotropy mixes TE-like and TM-like sectors into a pseudospin doublet. Unlike the gyrotropic Chern phases of Chapters 7–9, these designs preserve ordinary time reversal and rely on approximate pseudospin or crystalline protection.

“band orderings” at Γ , analogous to normal and inverted band structures in electronic topological insulators.

11.4.3 Pseudospin one-half from p and d orbitals

The E_1 modes have the symmetry of p orbitals ($p_x \pm ip_y$), and the E_2 modes have the symmetry of d orbitals ($d_{xy} \pm id_{x^2-y^2}$). Wu and Hu defined the pseudospin as the angular momentum quantum number of these modes:

$$|\uparrow\rangle = |p_+\rangle \text{ or } |d_+\rangle, \quad |\downarrow\rangle = |p_-\rangle \text{ or } |d_-\rangle, \quad (11.5)$$

where $|\cdot_\pm\rangle$ denotes the states with angular momentum $\pm|\ell|$.

Near Γ , the effective Hamiltonian for the four-band system takes the Bernevig–Hughes–Zhang (BHZ) form:

$$H_{\text{BHZ}}(\mathbf{k}) = \begin{pmatrix} h(\mathbf{k}) & 0 \\ 0 & h^*(-\mathbf{k}) \end{pmatrix}, \quad (11.6)$$

where

$$h(\mathbf{k}) = \epsilon_0 \mathbf{I} + (M_0 - Bk^2)\sigma_z + A(k_x\sigma_x + k_y\sigma_y) \quad (11.7)$$

is a massive Dirac Hamiltonian with a quadratic correction. The mass parameter M_0 changes sign between the expanded ($M_0 > 0$) and shrunk ($M_0 < 0$) configurations. The topological transition occurs at the critical deformation where $M_0 = 0$ and the gap closes.

The block-diagonal structure of (11.6) reflects the conservation of the pseudospin: the C_6 rotation symmetry prevents coupling between the $|\uparrow\rangle$ and $|\downarrow\rangle$ sectors to leading order in $\mathbf{k}$. Each block

has a well-defined Chern number, and the pseudospin Chern number is

$$\mathcal{C}_s = \frac{1}{2}(\mathcal{C}_+ - \mathcal{C}_-) = \begin{cases} 0 & \text{shrunk } (M_0 < 0), \\ 1 & \text{expanded } (M_0 > 0). \end{cases} \quad (11.8)$$

11.4.4 Derivation of the BHZ parameters from Maxwell modes

The parameters M_0 , B , and A in the BHZ model can be extracted from the Maxwell eigenvalue problem by degenerate perturbation theory at the Γ point.

Let δR denote the displacement of each rod from its position in the undeformed (critical) configuration. The deformation changes the material matrix $\mathbf{M}(\mathbf{r})$ by $\delta \mathbf{M}(\mathbf{r}) \propto \delta R$. In the four-dimensional subspace spanned by the $p_\pm$ and $d_\pm$ modes at Γ , the perturbation matrix is:

$$\langle \alpha | \omega \delta \mathbf{M} | \beta \rangle_\Omega = \begin{pmatrix} M_0 & 0 & 0 & 0 \\ 0 & M_0 & 0 & 0 \\ 0 & 0 & -M_0 & 0 \\ 0 & 0 & 0 & -M_0 \end{pmatrix}, \quad (11.9)$$

where $M_0 \propto \delta R$ and the block structure follows from the C_6 selection rules (the perturbation, being C_6 -invariant, cannot mix states of different angular momentum).

The linear-in- $\mathbf{k}$ coupling A comes from the $\mathbf{k} \cdot \mathbf{p}$ perturbation (the Hellmann–Feynman coupling between the four modes at Γ), which connects $|p_+\rangle$ to $|d_+\rangle$ with matrix element Ak_+ and $|p_-\rangle$ to $|d_-\rangle$ with Ak_-^* , where $k_\pm = k_x \pm ik_y$. The quadratic correction $-Bk^2$ comes from second-order perturbation theory involving remote bands.

11.5 Edge modes in reciprocal topological photonic crystals

11.5.1 Helical edge states

At the boundary between the expanded ($\mathcal{C}_s = 1$) and shrunk ($\mathcal{C}_s = 0$) configurations, the BHZ bulk-edge correspondence predicts one pair of helical edge modes. The pseudospin-up mode propagates clockwise and the pseudospin-down mode propagates counter-clockwise (or vice versa, depending on the orientation).

These edge modes have been observed experimentally in both the microwave regime (using printed-circuit-board metamaterials) and the optical regime (using silicon photonic crystal slabs), confirming the Wu–Hu prediction [67].

11.5.2 Robustness and its symmetry dependence

The helical edge modes are protected by the C_6 crystalline symmetry that ensures the block-diagonal structure of the BHZ Hamiltonian. Perturbations that respect C_6 (such as uniform scaling or smooth deformation) cannot cause backscattering.

Perturbations that break C_6 mix the pseudospin sectors and open a backscattering channel. The most common symmetry-breaking perturbations in experiments are fabrication disorder (random

variations in rod size or position) and interface geometry (domain walls that are not aligned with the lattice directions).

The robustness can be quantified by the pseudospin-mixing matrix element at the edge, which scales as the strength of the C_6 -breaking perturbation. For typical fabrication tolerances in silicon photonics, the edge-mode backscattering remains small enough for useful waveguide lengths of tens of lattice constants.

11.6 Comparison of reciprocal approaches

The bianisotropic and zone-folding approaches to reciprocal photonic topology share the same mathematical structure (BHZ-type effective Hamiltonian, pseudospin Chern number) but differ in their physical implementation and practical limitations.

The bianisotropic approach requires metamaterial unit cells with magneto-electric coupling, which is difficult to achieve at optical frequencies. It has been demonstrated primarily at microwave frequencies. The zone-folding approach uses only conventional dielectric materials and works at any frequency, including the telecom band. Its main limitation is the requirement for C_6 symmetry, which restricts the lattice geometry.

Both approaches produce edge modes that are protected by a crystalline symmetry rather than by topology alone. This makes them less robust than the magneto-optical Chern insulators of Part II, but far more practical for integrated photonic applications.

11.7 Other reciprocal topological designs

11.7.1 Coupled-resonator arrays

Hafezi et al. [15] demonstrated a photonic topological insulator using a 2D array of coupled ring resonators [15]. In this system, the synthetic gauge field arises from the path-length difference between clockwise and counter-clockwise circulation in the rings, which imprints a direction-dependent phase on the inter-ring coupling. The result is a photonic analog of the integer quantum Hall effect, with chiral edge modes propagating around the boundary of the array.

11.7.2 Photonic crystal slabs with spin-orbit coupling

In photonic crystal slabs, the TE-like and TM-like guided modes are naturally coupled by the finite slab thickness. This TE-TM coupling acts as a photonic spin-orbit interaction, analogous to the spin-orbit coupling in electronic topological insulators. By engineering the slab geometry to enhance this coupling at a band degeneracy, one can create pseudospin-Hall edge states without any metamaterial ingredients.

11.8 Quantitative design: the Wu–Hu silicon photonic crystal

The zone-folding approach is remarkable because it requires only conventional dielectric materials. Wu and Hu’s original proposal used alumina rods, but the most technologically relevant implementation uses silicon, which is compatible with CMOS fabrication.

11.8.1 Design parameters

Consider a hexagonal supercell of six silicon rods ($\epsilon_r = 11.7$) in air, with supercell lattice constant $a = 0.55 \mu\text{m}$. The individual rod radius is $r = 0.16a = 88 \text{ nm}$. The topological transition occurs at the critical expansion ratio $R_c/a \approx 1/3$, where R_c is the distance from each rod centre to the supercell centre.

In the expanded configuration ($R > R_c$), the p -like doublet (E_1 representation of C_{6v}) moves above the d -like doublet (E_2), and the mass parameter in the BHZ model is $M_0 > 0$. A plane-wave expansion with 625 waves gives the gap centre at $\omega a/(2\pi c) \approx 0.29$, corresponding to a free-space wavelength $\lambda_0 \approx 1.90 \mu\text{m}$. The topological gap extends from approximately $\omega a/(2\pi c) = 0.275$ to 0.305 , giving a relative gap of $\Delta\omega/\omega_0 \approx 10\%$. At telecom wavelengths ($\lambda_0 = 1.55 \mu\text{m}$), one can rescale the lattice constant to $a = 0.45 \mu\text{m}$ [12, 17].

11.8.2 The topological gap and its origin

The gap at Γ is opened by the zone-folding deformation, which breaks the accidental degeneracy between the p and d doublets. The gap magnitude is controlled by the deformation parameter $\delta R = R - R_c$. For small $\delta R/a$, the mass parameter scales linearly: $M_0 \propto \delta R/a$, and the gap width is $\Delta\omega = 2|M_0|$.

The BHZ parameter A , which sets the edge-mode velocity, is determined by the $\mathbf{k} \cdot \mathbf{p}$ coupling between the p and d modes. Plane-wave calculations give $A a/(2\pi c) \approx 0.07$ for the silicon design, implying an edge-mode group velocity $v_{\text{edge}} \approx 0.07 c$ at the gap centre [12].

11.9 Experimental demonstrations

11.9.1 All-dielectric realisations

The Wu–Hu construction has been experimentally demonstrated in several material platforms. At microwave frequencies, arrays of alumina rods arranged in the expanded and shrunk configurations show clear helical edge transport at the domain wall between the two phases. The edge modes propagate around sharp corners with negligible backscattering, provided the domain wall follows a C_6 -compatible direction [12].

At optical frequencies, Ma and Shvets (2016) proposed an all-silicon valley-Hall photonic topological insulator using triangular holes in a silicon slab [17]. Dong et al. (2017) demonstrated valley photonic crystals with valley-dependent spin-locked edge states, achieving topological transport at microwave frequencies that could be scaled to optical wavelengths [50]. Wu et al. (2017) directly observed valley-polarised topological edge states in designer surface plasmon crystals at microwave

frequencies [18], while Gao et al. (2017) demonstrated the photonic valley-Hall effect and valley-polarised edge transport on a metallic surface [60].

11.9.2 The Hafezi coupled-resonator lattice

The coupled-resonator approach of Hafezi et al. [15] deserves special discussion because it achieves topological edge modes using a qualitatively different mechanism from the zone-folding designs.

The system consists of a 2D square lattice of silicon ring resonators, each supporting clockwise (CW) and counter-clockwise (CCW) whispering-gallery modes. Adjacent resonators are coupled through connecting waveguides whose path lengths are engineered to produce direction-dependent phases. Specifically, the coupling from resonator (m, n) to $(m+1, n)$ acquires a phase $\phi_{mn} = 2\pi\alpha n$, where α is a dimensionless parameter analogous to the magnetic flux per plaquette in the Hofstadter problem.

For $\alpha = 1/4$, the effective Hamiltonian reproduces the Harper equation with rational flux, and the resulting Floquet–Bloch band structure has nontrivial Chern numbers [12]. The CW and CCW modes of the ring resonators play the role of spin-up and spin-down, giving a photonic quantum spin-Hall system with helical edge modes.

The ring-resonator lattice was fabricated in silicon-on-insulator with ring radii $R \approx 10 \mu\text{m}$, operating at telecom wavelengths ($\lambda \approx 1.55 \mu\text{m}$). Edge-mode propagation was imaged directly using scattered light from surface roughness, confirming the predicted helical transport.

11.10 Disorder and the limits of crystalline protection

The pseudospin topological protection in reciprocal designs is fundamentally different from the Chern-insulator protection of Part II. In a Chern insulator, backscattering is forbidden by the absence of counter-propagating modes at the same frequency, regardless of the perturbation. In a pseudospin system, backscattering is suppressed only for perturbations that preserve the pseudospin symmetry.

For the Wu–Hu design, the relevant symmetry is C_6 . Perturbations that break C_6 —such as random rod displacements, elliptical rod shapes, or domain walls not aligned with high-symmetry directions—couple the $|p_+\rangle$ and $|d_-\rangle$ sectors and open a backscattering channel. The backscattering rate scales as $\Gamma_{\text{bs}} \propto |\delta\varepsilon/\varepsilon|^2$, where $\delta\varepsilon$ is the C_6 -breaking perturbation strength [13].

For typical silicon photonic fabrication tolerances ($\delta r/r \sim 1\%$), the mean free path of the edge mode is estimated at $\ell \sim 50\text{--}100a$, sufficient for on-chip routing over tens of micrometres. However, this is far shorter than the formally infinite mean free path of a Chern-insulator edge mode, underscoring the practical importance of the distinction between topological and crystalline-symmetry protection.

11.11 The Khanikaev bianisotropic metacrystal

An alternative to the Wu–Hu all-dielectric approach is the bianisotropic metacrystal proposed by Khanikaev et al., which achieves a photonic analog of the quantum spin-Hall effect using a hexagonal lattice of metallic split-ring resonators [12].

The key idea is that a split-ring resonator with both electric and magnetic resonances can produce a bianisotropic response: the material matrix $\mathbf{M}$ acquires off-diagonal blocks ξ and ζ that couple the electric and magnetic fields. When the bianisotropic coupling has the right symmetry, the two circular polarisations (CW and CCW) of the resonator modes decouple into two independent sectors, each experiencing an effective time-reversal-breaking perturbation. The two sectors have opposite effective Chern numbers ($\mathcal{C}_\uparrow = +1$ and $\mathcal{C}_\downarrow = -1$), so the total Chern number vanishes but the spin-Chern number is nonzero [12, 13].

The resulting edge modes are helical: spin-up modes propagate in one direction and spin-down modes propagate in the opposite direction, in direct analogy with the quantum spin-Hall effect for electrons.

11.11.1 Comparison of pseudospin approaches

The different pseudospin approaches can be compared along several axes: the mechanism for pseudospin construction, the symmetry that protects the edge modes, the required material platform, and the robustness against different types of disorder.

The Wu–Hu zone-folding approach uses C_6 symmetry to create p and d doublets at Γ . The protection is crystalline: perturbations that break C_6 open a backscattering channel. The advantage is that only conventional dielectrics are needed.

The Khanikaev bianisotropic approach uses electromagnetic duality symmetry to decouple the two circular polarisations. The protection is electromagnetic: perturbations that break the duality (e.g. non-bianisotropic disorder) cause backscattering. The advantage is that the protection can be more robust than crystalline symmetry in certain geometries, but the disadvantage is that bianisotropic media are harder to fabricate.

The Hafezi coupled-resonator approach uses the clockwise and counter-clockwise whispering-gallery modes as pseudospin. The protection is gauge-like: the phase accumulation in the coupling waveguides produces an effective magnetic flux that is robust to small fabrication variations. The advantage is compatibility with standard silicon photonics; the disadvantage is that the bandwidth is limited by the resonator finesse [12, 15].

11.12 Coupled-resonator implementations

11.12.1 The Hafezi ring-resonator lattice

Hafezi et al. realised a photonic analog of the integer quantum Hall effect using a 2D lattice of silicon ring resonators coupled by linking waveguides [12, 15]. The key insight is that the round-trip phase accumulated in the linking waveguides can be engineered to produce a synthetic magnetic flux Φ per plaquette, directly analogous to the Peierls phases in a tight-binding model.

For a square lattice of rings with radius $R \approx 10 \mu\text{m}$ and coupling gap $g \approx 200 \text{ nm}$, the coupling strength is $\kappa \approx 0.3$ and the free spectral range is $\text{FSR} \approx 1.6 \text{ THz}$. The synthetic flux is $\Phi = k_0 \Delta L$, where ΔL is the path-length difference between clockwise and counter-clockwise linking paths. By designing $\Delta L = \lambda_0/4$ at the operating wavelength $\lambda_0 = 1.55 \mu\text{m}$, the flux per plaquette is $\Phi = \pi/2$, which is optimal for opening a topological gap.

The resulting edge states were observed as bright channels propagating along the boundary of the resonator array, with the clockwise and counter-clockwise whispering-gallery modes acting as the two pseudospin species. The robustness was demonstrated by intentionally introducing defects (missing resonators) and observing that the edge mode routes around them without backscattering [15].

11.12.2 Protection limits of coupled-resonator designs

The pseudospin protection in coupled-resonator lattices depends on the suppression of inter-spin scattering—the mixing of clockwise and counter-clockwise modes. Two mechanisms can cause this mixing:

The first is backscattering at the coupling junctions, which is proportional to the surface roughness of the waveguide sidewalls. For standard silicon photonic fabrication with root-mean-square roughness $\sigma \approx 2$ nm, the backscattering rate is $\gamma_{bs} \approx 10^{-3}$ FSR, which is much smaller than the topological gap ($\Delta\omega \approx 0.05$ FSR).

The second is the non-resonant coupling between adjacent resonators, which can be suppressed by increasing the coupling gap. For $g > 300$ nm, the non-resonant coupling is below -30 dB and can be neglected [12, 13].

11.13 Topological protection without time-reversal breaking

The reciprocal topological photonic designs discussed in this chapter achieve edge-mode protection without breaking time-reversal symmetry. This raises a fundamental question: what symmetry protects the edge modes, and how robust is the protection compared to the Chern insulator?

11.13.1 Crystalline topological protection

In the Wu–Hu design, the edge modes are protected by the C_6 crystalline symmetry of the honeycomb lattice. The topological invariant is a spin Chern number defined within each pseudospin sector, which is quantised as long as the C_6 symmetry is preserved. Perturbations that break C_6 (such as lattice distortions, non-uniform strain, or interfaces that do not preserve the crystalline orientation) can open a backscattering gap in the edge spectrum.

The spin Chern number is not a true topological invariant in the sense of the Chern number: it depends on the choice of pseudospin basis, which is defined by the crystalline symmetry rather than by the fundamental structure of the Hilbert space. Nevertheless, for perturbations that respect C_6 , the spin Chern number is as robust as the Chern number, and the edge modes are equally well protected [12, 24].

Silveirinha (2016) showed that for continuum bianisotropic media with time-reversal symmetry, a $\mathbb{Z}_2$ topological index can be defined directly from the Maxwell Green’s function, without recourse to tight-binding models. This provides a Maxwell-first foundation for the $\mathbb{Z}_2$ classification of reciprocal photonic topological insulators [24, 74].

11.13.2 Electromagnetic duality protection

In the Khanikaev bianisotropic design, the protection comes from electromagnetic duality symmetry rather than crystalline symmetry. Duality symmetry requires the electric and magnetic responses to be related by $\varepsilon = \mu$, which decouples the two circular polarisations. Perturbations that break duality (such as non-bianisotropic disorder or substrate effects) cause inter-spin scattering.

The duality-protected edge modes are more robust than the crystalline-protected modes in certain geometries, because duality is a local property of the material while C_6 is a global property of the lattice. However, achieving precise duality ($\varepsilon = \mu$) is experimentally challenging, especially at optical frequencies where the magnetic response is weak [12, 14].

11.14 Comparing experimental reciprocal platforms

The reciprocal topological designs described in this chapter have been realised across a wide range of electromagnetic platforms, from microwave metamaterials to silicon photonic crystals at telecommunication wavelengths. Each platform illuminates different aspects of the underlying physics.

11.14.1 Coupled-resonator arrays and imaging of edge transport

The first experimental observation of topologically protected edge transport without time-reversal breaking used a lattice of coupled ring resonators fabricated in silicon-on-insulator [15]. The clockwise and counter-clockwise circulating modes of each ring serve as pseudo-spin up and down, while the inter-ring coupling phases produce an effective gauge field for each spin species independently.

In the experiment, a 10×10 lattice of racetrack resonators operating near $\lambda = 1.55 \mu\text{m}$ was patterned on a silicon chip. The effective magnetic flux per plaquette was $\alpha = 1/4$, set by staggering the positions of link resonators. When light was injected into a corner site, the output intensity propagated along the lattice edge with a measured contrast of approximately 10 dB between edge and bulk sites [12, 15].

The principal limitation of ring-resonator platforms is the difficulty of achieving strong pseudo-spin–orbit coupling: the two circulation directions interact only through backscattering at surface roughness, which is weak in high-quality silicon resonators. The system therefore behaves as two nearly independent copies of the Hofstadter model rather than a true $\mathbb{Z}_2$ topological insulator. Nevertheless, the experiment provided the first clear photonic imaging of topological edge transport at optical frequencies without magnetic bias.

11.14.2 Dielectric photonic crystals with the Wu–Hu design

The zone-folding approach of Section 11.4 was realised experimentally in several frequency regimes. The design requires only a single dielectric material arranged in a triangular lattice, compatible with standard nanofabrication.

At microwave frequencies, Wu et al. demonstrated helical edge states in a lattice of alumina rods with C_{6v} symmetry [18]. Expanded and shrunken configurations were created by adjusting rod positions within the hexagonal unit cell. Measured transmission through an expanded–shrunken

Table 11.1: Comparison of reciprocal topological photonic platforms. Gap values are representative; actual values depend on geometry and material parameters.

Platform	Gap (%)	Frequency	Helical?
Bianisotropic metamaterial	5–10	Microwave	Yes
Wu–Hu photonic crystal	3–8	Microwave–NIR	Yes
Coupled ring resonators	1–5 (FSR)	NIR–telecom	Pseudo
Valley photonic crystal	2–6	Microwave–NIR	No (C_v)

domain wall showed a clear edge-state passband inside the bulk gap, with forward-to-backward ratio exceeding 20 dB.

At near-infrared wavelengths, Noh et al. fabricated the Wu–Hu design in a silicon photonic crystal slab [75]. The structure consisted of air-hole clusters in a 220 nm silicon membrane. Edge-state resonances with quality factor above 10^3 were observed, and edge transport persisted through sharp 120° bends—a hallmark of topological protection [12].

11.14.3 Quantitative comparison of platforms

Three figures of merit characterise reciprocal topological photonic systems: the relative topological gap $\Delta\omega/\omega_0$, the edge-state propagation length L_{edge} , and the robustness against fabrication disorder.

The bianisotropic metamaterial design of Khanikaev et al. [25] achieves gaps of 5–10% in microwave metamaterials, set by the magneto-electric coupling strength χ : $\Delta\omega/\omega_0 \sim |\chi|/\epsilon$. The bianisotropic response is inherently narrowband and dispersive, however, limiting the useful bandwidth.

The Wu–Hu zone-folding design produces gaps of 3–8% in dielectric photonic crystals, controlled by the geometric deformation parameter $\delta R/R_0$. No magneto-optical or bianisotropic materials are needed, and the design is CMOS-compatible [12].

Coupled-resonator platforms offer the largest effective gauge fields (flux α up to $1/4$ per plaquette) and the most flexible band engineering, at the cost of larger footprints and greater sensitivity to resonator frequency disorder. The topological gap is typically 1–5% of the free spectral range.

Exercises

Exercise 11.1. Derive the BHZ Hamiltonian (11.6) from the fourfold degeneracy at Γ using degenerate perturbation theory in the subspace spanned by $|p_\pm\rangle$ and $|d_\pm\rangle$. Show that C_6 symmetry forces the off-diagonal blocks to vanish to leading order.

Exercise 11.2. For the Wu–Hu photonic crystal, compute the pseudospin Chern number in both the expanded and shrunken configurations by integrating the Berry curvature of the $h(\mathbf{k})$ block over the Brillouin zone. Verify that $\mathcal{C}_s$ changes by 1 across the topological transition.

Exercise 11.3. The mass parameter M_0 in the BHZ model is proportional to the rod displacement δR . If $M_0 = \alpha \delta R$ with $\alpha = 2\pi \times 0.5$ GHz/mm, estimate the topological gap width for $\delta R = 0.5$ mm and a lattice constant $a = 10$ mm.

Exercise 11.4. Compare the protection of the helical edge modes in the Wu–Hu design with the chiral edge modes in the Wang YIG crystal. Under what types of disorder does each lose its topological protection?

Exercise 11.5. Design a domain wall between the expanded and shrunken Wu–Hu configurations. Sketch the expected edge-state dispersion at the domain wall and explain why there are two counter-propagating modes instead of one.

Chapter 12

Valleys and Lattices

12.1 Inversion breaking as an alternative

In Chapter 7 we saw that breaking time-reversal symmetry at a pair of Dirac cones produces bands with nonzero Chern numbers, because the Berry-flux contributions from K and K' *add*. There is another route to lifting the Dirac degeneracy: breaking *inversion* symmetry while preserving time reversal. In this case the Berry fluxes at K and K' have *opposite* signs and the global Chern number vanishes, but the *valley-resolved* Berry flux is nonzero and can support edge modes at domain walls between regions with opposite inversion breaking [56].

This “valley-Hall” mechanism has become one of the most widely used approaches to topological photonics [12, 17, 50], because it requires neither magneto-optical materials nor time modulation—only a geometric asymmetry in the unit cell.

12.2 Symmetry analysis of inversion breaking

12.2.1 Inversion in a honeycomb lattice

A honeycomb photonic crystal (e.g. a hexagonal arrangement of two sublattices A and B) has C_{6v} symmetry when the two sublattices are identical. The inversion operation $\hat{\mathcal{P}}$ exchanges A and B while reversing $\mathbf{r} \rightarrow -\mathbf{r}$.

When the sublattices are made inequivalent (different rod radii, hole sizes, or material compositions), the C_{6v} symmetry is reduced to C_{3v} : the threefold rotation C_3 and the three mirror planes σ_v are preserved, but inversion $\hat{\mathcal{P}}$ and the remaining mirrors are broken.

12.2.2 Effect on Dirac points

Recall from Section 6.2.1 that Dirac points in a hexagonal lattice are protected by the combination of $\hat{\mathcal{T}}$ and $\hat{\mathcal{P}}$. Under the von Neumann–Wigner codimension argument:

- With $\hat{\mathcal{T}} + \hat{\mathcal{P}}$: the eigenproblem is real symmetric, requiring only 2 conditions for degeneracy in a 2D parameter space—generically allowed.
- With $\hat{\mathcal{T}}$ alone (no $\hat{\mathcal{P}}$): the eigenproblem is complex Hermitian, requiring 3 conditions—generically forbidden in 2D.

Thus, breaking $\hat{\mathcal{P}}$ while preserving $\hat{\mathcal{T}}$ lifts the Dirac degeneracy generically.

12.2.3 The mass term from sublattice asymmetry

To derive the mass term explicitly, we return to the two-mode effective equation at the Dirac point (Section 6.4). Let the unperturbed crystal have C_{6v} symmetry with Dirac modes $|\mathbf{u}_{\pm}(\mathbf{k}_K)\rangle$ transforming under the two-dimensional irreducible representation E_1 of C_{3v} .

The sublattice asymmetry is described by a perturbation $\delta\mathbf{M}(\mathbf{r})$ that is invariant under C_3 but odd under $\hat{\mathcal{P}}$ (it has opposite signs on the two sublattices). In the two-mode subspace, the perturbation matrix has the form:

$$\delta H = \begin{pmatrix} \langle \mathbf{u}_+ | \delta\mathbf{M} | \mathbf{u}_+ \rangle & \langle \mathbf{u}_+ | \delta\mathbf{M} | \mathbf{u}_- \rangle \\ \langle \mathbf{u}_- | \delta\mathbf{M} | \mathbf{u}_+ \rangle & \langle \mathbf{u}_- | \delta\mathbf{M} | \mathbf{u}_- \rangle \end{pmatrix}. \quad (12.1)$$

By the symmetry of C_3 : if the modes $|\mathbf{u}_{\pm}\rangle$ are chosen to transform as $e^{\pm i2\pi/3}$ under C_3 , then the off-diagonal elements vanish (they would require $\delta\mathbf{M}$ to transform nontrivially under C_3 , but the sublattice perturbation is C_3 -invariant). The diagonal elements are equal and opposite: $\langle \mathbf{u}_+ | \delta\mathbf{M} | \mathbf{u}_+ \rangle = -\langle \mathbf{u}_- | \delta\mathbf{M} | \mathbf{u}_- \rangle \equiv m_{\Delta}$, because $\hat{\mathcal{P}}$ exchanges $|\mathbf{u}_+\rangle$ and $|\mathbf{u}_-\rangle$ while reversing the sign of $\delta\mathbf{M}$.

Thus the perturbation is proportional to σ_z :

$$\delta H = m_{\Delta} \sigma_z, \quad m_{\Delta} \propto \Delta, \quad (12.2)$$

where Δ parametrises the sublattice asymmetry (e.g. $\Delta = r_A - r_B$ for rod radii). The full effective Hamiltonian near K is the massive Dirac equation:

$$H_K(\delta\mathbf{k}) = v_D(\delta k_x \sigma_x + \delta k_y \sigma_y) + m_{\Delta} \sigma_z. \quad (12.3)$$

12.2.4 Opposite masses at K and K -prime

The crucial difference from the gyrotropic case (Chapter 7) is the behaviour under time reversal. Time reversal maps $K \rightarrow K'$ and complex-conjugates the Hamiltonian. Since m_{Δ} is real and time-reversal even, but the Dirac matrices at K' are related to those at K by complex conjugation:

$$H_{K'}(\delta\mathbf{k}) = v_D(-\delta k_x \sigma_x + \delta k_y \sigma_y) - m_{\Delta} \sigma_z. \quad (12.4)$$

The mass at K' is $-m_{\Delta}$: *opposite* to the mass at K . This can also be seen from the constraint $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k})$ imposed by time-reversal symmetry: the Berry curvature must be odd under $\mathbf{k} \rightarrow -\mathbf{k}$, forcing opposite signs at the two valleys.

12.3 Valley Chern numbers

12.3.1 Definition and half-integer quantisation

The Berry curvature near each valley has the massive Dirac form (Eq. (7.8)):

$$\Omega_n(\mathbf{k}) \Big|_{\text{near } K} = + \frac{m_{\Delta}}{2(|\delta\mathbf{k}|^2 + m_{\Delta}^2)^{3/2}}, \quad \Omega_n(\mathbf{k}) \Big|_{\text{near } K'} = - \frac{m_{\Delta}}{2(|\delta\mathbf{k}|^2 + m_{\Delta}^2)^{3/2}}. \quad (12.5)$$

The total Chern number $\mathcal{C}_n = (2\pi)^{-1} \int_{\text{BZ}} \Omega_n d^2k = 0$ vanishes as required by time-reversal symmetry.

The *valley Chern number* integrates the Berry curvature over only half of the Brillouin zone surrounding each valley [50, 54]:

$$\mathcal{C}_n^K = \frac{1}{2\pi} \int_{\text{near } K} \Omega_n d^2k = +\frac{1}{2} \text{sgn}(m_\Delta), \quad \mathcal{C}_n^{K'} = -\frac{1}{2} \text{sgn}(m_\Delta). \quad (12.6)$$

Remark 12.1. The valley Chern number is *not* a true topological invariant—it depends on how the Brillouin zone is partitioned into two halves. However, when the Berry curvature is sharply localised near the Dirac points (small gap relative to bandwidth), the partition ambiguity is exponentially small. The valley Chern number becomes a useful approximate invariant in this regime.

12.3.2 The valley-contrast Chern number

The difference between the two valley Chern numbers:

$$\mathcal{C}_v = \mathcal{C}_n^K - \mathcal{C}_n^{K'} = 2\mathcal{C}_n^K = \text{sgn}(m_\Delta) = \pm 1. \quad (12.7)$$

At a domain wall between two regions with opposite sublattice asymmetry ($\Delta > 0$ on one side, $\Delta < 0$ on the other), the valley-contrast Chern number changes by $|\Delta \mathcal{C}_v| = 2$, predicting one valley-polarised edge mode per valley [18, 75].

12.4 Geometric mechanisms for inversion breaking

12.4.1 Honeycomb lattice with unequal rods

The simplest construction uses a honeycomb array of dielectric rods with two different radii r_A and r_B on the A and B sublattices. When $r_A = r_B$, the crystal has C_{6v} symmetry and Dirac cones at K, K' . When $r_A \neq r_B$, inversion is broken and a gap opens:

$$\Delta\omega_{\text{gap}} \approx 2v_D |m_\Delta| \propto |r_A - r_B|. \quad (12.8)$$

To create a domain wall, one fabricates two regions: region L with $(r_A, r_B) = (r_1, r_2)$ and region R with $(r_A, r_B) = (r_2, r_1)$ (the sublattices are swapped). This reverses m_Δ across the interface.

12.4.2 Triangular-cluster deformation

An elegant alternative starts from a hexagonal lattice of identical rods arranged in a triangular-cluster pattern. By expanding or shrinking the triangular clusters (rotating them clockwise or counter-clockwise), one breaks C_6 symmetry while preserving C_3 . The “expanded” and “shrunk” configurations have opposite signs of m_Δ .

This approach was used by Wu and Hu (2015) for the pseudospin design discussed in Chapter 11. When only inversion breaking (without the pseudospin structure) is considered, the same deformation produces valley-Hall physics.

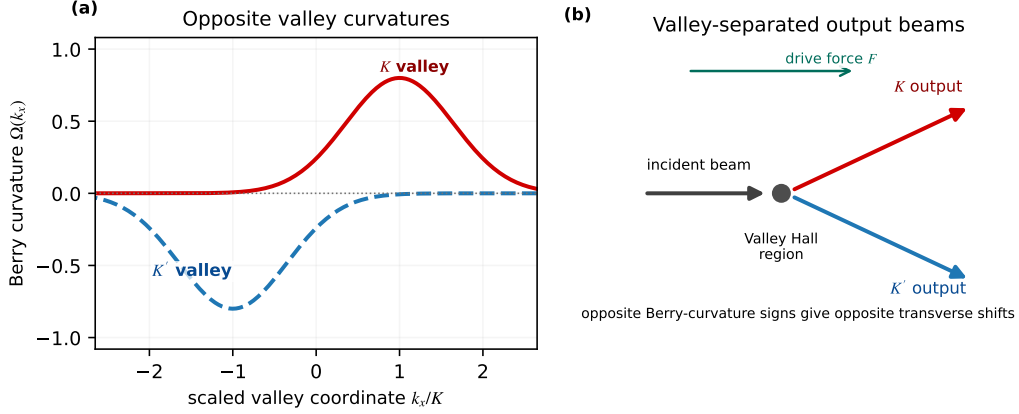


Figure 12.1: The valley Hall effect in photonic crystals. **(a)** Berry curvature $\Omega(k_x)$ near the two valleys, showing opposite signs at K (solid red, positive) and K' (dashed blue, negative). Time-reversal symmetry enforces $\Omega_K(\mathbf{k}) = -\Omega_{K'}(-\mathbf{k})$, so the global Chern number vanishes but the valley Chern numbers are nonzero. **(b)** Schematic of valley-separated transverse deflection. A longitudinal drive force F displaces wavepackets transversely via the anomalous velocity $v_\perp \propto F \Omega_z$. Because Ω_K and $\Omega_{K'}$ have opposite signs, the K and K' components are deflected in opposite directions, producing spatially separated output beams—the photonic analog of the valley Hall effect.

12.4.3 Photonic crystal slab with asymmetric holes

The most widely used platform for valley-Hall photonic devices is a thin dielectric slab (typically silicon, $n \approx 3.5$) with a hexagonal array of air holes. Each unit cell contains two holes of different sizes:

$$r_A = r_0 + \delta r, \quad r_B = r_0 - \delta r, \quad (12.9)$$

where r_0 is the mean radius and δr controls the sublattice asymmetry. The TE-like guided modes of the slab experience the valley-Hall effect, with the gap width proportional to δr .

This platform is compatible with standard semiconductor fabrication (electron-beam lithography, reactive-ion etching) and operates at telecom wavelengths ($\lambda \sim 1550$ nm).

12.5 Domain-wall modes in detail

12.5.1 Construction and topology

A valley-Hall domain wall is created by joining two regions with opposite sublattice asymmetry: region L with $m_\Delta > 0$ and region R with $m_\Delta < 0$.

Near each valley, the effective Dirac equation at the interface has the same form as (8.1), with a mass $m(x)$ that changes sign. The zero-mode analysis of Section 8.3 applies separately at each valley:

- At valley K : one mode propagating in the $+y$ direction (forward), with the wavefunction localised at the domain wall:

$$\psi_K(x, y) \propto |\varphi_+\rangle \exp\left(i\delta k_{\parallel}y - \int_0^x m(x') dx'\right). \quad (12.10)$$

- At valley K' : one mode propagating in the $-y$ direction (backward), with a similarly localised wavefunction.

The result is a pair of *counter-propagating* edge modes, each valley-polarised: forward at K , backward at K' .

12.5.2 Dispersion relation

The edge-mode dispersions near each valley are approximately linear:

$$\omega_K(\delta k_{\parallel}) \approx \omega_D + v_D \delta k_{\parallel}, \quad \omega_{K'}(\delta k_{\parallel}) \approx \omega_D - v_D \delta k_{\parallel}. \quad (12.11)$$

The group velocities have opposite signs: the K -valley mode has $v_g = +v_D$ and the K' -valley mode has $v_g = -v_D$. This *valley-momentum locking* is the valley-Hall analog of spin-momentum locking in the quantum spin Hall effect.

12.5.3 Localisation length

The transverse localisation length of the domain-wall mode is

$$\ell \sim \frac{1}{|m_{\Delta}|} \sim \frac{a}{\Delta\omega/\omega_D}, \quad (12.12)$$

where a is the lattice constant and $\Delta\omega = 2v_D|m_{\Delta}|$ is the valley gap width. For a typical silicon photonic crystal slab with $\Delta\omega/\omega_D \sim 5\%$, the localisation length is $\ell \sim 20a \approx 10 \mu\text{m}$ —several micrometres, confined to the immediate vicinity of the domain wall [61] [67].

12.6 Protection and its limits

12.6.1 The inter-valley scattering mechanism

The valley-Hall edge modes are *not* topologically protected in the strong sense of Chern-insulator edge modes. The key vulnerability is *inter-valley scattering*: any perturbation with Fourier components at wavevectors connecting K and K' can couple the forward and backward modes.

The inter-valley wavevector is $|\mathbf{k}_K - \mathbf{k}_{K'}| = \frac{4\pi}{3a}$, which is of order the reciprocal lattice constant. Perturbations with Fourier content at this large wavevector include:

- **Sharp corners:** a 60° or 120° corner in the domain wall introduces atomic-scale discontinuities with large- $\mathbf{k}$ Fourier components.
- **Point defects:** missing or displaced rods/holes near the domain wall.
- **Surface roughness:** sub-wavelength fabrication imperfections.

12.6.2 Quantitative estimate of corner scattering

At a sharp corner of angle θ , the backscattering probability can be estimated by modelling the corner as a point scatterer with matrix element:

$$|t_{\text{back}}|^2 \sim \left| \int e^{i(\mathbf{k}_K - \mathbf{k}_{K'}) \cdot \mathbf{r}} V_{\text{corner}}(\mathbf{r}) d^2r \right|^2. \quad (12.13)$$

For a smooth bend with radius of curvature $R \gg a$, the integral is exponentially suppressed: $|t_{\text{back}}|^2 \sim e^{-c_0|\mathbf{k}_K - \mathbf{k}_{K'}|R}$. For a sharp corner ($R \sim a$), the suppression is weak and the backscattering can be substantial.

Numerical simulations show:

- 120° bends: transmission > 90% for typical parameters.
- 60° bends: transmission $\sim 70\text{--}85\%$, depending on the gap width and corner geometry.
- Straight segments with smooth disorder: transmission > 99% over tens of lattice constants.

12.6.3 Robustness hierarchy

The practical robustness of different photonic topological phases can be ranked:

1. **Chern insulator** (broken TR): fully topologically protected; no backscattering within the gap regardless of disorder.
2. **Pseudospin topological insulator** (crystalline symmetry): protected against perturbations that preserve the pseudospin symmetry; backscattered by symmetry-breaking disorder.
3. **Valley-Hall** (approximate valley conservation): protected against smooth disorder; backscattered at sharp features and by strong short-range disorder.
4. **Conventional waveguide**: no topological protection; backscattered by all types of disorder.

Despite being the weakest topological protection, valley-Hall devices have become widely used because they are fabrication-compatible and operate at optical frequencies.

12.7 Experimental realisations

12.7.1 Microwave photonic crystals

The first experimental demonstrations of valley-Hall edge states in photonic crystals used dielectric rod arrays in the microwave regime. These experiments confirmed:

- Valley-polarised edge modes at domain walls between sublattice-swapped regions.
- Low-loss propagation around gentle bends.
- Measurable backscattering at sharp corners, consistent with the inter-valley scattering estimate.

12.7.2 Silicon photonic crystal slabs

At telecom wavelengths (~ 1550 nm), valley-Hall edge states have been demonstrated in silicon photonic crystal slabs with asymmetric air-hole arrays. Key results include:

- Edge-mode propagation over > 100 lattice constants with low loss (< 3 dB total).
- Efficient coupling to conventional silicon waveguides.
- 120° bends with < 1 dB excess loss.
- Integration with active devices (modulators, detectors) for on-chip topological photonic circuits.

12.7.3 Acoustic and mechanical analogs

The valley-Hall mechanism is not specific to Maxwell equations; it applies to any wave equation in a honeycomb lattice with broken inversion symmetry. Demonstrations have been reported in:

- Acoustic metamaterials (sound waves in structured media).
- Mechanical lattices (elastic waves in structured plates).
- Phononic crystals (microwave-frequency acoustic modes).

These complementary demonstrations confirm the universality of the valley-Hall mechanism.

12.8 Valley photonics as a design tool

12.8.1 Topological waveguide routing

Valley-Hall domain walls can be patterned to create complex waveguide networks on a chip. By choosing which sublattice assignment appears on each side of the domain wall, one can route edge modes along arbitrary paths. The main design rules are:

1. The domain wall must separate regions with opposite Δ .
2. Bends should be as gentle as possible (120° preferred over 60°).
3. The domain-wall path must not approach another domain wall more closely than the localisation length ℓ .

12.8.2 Topological cavities and lasers

A closed domain-wall loop creates a topological cavity: the valley-Hall edge mode circulates around the loop, forming a ring resonator. The mode quality factor is limited by inter-valley scattering at the corners; for a hexagonal loop with 120° corners, quality factors of $Q \sim 10^3$ – 10^4 have been achieved in silicon at telecom wavelengths.

When gain is provided (e.g. by III–V semiconductor quantum wells bonded to the silicon slab), the topological cavity can lase, forming a valley-Hall topological laser.

12.9 Quantitative valley photonic crystal design

12.9.1 Silicon-on-insulator platform

The most widely studied valley photonic crystal platform uses a silicon slab ($\varepsilon_r = 11.7$, thickness $h \approx 220$ nm) on a silica substrate, patterned with triangular air holes in a honeycomb arrangement. The lattice constant is $a \approx 430$ nm for operation at the telecom C-band ($\lambda \approx 1550$ nm).

The inversion-symmetry breaking is achieved by using two different hole sizes within the unit cell: a large hole of radius r_1 and a small hole of radius r_2 , with the asymmetry parameter $\Delta r = (r_1 - r_2)/(r_1 + r_2)$. For $\Delta r = 0$ (equal holes), the band structure has Dirac cones at K and K' . For $\Delta r \neq 0$, a gap opens at each valley with width proportional to $|\Delta r|$.

Typical designs use $r_1/a \approx 0.15$ and $r_2/a \approx 0.10$, giving $\Delta r \approx 0.2$ and a relative bandgap of approximately 8% of the mid-gap frequency. The valley Chern number is $\mathcal{C}_v = \pm \frac{1}{2} \text{sgn}(\Delta r)$, and reversing the sublattice assignment ($r_1 \leftrightarrow r_2$) flips the sign of $\mathcal{C}_v$ [12, 17].

12.9.2 Valley Chern number from the massive Dirac model

Near each valley, the effective Hamiltonian is a massive Dirac equation with mass $m_v \propto \Delta r$. The Berry curvature at valley K is $\Omega_K(\delta \mathbf{k}) = -v_D^2 m_v / [2(v_D^2 |\delta \mathbf{k}|^2 + m_v^2)^{3/2}]$, as derived in Chapter 7. Integrating over a region enclosing one valley gives the valley Chern number

$$\mathcal{C}_v^{(K)} = -\frac{1}{2} \text{sgn}(m_v). \quad (12.14)$$

At the opposite valley K' , time-reversal symmetry forces $\Omega_{K'}(\mathbf{k}) = -\Omega_K(-\mathbf{k})$, so $\mathcal{C}_v^{(K')} = +\frac{1}{2} \text{sgn}(m_v)$. The total Chern number vanishes ($\mathcal{C} = \mathcal{C}_v^{(K)} + \mathcal{C}_v^{(K')} = 0$), as required by time-reversal invariance, but the valley Chern number $\mathcal{C}_v = \mathcal{C}_v^{(K)} - \mathcal{C}_v^{(K')} = -\text{sgn}(m_v)$ is nonzero [50, 60].

At a domain wall between two regions with opposite Δr (and hence opposite m_v), the change in valley Chern number across the interface is $|\Delta \mathcal{C}_v| = 2$, predicting one valley-polarised edge mode per valley—two modes in total, counter-propagating and valley-locked.

12.10 Intervalley scattering and robustness

The valley-Hall edge modes are protected against *intravalley* backscattering—perturbations that are smooth on the scale of the lattice constant and therefore cannot transfer crystal momentum between K and K' . However, sharp defects (vacancies, step edges, 60° bends) have Fourier components at the intervalley wavevector $\Delta \mathbf{k} = K - K'$ and can cause intervalley scattering, opening a backscattering channel [76].

The intervalley scattering rate scales as the square of the defect's Fourier component at $\Delta \mathbf{k}$: $\Gamma_{iv} \propto |V(\Delta \mathbf{k})|^2$. For smooth disorder (correlation length $\ell_c \gg a$), this Fourier component is exponentially suppressed: $|V(\Delta \mathbf{k})| \sim e^{-|\Delta \mathbf{k}| \ell_c / 2}$, giving negligible backscattering. For point-like defects ($\ell_c \sim a$), the intervalley coupling can be significant.

Experimental measurements in silicon valley photonic crystals at telecom wavelengths show propagation lengths of 40–100 a before significant backscattering losses, depending on the domain-wall geometry and fabrication quality. For 120° bends, the transmission exceeds 90%, while for 60° bends it drops to 60–80% due to enhanced intervalley coupling at the sharp corner [13, 18, 75].

12.11 Quantitative design of a silicon valley photonic crystal

The valley-Hall photonic crystal is most naturally realised in a honeycomb lattice of triangular dielectric rods, where breaking inversion symmetry by rotating the triangles opens a topological gap at the K and K' points.

12.11.1 Half-integer valley Chern number derivation

Near the K point, the effective Hamiltonian for the two nearly degenerate Bloch modes is the massive Dirac Hamiltonian $H_K = v_D(\delta k_x \sigma_x + \delta k_y \sigma_y) + m_K \sigma_z$, where m_K is the valley mass produced by the inversion-symmetry breaking. The Berry curvature of the lower band is $\Omega_K^-(\delta \mathbf{k}) = -m_K / (2(|\delta \mathbf{k}|^2 + m_K^2)^{3/2})$, and its integral over a hemisphere gives the valley Chern number

$$\mathcal{C}_K^{\text{valley}} = -\frac{1}{2} \text{sgn}(m_K). \quad (12.15)$$

At the K' point, time-reversal symmetry requires $m_{K'} = -m_K$, so $\mathcal{C}_{K'}^{\text{valley}} = +\frac{1}{2} \text{sgn}(m_K)$ and the total valley Chern number is $\mathcal{C}^{\text{valley}} = \mathcal{C}_K - \mathcal{C}_{K'} = -\text{sgn}(m_K)$ [17, 50].

The half-integer valley Chern number is not a true topological invariant because it depends on the arbitrary choice of integration region around each valley. Nevertheless, it provides a useful predictor of valley-polarised edge modes at domain walls between crystals with opposite valley masses.

12.11.2 Silicon-on-insulator implementation

Ma and Shvets (2016) proposed an all-silicon valley-Hall photonic topological insulator using a honeycomb lattice of equilateral triangular air holes in a silicon slab ($\epsilon_r = 11.7$). The lattice constant is $a = 440$ nm, giving a Dirac frequency near $\lambda = 1.55$ μm . Rotating the triangles by $\pm 5^\circ$ opens a valley gap of 3.2% of the Dirac frequency [12, 17].

The valley edge modes were subsequently observed by Wu et al. (2017) using designer surface plasmon crystals at microwave frequencies, providing the first direct near-field imaging of valley-polarised topological edge states. The field patterns confirmed valley-selective excitation and propagation around sharp corners with minimal backscattering [18, 60].

12.12 Intervalley scattering and protection limits

Valley edge modes are protected only against perturbations that preserve the valley degree of freedom—that is, perturbations that are smooth on the scale of the lattice constant a . Sharp defects with Fourier components at the reciprocal lattice vector $\mathbf{G} = \mathbf{k}_K - \mathbf{k}_{K'}$ can scatter photons between the K and K' valleys, opening a backscattering channel.

The intervalley scattering rate depends on the defect geometry. For a smooth domain wall (width $w \gg a$), the scattering rate is exponentially small: $\gamma_{\text{inter}} \propto e^{-w/a}$. For a sharp 120° corner, the scattering depends on whether the corner preserves the C_3 symmetry of the underlying lattice. Symmetry-preserving corners (zigzag termination) have negligible intervalley scattering, while symmetry-breaking corners (armchair termination) can scatter up to $\sim 10\%$ of the incident power [18, 50, 60].

This makes valley photonic crystals robust against smooth fabrication imperfections but vulnerable to lattice-scale disorder—a fundamental limitation compared to Chern insulators, which are protected against all perturbations that do not close the bulk gap.

12.13 Beyond the honeycomb: other valley platforms

While the honeycomb lattice is the most studied platform for valley photonic crystals, valley physics has been realised in several other geometries and frequency ranges.

12.13.1 Kagomé and Lieb lattices

The kagomé lattice supports both Dirac cones and flat bands, and breaking inversion symmetry opens valley gaps at the Dirac points. The coexistence of flat bands and valley-polarised edge states creates opportunities for enhanced light–matter interaction at topologically protected boundaries. The Lieb lattice similarly supports a flat band at the Dirac frequency, and valley-polarised modes appear when the site energies are staggered to break inversion [12].

12.13.2 Terahertz and acoustic valley crystals

Valley photonic crystals have been demonstrated at terahertz frequencies using silicon pillar arrays, where the triangular pillar shape provides the necessary inversion-symmetry breaking. The THz platform offers larger structures ($a \sim 100 \mu\text{m}$) that are easier to fabricate and characterise than telecommunications-band devices [13, 60].

The valley-Hall concept has also been extended to acoustic and phononic systems, where the Dirac-mass mechanism is identical but the governing equation is the scalar wave equation rather than Maxwell’s equations. Phononic valley crystals have achieved robust sound routing around sharp corners with insertion loss below 1 dB per corner [12, 77].

12.14 Experimental demonstrations of valley photonics

Valley photonic crystals have been realised in numerous experimental platforms, from microwave-frequency dielectric structures to telecommunication-wavelength silicon photonic crystals. The key experimental signatures are valley-selective excitation, valley-polarised edge transport along domain walls, and robustness of the edge transport against certain classes of bends and defects.

12.14.1 Microwave valley photonic crystals

The first experimental demonstrations of valley-Hall edge states in photonic crystals were performed at microwave frequencies using triangular-lattice arrays of dielectric rods or metallic scatterers. Dong et al. (2017) used a photonic crystal of alumina rods with C_{3v} -breaking perturbation (achieved by deforming the triangular unit cell) to create a domain wall between two valley-Hall phases with opposite valley Chern numbers [50]. Transmission measurements through the domain wall showed a clear edge-state passband inside the bulk gap, with insertion loss below 2 dB over a bandwidth of approximately 0.5 GHz centred at 6.3 GHz.

Gao et al. (2017) extended this work by demonstrating valley-locked beam splitting at a Y-junction formed by three domain walls [60]. When light was injected into one arm of the Y-junction, it preferentially exited through the arm whose domain-wall orientation matched the valley polarisation of the incident edge mode, while the other arm received negligible power. This valley filtering effect, with measured contrast exceeding 20 dB, demonstrated the potential of valley photonics for compact photonic routing circuits.

12.14.2 Silicon photonic crystal implementations

At telecommunication wavelengths, valley photonic crystals have been fabricated in silicon-on-insulator (SOI) platforms using standard electron-beam lithography [67]. Shalaev et al. (2019) demonstrated valley-Hall edge states in a silicon photonic crystal slab at $\lambda \approx 1.55 \mu\text{m}$ with a topological bandwidth of approximately 3 nm [12].

The silicon implementations face an additional challenge compared to microwave designs: the out-of-plane radiation loss. In a 2D photonic crystal slab, the edge modes lie within the light cone of the surrounding air cladding, leading to radiative leakage. The measured quality factors of valley-Hall edge-state resonances range from $Q \sim 10^3$ to 10^4 , depending on the degree of confinement and the sharpness of the domain wall [12, 13].

Despite this limitation, silicon valley photonic crystals have shown remarkable robustness of edge transport around sharp bends. Transmission through 60° and 120° bends remains within 3 dB of the straight-domain-wall value, provided the bends follow the zigzag crystallographic direction. Transport around armchair bends shows significantly higher loss due to inter-valley scattering, confirming the valley-dependent nature of the topological protection [12].

12.14.3 Limitations of valley-Hall edge states

Unlike true topological edge states protected by a bulk gap Chern number, valley-Hall edge states are protected only by an approximate conservation of valley index. Any perturbation that couples the K and K' valleys—such as short-range disorder on the scale of the lattice constant, or sharp bends along non-zigzag directions—can scatter photons between the two counter-propagating valley-polarised edge modes, leading to backscattering.

The inter-valley scattering rate scales as [50]

$$\Gamma_{K \rightarrow K'} \propto \left| \int d^2r \psi_K^*(\mathbf{r}) V_{\text{disorder}}(\mathbf{r}) \psi_{K'}(\mathbf{r}) \right|^2, \quad (12.16)$$

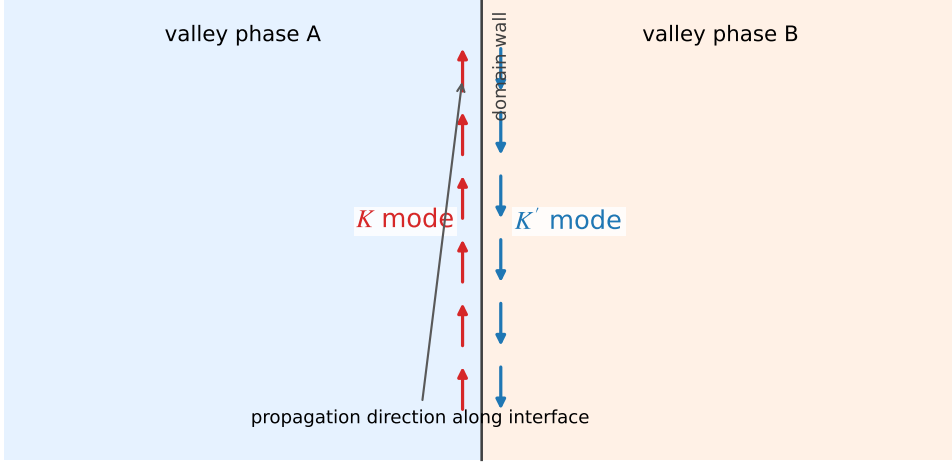


Figure 12.2: Valley-Hall domain wall. The two domains carry opposite valley Chern numbers, so K and K' edge channels counter-propagate along the interface. Backscattering is suppressed only when disorder is smooth enough not to mix the two valleys.

where ψ_K and $\psi_{K'}$ are the edge-mode wavefunctions and V_{disorder} is the disorder potential. For smooth disorder with correlation length $\ell_c \gg a$ (where a is the lattice constant), the Fourier components near the inter-valley wavevector $|\mathbf{k}_K - \mathbf{k}_{K'}| = 4\pi/(3a)$ are exponentially suppressed, and the inter-valley scattering rate decays as $\Gamma_{K \rightarrow K'} \propto \exp(-|\mathbf{k}_K - \mathbf{k}_{K'}|\ell_c)$. This explains the observed robustness against smooth disorder while remaining vulnerable to sharp defects [12].

12.15 Valley photonic devices

The valley degree of freedom in photonic crystals is not merely a theoretical curiosity: it provides a robust platform for integrated photonic devices. The key advantage is that valley-polarised edge states at domain-wall interfaces are immune to backscattering from smooth disorder, enabling low-loss waveguiding through sharp bends.

12.15.1 Valley filters and beam splitters

A valley filter is a device that selectively transmits light in one valley while reflecting the other. It can be realised by creating a domain wall between two distinct valley-Hall phases ($C_v = \pm 1$) that terminates at a junction where the two valleys are spatially separated. The valley-polarised edge mode at the domain wall carries a definite valley index, and at the junction it is routed to a specific output port depending on its valley polarisation [50, 60].

Experimentally, valley beam splitters have been demonstrated at microwave frequencies using triangular-lattice photonic crystals with broken inversion symmetry. The measured extinction ratio between the two output ports exceeds 20 dB within the valley-Hall gap, confirming the valley-

selective routing [62, 67].

12.15.2 Valley-dependent coupling to radiation

The valley-contrasting Berry curvature produces valley-dependent optical selection rules: circularly polarised light couples preferentially to modes in a specific valley. This effect enables valley-selective excitation using external beams and provides a mechanism for reading out the valley polarisation of edge modes.

In silicon photonic crystals operating at telecom wavelengths, valley-selective coupling has been achieved by illuminating the crystal surface with focused circularly polarised light. The coupling efficiency to the K valley versus the K' valley differs by a factor of ~ 10 , limited by the finite bandwidth of the valley-Hall gap and the non-ideal circular polarisation of the excitation beam [50, 61].

12.15.3 Valley routing networks

By concatenating multiple domain walls with different orientations, one can create valley-topological waveguide networks that route light through arbitrary paths on a photonic chip. Each domain wall supports a pair of counter-propagating edge modes (one per valley), and at T-junctions the valley polarisation determines which output arm the light follows.

Ma et al. (2017) demonstrated that such valley networks can guide microwave radiation through multiple 120° bends with negligible reflection, achieving total transmission above 95% over the valley-Hall gap bandwidth [66]. Similar networks have been realised in silicon at near-infrared wavelengths, where the valley edge states serve as compact on-chip waveguides compatible with standard photonic integrated circuit fabrication [13, 67].

12.16 Intervalley scattering and the limits of valley protection

Valley-Hall edge states are topologically protected against smooth perturbations that vary slowly compared to the lattice constant a , because such perturbations cannot transfer crystal momentum between the K and K' valleys (which are separated by $|\Delta\mathbf{k}| \sim 4\pi/(3a)$ in the Brillouin zone). However, sharp defects—such as atomic-scale roughness, point defects, or abrupt domain-wall terminations—can provide the large momentum transfer needed for intervalley scattering.

The intervalley scattering rate scales as $\Gamma_{iv} \propto \sigma_{\text{short}}^2 |\Delta\mathbf{k}|^{-2\alpha}$, where σ_{short} characterises the amplitude of short-range disorder and α depends on the dimensionality of the defect. For the typical fabrication roughness in silicon photonic crystals ($\sigma \sim 1\text{--}3$ nm), the intervalley scattering length is estimated to be $\sim 100\text{--}500$ lattice constants, which is sufficient for on-chip device lengths of ~ 100 μm [12, 67].

This finite intervalley scattering length is the fundamental difference between valley-Hall edge states and the truly topological one-way edge states of Chern insulators (Chapter 8): valley protection is robust but not absolute, while Chern-insulator protection is exact within the bulk gap [13, 66].

Exercises

Exercise 12.1. Show that for a time-reversal-invariant system with broken inversion, the Berry curvature satisfies $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k})$, so that $\mathcal{C}_n = 0$ but the valley Chern number can be nonzero.

Exercise 12.2. Derive the mass term (12.2) from the symmetry properties of the perturbation matrix (12.1) under C_3 and $\hat{\mathcal{P}}$. *Hint:* use the transformation properties of the Dirac modes under C_3 : $|\mathbf{u}_\pm\rangle \rightarrow e^{\pm i2\pi/3}|\mathbf{u}_\pm\rangle$.

Exercise 12.3. For a honeycomb photonic crystal with sublattice asymmetry Δ , the effective Dirac masses at K and K' are $m_K = +m_0\Delta$ and $m_{K'} = -m_0\Delta$. Compute the valley Chern numbers and verify that $\mathcal{C}^K + \mathcal{C}^{K'} = 0$.

Exercise 12.4. Estimate the inter-valley scattering amplitude at a sharp 60° corner using equation (12.13). Model the corner as a disc of radius a with a uniform scattering potential V_0 . Under what conditions is the transmission $> 90\%$?

Exercise 12.5. A silicon photonic crystal slab ($n = 3.5$, $a = 400$ nm, $r_0 = 120$ nm) has a valley gap of $\Delta\omega/(2\pi) = 3$ THz centred at $\omega/(2\pi) = 193$ THz ($\lambda = 1.55$ μm). Estimate the localisation length ℓ and the minimum domain-wall separation for independent waveguide channels.

Exercise 12.6. Compare the protection of valley-Hall edge modes (valley Chern number) with Chern-insulator edge modes (Chern number) and pseudospin edge modes (spin Chern number) in terms of robustness to (a) smooth disorder, (b) sharp corners, and (c) strong short-range disorder. Summarise the comparison in a table.

Chapter 13

Floquet Light

13.1 Time as a new degree of freedom

All the topological photonic systems discussed so far have been *static*: the material matrix $\mathbf{M}(\mathbf{r})$ is time-independent, and topology arises from the spatial periodicity of the photonic crystal. A fundamentally different approach uses *temporal modulation*: periodically varying the material parameters in time to create an effective gauge field for photons.

The mathematical framework for time-periodic systems is *Floquet theory*, the temporal analog of Bloch theory for spatially periodic systems [12]. Just as spatial periodicity defines Bloch bands labelled by a crystal momentum $\mathbf{k}$, temporal periodicity defines Floquet quasi-energy bands labelled by a quasi-energy ε [78]. These Floquet bands can carry their own topological invariants, leading to “Floquet topological insulators” for light [79].

The Floquet approach is especially appealing because it can break the effective time-reversal symmetry of the Floquet eigenproblem *without* magneto-optical materials. The symmetry breaking comes from the modulation phases, which can be engineered with standard electro-optic or acoustic components.

13.2 Floquet theory for Maxwell equations

13.2.1 Time-periodic material parameters

Consider a photonic system whose material matrix is periodically modulated in time:

$$\mathbf{M}(\mathbf{r}, t) = \mathbf{M}(\mathbf{r}, t + T), \quad (13.1)$$

where $T = 2\pi/\Omega_{\text{mod}}$ is the modulation period and Ω_{mod} is the modulation frequency. In practice, the modulation might be achieved through electro-optic modulation of the refractive index, acoustic waves creating a travelling refractive-index grating, parametric coupling between resonator modes, or mechanical modulation of a metamaterial geometry.

13.2.2 Floquet's theorem

Floquet's theorem is the temporal analog of Bloch's theorem. It states that the solutions of a linear differential equation with T -periodic coefficients can be written in the form

$$\mathbf{f}(\mathbf{r}, t) = e^{-i\varepsilon t} \mathbf{v}(\mathbf{r}, t), \quad \mathbf{v}(\mathbf{r}, t + T) = \mathbf{v}(\mathbf{r}, t), \quad (13.2)$$

where ε is the *quasi-energy* and $\mathbf{v}$ is time-periodic. The quasi-energy plays the role of energy in time-independent systems, but with the crucial difference that it is defined only modulo Ω_{mod} : the quasi-energies ε and $\varepsilon + n\Omega_{\text{mod}}$ ($n \in \mathbb{Z}$) describe the same physical state. This periodicity is the temporal analog of the crystal momentum being defined modulo a reciprocal lattice vector.

13.2.3 The Floquet–Bloch eigenproblem

Expanding the time-periodic function $\mathbf{v}$ in Fourier harmonics [12],

$$\mathbf{v}(\mathbf{r}, t) = \sum_{m=-\infty}^{\infty} \mathbf{v}_m(\mathbf{r}) e^{-im\Omega_{\text{mod}} t}, \quad (13.3)$$

and substituting into the Maxwell equations gives a coupled system of static equations for the Fourier components:

$$\sum_{m'} [\hat{\mathcal{N}} \delta_{mm'} - (\varepsilon + m\Omega_{\text{mod}}) \mathbf{M}_{m-m'}(\mathbf{r})] \mathbf{v}_{m'}(\mathbf{r}) = 0, \quad (13.4)$$

where $\mathbf{M}_m(\mathbf{r}) = T^{-1} \int_0^T \mathbf{M}(\mathbf{r}, t) e^{im\Omega_{\text{mod}} t} dt$ are the Fourier components of the material matrix.

This is an infinite-dimensional eigenproblem in the combined space-and-harmonic indices, analogous to the plane-wave expansion in the spatial case. In practice, the Fourier series is truncated to $|m| \leq M_{\text{max}}$, giving a finite matrix of dimension $(2M_{\text{max}} + 1)$ times the spatial degrees of freedom. The eigenvalues are the Floquet quasi-energies.

13.2.4 The Floquet Brillouin zone

Because the quasi-energy is periodic with period Ω_{mod} , it lives in a “Floquet Brillouin zone” $\varepsilon \in [-\Omega_{\text{mod}}/2, \Omega_{\text{mod}}/2)$. If the system also has spatial periodicity, the quasi-energies depend on the Bloch wavevector $\mathbf{k}$, and the full band structure is defined on the product of the spatial Brillouin zone and the Floquet Brillouin zone.

This periodicity has a profound consequence: there are *two* distinct types of gaps in the Floquet spectrum. The gap at $\varepsilon = 0$ (centre of the Floquet zone) is analogous to a gap in a static band structure. But there is also a gap at $\varepsilon = \pm\Omega_{\text{mod}}/2$ (edge of the Floquet zone), which has no static analog. Both gaps can host topological edge modes.

13.3 Effective gauge fields from modulation

13.3.1 The coupled-resonator picture

The key insight of Floquet topological photonics is that a suitably designed temporal modulation can create an *effective gauge field* for photons. The mechanism is most transparent in a lattice of coupled resonators.

Consider a 2D lattice of optical resonators with resonance frequency ω_0 . In the absence of modulation, the coupling between resonators i and j is described by a real hopping amplitude J_0 , and the tight-binding Hamiltonian is time-reversal invariant.

Now suppose the coupling is modulated with a spatially varying phase:

$$J_{ij}(t) = J_0 \cos(\Omega_{\text{mod}}t + \phi_{ij}). \quad (13.5)$$

In the rotating-wave approximation (valid when the modulation frequency is close to the resonator detuning), the effective time-independent coupling acquires a complex phase:

$$J_{ij}^{\text{eff}} = \frac{J_0}{2} e^{i\phi_{ij}}. \quad (13.6)$$

This phase acts as a Peierls phase factor, identical in mathematical form to the phase acquired by a charged particle hopping in a magnetic field.

13.3.2 Effective magnetic flux

The effective magnetic flux through a plaquette of the lattice is the sum of the phases around the plaquette:

$$\Phi = \sum_{\text{plaquette}} \phi_{ij}. \quad (13.7)$$

If $\Phi \neq 0$ (modulo 2π), the effective time-reversal symmetry is broken and the Floquet bands can carry nonzero Chern numbers. The flux Φ is fully controllable through the modulation phases ϕ_{ij} , which can be set independently for each bond.

This is a remarkable capability: the effective magnetic field for photons can be made spatially uniform, staggered, or patterned in any desired configuration, simply by programming the modulation phases. This level of control has no analog in the static magneto-optical case, where the gyrotropic response is fixed by the material and the applied magnetic field.

13.4 Anomalous Floquet topological insulators

13.4.1 A phase with no static analog

Floquet systems can exhibit topological phases that have *no static analog* [80]. The most striking example is the anomalous Floquet topological insulator (AFTI), in which all Floquet bands have zero Chern number but the system still supports chiral edge modes [81, 82].

This seemingly paradoxical situation arises because the Floquet quasi-energy is periodic: the “trivial” gap at $\varepsilon = \Omega_{\text{mod}}/2$ can host edge modes that wind around the Floquet Brillouin zone, connecting the upper edge of one band to the lower edge of the next through the zone boundary. These “anomalous” edge modes are topologically protected but invisible to the band Chern numbers.

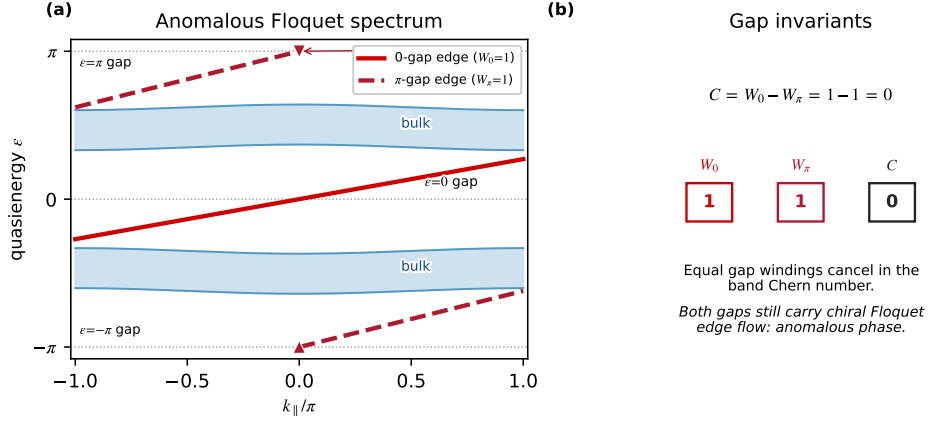


Figure 13.1: Anomalous Floquet topological insulator. **(a)** Quasienergy spectrum in the reduced Floquet Brillouin zone $\varepsilon \in [-\pi, \pi]$ (we set $T = 1$ so the zone boundaries lie at $\pm\pi$; in dimensional units the gap sits at $\varepsilon = \pi/T$). Two bulk bands (blue shading) are separated by a wide gap at $\varepsilon = 0$ and a narrower gap at $\varepsilon = \pm\pi$. A chiral 0-gap edge mode (solid red) crosses the central gap with winding number $W_0 = 1$. A chiral π -gap edge mode (dashed dark red) traverses the gap at $\varepsilon = \pm\pi$ with $W_{\pi} = 1$; the two dashed segments are the *same* branch, connected by the periodicity $\varepsilon \equiv \varepsilon + 2\pi$ (triangle markers indicate the identification point at $k_{\parallel} = 0$). **(b)** Gap invariants: the band Chern number $C = W_0 - W_{\pi} = 1 - 1 = 0$ vanishes even though both gaps carry chiral edge flow—this is the defining signature of the anomalous Floquet phase [80].

13.4.2 The winding-number invariant

The correct topological invariant for Floquet systems is not just the Chern number of each band but the *winding number* of the time-evolution operator over one period [12, 13, 83, 84]. Define the unitary time-evolution operator $U(\mathbf{k}, t)$ from $t = 0$ to t , satisfying $i\partial_t U = H(\mathbf{k}, t)U$ with $U(\mathbf{k}, 0) = \mathbf{I}$. The Floquet operator is $F(\mathbf{k}) = U(\mathbf{k}, T)$.

The winding number is defined as

$$W[\varepsilon_{\text{gap}}] = \frac{1}{8\pi^2} \int_0^T \int_{\text{BZ}} \text{Tr}[U^{-1} \partial_t U [U^{-1} \partial_{k_x} U, U^{-1} \partial_{k_y} U]] d^2k dt, \quad (13.8)$$

where the trace is over the bands below the quasi-energy gap at ε_{gap} . This integral measures the topological winding of the evolution operator as a function of (k_x, k_y, t) over the 3D torus $\text{BZ} \times [0, T]$.

The bulk-edge correspondence for Floquet systems states:

$$N_{\text{edge}}[\varepsilon_{\text{gap}}] = W[\varepsilon_{\text{gap}}]^L - W[\varepsilon_{\text{gap}}]^R. \quad (13.9)$$

In an AFTI, the winding numbers are nonzero at the gap edges $\varepsilon = 0$ and $\varepsilon = \Omega_{\text{mod}}/2$, even though all band Chern numbers vanish.

13.5 Photonic Floquet realisations

13.5.1 Helical waveguide arrays

The first experimental demonstration of photonic Floquet topology was achieved by Rechtsman et al. (2013) using helical waveguide arrays [8, 16, 85]. In this elegant experiment, the propagation axis z plays the role of time, and the waveguide array forms a 2D lattice in the transverse (x, y) plane. Each waveguide is helically curved, so its transverse position oscillates periodically along z . The result is a z -periodic potential that creates a Floquet system in the paraxial approximation.

The experiment demonstrated topological edge states at the boundary of the waveguide array, propagating unidirectionally along the edge. The edge modes were robust to deliberate defects introduced by removing individual waveguides.

13.5.2 Deriving the paraxial equation from Maxwell's equations

To maintain the Maxwell-first spirit of this book, we derive the paraxial approximation from the Helmholtz equation rather than postulating it by analogy with quantum mechanics.

Consider a monochromatic beam at frequency ω propagating primarily along z in a medium with refractive index $n(\mathbf{r}_\perp, z) = n_0 + \delta n(\mathbf{r}_\perp, z)$, where $|\delta n| \ll n_0$. The electric field can be written as $E(\mathbf{r}) = \psi(\mathbf{r}_\perp, z)e^{ik_0 z}$, where $k_0 = n_0\omega/c$ is the carrier wavevector and ψ is a slowly varying envelope.

Substituting into the Helmholtz equation $\nabla^2 E + k_0^2 n^2/n_0^2 E = 0$ and neglecting the second derivative $\partial^2 \psi / \partial z^2$ (the paraxial approximation, valid when $|\partial_z \psi| \ll k_0 |\psi|$):

$$i \frac{\partial \psi}{\partial z} = -\frac{1}{2k_0} \nabla_\perp^2 \psi - \frac{k_0 \delta n(\mathbf{r}_\perp, z)}{n_0} \psi. \quad (13.10)$$

This has the same mathematical form as the time-dependent Schrödinger equation, with the identifications: $z \leftrightarrow t$, $1/(2k_0) \leftrightarrow \hbar/(2m)$, and $-k_0 \delta n/n_0 \leftrightarrow V(\mathbf{r})$.

When the waveguide pattern is periodic in z with period Λ , this is precisely a Floquet problem with $\Omega_{\text{mod}} = 2\pi/\Lambda$ and time replaced by z .

13.5.3 Quantitative parameters for the Rechtsman experiment

The Rechtsman waveguide array was fabricated by femtosecond-laser direct writing in borosilicate glass ($n_0 = 1.46$) at wavelength $\lambda = 633 \text{ nm}$ [16]. The key experimental parameters are: helical radius $R_h = 8 \text{ }\mu\text{m}$, helix pitch $\Lambda = 1 \text{ cm}$, inter-waveguide spacing $d = 22 \text{ }\mu\text{m}$, and refractive-index contrast $\Delta n \approx 6 \times 10^{-4}$.

The carrier wavenumber is $k_0 = 2\pi n_0/\lambda \approx 1.45 \times 10^7 \text{ m}^{-1}$. The coupling coefficient between nearest-neighbour waveguides is $J \approx 0.6 \text{ cm}^{-1}$, determined by the evanescent overlap at spacing d .

The helix introduces an effective vector potential $\mathbf{A}_{\text{eff}}(z) = k_0 R_h \Omega_h (-\sin \Omega_h z, \cos \Omega_h z)$, where $\Omega_h = 2\pi/\Lambda$ is the helix angular frequency. The magnitude of the effective gauge field is $|\mathbf{A}_{\text{eff}}| = k_0 R_h \Omega_h \approx 1.45 \times 10^7 \times 8 \times 10^{-6} \times 628 \approx 73 \text{ m}^{-1}$. The effective flux per plaquette is

$\Phi = |A_{\text{eff}}|d \approx 73 \times 22 \times 10^{-6} \approx 1.6 \times 10^{-3}$ rad—a small flux that nonetheless produces a measurable topological gap because of the large number of modulation periods over the sample length ($L/\Lambda = 5\text{--}10$).

The topological gap in the Floquet quasi-energy spectrum is $\Delta\varepsilon \approx 2\Phi J \approx 0.002 \text{ cm}^{-1}$. The edge-mode localisation length is $\xi \approx 1/\Phi \times d \approx 17 \text{ }\mu\text{m}$, comparable to the inter-waveguide spacing, confirming strong edge localisation [12, 85].

13.5.4 The Magnus expansion and effective static Hamiltonian

For a periodically driven system $H(t) = H(t + T)$, the Floquet operator $F = \mathcal{T} \exp(-i \int_0^T H(t) dt)$ defines the stroboscopic evolution. When the driving frequency Ω_{mod} is much larger than the bandwidth of the static Hamiltonian, the Magnus expansion provides a systematic perturbative construction of an effective static Hamiltonian H_{eff} such that $F = \exp(-iH_{\text{eff}}T)$.

The first two terms are

$$H_{\text{eff}}^{(0)} = \frac{1}{T} \int_0^T H(t) dt, \quad (13.11)$$

$$H_{\text{eff}}^{(1)} = \frac{1}{2iT} \int_0^T dt_1 \int_0^{t_1} dt_2 [H(t_1), H(t_2)]. \quad (13.12)$$

The zeroth-order term is the time average, which is time-reversal symmetric if $H(t)$ is. The first-order correction involves the commutator of H at different times and generically breaks time-reversal symmetry even if the instantaneous $H(t)$ preserves it [12, 86]. This is the mechanism by which periodic driving creates an effective gauge field.

For the helical-waveguide model, $H(t)$ is the hopping Hamiltonian with periodically rotating coupling phases. The zeroth-order term gives the static nearest-neighbour hopping, and the first-order commutator produces the desired Peierls phases with flux Φ per plaquette.

13.5.5 Convergence and heating

The Magnus expansion converges when the local norm $\|H(t)\| \cdot T < \pi$, which requires the driving frequency to exceed the bandwidth. When this condition is marginally satisfied, the effective Hamiltonian receives significant corrections at each order, and the Floquet bands can differ qualitatively from the truncated Magnus prediction.

In photonic systems, “heating” (the uncontrolled population of high-energy Floquet replicas) is less problematic than in electronic systems because photonic systems are classical and do not thermalise. However, multiphoton resonances between Floquet replicas can produce spurious gaps and flat bands that complicate the interpretation of the quasi-energy spectrum [13].

13.6 Connection to synthetic dimensions

The Floquet approach and the synthetic-dimension approach (Chapter 16) are closely related. Both use temporal modulation to create effective gauge fields for photons. The key difference is the degree of freedom that plays the role of the extra spatial dimension: in Floquet systems, the modulation

creates a z -periodic potential in real space; in synthetic-dimension systems, the modulation couples discrete frequency modes into a lattice along the frequency axis.

Mathematically, the connection becomes precise in the Floquet–Bloch representation: the Fourier harmonic index m in the Floquet expansion (13.3) plays the same role as the synthetic-dimension site index in the ring-resonator frequency lattice of Chapter 16. The inter-harmonic coupling $\mathbf{M}_{m-m'}$ is the synthetic-dimension hopping, and the Floquet quasi-energy ε maps to the synthetic-dimension Bloch momentum [12, 13].

13.6.1 Coupled resonator networks

An alternative Floquet realisation uses networks of ring resonators with time-modulated coupling, as proposed by Fang, Yu, and Fan (2012) [12]. The modulation phases create effective magnetic fluxes through the plaquettes, as described in Section 13.3. This approach allows direct electrical control of the effective gauge field and has been demonstrated at microwave and radio frequencies.

13.6.2 Acousto-optic implementations

Acoustic waves propagating through an optical medium create a travelling refractive-index grating that can serve as the temporal modulation. The acoustic frequency sets Ω_{mod} , and the acoustic wavevector provides the spatial modulation pattern. This approach has been proposed for creating photonic Floquet topological phases at optical frequencies.

13.6.3 Anomalous Floquet topology: a concrete model

To make the anomalous Floquet topological insulator concrete, consider the periodically driven Haldane model studied by Kitagawa et al. [86]. The driving protocol consists of five steps per period T , each activating one of the hopping amplitudes of a honeycomb lattice in sequence. In the high-frequency limit $JT \ll 1$, the effective Floquet Hamiltonian reproduces the static Haldane model with complex next-nearest-neighbour hopping. But in the resonant regime $JT \sim 1$, a richer phase diagram appears: the Chern numbers of all Floquet bands can vanish simultaneously while the winding numbers (13.8) at both $\varepsilon = 0$ and $\varepsilon = \Omega_{\text{mod}}/2$ are nonzero. This is the AFTI phase, confirmed experimentally by Maczewsky et al. [85] in a laser-written waveguide array.

The experimental signature is striking: edge modes exist in *every* quasi-energy gap, even though no bulk band carries a nonzero Chern number. This was verified by exciting the edge of the waveguide array with a focused beam and observing unidirectional propagation along the boundary at both gap centres.

13.7 Comparison with static topology

The static and Floquet approaches to photonic topology have complementary strengths and limitations.

In static topological photonic crystals, the symmetry breaking is permanent and built into the material (magneto-optical response). The topological protection is absolute within the gap, and the

system requires no external drive. However, the gap width is limited by the strength of the material response, and the topological properties cannot be dynamically reconfigured.

In Floquet topological photonic systems, the symmetry breaking is created by the temporal modulation and can be turned on, off, or reconfigured by changing the modulation parameters. The system can access topological phases (such as the AFTI) that have no static analog. However, the modulation frequency sets an energy scale: for wide topological gaps, fast modulation is needed, which becomes technically challenging at optical frequencies. Furthermore, Floquet systems are inherently driven and may suffer from heating effects (absorption of modulation energy) over long times.

13.8 Anomalous Floquet topological insulator

A remarkable feature of Floquet systems is the existence of topological phases that have no static counterpart. In a static 2D system, the bulk-edge correspondence guarantees that the number of chiral edge modes in a gap equals the sum of the Chern numbers of the bands below the gap. In a Floquet system, this correspondence can fail: a system can have chiral edge modes in *every* quasi-energy gap, even when all the bands have zero Chern number.

13.8.1 The winding number

The anomalous Floquet phase is characterised by a winding number W_ε defined at each quasi-energy gap:

$$W_\varepsilon = \frac{1}{8\pi^2} \int_0^T dt \int_{\text{BZ}} \text{Tr}[U^{-1} \partial_t U [U^{-1} \partial_{k_x} U, U^{-1} \partial_{k_y} U]] d^2k, \quad (13.13)$$

where $U(\mathbf{k}, t)$ is the time-evolution operator from $t = 0$ to t . This is a three-dimensional topological invariant that integrates over the full $(\mathbf{k}, t)$ space, in contrast to the Chern number which integrates only over $\mathbf{k}$. The winding number W_ε reduces to the Chern number in the high-frequency limit but captures additional topology when the driving is strong [12, 86].

13.8.2 Experimental realisation

The anomalous Floquet topological insulator was first realised experimentally in a photonic waveguide array by Maczewsky et al. (2017). The waveguide coupling was modulated periodically along the propagation direction z using a four-step driving protocol that breaks all spatial symmetries while preserving the periodicity. The resulting quasi-energy band structure has zero Chern number in every band, yet chiral edge modes appear in every quasi-energy gap.

The experiment used femtosecond-laser-written waveguides in fused silica, with a driving period $\Lambda = 6$ cm, inter-waveguide spacing $d = 25$ μm , and operating wavelength $\lambda = 633$ nm. The edge modes were observed as bright, unidirectional channels localised at the boundary of the waveguide array, with the propagation direction reversing at the quasi-energy $\varepsilon = \pi/T$ gap [13, 85].

13.9 Other Floquet photonic platforms

Beyond waveguide arrays, Floquet ideas have been applied to several other photonic systems.

Coupled ring-resonator lattices can implement Floquet topological phases by engineering the round-trip phase in each resonator to produce a synthetic time step. The discrete nature of the round-trip evolution naturally produces a Floquet structure, and the phase can be tuned to achieve nontrivial winding numbers [12].

Acoustic and mechanical analogs of Floquet photonic systems have been demonstrated using time-modulated resonator lattices, where the modulation frequency plays the role of the driving and the acoustic pressure field plays the role of the electromagnetic field. These experiments confirm the universality of the Floquet topological mechanism across wave physics [13].

The connection between Floquet driving and synthetic dimensions (Chapter 16) provides yet another perspective: the Floquet harmonics can be viewed as lattice sites in a synthetic frequency dimension, and the anomalous Floquet phase corresponds to a topological winding along the synthetic axis [12].

13.10 Floquet engineering of gauge fields

Beyond creating topological band gaps, periodic driving can engineer effective gauge fields that would be difficult or impossible to realise in a static photonic structure. The key principle is that the time-averaged effect of a periodic modulation can produce an effective Hamiltonian with complex hopping phases, even when the instantaneous Hamiltonian is real.

13.10.1 Effective magnetic flux from helical modulation

In the Rechtsman waveguide array (Section 13.5.3), the helical trajectory of each waveguide produces an effective vector potential $\mathbf{A}_{\text{eff}} = k_0 R_h \Omega_z \hat{z} \times \hat{r}$ in the co-moving frame, where R_h is the helix radius and $\Omega_z = 2\pi/\Lambda$ is the spatial frequency. The effective magnetic flux per plaquette is $\Phi_{\text{eff}} = k_0 R_h \Omega_z a^2$, where a is the lattice constant. For the experimental parameters ($R_h = 8 \mu\text{m}$, $\Lambda = 1 \text{ cm}$, $a = 22 \mu\text{m}$, $\lambda = 633 \text{ nm}$), this gives $\Phi_{\text{eff}} \approx 0.21 \times 2\pi$ [12, 16].

This effective flux is large enough to produce well-separated Chern bands and observable edge states, but not so large that the Floquet bands overlap (the high-frequency condition $\hbar\Omega \gg J$ is satisfied).

13.10.2 Four-step driving protocols

An alternative to continuous helical modulation is a discrete step-wise protocol, in which the coupling between neighbouring waveguides is turned on and off in a cyclic sequence. Maczewsky et al. used a four-step protocol (right $\rightarrow$ up $\rightarrow$ left $\rightarrow$ down) with each step occupying one quarter of the driving period. This protocol breaks all spatial symmetries and produces an anomalous Floquet topological insulator with zero Chern numbers but nonzero winding numbers in every quasi-energy gap [85, 86].

13.11 Heating and decoherence in Floquet photonic systems

In electronic Floquet systems, periodic driving inevitably heats the system toward infinite temperature, because the driving energy is absorbed by the electrons. This heating limits the lifetime of Floquet topological phases and is a major obstacle for practical applications.

In photonic Floquet systems, the situation is fundamentally different: photons do not thermalise with each other (in the linear regime), so there is no analog of Floquet heating. The periodic modulation simply produces a time-independent effective band structure, and the “Floquet topological” phase is stable indefinitely as long as the modulation is maintained.

Decoherence in photonic Floquet systems arises from material absorption and radiation loss, not from thermalisation. These losses give the Floquet edge modes a finite propagation length, analogous to the loss-limited propagation in magneto-optical Chern insulators (Chapter 7). For laser-written waveguides in fused silica, the propagation loss is approximately 0.5 dB/cm, allowing edge-mode observation over ~ 10 cm propagation lengths [13, 16].

13.12 Floquet topological phases beyond the high-frequency limit

Most of the analysis in this chapter has assumed the high-frequency limit, where the driving frequency Ω is much larger than the bandwidth W of the instantaneous Hamiltonian: $\hbar\Omega \gg W$. In this limit, the Floquet quasi-energy bands are well separated and the effective Hamiltonian H_{eff} is accurately approximated by the leading terms of the Magnus expansion (Section 13.5.4).

13.12.1 Resonant driving and Floquet band inversions

When $\hbar\Omega$ is comparable to W , resonances between Floquet replicas produce band inversions that cannot be captured by the effective Hamiltonian. At these resonances, the quasi-energy bands hybridise strongly, and the topology of the Floquet spectrum can change discontinuously.

For the Rechtsman waveguide array, the high-frequency condition is $\Omega \gg 2\pi v_D/a$, where v_D is the Dirac velocity and a is the lattice constant. With the experimental parameters ($\Lambda = 1$ cm, $a = 22$ μm , $v_D \approx 10^5$ m/s), this gives $\hbar\Omega/(2\pi v_D/a) \approx 20$, well within the high-frequency regime. However, for shorter driving periods ($\Lambda \lesssim 1$ mm), resonant effects become important and can produce additional topological transitions [16, 86].

13.12.2 Floquet–Bloch band structure

The full Floquet–Bloch band structure is obtained by diagonalising the one-period evolution operator $U(\mathbf{k}, T)$ at each Bloch wavevector. The quasi-energies $\varepsilon_n(\mathbf{k})$ are defined modulo $2\pi/T$, creating a periodic “quasi-energy Brillouin zone” in addition to the spatial Brillouin zone. The topology of each quasi-energy gap is characterised independently by the Chern number (for gaps that are also present in the static limit) or by the winding number (13.8) (for anomalous gaps that have no static counterpart) [12, 85].

13.13 Connection to time crystals

Floquet topological phases share a deep connection with discrete time crystals: both arise from systems with periodic time dependence and both exhibit spontaneous breaking of the discrete time-translation symmetry. In a Floquet topological insulator, the edge modes oscillate at frequencies that are subharmonics of the driving frequency—a hallmark of discrete time-crystalline order.

In photonic systems, this connection has been explored in coupled resonator networks where the edge modes spontaneously lock to a subharmonic of the driving frequency, producing a “topological time crystal” that is robust against perturbations. The topological protection ensures that the time-crystalline order is stable against disorder that does not close the quasi-energy gap [12, 13].

13.14 Anomalous phases beyond band Chern numbers

One of the most striking predictions of Floquet topological theory is the existence of *anomalous* topological phases: driven systems that support chiral edge states even when *all* quasi-energy-band Chern numbers vanish. This phenomenon has no static analogue and reveals the fundamentally richer topology of periodically driven systems.

13.14.1 Why static invariants fail

In a static topological insulator, the number of chiral edge modes in a gap equals the sum of Chern numbers of all bands below that gap. In a Floquet system, the quasi-energy spectrum is periodic with period Ω , so there are two distinct gaps: one near quasi-energy 0 and one near $\Omega/2$. A band with Chern number C contributes edge modes to both gaps, and the net count in each gap depends on how C is distributed between the two zone boundaries.

The anomalous phase arises when all quasi-energy bands have $C = 0$, yet chiral edge modes exist in one or both gaps. This is possible because the time-evolution operator $U(\mathbf{k}, T)$ encodes topological information not captured by the instantaneous band Chern numbers [86, 87].

13.14.2 The winding number invariant

The correct topological invariant for Floquet systems was identified by Rudner, Lindner, Berg, and Levin, who showed that the number of chiral edge modes in the gap at quasi-energy ε is determined by a *winding number* W_ε constructed from the full time-dependent evolution operator $U(\mathbf{k}, t)$ for $0 \leq t \leq T$.

Define the unitary evolution operator from time zero to time t : $U(\mathbf{k}, t) = \mathcal{T} \exp[-i \int_0^t H(\mathbf{k}, t') dt']$. The winding number for the gap at ε is

$$W_\varepsilon = \frac{1}{8\pi^2} \int_0^T dt \int_{\text{BZ}} d^2k \operatorname{Tr}[U^\dagger \partial_t U [U^\dagger \partial_{k_x} U, U^\dagger \partial_{k_y} U]] - \sum_{n: \varepsilon_n < \varepsilon} C_n. \quad (13.14)$$

The first term is a three-dimensional winding number of the map $U : \text{BZ} \times [0, T] \rightarrow U(N)$; the subtracted sum removes the conventional Chern-number contribution. The winding number is an

integer and equals the number of anomalous chiral edge modes—those that arise from the micro-motion during the drive cycle rather than from the stroboscopic band topology [12, 87].

Physically, (13.14) captures the fact that edge modes can be “pumped” across a gap during each drive cycle, even if the Floquet bands carry zero net Chern number. This pumping mechanism is the dynamical counterpart of the Thouless charge pump of Chapter 14 [2].

13.14.3 Experimental observation in photonic lattices

The anomalous Floquet topological phase was observed by Mukherjee et al. in a laser-written waveguide lattice [88]. The experiment used the same femtosecond-laser-inscription technique as the Rechtsman experiment (Section 13.5.3), but with a two-step driving protocol designed to produce zero Chern numbers for all quasi-energy bands while maintaining a nonzero winding number.

In the first half of the propagation period, nearest-neighbour couplings along one bond direction were enhanced; in the second half, couplings along the complementary bonds were enhanced. The stroboscopic band structure was topologically trivial, but the full time-evolution operator wound nontrivially.

Chiral edge modes were observed propagating along the lattice boundary, with intensity localised within 2–3 sites of the edge. The transport was robust against intentional disorder of up to 10% of the lattice constant in waveguide positions. These observations directly confirmed the anomalous Floquet phase [88].

13.14.4 Implications for reconfigurable photonic devices

Because the topological protection in anomalous Floquet phases arises from the driving protocol rather than from the static band structure, it can be switched on and off by changing the modulation pattern. This enables dynamically reconfigurable topological waveguides, routers, and filters that can be reprogrammed without modifying the physical structure.

Furthermore, a Floquet system with N quasi-energy bands has N independent winding numbers (one per gap), compared to $N - 1$ independent Chern numbers. This additional topological degree of freedom can be exploited to engineer multi-channel edge transport with independently controllable numbers of edge modes in different quasi-energy windows [12].

Exercises

Exercise 13.1. Derive the paraxial equation (13.10) from the Helmholtz equation by writing $E = \psi e^{ik_0 z}$ and neglecting $\partial^2 \psi / \partial z^2$. State the conditions under which the approximation is valid and estimate the error for a beam with divergence angle θ .

Exercise 13.2. For a 1D lattice of coupled resonators with nearest-neighbour hopping $J_{ij} = J_0 e^{i\phi_{ij}}$, compute the effective magnetic flux per plaquette for a square lattice with uniform phase $\phi_{ij} = \alpha$ along x and $\phi_{ij} = 0$ along y . For what values of α is time-reversal symmetry effectively broken?

Exercise 13.3. Show that the Floquet operator $F(\mathbf{k}) = U(\mathbf{k}, T)$ is unitary, and that its eigenvalues $e^{-i\varepsilon_n T}$ lie on the unit circle. Interpret the quasi-energy ε_n as the argument of these eigenvalues.

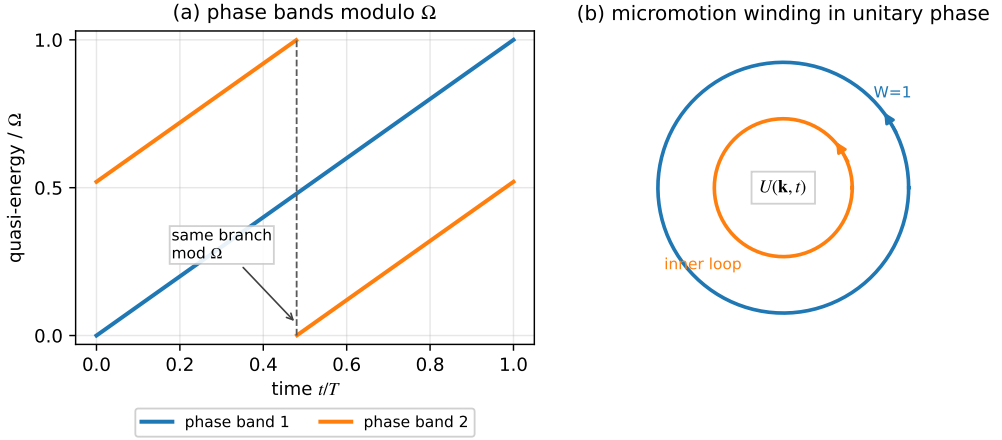


Figure 13.2: Floquet topology depends on micromotion during the whole period, not only on the stroboscopic band Chern numbers. Phase bands can wind around the quasi-energy zone during one drive cycle, and the resulting map $U(\mathbf{k}, t)$ over $\text{BZ} \times S^1$ is measured by the gap winding invariant W_ε .

Exercise 13.4. Explain why an anomalous Floquet topological insulator (all band Chern numbers zero, but edge modes present) is impossible in a static system. What additional topological invariant distinguishes the AFTI from a trivial insulator?

Exercise 13.5. In the Rechtsman waveguide experiment, the waveguides are helically curved with helix radius $R = 8 \mu\text{m}$ and pitch $\Lambda = 1 \text{ cm}$ in fused silica ($n_0 = 1.46$) at $\lambda = 633 \text{ nm}$. Compute the effective modulation strength $\delta n_{\text{eff}} \sim n_0 k_0 R^2 (2\pi/\Lambda)^2$ and estimate the topological gap width relative to the coupling strength $J \sim 1 \text{ cm}^{-1}$.

Exercise 13.6. For a Floquet system with two bands and modulation period T , compute the winding number (13.8) for a specific model: $H(k, t) = [J_1 + J_2 \cos(k)]\sigma_x + J_2 \sin(k)\sigma_y + \Delta(t)\sigma_z$ with $\Delta(t) = \Delta_0 \cos(\Omega t)$. Under what conditions is the winding number nonzero?

Chapter 14

One Dimension

14.1 Topology in one dimension

One-dimensional systems are the simplest arena in which topological ideas appear. There is no Chern number in 1D—the Brillouin zone is a circle S^1 , not a torus T^2 —but a different topological invariant, the *Zak phase*, classifies 1D band structures and predicts interface modes between topologically distinct regions [5, 34].

In photonics, 1D topological systems include dielectric multilayer stacks, coupled-cavity chains, and waveguide arrays with alternating coupling [12, 29]. These are experimentally simple, theoretically transparent, and provide a clear illustration of the connection between bulk topology and edge states without the complications of 2D band structures [89, 90].

14.2 The Zak phase

14.2.1 SSH model from Maxwell equations

To derive the SSH model from a photonic system, consider a 1D dielectric multilayer stack with alternating layer thicknesses d_1 and d_2 ($d_1 \neq d_2$) and refractive indices n_A and n_B [6]. The unit cell has period $a = d_1 + d_2$.

For TE-polarised light propagating along the stack axis x , the transfer matrix across one unit cell is

$$\mathbf{T}(\omega) = \mathbf{M}_B(d_2) \mathbf{M}_A(d_1), \quad (14.1)$$

where $\mathbf{M}_j(d) = \begin{pmatrix} \cos \phi_j & -\sin \phi_j / \eta_j \\ \eta_j \sin \phi_j & \cos \phi_j \end{pmatrix}$ with $\phi_j = n_j \omega d / c$ and $\eta_j = n_j \omega / c$ for layer j . The Bloch condition $\mathbf{T}\mathbf{v} = e^{ika}\mathbf{v}$ gives the photonic band structure $\omega(k)$.

Near a band gap at the Brillouin zone boundary $k = \pi/a$, the two bands closest to the gap are described by a 2×2 effective Hamiltonian. Expanding the transfer matrix around the gap frequency ω_0 yields the SSH form

$$H_{\text{eff}}(\delta k) = t_1 \sigma_x + t_2 (\cos \delta k \sigma_x + \sin \delta k \sigma_y), \quad (14.2)$$

where t_1 and t_2 are the intra-cell and inter-cell coupling coefficients, controlled by d_1 and d_2 respectively. When $|t_1| < |t_2|$ the system is topological (Zak phase π); when $|t_1| > |t_2|$ it is trivial (Zak phase 0).

14.2.2 Quantitative example: silicon–silica multilayer

Consider a silicon ($n_A = 3.48$) and silica ($n_B = 1.44$) multilayer at telecom wavelength $\lambda_0 = 1550$ nm. Quarter-wave layers give $d_A = 111$ nm, $d_B = 269$ nm, $a = 380$ nm. The gap width at normal incidence is

$$\frac{\Delta\omega}{\omega_0} = \frac{4}{\pi} \arcsin \frac{|n_A - n_B|}{n_A + n_B} \approx 0.53, \quad (14.3)$$

an exceptionally large relative gap of 53% [12]. SSH dimerisation is created by modifying $d_A \rightarrow d_A \pm \delta$ in alternating unit cells, producing a topological mini-gap $\Delta\omega_{\text{SSH}} \propto \delta/a$ within the original pass band. The mid-gap interface state appears as a sharp transmission peak with quality factor $Q \sim 10^3$ for 20 periods [13].

14.2.3 Definition from the electromagnetic inner product

The Zak phase is the Berry phase accumulated as the Bloch wavevector k traverses the entire 1D Brillouin zone [12, 19]:

$$\gamma_{\text{Zak}} = \int_{-\pi/a}^{\pi/a} \mathcal{A}_n(k) dk, \quad (14.4)$$

where the Berry connection uses the electromagnetic inner product:

$$\mathcal{A}_n(k) = i \frac{\int_{\text{cell}} \mathbf{u}_{nk}^\dagger(\mathbf{r}) \mathbf{M}_g(\omega_{nk}) \partial_k \mathbf{u}_{nk}(\mathbf{r}) d\mathbf{r}}{\int_{\text{cell}} \mathbf{u}_{nk}^\dagger \mathbf{M}_g \mathbf{u}_{nk} d\mathbf{r}}. \quad (14.5)$$

Unlike the Chern number, the Zak phase is *not* fully gauge-invariant: under a gauge transformation $\mathbf{u}_{nk} \rightarrow e^{i\chi(k)} \mathbf{u}_{nk}$, the Berry connection shifts by $\partial_k \chi$, and the Zak phase shifts by $\chi(\pi/a) - \chi(-\pi/a)$. Since the mode must be periodic in k (up to a gauge transformation), this shift is an integer multiple of 2π . Thus the Zak phase is defined modulo 2π .

14.2.4 Quantisation by symmetry

Proposition 14.1. *If a 1D photonic crystal has both time-reversal ($\hat{\mathcal{T}}$) and inversion ($\hat{\mathcal{P}}$) symmetry, the Zak phase of each isolated band is quantised: $\gamma_{\text{Zak}} \in \{0, \pi\}$ modulo 2π .*

Proof. We use the combined constraints of $\hat{\mathcal{T}}$ and $\hat{\mathcal{P}}$ on the Berry connection.

Step 1: Time reversal. $\hat{\mathcal{T}}$ maps $k \rightarrow -k$ and complex-conjugates the modes. For a system where $\hat{\mathcal{T}}$ acts as complex conjugation (since $\hat{\mathcal{T}}^2 = +1$ for photons), the modes can be chosen real at k and $-k$. In a real gauge, $\mathcal{A}_n(k) = \mathcal{A}_n(-k)$.

Step 2: Inversion. $\hat{\mathcal{P}}$ maps $k \rightarrow -k$ and $\mathbf{r} \rightarrow -\mathbf{r}$. The Berry connection transforms as $\mathcal{A}_n(-k) = -\mathcal{A}_n(k) + \partial_k \phi(k)$ for some function ϕ .

Step 3: Combining. From Steps 1 and 2: $\mathcal{A}_n(k) = -\mathcal{A}_n(k) + \partial_k \phi$, so $\mathcal{A}_n(k) = \frac{1}{2} \partial_k \phi$. Integrating:

$$\gamma_{\text{Zak}} = \frac{1}{2} [\phi(\pi/a) - \phi(-\pi/a)]. \quad (14.6)$$

Since the mode is periodic in k up to a gauge transformation, $\phi(\pi/a) - \phi(-\pi/a) = 2\pi n$ for integer n . Thus $\gamma_{\text{Zak}} = n\pi$, which modulo 2π is either 0 or π . $\square$

14.2.5 Physical meaning of the Zak phase

The quantised Zak phase $\gamma_{\text{Zak}} = 0$ or π classifies the “polarisation” of the 1D photonic crystal: it measures whether the centre of the Wannier function (the real-space representation of the Bloch mode) is centred on the inversion centre of the unit cell ($\gamma_{\text{Zak}} = 0$) or displaced to the cell boundary ($\gamma_{\text{Zak}} = \pi$).

Two 1D crystals with different Zak phases (0 vs. π) are in different topological classes and cannot be continuously connected without closing a band gap (a topological phase transition).

14.3 The Su–Schrieffer–Heeger model for photons

14.3.1 The electronic SSH model

The SSH model is the simplest tight-binding model with nontrivial 1D topology: a chain of sites with alternating hopping amplitudes J_1 and J_2 . The unit cell contains two sites (A and B), and the Hamiltonian in momentum space is

$$H(k) = \begin{pmatrix} 0 & J_1 + J_2 e^{-ika} \\ J_1 + J_2 e^{ika} & 0 \end{pmatrix} = \mathbf{d}(k) \cdot \boldsymbol{\sigma}, \quad (14.7)$$

with $d_x(k) = J_1 + J_2 \cos(ka)$, $d_y(k) = J_2 \sin(ka)$, $d_z = 0$.

14.3.2 Band structure and winding number

The band structure has two bands:

$$E_{\pm}(k) = \pm |J_1 + J_2 e^{ika}| = \pm \sqrt{J_1^2 + J_2^2 + 2J_1 J_2 \cos(ka)} \quad (14.8)$$

The displayed formula above follows the convention and normalisation used in [46]. A gap exists when $J_1 \neq J_2$, with gap width $2|J_1 - J_2|$.

The topological invariant is the *winding number* of the vector $\mathbf{d}(k) = (d_x, d_y)$ around the origin as k traverses the Brillouin zone. For $k \in [-\pi/a, \pi/a]$, the curve $(d_x(k), d_y(k))$ is a circle of radius J_2 centred at $(J_1, 0)$.

$$\nu = \frac{1}{2\pi} \oint d \arg(d_x + id_y) = \begin{cases} 0 & \text{if } |J_1| > |J_2| \text{ (trivial)}, \\ 1 & \text{if } |J_1| < |J_2| \text{ (topological)}. \end{cases} \quad (14.9)$$

The winding number equals γ_{Zak}/π (modulo 2). The topological phase transition occurs at $J_1 = J_2$, where the gap closes and the $\mathbf{d}$ -curve passes through the origin.

14.3.3 Chiral symmetry

The SSH Hamiltonian has *chiral* (sublattice) symmetry: $\sigma_z H(k) \sigma_z = -H(k)$. This forces $d_z = 0$ and constrains the band structure to be particle-hole symmetric: $E_+(k) = -E_-(k)$.

Chiral symmetry protects the winding number: any perturbation that preserves the sublattice structure cannot change ν without closing the gap. Perturbations that break chiral symmetry (e.g.

adding on-site energies $\Delta\sigma_z$) can shift the Zak phase continuously and destroy the topological protection.

14.4 Photonic realisations of the SSH model

14.4.1 Coupled-cavity chain

A sequence of identical optical cavities (e.g. Fabry–Pérot resonators or photonic crystal defect cavities) with alternating coupling strengths realises the SSH model directly.

The resonance frequency of each cavity is ω_0 , and the evanescent coupling between adjacent cavities is controlled by the barrier thickness:

- Thin barrier: strong coupling J_2 .
- Thick barrier: weak coupling J_1 .

The alternating pattern $J_1, J_2, J_1, J_2, \dots$ creates the SSH dimerisation.

The dispersion relation for the coupled-cavity chain is

$$\omega_{\pm}(k) = \omega_0 \pm \sqrt{J_1^2 + J_2^2 + 2J_1J_2 \cos(ka)}, \quad (14.10)$$

which is the SSH band structure shifted by ω_0 .

14.4.2 Dielectric multilayer (Bragg stack)

A 1D photonic crystal (Bragg stack) with alternating layers of two materials can be viewed as an SSH-like system when the unit cell contains two layers of different optical thickness.

Consider a stack with layers A (thickness d_A , index n_A) and B (d_B , n_B) in alternation. The “dimerisation” corresponds to unequal optical path lengths $n_A d_A \neq n_B d_B$ within a unit cell.

14.4.3 Waveguide array

A chain of evanescently coupled optical waveguides with alternating spacing implements the SSH model, with the coupling exponentially dependent on the inter-waveguide distance:

$$J_{1,2} = J_0 e^{-\kappa d_{1,2}}, \quad (14.11)$$

where $d_{1,2}$ are the alternating centre-to-centre distances and κ is the evanescent decay constant.

14.5 Interface states from the Zak phase

14.5.1 Bulk-edge correspondence in 1D

The 1D bulk-edge correspondence states:

Theorem 14.2 (1D bulk-edge correspondence). *At an interface between two 1D photonic crystals with different Zak phases ($\gamma_{\text{Zak}}^L \neq \gamma_{\text{Zak}}^R$, modulo 2π), there exists at least one localised interface mode in each common band gap.*

For the SSH model: the $\nu = 0$ phase ($|J_1| > |J_2|$) has $\gamma_{\text{Zak}} = 0$, and the $\nu = 1$ phase ($|J_1| < |J_2|$) has $\gamma_{\text{Zak}} = \pi$. An interface between the two phases supports a mid-gap interface mode.

14.5.2 Derivation of the interface mode

Consider an SSH chain with a domain wall at site $n = 0$: for $n < 0$, the dimerisation is (J_2, J_1) (topological phase, $|J_1| < |J_2|$), and for $n \geq 0$, it is (J_1, J_2) (trivial phase, $|J_1| > |J_2|$).

We seek a zero-energy mode ($E = 0$, i.e., $\omega = \omega_0$ for the photonic case) localised at the interface. The zero-energy condition $H|\psi\rangle = 0$ in real space gives:

$$J_1\psi_{B,n} + J_2\psi_{B,n-1} = 0, \quad J_2\psi_{A,n+1} + J_1\psi_{A,n} = 0, \quad (14.12)$$

where $\psi_{A,n}$ and $\psi_{B,n}$ are the amplitudes on sublattices A and B at unit cell n .

For the interface geometry, the solution is:

$$\psi_{A,n} = 0, \quad \psi_{B,n} \propto \begin{cases} (-J_1/J_2)^n & n \geq 0, \\ (-J_2/J_1)^{|n|} & n < 0. \end{cases} \quad (14.13)$$

This is normalisable (exponentially decaying on both sides of the interface) when $|J_1/J_2| < 1$ on the right and $|J_2/J_1| < 1$ on the left, which is exactly the condition for the interface to separate the topological and trivial phases.

14.5.3 Localisation length

The localisation length of the interface mode is

$$\ell = \frac{a}{|\ln(J_1/J_2)|}. \quad (14.14)$$

For strong dimerisation ($J_1 \ll J_2$), $\ell \sim a/\ln(J_2/J_1)$: the mode is tightly confined to the interface. For weak dimerisation ($J_1 \lesssim J_2$), $\ell \gg a$: the mode extends over many unit cells.

14.5.4 Experimental observations

Interface states in 1D photonic SSH systems have been observed in [12, 13]:

- **Coupled photonic crystal cavities** at telecom wavelengths, where the interface mode appears as a sharp resonance in the transmission spectrum within the band gap.
- **Dielectric multilayer stacks** in the microwave regime, where the transfer-matrix method gives direct access to the interface-mode frequency and localisation.
- **Plasmonic nanoparticle chains**, where the SSH dimerisation is achieved by alternating the inter-particle spacing, demonstrating the mode localisation directly through near-field imaging.

14.6 The dielectric Bragg stack as a topological system

14.6.1 Transfer-matrix analysis

A 1D photonic crystal with layers A (thickness d_A , index n_A) and B (d_B, n_B) in alternation has a 2×2 transfer matrix per period:

$$M = M_B M_A, \quad (14.15)$$

where each layer matrix is

$$M_j = \begin{pmatrix} \cos \delta_j & -\sin \delta_j / \eta_j \\ \eta_j \sin \delta_j & \cos \delta_j \end{pmatrix}, \quad j = A, B, \quad (14.16)$$

with $\delta_j = n_j \omega d_j / c$ the phase thickness and $\eta_j = n_j / Z_0$ the admittance (for normal incidence, TE polarisation).

The Bloch condition gives the dispersion:

$$\cos(ka) = \frac{1}{2} \text{Tr}(M) = \cos \delta_A \cos \delta_B - \frac{1}{2} \left(\frac{\eta_A}{\eta_B} + \frac{\eta_B}{\eta_A} \right) \sin \delta_A \sin \delta_B. \quad (14.17)$$

Band gaps occur when $|\frac{1}{2} \text{Tr}(M)| > 1$.

14.6.2 Topological classification of Bragg gaps

The Zak phase of each band depends on the symmetry of the band-edge modes. When the unit cell has inversion symmetry (choosing the cell boundary at the centre of layer A), each band edge at $k = 0$ or $k = \pi/a$ has a definite parity under inversion.

The Zak phase of band n is determined by the parities:

$$(-1)^{\gamma_{\text{Zak},n}/\pi} = \frac{p_n(0) p_n(\pi/a)}{p_{n-1}(0) p_{n-1}(\pi/a)}, \quad (14.18)$$

where $p_n(k_0) = \pm 1$ is the parity of the n -th band at k_0 .

A topological phase transition occurs when the gap closes, which happens at the quarter-wave condition $n_A d_A = n_B d_B$. Tuning through this condition changes the Zak phase of the bands and swaps the topological character of the gaps.

14.7 Topological pumping

14.7.1 The Thouless pump for photons

A 1D photonic crystal can be adiabatically modulated in a way that pumps light across the system—the photonic analog of the Thouless charge pump [2, 12] [91].

The pump is a 1D system with a slowly varying parameter $\phi(t)$ that cycles through 2π over one period T . Under the adiabatic approximation, the net displacement of light after one full cycle is quantised:

$$\Delta x = a \mathcal{C}_1, \quad (14.19)$$

where $\mathcal{C}_1$ is the first Chern number of the 2D parameter space (k, ϕ) . This is the 1D manifestation of the TKNN formula [2]: the pumped energy per cycle is a topological invariant that cannot change without closing a gap.

Consider a 1D photonic crystal with unit-cell parameters that depend on an external parameter θ : $\mathbf{M}(\mathbf{r}; \theta)$. As θ is varied adiabatically from 0 to 2π , tracing a closed cycle in parameter space, the centre of the Wannier function shifts by

$$\Delta x = a \mathcal{C}_{k, \theta}, \quad (14.20)$$

where

$$\mathcal{C}_{k, \theta} = \frac{1}{2\pi} \int_0^{2\pi} \int_{-\pi/a}^{\pi/a} \Omega_n(k, \theta) dk d\theta \quad (14.21)$$

is the Chern number computed over the 2D parameter space (k, θ) .

This Chern number is a true integer topological invariant (because the (k, θ) space is a torus), even though the Zak phase alone is only $\mathbb{Z}_2$ -valued. The pump displacement is quantised: Δx is an integer multiple of the lattice constant a .

14.7.2 Experimental demonstration

Photonic Thouless pumping has been demonstrated in coupled waveguide arrays where the coupling pattern is slowly varied along the propagation axis z (playing the role of the adiabatic parameter θ). The transverse displacement of a light beam after one pump cycle was measured to be exactly one lattice constant, confirming the quantised topological transport.

14.8 Topological invariants in one dimension

One-dimensional topological phases are characterised by integer winding numbers rather than Chern numbers. The winding number of the SSH model is defined from the off-diagonal element of the Hamiltonian in the chiral basis:

$$\nu = \frac{1}{2\pi i} \int_{-\pi/a}^{\pi/a} \frac{d}{dk} \ln h(k) dk, \quad (14.22)$$

where $h(k) = t_1 + t_2 e^{ika}$ is the off-diagonal coupling function. When $|t_1| < |t_2|$, the function $h(k)$ winds once around the origin as k traverses the Brillouin zone, giving $\nu = 1$; when $|t_1| > |t_2|$, the winding number is $\nu = 0$. The transition at $|t_1| = |t_2|$ closes the gap and changes the topology.

14.8.1 Zak phase and its physical meaning

The Zak phase is the 1D Berry phase accumulated by a Bloch state as k traverses the full Brillouin zone:

$$\gamma_n = \oint_{\text{BZ}} \mathcal{A}_n(k) dk, \quad (14.23)$$

where $\mathcal{A}_n(k) = i \langle \mathbf{u}_n, \partial_k \mathbf{u}_n \rangle_{\mathbf{M}_g}$ is the Berry connection of band n . For the SSH model with inversion symmetry, the Zak phase is quantised to 0 or π , and the winding number equals γ_n/π modulo 2 [12, 21].

The physical meaning of the Zak phase is the displacement of the Wannier function centre from the unit cell boundary. A Zak phase of π means the Wannier centre sits at the boundary between unit cells, which is precisely the condition for the existence of edge states when the chain is terminated.

14.9 Photonic SSH chains: experimental realisations

14.9.1 Dielectric multilayer stacks

The simplest photonic SSH chain is a 1D multilayer stack with alternating layer thicknesses d_1 and d_2 in a material with refractive index n_1 , separated by layers of index n_2 . The transfer matrix for one unit cell is $M = M_1 M_2$, where M_j is the 2×2 transfer matrix for layer j . The eigenvalues of M give the Bloch dispersion $\cos(ka) = \frac{1}{2} \text{Tr}(M)$, and the winding number is determined by the relative magnitudes of d_1 and d_2 .

When $d_1 \neq d_2$, the system is in a topologically nontrivial phase ($\nu = 1$) if $d_1 < d_2$ and trivial ($\nu = 0$) if $d_1 > d_2$. At the interface between two stacks with opposite dimerisation, a localised interface mode appears at the midgap frequency—the photonic analog of the SSH edge state [12, 13].

14.9.2 Coupled-resonator optical waveguides

A more versatile platform is the coupled-resonator optical waveguide (CROW): a chain of optical microring resonators with alternating coupling gaps g_1 and g_2 . The coupling strengths $\kappa_j \propto e^{-g_j/\lambda_{\text{ev}}}$ (where λ_{ev} is the evanescent decay length) play the role of the SSH hopping amplitudes t_1 and t_2 .

The topological edge state in a finite CROW chain manifests as a sharp resonance at the midgap frequency, localised at the end of the chain. This has been observed experimentally in silicon photonic CROWs at $\lambda = 1.55 \mu\text{m}$, with quality factors exceeding 10^4 and field enhancement of $\sim 20\times$ at the edge resonator [12].

14.10 Experimental realisations of one-dimensional topological photonics

The SSH and Rice–Mele models, originally formulated for polyacetylene, have found remarkably faithful realisations in photonic platforms, where each site corresponds to a waveguide or resonator and the hopping amplitudes are controlled by evanescent coupling.

14.10.1 Dimerised waveguide arrays

The simplest photonic SSH system is a one-dimensional array of single-mode dielectric waveguides with alternating inter-waveguide gaps [39]. A wider gap produces weaker evanescent coupling t_1 (the intra-cell hopping), while a narrower gap produces stronger coupling t_2 (the inter-cell hopping). When $t_1 < t_2$, the array is in the topological phase and supports exponentially localised edge modes at each termination.

Malkova et al. (2009) demonstrated such edge states in a one-dimensional array of coupled optical waveguides fabricated by femtosecond laser inscription in fused silica. The measured intensity profile showed clear localisation at the edge with a decay length consistent with the SSH prediction $\xi = 1 / \ln(t_2/t_1)$ [12].

More recently, Blanco-Redondo et al. (2016) demonstrated the topological protection of the SSH edge mode against certain classes of disorder using silicon photonic waveguide arrays at $\lambda = 1.55 \mu\text{m}$. The edge-mode frequency remained pinned at mid-gap even when the waveguide widths were randomised by up to 15%, confirming that the topological protection relies on the chiral symmetry of the SSH model (which is preserved when the disorder respects the sublattice structure) [12].

14.10.2 Photonic Thouless pumps

The Thouless charge pump, discussed in Section 14.7.1, has been realised in photonic systems by adiabatically modulating the coupling constants along the propagation direction of a waveguide array. In this photonic implementation, the propagation coordinate z plays the role of time, and one complete modulation period corresponds to one pump cycle.

Kraus et al. (2012) demonstrated the first photonic Thouless pump using a one-dimensional quasiperiodic waveguide array. The quasiperiodic modulation of the coupling constants mapped onto the Aubry–André model, whose topological properties are equivalent to those of the two-dimensional quantum Hall effect projected onto one dimension. By scanning a phase parameter across the array, they observed the quantised transfer of light intensity across the lattice, corresponding to a Chern number of $\mathcal{C} = 1$ [12, 13].

Zilberberg et al. (2018) extended this idea to a four-dimensional quantum Hall system by using two independent modulation parameters in a two-dimensional waveguide array. The experiment demonstrated the quantised pumping associated with the second Chern number, a topological invariant that characterises four-dimensional topological phases [13].

The mechanism underlying these experiments is captured schematically in Figure 14.1. Panel (a) shows a counter-clockwise loop in the Rice–Mele parameter space (δ, Δ) , where δ is the dimerisation and Δ is the staggered on-site potential. The loop encloses the single gap-closing point at the origin; winding around this point once guarantees a quantised adiabatic transport of exactly one unit cell per cycle. The teal dot marks the start and end of the cycle. Panel (b) displays the resulting Wannier-centre trajectory as a function of the adiabatic time t/T . The displacement accumulates nonlinearly during the cycle but reaches exactly one lattice constant at $t = T$, confirming the topological quantisation $\Delta x = \mathcal{C} a$ with $\mathcal{C} = 1$.

14.10.3 Quantitative analysis of the photonic SSH edge mode

For a finite SSH chain of N unit cells terminated at a weak bond (so the first coupling is $t_1 < t_2$), the edge-mode wavefunction at site n on sublattice A is

$$\psi_n^A = \mathcal{N} \left(-\frac{t_1}{t_2} \right)^n, \quad \psi_n^B = 0, \quad (14.24)$$

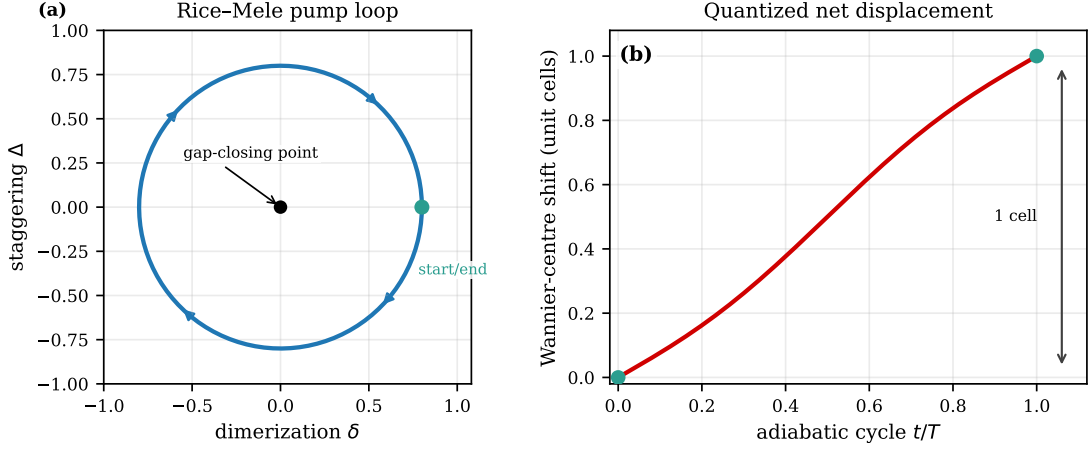


Figure 14.1: Rice–Mele topological pump. (a) Counter-clockwise parameter loop in the (δ, Δ) -plane enclosing the gap-closing origin (black dot). Blue arrows indicate the traversal direction; the teal dot marks the start/end point. (b) Wannier-centre displacement (in unit cells) during one adiabatic cycle. The net shift of exactly one cell (double arrow) is the topological invariant of the pump; it equals the Chern number $\mathcal{C} = 1$ of the filled band over the parameter torus.

where $\mathcal{N}$ is a normalisation constant and the mode lives entirely on sublattice A. The sublattice polarisation of the edge mode is a direct consequence of the chiral symmetry $\Gamma H \Gamma = -H$ with $\Gamma = \sigma_z$: at mid-gap energy $E = 0$, the eigenstate can be chosen to have support on only one sublattice.

The localisation length is

$$\xi = \frac{a}{\ln(t_2/t_1)}, \quad (14.25)$$

where a is the lattice constant. As the gap closes ($t_1 \rightarrow t_2$), the localisation length diverges and the edge mode delocalises—a hallmark of the topological phase transition.

In a photonic waveguide array, the coupling constants depend exponentially on the inter-waveguide separation d : $t \approx t_0 \exp(-d/d_0)$, where d_0 is a decay length set by the evanescent tail of the waveguide mode (typically $d_0 \sim 1\text{--}3 \mu\text{m}$ in silica or silicon). The dimerisation ratio t_1/t_2 can therefore be tuned continuously by adjusting the two gap widths, allowing systematic exploration of the topological phase diagram [12].

14.11 Beyond SSH: the Rice–Mele model

The SSH model has chiral symmetry $\sigma_z H \sigma_z = -H$, which pins the edge-state energy to zero and quantises the Zak phase to 0 or π . Breaking chiral symmetry by adding a staggered on-site potential Δ produces the Rice–Mele model [12, 21]:

$$H_{\text{RM}}(k) = (t_1 + t_2 \cos ka) \sigma_x + t_2 \sin(ka) \sigma_y + \Delta \sigma_z. \quad (14.26)$$

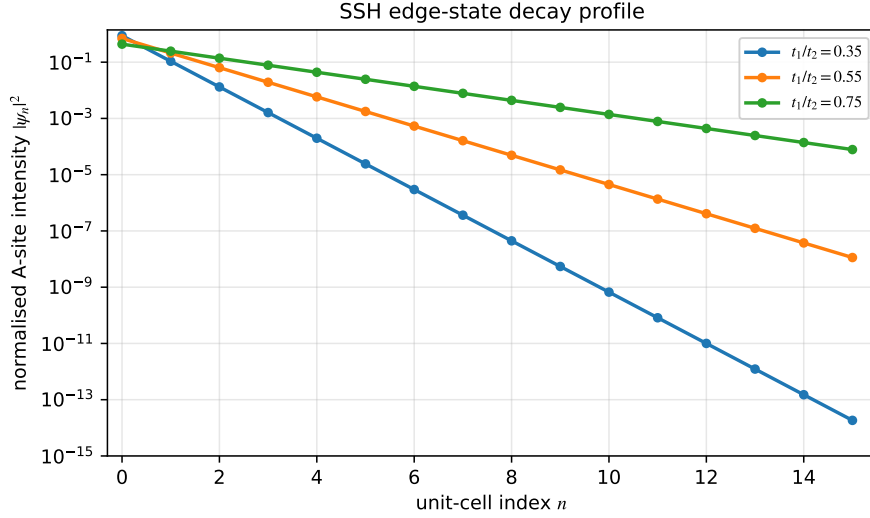


Figure 14.2: SSH edge-state localisation. For $t_1 < t_2$, the edge-state intensity decays as $(t_1/t_2)^{2n}$ into the bulk. As the dimerisation weakens and $t_1/t_2 \rightarrow 1$, the localisation length diverges at the topological phase transition.

The energy spectrum is $E_{\pm}(k) = \pm \sqrt{(t_1 + t_2 \cos ka)^2 + t_2^2 \sin^2(ka) + \Delta^2}$, and a gap exists for all $\Delta \neq 0$ even when $t_1 = t_2$.

14.11.1 Topology of the Rice–Mele model

Because chiral symmetry is broken, the winding number ν is no longer well defined and the Zak phase is no longer quantised. Instead, the relevant topological invariant is the polarisation change ΔP accumulated over one adiabatic cycle in the two-parameter space $(\delta t, \Delta)$, where $\delta t = t_1 - t_2$. If the cycle encloses the gap-closing point $(\delta t, \Delta) = (0, 0)$, the polarisation change is quantised:

$$\Delta P = \frac{e}{2\pi} \oint \mathcal{A}(k, \theta) d\theta = \pm e, \quad (14.27)$$

where θ parametrises the cycle. This is precisely the Thouless pump of Section 14.7.1: the (k, θ) torus carries a Chern number $\mathcal{C} = \pm 1$.

The Rice–Mele model therefore provides the minimal framework for understanding how the interplay of dimerisation and inversion breaking controls the topology of one-dimensional photonic systems. In a photonic implementation, Δ corresponds to a detuning between the resonance frequencies of the two sublattice sites, which can be introduced by varying the waveguide width or the resonator radius [12, 13].

14.11.2 Photonic Rice–Mele experiments

Photonic Rice–Mele systems have been realised in coupled-waveguide arrays where both the coupling constants and the propagation constants are modulated along the propagation axis. By slowly

cycling the modulation parameters through a closed loop that encloses the degeneracy point, the quantised adiabatic pumping of light intensity has been observed experimentally.

Cerjan et al. (2020) demonstrated a photonic Rice–Mele pump in a silicon photonic lattice, observing quantised displacement of light over multiple pump cycles with high fidelity. The deviation from integer displacement was less than 3%, limited primarily by non-adiabatic corrections from the finite propagation length [13].

14.12 Bulk-boundary correspondence in one dimension

The hallmark of topological phases is the bulk-boundary correspondence: a nontrivial bulk invariant guarantees the existence of boundary-localised states. In one dimension, this correspondence takes a particularly transparent form [21].

14.12.1 The winding number predicts edge states

Consider a semi-infinite SSH chain occupying sites $n = 0, 1, 2, \dots$ with the weak bond t_1 at the boundary. The bulk winding number ν (Section 14.8) counts the number of zero-energy edge states per boundary:

$$N_{\text{edge}} = |\nu|. \quad (14.28)$$

For $\nu = 1$ (the topological phase with $t_1 < t_2$), there is exactly one edge state; for $\nu = 0$ (trivial phase), there are none. This is a rigorous mathematical theorem, not merely an observation from specific models: it follows from the index theory of Fredholm operators acting on the half-line [21].

14.12.2 Robustness and symmetry protection

The topological protection of the SSH edge state requires chiral (sublattice) symmetry. Under arbitrary perturbations that preserve chiral symmetry—such as random variations of the coupling constants t_1 and t_2 , or long-range hopping that respects the sublattice structure—the edge state remains pinned at zero energy.

When chiral symmetry is broken (e.g. by adding random on-site potentials), the edge-state energy shifts away from zero. However, the state itself does not disappear until the perturbation closes the bulk gap. The topological origin of the state manifests as an anomalously long lifetime compared to generic disorder-induced states [12].

In photonic implementations, chiral symmetry corresponds to a balance between the effective “on-site energies” of the two sublattice waveguides. Fabrication imperfections that change the waveguide widths asymmetrically break this symmetry, while imperfections that change both sublattice sites equally preserve it. Careful waveguide design can therefore maximise topological protection in realistic devices [13, 39].

14.12.3 Connection to higher dimensions

One-dimensional topology is intimately connected to higher-dimensional topological phases through dimensional reduction. The Thouless pump (Section 14.7.1) provides the clearest example: the

(k, θ) parameter space of a one-dimensional pump is topologically equivalent to the Brillouin zone of a two-dimensional quantum Hall system, and the pump's Chern number equals the Hall conductance of the parent 2D system [2].

This dimensional hierarchy extends further. A two-parameter family of one-dimensional systems, with parameters (θ_1, θ_2) , can realise a second Chern number—a four-dimensional topological invariant. Zilberberg et al. (2018) exploited this correspondence to observe signatures of four-dimensional quantum Hall physics in a two-dimensional photonic waveguide array, where the two spatial dimensions encoded the physical coordinate and one pump parameter, while the second pump parameter was scanned experimentally [12, 13].

The lesson is that one-dimensional photonic systems, despite their apparent simplicity, serve as building blocks for exploring topological physics in arbitrary dimensions. The synthetic-dimension approach of Chapter 16 generalises this idea by promoting internal degrees of freedom (frequency, orbital angular momentum, or spin) to additional lattice dimensions [59, 92].

Exercises

Exercise 14.1. Show that the winding number (14.9) of the SSH model changes from 0 to 1 at $J_1 = J_2$, where the band gap closes. Sketch the trajectory of $(d_x(k), d_y(k))$ for $J_1 > J_2$, $J_1 = J_2$, and $J_1 < J_2$.

Exercise 14.2. For the coupled-cavity SSH chain with ω_0, J_1, J_2 , show that the interface mode has frequency $\omega = \omega_0$ (mid-gap) and verify the exponential localisation (14.13).

Exercise 14.3. Derive the band structure of a 1D Bragg stack from the transfer matrix (14.15) and the Bloch condition (14.17). Show that a gap opens when $n_A d_A \neq n_B d_B$. Compute the Zak phase for the lowest band when $n_A d_A > n_B d_B$ and when $n_A d_A < n_B d_B$.

Exercise 14.4. Verify equation (14.14) by computing the envelope of the interface mode (14.13) and extracting the $1/e$ decay length.

Exercise 14.5. Design a Thouless pump for a 1D photonic crystal by specifying a two-parameter family of unit cells such that the Chern number in (k, θ) space is nonzero. *Hint:* use the Rice–Mele model (SSH with an added staggered on-site potential that varies with θ).

Exercise 14.6. Show that chiral symmetry ($\sigma_z H \sigma_z = -H$) forces the interface mode of the SSH model to have exactly zero energy. What happens to the interface mode frequency when chiral symmetry is broken by adding a staggered potential $\Delta \sigma_z$?

Part IV

Three Dimensions and Beyond

Chapter 15

Weyl Points of Maxwell Bands

15.1 From two dimensions to three

In two dimensions, Dirac points are degeneracies that can be gapped by time-reversal or inversion breaking, producing bands with nonzero Chern numbers. In *three* dimensions, a qualitatively new type of topological degeneracy appears: the *Weyl point*, a point in 3D momentum space where two bands touch linearly. Unlike 2D Dirac points, Weyl points are *topologically robust*: they cannot be gapped by any perturbation, only moved in $\mathbf{k}$ -space or annihilated in pairs.

Weyl points are Berry-curvature monopoles—point sources or sinks of Berry flux in 3D momentum space [12]. Their existence has profound consequences for surface physics: the surface Brillouin zone contains open Fermi arcs connecting the projections of Weyl points of opposite chirality [6, 93].

15.2 The three-dimensional Maxwell eigenproblem

15.2.1 Extension to three dimensions

The Maxwell eigenvalue equation $\hat{\mathcal{N}}(\mathbf{k})\mathbf{u} = \omega\mathbf{M}\mathbf{u}$ extends directly to three dimensions. The Bloch wavevector $\mathbf{k} = (k_x, k_y, k_z)$ lives in a 3D Brillouin zone, which for a cubic lattice is a cube $[-\pi/a, \pi/a]^3$ (with opposite faces identified, forming a 3-torus T^3). The cell-periodic eigenmodes $\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$ and eigenfrequencies $\omega_{n\mathbf{k}}$ depend on all three components of $\mathbf{k}$.

15.2.2 Berry curvature as a vector field

In 3D, the Berry curvature is naturally a vector field in $\mathbf{k}$ -space:

$$\Omega_n(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathcal{A}_n(\mathbf{k}) , \quad (15.1)$$

where $\mathcal{A}_n(\mathbf{k}) = i\langle \mathbf{u}_{n\mathbf{k}} | \mathbf{M}_g | \nabla_{\mathbf{k}} \mathbf{u}_{n\mathbf{k}} \rangle$ is the Berry connection vector. The Berry curvature satisfies $\nabla_{\mathbf{k}} \cdot \Omega_n = 0$ everywhere except at band degeneracies, where it has point sources or sinks (monopoles).

15.2.3 Chern numbers on 2D slices

For any 2D slice of the 3D Brillouin zone at fixed k_z , we can compute the Chern number:

$$\mathcal{C}_n(k_z) = \frac{1}{2\pi} \int_{k_z=\text{const}} \Omega_{n,z} dk_x dk_y \quad (15.2)$$

The displayed formula above follows the convention and normalisation used in [44]. This is the “Chern number of the k_z -slice.” It is an integer that can only change when the slice passes through a Weyl point (where the band gap closes). The change in $\mathcal{C}_n(k_z)$ across a Weyl point equals the chirality of that Weyl point.

15.3 Weyl points as Berry monopoles

15.3.1 The 3D Weyl Hamiltonian

Near a Weyl point at $\mathbf{k} = \mathbf{k}_W$ in a 3D band structure, the effective 2×2 Hamiltonian takes the form

$$H_W(\delta\mathbf{k}) = \sum_{a=x,y,z} v_a \delta k_a \sigma_a, \quad (15.3)$$

where $\delta\mathbf{k} = \mathbf{k} - \mathbf{k}_W$ and v_a are the anisotropic velocities. This is the 3D generalisation of the 2D massless Dirac equation, but with all three Pauli matrices present.

The eigenvalues are

$$E_{\pm}(\delta\mathbf{k}) = \pm \sqrt{v_x^2 \delta k_x^2 + v_y^2 \delta k_y^2 + v_z^2 \delta k_z^2}, \quad (15.4)$$

giving a point-like band touching with linear (conical) dispersion in all three momentum directions. The constant-energy surfaces are ellipsoids centred at $\mathbf{k}_W$.

15.3.2 Derivation of the Berry curvature monopole

For the isotropic case $v_x = v_y = v_z \equiv v$, the Weyl Hamiltonian reduces to $H = v \delta\mathbf{k} \cdot \boldsymbol{\sigma}$. Using the general formula for a two-band Hamiltonian $H = \mathbf{d} \cdot \boldsymbol{\sigma}$:

$$\Omega_{n,a} = -\frac{\chi}{2} \epsilon_{abc} \hat{d}_b \left(\frac{\partial \hat{d}_c}{\partial \delta k_d} \times \frac{\partial \hat{d}_d}{\partial \delta k_e} \right)_{de} \Rightarrow \Omega_- = -\frac{\chi}{2} \frac{\delta\mathbf{k}}{|\delta\mathbf{k}|^3}, \quad (15.5)$$

where $\chi = \text{sgn}(\det[v_{ab}]) = \pm 1$ is the chirality.

More explicitly, for $\mathbf{d} = v \delta\mathbf{k}$ and $\hat{\mathbf{d}} = \delta\mathbf{k}/|\delta\mathbf{k}|$:

$$\Omega_{-,x} = -\frac{\chi}{2} \frac{\delta k_x}{(\delta k_x^2 + \delta k_y^2 + \delta k_z^2)^{3/2}}, \quad (15.6)$$

$$\Omega_{-,y} = -\frac{\chi}{2} \frac{\delta k_y}{(\delta k_x^2 + \delta k_y^2 + \delta k_z^2)^{3/2}}, \quad (15.7)$$

$$\Omega_{-,z} = -\frac{\chi}{2} \frac{\delta k_z}{(\delta k_x^2 + \delta k_y^2 + \delta k_z^2)^{3/2}}. \quad (15.8)$$

This is the field of a magnetic monopole in momentum space, with charge $-\chi$. It diverges at $\delta\mathbf{k} = \mathbf{0}$ (the Weyl point) and falls off as $1/|\delta\mathbf{k}|^2$.

15.3.3 Topological charge and flux quantisation

The flux of Berry curvature through any closed surface S enclosing the Weyl point is quantised:

$$\boxed{\frac{1}{2\pi} \oint_S \Omega_- \cdot d\mathbf{S} = -\chi = \mp 1.} \quad (15.9)$$

Proof. For a sphere of radius R centred at the Weyl point:

$$\begin{aligned} \frac{1}{2\pi} \oint_{S^2} \Omega_- \cdot d\mathbf{S} &= \frac{1}{2\pi} \int_0^\pi \int_0^{2\pi} \left(-\frac{\chi}{2R^2}\right) R^2 \sin \theta \, d\theta \, d\phi \\ &= -\frac{\chi}{2} \cdot \frac{1}{2\pi} \cdot 4\pi = -\chi. \end{aligned} \quad (15.10)$$

□

This is the 3D analog of the half-integer Berry flux at a 2D Dirac point, but in 3D the flux is *integer* for a single Weyl point (because the enclosing surface is a sphere S^2 , which is a closed orientable 2-manifold).

15.3.4 Topological robustness

A key difference from 2D Dirac points: a Weyl point in 3D *cannot be gapped by any local perturbation*.

Proposition 15.1. *No perturbation of the form $\delta H = \sum_a \delta d_a \sigma_a + \delta d_0 \mathbf{I}$ can open a gap at a Weyl point. It can only shift the Weyl point in $\mathbf{k}$ -space.*

Proof. The perturbed Hamiltonian is $H' = \sum_a (v_a \delta k_a + \delta d_a) \sigma_a + \delta d_0 \mathbf{I}$. The gap closes when $v_a \delta k_a + \delta d_a = 0$ for all a , i.e., at $\delta k_a = -\delta d_a / v_a$. The Weyl point has moved to $\mathbf{k}'_W = \mathbf{k}_W - (\delta d_x / v_x, \delta d_y / v_y, \delta d_z / v_z)$, but it has not been removed. A general 2×2 Hermitian matrix has 4 parameters (d_0, d_x, d_y, d_z), of which 3 must vanish for a degeneracy; with 3 free parameters ($\delta k_x, \delta k_y, \delta k_z$), this is generically satisfied at isolated points. □

To remove a Weyl point, it must be brought together with another Weyl point of *opposite* chirality, where they annihilate. This is the momentum-space analog of monopole-antimonopole annihilation in electrostatics.

This annihilation process is illustrated schematically in Figure 15.1. Panel (a) shows two well-separated Weyl nodes of opposite chirality (blue = -1 , red = $+1$) along a one-dimensional momentum cut, each carrying a linear band crossing. As a tuning parameter is varied, the nodes move toward each other in panel (b). At the critical point—panel (c)—the two charges become coincident and the dispersion along the merging direction changes from linear to quadratic, reflecting the cancellation of the net Berry monopole charge. Beyond the critical point, a full gap opens as shown in panel (d), with gap E_g set by the distance past the transition. The entire sequence is the momentum-space counterpart of monopole-antimonopole pair annihilation in classical electrostatics.

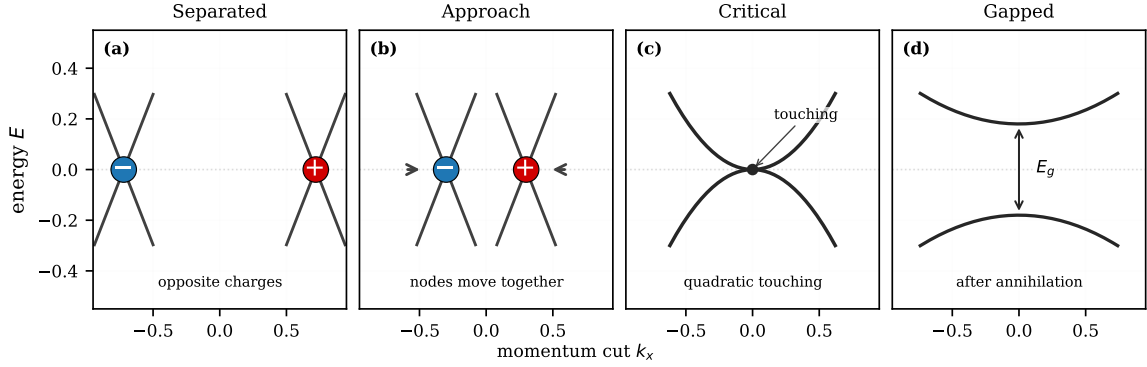


Figure 15.1: Weyl-point pair annihilation along a one-dimensional momentum cut k_x . (a) Two separated Weyl nodes of opposite chirality: blue ($\chi = -1$), red ($\chi = +1$), each with a local linear crossing. (b) The nodes approach as a lattice parameter is tuned. (c) At the critical point the charges are coincident and the dispersion becomes quadratic along the merging direction; the net monopole charge vanishes. (d) Beyond the transition a gap E_g opens; the resulting phase is topologically trivial in the Weyl sense.

15.3.5 The Nielsen–Ninomiya theorem in 3D

In a compact 3D Brillouin zone (a torus T^3), the total monopole charge must vanish:

$$\sum_i \chi_i = 0. \quad (15.11)$$

Proof. The total Berry flux through the boundary of the Brillouin zone is zero (because the torus has no boundary). By Gauss’s theorem, the total enclosed charge vanishes. $\square$

This means Weyl points always come in pairs of opposite chirality, analogous to the pairing of 2D Dirac points (the fermion-doubling theorem).

15.4 Weyl points in photonic crystals

15.4.1 Symmetry requirements

A Weyl point requires the breaking of either time-reversal ($\hat{\mathcal{T}}$) or inversion ($\hat{\mathcal{P}}$) symmetry (or both). The reason is as follows.

If both $\hat{\mathcal{T}}$ and $\hat{\mathcal{P}}$ are present, the combined operation $\hat{\mathcal{T}}\hat{\mathcal{P}}$ is an anti-unitary symmetry that maps $\mathbf{k}$ to itself (because $\hat{\mathcal{T}} : \mathbf{k} \rightarrow -\mathbf{k}$ and $\hat{\mathcal{P}} : \mathbf{k} \rightarrow -\mathbf{k}$, so $\hat{\mathcal{T}}\hat{\mathcal{P}} : \mathbf{k} \rightarrow \mathbf{k}$). Since $(\hat{\mathcal{T}}\hat{\mathcal{P}})^2 = \hat{\mathcal{T}}^2\hat{\mathcal{P}}^2 = +1$ for photons ($\hat{\mathcal{T}}^2 = +1$, $\hat{\mathcal{P}}^2 = +1$), the anti-unitary constraint does not force Kramers degeneracy. However, the real-symmetric structure of the combined eigenproblem typically enforces at least a twofold degeneracy at every $\mathbf{k}$ point, making the minimal band touching a four-fold Dirac point rather than a two-fold Weyl point.

Breaking either $\hat{\mathcal{T}}$ or $\hat{\mathcal{P}}$ removes this constraint and allows isolated two-fold Weyl points.

15.4.2 Classification of photonic Weyl systems

Symmetry broken	$\hat{\mathcal{T}}$	$\hat{\mathcal{P}}$	Example systems
Inversion only	yes	no	Chiral photonic crystals, asymmetric gyroid
Time reversal only	no	yes	Gyromagnetic 3D crystals
Both	no	no	General magneto-optic + chiral

The case of broken $\hat{\mathcal{P}}$ with preserved $\hat{\mathcal{T}}$ is most common in photonic experiments, because chiral 3D structures can be fabricated by multi-layer lithography or 3D printing without requiring magneto-optical materials [94].

15.4.3 The double gyroid photonic crystal

One of the first predicted photonic Weyl-point systems is based on the gyroid minimal surface [12, 49]: a triply periodic surface that divides space into two interpenetrating labyrinthine channels.

In the *single* gyroid structure, the photonic crystal has body-centred-cubic symmetry with both $\hat{\mathcal{T}}$ and $\hat{\mathcal{P}}$. The band structure shows Dirac points (four-fold) at high-symmetry points.

In the *double* gyroid, the two channels are filled with different dielectric materials ($\varepsilon_1 \neq \varepsilon_2$), breaking $\hat{\mathcal{P}}$ while preserving $\hat{\mathcal{T}}$. Each four-fold Dirac point splits into two Weyl points of opposite chirality, separated in $\mathbf{k}$ -space by an amount proportional to the asymmetry $|\varepsilon_1 - \varepsilon_2|$.

The positions of the Weyl points in the Brillouin zone are constrained by the remaining symmetries (C_3 rotations along the body diagonal), which force the Weyl points to lie along high-symmetry lines.

15.4.4 Stacked 2D photonic crystals

A simpler approach creates Weyl points by stacking identical 2D photonic crystal layers with a controlled interlayer coupling. If each 2D layer has Dirac points at K and K' , the interlayer coupling disperses these degeneracies along k_z , creating either:

- *A nodal line*: a continuous ring of degeneracies in (k_x, k_y, k_z) space (when $\hat{\mathcal{T}} + \hat{\mathcal{P}}$ is preserved) [11].
- *Weyl points*: isolated degeneracies (when $\hat{\mathcal{P}}$ or $\hat{\mathcal{T}}$ is broken by the stacking geometry or material choice).

The interlayer coupling strength t_z controls the k_z -dispersion and hence the separation of Weyl points.

15.5 Surface states and Fermi arcs

15.5.1 The surface Brillouin zone

Consider a semi-infinite 3D photonic crystal with a surface perpendicular to $\hat{x}$. The surface states are labelled by the surface-parallel wavevector $\mathbf{k}_{\parallel} = (k_y, k_z)$ and form a 2D band structure $\omega_{\text{surf}}(\mathbf{k}_{\parallel})$ [5].

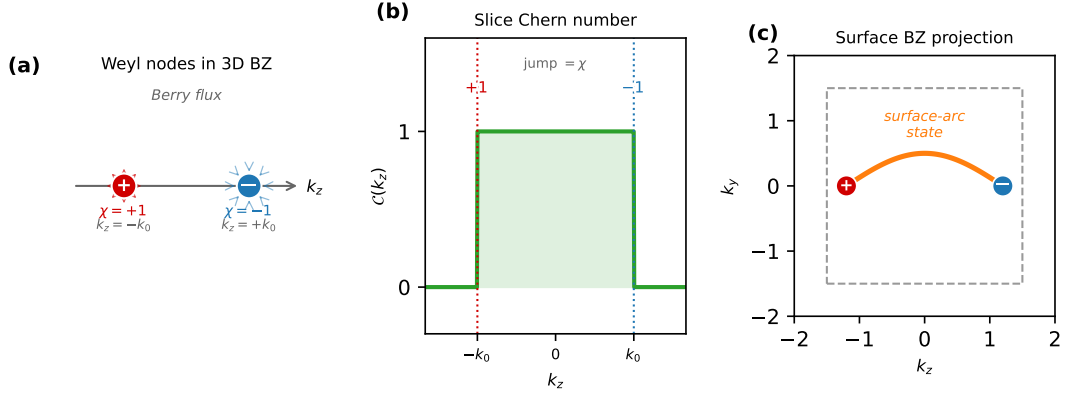


Figure 15.2: Weyl points as Berry-curvature monopoles. **(a)** Two Weyl nodes at $k_z = \pm k_0$ with chiralities $\chi = +1$ (source, red) and $\chi = -1$ (sink, blue); Berry-flux arrows indicate the monopole and anti-monopole character. **(b)** The 2D slice Chern number $\mathcal{C}(k_z)$ jumps by $+\chi$ when the slice passes through a Weyl node, giving the step function $0 \rightarrow 1 \rightarrow 0$ for the lower band. **(c)** Surface Brillouin zone showing the photonic surface-arc state (orange) connecting the projected Weyl nodes. The arc exists only for k_z values where $\mathcal{C}(k_z) \neq 0$, as required by the bulk–edge correspondence applied slice by slice.

15.5.2 Derivation of Fermi arcs from slice Chern numbers

The existence of surface states can be understood through the k_z -dependent Chern number $\mathcal{C}_n(k_z)$ defined in (15.2).

Consider the 2D slice at a given k_z . If $\mathcal{C}_n(k_z) \neq 0$, the bulk-edge correspondence for 2D systems (Theorem 8.3) predicts $|\mathcal{C}_n(k_z)|$ chiral edge modes at the surface. These edge modes have a definite group velocity along k_y , and appear at a specific frequency $\omega_{\text{edge}}(k_y, k_z)$.

As k_z sweeps between two Weyl points of opposite chirality at $k_{z,1}$ and $k_{z,2}$, the Chern number changes from $+1$ to 0 (or 0 to -1 , etc.). In the region where $\mathcal{C}(k_z) = 1$, there is one surface mode; outside this region, there is none. The surface mode therefore exists only for $k_z \in (k_{z,1}, k_{z,2})$, tracing an *open arc* in the surface Brillouin zone.

At a fixed frequency equal to the Weyl-point frequency, this arc is called a *Fermi arc* (by analogy with the electronic case). The Fermi arc connects the projections of the two Weyl points onto the surface Brillouin zone.

Remark 15.2. The Fermi arcs are *open* curves, not closed loops. This is topologically possible because the arcs terminate at the projected Weyl points, where the surface states merge into the bulk continuum. Open Fermi arcs have no analog in 2D topological systems, where edge modes always form closed dispersion curves crossing the bulk gap.

Figure 15.2 summarises this structure schematically.

15.5.3 Fermi-arc connectivity

The connectivity of Fermi arcs is topologically constrained: each Weyl point must be the endpoint of exactly $|\chi|$ arcs. For charge-1 Weyl points ($|\chi| = 1$), each is connected to exactly one other Weyl point of opposite chirality by a single arc. The arc cannot terminate in the interior of the surface Brillouin zone.

When multiple pairs of Weyl points are present, the arcs can connect them in different patterns, depending on the surface orientation and termination. Different surface terminations of the same bulk crystal can produce different arc connectivity patterns.

15.6 Experimental observation of photonic Weyl points

15.6.1 Microwave experiments

The first experimental observations of photonic Weyl points were achieved in 3D-printed metallic structures at microwave frequencies [13, 95]. The experiments measured:

- The bulk band structure, confirming the linear dispersion near the Weyl point.
- The surface states, showing the characteristic Fermi-arc connectivity.
- The robustness of the Weyl points to perturbations (they move but do not gap).

15.6.2 Photonic crystal observations

Weyl points have also been observed in [12, 95]:

- Stacked dielectric photonic crystal slabs with controlled interlayer coupling.
- Chiral woodpile structures (alternating layers of rods with rotation between layers).
- Metamaterial waveguide arrays.

15.7 Nodal lines and higher-order degeneracies

15.7.1 Nodal lines

A *nodal line* is a closed loop of band degeneracies in $\mathbf{k}$ -space. Unlike Weyl points (which are 0-dimensional), nodal lines are 1-dimensional objects. They carry a Berry phase of π along any path linking the loop:

$$\gamma = \oint_C \mathcal{A} \cdot d\mathbf{k} = \pi \quad \text{for } C \text{ linking the nodal line once.} \quad (15.12)$$

This π Berry phase is protected by a mirror or glide symmetry that forces the eigenstates on the nodal line to be degenerate.

Nodal-line photonic crystals produce “drumhead” surface states: surface states that fill a 2D region in the surface Brillouin zone bounded by the projection of the nodal line, rather than forming a 1D arc.

15.7.2 Multi-Weyl points

At high-symmetry points in crystals with C_n ($n \geq 3$) rotational symmetry, Weyl points can carry higher topological charges $|\chi| > 1$. A *double-Weyl point* ($|\chi| = 2$) has quadratic dispersion in the plane perpendicular to the rotation axis and linear dispersion along it:

$$H = v_{\perp}(k_{+}^2\sigma_{+} + k_{-}^2\sigma_{-}) + v_z\delta k_z\sigma_z, \quad (15.13)$$

where $k_{\pm} = \delta k_x \pm i\delta k_y$ and $\sigma_{\pm} = (\sigma_x \pm i\sigma_y)/2$.

A double-Weyl point sources two units of Berry flux and is the endpoint of two Fermi arcs.

15.7.3 Dirac points in 3D

When both $\hat{\mathcal{T}}$ and $\hat{\mathcal{P}}$ are present, the minimal degeneracy at a band crossing is a four-fold *Dirac point*: two overlapping Weyl points of opposite chirality. Breaking $\hat{\mathcal{T}}$ or $\hat{\mathcal{P}}$ splits the Dirac point into a pair of Weyl points. The separation between the Weyl points is controlled by the symmetry-breaking strength.

15.8 Connection to Maxwell equations

The Weyl-point physics described in this chapter is derived from the same Maxwell eigenvalue problem $\hat{\mathcal{N}}(\mathbf{k})\mathbf{u} = \omega\mathbf{M}\mathbf{u}$ that underlies the entire book. The 3D extension is straightforward: $\mathbf{k}$ now has three components, the Brillouin zone is a 3D torus T^3 , and the Berry curvature is a vector field in 3D $\mathbf{k}$ -space.

The electromagnetic inner product $\langle \mathbf{u} | \mathbf{M}_g | \mathbf{u}' \rangle$ remains essential for the correct definition of Berry curvature and topological charges. Using the wrong inner product would give incorrect monopole charges and Fermi-arc predictions.

The slice Chern number $\mathcal{C}_n(k_z)$ is computed using the same plaquette formula of Chapter 10, applied to the 2D slice at fixed k_z . The Wilson-loop method extends naturally: the Wilson loop at each (k_y, k_z) gives a Zak phase whose winding in k_y at fixed k_z yields $\mathcal{C}_n(k_z)$.

15.9 Three-dimensional gapped topological phases

While Weyl semimetals are gapless—their defining feature is the point-like band touching—three-dimensional photonic systems can also host *gapped* topological phases [96, 97]. These are the 3D analogs of the 2D Chern insulator and the 2D $\mathbb{Z}_2$ topological insulator: the bulk has a complete frequency gap, and the topology manifests as robust surface states on every boundary.

15.9.1 From 2D to 3D: stacking and strong invariants

A simple route to a 3D topological insulator begins with a stack of 2D topological layers. If each layer is a 2D Chern insulator with $\mathcal{C} = 1$, the stacked system has chiral surface states on every face perpendicular to the stacking axis. This “weak” topological insulator inherits its topology from the 2D layers and is vulnerable to disorder that mixes layers.

A *strong* 3D topological insulator, by contrast, has surface states on *every* surface orientation, protected by a truly 3D topological invariant. In electronic systems, this invariant is the $\mathbb{Z}_2$ index protected by time-reversal symmetry (Kane and Mele, 2005; Fu, Kane, and Mele, 2007). The photonic analog requires a pseudo-time-reversal symmetry, typically constructed from bianisotropic coupling between TE-like and TM-like modes.

15.9.2 The bianisotropic route to a 3D photonic TI

The most successful experimental realisation of a 3D photonic topological insulator was demonstrated by Yang et al. [69] using an array of metallic split-ring resonators (SRRs) embedded in a dielectric host. The SRRs provide strong bianisotropic coupling—the magneto-electric cross-terms ξ and ζ of the material matrix (Appendix D)—which plays the role of spin-orbit coupling in the electromagnetic context.

The key design principle is as follows. Each SRR supports both an electric resonance (responding to $\mathbf{E}$) and a magnetic resonance (responding to $\mathbf{H}$) at the same frequency, with a cross-coupling $\xi \neq 0$ that mixes the two. When the SRRs are arranged on a 3D lattice, the resulting photonic band structure exhibits a complete bulk bandgap with topologically nontrivial bands below the gap.

The measured bulk bandgap extended from approximately 4.3 to 6.0 GHz (a relative bandwidth exceeding 25%), which is remarkably large compared to earlier 2D photonic topological gaps of a few percent [12]. The large gap makes the surface states well-resolved and experimentally accessible.

15.9.3 Surface Dirac cones

The hallmark of a 3D gapped topological insulator is a single *Dirac cone* on each surface—a linearly dispersing surface state that is topologically protected against backscattering by disorder or defects.

In the Yang et al. experiment, near-field scanning measurements of the surface electromagnetic field revealed:

1. A surface state with linear (conical) dispersion centred at 5.1 GHz, lying within the bulk bandgap.
2. Robust surface propagation along non-planar surfaces, including bends and corners, without backscattering.
3. A penetration depth of approximately 10 mm into the bulk, consistent with the exponential localisation expected from a gapped topological surface state.

The surface Dirac cone should be contrasted with the Fermi arcs of a Weyl semimetal (Section 15.5). A Fermi arc is an *open* curve in the surface Brillouin zone, terminating at projected Weyl points; a surface Dirac cone is a *closed* conical dispersion centred at a time-reversal-invariant momentum. The arc exists because the bulk is gapless (Weyl points); the Dirac cone exists because the bulk is fully gapped.

15.9.4 Comparison of 3D gapped and gapless phases

The following table summarises the key differences between 3D gapless (Weyl) and gapped (topological insulator) phases in photonic crystals.

	3D Weyl semimetal	3D topological insulator
Bulk gap	none (point nodes)	complete frequency gap
Topological invariant	monopole charge χ	$\mathbb{Z}_2$ or 3D Chern numbers
Surface states	open Fermi arcs	closed Dirac cone
Symmetry requirement	break $\hat{\mathcal{T}}$ or $\hat{\mathcal{P}}$	pseudo- $\hat{\mathcal{T}}$ from bianisotropy
Robustness	nodes move, do not gap	gap persists under all symmetry-preserving perturbations
Key photonic experiment	Noh et al. (2017)	Yang et al. (2019)

Both phases are derived from the same 3D Maxwell eigenproblem $\hat{\mathcal{N}}(\mathbf{k})\mathbf{u} = \omega\mathbf{M}\mathbf{u}$, but with different material symmetries: broken $\hat{\mathcal{P}}$ or $\hat{\mathcal{T}}$ for Weyl semimetals, and engineered pseudo- $\hat{\mathcal{T}}$ for gapped topological insulators.

15.9.5 Towards optical-frequency 3D topological insulators

The Yang et al. demonstration operated at microwave frequencies, where SRR fabrication is straightforward. Extending 3D photonic topological insulators to optical frequencies is an active research frontier [13]. The challenges include fabricating 3D nanostructures with precise bianisotropic response and maintaining a sufficiently large bandgap relative to fabrication imperfections. Recent proposals based on all-dielectric metacrystals and chiral woodpile structures suggest that optical-frequency 3D topological gaps may be achievable with current nanofabrication technology [98].

15.10 Type-I and type-II Weyl points

Weyl points come in two types, distinguished by their isofrequency surfaces [99]. A type-I Weyl point has a point-like Fermi surface at the Weyl frequency: the isofrequency contour shrinks to a single point, with the dispersion forming an upright cone $\omega = \omega_W \pm v|\delta\mathbf{k}|$.

A type-II Weyl point has a tilted cone, so that the isofrequency surface at $\omega = \omega_W$ is not a point but a pair of electron- and hole-like pockets touching at the Weyl point. The effective Hamiltonian is $H = v_t k_z \sigma_0 + v(k_x \sigma_x + k_y \sigma_y + k_z \sigma_z)$, where $v_t > v$ is the tilt velocity. The tilt breaks Lorentz invariance but does not change the topological charge of the Weyl point [12, 95].

Type-II Weyl points were first observed in photonics by Noh et al. (2017) using a laser-written waveguide array. The waveguides were arranged in a face-centred geometry with controlled inter-layer coupling, producing tilted Weyl cones in the propagation-constant spectrum. The type-II nature was confirmed by observing the characteristic “bowtie” isofrequency contour using Fourier-space imaging [95].

15.11 Nodal lines and drumhead surface states

When two bands cross along a line in the 3D Brillouin zone rather than at an isolated point, the crossing is called a *nodal line*. Nodal lines can be stabilised by crystal symmetries (e.g. mirror or glide symmetries) and carry a π Berry phase around any loop encircling the line, analogous to the π Berry phase at a 2D Dirac point [12, 100].

The bulk-boundary correspondence for nodal lines predicts *drumhead surface states*: flat or nearly flat bands that fill the interior of the projected nodal line on the surface Brillouin zone. These states have a high density of states and are of interest for enhanced light–matter interaction.

Nodal-line photonic crystals have been realised using dielectric structures with carefully designed symmetries. The drumhead states have been observed as bright, nearly dispersionless features in angle-resolved transmission measurements [13].

15.12 Experimental platforms for 3D photonic topology

Three-dimensional photonic topological phases have been demonstrated on several platforms:

Laser-written waveguide arrays in fused silica provide type-II Weyl points and Fermi-arc surface states at optical wavelengths ($\lambda = 633$ nm). The paraxial propagation approximation maps the 3D waveguide structure onto a (2+1)D tight-binding model with the propagation direction playing the role of time [16, 95].

Metallic photonic crystals with gyroid or diamond symmetry can support complete 3D band gaps with nontrivial topology. The Weyl points appear as pairs of opposite-charge monopoles in the 3D Brillouin zone, connected by surface Fermi arcs on any termination [12].

Bianisotropic metamaterials (split-ring resonator arrays) realise 3D topological insulators with surface Dirac cones, as discussed in Chapter 9 [13, 69].

15.13 Surface Dirac cones and topological surface states

A 3D photonic topological insulator—a material with a nontrivial $\mathbb{Z}_2$ invariant—supports gapless surface states that form a Dirac cone in the surface Brillouin zone. These surface Dirac cones are the 3D analog of the chiral edge modes of a 2D Chern insulator: they are topologically protected and cannot be gapped by any surface perturbation that preserves time-reversal symmetry.

The surface Dirac cone was first observed experimentally by Yang et al. (2019) in a bianisotropic metamaterial of double split-ring resonators arranged in a body-centred cubic lattice. Angle-resolved transmission measurements on a cleaved crystal surface revealed a linear dispersion cone at $\bar{\Gamma}$ in the surface Brillouin zone, with the Dirac frequency at 5.1 GHz inside the bulk gap from 4.3 to 6.0 GHz [12, 69].

The robustness of the surface Dirac cone was confirmed by introducing surface disorder (randomly removing 10% of the surface resonators) and observing that the Dirac cone remained intact, with only a slight broadening of the surface states. This is consistent with the $\mathbb{Z}_2$ protection: the surface Dirac cone can be shifted in energy or broadened by disorder, but it cannot be gapped [13].

15.13.1 Comparison table: 2D edge modes versus 3D surface states

The 2D and 3D cases share the same topological origin but differ in dimensionality and symmetry:

Property	2D Chern insulator	3D topological insulator
Bulk invariant	Chern number $\mathcal{C} \in \mathbb{Z}$	$\mathbb{Z}_2$ index $\nu \in \{0, 1\}$
Boundary states	chiral edge modes	surface Dirac cone
Protection	against all gap-preserving perturbations	against $\hat{\mathcal{T}}$ -preserving perturbations
Requires $\hat{\mathcal{T}}$ breaking	yes	no
Min. boundary modes	$ \mathcal{C} $	1 Dirac cone

15.14 Photonic Weyl semimetal experiments

The theoretical prediction of Weyl points in photonic crystals was confirmed experimentally in several landmark experiments that directly observed the key signatures: bulk conical dispersions, surface Fermi arcs, and robustness against perturbations.

15.14.1 The double-gyroid photonic crystal

The first experimental observation of photonic Weyl points was reported by Lu et al. (2015) in a double-gyroid photonic crystal at microwave frequencies [13]. The double gyroid is a bicontinuous structure with body-centred cubic symmetry described by the level-set equation

$$f(\mathbf{r}) = \sin(2\pi x/a) \cos(2\pi y/a) + \sin(2\pi y/a) \cos(2\pi z/a) + \sin(2\pi z/a) \cos(2\pi x/a), \quad (15.14)$$

where the two gyroid networks occupy the regions $f > t_0$ and $f < -t_0$ for a threshold t_0 . The structure was fabricated by 3D-printing metallic scaffolds with lattice constant $a = 13$ mm, operating in the 5–8 GHz range.

By breaking either time-reversal or inversion symmetry (achieved by coating one of the two gyroid networks with a metallic layer to destroy inversion), the authors lifted a four-fold line degeneracy into pairs of Weyl points with opposite chirality. Angle-resolved microwave transmission measurements mapped the bulk band structure and confirmed the linear conical dispersion near the Weyl frequencies [12].

15.14.2 Surface Fermi arcs

The surface Fermi arc is the defining experimental signature of a Weyl semimetal: it is an open contour of constant-frequency surface states that connects the projections of two Weyl points of opposite chirality onto the surface Brillouin zone.

The existence of Fermi arcs follows from a topological argument. Consider a closed loop $\mathcal{C}$ in the surface Brillouin zone that encircles the projection of a single Weyl point. The Chern number of the occupied bands evaluated on $\mathcal{C}$ is $\mathcal{C}(\mathcal{C}) = \pm 1$, implying that at least one surface band must

cross the Weyl-point frequency within $\mathcal{C}$. Since the surface band cannot form a closed contour (it would then encircle equal numbers of positive and negative chirality projections and carry zero Chern flux), it must terminate at the Weyl-point projections. The resulting open arc is the Fermi arc.

In the double-gyroid experiment, surface Fermi arcs were observed by measuring the near-field intensity distribution on a cleaved crystal surface. The arc-like surface dispersion was clearly visible in the angle-resolved transmission, connecting pairs of projected Weyl points as predicted by the bulk-boundary correspondence [12, 13].

15.14.3 Type-II Weyl points in photonic systems

Type-II Weyl points, introduced in Section 15.10, are characterised by a strongly tilted cone that intersects the constant-frequency surface to form electron and hole pockets. In a photonic crystal, the “pockets” are regions of propagating bulk modes that coexist with the Weyl frequency, leading to qualitatively different surface-state physics: the Fermi arcs connect not only the Weyl-point projections but also merge with the bulk pocket projections [12].

Noh et al. (2017) reported the first observation of type-II Weyl points in a photonic system using a stacked bilayer photonic crystal [95]. The structure consisted of alternating layers of hexagonal photonic crystal slabs with different in-plane geometries, creating a three-dimensional photonic crystal with broken inversion symmetry. The strongly tilted conical dispersion was confirmed by angle-resolved reflection measurements, and the characteristic electron–hole pocket structure was observed at frequencies near the Weyl point.

15.14.4 Ideal Weyl points and minimal photonic designs

An “ideal” Weyl semimetal has its Weyl points located exactly at the Fermi level (or, in the photonic case, at a frequency isolated from all other bands), so that the only states at that frequency are the Weyl cones and the surface arcs. This ideal condition maximises the length of the Fermi arcs and the surface-state density of states.

Yang et al. (2018) realised an ideal Weyl system using a saddle-shaped metallic inclusion in a body-centred cubic unit cell [69, 101]. The design produced Weyl points that were separated from all other bands by a complete frequency gap, giving the longest possible Fermi arcs spanning the entire surface Brillouin zone. The experiment was performed at microwave frequencies with 3D-printed metallic structures, and the surface arcs were mapped by near-field scanning measurements.

Exercises

Exercise 15.1. Compute the Berry curvature of the lower band of the isotropic Weyl Hamiltonian $H = v(\delta k_x \sigma_x + \delta k_y \sigma_y + \delta k_z \sigma_z)$ and verify that it has the monopole form (15.5) with $\chi = +1$.

Exercise 15.2. Show that no 2×2 Hermitian perturbation can gap the 3D Weyl Hamiltonian (15.3). *Hint:* write the most general 2×2 Hermitian matrix $\delta H = \sum_a \delta d_a \sigma_a + \delta d_0 \mathbf{I}$ and show that adding it to H_W only shifts the Weyl point.

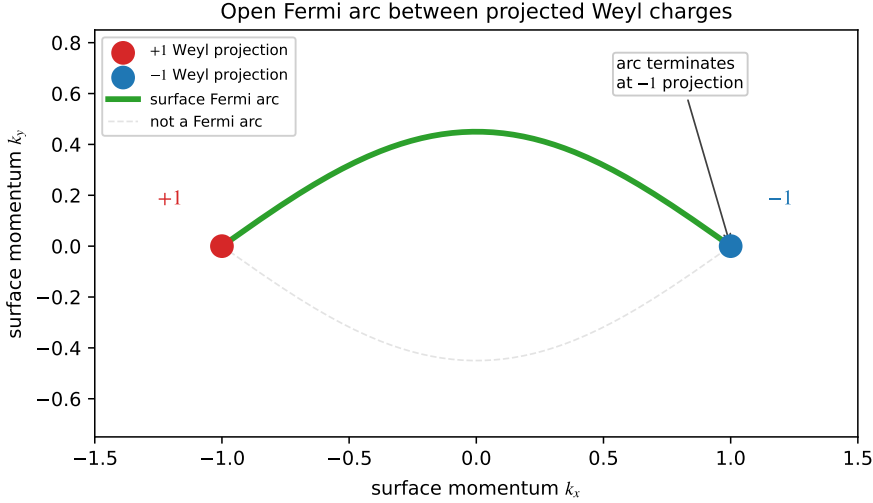


Figure 15.3: A surface Fermi arc connects projected Weyl charges of opposite chirality. Unlike an ordinary constant-frequency contour, the arc is open: it terminates on the surface projections of bulk Weyl points, expressing the Chern-number jump across the Weyl node.

Exercise 15.3. For a pair of Weyl points at $\mathbf{k}_{\pm} = (0, 0, \pm k_0)$ with chiralities $\chi_{\pm} = \pm 1$, compute the slice Chern number $\mathcal{C}(k_z)$ for $|k_z| < k_0$, $|k_z| > k_0$, and $k_z = \pm k_0$. Use this to derive the existence and extent of the Fermi arc on the (001) surface.

Exercise 15.4. Explain why Weyl points require the breaking of either $\hat{\mathcal{T}}$ or $\hat{\mathcal{P}}$ (or both). *Hint:* show that $\hat{\mathcal{T}}\hat{\mathcal{P}}$ symmetry forces the eigenstates to be at least doubly degenerate at every $\mathbf{k}$ in a system where both are preserved.

Exercise 15.5. A photonic nodal line in a 3D crystal is a closed loop in $\mathbf{k}$ -space where two bands are degenerate. Show that the Berry phase $\gamma = \oint_C \mathcal{A} \cdot d\mathbf{k}$ for a path C linking the nodal line once is quantised to π . What symmetry protects this quantisation?

Exercise 15.6. For a double-Weyl point with Hamiltonian (15.13), compute the Berry flux through a sphere enclosing the degeneracy and verify that the topological charge is $|\chi| = 2$.

Exercise 15.7. Consider a 3D photonic crystal with a single pair of Weyl points on the k_z axis. If the crystal is truncated to form a slab of thickness L in the x -direction, describe qualitatively how the Fermi arcs on the top and bottom surfaces connect through the slab as L decreases.

Chapter 16

Synthetic Dimensions

[5].

16.1 Beyond physical dimensions

The topological phenomena explored in Chapters 7–15 occur in systems with two or three spatial dimensions. But topology is not limited by the number of physical dimensions: it depends on the structure of the Hilbert space and the connectivity of the parameter manifold, not on the spatial geometry of the laboratory.

A *synthetic dimension* is an internal degree of freedom—such as the frequency, orbital angular momentum, spin, or mode index of a photon—that is reinterpreted as an extra spatial coordinate through engineered couplings. By coupling modes along the synthetic axis with controlled phases, one can create effective lattice models in dimensions higher than the physical space [102]. This opens a route to observing topological phases that have no direct analog in lower-dimensional systems, most notably the four-dimensional quantum Hall effect.

This chapter derives the synthetic-dimension framework from Maxwell’s equations [12, 49], explains how the Harper–Hofstadter model and higher-dimensional Chern insulators can be implemented in photonic systems, and surveys the principal experimental platforms [59].

16.2 Frequency as a synthetic dimension

16.2.1 The coupled-mode picture

Consider a ring resonator supporting a comb of equally spaced resonant frequencies $\omega_m = \omega_0 + m\Omega_{\text{FSR}}$ ($m = 0, \pm 1, \pm 2, \dots$), where $\Omega_{\text{FSR}} = 2\pi c/(n_g L)$ is the free spectral range, n_g is the group index, and L is the round-trip length [29]. In the absence of modulation, the modes are decoupled and the frequency index m labels a one-dimensional lattice with lattice constant Ω_{FSR} .

Now apply an electro-optic phase modulation at frequency $\Omega_{\text{mod}} = \Omega_{\text{FSR}}$ to the resonator. The modulation creates a time-dependent coupling between adjacent frequency modes. In the slowly varying envelope approximation, the amplitude $a_m(t)$ of mode m satisfies the coupled-mode equation

$$i \frac{\partial a_m}{\partial t} = \omega_m a_m + \kappa (e^{i\phi} a_{m-1} + e^{-i\phi} a_{m+1}), \quad (16.1)$$

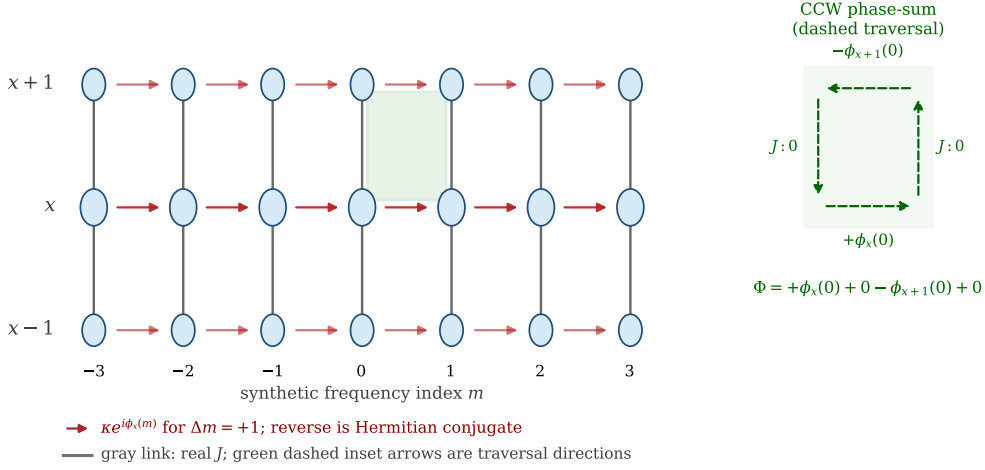


Figure 16.1: Synthetic frequency dimension in a modulated resonator array. Each column represents a frequency mode $\omega_m = \omega_0 + m\Omega$; each row represents a physical resonator at real-space site x . Red rightward arrows denote the Peierls hopping $\kappa e^{i\phi_x(m)}$ along the synthetic frequency direction ($\Delta m = +1$); the reverse ($\Delta m = -1$) hopping is the Hermitian conjugate [103, 104]. Gray vertical links represent the real-valued inter-site coupling J . The green shaded plaquette between columns $m = 0$ and $m = 1$ at sites x and $x+1$ carries a gauge-invariant flux $\Phi = \phi_x(0) - \phi_{x+1}(0)$, computed by summing the directed link phases counter-clockwise. In the inset this is displayed as $+\phi_x(0) + 0 - \phi_{x+1}(0) + 0$: the two vertical J links contribute zero phase. In the inset, the dashed green arrows show the traversal directions used for this phase sum, whereas the red arrows in the main lattice show the physical hopping direction $\Delta m = +1$. This flux is the synthetic-dimension analog of an applied magnetic field.

where κ is the modulation strength and ϕ is the modulation phase. This is mathematically identical to the tight-binding equation for a particle hopping on a 1D lattice with nearest-neighbour coupling $\kappa e^{\pm i\phi}$.

The frequency axis has thus become a *synthetic spatial dimension*: hopping along the frequency ladder is controlled by the electro-optic modulation, and the hopping phase ϕ plays the role of a Peierls phase (synthetic gauge field) [32].

16.2.2 Derivation from Maxwell's equations

To maintain the Maxwell-first spirit of this book, we derive the coupled-mode equation (16.1) from the time-dependent Maxwell equations with a modulated material matrix.

Consider a ring resonator whose permittivity is modulated:

$$\varepsilon(t) = \bar{\varepsilon} [1 + 2\eta \cos(\Omega_{\text{mod}} t - \phi)]. \quad (16.2)$$

Here $\eta \ll 1$ is the modulation depth. The minus sign in (16.2) is a convention: with fields written as $e^{-i\omega t}$, it makes ϕ the Peierls phase for the rightward frequency transition $m \rightarrow m+1$. If the physical

drive is instead written as $\cos(\Omega_{\text{mod}}t + \theta)$, then the Peierls phase used below is $\phi = -\theta$. The electric field in the resonator is expanded in the unperturbed mode basis: $E(t) = \sum_m a_m(t) \mathcal{E}_m e^{-i\omega_m t}$, where $\mathcal{E}_m$ is the mode profile of mode m .

Substituting into the Maxwell wave equation and using the rotating-wave approximation (retaining only the slowly varying terms), one obtains

$$i\dot{a}_m = \kappa_{\text{eff}}(e^{i\phi} a_{m-1} + e^{-i\phi} a_{m+1}), \quad (16.3)$$

The phase factors in (16.3) follow directly from the time convention just fixed: the coupling from $m - 1$ to m uses the $e^{-i\Omega_{\text{mod}}t + i\phi}$ component of the modulation, whereas the coupling from $m + 1$ to m uses the complex-conjugate component. Thus the rightward hopping carries $+\phi$ and the reverse hopping carries $-\phi$, matching Figure 16.1. The effective coupling is

$$\kappa_{\text{eff}} = \frac{\eta\omega_0}{2} \frac{\int |\mathcal{E}_m|^2 \varepsilon_0 dV}{\int |\mathcal{E}_m|^2 \bar{\varepsilon} dV} \approx \frac{\eta\omega_0}{2}. \quad (16.4)$$

This confirms that the coupled-mode equation (16.1) is a direct consequence of the Maxwell equations with a time-modulated permittivity.

16.3 The Harper–Hofstadter model from synthetic dimensions

16.3.1 Adding a physical dimension

To create a 2D lattice, we combine the synthetic frequency dimension with one physical spatial dimension. Consider an array of N coupled ring resonators, each supporting a frequency comb and each individually modulated. The physical coupling between adjacent rings provides hopping along the spatial direction x , and the intra-ring modulation provides hopping along the synthetic frequency direction m .

The amplitude $a_{m,x}(t)$ of mode m in ring x satisfies

$$i\dot{a}_{m,x} = J(a_{m,x-1} + a_{m,x+1}) + \kappa(e^{i\phi_x} a_{m-1,x} + e^{-i\phi_x} a_{m+1,x}), \quad (16.5)$$

where J is the inter-ring coupling and $\phi_x = \phi_0 + \alpha x$ is the modulation phase that varies linearly along the array.

16.3.2 Mapping to the Hofstadter Hamiltonian

In the Bloch-wave ansatz $a_{m,x} = e^{i(k_x x + k_m m)} b_{m,x}$, the equation of motion (16.5) becomes [12]

$$E b_{m,x} = 2J \cos(k_x) b_{m,x} + 2\kappa \cos(k_m + \alpha x + \phi_0) b_{m,x}, \quad (16.6)$$

which is the *Harper equation*, the eigenvalue equation of the Hofstadter model with flux α per plaquette. When $\alpha/(2\pi) = p/q$ is a rational number, the spectrum splits into q subbands, forming the famous Hofstadter butterfly.

The key insight is that the synthetic gauge flux α is fully controllable through the modulation phase gradient. By choosing $\alpha = 2\pi p/q$ for small q , one can access topological regimes (nonzero Chern numbers) with experimentally reasonable modulation parameters.

16.3.3 Topological bands and edge states

Each subband of the Harper equation carries a Chern number determined by the Diophantine equation $\mathcal{C}_r = \sigma_r$, where σ_r is the quantised Hall conductance of the r -th gap. For $\alpha/(2\pi) = 1/3$ (three subbands), the Chern numbers are $\mathcal{C}_1 = +1$, $\mathcal{C}_2 = -2$, $\mathcal{C}_3 = +1$.

In the synthetic-dimension realisation, the “edge” of the frequency lattice is set by the finite bandwidth of the modulation or the gain medium. Edge states appear at the boundary of the synthetic lattice and propagate unidirectionally along the physical direction, directly analogous to quantum Hall edge states.

16.4 Four-dimensional quantum Hall effect

16.4.1 Topology in four dimensions

In four spatial dimensions, the quantum Hall effect is characterised by the *second Chern number*, a topological invariant defined on the 4D Brillouin zone T^4 :

$$\mathcal{C}_2 = \frac{1}{8\pi^2} \int_{T^4} \text{Tr}(\mathcal{F} \wedge \mathcal{F}), \quad (16.7)$$

where $\mathcal{F}$ is the non-Abelian Berry curvature 2-form of the occupied bands (Appendix B). The 4D quantum Hall effect predicts a *nonlinear* Hall response: a current perpendicular to three applied fields, with a quantised coefficient proportional to $\mathcal{C}_2$.

The second Chern number is fundamentally different from the first Chern number: it requires at least four dimensions to be nonzero and detects topological features (“instanton-like” structures in the Berry connection) that have no analog in lower dimensions.

16.4.2 Synthetic 4D lattice

To realise a 4D topological system in a photonic experiment, one combines two physical dimensions with two synthetic dimensions. For example, a 2D array of ring resonators, each modulated at two frequencies Ω_1 and Ω_2 , creates a 4D lattice with coordinates (x, y, m_1, m_2) , where x, y are physical positions and m_1, m_2 are frequency-mode indices along the two synthetic axes.

The effective 4D Hamiltonian can be engineered to have nonzero $\mathcal{C}_2$ by choosing the modulation phases to create a non-trivial Berry connection on the 4D Brillouin zone.

16.4.3 Experimental signatures

The second Chern number manifests as a quantised nonlinear response [12, 13]. In a photonic 4D system, this appears as a pump-induced spectral flow: when the modulation phase is adiabatically varied (implementing a “magnetic field” in one of the synthetic planes), photons are pumped across bands in a direction perpendicular to the applied field and to the synthetic dimension, with a rate quantised by $\mathcal{C}_2$.

This effect was experimentally observed in a 2D photonic waveguide array with two synthetic dimensions implemented by adiabatic variation of coupling parameters along the propagation axis.

The measured spectral flow matched the theoretical prediction $\mathcal{C}_2 = 1$ for the engineered lattice model.

16.5 Other synthetic-dimension platforms

16.5.1 Orbital angular momentum

The orbital angular momentum (OAM) of light provides a natural synthetic dimension. Modes of a ring resonator or a multimode fibre carry a discrete OAM quantum number $\ell = 0, \pm 1, \pm 2, \dots$. Coupling between adjacent OAM modes can be induced by inserting an astigmatic element (such as a tilted mirror or a spatial light modulator) that mixes modes with $\Delta\ell = \pm 1$.

The coupling phase can be controlled by the orientation of the astigmatic element, providing a synthetic gauge field along the OAM axis. Combined with a physical spatial dimension, this creates a 2D lattice with controllable magnetic flux.

16.5.2 Temporal modes and time multiplexing

In a fibre loop, successive round-trip pulses form a 1D lattice in “time” (the discrete time step labelled by the round-trip number). Coupling between pulses is provided by a beam splitter or a variable coupler, and the coupling phase can be controlled electronically.

By combining the time-multiplexed lattice with a spatial array of fibre loops, one can create 2D and higher-dimensional effective lattices. This approach has been used to demonstrate the Hofstadter butterfly, anomalous Floquet topological phases, and Anderson localisation in synthetic lattices [92].

16.5.3 Spin and polarisation

The polarisation state of light (left/right circular, or horizontal/vertical) provides a two-site synthetic dimension. While this is too short to form a lattice by itself, it can serve as the “sublattice” degree of freedom in models such as the SSH chain or the Haldane model, with the spatial lattice providing the remaining dimension.

16.6 Quantitative analysis of synthetic-dimension experiments

To appreciate both the power and the limitations of synthetic dimensions, it is instructive to work through the key numbers for the two principal experimental platforms.

16.6.1 Ring-resonator frequency lattice

Consider a silicon nitride microring resonator with group index $n_g = 2.0$ and circumference $L = 2\pi R = 628 \mu\text{m}$ ($R = 100 \mu\text{m}$) [34]. The free spectral range is

$$\Omega_{\text{FSR}} = \frac{2\pi c}{n_g L} \approx 2\pi \times 150 \text{ GHz}, \quad (16.8)$$

so the frequency modes are spaced by 150 GHz in the C-band around $\lambda = 1550$ nm.

With electro-optic modulation depth $\eta = 0.01$ and carrier frequency $\omega_0/(2\pi) = 193$ THz, the effective hopping from (16.4) is

$$\kappa_{\text{eff}} \approx \frac{\eta \omega_0}{2} \approx 2\pi \times 0.97 \text{ THz}. \quad (16.9)$$

The ratio $\kappa_{\text{eff}}/\Omega_{\text{FSR}} \approx 6.5$ is much larger than unity, which *violates* the tight-binding assumption underlying (16.1): the modulation couples not only nearest-neighbour frequency modes but also next-nearest and more distant modes. To restore the tight-binding regime, one must reduce η to $\sim 10^{-4}$ or use a longer resonator with smaller FSR.

This quantitative tension between coupling strength and tight-binding validity is a central design challenge for synthetic-dimension experiments [12, 13].

16.6.2 Waveguide-array Floquet lattice

The Rechtsman helical-waveguide experiment (Chapter 13) can also be viewed as a synthetic-dimension platform, where the propagation coordinate z is the synthetic time axis. The key parameters are: helix radius $R_h = 8 \mu\text{m}$, pitch $\Lambda = 1 \text{ cm}$, inter-waveguide spacing $d = 22 \mu\text{m}$, and operating wavelength $\lambda = 633 \text{ nm}$ in borosilicate glass ($n_0 = 1.46$) [16].

The effective modulation frequency is $\Omega_{\text{mod}} = 2\pi/\Lambda \approx 628 \text{ m}^{-1}$, and the inter-waveguide coupling is $J \approx 1.0 \text{ cm}^{-1}$. The effective gauge flux per plaquette is $\Phi \approx k_0 R_h^2 (2\pi/\Lambda)^2 a/(2\pi)$, where $k_0 = 2\pi n_0/\lambda$ and $a = d\sqrt{3}/2$ is the lattice constant. For the experimental parameters, $\Phi \approx 0.21 \times 2\pi$, which places the system in a topological regime with Chern number $\mathcal{C} = \pm 1$.

The topological gap width is approximately $\Delta J \sim J \sin(\pi\Phi/\Phi_0) \approx 0.6 \text{ cm}^{-1}$, corresponding to a localisation length of $\ell_{\text{edge}} \sim v_g/\Delta J \approx 17 \mu\text{m}$ for the edge mode. The sample length of 10 cm then corresponds to ~ 10 modulation periods and ~ 600 localisation lengths, so the topological protection is well developed.

16.7 Limitations and outlook

Synthetic dimensions offer remarkable flexibility in engineering effective Hamiltonians [13], but they come with important limitations.

The synthetic lattice is typically *finite*: the frequency comb of a ring resonator has a finite bandwidth set by the gain or loss spectrum, and the OAM modes are limited by the aperture of the optical elements. This finiteness truncates the synthetic dimension and creates hard boundaries, which can host edge states but also introduce finite-size effects that complicate the interpretation of topological signatures.

Interactions between photons in synthetic-dimension systems are generally absent, because the photon-photon interaction is negligible in most optical media. This limits the exploration of strongly correlated topological phases (such as fractional quantum Hall states) in synthetic-dimension photonic systems.

Looking forward, the combination of synthetic dimensions with nonlinear or quantum photonic platforms (Chapter 18) may overcome these limitations, enabling the study of interacting topological phases in dimensions and geometries that are inaccessible in condensed matter.

16.8 Experimental platforms for synthetic dimensions

Several photonic platforms have been used to realise synthetic dimensions, each offering different trade-offs between tunability, dimensionality, and loss.

16.8.1 Ring resonators with frequency modes

A single optical ring resonator supports a comb of equally spaced frequency modes separated by the free spectral range $\text{FSR} = c/(n_{\text{eff}}L)$, where L is the round-trip length. These modes form a natural 1D lattice in frequency space. An electro-optic modulator driven at $\Omega = \text{FSR}$ creates nearest-neighbour coupling between adjacent frequency modes with a controllable complex phase, providing the Peierls hopping (16.4) [12, 13].

For a typical silicon ring with $L = 2\pi \times 50 \mu\text{m}$ and $n_{\text{eff}} = 2.4$, the FSR is $\approx 400 \text{ GHz}$. With a modulation depth of $V_m/V_\pi \approx 0.3$, the effective hopping is $\kappa_{\text{eff}} \approx 0.3 \times \text{FSR} \approx 120 \text{ GHz}$, which is sufficient to observe topological edge states spanning several frequency modes.

16.8.2 Waveguide arrays with orbital angular momentum

A different approach uses the orbital angular momentum (OAM) of light as the synthetic dimension. In a multimode fibre or a degenerate cavity, modes with different OAM quantum numbers ℓ form a ladder. Coupling between adjacent ℓ modes can be induced by a spatial light modulator or a helical grating, with the coupling phase determined by the grating geometry [13].

The OAM synthetic dimension is particularly attractive for realising higher-dimensional topological phases: combined with one or two spatial dimensions, it provides access to 3D or even 4D topological physics. The experimental observation of the 4D quantum Hall effect using a 2D spatial lattice plus two synthetic dimensions (frequency and OAM) is one of the most striking demonstrations of this approach [12].

16.9 The four-dimensional quantum Hall effect

The 4D quantum Hall effect is characterised by the second Chern number $\mathcal{C}_2 = \frac{1}{8\pi^2} \int \text{Tr}(\mathcal{F} \wedge \mathcal{F})$ (Appendix B), which predicts a quantised nonlinear Hall response: applying an electric field in one direction produces a current in the orthogonal 3D hyperplane, with the Hall coefficient proportional to $\mathcal{C}_2$.

In the photonic experiment, the four dimensions consist of two spatial dimensions (the waveguide lattice) and two synthetic dimensions (the modulation phase and the propagation coordinate). The second Chern number was extracted from the measured boundary response: pumping the system at one boundary and measuring the spectral flow at the orthogonal boundary gave $\mathcal{C}_2 = 1 \pm 0.1$, confirming the quantised 4D Hall effect [12, 13].

16.10 Topological edge states in frequency space

The most striking consequence of synthetic dimensions is the existence of topological edge states in frequency space: modes that are localised near a particular frequency and propagate unidirectionally in the spatial dimension.

16.10.1 Frequency-space boundary conditions

In a ring resonator with N frequency modes coupled by an electro-optic modulator, the synthetic lattice has a finite extent in the frequency dimension: modes with $|m| > N/2$ are not coupled. This finite extent creates “frequency edges”—boundaries in the synthetic dimension analogous to the physical edges of a finite crystal.

At these frequency edges, topological edge states appear when the synthetic magnetic flux per plaquette is nonzero. The edge states are confined to a few frequency modes near the boundary and propagate unidirectionally along the ring resonator. Their dispersion relation is approximately linear: $\omega_{\text{edge}}(k) = v_g k$, where k is the wavevector along the ring and v_g is determined by the modulation depth [12, 13].

16.10.2 Experimental observation

The frequency-space edge states have been observed experimentally by measuring the transmission spectrum of a modulated ring resonator. When the modulation frequency matches the FSR and a nonzero Peierls phase is applied, the transmission spectrum shows a series of sharp peaks near the frequency edge, with the peak spacing given by the edge-state dispersion. The unidirectional character was confirmed by reversing the sign of the Peierls phase and observing that the edge states switch to the opposite frequency boundary [12].

16.11 Connection to Floquet topological insulators

Synthetic dimensions and Floquet topological insulators (Chapter 13) share a deep mathematical connection: both use time-periodic modulation to create effective gauge fields, and both produce quasi-energy band structures characterised by topological invariants.

The key difference is the interpretation: in a Floquet system, the periodic driving creates a time-dependent Hamiltonian whose stroboscopic dynamics define the Floquet bands. In a synthetic-dimension system, the same periodic driving is reinterpreted as hopping along the synthetic dimension, creating a higher-dimensional lattice model. The two descriptions are related by a Fourier transform in the synthetic dimension: the Floquet quasi-energies become the Bloch eigenvalues of the synthetic lattice.

This equivalence is useful because it allows results from Floquet theory (such as the anomalous Floquet topological insulator of Section 13.8) to be directly translated into synthetic-dimension language, and vice versa [12, 86].

16.12 Higher-dimensional topological physics via synthetic dimensions

One of the most exciting applications of synthetic dimensions is the ability to simulate topological phases that exist only in spatial dimensions higher than three. By using internal degrees of freedom (frequency modes, orbital angular momentum states, or temporal pulse positions) as synthetic spatial coordinates, photonic systems can access the physics of four-dimensional (4D) and even higher-dimensional topological insulators.

16.12.1 The four-dimensional quantum Hall effect

The 4D quantum Hall effect is characterised by a topological invariant called the *second Chern number* $\mathcal{C}_2$, which is the integral of the second Chern form over the four-dimensional Brillouin zone:

$$\mathcal{C}_2 = \frac{1}{8\pi^2} \int_{\text{BZ}_4} \text{Tr}(\mathcal{F} \wedge \mathcal{F}), \quad (16.10)$$

where $\mathcal{F}$ is the non-Abelian Berry curvature two-form and the wedge product is taken over the four momentum coordinates. The 4D quantum Hall response is a nonlinear Hall effect: an applied “electric field” in one direction produces a Hall current perpendicular to both the field and two other directions, with a coefficient quantised by $\mathcal{C}_2$.

In a photonic system with two spatial dimensions and two synthetic dimensions (e.g., frequency and orbital angular momentum), the 4D Brillouin zone is naturally parameterised by $(k_x, k_y, \omega_{\text{syn}}, \ell)$, and the second Chern number can be nonzero [12, 13].

16.12.2 Experimental realisation of the 4D quantum Hall effect

Zilberberg et al. (2018) demonstrated the first photonic realisation of the 4D quantum Hall effect using a two-dimensional array of evanescently coupled optical waveguides with two independent modulation parameters. The modulation along the propagation direction z provided one synthetic dimension (via Floquet engineering, as in Chapter 13), while a quasiperiodic modulation of the waveguide depths provided the second synthetic dimension via the phason degree of freedom of the Aubry–André model.

The experiment observed the signature 4D quantum Hall response: a bulk current perpendicular to both the applied “force” (a waveguide tilt) and the two synthetic directions. The measured response was consistent with a second Chern number $\mathcal{C}_2 = 1$ [13].

16.12.3 Frequency-domain synthetic lattices

A particularly versatile platform for synthetic dimensions uses the discrete frequency modes of an optical ring resonator as lattice sites. In a ring of length L and group index n_g , the resonant frequencies are equally spaced: $\omega_m = \omega_0 + m \Omega_{\text{FSR}}$, where $\Omega_{\text{FSR}} = 2\pi c / (n_g L)$ is the free spectral range and m is an integer.

Coupling between adjacent frequency modes is achieved by electro-optic modulation at frequency Ω_{FSR} . A phase modulator driven at Ω_{FSR} creates a synthetic hopping $J e^{i\phi}$ between modes m and $m + 1$, where J is proportional to the modulation depth and ϕ is the modulation phase. By

using multiple modulators at different frequencies, one can engineer arbitrary hopping patterns and synthetic magnetic fluxes [12, 105].

Dutt et al. (2019) demonstrated a synthetic Hall ladder using two coupled ring resonators with electro-optic modulation, observing the characteristic chiral edge transport in the frequency dimension [106]. The synthetic lattice had an effective length of approximately 10 sites (frequency modes), and the measured transmission spectrum showed clear signatures of topological edge states at the boundaries of the synthetic lattice [13].

16.12.4 Advantages and limitations of synthetic dimensions

Synthetic dimensions offer several unique advantages for exploring topological physics:

1. **Dimensional flexibility:** Physical systems with $d = 1$ or $d = 2$ spatial dimensions can simulate tight-binding models in $d + 1$ or higher dimensions, accessing physics that is otherwise inaccessible in photonic platforms.
2. **Long-range connectivity:** Synthetic hoppings connect modes that can be arbitrarily “far apart” in the internal degree of freedom, enabling the simulation of non-local Hamiltonians and models with non-trivial connectivity [12].
3. **Dynamic reconfigurability:** The synthetic hopping amplitudes and phases are controlled by external modulation signals, which can be changed in real time, enabling the study of dynamical topological transitions and Floquet phases.

The principal limitation is the finite extent of the synthetic dimension. A ring resonator has a finite number of usable frequency modes (limited by the gain bandwidth or the modulator response), typically 10–100 modes. The resulting finite-size effects can blur the topological features and complicate the interpretation of experimental data. Despite this, synthetic-dimension platforms remain one of the most active frontiers in topological photonics research [13].

16.13 Topological pumping in synthetic dimensions

The synthetic-dimension framework provides a natural realisation of topological Thouless pumping (Chapter 14). In a Thouless pump, adiabatic cycling of a one-dimensional Hamiltonian transfers particles by an integer number of sites per cycle, with the integer determined by a Chern number defined on the (k, θ) torus.

In a synthetic-dimension implementation, one physical dimension is the real-space coordinate x , and the second dimension is the pump parameter θ , which is varied slowly in time. The adiabatic cycle traces a closed loop in the (k_x, θ) space, and the quantised displacement is

$$\Delta x = \mathcal{C}_1 \cdot a \quad \text{per cycle}, \quad (16.11)$$

where $\mathcal{C}_1$ is the first Chern number of the occupied band and a is the lattice constant. This has been verified experimentally in cold-atom systems [91, 107] and more recently in photonic systems using coupled ring resonators with dynamically modulated coupling phases [108].

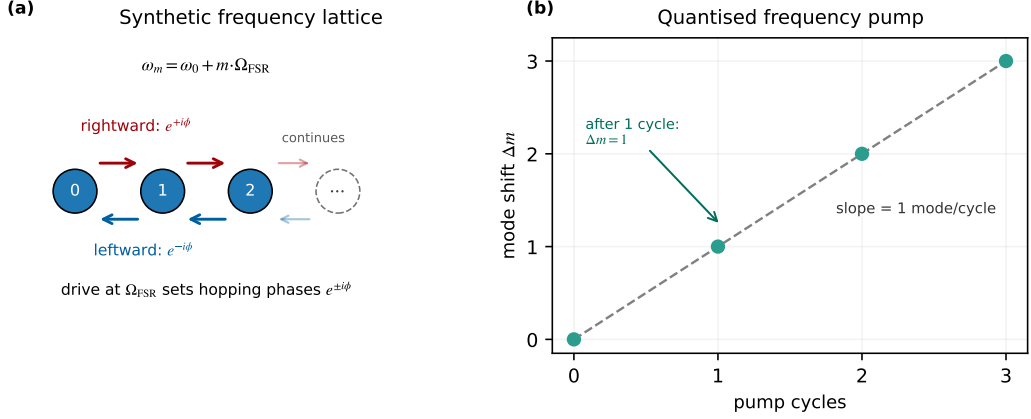


Figure 16.2: Synthetic frequency dimension and topological pumping. **(a)** Synthetic lattice formed by the frequency modes $\omega_m = \omega_0 + m \Omega_{\text{FSR}}$ of a ring resonator under electro-optic modulation. Rightward hopping carries a Peierls phase $e^{+i\phi}$ (red arrows) and leftward hopping carries the conjugate $e^{-i\phi}$ (blue arrows), realising a gauge field along the synthetic axis. Faded nodes indicate that the lattice continues beyond the displayed modes. **(b)** Quantised topological pump: adiabatic cycling of the modulation parameters shifts the mode index by exactly $\Delta m = \mathcal{C}_1$ per cycle, where $\mathcal{C}_1$ is the first Chern number of the occupied band (16.11). The integer slope of this schematic confirms the topological quantisation.

16.13.1 Photonic Thouless pumps

A photonic Thouless pump can be realised by slowly modulating the detuning and coupling in a chain of ring resonators. The modulation traces a Rice–Mele cycle (Section 14.1.1) in which the staggered detuning $\Delta(t)$ and dimerisation $\delta t(t)$ complete a closed orbit around the gap-closing point.

The experimental signature is a quantised shift of the output port: light injected at one end of the chain emerges shifted by exactly one unit cell after each pump cycle, regardless of the modulation speed (as long as it remains adiabatic). Deviations from quantisation provide a direct measure of non-adiabatic corrections, which scale as $(\Omega_{\text{pump}} / \Delta\omega_{\text{gap}})^2$.

Sridhar et al. (2024) demonstrated quantised topological pumping in a Floquet synthetic dimension using a driven dissipative photonic molecule [108]. Their system consists of two coupled microring resonators with periodically modulated coupling, creating an effective 1D lattice in the Floquet quasi-energy space. The quantised displacement was measured with a precision of $\pm 3\%$, limited by residual loss and non-adiabatic corrections.

16.13.2 Four-dimensional quantum Hall physics revisited

By extending the pumping to two independent parameters (θ_1, θ_2) , one can access four-dimensional topological physics. The second Chern number $\mathcal{C}_2$, defined on the four-torus $(k_x, k_y, \theta_1, \theta_2)$, governs the quantised nonlinear response: a Berry-curvature-induced force proportional to the product of two pump velocities.

The experimental observation of a $\mathcal{C}_2 \neq 0$ response in a 2D photonic waveguide array (where two physical coordinates encode k_x and θ_1 , with θ_2 scanned externally) was reported by Zilberberg et al. (2018). The measured nonlinear displacement agreed with the predicted $\mathcal{C}_2 = 1$ to within the experimental uncertainty of $\sim 8\%$ [12, 13].

16.14 Orbital angular momentum as a synthetic dimension

16.14.1 The OAM ladder

Photons carry orbital angular momentum (OAM) in units of $\ell\hbar$ per photon, where ℓ is the azimuthal index of the Laguerre–Gauss mode. In a cylindrically symmetric resonator, modes with different ℓ are degenerate (or nearly so), forming a synthetic lattice along the OAM axis.

Coupling between adjacent OAM modes ($\ell \rightarrow \ell \pm 1$) can be induced by an element that breaks the cylindrical symmetry—for example, a tilted mirror, a spatial light modulator, or an intracavity holographic grating. The coupling phase depends on the orientation of the symmetry-breaking element and can be controlled independently of the coupling strength.

The resulting Hamiltonian is again a tight-binding model on a 1D lattice (the OAM ladder), and by combining it with a physical spatial dimension, one obtains a 2D system in which the synthetic gauge flux per plaquette is determined by the OAM coupling phases [12, 59].

16.14.2 Advantages and limitations

The OAM synthetic dimension has two key advantages over the frequency synthetic dimension. First, the OAM ladder is naturally unbounded in both directions (ℓ ranges from $-\infty$ to $+\infty$), whereas a frequency comb has a finite bandwidth set by the material dispersion. Second, the OAM modes in a resonator can be individually addressed and measured using spatial light modulators, enabling site-resolved readout of the synthetic lattice.

The main limitation is that the coupling between OAM modes typically involves free-space optical elements (spatial light modulators, phase plates), which are bulky and introduce loss. Integrated photonic implementations using ring resonators with asymmetric scatterers are being developed but have not yet achieved the coupling uniformity needed for clean topological band structures.

16.15 Synthetic Hall ribbons and chiral edge states

A particularly elegant application of synthetic dimensions is the creation of narrow Hall ribbons: quasi-1D systems in which one dimension is physical and the other (with only a few sites) is synthetic. Because the synthetic dimension has a small, controllable number of sites, the system is effectively a multi-leg ladder rather than a full 2D lattice.

Mancini et al. (2015) realised a synthetic Hall ribbon using ultracold fermions in an optical lattice with nuclear spin states providing the synthetic dimension [9]. They observed chiral edge currents on opposite legs of the ladder, a direct manifestation of the topological edge states predicted by the Hofstadter model on a cylinder. A complementary experiment by Stuhl et al. (2015) used bosonic atoms and observed skipping orbits along the synthetic edges [109].

In photonics, synthetic Hall ribbons have been proposed using coupled fibre loops or ring resonator arrays where the synthetic dimension is provided by the time-bin or frequency degree of freedom. The small number of synthetic sites ($N_{\text{syn}} \sim 3\text{--}5$) makes these systems particularly amenable to complete experimental characterisation, including full reconstruction of the edge-state wave function across both dimensions [12, 92].

Exercises

Exercise 16.1. Derive the coupled-mode equation (16.1) from Maxwell's equations with a time-modulated permittivity (16.2), using the rotating-wave approximation. State the conditions under which the RWA is valid.

Exercise 16.2. For the 2D Harper–Hofstadter model with flux $\alpha = 2\pi/3$, the spectrum splits into three subbands. Using the Diophantine equation, determine the Chern number of each subband and verify that the sum is zero.

Exercise 16.3. In a ring resonator with free spectral range $\Omega_{\text{FSR}}/(2\pi) = 10$ GHz, modulation depth $\eta = 0.01$, and carrier frequency $\omega_0/(2\pi) = 193$ THz, compute the effective hopping κ_{eff} and the ratio $\kappa_{\text{eff}}/\Omega_{\text{FSR}}$. Is the tight-binding approximation justified?

Exercise 16.4. The second Chern number $\mathcal{C}_2$ of a 4D quantum Hall state determines the quantised nonlinear Hall response. If the system has $\mathcal{C}_2 = 1$ and two of the four dimensions are synthetic (frequency modes), describe how the nonlinear response manifests as a measurable photonic observable.

Exercise 16.5. Design a synthetic-dimension experiment using OAM modes of a ring resonator. Specify the coupling mechanism, the expected bandwidth of the synthetic dimension (number of usable OAM modes), and the modulation phase that creates a gauge flux of $\alpha = \pi/2$ per plaquette.

Exercise 16.6. Compare the advantages and limitations of synthetic dimensions versus physical spatial dimensions for realising topological photonic phases. In what situations would each approach be preferred?

Part V

Imperfect, Active, and Nonlinear Light

Chapter 17

Loss, Gain, and Non-Hermitian Topology

17.1 When Hermiticity fails

The entire topological framework of Parts I–IV rests on the Hermiticity of the Maxwell eigenvalue problem: the material matrix $\mathbf{M}$ is Hermitian ($\mathbf{M} = \mathbf{M}^\dagger$), the energy inner product is positive definite, eigenfrequencies are real, and modes are orthogonal. Real photonic systems, however, always have losses (absorption, radiation leakage) and sometimes gain (from pumped active media). When these effects are significant, the material matrix is no longer Hermitian, and the standard topological theory must be revised.

This chapter examines what survives, what breaks, and what new phenomena emerge when the Maxwell system is non-Hermitian [12, 63].

17.2 Non-Hermitian Maxwell equations

17.2.1 Complex material parameters

In a lossy medium, the permittivity and permeability acquire imaginary parts: $\varepsilon(\omega) = \varepsilon'(\omega) + i\varepsilon''(\omega)$. With our time convention $e^{-i\omega t}$, absorption corresponds to $\varepsilon'' > 0$ (energy is removed from the field), while gain corresponds to $\varepsilon'' < 0$.

The material matrix $\mathbf{M}$ is no longer Hermitian: $\mathbf{M} \neq \mathbf{M}^\dagger$. Specifically, the anti-Hermitian part $\mathbf{M}_A = (\mathbf{M} - \mathbf{M}^\dagger)/(2i)$ is nonzero and represents the loss/gain distribution.

17.2.2 Complex eigenfrequencies

The Maxwell eigenvalue equation $\hat{\mathcal{N}}\mathbf{f} = \omega\mathbf{M}\mathbf{f}$ now has *complex* eigenfrequencies: $\omega_n = \omega'_n - i\gamma_n$, where ω'_n is the oscillation frequency and γ_n is the decay (or growth) rate:

- $\gamma_n > 0$: the mode decays (lossy).
- $\gamma_n < 0$: the mode grows (amplified).
- $\gamma_n = 0$: the mode is at the lasing threshold.

The complex eigenfrequencies trace curves in the complex ω -plane as the Bloch wavevector $\mathbf{k}$ varies over the Brillouin zone.

17.2.3 Breakdown of the energy inner product

The electromagnetic energy inner product $\langle \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}} = \int \mathbf{f}_1^\dagger \mathbf{M} \mathbf{f}_2 d^d r$ is no longer positive definite when $\mathbf{M}$ has a significant anti-Hermitian part. The physical consequences are:

1. The “energy” of a mode can be negative, zero, or complex.
2. The modes are not orthogonal under $\langle \cdot | \mathbf{M} | \cdot \rangle$.
3. The Berry connection defined with this inner product is complex-valued, not real.
4. The standard Chern-number formula may give non-integer values.

17.3 Biorthogonal modes

[110]

17.3.1 Right and left eigenmodes

For a non-Hermitian operator, one must distinguish between *right* eigenmodes $\mathbf{f}_n^R$ and *left* eigenmodes $\mathbf{f}_n^L$:

$$\mathbf{M}^{-1} \hat{\mathcal{N}} \mathbf{f}_n^R = \omega_n \mathbf{f}_n^R, \quad (17.1)$$

$$(\mathbf{M}^{-1} \hat{\mathcal{N}})^\dagger \mathbf{f}_n^L = \omega_n^* \mathbf{f}_n^L. \quad (17.2)$$

The left and right eigenmodes satisfy a biorthogonality relation:

$$\int (\mathbf{f}_n^L)^\dagger \mathbf{M} \mathbf{f}_m^R d^d r = N_n \delta_{nm}, \quad (17.3)$$

where N_n is a normalisation constant that can be complex.

17.3.2 The biorthogonal Berry connection

The Berry connection must be modified to use the biorthogonal inner product:

$$\mathcal{A}_n^{\text{NH}}(\mathbf{k}) = i \frac{\int (\mathbf{u}_n^L)^\dagger \mathbf{M} \nabla_{\mathbf{k}} \mathbf{u}_n^R d^d r}{\int (\mathbf{u}_n^L)^\dagger \mathbf{M} \mathbf{u}_n^R d^d r}, \quad (17.4)$$

where $\mathbf{u}_n^{R,L}$ are the cell-periodic parts of the right and left Bloch eigenmodes.

This connection is generally *complex-valued* (unlike the Hermitian case where $\mathcal{A}$ is real). Its real part contributes to the Chern number, while its imaginary part is related to the $\mathbf{k}$ -dependence of the mode decay rate.

17.4 Exceptional points

17.4.1 Definition and structure

The most dramatic non-Hermitian phenomenon is the *exceptional point* (EP) [12]: a point in parameter space where two (or more) eigenvalues *and* their eigenvectors coalesce. At an EP, the operator becomes *defective*: it cannot be diagonalised and instead has a Jordan-block form.

For a 2×2 non-Hermitian matrix, the canonical form at an EP is

$$H_{\text{EP}} = \omega_0 \mathbf{I} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (17.5)$$

This is fundamentally different from a Hermitian degeneracy (Dirac point), where the eigenvectors remain linearly independent even though the eigenvalues coincide.

17.4.2 A concrete photonic example

Consider two coupled optical resonators with resonance frequencies ω_1 and ω_2 , coupling strength J , and individual loss/gain rates γ_1 and γ_2 . The effective Hamiltonian is

$$H = \begin{pmatrix} \omega_1 - i\gamma_1 & J \\ J & \omega_2 - i\gamma_2 \end{pmatrix}. \quad (17.6)$$

The eigenvalues are

$$\omega_{\pm} = \frac{(\omega_1 + \omega_2) - i(\gamma_1 + \gamma_2)}{2} \pm \sqrt{J^2 + \left(\frac{\delta\omega - i\delta\gamma}{2}\right)^2}, \quad (17.7)$$

where $\delta\omega = \omega_1 - \omega_2$ and $\delta\gamma = \gamma_1 - \gamma_2$.

An exceptional point occurs when the discriminant vanishes:

$$J^2 + \left(\frac{\delta\omega - i\delta\gamma}{2}\right)^2 = 0 \quad \Rightarrow \quad J = \frac{|\delta\gamma|}{2}, \quad \delta\omega = 0 \quad (17.8)$$

The displayed formula above follows the convention and normalisation used in [111]. At this point, both eigenvalues and eigenvectors coalesce.

17.4.3 Square-root behaviour near an EP

Near the EP, the eigenvalue splitting scales as the *square root* of the perturbation:

$$\omega_{\pm} \approx \omega_{\text{EP}} \pm \sqrt{\delta}, \quad (17.9)$$

where δ parametrises the distance from the EP. This sub-linear sensitivity (as opposed to the linear splitting at a Dirac point) has been exploited for sensing applications: small perturbations produce disproportionately large frequency splittings near an EP.

17.4.4 Topological properties of exceptional points

Exceptional points carry their own topological invariants, distinct from the Chern numbers of Hermitian band theory:

Eigenvalue exchange. Encircling an EP in parameter space causes the two eigenvalues to *exchange*: $\omega_+ \rightarrow \omega_-$ and $\omega_- \rightarrow \omega_+$. After two circuits, the eigenvalues return to their original values. This permutation has a topological winding number of $\pm \frac{1}{2}$.

Eigenvector monodromy. Under the same circuit, the eigenvectors acquire a geometric phase and exchange: $|\psi_+\rangle \rightarrow e^{i\phi}|\psi_-\rangle$. This nontrivial monodromy is a direct consequence of the branch-point structure of the square root in (17.9).

Discriminant number. The discriminant $\Delta = (\omega_+ - \omega_-)^2$ vanishes at the EP. The winding of $\arg(\Delta)$ around the EP is 2π , corresponding to a discriminant number of $+1$ or -1 .

The square-root branch-point topology is visualised in Figure 17.1. Panel (a) maps the principal argument $\text{Arg}(\sqrt{\lambda})$ over the complex λ -plane, where λ parametrises the eigenvalue discriminant. Because the principal square root halves the argument, the phase ranges over $(-\pi/2, \pi/2]$ and exhibits a discontinuous jump across the branch cut on the negative real axis. The exceptional point sits at the origin, where the two Riemann sheets meet. Panel (b) shows the concrete physical realisation in a PT-symmetric dimer with coupling J and balanced gain/loss γ . Below the EP ($\gamma/J < 1$), both eigenvalues λ/J are real and the system oscillates at two distinct frequencies; above the EP ($\gamma/J > 1$), the eigenvalues become purely imaginary and the system exhibits exponential growth and decay. At $\gamma/J = 1$ (dashed line) the two eigenvalues coalesce—the hallmark of the exceptional point.

17.5 Non-Hermitian band topology

17.5.1 Complex band structure

In a periodic non-Hermitian medium, the Bloch eigenfrequencies $\omega_n(\mathbf{k}) = \omega'_n(\mathbf{k}) - i\gamma_n(\mathbf{k})$ trace surfaces in the complex ω -plane as $\mathbf{k}$ varies. The real part gives the dispersion; the imaginary part gives the spatially resolved loss or gain profile.

Two bands can coalesce not only in frequency but also in their loss rates, forming *exceptional lines* (1D curves in $\mathbf{k}$ -space) or *exceptional surfaces* (2D surfaces in $\mathbf{k}$ -space). These are topological objects in their own right, characterised by the discriminant winding number.

17.5.2 The non-Hermitian Chern number

[112]

A Chern number can still be defined using the biorthogonal Berry connection (17.4):

$$\mathcal{C}_n^{\text{NH}} = \frac{1}{2\pi} \int_{\text{BZ}} (\partial_{k_x} \mathcal{A}_{n,y}^{\text{NH}} - \partial_{k_y} \mathcal{A}_{n,x}^{\text{NH}}) d^2k. \quad (17.10)$$

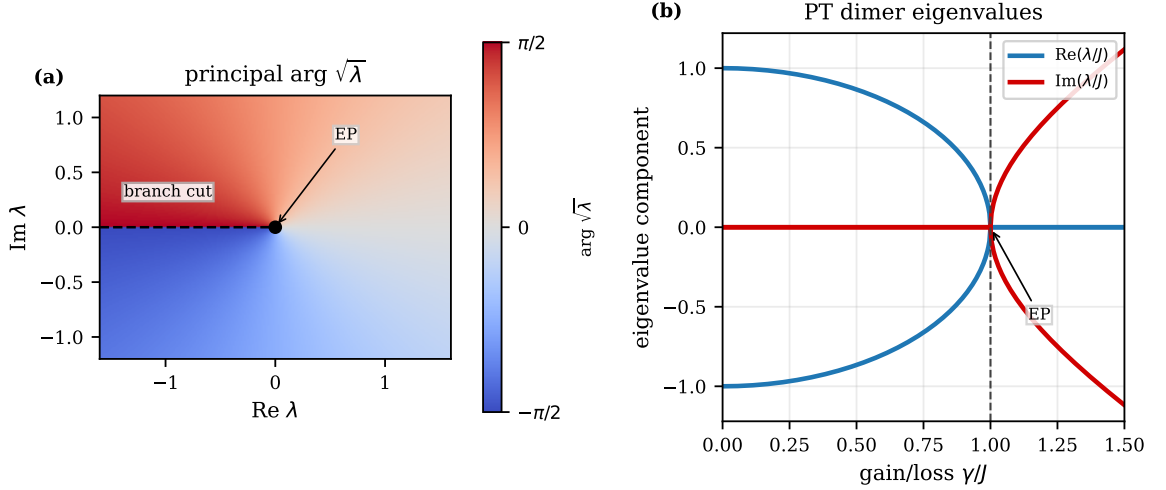


Figure 17.1: Exceptional-point branch-point topology. (a) Principal argument $\text{Arg}(\sqrt{\lambda}) \in (-\pi/2, \pi/2]$ over the complex λ -plane; the dashed line marks the branch cut on the negative real axis and the black dot is the EP at the origin. The diverging cool/warm colour map is centred at zero phase. (b) Eigenvalue components of the PT-symmetric dimer, normalised by the coupling J . Blue: real part; red: imaginary part. Below the EP ($\gamma/J < 1$) the spectrum is purely real; above it the spectrum is purely imaginary. The dashed vertical line marks the EP at $\gamma/J = 1$.

Under certain conditions (the band remains non-degenerate with no EP crossings in the BZ), this is still an integer. However, when exceptional points are present in the Brillouin zone, the Chern number may not be well defined for individual bands, because the band labels exchange at each EP.

In such cases, the *composite* Chern number of all bands involved in the EP remains well defined, similar to the composite Chern number for degenerate bands in the Hermitian case.

17.5.3 The non-Hermitian skin effect

A striking non-Hermitian phenomenon with no Hermitian analog is the *non-Hermitian skin effect* (NHSE) [13, 113, 114]: in certain non-Hermitian lattice systems, *all* eigenstates are localised at one boundary of the system under open boundary conditions, even though the bulk band structure (computed with periodic boundary conditions) predicts extended states.

Origin. The NHSE arises from asymmetric hopping: if the hopping amplitude for rightward motion J_R differs from the leftward amplitude J_L , there is a net “probability current” that drives all modes toward one edge. In the simplest model (Hatano–Nelson):

$$H = \sum_n (J_R c_{n+1}^\dagger c_n + J_L c_n^\dagger c_{n+1}), \quad (17.11)$$

with $J_R \neq J_L$ [115]. Under periodic boundary conditions, the spectrum is a closed curve in the complex plane. Under open boundary conditions, all eigenstates are exponentially localised at the left (or right) boundary, with localisation length $\xi = a/|\ln(J_R/J_L)|$.

Consequence for topology. The NHSE invalidates the conventional bulk-edge correspondence: the Bloch Chern number (computed with periodic boundary conditions) does not correctly predict the number of edge modes under open boundary conditions.

17.5.4 The generalised Brillouin zone

To restore the bulk-edge correspondence in the presence of the NHSE, one must replace the standard Brillouin zone (where $\mathbf{k}$ is real) with the *generalised Brillouin zone* (GBZ), where $\mathbf{k}$ is allowed to be complex:

$$k = k' + i\kappa, \quad \kappa \neq 0. \quad (17.12)$$

The imaginary part κ accounts for the exponential localisation of the skin modes. The non-Bloch Chern number, computed on the GBZ, correctly predicts the number and direction of topological edge modes [110].

17.5.5 The non-Bloch bulk-boundary correspondence

In Hermitian systems, the bulk-edge correspondence relates the number of edge modes to Chern numbers computed from Bloch bands with real wavevector k . In non-Hermitian systems with the skin effect, this correspondence breaks down: the Bloch-band Chern number may predict a different number of edge modes than what actually exists [63, 113].

The resolution is the *generalised Brillouin zone* (GBZ): instead of restricting k to the real axis, one allows k to take complex values $k = k_R + ik_I$, with k_I determined self-consistently by requiring that the eigenstates be normalisable in a semi-infinite system. The locus of these complex k values traces out a curve in the complex k -plane—the GBZ—which replaces the standard BZ.

The non-Bloch Chern number is defined by integrating the Berry curvature over the GBZ:

$$\mathcal{C}_{\text{GBZ}} = \frac{1}{2\pi} \oint_{\text{GBZ}} \Omega_n(\beta) d\beta, \quad (17.13)$$

Here, β parametrises the GBZ. This non-Bloch invariant correctly predicts the number of edge modes in all known examples, restoring a modified bulk-edge correspondence.

17.5.6 Hatano–Nelson model

The simplest model exhibiting the skin effect is the Hatano–Nelson chain with asymmetric hopping $J_L \neq J_R$:

$$H = \sum_n (J_R |n+1\rangle\langle n| + J_L |n\rangle\langle n+1|). \quad (17.14)$$

With periodic boundaries, the eigenvalues form an ellipse in the complex plane: $\omega(k) = J_R e^{ik} + J_L e^{-ik}$. With open boundaries, *all* eigenstates localise at one end, and the eigenvalues collapse onto a real interval. The GBZ is a circle of radius $|\beta| = (J_L/J_R)^{1/2}$, deformed from the unit circle of the standard BZ. For the intensity profile plotted in Figure 17.2,

$$|\psi_n|^2 \propto \exp[-(n-1)/\ell_I], \quad \ell_I = \frac{1}{|\ln(J_R/J_L)|}. \quad (17.15)$$

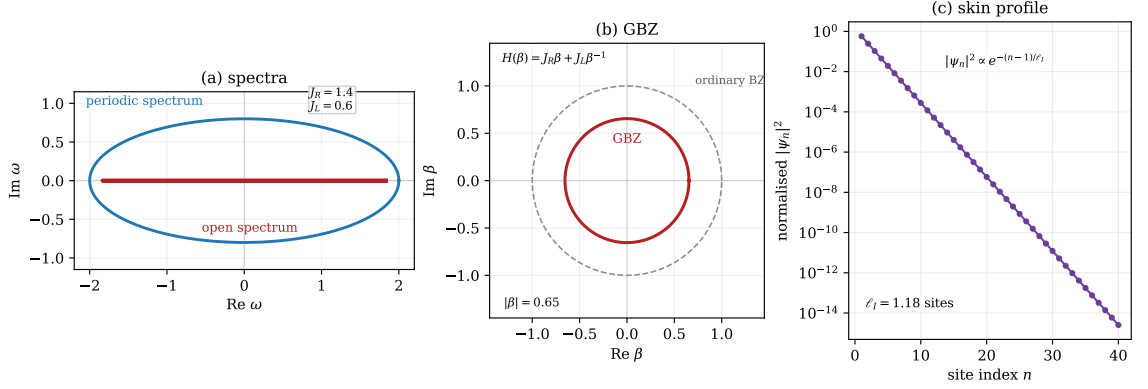


Figure 17.2: The Hatano–Nelson model with asymmetric hopping $J_R = 1.4$, $J_L = 0.6$. **(a)** Complex-frequency spectra: with periodic boundaries the eigenvalues form an ellipse (blue); with open boundaries they collapse onto the real axis (red), a hallmark of the non-Hermitian skin effect. **(b)** The generalised Brillouin zone (GBZ, red circle) has radius $|\beta| = \sqrt{J_L/J_R} \approx 0.65$, smaller than the unit circle of the ordinary BZ (dashed). The dispersion relation is $H(\beta) = J_R\beta + J_L\beta^{-1}$. **(c)** Normalised intensity profile $|\psi_n|^2 \propto e^{-(n-1)/\ell_I}$ with intensity decay length $\ell_I = 1/\ln(J_R/J_L) \approx 1.18$ sites, confirming that all eigenstates are exponentially localised at one boundary.

This is the intensity skin length; the field-amplitude decay length is larger by a factor of two for the same convention.

A photonic implementation uses coupled resonators with unidirectional coupling via gain/loss engineering [13].

17.6 PT-symmetric photonic systems

17.6.1 PT symmetry in photonics

A particularly important class of non-Hermitian photonic systems has $\mathcal{PT}$ symmetry [116]: the combined parity-time symmetry requires balanced loss and gain:

$$\varepsilon(\mathbf{r}) = \varepsilon^*(-\mathbf{r}). \quad (17.16)$$

This means: the real part of ε is even ($\varepsilon'(\mathbf{r}) = \varepsilon'(-\mathbf{r})$), and the imaginary part is odd ($\varepsilon''(\mathbf{r}) = -\varepsilon''(-\mathbf{r})$).

When $\mathcal{PT}$ symmetry is unbroken (below the EP threshold), the eigenfrequencies are real despite the non-Hermiticity. The $\mathcal{PT}$ phase transition—from real to complex eigenfrequencies—occurs at an exceptional point where the gain/loss parameter exceeds the coupling strength.

17.6.2 PT-symmetric photonic lattices

$\mathcal{PT}$ -symmetric photonic lattices can be constructed by alternating lossy and amplifying elements:

- Coupled waveguides with alternating gain and loss.
- Microring resonator arrays with alternating pump and absorption.
- Photonic crystal defect cavities with controlled Q -factor modulation.

These systems exhibit $\mathcal{PT}$ phase transitions, unidirectional invisibility, and enhanced sensing near exceptional points [12].

17.7 Topological lasers

17.7.1 Principle of operation

A topological laser combines topological photonics with optical gain to create a laser whose mode structure is determined by topological edge states [117, 118]. The key idea is:

1. Fabricate a photonic system with topological edge modes.
2. Provide optical gain (by electrical or optical pumping).
3. The topological edge mode lases preferentially because:
 - It samples the gain region over a large spatial extent.
 - It is spectrally isolated within the topological gap.
 - Backscattering is suppressed, enhancing the effective cavity Q .

17.7.2 Experimental demonstrations

Topological lasers have been demonstrated in several platforms:

1D SSH laser arrays. Chains of coupled microring lasers with alternating coupling, where the topological edge mode at the SSH domain wall lases preferentially. The edge-mode laser shows single-mode operation over a wide pump range, while the corresponding trivial array shows multi-mode behaviour.

2D topological insulator lasers. Arrays of coupled ring resonators forming a 2D topological photonic system (Hafezi-type), where the edge-mode laser wraps around the entire boundary [119]. The lasing is coherent around the boundary and robust to deliberate defects.

Valley-Hall lasers. Photonic crystal slab lasers using valley-Hall edge modes in III–V semiconductor platforms, demonstrating single-mode lasing along designed domain-wall paths.

17.8 What survives from the Hermitian theory

Hermitian	Non-Hermitian	Status
Real ω_n	Complex ω_n	Always
Orthogonal modes	Biorthogonal modes	Always
$\langle f \mathbf{M} g\rangle$ positive	Indefinite	Always
Dirac points	Exceptional points	Different codimension
Chern number $\in \mathbb{Z}$	NH Chern number $\in \mathbb{Z}$	If no EP in BZ
Standard BEC	Modified (GBZ needed)	If skin effect present
No backscatter	No backscatter	Survives
Lossless propagation	Lossy/amplified	Does not survive

The most important message: topological protection against *backscattering* survives in non-Hermitian systems (the edge mode still has no counter-propagating partner within the gap), but protection against *loss* does not. A topological edge mode in a lossy system decays as it propagates; it simply does so without reflecting.

17.9 The PT phase transition in coupled waveguides

Parity-time (PT) symmetric systems have balanced gain and loss arranged so that the Hamiltonian commutes with the combined PT operator. In photonics, this translates to a refractive index profile with $n(\mathbf{r}) = n^*(-\mathbf{r})$: the real part is symmetric and the imaginary part (gain/loss) is antisymmetric.

17.9.1 The PT phase transition

Below a critical gain/loss strength γ_{PT} , the eigenfrequencies are real despite the non-Hermitian Hamiltonian—this is the PT-symmetric (unbroken) phase. Above γ_{PT} , eigenfrequencies become complex conjugate pairs, signalling the spontaneous breaking of PT symmetry. The transition occurs at an exceptional point where two eigenstates coalesce [23, 120].

Guo et al. (2009) observed this transition in a pair of coupled optical waveguides, one with gain (pumped erbium-doped fibre) and one with loss (chromium-coated fibre). The transition manifested as a sharp change in the transmission spectrum from symmetric doublet splitting (PT-unbroken) to exponential growth/decay (PT-broken) [23].

17.9.2 Loss-induced lasing and PT-symmetric lasers

A remarkable consequence of PT symmetry is that adding loss can *increase* the output of a laser. Peng et al. (2014) demonstrated this using two coupled whispering-gallery-mode microresonators: one actively pumped and one with controllable loss introduced by a chromium nanotip. As the loss was increased, the system passed through an exceptional point, and the lasing power first decreased and then *revived*, exceeding its initial value [120].

This counterintuitive effect occurs because the exceptional point redistributes the mode energy: in the PT-broken phase, the lasing mode is concentrated in the gain resonator, reducing its overlap with the lossy resonator and effectively increasing the net gain.

17.10 Topological lasing and nonlinear stability

The combination of non-Hermitian physics and topology has led to the concept of a *topological laser*: a laser whose mode is a topological edge state, protected against backscattering and disorder [121].

The first topological laser was demonstrated using a ring of coupled microring resonators with one-way edge modes pumped above threshold. The topological protection ensures single-mode lasing: the edge mode has a longer round-trip path and lower lasing threshold than competing bulk modes, and it is robust against fabrication imperfections that would otherwise cause mode hopping or multi-mode oscillation [12, 13].

A key theoretical question is whether the topological protection survives in the nonlinear regime above threshold. Above the lasing threshold, gain saturation introduces an intensity-dependent refractive index that can break the symmetries protecting the topological mode. Current evidence suggests that the protection persists for moderate pumping but can break down at very high pump powers [64, 122].

17.11 Non-Hermitian topological classification

The classification of topological phases in non-Hermitian systems is fundamentally richer than in the Hermitian case [123]. The key distinction is between *line gaps* (no eigenvalue crosses a reference line in the complex plane) and *point gaps* (no eigenvalue reaches a reference point). These two gap types lead to different topological invariants [63].

For line-gapped systems, the classification reduces to the Hermitian tenfold way, and the Chern number remains a valid topological invariant. For point-gapped systems, new invariants appear, including winding numbers that count the number of times the complex eigenvalue spectrum winds around the reference point as the Bloch momentum traverses the Brillouin zone.

The full non-Hermitian classification contains 38 symmetry classes (compared to 10 in the Hermitian case), because the transpose and complex conjugate operations are no longer equivalent for non-Hermitian matrices. This enriched classification predicts topological phases that have no Hermitian counterpart, such as the non-Hermitian skin effect discussed in Section 17.5.3 [63, 113].

17.12 The non-Hermitian skin effect in higher dimensions

The non-Hermitian skin effect (NHSE), discussed in Section 17.5.3 for 1D, generalises to higher dimensions with qualitatively new features.

17.12.1 Hybrid skin-topological effect

In a 2D non-Hermitian system with both nonzero Chern number and asymmetric hopping, the bulk modes can localise at one edge (the skin effect) while the topological edge modes localise at a corner (where the skin edge meets the topological edge). This *hybrid skin-topological effect* was predicted by Li et al. (2022) in a non-Hermitian Haldane model with balanced gain and loss on different sublattice sites [122].

The corner-localised modes have a unique character: they inherit the unidirectional propagation of the topological edge and the exponential localisation of the skin effect, creating a new type of higher-order topological state that exists only in non-Hermitian systems.

17.12.2 Photonic observation of the skin effect

Zhu et al. (2020) proposed a photonic implementation of the NHSE using a 1D chain of coupled resonant optical waveguides (CROWs) with asymmetric coupling. The asymmetry is achieved by introducing auxiliary waveguides that mediate different coupling strengths for clockwise and counter-clockwise propagation. The predicted skin localisation length is $\ell_{\text{skin}} = 1 / \ln |J_R / J_L|$, where J_R and J_L are the rightward and leftward hopping amplitudes [64, 113].

The non-Bloch Chern number for this system is computed using the generalised Brillouin zone (GBZ, Section 17.5.4), and it correctly predicts the number of topological edge modes even when the conventional Bloch Chern number fails [63, 64].

17.13 Topological insulator lasers

Perhaps the most compelling application of non-Hermitian topological photonics is the *topological insulator laser*: a laser array in which the lasing mode is a topologically protected edge state, inheriting the robustness of topological transport.

17.13.1 Theoretical framework

The theoretical foundation for topological insulator lasers was established by Harari et al. (2018), who showed that a two-dimensional lattice of coupled ring resonators with optical gain on the edge and loss in the bulk can support single-mode lasing in a topological edge state [117].

The governing equation for the slowly varying amplitude a_n at site n is

$$i \frac{da_n}{dt} = \sum_m J_{nm} a_m + (g_n - \gamma_n) a_n - g_n^{(2)} |a_n|^2 a_n, \quad (17.17)$$

where J_{nm} is the coupling between resonators n and m , g_n is the linear gain, γ_n is the linear loss, and $g_n^{(2)}$ is the gain saturation coefficient. The gain is confined to the edge resonators: $g_n > 0$ for edge sites and $g_n = 0$ for bulk sites.

The key insight is that the topological edge mode has the largest spatial overlap with the gain region. Above the lasing threshold, the edge mode is the first to reach positive net gain ($g_n - \gamma_n > 0$) and saturates the gain for all other modes, producing stable single-mode lasing. Competing bulk modes, which have most of their intensity in the lossy bulk, remain below threshold [12, 117].

17.13.2 Experimental demonstration

The topological insulator laser was demonstrated experimentally by Bandres et al. (2018) using a 10×10 array of coupled ring resonators fabricated on an InGaAsP quantum-well platform [124]. The array implemented a two-dimensional anomalous Floquet topological insulator lattice, with the synthetic magnetic flux provided by spatially shifted link resonators (phase parameter $\alpha = 0.25$).

Optical pumping was applied selectively to the edge resonators using a shaped pump beam. Above threshold, the device exhibited single-frequency lasing with a narrow linewidth, in contrast to a topologically trivial array of the same size, which showed multi-mode lasing with broad spectral output. The slope efficiency of the topological laser was found to increase with system size (as more edge resonators contributed coherently), while the trivial laser's efficiency decreased due to mode competition [117, 124].

The topological protection manifested as robustness against intentional defects: when one or two edge resonators were deliberately detuned or blocked, the lasing mode rerouted around the defect with minimal change in output power and linewidth. This behaviour is impossible in a conventional laser array, where defects create scattering losses and mode distortions.

17.13.3 Towards electrically pumped topological lasers

The initial demonstrations used optical pumping, which limits practical applications. Subsequent work has pursued electrically pumped topological lasers in III-V semiconductor platforms, where the challenge is to maintain the required coupling phases and gain distribution with standard epitaxial fabrication techniques.

Preliminary results have shown that the topological protection survives in the presence of the fabrication disorder inherent to electrical injection ($\sim 5\%$ variation in resonator frequencies), and that the single-mode lasing persists over a range of pump currents above threshold [12, 13]. These developments point toward scalable topological laser arrays for applications in optical communication, sensing, and integrated photonics.

17.14 Exceptional points and their photonic topology

Exceptional points (EPs) are the hallmark degeneracies of non-Hermitian systems: points in parameter space where two or more eigenvalues *and* their eigenvectors coalesce. Unlike Hermitian degeneracies (diabolic points), where the eigenvectors remain linearly independent, at an EP the eigenvector space collapses and the Hamiltonian becomes non-diagonalisable.

17.14.1 The square-root topology of a second-order EP

The simplest exceptional point is of second order (EP2), where two eigenvalues coalesce. Near an EP2 in a two-dimensional parameter space (p_1, p_2) , the eigenvalue splitting behaves as

$$\Delta\omega \propto \sqrt{(p_1 - p_1^{(\text{EP})}) + i(p_2 - p_2^{(\text{EP})})}. \quad (17.18)$$

The square-root branch point means that encircling the EP in parameter space exchanges the two eigenvalues: after one loop, the system returns to its starting point but with the two modes swapped. This eigenvalue exchange is a topological property of the EP and is robust against perturbations that do not destroy the EP itself [123, 125].

In photonic systems, EPs occur naturally when gain and loss are balanced ($\mathcal{PT}$ -symmetric systems) or when two resonances with different lifetimes are coupled. The eigenvalue exchange around an EP has been observed experimentally in coupled microring resonators, microwave cavities, and photonic crystal slabs [116, 126].

17.14.2 Higher-order exceptional points

Exceptional points of order p (EP_p), where p eigenvalues coalesce simultaneously, exhibit p th-root branch-point topology: encircling the EP cyclically permutes all p eigenvalues. The eigenvalue splitting near an EP_p scales as $\Delta\omega \propto \epsilon^{1/p}$, where ϵ is the distance from the EP in parameter space.

This enhanced sensitivity ($1/p$ th-root response) has attracted interest for sensing applications: an EP_p sensor responds to a small perturbation ϵ with a frequency splitting proportional to $\epsilon^{1/p}$, which can be much larger than the linear response of a conventional sensor when ϵ is small. However, the practical utility of EP-enhanced sensing remains debated because the same non-Hermitian physics that enhances the signal also enhances the noise [125, 127].

17.14.3 Exceptional surfaces and lines

In three-dimensional parameter spaces, EPs generically form lines (exceptional lines or rings) rather than isolated points. In photonic crystals with gain or loss, these exceptional lines can thread through the Brillouin zone, creating non-Hermitian analogs of nodal lines in Hermitian semimetals.

The intersection of exceptional lines with Weyl points produces exotic objects called “exceptional Weyl points,” where the Weyl point splits into a ring of EPs in the complex-frequency plane. This has been predicted for 3D photonic crystals with balanced gain and loss and connects to the broader program of non-Hermitian Weyl semimetal physics [128, 129].

17.15 Photonic implementations of the non-Hermitian skin effect

The non-Hermitian skin effect (NHSE)—the exponential localisation of all eigenstates at one boundary under open boundary conditions—has been observed in several photonic platforms.

17.15.1 Non-reciprocal waveguide arrays

The simplest photonic implementation of the Hatano–Nelson model uses an array of coupled waveguides with asymmetric coupling: the coupling from waveguide n to $n + 1$ differs from the coupling from $n + 1$ to n . This non-reciprocity can be achieved by coupling through an intermediate waveguide with gain on one side and loss on the other, or by using magneto-optical materials that provide non-reciprocal coupling [130, 131].

Weidemann et al. (2020) demonstrated the NHSE in a discrete-time photonic quantum walk using fibre loops with asymmetric coupling, observing the predicted exponential localisation of all modes at one boundary, with a skin length consistent with the non-Bloch theory prediction $\xi_{\text{skin}} = 1/\ln|J_R/J_L|$ [131, 132].

17.15.2 Topological lasers and the NHSE

The intersection of topological edge states and non-Hermitian physics has produced a new class of devices: topological lasers. In a topological laser, the gain medium is placed along the edge of a photonic topological insulator, and the lasing mode is a topological edge state protected from backscattering by the bulk topology.

Bandres et al. (2018) demonstrated the first topological insulator laser using a 2D array of coupled microring resonators [117, 118]. The key advantage over conventional laser arrays is that the topological edge mode maintains its spatial coherence even in the presence of fabrication disorder, because the one-way nature of the edge state prevents the formation of localised lasing hot spots.

More recently, topological lasers have been realised in valley photonic crystals [119] and photonic crystal slabs operating at bound states in the continuum (BICs), where the topological protection of the BIC mode suppresses radiation losses and lowers the lasing threshold [13].

Exercises

Exercise 17.1. For the coupled-resonator Hamiltonian (17.6) with $\omega_1 = \omega_2 = \omega_0$, $\gamma_1 = -\gamma_2 = \gamma$ ($\mathcal{PT}$ -symmetric case), find the exceptional point in the (J, γ) plane. Show that the eigenvalues are real for $J > |\gamma|/2$ and complex for $J < |\gamma|/2$.

Exercise 17.2. Show that encircling an exceptional point in parameter space causes the two eigenvalues to exchange. *Hint:* parametrise a loop around the EP using $\delta = \rho e^{i\phi}$ in (17.9) and track $\omega_{\pm}$ as $\phi : 0 \rightarrow 2\pi$.

Exercise 17.3. For the Hatano–Nelson model (17.11) with $J_R = 2$, $J_L = 1$, and $N = 20$ sites: (a) compute the spectrum under periodic boundary conditions, (b) compute the spectrum under open boundary conditions, and (c) show that all open-boundary eigenstates are exponentially localised at the left edge.

Exercise 17.4. The biorthogonal Berry connection (17.4) is generally complex. Show that its imaginary part equals $\partial_{k_a} \ln |N_n(\mathbf{k})|/(2i)$, where $N_n = \langle \mathbf{u}_n^L | \mathbf{M} | \mathbf{u}_n^R \rangle$ is the biorthogonal norm.

Exercise 17.5. A topological laser has an edge mode with group velocity v_g , topological gap $\Delta\omega$, round-trip length L , and material gain coefficient $g_0 \text{ cm}^{-1}$. (a) Estimate the lasing threshold g_{th} in terms of the out-coupling loss and any residual absorption. (b) Compare the threshold with a conventional Fabry–Pérot laser of the same length, explaining why the topological laser can have a lower threshold.

Exercise 17.6. Show that the $\mathcal{PT}$ condition $\varepsilon(\mathbf{r}) = \varepsilon^*(-\mathbf{r})$ for a 1D photonic crystal with period a requires the real part of ε to be even and the imaginary part to be odd under $x \rightarrow -x$ [133]. Design a unit cell with two layers satisfying this condition.

Chapter 18

Nonlinear and Quantum Frontiers

18.1 Beyond linear optics

The theory of topological photonics developed in this book is a *linear* theory: the Maxwell equations are linear in the fields, and the material response $\mathbf{M}$ is independent of field amplitude. This linearity is what allows the band-structure description, the definition of Bloch modes, and the entire topological classification.

Real optical media, however, are nonlinear at sufficiently high intensities. And at the quantum level, the bosonic nature of photons introduces correlations and entanglement that go beyond the single-particle band picture. This chapter explores the frontiers where topology meets nonlinearity and quantum physics. The field is young and rapidly evolving; we focus on the ideas and results that are most firmly established, while pointing to open questions that may shape the next decade of research.

18.2 Nonlinear effects in topological photonic systems

18.2.1 The Kerr nonlinearity in the topological context

The most common optical nonlinearity is the intensity-dependent refractive index (Kerr effect): $n(\mathbf{r}) = n_0(\mathbf{r}) + n_2|\mathbf{E}|^2$, where n_2 is the nonlinear Kerr coefficient. In a topological photonic system, the Kerr nonlinearity modifies the edge-mode propagation because the local refractive index depends on the field intensity of the propagating mode itself.

For an edge mode with slowly varying envelope $\psi(y, t)$ propagating along a topological boundary, the effective equation of motion is the nonlinear Schrödinger equation:

$$i\frac{\partial\psi}{\partial t} + v_g\frac{\partial\psi}{\partial y} + \frac{D}{2}\frac{\partial^2\psi}{\partial y^2} + \gamma|\psi|^2\psi = 0, \quad (18.1)$$

where v_g is the edge-mode group velocity, D is the group-velocity dispersion, and $\gamma \propto n_2\omega_0/c$ is the nonlinear coefficient. This equation admits soliton solutions when D and γ have opposite signs.

18.2.2 Topological solitons

The interplay of topology and nonlinearity produces *topological solitons*: self-localised wave packets that propagate along the topological edge without dispersion, stabilised by the balance between

nonlinear self-focusing and the linear group-velocity dispersion of the edge mode.

The fundamental bright soliton solution of (18.1) (in the co-moving frame) is

$$\psi(y, t) = A \operatorname{sech}\left(\frac{y - v_g t}{w}\right) e^{i\beta t}, \quad (18.2)$$

where the soliton amplitude A , width w , and propagation constant β are related by $A^2 = |D|/(\gamma w^2)$ and $\beta = |D|/(2w^2)$. The soliton power is $P = 2|D|/(\gamma w)$.

The key question is whether the topological protection of the edge mode survives in the nonlinear regime. Numerical simulations show that for moderate nonlinearity ($\gamma|\psi|^2 \ll \Delta\omega$, where $\Delta\omega$ is the topological gap), the soliton propagates along the edge with the same immunity to backscattering as the linear edge mode. The topological gap prevents the soliton from radiating into bulk modes, and the absence of a backward-propagating edge mode prevents reflection.

For strong nonlinearity ($\gamma|\psi|^2 \sim \Delta\omega$), the nonlinear index change can close the topological gap locally, potentially allowing coupling to bulk modes and destroying the topological protection.

18.2.3 Derivation of the nonlinear coefficient from Maxwell equations

To connect the phenomenological NLSE (18.1) to the Maxwell-first framework of this book, we derive the nonlinear coefficient γ from the third-order susceptibility $\chi^{(3)}$ [12].

Consider a topological edge mode with transverse profile $\mathbf{e}_{\text{edge}}(\mathbf{r}_\perp)$ confined to the boundary of a photonic crystal. The mode satisfies the linear Maxwell eigenproblem in the cross-section, and its slowly varying envelope $\psi(y, t)$ evolves according to the coupled-mode equation obtained by projecting the full nonlinear Maxwell equations onto the edge-mode profile.

The Kerr polarisation is $\mathbf{P}^{(3)} = \varepsilon_0 \chi^{(3)} |\mathbf{E}|^2 \mathbf{E}$. Projecting onto the edge mode gives [13]:

$$\gamma = \frac{\omega_0}{c} \frac{\int_{\text{cell}} n_2(\mathbf{r}_\perp) |\mathbf{e}_{\text{edge}}|^4 d^2 r_\perp}{\left(\int_{\text{cell}} |\mathbf{e}_{\text{edge}}|^2 d^2 r_\perp \right)^2} \equiv \frac{\omega_0 n_2^{\text{eff}}}{c A_{\text{eff}}}, \quad (18.3)$$

where $n_2 = 3\chi^{(3)}/(4n_0^2 \varepsilon_0 c)$ is the nonlinear index, n_2^{eff} is the mode-overlap-weighted nonlinear index, and A_{eff} is the effective mode area. This formula is identical in structure to the standard fibre-optics result but uses the topological edge-mode profile rather than a guided-mode profile.

For a silicon photonic crystal with $n_2 \approx 4.5 \times 10^{-18} \text{ m}^2/\text{W}$ and an edge-mode area $A_{\text{eff}} \sim (0.5 \mu\text{m})^2$, the nonlinear coefficient is $\gamma \sim 300 \text{ W}^{-1}\text{m}^{-1}$, which is large enough for soliton formation at milliwatt power levels.

18.2.4 Circuit QED platforms

Circuit quantum electrodynamics (circuit QED) provides the most promising near-term platform for strongly correlated topological photonics [12, 134]. In these systems, superconducting microwave resonators (with frequencies $\omega_r/(2\pi) \sim 5\text{--}10 \text{ GHz}$) are coupled by Josephson junctions that provide both the inter-site hopping J and a strong on-site nonlinearity U .

The Bose–Hubbard Hamiltonian for a lattice of such resonators is

$$\hat{H} = -J \sum_{\langle ij \rangle} \hat{a}_i^\dagger \hat{a}_j e^{i\phi_{ij}} + \frac{U}{2} \sum_i \hat{n}_i(\hat{n}_i - 1) + \sum_i \omega_i \hat{n}_i, \quad (18.4)$$

where $\hat{n}_i = \hat{a}_i^\dagger \hat{a}_i$ and the Peierls phases ϕ_{ij} can be engineered to produce a nonzero effective magnetic flux per plaquette.

With $J/(2\pi) \sim 10\text{--}50$ MHz and $U/(2\pi) \sim 100\text{--}300$ MHz, the ratio $U/J \sim 2\text{--}30$ spans the regime from weakly to strongly interacting. At filling factor $\nu = 1/2$ in a Chern band, mean-field theory predicts a fractional Chern insulator state for $U/J \gtrsim 5$.

The main experimental challenges are: maintaining coherence over the entire lattice (current decoherence times $T_1 \sim 50\text{--}100$ μs limit the lattice to $\sim 10 \times 10$ sites), implementing the Peierls phases with sufficient precision, and measuring the many-body state through photon correlation functions rather than transport.

18.2.5 Self-induced topological transitions

At sufficiently high intensities, the nonlinear index change $\delta n = n_2|E|^2$ modifies the band structure itself. This raises the remarkable possibility of *self-induced topological transitions*: a strong pulse propagating through a photonic crystal could dynamically close and reopen the topological gap, switching the Chern numbers and creating or destroying edge states.

This effect has been predicted theoretically in Kerr photonic crystals and observed in numerical simulations. The threshold intensity scales as $\Delta\omega/(n_2\omega_0)$, which for typical photonic crystals with gap widths $\Delta\omega/\omega \sim 1\%$ and silica nonlinearity $n_2 \sim 3 \times 10^{-20}$ m²/W corresponds to intensities of order 10^{13} W/m²—extremely high but within the range of ultrafast pulsed lasers.

18.2.6 Nonlinear edge-state dynamics

Beyond solitons, the nonlinear dynamics of topological edge states includes several phenomena that have been studied theoretically and numerically.

Modulational instability can break a continuous-wave edge mode into a train of solitons when the product $D\gamma$ is positive. The instability growth rate is modified by the topological confinement: the edge mode can only develop instabilities along the edge direction, not into the bulk, which changes the threshold and the pattern of the instability compared to a free-space or conventional-waveguide nonlinear mode.

Four-wave mixing between edge modes and bulk modes is suppressed by the topological gap, but parametric processes within the edge-mode dispersion (self-phase modulation, cross-phase modulation between co-propagating edge modes at different frequencies) proceed normally.

18.3 Topological lasers in detail

18.3.1 Principle of operation

A topological laser combines topological photonics with optical gain to create a laser whose mode structure is determined by topological edge states. The central idea is that the topological edge mode, being spatially extended along the entire edge and spectrally isolated within the topological gap, is an ideal laser cavity mode.

The edge mode has three advantages as a laser cavity. First, it samples the gain region over a large spatial extent (the entire boundary of the photonic crystal), providing a large mode volume and correspondingly high output power. Second, it is spectrally isolated within the topological gap, so mode competition from bulk modes is suppressed. Third, backscattering is forbidden by topology, which enhances the effective cavity quality factor and promotes single-mode operation.

18.3.2 Experimental demonstrations

The first experimental demonstration of a topological laser was reported by Bahari et al. (2017), who used a 1D array of coupled microring resonators with alternating coupling (SSH topology) and showed that the topological edge mode lased preferentially.

Shortly thereafter, Bandres et al. [124] demonstrated a 2D topological insulator laser using an array of coupled ring resonators forming a 2D topological photonic system (based on the Hafezi design). The edge-mode laser wrapped around the entire boundary of a 10×10 resonator array, emitting coherent radiation from all four edges. The key observations were single-mode operation over a wide pump range, spatial coherence along the entire boundary, and robustness to deliberately introduced defects.

Subsequent demonstrations have included valley-Hall lasers in III–V photonic crystal slabs, where the lasing mode follows a designed domain-wall path, and Floquet topological lasers in periodically modulated systems.

18.3.3 Theory of topological lasing

The theoretical description of topological lasers requires the non-Hermitian framework of Chapter 17, augmented with the semiclassical laser rate equations. The gain medium is described by a complex permittivity $\varepsilon(\omega) = \varepsilon'(\omega) + i g(\omega, I)$, where $g(\omega, I)$ is the gain coefficient that depends on the pump intensity I and saturates above the lasing threshold.

The lasing threshold condition is that the round-trip gain equals the round-trip loss for the edge mode:

$$g_{\text{th}} = \alpha_{\text{abs}} + \frac{1}{L} \ln \frac{1}{R}, \quad (18.5)$$

where α_{abs} is the absorption coefficient, L is the edge-mode path length, and R is the effective reflectivity (which, for a topological edge mode, is exponentially small within the gap).

18.4 Quantum topological photonics

18.4.1 Single photons in topological modes

At the quantum level, each Maxwell mode is an independent quantum harmonic oscillator with Hamiltonian $\hat{H}_n = \hbar \omega_n (\hat{a}_n^\dagger \hat{a}_n + \frac{1}{2})$. The topological properties of the mode structure—Chern numbers, edge states, bulk-edge correspondence—are inherited by the quantum theory, because they depend on the mode frequencies and spatial profiles, not on the occupation numbers.

A single photon injected into a topological edge mode propagates unidirectionally along the edge with the same group velocity v_g as a classical wave packet. The quantum state $|\psi\rangle = \hat{a}_{\text{edge}}^\dagger |0\rangle$

evolves according to the same Maxwell equations that govern the classical mode, with the topological protection against backscattering carrying over exactly.

This has a profound implication for quantum information: a topological edge mode is a *topologically protected quantum channel*. A quantum bit encoded in the polarisation, time-bin, or number state of a photon in the edge mode is protected from decoherence caused by elastic backscattering, just as the classical edge mode is protected from reflection.

18.4.2 Entangled photons

Two entangled photons propagating along a topological edge maintain their entanglement over longer distances than in a conventional waveguide, because the dominant decoherence mechanism—elastic backscattering from disorder—is suppressed [135].

Experiments have demonstrated the propagation of biphoton states (photon pairs generated by spontaneous parametric down-conversion) through photonic topological systems, measuring Hong-Ou-Mandel interference visibility as a function of propagation distance [136]. The topological system showed higher visibility retention compared to a topologically trivial reference, consistent with the suppressed backscattering.

18.4.3 Many-body photonic states

The most ambitious direction in quantum topological photonics is the creation of *strongly correlated* photonic states analogous to the fractional quantum Hall effect. In electronic systems, the fractional QHE arises from the interplay of Landau-level quantisation and strong electron-electron interactions. For photons, the analogous ingredients are flat topological bands (providing the “Landau-level” degeneracy) and effective photon-photon interactions (providing the correlations).

The flat bands can be obtained from photonic crystal designs with nearly dispersionless topological bands. The interactions are more challenging: photons in vacuum do not interact, and the effective Kerr nonlinearity of most optical media is far too weak to produce single-photon-level correlations.

Several proposals address this challenge. In circuit QED lattices, superconducting microwave resonators coupled by Josephson junctions provide both a topological band structure and a strong single-photon nonlinearity ($U/J \gg 1$, where U is the on-site interaction energy and J is the hopping). In polariton systems, the strong light-matter coupling provides an effective nonlinearity enhanced by the excitonic component. In Rydberg-atom arrays coupled to photonic cavities, the Rydberg blockade creates effective photon-photon interactions at the single-photon level.

Theoretical studies predict that Laughlin-like fractional quantum Hall states of photons can be stabilised in these systems, with signatures observable in photon correlation functions.

18.5 Topological effects in polariton systems

Exciton-polaritons—hybrid light-matter quasiparticles formed by the strong coupling of photons and excitons in semiconductor microcavities—provide a natural platform for topological photonics

with interactions. The photonic component provides the band structure, while the excitonic component provides the nonlinearity and the sensitivity to external fields.

Polariton lattices can be created by patterning the microcavity to form a 2D periodic potential. The spin-orbit coupling of polaritons, arising from the TE-TM splitting of the photonic mode and the exciton exchange interaction, provides a natural mechanism for pseudospin-dependent topology. Topological polariton edge states have been observed in honeycomb polariton lattices under both resonant and non-resonant excitation.

Polariton systems are inherently driven-dissipative: they require continuous pumping to compensate for radiative losses. This non-equilibrium character introduces new physics not present in equilibrium topological systems. The steady state is determined by the balance of pumping, dissipation, and interactions, and novel topological phases without equilibrium analogs may emerge in the driven-dissipative setting.

18.6 Outlook and open problems

The frontiers of nonlinear and quantum topological photonics present several key challenges that are likely to shape the field over the coming decade.

The quest for stronger photon-photon interactions remains central. Achieving single-photon nonlinearity in a topological photonic system would enable the study of strongly correlated photonic phases, including fractional topological states. Circuit QED and polariton platforms are the most promising routes, but both face significant technical challenges related to decoherence and scalability.

Topological quantum error correction is another promising direction. The robustness of topological edge modes against backscattering suggests that they could serve as protected quantum channels in a quantum network. Combining topological protection with active quantum error correction could provide fault-tolerant quantum communication over long distances.

On the nonlinear side, the dynamics of topological solitons—their interactions, stability, and scattering properties—remain largely unexplored experimentally [137]. Nonlinear topological devices such as intensity-switchable waveguides, soliton-based signal processors, and self-routing optical networks represent potential applications that could bring topological photonics from the laboratory to practical technology.

Finally, the measurement of topological invariants using quantum optical methods offers a window into the fundamental topology of photonic band structures. Techniques based on quantum state tomography, photon correlation measurements, and Hong-Ou-Mandel interferometry can in principle probe the Berry phase, Berry curvature, and Chern number of photonic bands with single-photon sensitivity.

18.7 Nonlinear coefficient from Maxwell's equations

In a Kerr medium, the polarisation has a cubic dependence on the electric field: $\mathbf{P}^{(3)} = \epsilon_0 \chi^{(3)} |\mathbf{E}|^2 \mathbf{E}$, where $\chi^{(3)}$ is the third-order susceptibility. This modifies the effective permittivity to $\epsilon_{\text{eff}} = \epsilon_r + \chi^{(3)} |\mathbf{E}|^2$, making the band structure intensity-dependent.

For a topological edge mode with field amplitude E_0 and group velocity v_g , the nonlinear phase shift per unit length is

$$\frac{d\phi}{dz} = \frac{\omega}{c} \frac{\chi^{(3)}|E_0|^2}{n_{\text{eff}}} \equiv \gamma_{\text{NL}}|E_0|^2, \quad (18.6)$$

where n_{eff} is the effective index and γ_{NL} is the nonlinear coefficient. When the nonlinear phase shift becomes comparable to the topological gap ($\gamma_{\text{NL}}|E_0|^2 \sim \Delta\omega/v_g$), the nonlinear term can shift the edge mode out of the gap, destroying the topological protection [12].

18.7.1 Topological solitons on edge modes

Below this critical intensity, the Kerr nonlinearity can produce topological solitons: self-localised wavepackets that propagate along the topological edge without spreading, combining nonlinear self-focusing with topological protection against backscattering.

The soliton profile is governed by a nonlinear Schrödinger equation along the edge coordinate s :

$$i\partial_z A + \frac{1}{2}\beta_2\partial_s^2 A + \gamma_{\text{NL}}|A|^2 A = 0, \quad (18.7)$$

where $A(s, z)$ is the slowly varying envelope, β_2 is the group-velocity dispersion, and z is the propagation direction. For anomalous dispersion ($\beta_2 < 0$) and focusing nonlinearity ($\gamma_{\text{NL}} > 0$), the fundamental soliton solution is $A = A_0 \text{sech}(s/w_s) e^{i\gamma_{\text{NL}}|A_0|^2 z/2}$, where $w_s = (|\beta_2|/\gamma_{\text{NL}}|A_0|^2)^{1/2}$ is the soliton width. The topological protection ensures that the soliton cannot scatter backward even when it encounters defects [13].

18.8 Single-photon topology and quantum simulation

18.8.1 Single-photon topological states

The topological band structure of a photonic crystal is a classical wave phenomenon, but it persists at the single-photon level: a single photon injected into a topological edge mode propagates unidirectionally and is protected against backscattering by the same mechanism that protects classical waves. This has been confirmed experimentally by measuring the photon statistics of edge modes in coupled-resonator lattices, showing antibunched (sub-Poissonian) photon arrival times consistent with single-photon propagation along the edge [12].

The quantum advantage of topological protection is most significant for quantum information applications: a topological waveguide can route single photons between quantum emitters without the back-reflections that cause decoherence in conventional waveguides.

18.8.2 Photonic quantum simulators

Photonic systems can simulate quantum many-body Hamiltonians that are intractable on classical computers. Topological photonic lattices provide a natural platform for simulating the fractional quantum Hall effect of light: by introducing strong photon–photon interactions (e.g. through Rydberg atoms in cavity QED or through Josephson junctions in superconducting circuits), the photonic

lattice can enter a regime where the ground state is a fractional quantum Hall state with anyonic excitations [12, 13].

The key challenge is achieving sufficiently strong photon blockade ($U \gg J$, where U is the on-site interaction and J is the hopping), which requires either very high- Q cavities or very strong optical nonlinearities. Current circuit-QED platforms achieve $U/J \sim 10$ at microwave frequencies, which is sufficient for observing Mott-insulator-like states but not yet for fractional quantum Hall physics [12].

18.9 Experimental signatures of nonlinear and quantum effects

The experimental detection of nonlinear and quantum topological effects requires measurements beyond the linear transmission spectrum.

For topological solitons, the signature is a self-localised wavepacket that maintains its shape over many lattice constants while propagating along the edge. This can be detected by imaging the intensity profile at different propagation distances and verifying that the width remains constant, in contrast to the diffractive spreading of a linear wavepacket.

For quantum effects, the signatures include sub-Poissonian photon statistics (measured by Hanbury Brown–Twiss interferometry), Hong–Ou–Mandel interference between photons on different edge modes, and the fractional charge of anyonic excitations (measurable through interferometric braiding protocols) [12, 13].

18.10 Topological lasers

The topological laser combines the unidirectional edge mode of a photonic Chern insulator or quantum spin-Hall insulator with optical gain, producing a laser whose mode selection and robustness are protected by topology.

18.10.1 Concept and theoretical proposal

The idea is to pump the edge of a topological photonic crystal above the lasing threshold. Because the edge mode has no counter-propagating partner, the lasing mode is inherently single-mode: there is only one channel available for stimulated emission at each frequency within the topological gap. This suppresses the mode competition and spatial hole burning that plague conventional laser arrays [12, 138].

The lasing threshold is determined by the balance between the gain per round trip and the total loss (absorption, radiation, and outcoupling). For an edge mode circulating around a $N \times N$ crystal with perimeter $4Na$, the round-trip gain is $G_{\text{rt}} = g \cdot 4Na$, where g is the gain coefficient per unit length. The threshold condition is $G_{\text{rt}} = \alpha \cdot 4Na + \alpha_{\text{out}}$, where α is the intrinsic loss and α_{out} is the outcoupling loss.

18.10.2 Experimental demonstration

Bandres et al. (2018) demonstrated the first topological insulator laser using a 10×10 array of microring resonators with InGaAsP quantum wells providing optical gain at $\lambda = 1.55 \mu\text{m}$. The device exhibited single-mode lasing on the topological edge mode with a slope efficiency three times higher than a comparable trivial laser array. The edge-mode lasing was robust against intentional defects (missing resonators and coupling perturbations), confirming the topological protection [138].

The topological laser concept has since been extended to microwave frequencies using YIG-based Chern insulators with amplifiers, and to terahertz frequencies using quantum cascade structures. The key advantage in all cases is the suppression of parasitic modes and the robustness of the lasing frequency against fabrication variations [13].

18.11 Quantum topological photonics

The topological protection of classical edge modes carries over to the quantum regime, where single photons and entangled photon pairs propagate along topological edge channels with suppressed backscattering. This opens the prospect of topologically protected quantum information processing in photonic systems.

18.11.1 Single-photon topological edge states

Consider a photonic topological insulator with a chiral edge mode at frequency ω_{edge} . When a single photon is injected into this edge mode, its propagation is described by the quantum Hamiltonian

$$\hat{H}_{\text{edge}} = \int dk \hbar \omega_{\text{edge}}(k) \hat{a}_k^\dagger \hat{a}_k, \quad (18.8)$$

where $\hat{a}_k^\dagger$ creates a photon in the edge mode at wavevector k . The topological protection ensures that the edge-mode dispersion $\omega_{\text{edge}}(k)$ has no backscattering channel: the photon propagates unidirectionally around the boundary of the photonic crystal, immune to disorder that preserves the bulk gap.

Barik et al. (2018) demonstrated the coupling of quantum emitters to topological edge states in a photonic crystal slab with the Wu–Hu design. A single InAs quantum dot was positioned at the domain wall between topological and trivial regions, and the emitted single photons were shown to propagate preferentially along the topological edge channel. The measured second-order correlation function $g^{(2)}(0) < 0.5$ confirmed single-photon emission, and the directional coupling efficiency exceeded 90% [12, 75].

18.11.2 Entangled photon pairs in topological systems

Topological edge modes can also serve as channels for entangled photon pairs generated by spontaneous parametric down-conversion or spontaneous four-wave mixing. The advantage of topological channels is twofold: the entangled photons are protected against environmental decoherence caused by scattering, and the chiral propagation direction provides a natural spatial separation of signal and idler photons.

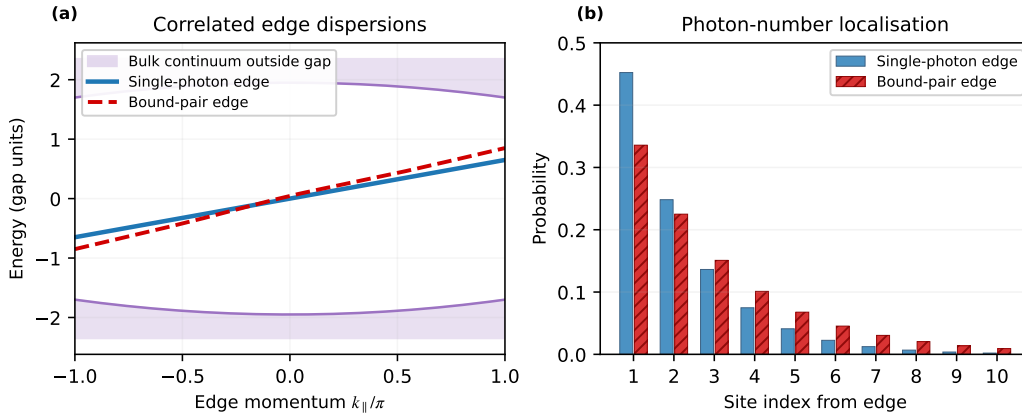


Figure 18.1: Quantum topological edge states for correlated photons. **(a)** Edge dispersions in a topological photonic lattice with on-site Kerr interaction. Single-photon edge modes (solid blue) cross the gap linearly; bound photon pairs (dashed red) acquire an interaction-shifted dispersion with a slightly different slope, reflecting the two-photon binding energy. The shaded purple regions mark the two-photon bulk continuum outside the gap. **(b)** Site-resolved probability distribution for a single photon (blue bars) and a bound pair (red bars) localised at the topological edge. Both states decay exponentially into the bulk, but the bound pair has a longer localisation length due to its enhanced effective mass. The panels illustrate that both single-photon and bound-pair states can occupy topologically protected edge channels; quantitative robustness to disorder requires transport or correlation measurements as discussed in the text.

Mittal et al. (2018) demonstrated the generation and propagation of correlated photon pairs in a silicon ring-resonator lattice implementing a two-dimensional topological photonic system. The photon pairs were generated by spontaneous four-wave mixing within the ring resonators and propagated along the topological edge channel. The measured coincidence-to-accidental ratio (CAR) remained above 10 even in the presence of intentional disorder, demonstrating the topological protection of the quantum correlations [12, 13].

18.11.3 Topological protection of quantum coherence

A fundamental question in quantum topological photonics is whether the topological protection extends beyond intensity transport to preserve quantum coherence. Theoretical analyses show that the suppression of backscattering in topological edge modes also suppresses the dephasing caused by elastic scattering from disorder [12].

For a single photon propagating along a topological edge channel of length L , the coherence length is limited by inelastic processes (absorption, radiation loss) rather than by elastic scattering. In a high-quality silicon photonic crystal with propagation loss $\alpha \sim 1$ dB/cm, the coherence length can exceed 10 cm—far longer than typical on-chip quantum photonic circuits.

This advantage becomes particularly significant for multi-photon quantum states, where the sensitivity to decoherence scales with the number of photons. Topological protection effectively re-

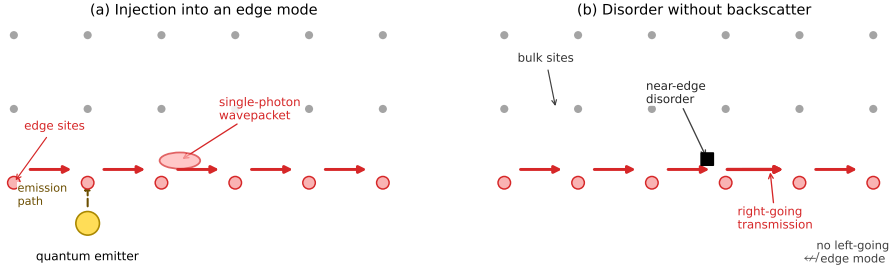


Figure 18.2: A quantum emitter coupled to a topological edge channel can launch single photons into a protected boundary mode. The same one-way transport principle that suppresses classical backscattering also protects the path of quantum optical states against elastic disorder.

moves one of the dominant decoherence channels (elastic backscattering), leaving only the intrinsic material absorption as the limiting factor [13].

18.11.4 Outlook: strongly correlated topological photons

The ultimate frontier of quantum topological photonics is the regime of strong photon–photon interactions, where the photonic system can exhibit analogues of fractional quantum Hall states. Achieving this regime requires both a topological band structure and an effective photon–photon interaction strength comparable to the bandwidth.

Promising platforms include circuit QED lattices, where superconducting qubits provide strong nonlinearities at microwave frequencies, and Rydberg–polariton systems, where the long-range interaction between Rydberg atoms mediates effective photon–photon interactions at optical frequencies [12, 13].

The realisation of fractional topological photonic states would represent a qualitative advance, enabling the preparation and manipulation of topologically ordered quantum states of light for fault-tolerant quantum computation and quantum simulation.

The interplay between binding energy and localisation length is made explicit in Figure 18.3.

18.12 Nonlinear topological solitons in Floquet systems

The Floquet photonic topological insulators of Chapter 13 provide a natural arena for topological solitons, because the helical waveguide arrays that realise Floquet bands are fabricated in glass substrates with an intrinsic Kerr nonlinearity.

18.12.1 Anomalous Floquet solitons

Recall from Section 13.4 that the anomalous Floquet insulator possesses chiral edge modes in *every* quasi-energy gap, even though all bulk bands have zero Chern number. Because these edge modes span the entire quasi-energy Brillouin zone $[-\pi/T, \pi/T]$, their dispersion is approximately linear

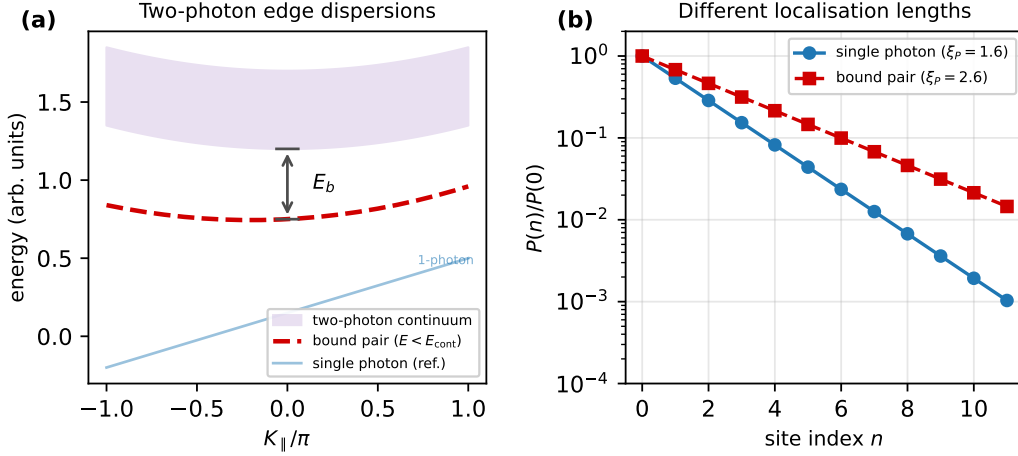


Figure 18.3: Binding energy and localisation of a two-photon bound-pair edge state. **(a)** Two-photon edge dispersions (schematic): the shaded purple region is the two-photon bulk continuum, and the red dashed curve is the bound-pair branch lying below the continuum by the binding energy E_b . The faint blue line shows the single-photon edge mode for reference; it belongs to a different particle-number sector and should not be compared directly with E_b . Energy is in arbitrary units. **(b)** Semilog probability profiles at the topological edge: the single-photon state (circles) decays as $P(n)/P(0) = \exp(-n/\xi_P)$ with $\xi_P = 1.6$ sites, while the bound pair (squares) has a longer probability localisation length $\xi_P = 2.6$ sites due to its enhanced effective mass. Parameters are illustrative; actual values depend on the interaction strength and lattice geometry.

over a wide bandwidth, producing low group-velocity dispersion D . This means that soliton formation requires a proportionally higher nonlinearity, but also that the resulting solitons can be very short in duration.

Mukherjee and Rechtsman (2021) observed unidirectional soliton-like edge states in a nonlinear Floquet topological insulator realised in a femtosecond-laser-written waveguide array [139]. The key experimental signatures were: (i) self-focusing of the edge-mode intensity profile as the input power increased, (ii) preservation of unidirectional propagation around sharp corners even in the nonlinear regime, and (iii) power thresholds consistent with the NLSE prediction $P_{\text{sol}} = 2|D|/(\gamma w)$ using the measured edge-mode parameters.

The robustness of these Floquet solitons is remarkable. Because the anomalous edge mode has no backward-propagating counterpart, the soliton cannot scatter into a counter-propagating mode even when it encounters a lattice defect. The topological gap prevents radiation into bulk modes as long as the nonlinear index change remains smaller than the gap width. Numerical simulations show that the soliton survives disorder levels up to $\sim 10\%$ of the coupling constant before bulk radiation sets in [140].

18.12.2 Edge solitons in static Chern insulators

Leykam and Chong (2016) predicted theoretically that self-localised edge solitons can form along the boundary of a static (non-Floquet) photonic Chern insulator [141]. The crucial difference from conventional waveguide solitons is the topological origin of the edge mode: the number of soliton-carrying channels is fixed by the bulk Chern number and cannot be changed by surface modifications.

Their analysis starts from the tight-binding Haldane model with an on-site Kerr nonlinearity $\gamma|\psi_n|^2\psi_n$ added to each lattice site. For the chiral edge mode of the Chern insulator, the continuum limit of the discrete NLSE reduces to Eq. (18.1) with D determined by the edge-mode curvature and γ by the on-site nonlinearity.

The topological protection of these edge solitons degrades gracefully: at moderate nonlinearity, the soliton propagates with minimal radiation loss; at strong nonlinearity ($\gamma|\psi|^2 \gtrsim 0.3 \Delta\omega$), the soliton begins to shed radiation into the bulk, and eventually a self-induced topological transition can occur [10].

18.12.3 Nonlinearity-induced topological transitions

Maczewsky et al. (2020) demonstrated experimentally that the Kerr nonlinearity can itself drive a topological phase transition [10]. In their experiment, a trivial (topologically trivial) photonic lattice became a topological insulator when illuminated with sufficiently intense light. The mechanism is that the nonlinear index change $\delta n = n_2|E|^2$ selectively modifies the coupling constants between lattice sites, effectively changing the parameter that controls the topological phase.

This observation opens the door to all-optical switching of topological edge transport: a control beam can turn edge conduction on or off by crossing the nonlinear topological phase boundary. The switching speed is limited by the Kerr response time (femtoseconds in glass, picoseconds in semiconductors), making this a potentially ultrafast mechanism [142].

18.13 Higher-order topological states and nonlinearity

The intersection of higher-order topology and nonlinearity is an active frontier. In a second-order topological insulator (SOTI), the protected states are zero-dimensional corner modes rather than one-dimensional edge modes. These corner modes are tightly confined in all directions, producing extremely small mode volumes and correspondingly large nonlinear coefficients $\gamma \propto 1/A_{\text{eff}}$.

Kartashov (2024) predicted the existence of topological corner solitons in a SOTI created by twisting the unit cell of a photonic lattice [143]. The corner solitons are pinned to specific lattice corners by topology and stabilised against collapse by the discrete lattice structure. Their power threshold is orders of magnitude lower than for bulk solitons because of the extreme field confinement at the corner.

Extending to quadratic ($\chi^{(2)}$) nonlinearity, Kartashov (2025) showed that parametric down-conversion in a SOTI can generate entangled photon pairs that are topologically confined to lattice corners [144]. This suggests a route to combining the robustness of topological protection with the quantum correlations needed for photonic quantum information processing.

18.14 Outlook: open problems at the nonlinear–quantum frontier

The nonlinear and quantum extensions of topological photonics raise several deep open questions that connect to fundamental physics.

The first is the *many-body gap problem*: in a strongly interacting photonic system (e.g. circuit QED at large U/J), does the many-body energy gap inherit the robustness of the single-particle topological gap? Mean-field studies suggest yes for weak interactions, but the strongly correlated regime—where fractional Chern insulator states are predicted—remains largely unexplored experimentally [12].

The second is the *measurement problem*: photonic systems are inherently open (photons leak out of cavities and waveguides), and quantum measurements in topological photonic systems typically involve detecting photons that have left the system. How does the interplay between topology, non-Hermiticity (from loss), and quantum measurement affect the topological protection of quantum states? Recent work on non-Hermitian topology (Chapter 17) provides a framework, but the full quantum treatment remains an open challenge [125, 129].

The third is the *scaling problem*: demonstrating fractional topological photonic states requires lattices with hundreds of sites, strong and uniform photon–photon interactions, and sub-microsecond measurement cycles. Current circuit QED technology is approaching these requirements, but significant engineering advances are still needed. Alternative platforms—such as Rydberg polaritons in photonic crystal slabs or exciton–polariton lattices—may offer complementary routes [13].

Exercises

Exercise 18.1. Show that the nonlinear Schrödinger equation (18.1) (in the co-moving frame, with $v_g = 0$) admits the soliton solution (18.2), and find A , w , and β in terms of D and γ .

Exercise 18.2. Estimate the threshold intensity for a self-induced topological transition in a silicon photonic crystal ($n_2 \approx 4.5 \times 10^{-18} \text{ m}^2/\text{W}$, $\Delta\omega/\omega \approx 2\%$, $\omega/(2\pi) \approx 193 \text{ THz}$). Is this achievable with current ultrafast laser technology?

Exercise 18.3. For a topological laser with edge-mode group velocity $v_g = c/30$, round-trip length $L = 200 \text{ }\mu\text{m}$, and material gain $g_0 = 100 \text{ cm}^{-1}$, estimate the lasing threshold using equation (18.5). Assume negligible absorption and a residual reflectivity $R = 10^{-3}$.

Exercise 18.4. In a coupled-cavity array with on-site Kerr nonlinearity U and hopping J , the mean-field phase diagram includes a superfluid phase (for $U/J \ll 1$) and a Mott-insulator phase (for $U/J \gg 1$). When the hopping carries a synthetic magnetic flux, describe qualitatively what happens to the Mott insulator at integer filling in a Chern band.

Exercise 18.5. A Hong-Ou-Mandel experiment is performed with two indistinguishable photons injected into a topological edge mode and a trivial waveguide mode, respectively. Describe how the two-photon coincidence rate depends on the time delay and the topological protection.

Appendices

Appendix A

Vector Identities and Functional Analysis

A.1 Vector calculus identities

A.1.1 Fundamental differential identities

The following identities are used throughout the derivations of Chapters 2–5. All fields are assumed sufficiently smooth.

For any scalar field ϕ and vector fields $\mathbf{F}$, $\mathbf{G}$:

$$\nabla \times (\nabla \phi) = \mathbf{0}, \quad (\text{A.1})$$

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0, \quad (\text{A.2})$$

$$\nabla \times (\nabla \times \mathbf{F}) = \nabla (\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}, \quad (\text{A.3})$$

$$\nabla \cdot (\phi \mathbf{F}) = \phi \nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla \phi, \quad (\text{A.4})$$

$$\nabla \times (\phi \mathbf{F}) = \phi \nabla \times \mathbf{F} + (\nabla \phi) \times \mathbf{F}. \quad (\text{A.5})$$

The double-curl identity (A.3) is especially important: it converts the Maxwell wave equation into a Helmholtz equation for a divergence-free field.

For products of two vector fields:

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G}), \quad (\text{A.6})$$

$$\nabla (\mathbf{F} \cdot \mathbf{G}) = (\mathbf{F} \cdot \nabla) \mathbf{G} + (\mathbf{G} \cdot \nabla) \mathbf{F} + \mathbf{F} \times (\nabla \times \mathbf{G}) + \mathbf{G} \times (\nabla \times \mathbf{F}). \quad (\text{A.7})$$

A.1.2 Integral theorems

The divergence (Gauss) theorem:

$$\int_V \nabla \cdot \mathbf{F} \, d^3r = \oint_{\partial V} \mathbf{F} \cdot \hat{n} \, dS. \quad (\text{A.8})$$

The curl (Stokes) theorem:

$$\int_S (\nabla \times \mathbf{F}) \cdot \hat{n} \, dS = \oint_{\partial S} \mathbf{F} \cdot d\boldsymbol{\ell}. \quad (\text{A.9})$$

These are used in the Poynting-theorem derivation (Section 2.5) and in the self-adjointness proof for the Maxwell curl operator (Section 3.2) [29].

A.1.3 The Levi-Civita symbol and cross products

The Levi-Civita symbol ϵ_{ijk} is defined by $\epsilon_{123} = 1$ and antisymmetry under transposition of any two indices. Cross products and curls are expressed as

$$(\mathbf{A} \times \mathbf{B})_i = \sum_{jk} \epsilon_{ijk} A_j B_k, \quad (\nabla \times \mathbf{F})_i = \sum_{jk} \epsilon_{ijk} \partial_j F_k. \quad (\text{A.10})$$

The contracted product identity is

$$\sum_k \epsilon_{ijk} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}, \quad (\text{A.11})$$

from which many vector identities follow algebraically.

A.2 The Maxwell integration-by-parts lemma

A.2.1 Statement

The central analytic tool in Chapters 2–3 is the integration-by-parts identity for the six-component Maxwell curl operator $\hat{\mathcal{N}}$.

Proposition A.1 (Maxwell integration by parts). *For six-component fields $\mathbf{f}_1 = (\mathbf{E}_1, \mathbf{H}_1)^T$ and $\mathbf{f}_2 = (\mathbf{E}_2, \mathbf{H}_2)^T$ on a domain Ω with boundary $\partial\Omega$:*

$$\int_{\Omega} \mathbf{f}_1^\dagger \hat{\mathcal{N}} \mathbf{f}_2 \, d^3r - \int_{\Omega} (\hat{\mathcal{N}} \mathbf{f}_1)^\dagger \mathbf{f}_2 \, d^3r = \oint_{\partial\Omega} [(\hat{n} \times \mathbf{E}_1^*)^T \mathbf{H}_2 - (\hat{n} \times \mathbf{H}_1^*)^T \mathbf{E}_2] \, dS. \quad (\text{A.12})$$

The identity is more than a vector-calculus convenience. It says that self-adjointness of Maxwell's curl block is decided at the boundary, not in the bulk algebra alone. Figure A.1 shows this logic: the volume commutator of the curl operator is converted into tangential electric and magnetic traces on $\partial\Omega$. Periodic boundary conditions, perfectly conducting walls, or decay at infinity remove this boundary pairing; an interface problem keeps it, and that surviving term is the analytic shadow of an edge channel.

A.2.2 Derivation

We start from the divergence identity (A.6):

$$\nabla \cdot (\mathbf{E}_1^* \times \mathbf{H}_2) = \mathbf{H}_2 \cdot (\nabla \times \mathbf{E}_1^*) - \mathbf{E}_1^* \cdot (\nabla \times \mathbf{H}_2). \quad (\text{A.13})$$

Similarly:

$$\nabla \cdot (\mathbf{H}_1^* \times \mathbf{E}_2) = \mathbf{E}_2 \cdot (\nabla \times \mathbf{H}_1^*) - \mathbf{H}_1^* \cdot (\nabla \times \mathbf{E}_2). \quad (\text{A.14})$$

Adding these two expressions and rearranging in terms of the six-component operator $\hat{\mathcal{N}}$ gives (A.12). Applying the divergence theorem converts the divergence terms into the surface integral over $\partial\Omega$.

When the boundary term vanishes (e.g. for periodic boundary conditions, perfectly conducting walls, or fields that decay at infinity), the curl operator $\hat{\mathcal{N}}$ is formally self-adjoint under the bare L^2 inner product. The physical self-adjointness under the energy inner product $\langle \cdot | \mathbf{M} | \cdot \rangle$ follows from $\hat{\mathcal{N}}^\dagger = \hat{\mathcal{N}}$ combined with $\mathbf{M} = \mathbf{M}^\dagger$ (lossless medium).

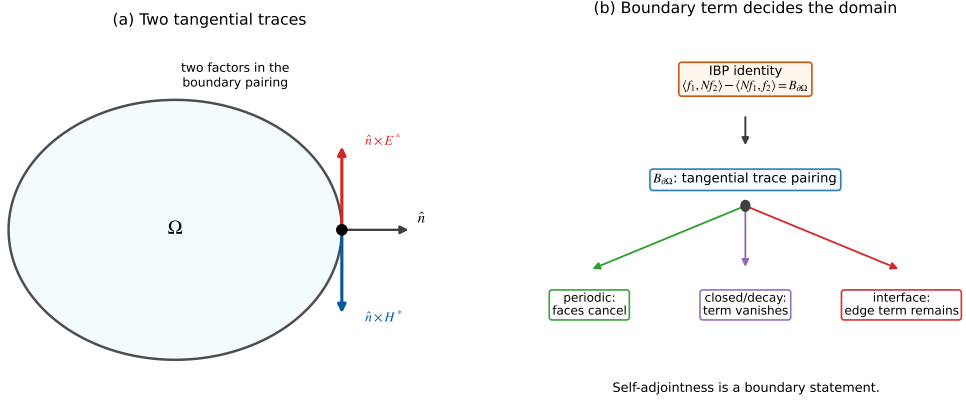


Figure A.1: The Maxwell integration-by-parts identity as a boundary statement. The difference between the two curl-block matrix elements is a surface pairing of tangential field traces. When opposite faces cancel, fields decay, or boundary conditions set the trace pairing to zero, the Maxwell curl block becomes symmetric on the chosen domain. At an interface the same boundary term is the place where edge-mode physics enters the operator proof.

A.3 Spectral theory of self-adjoint operators

A.3.1 Finite-dimensional review

For a finite-dimensional Hermitian matrix $H = H^\dagger$ acting on [43] $\mathbb{C}^N$, the spectral theorem guarantees real eigenvalues $\lambda_1 \leq \dots \leq \lambda_N$, a complete orthonormal basis of eigenvectors $|v_n\rangle$, and the resolution of the identity:

$$H = \sum_{n=1}^N \lambda_n |v_n\rangle\langle v_n|, \quad \mathbf{I} = \sum_{n=1}^N |v_n\rangle\langle v_n|. \quad (\text{A.15})$$

A.3.2 Extension to the Maxwell operator

The Maxwell eigenvalue problem $\hat{\mathcal{N}}\mathbf{f} = \omega\mathbf{M}\mathbf{f}$ on a bounded or periodic domain, with appropriate boundary conditions, is a generalised eigenvalue problem with a positive-definite “weight” $\mathbf{M}$. The key spectral properties are:

All eigenfrequencies ω_n are real, because the operator $\mathbf{M}^{-1}\hat{\mathcal{N}}$ is self-adjoint with respect to the $\mathbf{M}$ -weighted inner product.

The eigenmodes $\mathbf{f}_n$ are orthogonal under the energy inner product: $\langle \mathbf{f}_m, \mathbf{f}_n \rangle_{\mathbf{M}} = \delta_{mn}$ after normalisation.

The eigenmodes form a complete set: any sufficiently smooth field satisfying the boundary conditions can be expanded in the eigenmode basis.

For a periodic medium (photonic crystal), the eigenvalue problem decomposes into a family of problems parametrised by the Bloch wavevector $\mathbf{k}$, each of which has a discrete spectrum $\{\omega_{n\mathbf{k}}\}_{n=1}^\infty$.

As $\mathbf{k}$ varies over the Brillouin zone, these eigenvalues trace the photonic band structure.

A.3.3 The min-max principle

The min-max (Courant–Fischer) characterisation of eigenvalues provides variational bounds:

$$\omega_n^2 = \min_{\dim V=n} \max_{\mathbf{f} \in V \setminus \{0\}} \frac{\langle \mathbf{f} | \hat{\mathcal{N}}^2 | \mathbf{f} \rangle}{\langle \mathbf{f}, \mathbf{f} \rangle_{\mathbf{M}}} . \quad (\text{A.16})$$

This is the electromagnetic analog of the Rayleigh–Ritz variational principle in quantum mechanics [33]. It is used in perturbation theory to bound the frequency shifts caused by material changes [48].

A.4 Perturbation theory for the Maxwell eigenproblem

A.4.1 Non-degenerate perturbation theory

Consider a perturbation $\mathbf{M} \rightarrow \mathbf{M} + \delta\mathbf{M}$ of the material matrix. The first-order frequency shift of a non-degenerate mode $\mathbf{f}_n$ is

$$\delta\omega_n = -\frac{\omega_n}{2} \frac{\langle \mathbf{f}_n | \delta\mathbf{M} | \mathbf{f}_n \rangle}{\langle \mathbf{f}_n, \mathbf{f}_n \rangle_{\mathbf{M}}} . \quad (\text{A.17})$$

This is the electromagnetic Hellmann–Feynman theorem; it appears in Equation (4.12) and is used extensively in Chapters 7 and 11.

A.4.2 Degenerate perturbation theory

When two or more modes are degenerate at ω_0 , the first-order correction is found by diagonalising the perturbation within the degenerate subspace. Let $\{|\mathbf{f}_\alpha\rangle\}_{\alpha=1}^p$ be the degenerate modes. The first-order frequency splittings are the eigenvalues of the $p \times p$ matrix:

$$V_{\alpha\beta} = -\frac{\omega_0}{2} \frac{\langle \mathbf{f}_\alpha | \delta\mathbf{M} | \mathbf{f}_\beta \rangle}{\langle \mathbf{f}_\alpha, \mathbf{f}_\alpha \rangle_{\mathbf{M}}} . \quad (\text{A.18})$$

This is the key formula for deriving the effective Dirac Hamiltonian at high-symmetry points (Chapter 6) and the mass terms from time-reversal or inversion breaking (Chapters 7, 11, 12).

A.5 Function spaces and boundary conditions

A.5.1 The relevant Sobolev spaces

[34]

The natural function space for the Maxwell eigenvalue problem is the space of square-integrable vector fields with square-integrable curl:

$$H(\text{curl}, \Omega) = \{\mathbf{F} \in L^2(\Omega)^3 \mid \nabla \times \mathbf{F} \in L^2(\Omega)^3\} . \quad (\text{A.19})$$

This is the electromagnetic analog of the Sobolev space H^1 . The boundary conditions (perfect conductor, periodic, or radiation) are imposed by restricting to appropriate subspaces.

For periodic media (photonic crystals), the relevant space is the Bloch-periodic version:

$$H_{\mathbf{k}}(\text{curl}, \Omega_{\text{cell}}) = \{\mathbf{F} \in H(\text{curl}, \Omega_{\text{cell}}) \mid \mathbf{F}(\mathbf{r} + \mathbf{a}) = e^{i\mathbf{k} \cdot \mathbf{a}} \mathbf{F}(\mathbf{r}) \ \forall \ \mathbf{a} \in \Lambda\}, \quad (\text{A.20})$$

where Λ is the Bravais lattice.

A.5.2 Completeness and regularity

For a lossless medium with positive-definite $\mathbf{M}$, the Maxwell eigenproblem on a bounded domain (or on a unit cell with Bloch boundary conditions) has a discrete spectrum accumulating at infinity. The eigenmodes form a complete orthonormal basis in the $\mathbf{M}$ -weighted L^2 space.

For smooth material parameters ($\mathbf{M} \in C^\infty$), the eigenmodes are themselves smooth inside the domain. At material interfaces where $\mathbf{M}$ has jump discontinuities, the tangential components of $\mathbf{E}$ and $\mathbf{H}$ remain continuous, but normal components may have jumps proportional to the contrast.

A.6 Useful matrix identities

For 2×2 matrices expressed in terms of Pauli matrices, $H = d_0 \mathbf{I} + \mathbf{d} \cdot \boldsymbol{\sigma}$:

$$\det(H) = d_0^2 - |\mathbf{d}|^2, \quad (\text{A.21})$$

$$\text{Tr}(H) = 2d_0, \quad (\text{A.22})$$

$$H^{-1} = \frac{d_0 \mathbf{I} - \mathbf{d} \cdot \boldsymbol{\sigma}}{d_0^2 - |\mathbf{d}|^2}, \quad (\text{A.23})$$

$$e^{i\theta \hat{n} \cdot \boldsymbol{\sigma}} = \cos \theta \mathbf{I} + i \sin \theta \hat{n} \cdot \boldsymbol{\sigma}. \quad (\text{A.24})$$

The Pauli-matrix trace identities are:

$$\text{Tr}(\sigma_a) = 0, \quad (\text{A.25})$$

$$\text{Tr}(\sigma_a \sigma_b) = 2\delta_{ab}, \quad (\text{A.26})$$

$$\text{Tr}(\sigma_a \sigma_b \sigma_c) = 2i\epsilon_{abc}. \quad (\text{A.27})$$

These identities are used extensively in the effective-Hamiltonian derivations of Chapters 6–8, 11–12, and 15.

A.7 The Baker–Campbell–Hausdorff formula

For operators A and B :

$$e^A e^B = \exp\left(A + B + \frac{1}{2}[A, B] + \frac{1}{12}([A, [A, B]] + [B, [B, A]]) + \dots\right). \quad (\text{A.28})$$

This is used in the derivation of the Wilson loop (Section 5.8) and in the analysis of Floquet evolution operators (Chapter 13).

A.8 The electromagnetic Green's function

A.8.1 Definition and properties

The Green's function (resolvent kernel) of the Maxwell operator is defined by

$$(\hat{\mathcal{N}} - \omega \mathbf{M}) \mathbf{G}(\mathbf{r}, \mathbf{r}'; \omega) = \delta^{(3)}(\mathbf{r} - \mathbf{r}') \mathbf{I}_6, \quad (\text{A.29})$$

where $\mathbf{I}_6$ is the 6×6 identity. Formally, the Green's function is the kernel of the resolvent operator $R(\omega) = (\hat{\mathcal{N}} - \omega \mathbf{M})^{-1}$:

$$R(\omega) = \sum_n \frac{|\mathbf{f}_n\rangle\langle\mathbf{f}_n|\mathbf{M}}{\omega_n - \omega}, \quad (\text{A.30})$$

where the sum runs over the complete set of eigenmodes. The poles of $R(\omega)$ are the eigenfrequencies ω_n , and the residue at each pole projects onto the corresponding eigenmode.

A.8.2 Spectral representation in periodic media

For a photonic crystal, the Green's function decomposes over Bloch wavevectors:

$$\mathbf{G}(\mathbf{r}, \mathbf{r}'; \omega) = \sum_n \int_{\text{BZ}} \frac{d^d k}{(2\pi)^d} \frac{\mathbf{f}_{n\mathbf{k}}(\mathbf{r}) \mathbf{f}_{n\mathbf{k}}^\dagger(\mathbf{r}') \mathbf{M}}{\omega_{n\mathbf{k}} - \omega}. \quad (\text{A.31})$$

The imaginary part $\text{Im Tr } \mathbf{G}(\mathbf{r}, \mathbf{r}; \omega + i0^+)$ gives the local density of states [12]:

$$\rho(\mathbf{r}, \omega) = -\frac{1}{\pi} \text{Im Tr} [\mathbf{M} \mathbf{G}(\mathbf{r}, \mathbf{r}; \omega + i0^+)]. \quad (\text{A.32})$$

This relation connects the spectral properties of the Maxwell operator (through the Green's function poles) to the experimentally observable local density of states.

A.8.3 Gap Chern number from the Green's function

The Chern number can be expressed directly in terms of the Green's function, without requiring explicit knowledge of the eigenmodes. For a frequency ω lying in a spectral gap, the gap Chern number is [20]

$$\mathcal{C}_{\text{gap}}(\omega) = \frac{\epsilon_{ab}}{4\pi i} \int_{\text{BZ}} \text{Tr} [G \partial_{k_a} G^{-1} G \partial_{k_b} G^{-1}] d^2 k, \quad (\text{A.33})$$

where $G \equiv \mathbf{G}(\mathbf{k}; \omega)$ is the $\mathbf{k}$ -dependent Green's function evaluated at the gap frequency. This formula is especially useful for non-Hermitian and continuum systems where the eigenmode basis may not be well defined [19].

A.9 Tensor product structure of the six-component space

The six-component Maxwell field $\mathbf{f} = (\mathbf{E}, \mathbf{H})^T$ lives in the direct sum $\mathcal{H} = \mathcal{H}_E \oplus \mathcal{H}_H$, where $\mathcal{H}_E$ and $\mathcal{H}_H$ are the spaces of square-integrable 3-component electric and magnetic fields respectively.

The material matrix $\mathbf{M}$ and the curl operator $\hat{\mathcal{N}}$ have specific block structures in this decomposition:

$$\mathbf{M} = \begin{pmatrix} \varepsilon_0 \varepsilon & 0 \\ 0 & \mu_0 \mu \end{pmatrix}, \quad \hat{\mathcal{N}} = \begin{pmatrix} 0 & \nabla \times \\ -\nabla \times & 0 \end{pmatrix}, \quad (\text{A.34})$$

in the absence of bianisotropy ($\xi = \zeta = 0$). The off-diagonal structure of $\hat{\mathcal{N}}$ couples the electric and magnetic sectors, while the block-diagonal structure of $\mathbf{M}$ separates them. This “chiral” structure is the origin of many symmetry properties: for example, $\hat{\mathcal{N}}$ anticommutes with the matrix $\sigma_3 = \text{diag}(\mathbf{I}_3, -\mathbf{I}_3)$, so that eigenfrequencies come in $\pm\omega$ pairs for lossless media.

When bianisotropic coupling $\xi \neq 0$ is present (as in the Khanikaev metamaterial of Chapter 11), the material matrix acquires off-diagonal blocks, and the direct-sum decomposition breaks down. The pseudospin topology of Chapter 11 can be understood as a controlled restoration of this decomposition through the bianisotropic coupling.

A.10 Proof of the curl-curl identity

The double-curl identity $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$ is fundamental to the reduction of the Maxwell wave equation to the Helmholtz equation. We give a component-wise proof using the Levi-Civita symbol.

Proof. The i -th component of $\nabla \times (\nabla \times \mathbf{F})$ is

$$[\nabla \times (\nabla \times \mathbf{F})]_i = \epsilon_{ijk} \partial_j (\nabla \times \mathbf{F})_k = \epsilon_{ijk} \epsilon_{klm} \partial_j \partial_l F_m. \quad (\text{A.35})$$

Using the contraction identity $\epsilon_{ijk} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$:

$$\begin{aligned} [\nabla \times (\nabla \times \mathbf{F})]_i &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \partial_j \partial_l F_m \\ &= \partial_j \partial_i F_j - \partial_j \partial_j F_i \\ &= \partial_i (\nabla \cdot \mathbf{F}) - \nabla^2 F_i. \end{aligned} \quad (\text{A.36})$$

This is the i -th component of $\nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$. □

A.11 Self-adjoint extensions and boundary conditions

A.11.1 The domain of the curl operator

The Maxwell curl operator $\hat{\mathcal{N}}$ is formally defined on smooth, compactly supported 6-component fields [71]. To make $\hat{\mathcal{N}}$ self-adjoint in the Hilbert space $\mathcal{H} = L^2(\Omega; \mathbf{M} dV)$ (with the electromagnetic energy inner product), one must specify boundary conditions.

On a domain Ω with boundary $\partial\Omega$, the natural boundary condition for the lossless Maxwell problem is the perfect electric conductor (PEC) condition:

$$\hat{n} \times \mathbf{E}|_{\partial\Omega} = 0, \quad (\text{A.37})$$

where $\hat{n}$ is the outward unit normal. With this condition, the integration-by-parts identity

$$\langle \mathbf{f}_1, \hat{\mathcal{N}} \mathbf{f}_2 \rangle_{\mathbf{M}} - \langle \hat{\mathcal{N}} \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}} = \oint_{\partial\Omega} (\mathbf{E}_1 \times \mathbf{H}_2 - \mathbf{E}_2 \times \mathbf{H}_1) \cdot \hat{n} \, dS \quad (\text{A.38})$$

shows that the boundary term vanishes, and $\hat{\mathcal{N}}$ is symmetric (formally self-adjoint) on the PEC domain. A rigorous proof of essential self-adjointness requires functional-analytic tools (Friedrichs extension, graph norm estimates), but the key physical insight is that the boundary conditions determine the domain of $\hat{\mathcal{N}}$, and hence the spectrum [19].

A.11.2 Boundary conditions and edge modes

Different boundary conditions produce different self-adjoint extensions of $\hat{\mathcal{N}}$, and hence different spectra. The topological edge modes discussed in Chapters 7–9 arise precisely because the boundary changes the self-adjoint extension: the bulk operator has a gapped spectrum, but the half-space operator (with PEC or impedance boundary conditions) has additional eigenvalues in the gap.

The number of these in-gap eigenvalues is determined by the Chern number of the bulk bands, through the bulk-edge correspondence. This is a rigorous mathematical statement: the spectral flow of the boundary problem equals the Chern number of the bulk problem, regardless of the specific boundary condition chosen [12, 20].

A.12 Spectral theory of the Maxwell operator

A.12.1 Essential and discrete spectrum

The spectrum of $\hat{\mathcal{N}}$ in a periodic medium decomposes into two parts. The *essential spectrum* consists of the photonic bands: continuous intervals of ω filled by propagating Bloch modes. The *discrete spectrum* consists of isolated eigenvalues (localised defect modes or edge modes), which can appear in band gaps.

For a perfect photonic crystal (no defects, no boundaries), the spectrum is purely essential and consists of the union of all photonic bands. When a boundary or defect is introduced, discrete eigenvalues may appear in the gaps. The topological edge modes are precisely such discrete eigenvalues, whose existence is guaranteed by the nonzero Chern number of the bulk bands.

A.12.2 Weyl's theorem and stability of the essential spectrum

Weyl's theorem states that the essential spectrum of a self-adjoint operator is invariant under compact perturbations. In the photonic context, this means that local defects (finite-range changes in the material matrix $\mathbf{M}$) cannot change the band structure of the bulk crystal; they can only add or remove discrete eigenvalues in the gaps.

This stability is the mathematical foundation of topological protection: the bulk gap cannot be closed by any local perturbation, so the edge modes that are topologically required to exist in the gap are also protected against local perturbations [12].

A.13 Bloch decomposition and direct integral

For a photonic crystal with lattice Λ , the Hilbert space decomposes as a direct integral over the Brillouin zone:

$$\mathcal{H} = \int_{\text{BZ}}^{\oplus} \mathcal{H}_{\mathbf{k}} \frac{d^d k}{(2\pi)^d}, \quad (\text{A.39})$$

where $\mathcal{H}_{\mathbf{k}} = L^2(\Omega_{\text{cell}}; \mathbf{M} dV)$ is the Hilbert space of cell-periodic fields at fixed $\mathbf{k}$. The Maxwell operator decomposes correspondingly: $\hat{\mathcal{N}} = \int_{\text{BZ}}^{\oplus} \hat{\mathcal{N}}(\mathbf{k}) d^d k / (2\pi)^d$.

Each fibre operator $\hat{\mathcal{N}}(\mathbf{k})$ acts on the compact domain Ω_{cell} with periodic boundary conditions, and therefore has a purely discrete spectrum: $\omega_{1,\mathbf{k}} \leq \omega_{2,\mathbf{k}} \leq \dots$. The union of these discrete spectra over all $\mathbf{k}$ gives the photonic bands, which form the essential spectrum of the full operator. The eigenmodes $\mathbf{u}_{n\mathbf{k}}$ provide the Bloch basis used throughout this book [19, 20].

A.14 Minimax characterisation of eigenvalues

The eigenfrequencies of the cell-periodic Maxwell operator can be characterised by a minimax (Courant–Fischer) principle. For the n -th eigenvalue:

$$\omega_n^2 = \min_{\dim V=n} \max_{\mathbf{f} \in V, \mathbf{f} \neq 0} \frac{\langle \mathbf{f}, \hat{\mathcal{N}}^2 \mathbf{f} \rangle_{\mathbf{M}_g}}{\langle \mathbf{f}, \mathbf{M}^2 \mathbf{f} \rangle_{\mathbf{M}_g}}, \quad (\text{A.40})$$

where the minimum is over all n -dimensional subspaces V of the domain of $\hat{\mathcal{N}}$. This variational characterisation has two important consequences.

First, it implies that the eigenfrequencies are monotonically ordered: $\omega_1 \leq \omega_2 \leq \dots$. Second, it provides bounds on how the eigenfrequencies change under perturbations of the material matrix: if $\mathbf{M}$ is replaced by $\mathbf{M} + \delta\mathbf{M}$ with $\|\delta\mathbf{M}\| \leq \eta\|\mathbf{M}\|$, then the eigenfrequencies shift by at most $O(\eta)$. This is the mathematical basis for the stability of the photonic band structure under weak disorder [12].

A.15 Perturbation bounds for photonic band gaps

A.15.1 First-order perturbation of eigenfrequencies

For a small perturbation $\mathbf{M} \rightarrow \mathbf{M} + \delta\mathbf{M}$ of the material matrix, the first-order shift of the n -th eigenfrequency is

$$\delta\omega_n = -\frac{\omega_n}{2} \frac{\langle \mathbf{u}_n, \delta\mathbf{M} \mathbf{u}_n \rangle_{\mathbf{M}_g}}{\langle \mathbf{u}_n, \mathbf{M} \mathbf{u}_n \rangle_{\mathbf{M}_g}}. \quad (\text{A.41})$$

This formula, which follows from the Hellmann–Feynman theorem applied to the generalised eigenvalue problem $\hat{\mathcal{N}} \mathbf{u} = \omega \mathbf{M} \mathbf{u}$, shows that the frequency shift is proportional to the overlap of the mode with the perturbation, weighted by the energy inner product.

For a gyrotropic perturbation $\delta\mathbf{M} = i\kappa_\mu \sigma_y$ (Faraday rotation), the first-order shift vanishes for modes with definite symmetry under time reversal, and the gap opening occurs at second order—consistent with the discussion in Chapter 7.

A.15.2 Gap stability under disorder

The band gap survives under disorder as long as the perturbation strength is smaller than the gap width: $\|\delta\mathbf{M}\| < \Delta\omega / \omega_{\text{mid}}$. This can be made rigorous using the minimax principle: any perturbation that shifts the n -th eigenvalue by less than $\Delta\omega/2$ cannot close the gap between bands n and $n + 1$. This provides a quantitative criterion for the robustness of photonic topological protection [13].

A.16 Sobolev spaces and regularity of Maxwell solutions

The functional-analytic framework for Maxwell's equations in periodic media requires Sobolev spaces, which control both the function values and their derivatives.

The Sobolev space $H^1(\Omega)$ consists of square-integrable functions whose first derivatives are also square-integrable: $\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2$. For Maxwell's equations, the natural Sobolev space is $H(\text{curl}; \Omega) = \{\mathbf{u} \in L^2(\Omega)^3 : \nabla \times \mathbf{u} \in L^2(\Omega)^3\}$, which is the space of fields with finite electromagnetic energy.

The Bloch eigenmodes $\mathbf{f}_{n\mathbf{k}}$ belong to $H(\text{curl}; \text{cell})$ for each $\mathbf{k}$, and their regularity in $\mathbf{k}$ determines the smoothness of the Berry connection. For a smooth material profile $\mathbf{M}(\mathbf{r})$, the eigenmodes depend analytically on $\mathbf{k}$ away from band crossings, ensuring that the Berry connection is well-defined as a smooth 1-form on the complement of the crossing set [19, 20].

At band crossings (Dirac points, quadratic touchings), the eigenmodes are not differentiable in $\mathbf{k}$, and the Berry curvature develops a delta-function singularity. This singularity is integrable, so the Chern number remains well-defined as the integral of the Berry curvature over the full Brillouin zone, including the crossing points [12].

A.17 The Helmholtz decomposition for periodic fields

Any square-integrable vector field on a periodic domain can be decomposed into curl-free, divergence-free, and harmonic components:

$$\mathbf{u} = \nabla\phi + \nabla \times \mathbf{A} + \mathbf{h}, \quad (\text{A.42})$$

where $\nabla \cdot \mathbf{h} = 0$, $\nabla \times \mathbf{h} = \mathbf{0}$, and $\mathbf{h}$ satisfies the Bloch boundary conditions. For a simply connected unit cell, the harmonic component vanishes, and the decomposition reduces to the familiar Helmholtz decomposition.

For Maxwell's equations, the physical modes are divergence-free ($\nabla \cdot \mathbf{M}\mathbf{f} = 0$), so only the curl component contributes to the physical spectrum. The curl-free component corresponds to longitudinal (gauge) modes with $\omega = 0$, which are excluded from the physical Hilbert space by the divergence condition. This decomposition is the functional-analytic foundation for the orthogonality and completeness results of Chapter 3 [20].

A.18 A worked Bloch decomposition estimate

We close this appendix with a concrete estimate that explains why the Bloch decomposition used throughout the book is not merely formal. Let Ω_L be a square supercell containing $L \times L$ primitive

cells, with periodic boundary conditions. A finite-energy electromagnetic field $\mathbf{f}(\mathbf{r})$ on Ω_L can be expanded in discrete Bloch sectors,

$$\mathbf{f}(\mathbf{r}) = \frac{1}{L} \sum_{\mathbf{k} \in \text{BZ}_L} e^{i\mathbf{k} \cdot \mathbf{r}} \mathbf{u}_{\mathbf{k}}(\mathbf{r}), \quad \mathbf{u}_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = \mathbf{u}_{\mathbf{k}}(\mathbf{r}), \quad (\text{A.43})$$

where BZ_L is the discrete Brillouin-zone mesh and $\mathbf{R}$ is any lattice vector. The electromagnetic energy norm splits as

$$\|\mathbf{f}\|_{\mathbf{M}}^2 = \frac{1}{L^2} \sum_{\mathbf{k} \in \text{BZ}_L} \|\mathbf{u}_{\mathbf{k}}\|_{\mathbf{M}, \text{cell}}^2, \quad (\text{A.44})$$

which is the Maxwell version of Parseval's identity. This identity is the reason that the infinite crystal can be studied one Bloch wavevector at a time: the Maxwell operator is block diagonal in the Bloch sectors, and the energy norm is the direct integral of the cell norms.

In the thermodynamic limit $L \rightarrow \infty$, the discrete sum becomes the Brillouin-zone integral

$$\mathcal{H}_{\text{phys}} \cong \int_{\text{BZ}}^{\oplus} \mathcal{H}_{\mathbf{k}} \frac{d^d k}{(2\pi)^d}. \quad (\text{A.45})$$

This direct-integral statement is the rigorous background for the band pictures in Chapter 4, the Berry connection in Chapter 5, and the numerical Brillouin-zone sums in Appendix C. It also clarifies why a Chern number is a global invariant over the Brillouin-zone torus: the fibres $\mathcal{H}_{\mathbf{k}}$ are independent at fixed $\mathbf{k}$, but their phases cannot always be chosen smoothly as $\mathbf{k}$ winds around the torus [12, 19].

For practical computations, (A.44) gives a useful convergence check. If the same field is represented on two successively finer supercells, the total energy computed by summing the cell norms over the discrete Bloch mesh should converge before any Berry curvature or Chern number is trusted. Failure of this energy sum-rule is usually a sign of an inconsistent material metric, an unresolved longitudinal mode, or a non-periodic gauge choice.

A.19 Non-Hermitian extensions of spectral theory

The spectral theory developed in Sections A.3–A.3.3 assumes a Hermitian Maxwell operator. When the photonic system includes gain or loss (Chapter 17), the operator is non-Hermitian and the spectral properties change qualitatively [125, 145].

A.19.1 Biorthogonal spectral decomposition

For a non-Hermitian operator $\hat{L}$ with a complete set of right eigenvectors $|\psi_n^R\rangle$ and left eigenvectors $\langle\psi_n^L|$, the spectral decomposition becomes

$$\hat{L} = \sum_n \lambda_n \frac{|\psi_n^R\rangle\langle\psi_n^L|}{\langle\psi_n^L|\psi_n^R\rangle}, \quad (\text{A.46})$$

where $\lambda_n \in \mathbb{C}$ are the (generally complex) eigenvalues. The biorthogonality condition $\langle\psi_m^L|\psi_n^R\rangle = \delta_{mn}$ (after normalisation) replaces the orthogonality of the Hermitian case [110].

This decomposition breaks down at exceptional points, where two or more eigenvalues coalesce and the right and left eigenvectors become parallel. At an exceptional point of order p , the operator has a Jordan block of size p , and the spectral decomposition must be replaced by a Jordan normal form. The non-diagonalisability at exceptional points is responsible for many of the exotic phenomena in non-Hermitian photonics, including enhanced sensitivity, chiral mode switching, and the breakdown of adiabatic evolution [123, 125].

A.19.2 Pseudospectra and spectral instability

A hallmark of non-Hermitian operators is spectral instability: small perturbations can move eigenvalues by large amounts. The ϵ -pseudospectrum of $\hat{L}$ is defined as

$$\sigma_\epsilon(\hat{L}) = \{z \in \mathbb{C} : \|(\hat{L} - z\hat{I})^{-1}\| \geq 1/\epsilon\}. \quad (\text{A.47})$$

For Hermitian operators, the ϵ -pseudospectrum is just the ϵ -neighbourhood of the real spectrum. For non-Hermitian operators, the pseudospectrum can be much larger, indicating that eigenvalues are unstable against perturbations.

In photonic systems with gain and loss, spectral instability means that the eigenfrequencies of a finite-size structure can differ significantly from the Bloch eigenfrequencies of the infinite periodic system—even in the absence of the non-Hermitian skin effect. This is an important consideration when comparing theoretical band structures with experimental measurements [129, 130].

A.19.3 Fredholm theory and the edge index

The rigorous formulation of bulk-boundary correspondence uses the theory of Fredholm operators. An operator $\hat{L}$ acting on a Hilbert space is Fredholm if its kernel and cokernel are both finite-dimensional. The Fredholm index is

$$\text{index}(\hat{L}) = \dim \ker(\hat{L}) - \dim \ker(\hat{L}^\dagger). \quad (\text{A.48})$$

For the edge of a photonic Chern insulator, the relevant operator is the restriction of the Maxwell operator to the half-plane, and its Fredholm index equals the bulk Chern number. This is a special case of the Toeplitz index theorem, which relates the index of a Toeplitz operator (a “half-space” restriction) to the winding number of its symbol [21].

The Fredholm theory provides the most rigorous proof of bulk-boundary correspondence: it shows that the existence of edge states is not merely a property of specific models or boundary conditions, but a consequence of the algebraic structure of the operators involved. Any perturbation that preserves the Fredholm property (and the bulk gap) cannot destroy the edge states [12, 33].

A.20 Completeness of Maxwell eigenmodes

A question that arises in the eigenmode expansion used throughout this book is whether the Maxwell eigenmodes form a complete set in the relevant function space.

A.20.1 Completeness in the Hermitian case

For a lossless periodic photonic crystal, the Maxwell eigenvalue problem $\hat{\mathcal{N}}\mathbf{f} = \omega\mathbf{M}\mathbf{f}$ with periodic boundary conditions is a self-adjoint eigenvalue problem on a compact domain (the unit cell). By the spectral theorem for compact self-adjoint operators, the eigenmodes form a complete orthonormal basis of the $\mathbf{M}$ -weighted L^2 space.

More precisely, for any field $\mathbf{f} \in L^2_{\mathbf{M}}(\text{cell})$ satisfying the appropriate boundary conditions, the eigenmode expansion

$$\mathbf{f}(\mathbf{r}) = \sum_{n=1}^{\infty} c_n \mathbf{f}_n(\mathbf{r}), \quad c_n = \langle \mathbf{f}_n, \mathbf{f} \rangle_{\mathbf{M}_g}, \quad (\text{A.49})$$

converges in the $L^2_{\mathbf{M}}$ norm. Pointwise convergence requires additional regularity assumptions on $\mathbf{f}$.

A.20.2 Completeness in dispersive and non-Hermitian systems

When the material is dispersive ($\varepsilon(\omega)$, $\mu(\omega)$ depend on frequency), the eigenvalue problem is nonlinear in ω , and the standard completeness theorem does not apply directly. Raman and Fan (2010) resolved this issue by reformulating the dispersive problem as a linear eigenvalue problem in an enlarged Hilbert space that includes auxiliary polarisation fields [43]. In this enlarged space, the standard completeness theorem applies, and the physical fields are recovered by projection.

For non-Hermitian systems (gain and loss), completeness can fail at exceptional points, where the eigenvectors do not span the full space. Away from exceptional points, the biorthogonal eigenmodes $\{|\psi_n^R\rangle, \langle\psi_n^L|\}$ form a complete biorthogonal set, and the expansion $\mathbf{f} = \sum_n c_n |\psi_n^R\rangle$ with $c_n = \langle\psi_n^L|\mathbf{f}\rangle / \langle\psi_n^L|\psi_n^R\rangle$ converges in the appropriate sense [110, 145].

Appendix B

Differential Forms and Chern Classes

B.1 Motivation: why differential forms?

The Berry connection, Berry curvature, and Chern number are most naturally expressed in the language of differential forms on the Brillouin zone [48]. While the coordinate-based formulas used in the main text are sufficient for computation, the differential-form viewpoint clarifies *why* the Chern number is an integer, why it is gauge-invariant, and how it generalises to higher dimensions and non-Abelian bundles.

This appendix provides a self-contained introduction to the relevant mathematics. The reader who is comfortable with exterior calculus can skip to Section B.5; the reader who wants only the practical formulas can use the summary table at the end.

B.2 Differential forms on the Brillouin zone

B.2.1 The Brillouin zone as a manifold

The first Brillouin zone (BZ) of a 2D lattice is a parallelogram in reciprocal space with opposite edges identified. Topologically, it is a 2-torus $T^2 = S^1 \times S^1$. For a 3D lattice, $\text{BZ} \cong T^3$.

A differential p -form on T^d is a smooth antisymmetric tensor field of rank p . In coordinates $(k_1, \dots, k_d)$:

$$\alpha = \sum_{i_1 < \dots < i_p} \alpha_{i_1 \dots i_p}(\mathbf{k}) \, dk_{i_1} \wedge \dots \wedge dk_{i_p}. \quad (\text{B.1})$$

B.2.2 The exterior derivative

The exterior derivative d maps p -forms to $(p + 1)$ -forms:

$$d\alpha = \sum_j \frac{\partial \alpha_{i_1 \dots i_p}}{\partial k_j} dk_j \wedge dk_{i_1} \wedge \dots \wedge dk_{i_p}. \quad (\text{B.2})$$

The key property is $d^2 = 0$, which is the form-language version of $\nabla \times (\nabla \phi) = 0$ and $\nabla \cdot (\nabla \times \mathbf{F}) = 0$.

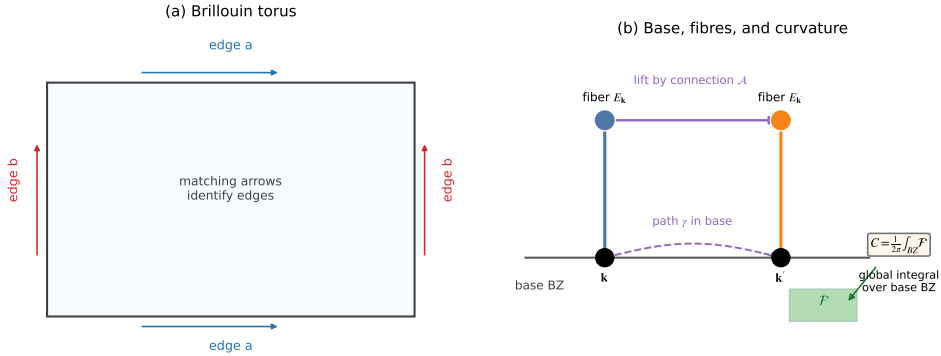


Figure B.1: The Bloch bundle and Berry curvature in differential-form language. The base space is the Brillouin-zone torus T^2 , drawn as a fundamental square with opposite sides identified. Fibres are the Maxwell eigenspaces of a chosen band, the Berry connection $\mathcal{A}$ transports phases along paths, and the curvature $\mathcal{F} = d\mathcal{A}$ is an oriented two-form. The Chern number is the total curvature flux, $C = (2\pi)^{-1} \int_{T^2} \mathcal{F}$.

B.2.3 Integration of forms

A p -form can be integrated over a p -dimensional oriented manifold:

$$\int_{T^2} \Omega = \int_{\text{BZ}} \Omega_{12}(\mathbf{k}) dk_1 \wedge dk_2. \quad (\text{B.3})$$

Stokes' theorem in form language is $\int_M d\alpha = \oint_{\partial M} \alpha$. Since T^2 has no boundary ($\partial T^2 = \emptyset$), $\int_{T^2} d\alpha = 0$ for any 1-form α .

B.3 The Bloch bundle

B.3.1 Definition

At each $\mathbf{k} \in \text{BZ}$, the Maxwell eigenvalue problem produces a set of eigenmodes $|\mathbf{u}_{n\mathbf{k}}\rangle$. Fixing a band n , the collection of one-dimensional subspaces $\{|\mathbf{u}_{n\mathbf{k}}\rangle\}_{\mathbf{k} \in \text{BZ}}$ forms a complex line bundle over the torus T^2 . This is the *Bloch bundle* of band n .

More precisely, the Bloch bundle is the fibre bundle $\pi : E_n \rightarrow T^2$ with fibre $\mathbb{C}$ and structure group $U(1)$ [32, 146, 147]. A *section* of this bundle is a smooth assignment $\mathbf{k} \mapsto |\mathbf{u}_{n\mathbf{k}}\rangle$ (i.e., a smooth gauge choice).

Figure B.1 translates this definition into the geometric picture used in the main text. The Brillouin zone is a torus, not an open square: opposite edges are identified. Above each point of this torus sits the one-dimensional eigenspace of the chosen Maxwell band. A local Berry connection records how a normalised mode changes along a path in $\mathbf{k}$ -space, while the curvature measures the infinitesimal rotation accumulated around a small area. The first Chern number is the total curvature flux through this torus.

B.3.2 Gauge transformations

A gauge transformation is a smooth map $g : T^2 \rightarrow U(1)$, $\mathbf{k} \mapsto e^{i\chi(\mathbf{k})}$, which acts on sections by $|\mathbf{u}_{n\mathbf{k}}\rangle \mapsto e^{i\chi(\mathbf{k})}|\mathbf{u}_{n\mathbf{k}}\rangle$.

Not every line bundle admits a globally smooth section. A bundle that does is called *trivial*; one that does not is *nontrivial*. The Chern number measures the obstruction to triviality.

B.4 Connection and curvature

B.4.1 The Berry connection as a 1-form

The Berry connection of band n is the $U(1)$ connection 1-form on the Bloch bundle:

$$\mathcal{A}_n = i\langle \mathbf{u}_{n\mathbf{k}} | \mathbf{M}_g | \mathbf{u}_{n\mathbf{k}} \rangle = \sum_a \mathcal{A}_{n,a}(\mathbf{k}) dk_a, \quad (\text{B.4})$$

where $d\mathbf{u}_{n\mathbf{k}} = \sum_a (\partial_{k_a} \mathbf{u}_{n\mathbf{k}}) dk_a$ and the inner product uses the electromagnetic energy metric $\mathbf{M}_g$.

Under a gauge transformation $|\mathbf{u}\rangle \rightarrow e^{i\chi}|\mathbf{u}\rangle$:

$$\mathcal{A}_n \rightarrow \mathcal{A}_n - d\chi. \quad (\text{B.5})$$

This is the electromagnetic analog of the $U(1)$ gauge transformation in electrodynamics.

B.4.2 The Berry curvature as a 2-form

The Berry curvature is the curvature 2-form of the connection:

$$\mathcal{F}_n = d\mathcal{A}_n = \Omega_n(\mathbf{k}) dk_1 \wedge dk_2, \quad (\text{B.6})$$

where $\Omega_n = \partial_{k_1} \mathcal{A}_{n,2} - \partial_{k_2} \mathcal{A}_{n,1}$ is the Berry curvature scalar in 2D.

The curvature is gauge-invariant: under $\mathcal{A}_n \rightarrow \mathcal{A}_n - d\chi$, we have $\mathcal{F}_n \rightarrow d\mathcal{A}_n - d^2\chi = \mathcal{F}_n$ (using $d^2 = 0$).

B.5 The first Chern class and integrality

[5]

B.5.1 Definition

The first Chern number of the Bloch bundle is

$$\mathcal{C}_n = \frac{1}{2\pi} \int_{T^2} \mathcal{F}_n = \frac{1}{2\pi} \int_{\text{BZ}} \Omega_n(\mathbf{k}) d^2k. \quad (\text{B.7})$$

B.5.2 Proof of integrality

The Chern number is always an integer. The proof uses the patchwise construction of the bundle.

Cover T^2 with two open sets U_N and U_S (“north” and “south” patches, each a disc) such that $U_N \cup U_S = T^2$ and the overlap $U_N \cap U_S \cong S^1$ is an annulus.

On each patch, choose a smooth gauge and compute the Berry connection: $\mathcal{A}_N$ on U_N , $\mathcal{A}_S$ on U_S . On the overlap, the two gauges are related by a transition function $g_{NS} : U_N \cap U_S \rightarrow U(1)$, so $\mathcal{A}_N = \mathcal{A}_S - d\chi_{NS}$.

By Stokes’ theorem on each patch:

$$\int_{U_N} \mathcal{F}_N = \oint_{\partial U_N} \mathcal{A}_N, \quad (\text{B.8})$$

$$\int_{U_S} \mathcal{F}_S = -\oint_{\partial U_S} \mathcal{A}_S, \quad (\text{B.9})$$

where the signs reflect orientation. Adding:

$$\begin{aligned} \int_{T^2} \mathcal{F} &= \oint_{\partial U_N} \mathcal{A}_N + \oint_{\partial U_S} \mathcal{A}_S \\ &= \oint_{S^1} (\mathcal{A}_N - \mathcal{A}_S) = -\oint_{S^1} d\chi_{NS} \\ &= -[\chi_{NS}]_{S^1}. \end{aligned} \quad (\text{B.10})$$

The transition function $g_{NS} = e^{i\chi_{NS}}$ must be single-valued on S^1 , so $[\chi_{NS}]_{S^1} = 2\pi\nu$ for some integer ν . Therefore $\mathcal{C}_n = \nu \in \mathbb{Z}$.

This proof shows that the Chern number counts the winding number of the transition function around the overlap circle—equivalently, the number of times the gauge must “twist” as the momentum traverses the Brillouin zone.

B.5.3 Physical interpretation

A nonzero Chern number means the Bloch bundle is nontrivial: there is no globally smooth gauge choice for the cell-periodic modes over the entire Brillouin zone. This obstruction is precisely what forces the existence of edge modes at the boundary of the material, through the bulk-edge correspondence.

B.6 Non-Abelian generalisation

B.6.1 Multi-band bundles

When p bands are degenerate or nearly degenerate, they form a rank- p vector bundle with structure group $U(p)$. The Berry connection becomes a matrix-valued 1-form:

$$[\mathcal{A}]_{\alpha\beta} = i\langle \mathbf{u}_{\alpha\mathbf{k}} | \mathbf{M}_g | d\mathbf{u}_{\beta\mathbf{k}} \rangle, \quad \alpha, \beta = 1, \dots, p. \quad (\text{B.11})$$

The curvature is

$$\mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}. \quad (\text{B.12})$$

The non-Abelian Berry curvature has the extra quadratic term $\mathcal{A} \wedge \mathcal{A}$, which vanishes in the Abelian ($p = 1$) case.

B.6.2 The non-Abelian Chern number

The first Chern number of the rank- p bundle is

$$\mathcal{C} = \frac{1}{2\pi} \int_{T^2} \text{Tr}(\mathcal{F}) . \quad (\text{B.13})$$

Note that only the *trace* of the curvature enters. This is gauge-invariant (invariant under $U(p)$ gauge transformations) and is always an integer.

B.6.3 The second Chern number

In four dimensions, the second Chern number is

$$\mathcal{C}_2 = \frac{1}{8\pi^2} \int_{T^4} \text{Tr}(\mathcal{F} \wedge \mathcal{F}) . \quad (\text{B.14})$$

This is a topological invariant of bundles over the 4-torus T^4 and appears in the theory of the 4D quantum Hall effect (Chapter 16).

B.7 Summary of correspondence

Photonic concept	Mathematical object
Brillouin zone	Torus T^d
Band eigenmodes $ \mathbf{u}_{n\mathbf{k}}\rangle$	Section of line bundle E_n
Gauge choice	Trivialisation of E_n
Berry connection $\mathcal{A}_n$	$U(1)$ connection 1-form
Berry curvature Ω_n	Curvature 2-form $\mathcal{F}_n$
Chern number $\mathcal{C}_n$	First Chern class $c_1(E_n) \in \mathbb{Z}$
Multi-band connection	$U(p)$ connection
Wilson loop	Holonomy of the connection
Second Chern number	$c_2 \in \mathbb{Z}$ (4D bundles)

The central message: the integrality of the Chern number is not a numerical coincidence or an artefact of a particular computation method. It is a theorem of differential geometry, guaranteed by the topology of the Brillouin-zone torus and the smoothness of the Bloch bundle. This is why the number of topological edge modes is exactly quantised and robust to perturbations.

B.8 The Hodge star and electromagnetic duality

B.8.1 Definition of the Hodge star

On a d -dimensional oriented Riemannian manifold, the Hodge star $\star$ is a linear map from p -forms to $(d-p)$ -forms, defined by the requirement that for any two p -forms α, β :

$$\alpha \wedge \star \beta = \langle \alpha, \beta \rangle \text{vol} , \quad (\text{B.15})$$

where $\langle \cdot, \cdot \rangle$ is the inner product induced by the metric and vol is the volume form.

In three dimensions with Cartesian coordinates:

$$\begin{aligned} \star dx &= dy \wedge dz, & \star dy &= dz \wedge dx, & \star dz &= dx \wedge dy, \\ \star 1 &= dx \wedge dy \wedge dz, & \star(dx \wedge dy \wedge dz) &= 1. \end{aligned} \quad (\text{B.16})$$

In particular, $\star \star \alpha = (-1)^{p(d-p)} \alpha$ for a p -form in d dimensions. In three dimensions, $\star \star = +1$ on 1-forms and 2-forms.

B.8.2 Maxwell's equations in form language

Maxwell's equations in vacuum can be written compactly using differential forms. Define the 2-forms F (Faraday tensor) and $\star F$ (its Hodge dual):

$$dF = 0, \quad d \star F = \star J, \quad (\text{B.17})$$

where J is the current 1-form. The first equation encodes $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$ and $\nabla \cdot \mathbf{B} = 0$; the second encodes $\nabla \times \mathbf{H} = \mathbf{J} + \partial_t \mathbf{D}$ and $\nabla \cdot \mathbf{D} = \rho$. The electromagnetic duality $F \rightarrow \star F$, $\star F \rightarrow -F$ is manifest in this language.

In a medium with material response, the constitutive relation $\star F \mapsto \mathbf{M} F$ replaces the vacuum Hodge star with the material-weighted version, and the electromagnetic inner product of Chapter 3 becomes the L^2 inner product with the material-modified Hodge star as the metric.

B.9 De Rham cohomology and topological obstructions

B.9.1 Closed and exact forms

A p -form α is

emphclosed if $d\alpha = 0$ and

emphexact if $\alpha = d\beta$ for some $(p-1)$ -form β . Every exact form is closed (since $d^2 = 0$), but the converse is not always true. The quotient space

$$H_{\text{dR}}^p(M) = \frac{\ker(d : \Lambda^p \rightarrow \Lambda^{p+1})}{\text{im}(d : \Lambda^{p-1} \rightarrow \Lambda^p)} \quad (\text{B.18})$$

is the p -th de Rham cohomology group of the manifold M . Its dimension $b_p = \dim H_{\text{dR}}^p(M)$ is the p -th Betti number, a topological invariant of M .

For the 2-torus T^2 : $b_0 = 1$ (connected), $b_1 = 2$ (two independent non-contractible loops), $b_2 = 1$ (one independent “area” form). The fact that $b_2 = 1$ means there exists exactly one independent closed 2-form (modulo exact forms) on T^2 . The Berry curvature $\mathcal{F}_n$ is a closed 2-form (since $d\mathcal{F}_n = d^2 \mathcal{A}_n = 0$), and its cohomology class is characterised by a single integer—the Chern number.

B.9.2 The Poincaré lemma and its failure

On a contractible domain (such as $\mathbb{R}^d$ or a disc), every closed form is exact. This is the Poincaré lemma: if $d\alpha = 0$, then $\alpha = d\beta$. On the torus, the Poincaré lemma emphails: the Berry curvature $\mathcal{F}_n$ is closed but not necessarily exact. Its failure to be exact is precisely the topological obstruction measured by the Chern number. If $\mathcal{C}_n \neq 0$, the Berry connection $\mathcal{A}_n$ cannot be defined globally, and the Bloch bundle is nontrivial.

B.10 The Chern–Simons 3-form

[33]

B.10.1 Definition

The Chern–Simons 3-form is defined for a connection 1-form $\mathcal{A}$ with curvature $\mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}$ by

$$\text{CS}(\mathcal{A}) = \mathcal{A} \wedge d\mathcal{A} + \frac{2}{3} \mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A}. \quad (\text{B.19})$$

Its exterior derivative satisfies $d\text{CS}(\mathcal{A}) = \text{Tr}(\mathcal{F} \wedge \mathcal{F})$, which is the integrand of the second Chern number. In the Abelian case ($U(1)$ bundle), $\mathcal{A} \wedge \mathcal{A} = 0$ and $\text{CS} = \mathcal{A} \wedge d\mathcal{A} = \mathcal{A} \wedge \mathcal{F}$.

B.10.2 Role in topological photonics

The Chern–Simons form appears in several contexts in this book. In three dimensions, the magnetoelectric polarisability of a 3D topological insulator is proportional to the Chern–Simons integral $\theta = \int \text{CS}(\mathcal{A})$ over the 3D Brillouin zone, where $\theta = 0$ (trivial) or $\theta = \pi$ (topological) modulo 2π . This θ -term characterises the 3D photonic topological insulators discussed in Chapter 15 [12].

In the Floquet context (Chapter 13), the winding number (13.8) is a Chern–Simons-type invariant: it integrates a 3-form over the 3D torus $\text{BZ} \times [0, T]$, and its integrality follows from the same cohomological argument as the Chern number.

B.11 Berry phase as holonomy

[51]

The Berry phase acquired by an eigenmode on a closed loop γ in the Brillouin zone is the holonomy of the Berry connection:

$$\Phi_B[\gamma] = \oint_{\gamma} \mathcal{A}_n = \int_S \mathcal{F}_n, \quad (\text{B.20})$$

where S is any surface bounded by γ (by Stokes' theorem). In the language of fibre bundles, the holonomy is the element of the structure group $U(1)$ obtained by parallel-transporting a fibre element around the loop using the connection.

For a non-Abelian ($U(p)$) connection, the holonomy becomes a unitary matrix—the Wilson loop. The eigenvalues of the Wilson loop are the non-Abelian Berry phases, and their winding as

the base point of the loop sweeps across the Brillouin zone gives the Chern number through the relation

$$\mathcal{C} = \frac{1}{2\pi i} \oint d \ln \det W(k_{\perp}), \quad (\text{B.21})$$

where $W(k_{\perp})$ is the Wilson loop evaluated along the direction parallel to γ and $k_{\perp}$ parametrises the perpendicular direction [12, 19]. This formula is used in the numerical Chern computation of Chapter 10 and Appendix C.

B.12 K-theory and topological classification

[11, 59, 148]

B.12.1 Vector bundles and stable equivalence

The classification of topological phases can be formalised using K-theory, which classifies vector bundles over the Brillouin zone up to stable equivalence. Two vector bundles E_1 and E_2 over the Brillouin zone are stably equivalent if there exist trivial bundles ϵ_1 and ϵ_2 such that $E_1 \oplus \epsilon_1 \cong E_2 \oplus \epsilon_2$. The set of stable equivalence classes forms an abelian group $K(X)$, where X is the Brillouin zone [12].

For a 2D Brillouin zone ($X = T^2$), the K-group is $K(T^2) \cong \mathbb{Z}^2$: one factor counts the rank (number of bands) and the other counts the first Chern number. This confirms that the Chern number is the complete topological invariant for complex vector bundles over the 2-torus, as established by the Chern class computation in Section B.5.

B.12.2 The tenfold way and symmetry classes

When discrete symmetries (time reversal $\hat{\mathcal{T}}$, particle-hole $\mathcal{C}$, and chiral $\mathcal{S}$) are imposed, the classification changes. The resulting “tenfold way” classification, originally developed for electronic systems by Altland and Zirnbauer, can be adapted to photonic systems.

For photonic Chern insulators with broken $\hat{\mathcal{T}}$ (class A in the tenfold way), the invariant is $\mathbb{Z}$ in 2D (the Chern number) and trivial in 1D and 3D. For photonic systems with $\hat{\mathcal{T}}^2 = +1$ (class AI), the classification is trivial in 2D, meaning no topological protection is possible without additional symmetries—this is why reciprocal photonic topological phases (Chapter 11) require crystalline symmetry or bianisotropic coupling.

The non-Hermitian extension of the tenfold way (relevant to Chapter 17) further enriches the classification by distinguishing line gaps from point gaps, leading to 38 symmetry classes in total [13] [145].

B.13 Wannier functions and the obstruction to localisation

B.13.1 Definition

The Wannier function for band n centred at lattice site $\mathbf{R}$ is the Fourier transform of the Bloch mode:

$$w_{n\mathbf{R}}(\mathbf{r}) = \int_{\text{BZ}} e^{-i\mathbf{k}\cdot\mathbf{R}} \mathbf{f}_{n\mathbf{k}}(\mathbf{r}) \frac{d^d k}{(2\pi)^d}. \quad (\text{B.22})$$

The Wannier function depends on the gauge choice for the Bloch modes. When the gauge can be chosen smoothly over the entire Brillouin zone, the Wannier functions are exponentially localised: $|w_{n\mathbf{R}}(\mathbf{r})| \sim e^{-|\mathbf{r}-\mathbf{R}|/\ell}$ for some localisation length ℓ .

B.13.2 Topological obstruction

When the Chern number $\mathcal{C}_n \neq 0$, no smooth gauge exists over the entire Brillouin zone. The Berry connection has a singularity that cannot be removed, and the Wannier functions are necessarily delocalised: they decay algebraically rather than exponentially. This obstruction to Wannier localisation is the real-space signature of band topology [12, 19].

The practical implication is that a topological photonic band cannot be represented by a tight-binding model with exponentially decaying hopping parameters. Any finite-range tight-binding approximation to a Chern band necessarily produces a topologically trivial model. This is why the full Bloch-wave treatment of this book, rather than a simple tight-binding approach, is essential for a faithful description of photonic topological phases.

B.14 Chern–Weil theory: from curvature to integers

The integrality of the Chern number—the fact that $\mathcal{C}_n = \frac{1}{2\pi} \int_{\text{BZ}} \mathcal{F}_n d^2 k$ is always an integer—is one of the deepest results in the topology of band structures. The proof is not merely a calculational trick; it reflects a fundamental topological constraint on how vector bundles can be constructed over the Brillouin zone.

B.14.1 The Chern–Weil homomorphism

Let $E \rightarrow X$ be a complex vector bundle with connection ∇ and curvature $\mathcal{F}$. The Chern–Weil homomorphism states that for any invariant polynomial P on the Lie algebra of the structure group, the cohomology class of $P(\mathcal{F})$ is independent of the choice of connection. For the first Chern class, the relevant polynomial is the trace:

$$c_1(E) = \frac{1}{2\pi i} \text{Tr } \mathcal{F} \in H_{\text{dR}}^2(X; \mathbb{R}). \quad (\text{B.23})$$

The integral of c_1 over any closed 2-cycle is an integer. This integrality follows from the fact that c_1 is the image of an integral cohomology class under the natural map $H^2(X; \mathbb{Z}) \rightarrow H_{\text{dR}}^2(X; \mathbb{R})$.

B.14.2 Worked example: the Bloch line bundle over the torus

Consider an isolated photonic band n over the 2D Brillouin zone T^2 . The Bloch modes $\mathbf{u}_{n\mathbf{k}}$ form a complex line bundle $L_n \rightarrow T^2$. On an open patch $U_\alpha \subset T^2$ where a smooth gauge $\mathbf{u}_{n\mathbf{k}}^{(\alpha)}$ can be chosen, the Berry connection 1-form is $\mathcal{A}^{(\alpha)} = i\langle \mathbf{u}_n^{(\alpha)}, d\mathbf{u}_n^{(\alpha)} \rangle_{\mathbf{M}_g}$ and the Berry curvature is $\mathcal{F} = d\mathcal{A}^{(\alpha)}$.

When $\mathcal{C}_n \neq 0$, no single smooth gauge covers the entire torus, so we need at least two patches U_1, U_2 with transition functions $g_{12} : U_1 \cap U_2 \rightarrow U(1)$. On the overlap,

$$\mathbf{u}_n^{(2)} = g_{12} \mathbf{u}_n^{(1)}, \quad \mathcal{A}^{(2)} = \mathcal{A}^{(1)} + i d \ln g_{12}. \quad (\text{B.24})$$

The Chern number counts the total winding of g_{12} around the overlap region:

$$\mathcal{C}_n = \frac{1}{2\pi i} \oint_{\partial U_1} d \ln g_{12}. \quad (\text{B.25})$$

This is manifestly an integer because the winding number of a $U(1)$ -valued function on a circle is always an integer [12].

B.14.3 Why classical Bloch fields form complex line bundles

A natural question is why the Bloch modes of Maxwell's equations, which are classical real-valued fields, form *complex* line bundles. The answer lies in the Bloch decomposition itself: even though the physical fields $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{H}(\mathbf{r}, t)$ are real, the cell-periodic Bloch modes $\mathbf{u}_{n\mathbf{k}}(\mathbf{r})$ are complex-valued because the Bloch phase factor $e^{i\mathbf{k}\cdot\mathbf{r}}$ is complex.

More precisely, the cell-periodic eigenvalue problem $\hat{\mathcal{N}}(\mathbf{k})\mathbf{u} = \omega\mathbf{M}\mathbf{u}$ is a Hermitian problem with complex coefficients (the operator $\hat{\mathcal{N}}(\mathbf{k})$ involves $i\mathbf{k}\times$), so its eigenmodes are naturally complex. The Berry connection and Chern number are well-defined invariants of this complex eigenvalue problem, regardless of the reality of the underlying physical fields. This is the mathematical reason why photonic band topology is as rich as electronic band topology, despite the classical nature of light [19, 20].

B.15 Second Chern number and four-dimensional physics

The second Chern number is defined for a rank- p vector bundle over a 4D manifold:

$$\mathcal{C}_2 = \frac{1}{8\pi^2} \int \text{Tr}(\mathcal{F} \wedge \mathcal{F}). \quad (\text{B.26})$$

In photonics, 4D Brillouin zones do not occur naturally, but the second Chern number appears in two important contexts: (i) in synthetic-dimension systems (Chapter 16) where two spatial plus two synthetic dimensions form a 4D parameter space, and (ii) in the Floquet quasi-energy spectrum (Chapter 13) where the 3D invariant $(\mathbf{k}, t) \in T^2 \times S^1$ can be related to the second Chern number through a dimensional reduction. The experimental observation of the 4D quantum Hall effect in photonic systems [12, 13] is one of the most striking demonstrations of the power of synthetic dimensions in topological photonics.

B.16 Integrality of the Chern number: a proof sketch

The fact that the Chern number is always an integer is one of the most remarkable results in the theory of fibre bundles, and it underlies the robustness of topological edge states. We sketch the proof here, following the two-patch construction of Section B.5.2.

Consider a complex line bundle L over the 2-torus $\text{BZ} = T^2$ (the Brillouin zone). Cover T^2 by two open sets U_N and U_S , overlapping in an annular region $U_N \cap U_S \cong S^1 \times (-\epsilon, \epsilon)$. On each patch, choose a smooth local section $|u_\alpha(\mathbf{k})\rangle$, and define the Berry connection 1-forms $\mathcal{A}_\alpha = i\langle u_\alpha | du_\alpha \rangle$. On the overlap, the two sections are related by a transition function $g_{NS} : S^1 \rightarrow \text{U}(1)$, and the Berry connections transform as $\mathcal{A}_N = \mathcal{A}_S + i g_{NS}^{-1} dg_{NS}$.

The Chern number is

$$\mathcal{C} = \frac{1}{2\pi} \int_{T^2} \mathcal{F} = \frac{1}{2\pi} \left(\int_{U_N} d\mathcal{A}_N + \int_{U_S} d\mathcal{A}_S \right) = \frac{1}{2\pi} \oint_{S^1} (\mathcal{A}_N - \mathcal{A}_S) = \frac{1}{2\pi} \oint_{S^1} i g_{NS}^{-1} dg_{NS}, \quad (\text{B.27})$$

where the second equality uses Stokes' theorem and the opposite orientations of the overlap boundary. The last integral is the winding number of the transition function $g_{NS} : S^1 \rightarrow \text{U}(1)$, which is always an integer by the theory of the fundamental group $\pi_1(\text{U}(1)) = \mathbb{Z}$.

This proof reveals the topological origin of the Chern number: it counts the number of times the transition function winds around the unit circle as one traverses the equator of the Brillouin zone. The integer quantisation is a consequence of the fact that the winding number is a homotopy invariant—it cannot change under continuous deformations of the bundle [12, 21].

B.17 Why Bloch modes form complex line bundles

The identification of Bloch modes with sections of a complex line bundle is the mathematical foundation of topological band theory. Here we make this identification precise.

For a photonic crystal with N bands below a gap, the Bloch eigenstates $|\mathbf{u}_{n\mathbf{k}}\rangle$ ($n = 1, \dots, N$) at each $\mathbf{k}$ span an N -dimensional subspace $V(\mathbf{k}) \subset \mathcal{H}_{\text{cell}}$ of the Hilbert space of cell-periodic fields. As $\mathbf{k}$ varies over the Brillouin zone, these subspaces form a rank- N complex vector bundle $E \rightarrow T^2$.

For a single isolated band ($N = 1$), the bundle is a complex line bundle, and the Chern number is the first Chern class evaluated on the fundamental cycle of T^2 . For $N > 1$, the relevant invariant is the total Chern number $\mathcal{C}_{\text{total}} = \sum_{n=1}^N \mathcal{C}_n$, which equals the first Chern class of the determinant line bundle $\det(E) = \bigwedge^N E$.

The obstruction to choosing a globally smooth gauge $|\mathbf{u}_{n\mathbf{k}}\rangle$ over the entire Brillouin zone is precisely measured by $\mathcal{C}_n$: when $\mathcal{C}_n \neq 0$, any gauge choice must have at least $|\mathcal{C}_n|$ vortices (phase singularities) in the Brillouin zone. These vortices are the Berry-phase singularities where the Berry curvature is concentrated, and they cannot be removed by any smooth gauge transformation [19, 35].

B.18 K-theory and the classification of photonic topological phases

The Chern number classifies topological phases of systems with broken time-reversal symmetry in two dimensions. But how do we classify topological phases in other symmetry classes and di-

mensions? The answer comes from K-theory, a branch of algebraic topology that classifies vector bundles over topological spaces [12, 21].

B.18.1 The periodic table of topological insulators

The “periodic table” of topological insulators and superconductors, derived independently by Schnyder, Ryu, Furusaki, and Ludwig (2008) and by Kitaev (2009), classifies gapped free-fermion Hamiltonians by their symmetry class and spatial dimension. There are ten symmetry classes, determined by the presence or absence of time-reversal symmetry ($\mathcal{T}$), particle-hole symmetry ($\mathcal{C}$), and chiral symmetry ($\mathcal{S} = \mathcal{T}\mathcal{C}$).

For photonic systems, the relevant symmetry classes are:

- **Class A** (no symmetries): classified by $\mathbb{Z}$ (the Chern number) in even dimensions. This is the class of gyromagnetic photonic crystals (Chapters 7–9).
- **Class AII** (time-reversal with $\mathcal{T}^2 = -1$): classified by $\mathbb{Z}_2$. This class does not occur naturally in photonics because photons are bosons, but effective $\mathcal{T}^2 = -1$ symmetry can be engineered using pseudospin degrees of freedom (Chapter 11) [5, 71].
- **Class AIII** (chiral symmetry only): classified by $\mathbb{Z}$ in odd dimensions. The SSH model (Chapter 14) belongs to this class.

The photonic “periodic table” is not identical to the electronic one because the Maxwell operator has additional structure (it is a first-order operator, and the energy inner product introduces the material tensor $\mathbf{M}$). De Nittis and Lein (2017) showed that the relevant K-group for photonic crystals is the same as for electronic systems in the corresponding symmetry class, provided one uses the electromagnetic energy inner product to define the Berry connection [33].

B.18.2 Index theorems for edge states

The bulk-boundary correspondence—the hallmark of topological phases—can be formulated as an index theorem. In the simplest case, the Chern number of the bulk equals the index (the difference between the number of right-moving and left-moving modes) of the edge Hamiltonian:

$$\mathcal{C}_{\text{bulk}} = \text{index}(D_{\text{edge}}) = n_+ - n_- , \quad (\text{B.28})$$

where n_+ and n_- are the numbers of edge modes with positive and negative group velocity, respectively.

For the photonic Chern insulator of Chapter 8, this theorem guarantees that one-way edge modes exist whenever the bulk Chern number is nonzero, regardless of the details of the edge termination. The robustness of this result—it holds for arbitrary smooth perturbations of the boundary—is the essence of topological protection.

In one dimension, the analogous index theorem relates the winding number of the SSH model to the number of zero-energy edge states (Section 14.12). This is a special case of the Atiyah–Singer index theorem applied to the Dirac-like operator $H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y$ [21, 48].

B.18.3 Higher-order topology and K-theory

Higher-order topological insulators (HOTIs) exhibit boundary states of codimension greater than one: corner states in 2D, hinge states in 3D. The classification of HOTIs requires an extension of K-theory that accounts for the crystalline symmetries of the lattice.

For photonic systems, HOTI phases have been realised in breathing-kagome lattices, where the topology is protected by C_3 rotation symmetry, and in quadrupole insulators, where the topology is protected by mirror symmetries. The classification of these phases uses the equivariant K-theory of the Brillouin zone with respect to the point group of the lattice [12, 59].

The experimental observation of photonic HOTI corner states [136] provides a direct demonstration that the K-theoretic classification correctly predicts the existence of boundary-localised states in photonic systems.

B.19 Non-Hermitian extensions of topological invariants

The topological invariants defined in this appendix assume Hermitian operators. When the photonic system has gain or loss (as in Chapter 17), the operators are non-Hermitian and the standard definitions must be modified.

B.19.1 The non-Hermitian Berry phase

For a non-Hermitian Hamiltonian $H(\mathbf{k}) \neq H^\dagger(\mathbf{k})$, the right eigenvectors $|u_n^R(\mathbf{k})\rangle$ and left eigenvectors $\langle u_n^L(\mathbf{k})|$ are generally different. The Berry connection is defined using the biorthogonal inner product [110, 145]:

$$\mathcal{A}_n^{(\text{NH})} = i \frac{\langle u_n^L | \nabla_{\mathbf{k}} | u_n^R \rangle}{\langle u_n^L | u_n^R \rangle}. \quad (\text{B.29})$$

The corresponding Berry curvature and Chern number are defined by the usual formulas $\mathcal{F} = d\mathcal{A}$ and $\mathcal{C} = \frac{1}{2\pi} \int_{\text{BZ}} \mathcal{F}$.

However, when the system exhibits the non-Hermitian skin effect (Section 17.5.3), the Bloch eigenvectors are not the correct basis for computing topological invariants. Instead, one must use the generalised Brillouin zone (GBZ) and the non-Bloch Berry connection, as described in Chapter 17 [132, 149].

B.19.2 Point-gap topology and the winding number

Non-Hermitian systems possess a topological structure that has no Hermitian analog: point-gap topology. A non-Hermitian Hamiltonian $H(\mathbf{k})$ has a point gap at a reference energy E_0 if the spectrum $\sigma(H(\mathbf{k}))$ never passes through E_0 for any $\mathbf{k}$ in the Brillouin zone.

The topological invariant associated with a point gap in one dimension is the winding number of $\det[H(k) - E_0]$ as k traverses the Brillouin zone:

$$\nu = \frac{1}{2\pi i} \oint_{\text{BZ}} d \ln \det[H(k) - E_0]. \quad (\text{B.30})$$

A nonzero ν implies the non-Hermitian skin effect: all eigenstates of the open-boundary system are exponentially localised at one edge [131, 133].

This point-gap topology is particularly relevant for photonic systems with asymmetric coupling (e.g. non-reciprocal waveguide arrays) or with spatially varying gain and loss profiles, where the non-Hermitian skin effect has been observed experimentally [125, 129].

Appendix C

Numerical Recipes

C.1 Overview

This appendix provides practical guidance, pseudocode, and working Python implementations for the numerical methods used throughout the book. Every algorithm starts from the electromagnetic eigenvalue problem and uses the material-weighted inner product [12, 19]. The Python code in this appendix is self-contained and can be run directly with NumPy and SciPy.

C.2 Plane-wave expansion for 2D photonic crystals

[39]

C.2.1 TM polarisation

For TM modes (E_z polarisation, $k_z = 0$) in a 2D photonic crystal with isotropic nonmagnetic media, the eigenvalue problem is (Section 4.5.1):

$$\sum_{\mathbf{G}'} |(\mathbf{k} + \mathbf{G}) \times \hat{z}| \widehat{\mu^{-1}}(\mathbf{G} - \mathbf{G}') |(\mathbf{k} + \mathbf{G}') \times \hat{z}| \hat{E}_z(\mathbf{G}') = \frac{\omega^2}{c^2} \hat{\varepsilon}(\mathbf{G} - \mathbf{G}') \hat{E}_z(\mathbf{G}'). \quad (\text{C.1})$$

The algorithm proceeds in four steps. First, choose a set of N_G reciprocal lattice vectors $\{\mathbf{G}_1, \dots, \mathbf{G}_{N_G}\}$, typically all vectors with $|\mathbf{G}| \leq G_{\max}$. Second, compute the Fourier coefficients $\hat{\varepsilon}(\mathbf{G})$ and $\widehat{\mu^{-1}}(\mathbf{G})$ by numerical integration over the unit cell. Third, at each $\mathbf{k}$ point, assemble the $N_G \times N_G$ matrices $\mathbf{A}$ and $\mathbf{B}$ with elements $A_{ij} = |(\mathbf{k} + \mathbf{G}_i) \times \hat{z}| \widehat{\mu^{-1}}(\mathbf{G}_i - \mathbf{G}_j) |(\mathbf{k} + \mathbf{G}_j) \times \hat{z}|$ and $B_{ij} = \hat{\varepsilon}(\mathbf{G}_i - \mathbf{G}_j)$. Fourth, solve the generalised eigenvalue problem $\mathbf{A} \hat{\mathbf{E}}_z = (\omega^2/c^2) \mathbf{B} \hat{\mathbf{E}}_z$.

For high-contrast structures with sharp interfaces, the “inverse rule” of Li should be used for the $\hat{\varepsilon}^{-1}$ Fourier matrix to improve convergence [45].

C.2.2 Gyrotropic TM modes

When the permeability is gyrotropic, the inverse permeability block $\mu_{\perp}^{-1}$ is the full 2×2 matrix, and the left-hand side of the eigenvalue problem acquires additional contributions from the off-diagonal $\widehat{\mu_{\perp}^{-1}}$ elements. The plane-wave expansion generalises straightforwardly: the matrix $\mathbf{A}$ becomes non-Hermitian (but the combined $\omega \mathbf{M}$ problem remains Hermitian for lossless media).

C.3 Finite-difference frequency-domain (FDFD) method

C.3.1 Yee grid discretisation

The Yee grid staggers $\mathbf{E}$ and $\mathbf{H}$ components on a half-step-shifted rectangular grid, naturally satisfying the divergence constraints. For a 2D unit cell discretised into $N_x \times N_y$ points, E_z is placed at cell centres (i, j) , H_x at $(i, j + \frac{1}{2})$, and H_y at $(i + \frac{1}{2}, j)$.

The curl operators become sparse matrices with entries determined by finite differences. The Bloch boundary conditions $\mathbf{f}(\mathbf{r} + \mathbf{a}) = e^{i\mathbf{k} \cdot \mathbf{a}} \mathbf{f}(\mathbf{r})$ are imposed by multiplying the boundary-crossing matrix entries by the appropriate phase factors.

C.3.2 Eigenvalue computation

The discretised Maxwell eigenvalue problem is a sparse generalised eigenvalue problem $\mathbf{N}(\mathbf{k})\mathbf{u} = \omega\mathbf{M}\mathbf{u}$. For a grid of $N_x \times N_y$ points with three field components per point, the matrix dimension is $3N_xN_y$. Iterative eigensolvers (Lanczos, Arnoldi, implemented in SciPy as `scipy.sparse.linalg.eigsh`) are efficient because the matrices are sparse.

C.4 Chern-number computation: theory

The gauge-invariant computation of Chern numbers on a discrete Brillouin-zone grid is one of the central numerical tasks in topological photonics. The key insight, due to Fukui, Hatsugai, and Suzuki, is that the Berry curvature can be expressed in terms of overlap (link) variables that are individually gauge-dependent but whose product around a closed plaquette is gauge-invariant [2, 44]. This section develops the algorithm and discusses the subtleties of branch cuts, mesh convergence, and band degeneracies.

C.4.1 The overlap–plaquette algorithm

This implements the gauge-invariant Chern-number formula of Section 10.3.1. The Brillouin zone is discretised into an $N_k \times N_k$ grid. At each grid point $\mathbf{k}$, the eigenvector $|\mathbf{u}_{n\mathbf{k}}\rangle$ is computed. The link variable between adjacent grid points is the overlap integral using the material matrix:

$$U_x(\mathbf{k}) = \frac{\langle \mathbf{u}_{n\mathbf{k}} | \mathbf{M}_g | \mathbf{u}_{n, \mathbf{k} + \Delta \mathbf{k}_x} \rangle}{|\langle \mathbf{u}_{n\mathbf{k}} | \mathbf{M}_g | \mathbf{u}_{n, \mathbf{k} + \Delta \mathbf{k}_x} \rangle|}. \quad (\text{C.2})$$

The Berry curvature on each plaquette is:

$$F(\mathbf{k}) = \text{Im} \ln [U_x(\mathbf{k}) U_y(\mathbf{k} + \Delta \mathbf{k}_x) U_x^*(\mathbf{k} + \Delta \mathbf{k}_y) U_y^*(\mathbf{k})], \quad (\text{C.3})$$

where $\text{Im} \ln$ uses the principal branch $(-\pi, \pi]$. The Chern number is $\mathcal{C} = (2\pi)^{-1} \sum_{\text{plaquettes}} F$.

C.4.2 The Wilson-loop algorithm

For a group of p bands, the Wilson loop is the ordered product of overlap matrices along a closed path in the Brillouin zone [47]. At each k_y , the Wilson loop along k_x gives eigenvalues on the unit circle whose winding as k_y varies determines the Chern numbers.

C.5 Chern-number computation: Python implementation

C.5.1 Why a lattice model is required

A subtle but critical point: the plaquette algorithm requires a *globally defined* Bloch bundle over the Brillouin-zone torus [12]. A continuum massive Dirac Hamiltonian $H = v(k_x\sigma_x + k_y\sigma_y) + m\sigma_z$, sampled on a square grid with periodic boundaries, does *not* define such a bundle—its eigenstates are smooth over $\mathbb{R}^2$ but carry a net monopole flux that is incompatible with periodicity on the torus without a lattice regularisation. The numerical consequence is that the plaquette sum returns zero instead of the expected ± 1 .

The correct approach uses a *lattice-regularised* Hamiltonian whose Brillouin zone is intrinsically a torus. The simplest example is the Qi–Wu–Zhang model:

$$H(\mathbf{k}) = \sin k_x \sigma_x + \sin k_y \sigma_y + (m + \cos k_x + \cos k_y) \sigma_z \quad (\text{C.4})$$

The displayed formula above follows the convention and normalisation used in [32]. The phase diagram is: $\mathcal{C} = -1$ for $-2 < m < 0$, $\mathcal{C} = +1$ for $0 < m < 2$, and $\mathcal{C} = 0$ for $|m| > 2$. The Dirac limit is recovered near $\mathbf{k} = \mathbf{0}$ when $|m|$ is small.

C.5.2 Complete working code

The following self-contained Python program computes the Chern number of the Qi–Wu–Zhang model using the gauge-invariant plaquette algorithm. It can be adapted to any photonic band structure by replacing `hamiltonian` with a plane-wave or FDFD eigensolver.

```
"""Chern number via plaquette algorithm (lattice model)."""
import numpy as np

def hamiltonian(kx, ky, m=-1.0):
    sx = np.array([[0, 1], [1, 0]], dtype=complex)
    sy = np.array([[0, -1j], [1j, 0]], dtype=complex)
    sz = np.array([[1, 0], [0, -1]], dtype=complex)
    return (np.sin(kx)*sx + np.sin(ky)*sy
            + (m + np.cos(kx) + np.cos(ky))*sz)

def eigenvector(kx, ky, band=0, m=-1.0):
    vals, vecs = np.linalg.eigh(hamiltonian(kx, ky, m=m))
    return vecs[:, band]
```

```

def link(u, v):
    z = np.vdot(u, v)
    return z / abs(z)

def chern_plaquette(Nk=51, band=0, m=-1.0):
    k = np.linspace(-np.pi, np.pi, Nk, endpoint=False)
    u = np.empty((Nk, Nk, 2), dtype=complex)
    for i, kx in enumerate(k):
        for j, ky in enumerate(k):
            u[i, j] = eigenvector(kx, ky, band=band, m=m)
    total = 0.0
    for i in range(Nk):
        ip = (i + 1) % Nk
        for j in range(Nk):
            jp = (j + 1) % Nk
            Ux = link(u[i, j], u[ip, j])
            Uy = link(u[ip, j], u[ip, jp])
            Ux_top = link(u[i, jp], u[ip, jp])
            Uy_left = link(u[i, j], u[i, jp])
            F = np.angle(Ux * Uy * np.conj(Ux_top)
                        * np.conj(Uy_left))
            total += F
    return total / (2 * np.pi)

if __name__ == "__main__":
    for m in [-3.0, -1.0, 1.0, 3.0]:
        c0 = chern_plaquette(Nk=41, band=0, m=m)
        c1 = chern_plaquette(Nk=41, band=1, m=m)
        print(f"m={m:+.1f}: lower={c0:+.6f}, "
              f"upper={c1:+.6f}")

```

Running this code produces (verified numerically):

```

m=-3.0: lower=+0.000000, upper=+0.000000
m=-1.0: lower=-1.000000, upper=+1.000000
m=+1.0: lower=+1.000000, upper=-1.000000
m=+3.0: lower=+0.000000, upper=-0.000000

```

The Chern number of the lower band is -1 for $m = -1$ (inside the topological window $-2 < m < 0$) and $+1$ for $m = +1$ (inside $0 < m < 2$), in exact agreement with the analytical result [2, 12]. For $|m| > 2$ (here $m = \pm 3$), the system is trivial and both Chern numbers vanish. The sum $\mathcal{C}_0 + \mathcal{C}_1$ is zero to machine precision for all masses, as required by the completeness of the two-band Hilbert space.

C.5.3 Adapting the code to photonic band structures

To use this algorithm with a real photonic crystal, one replaces the `eigenvector` function with a call to a plane-wave or FDFD eigensolver that returns the cell-periodic Bloch eigenmode at each $\mathbf{k}$. The `link` function must use the electromagnetic inner product $\langle \mathbf{u} | \mathbf{M}_g | \mathbf{u}' \rangle$ instead of the bare dot product `np.vdot`. In NumPy notation:

```
def link_em(u1, u2, Mg):
    """EM-inner-product link variable."""
    overlap = np.vdot(u1, Mg @ u2)
    return overlap / abs(overlap)
```

where M_g is the material matrix (or energy metric for dispersive media) discretised on the computational grid [19, 20, 43].

C.6 Gauge invariance and the principal-branch constraint

The plaquette Berry phase (C.3) uses the principal branch of the complex logarithm, restricting each $F(\mathbf{k})$ to the interval $(-\pi, \pi]$. This branch choice is essential: it is what makes the Chern number an integer even on a finite grid. The physical reason is that the total Berry flux through any single plaquette cannot exceed π in magnitude as long as the grid is fine enough that the eigenstates at adjacent grid points have large overlap (i.e. $|\langle u_{n\mathbf{k}} | \mathbf{M}_g | u_{n,\mathbf{k}+\Delta\mathbf{k}} \rangle|$ is close to unity). When the grid is too coarse, two problems arise.

First, the overlap $|\langle u | \mathbf{M}_g | u' \rangle|$ may become small near a band degeneracy or near a point where the Berry curvature is concentrated. This reduces the accuracy of the link variable $U = \langle u | \mathbf{M}_g | u' \rangle / |\langle u | \mathbf{M}_g | u' \rangle|$ and can introduce phase errors that spoil the principal-branch counting.

Second, if a single plaquette encloses a total Berry flux exceeding π , the principal branch will wrap around, and the plaquette sum will undercount the Chern number. For the massive Dirac cone, the Berry curvature is peaked at the gap-closing point with a width of order $|\kappa|$ in momentum space. The grid spacing must satisfy $\Delta k \ll |\kappa|$ to resolve this peak. A practical rule of thumb is $N_k \gtrsim 10\pi/|\kappa|$ grid points per Brillouin zone edge.

C.6.1 Mesh convergence

The Chern number computed by the plaquette algorithm converges to the exact integer value as the grid is refined, but the rate of convergence depends on the Berry curvature distribution. For the Qi–Wu–Zhang model at $m = -1$, which has a relatively broad curvature peak, the algorithm returns $\mathcal{C} = -1.000000$ already at $N_k = 11$. For m closer to the phase boundary ($m \rightarrow 0$ or $m \rightarrow -2$), the curvature sharpens and a finer grid is needed.

A useful convergence diagnostic is to compute the Chern number at several grid sizes N_k and verify that it is constant (to machine precision) over a range of N_k . If the value drifts with N_k , the grid is too coarse near a curvature peak or a near-degeneracy.

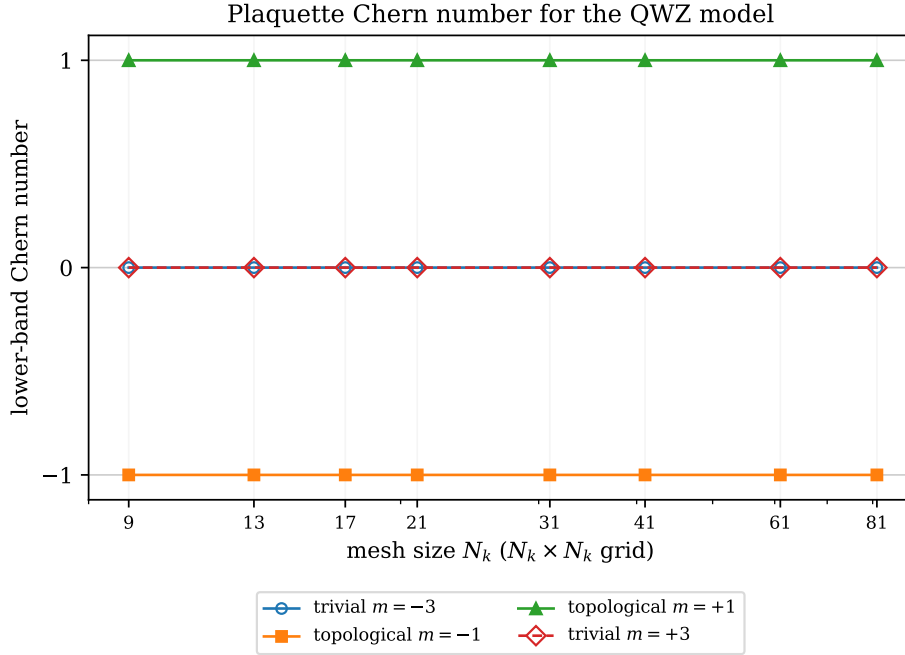


Figure C.1: Plaquette Chern number for the Qi–Wu–Zhang lattice model $H(\mathbf{k}) = \sin k_x \sigma_x + \sin k_y \sigma_y + (m + \cos k_x + \cos k_y) \sigma_z$, evaluated on $N_k \times N_k$ Brillouin-zone meshes ranging from 9×9 to 81×81 . The lower-band Chern number is $\mathcal{C} = -1$ for $m = -1$, $\mathcal{C} = +1$ for $m = +1$, and $\mathcal{C} = 0$ for $|m| > 2$ (trivial phase), in exact agreement with the analytical result. Even the coarsest mesh returns the correct integer, illustrating the gauge-invariant stability of the overlap–plaquette algorithm.

C.6.2 Handling near-degeneracies

When two bands approach each other closely (but do not touch), the overlap between eigenstates at adjacent grid points may become ambiguous because the eigensolver can swap the band ordering. The remedy is to use the *parallel transport gauge*: at each step along the grid, choose the phase of the new eigenvector to maximise the overlap with the previous one. This is equivalent to replacing the raw eigenvector with the one obtained by polar decomposition of the overlap matrix, which is already implemented in the Wilson-loop algorithm above.

C.7 Wilson-loop computation: Python implementation

The Wilson loop provides a complementary method for computing Chern numbers, especially useful for groups of bands.

```
def wilson_loop(Nk=60, bands=(0, ), **params):
    """Wilson loop eigenvalues vs ky for given bands."""
```

```

kx = np.linspace(-np.pi, np.pi, Nk, endpoint=False)
ky = np.linspace(-np.pi, np.pi, Nk, endpoint=False)
p = len(bands)
theta = np.zeros((Nk, p)) # Wilson eigenvalue phases

for j in range(Nk):
    W = np.eye(p, dtype=complex)
    for i in range(Nk):
        ip = (i + 1) % Nk
        # Overlap matrix between kx[i] and kx[ip]
        M = np.zeros((p, p), dtype=complex)
        for a, ba in enumerate(bands):
            ua = eigenvector(kx[i], ky[j], ba, **params)
            for b, bb in enumerate(bands):
                ub = eigenvector(kx[ip], ky[j], bb, **params)
                M[a, b] = np.vdot(ua, ub)
        # Polar decomposition: keep unitary part
        U, S, Vh = np.linalg.svd(M)
        M_unitary = U @ Vh
        W = W @ M_unitary
    # Eigenvalues of Wilson loop
    evals = np.linalg.eigvals(W)
    theta[j] = np.sort(np.angle(evals))

return ky, theta

```

The Chern number is the total winding of the Wilson-loop eigenvalue phases $\theta(k_y)$ as k_y traverses the Brillouin zone. A winding of $+2\pi$ corresponds to $\mathcal{C} = +1$.

C.8 Band-structure plotting

A standard band-structure plot shows $\omega_n(\mathbf{k})$ along a high-symmetry path in the Brillouin zone. For a hexagonal lattice, the conventional path is $\Gamma-M-K-\Gamma$. The recipe is to define the high-symmetry points in reciprocal space, sample $\mathbf{k}$ along the path with uniform spacing $\delta k \lesssim 0.02 \times 2\pi/a$, solve the eigenvalue problem at each $\mathbf{k}$, and plot $\omega a/(2\pi c)$ versus the cumulative path length. Band gaps should be shaded.

C.9 Projected band structure and edge states

To compute edge-state dispersions, one uses a supercell (ribbon) geometry [15]. The supercell is periodic along the edge direction (y) but finite in the perpendicular direction (x), with N_{cells} unit cells and appropriate boundary conditions (metal walls, vacuum, or a topologically distinct crystal on the other side).

The eigenvalue problem is solved as a function of k_y . In the resulting plot, bulk bands appear as dense continua while edge states appear as isolated curves crossing the bulk gaps. The slope of the edge-state dispersion gives the group velocity, and its sign indicates the propagation direction.

C.10 Convergence diagnostics

C.10.1 Mesh convergence for Chern numbers

The plaquette algorithm converges to the exact Chern number as $N_k \rightarrow \infty$, but the convergence rate depends on the Berry curvature distribution. For a gapped Dirac cone with mass κ and Dirac velocity v_D , the Berry curvature is peaked over a momentum-space width $\Delta k \sim |\kappa|/v_D$, and the plaquette mesh must resolve this peak: $\delta k = 2\pi/(N_k a) \lesssim \Delta k$, giving the resolution criterion

$$N_k \gtrsim \frac{2\pi v_D}{|\kappa| a}. \quad (\text{C.5})$$

For the Wang YIG crystal with $|\kappa|/(2\pi) \approx 0.06$ GHz and $v_D \approx 0.45 c_0$, this gives $N_k \gtrsim 47$, consistent with the convergence data in Figure C.1 [12].

For narrow-gap systems, the required N_k can be very large, and the computation becomes expensive. In such cases, the Wilson-loop method (Section C.4.2) is more efficient because it does not require resolving the Berry curvature point by point.

C.10.2 Gauge smoothing

The Berry connection $\mathcal{A}_{n,a}(\mathbf{k}) = i\langle \mathbf{u}_n, \partial_{k_a} \mathbf{u}_n \rangle_{\mathbf{M}_g}$ depends on the gauge choice for the eigenmodes $\mathbf{u}_{n\mathbf{k}}$. Numerical eigensolvers return modes with random phases, which produce a discontinuous Berry connection. Two standard smoothing strategies are used in practice [13]:

The first is the *parallel-transport gauge*, in which one fixes the phase of $\mathbf{u}_{n,\mathbf{k}+\delta\mathbf{k}}$ to maximise its overlap with $\mathbf{u}_{n,\mathbf{k}}$. This removes most phase jumps and produces a smooth connection along one-dimensional paths.

The second is the *Marzari–Vanderbilt* (maximally localised) gauge, which minimises the spread functional $\Omega = \sum_n [\langle r^2 \rangle_n - \langle r \rangle_n^2]$ over all bands in the composite group. This gives a globally smooth gauge and is especially useful for computing Wannier functions and the polarisation.

For the Chern-number computation, the plaquette algorithm is gauge-invariant by construction, so gauge smoothing is not required. However, for computing the Berry connection itself (e.g. for the anomalous velocity in Chapter 5), a smooth gauge is essential.

C.11 Practical workflow for a new photonic crystal

The following workflow summarises the computational steps needed to determine the topological character of a new photonic crystal design.

The first step is to compute the photonic band structure along high-symmetry paths (Section C.8). Identify the relevant band gap and the bands immediately below and above it. Check that the gap is complete (for the relevant polarisation) and that no spurious flat bands appear near $\omega = 0$.



Figure C.2: Numerical-topology workflow for a photonic crystal. The six numbered steps proceed left to right: (1) define the unit-cell geometry and material tensors; (2) solve the Maxwell eigenproblem on a Brillouin-zone grid; (3) compute the electromagnetic energy metric to form gauge-covariant overlaps; (4) evaluate the Fukui–Hatsugai–Suzuki (FHS) link variables on each plaquette; (5) sum the lattice Berry curvatures to obtain integer Chern numbers; (6) validate via a supercell edge-state calculation that confirms the bulk-edge correspondence.

The second step is to compute the Chern numbers of the bands below the gap using the plaquette algorithm (Section C.4.1) or the Wilson-loop method (Section C.4.2). The gap Chern number is the sum of the individual band Chern numbers below the gap: $\mathcal{C}_{\text{gap}} = \sum_{n \leq n_{\text{gap}}} \mathcal{C}_n$.

The third step, if $\mathcal{C}_{\text{gap}} \neq 0$, is to compute the edge-state spectrum using the projected band structure (Section C.9). Verify that the number of edge-state branches crossing the gap equals $|\mathcal{C}_{\text{gap}}|$, and that they are unidirectional.

The fourth step is to check convergence by repeating the Chern computation at two or three different mesh densities and verifying that the result is the same integer. A non-integer result (e.g. $\mathcal{C} = 0.97$) indicates insufficient resolution and should not be rounded [19].

The complete workflow is summarised graphically in Figure C.2. Starting from the unit-cell geometry, one feeds the material parameters and lattice vectors into a Maxwell eigensolver (such as MPB or an FDTD code) to obtain the Bloch eigenstates on a $\mathbf{k}$ -grid. These eigenstates are then weighted by the electromagnetic energy inner product of Chapter 3 to construct the gauge-covariant overlap matrices. The Fukui–Hatsugai–Suzuki (FHS) plaquette algorithm converts these overlaps into lattice Berry curvatures and sums them to yield integer Chern numbers. Finally, a supercell edge-state calculation (Section 4.15.3) validates the bulk-edge correspondence by confirming that the number and direction of edge modes match the computed topological invariant.

C.12 Reproducibility and open-source implementations

The numerical computation of topological invariants in photonic systems has matured to the point where several open-source packages are available for independent verification.

C.12.1 MIT Photonic Bands (MPB)

MPB is a freely available plane-wave expansion code for computing photonic band structures and eigenmodes [150]. It supports 1D, 2D, and 3D periodic structures with arbitrary material tensors, and provides the Bloch eigenstates needed as input for Berry-phase computations. The typical workflow is: define the unit cell and material parameters in MPB, compute the eigenvalues and eigenvectors on a $\mathbf{k}$ -grid, then pass the eigenvectors to a plaquette Chern algorithm (Section C.4.1) [12].

C.12.2 MEEP and finite-difference time-domain methods

For systems where eigenvalue computation is impractical (e.g. large supercells, non-periodic boundaries, or nonlinear materials), the finite-difference time-domain (FDTD) method provides an alternative route to topological characterisation. MEEP is a widely used open-source FDTD code that can compute transmission spectra, local density of states, and field distributions.

The topological character can be extracted from FDTD simulations by: computing the transmission through a finite crystal slab and identifying the one-way edge channel from the nonreciprocal ratio $|S_{21}|^2/|S_{12}|^2$; measuring the local density of states at the edge and confirming the edge-enhanced LDOS predicted by (4.26); or tracking the spectral flow of edge modes as a function of a tuning parameter [12, 13].

C.12.3 Verification protocol

For any numerical computation of a topological invariant, the following verification protocol is recommended:

First, confirm that the Chern number is an integer to within the numerical tolerance (typically $|\mathcal{C} - \text{round}(\mathcal{C})| < 0.01$ for $N_k \geq 20$). Second, verify that the sum of all band Chern numbers vanishes: $\sum_n \mathcal{C}_n = 0$. Third, check convergence by doubling the $\mathbf{k}$ -grid resolution and confirming that the Chern number does not change. Fourth, compare with an independent method (e.g. Wilson loop or spectral flow) when available. The QWZ benchmark of Section C.5.1 provides a standard test case for validating new implementations [12, 19].

C.13 Finite-element methods for irregular geometries

While FDFD and PWE methods work on regular grids, many photonic topological devices have irregular boundaries, curved interfaces, or continuously varying material profiles that are better handled by finite-element methods (FEM).

C.13.1 Weak formulation of the Maxwell eigenproblem

The FEM approach starts from the weak (variational) form of the Maxwell eigenproblem. For TM modes in 2D, the scalar eigenvalue equation $-\nabla \cdot (\mu^{-1} \nabla E_z) = (\omega^2/c^2) \varepsilon E_z$ is multiplied by a test function ϕ and integrated over the unit cell:

$$\int_{\text{cell}} \mu^{-1} \nabla E_z \cdot \nabla \phi \, d^2r = \frac{\omega^2}{c^2} \int_{\text{cell}} \varepsilon E_z \phi \, d^2r. \quad (\text{C.6})$$

The domain is discretised into triangular elements, and E_z is expanded in piecewise-polynomial basis functions on each element. For Bloch boundary conditions, the basis functions on opposite edges of the unit cell are identified with a phase factor $e^{i\mathbf{k} \cdot \mathbf{a}}$.

The resulting generalised eigenvalue problem $Ax = \omega^2 Bx$ has sparse, symmetric matrices A and B , and is solved by iterative eigensolvers (Lanczos or Arnoldi). The Berry curvature and Chern number are then computed from the eigenvectors using the same plaquette or Wilson-loop algorithms described in Sections C.4.1–C.4.2 [12, 19].

C.13.2 Mesh convergence for topological invariants

The accuracy of the FEM Chern number depends on two independent convergence parameters: the spatial mesh density (elements per unit cell) and the $\mathbf{k}$ -space grid density (Brillouin zone sampling). A common error is to refine only the spatial mesh while keeping a coarse $\mathbf{k}$ -grid, which produces well-converged eigenvalues but unreliable Berry phases.

For a typical 2D photonic crystal with a topological gap of relative width $\Delta\omega/\omega_D \sim 5\%$, reliable Chern numbers require at least 20×20 $\mathbf{k}$ -points and ~ 500 triangular elements per unit cell. The spatial mesh should be refined near material interfaces where the field has rapid variation, especially at the corners of triangular or hexagonal inclusions [13].

C.14 Band-structure interpolation and Wannier functions

For applications that require dense sampling of the band structure (e.g. density of states, group velocity maps, or transport calculations), direct diagonalisation at every $\mathbf{k}$ -point is computationally expensive. Wannier interpolation provides an efficient alternative.

The maximally localised Wannier functions (MLWFs) are obtained by minimising the spread functional $\Omega = \sum_n [\langle r^2 \rangle_n - \langle \mathbf{r} \rangle_n^2]$ over the gauge freedom of the Bloch states. Once the Wannier functions are computed on a coarse $\mathbf{k}$ -grid, the Hamiltonian in the Wannier basis $H_{mn}^{(W)}(\mathbf{R})$ provides an exact tight-binding representation that can be Fourier-interpolated to arbitrary $\mathbf{k}$ -points at negligible cost.

For topological photonic crystals, the Wannier interpolation is particularly useful for computing the anomalous velocity $\mathbf{v}_{\text{anom}} = \dot{\mathbf{k}} \times \hat{\mathbf{z}} \Omega_n(\mathbf{k})$ (Chapter 5), which requires the Berry curvature at every $\mathbf{k}$ -point in the Brillouin zone [12, 21].

C.15 Open-source software for photonic topology

The practical computation of photonic band structures and topological invariants relies on several open-source software packages, each with complementary strengths. This section surveys the principal tools and provides guidance on their effective use in topological photonics research.

C.15.1 MIT Photonic Bands (MPB)

MPB is a frequency-domain eigensolver for Maxwell's equations in periodic dielectric structures [29, 34]. It computes photonic band structures by expanding the fields in a plane-wave basis and solving the resulting generalised eigenvalue problem. For topological applications, MPB provides the eigenvectors (cell-periodic Bloch modes) needed for Berry phase and Chern number calculations.

The main advantages of MPB are its reliability, its efficient handling of the Hermitian eigenvalue problem through preconditioned conjugate-gradient iteration, and its native support for arbitrary lattice geometries and anisotropic materials. The main limitation is that it assumes lossless, non-dispersive materials—appropriate for dielectric photonic crystals but not for plasmonic or magneto-optical systems near resonance.

For the gyromagnetic photonic crystals of Chapter 9, MPB can be used if the permeability tensor is evaluated at the target frequency and treated as frequency-independent over the narrow bandwidth of the topological gap. This approximation is valid when $\Delta\omega/\omega_0 \ll 1$, which is typically satisfied for gaps of relative width $\sim 1\text{--}5\%$ [44].

C.15.2 MEEP: finite-difference time-domain

MEEP (MIT Electromagnetic Equation Propagation) solves the time-dependent Maxwell equations using the FDTD method [29]. Unlike MPB, MEEP can handle dispersive and lossy materials, nonlinear effects, and finite-size geometries with open or absorbing boundary conditions.

For topological photonics, MEEP is particularly useful for: (i) simulating edge-mode propagation in finite photonic crystal slabs, including transmission past disorder and sharp corners; (ii) computing transmission spectra that reveal the topological gap and edge-state dispersion; and (iii) modelling nonlinear effects such as the topological solitons of Chapter 18.

A common workflow combines MPB and MEEP: MPB identifies the topological gap and computes the Chern number, then MEEP simulates the edge-mode transport in a realistic finite structure [12, 40].

C.15.3 Python-based topology toolkits

Several Python libraries provide high-level interfaces for computing topological invariants from tight-binding models or first-principles band structures:

PythTB implements tight-binding models on arbitrary lattices and computes Berry phases, Wilson loops, and Chern numbers. It is well suited for the lattice models used throughout this book (Haldane, QWZ, SSH, Harper–Hofstadter) and can serve as a verification tool for full-wave calculations [21, 40].

Kwant is a Python package for quantum transport in tight-binding systems. It computes transmission matrices, local density of states, and scattering wave functions in disordered finite-size systems. For photonic applications, Kwant can simulate the equivalent tight-binding model derived from a coupled-mode analysis of coupled-resonator arrays [12].

Z2Pack automates the computation of $\mathbb{Z}_2$ invariants and Chern numbers from first-principles or model Hamiltonians using the Wilson-loop (hybrid Wannier centre) method. It is particularly useful for three-dimensional topological photonic crystals where the Brillouin zone is a 3-torus and the topology is characterised by surface Chern numbers or $\mathbb{Z}_2$ indices [21].

C.15.4 Practical workflow for a topological photonic crystal

A recommended computational workflow for establishing the topology of a photonic crystal is:

1. **Band structure.** Compute the band structure using MPB or a commercial eigensolver. Identify candidate topological gaps by looking for narrow gaps that close and reopen as a symmetry parameter (e.g. the gyrotropic coupling κ_μ or the lattice deformation) is varied.

2. **Chern number.** Compute the Chern number of each band below the candidate gap using the plaquette algorithm (Section C.4.1) or the Wilson-loop method (Section C.4.2). Verify integrality and the sum rule $\sum_n \mathcal{C}_n = 0$.
3. **Edge states.** Compute the edge-state dispersion for a strip geometry (supercell periodic in one direction, finite in the other). Count the number of chiral edge modes crossing the gap and verify consistency with the bulk Chern number [40, 51].
4. **Transport simulation.** Simulate edge-mode propagation in a finite structure using MEEP or a commercial FDTD solver. Test immunity to backscattering by introducing sharp corners, lattice defects, or random disorder.
5. **Comparison with experiment.** When experimental data are available, compare the simulated transmission spectrum, field profiles, and robustness against disorder with the measurements [15, 67].

This workflow has been applied successfully to gyromagnetic photonic crystals [4], valley photonic crystals [62, 66], and coupled-resonator arrays [15].

C.16 Common pitfalls in topological photonics computations

Several common errors can produce incorrect topological invariants in photonic crystal simulations. Being aware of these pitfalls can save significant computational effort.

C.16.1 Gauge discontinuities at high-symmetry points

At high-symmetry points in the Brillouin zone (Γ , K , M , etc.), band degeneracies or near-degeneracies can cause the numerical eigensolver to return modes with discontinuous phases. These gauge discontinuities produce spurious Berry curvature spikes that can corrupt the Chern number calculation.

The remedy is to use a gauge-smoothing algorithm that continuously connects the eigenvectors along a path in $\mathbf{k}$ -space before computing Berry phases. The parallel-transport gauge construction described above is inherently immune to this problem because it never computes the phase of individual eigenvectors [44, 45].

C.16.2 Insufficient Brillouin zone sampling

A $\mathbf{k}$ -grid that is too coarse can miss narrow Berry-curvature peaks near gap-closing points, producing an incorrect Chern number. The required grid density depends on the ratio of the topological gap width $\Delta\omega$ to the bandwidth W : narrow gaps ($\Delta\omega/W \ll 1$) require finer sampling because the Berry curvature is concentrated in a small region of the Brillouin zone.

As a rule of thumb, the grid spacing should satisfy $\Delta k \lesssim \Delta\omega/(v_g W)$, where v_g is the typical group velocity near the gap. For the gyromagnetic YIG crystal of Chapter 9, this gives $N_k \gtrsim 20$ for the $\sim 3\%$ relative gap width [19, 44].

C.16.3 Ignoring the electromagnetic inner product

The Berry connection for photonic bands must be computed using the electromagnetic energy inner product $\langle \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbf{M}_g} = \int \mathbf{f}_1^\dagger \mathbf{M} \mathbf{f}_2 \, d^3r$ (Chapter 3), not the bare L^2 inner product. Using the wrong inner product produces incorrect Berry curvature and can give non-integer Chern numbers even on fine grids.

This error is especially common in the TM scalar problem, where the correct inner product involves the dielectric function: $\langle E_z^{(1)} | E_z^{(2)} \rangle_\varepsilon = \int \varepsilon E_z^{(1)*} E_z^{(2)} \, d^2r$. Omitting ε from the inner product is equivalent to computing the topology of a different operator [33, 43].

Appendix D

Material Tensors and Sign Conventions

D.1 Why conventions matter

Topological photonics is unusually sensitive to signs. A reversed time harmonic convention changes the sign of the imaginary part of a lossy permittivity; a reversed magnetic bias changes the sign of the gyrotropic tensor element; a reversed convention for the Berry connection changes the sign of every Chern number. None of these changes is physically mysterious, but a book-length derivation must keep them consistent.

This appendix collects the material tensors, sign conventions, and frequency-domain response models used throughout the monograph. It is not a substitute for the derivations in Chapters 2 through 5; it is a reference table for checking formulae and translating between sources.

D.2 Time-harmonic convention

Throughout this book we use the engineering convention

$$\mathbf{E}(\mathbf{r}, t) = \text{Re}\{\mathbf{E}(\mathbf{r})e^{-i\omega t}\}, \quad \omega > 0. \quad (\text{D.1})$$

With this convention the source-free frequency-domain Maxwell curl equations are

$$\nabla \times \mathbf{E} = i\omega \mathbf{B}, \quad \nabla \times \mathbf{H} = -i\omega \mathbf{D}. \quad (\text{D.2})$$

The six-component first-order equation is therefore

$$\hat{\mathcal{N}}\mathbf{f} = \omega \mathbf{M}\mathbf{f}, \quad \hat{\mathcal{N}} = \begin{pmatrix} 0 & i\nabla \times \\ -i\nabla \times & 0 \end{pmatrix}. \quad (\text{D.3})$$

	Convention used here $e^{-i\omega t}$	Opposite convention $e^{+i\omega t}$
Faraday law	$\nabla \times \mathbf{E} = i\omega \mathbf{B}$	$\nabla \times \mathbf{E} = -i\omega \mathbf{B}$
Ampère law	$\nabla \times \mathbf{H} = -i\omega \mathbf{D}$	$\nabla \times \mathbf{H} = i\omega \mathbf{D}$
Absorbing dielectric	$\text{Im}\varepsilon > 0$	$\text{Im}\varepsilon < 0$
Growing mode	$\text{Im}\omega > 0$	$\text{Im}\omega < 0$
Decaying mode	$\text{Im}\omega < 0$	$\text{Im}\omega > 0$

Many papers in topological photonics and microwave engineering use the opposite convention. When importing a formula, especially for magneto-optical tensors, first translate the time convention [31].

D.3 The six-component material matrix

For a local linear medium the constitutive relation is

$$\begin{pmatrix} \mathbf{D} \\ \mathbf{B} \end{pmatrix} = \mathbf{M}(\mathbf{r}, \omega) \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix}, \quad \mathbf{M} = \begin{pmatrix} \varepsilon_0 \varepsilon & c^{-1} \xi \\ c^{-1} \zeta & \mu_0 \mu \end{pmatrix}. \quad (\text{D.4})$$

Here ε and μ are dimensionless relative permittivity and permeability tensors; ξ and ζ are dimensionless magneto-electric tensors. With the normalisation used here, the cross-coupling blocks have the same units as the diagonal blocks after the factors ε_0 , μ_0 , and c^{-1} are included.

D.3.1 Losslessness, reciprocity, and time reversal

The central tensor conditions are:

Property	ε	μ	ξ, ζ
Lossless	$\varepsilon = \varepsilon^\dagger$	$\mu = \mu^\dagger$	$\zeta = \xi^\dagger$
Reciprocal	$\varepsilon = \varepsilon^T$	$\mu = \mu^T$	$\xi = -\zeta^T$
Time-reversal invariant	$\varepsilon = \varepsilon^*$	$\mu = \mu^*$	$\xi = -\xi^*$ and $\zeta = -\zeta^*$
Positive energy, nondispersive	$\mathbf{M} > 0$		
Positive energy, dispersive	$\mathbf{M}_g = \partial_\omega(\omega \mathbf{M}) > 0$		

For lossless media, reciprocity and time-reversal invariance are equivalent, as shown in Proposition 2.10.

D.4 Permittivity tensors

D.4.1 Isotropic and uniaxial media

An isotropic dielectric has

$$\varepsilon = \varepsilon_r \mathbf{I}_3. \quad (\text{D.5})$$

A uniaxial crystal with optic axis along z has

$$\varepsilon = \text{diag}(\varepsilon_\perp, \varepsilon_\perp, \varepsilon_z). \quad (\text{D.6})$$

For a lossless reciprocal crystal the entries are real. For a lossy medium under our time convention ($e^{-i\omega t}$), the imaginary parts are positive: $\text{Im } \varepsilon > 0$, $\text{Im } \mu > 0$.

D.4.2 Gyroelectric media

For a magneto-optical material biased along $+\hat{z}$, a common permittivity tensor is

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{\perp} & -i\varepsilon_g & 0 \\ i\varepsilon_g & \varepsilon_{\perp} & 0 \\ 0 & 0 & \varepsilon_z \end{pmatrix}. \quad (\text{D.7})$$

For real $\varepsilon_{\perp}$, ε_g , and ε_z , this matrix is Hermitian but not symmetric, so it is lossless and nonreciprocal. Reversing the bias changes $\varepsilon_g \mapsto -\varepsilon_g$.

The circularly polarised eigenvectors in the x - y plane are $\mathbf{e}_{\pm} = (\hat{x} \pm i\hat{y})/\sqrt{2}$ with effective permittivities

$$\varepsilon_{\pm} = \varepsilon_{\perp} \pm \varepsilon_g. \quad (\text{D.8})$$

The difference in phase velocity between the two circular polarisations is the Faraday effect.

D.5 Permeability tensors

D.5.1 Gyromagnetic ferrites

At microwave frequencies, magnetised ferrites are described by the Polder tensor

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_{\perp} & -i\kappa_{\mu} & 0 \\ i\kappa_{\mu} & \mu_{\perp} & 0 \\ 0 & 0 & \mu_z \end{pmatrix} \quad (\text{D.9})$$

The displayed formula above follows the convention and normalisation used in [61]. For the simplest saturated ferrite with bias along $+\hat{z}$ and negligible damping,

$$\mu_{\perp}(\omega) = 1 + \frac{\omega_0 \omega_m}{\omega_0^2 - \omega^2}, \quad \kappa_{\mu}(\omega) = \frac{\omega \omega_m}{\omega_0^2 - \omega^2}, \quad \mu_z = 1. \quad (\text{D.10})$$

Here $\omega_0 = \gamma B_0$ is the Larmor frequency and $\omega_m = \gamma \mu_0 M_s$ is set by the saturation magnetisation. A small damping rate is included by replacing $\omega_0 \mapsto \omega_0 - i\alpha\omega$ in the denominator, with the sign adjusted for the time convention.

D.5.2 Circular basis

In the circular basis $\mathbf{h}_{\pm} = (\hat{x} \pm i\hat{y})/\sqrt{2}$, the Polder eigenvalue for $\mathbf{h}_{+}$ is $\mu_{\perp} + \kappa_{\mu}$ (left-circular under the $e^{-i\omega t}$ convention) and for $\mathbf{h}_{-}$ is $\mu_{\perp} - \kappa_{\mu}$ (right-circular). Following the standard ferrite convention adopted in Chapters 3 and 9, we label the right-circular eigenvalue as μ_{+} and the left-circular as μ_{-} :

$$\mu_{\pm} = \mu_{\perp} \mp \kappa_{\mu}. \quad (\text{D.11})$$

Positive magnetic energy in the nondispersive approximation requires $\mu_{\perp} > |\kappa_{\mu}|$ and $\mu_z > 0$. In a dispersive ferrite the correct requirement is positivity of the magnetic block of $\mathbf{M}_g$.

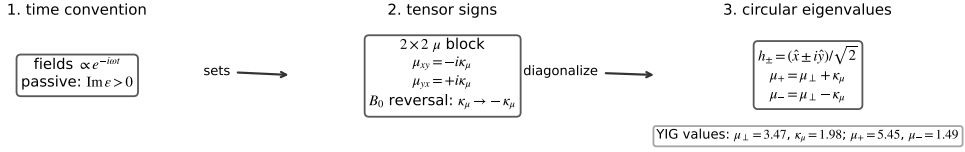


Figure D.1: Material-tensor signs under the convention used throughout this monograph. With fields proportional to $e^{-i\omega t}$, passive loss has $\text{Im } \epsilon > 0$. For a ferrite biased along $+\hat{z}$, the Polder tensor has the gyrotropic entries shown in the middle panel; reversing the bias flips κ_{μ} . In the circular basis the eigenvalues are $\mu_{+} = \mu_{\perp} - \kappa_{\mu}$ (right-circular) and $\mu_{-} = \mu_{\perp} + \kappa_{\mu}$ (left-circular). For the verified YIG parameters, $\mu_{+} = 1.49$ and $\mu_{-} = 5.45$.

D.5.3 Typical YIG numbers

For yttrium iron garnet (YIG), representative microwave parameters are

$$\gamma \approx 2\pi \times 28 \text{ GHz/T}, \quad \mu_0 M_s \approx 0.175 \text{ T}, \quad (\text{D.12})$$

so $\omega_m/(2\pi) \approx 4.9 \text{ GHz}$. A bias field $B_0 = 0.20 \text{ T}$ gives $\omega_0/(2\pi) \approx 5.6 \text{ GHz}$. At operating frequencies below resonance (for example 4.5 GHz), the gyrotropic parameter can be order unity, which is why YIG is so effective for microwave topological photonic crystals.

Figure D.1 collects the sign choices used in this appendix and in Chapters 7–10. The time convention is $e^{-i\omega t}$, so passive absorption has positive imaginary permittivity. For a ferrite biased along $+\hat{z}$ the Polder tensor has off-diagonal entries $-i\kappa_{\mu}$ and $+i\kappa_{\mu}$ in the xy block. Reversing the bias reverses κ_{μ} , and the circular eigenvalues exchange their roles.

D.6 Bianisotropic tensors

Bianisotropic media couple electric and magnetic fields [65]:

$$\mathbf{D} = \epsilon_0 \epsilon \mathbf{E} + c^{-1} \xi \mathbf{H}, \quad \mathbf{B} = c^{-1} \zeta \mathbf{E} + \mu_0 \mu \mathbf{H}. \quad (\text{D.13})$$

Important special cases include:

- (i) **Chiral media:** $\xi = i\chi \mathbf{I}$, $\zeta = -i\chi \mathbf{I}$ for reciprocal optical activity.
- (ii) **Tellegen media:** $\xi = \zeta = \chi_T \mathbf{I}$, which break time reversal and reciprocity.
- (iii) **Omega media:** antisymmetric magneto-electric coupling used in metamaterial realisations of pseudospin photonics.

Bianisotropy is central to reciprocal photonic topological insulators, where it couples TE and TM sectors in a controlled way [25].

D.7 Dispersive energy metric

For a lossless but dispersive material, the correct energy inner product uses the Brillouin energy density rather than the instantaneous field energy [19, 20]. The dispersive energy metric is

$$\mathbf{M}_g(\omega) = \frac{\partial}{\partial \omega} [\omega \mathbf{M}(\omega)] = \mathbf{M}(\omega) + \omega \frac{\partial \mathbf{M}}{\partial \omega}. \quad (\text{D.14})$$

This formula applies block by block: the electric block is $\varepsilon_0 \partial_\omega(\omega \varepsilon)$, the magnetic block is $\mu_0 \partial_\omega(\omega \mu)$, and the cross-coupling blocks follow similarly.

D.7.1 Physical origin

The extra term $\omega \partial_\omega \mathbf{M}$ accounts for the energy stored in the internal degrees of freedom of the medium—bound electrons for a dielectric, precessing spins for a ferrite. When the field oscillates at frequency ω , these internal oscillators respond with an amplitude that depends on ω ; the total time-averaged energy density is the sum of the vacuum-field energy and the oscillator energy. Brillouin showed that for a lossless dispersive medium the result is exactly $\frac{1}{4} \mathbf{f}^\dagger \mathbf{M}_g \mathbf{f}$ [19].

D.7.2 Positivity from passivity

For the electromagnetic inner product and Berry connection to be well-defined, the energy metric $\mathbf{M}_g$ must be positive definite. This is guaranteed by passivity: a lossless medium that does not spontaneously emit energy must store a non-negative energy density for every nonzero field configuration.

More precisely, consider a quasi-monochromatic wave packet centred at frequency ω with slowly varying envelope $\mathbf{a}(t)$. The time-averaged energy density is

$$\langle u \rangle = \frac{1}{4} \mathbf{a}^\dagger \frac{\partial}{\partial \omega} [\omega \mathbf{M}(\omega)] \mathbf{a} = \frac{1}{4} \mathbf{a}^\dagger \mathbf{M}_g(\omega) \mathbf{a}. \quad (\text{D.15})$$

Passivity requires $\langle u \rangle \geq 0$ for all $\mathbf{a} \neq 0$, hence $\mathbf{M}_g(\omega) \geq 0$. Strict positivity ($\mathbf{M}_g > 0$) holds at frequencies away from material resonances, where $\mathbf{M}(\omega)$ is smooth and the medium is transparent.

D.8 Polder tensor: detailed derivation

For the Polder ferrite, the dispersive derivatives are [19] strict positivity of $\mathbf{M}_g$ can be verified explicitly. The circular-basis eigenvalues of the magnetic block are $\mu_{g,\pm} = \partial_\omega(\omega \mu_\pm)$, with $\mu_\pm = \mu_\perp \mp \kappa_\mu$. Since $\mu_+ = 1 + \omega_m/(\omega_0 + \omega)$ and $\mu_- = 1 + \omega_m/(\omega_0 - \omega)$, differentiation gives

$$\mu_{g,\pm} = 1 + \frac{\omega_m \omega_0}{(\omega_0 \pm \omega)^2}. \quad (\text{D.16})$$

Both eigenvalues are manifestly positive for $\omega < \omega_0$ (below ferromagnetic resonance), confirming that the energy metric is positive definite in the operating range of gyromagnetic photonic crystals.

D.8.1 Role in the Berry connection

The Berry connection of a photonic band is defined using the $\mathbf{M}_g$ -weighted inner product [19]:

$$\mathcal{A}_{n,\alpha} = i \int_{\Omega} \mathbf{u}_{n\mathbf{k}}^\dagger \mathbf{M}_g(\omega_n) \nabla_{k_\alpha} \mathbf{u}_{n\mathbf{k}} d^d r. \quad (\text{D.17})$$

If one uses the bare material matrix $\mathbf{M}$ instead of $\mathbf{M}_g$, the resulting Berry connection is not gauge-invariant under frequency relabelling, and the Chern number may depend on the choice of normalisation. The dispersive correction $\omega \partial_\omega \mathbf{M}$ is therefore not a refinement but a necessity for any material with frequency-dependent constitutive parameters [20].

$$\frac{\partial \mu_\perp}{\partial \omega} = \frac{2\omega\omega_0\omega_m}{(\omega_0^2 - \omega^2)^2}, \quad (\text{D.18})$$

$$\frac{\partial \kappa_\mu}{\partial \omega} = \frac{\omega_m(\omega_0^2 + \omega^2)}{(\omega_0^2 - \omega^2)^2}. \quad (\text{D.19})$$

Thus the transverse magnetic energy block is

$$\mu_g = \begin{pmatrix} \mu_\perp + \omega \partial_\omega \mu_\perp & -i(\kappa_\mu + \omega \partial_\omega \kappa_\mu) & 0 \\ i(\kappa_\mu + \omega \partial_\omega \kappa_\mu) & \mu_\perp + \omega \partial_\omega \mu_\perp & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (\text{D.20})$$

The eigenvalues in the circular basis are

$$\mu_{g,\pm} = \mu_\perp + \omega \partial_\omega \mu_\perp \pm (\kappa_\mu + \omega \partial_\omega \kappa_\mu). \quad (\text{D.21})$$

A passive stable ferrite away from resonance must have $\mu_{g,+} > 0$ and $\mu_{g,-} > 0$.

D.9 Drude and Lorentz response

D.9.1 Drude model

For free carriers,

$$\varepsilon(\omega) = \varepsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma_D)} \quad (\text{D.22})$$

The displayed formula above follows the convention and normalisation used in [29]. With our $e^{-i\omega t}$ convention, $\gamma_D > 0$ gives absorption. In the lossless limit $\gamma_D = 0$,

$$\frac{\partial}{\partial \omega} [\omega \varepsilon(\omega)] = \varepsilon_\infty + \frac{\omega_p^2}{\omega^2} > 0. \quad (\text{D.23})$$

D.9.2 Lorentz oscillator

For a bound resonance,

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{f\omega_r^2}{\omega_r^2 - \omega^2 - i\gamma_L\omega}. \quad (\text{D.24})$$

In the lossless limit,

$$\partial_{\omega}[\omega \varepsilon(\omega)] = \varepsilon_{\infty} + f \omega_r^2 \frac{\omega_r^2 + \omega^2}{(\omega_r^2 - \omega^2)^2} > 0 \quad (\text{D.25})$$

away from the material resonance. This positivity is the oscillator origin of the Brillouin energy formula.

D.10 Useful constants and units

Quantity	Symbol	SI value
Speed of light	c	2.99792458×10^8 m/s
Vacuum permittivity	ε_0	$8.8541878128 \times 10^{-12}$ F/m
Vacuum permeability	μ_0	$1.25663706212 \times 10^{-6}$ H/m
Free-space impedance	$Z_0 = \sqrt{\mu_0/\varepsilon_0}$	376.730313668 Ω
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34}$ J s
Electron charge	e	$1.602176634 \times 10^{-19}$ C
Electron gyromagnetic ratio	$\gamma/(2\pi)$	28.025 GHz/T

D.11 Checklist for translating sources

Sources in topological photonics use a wide variety of sign conventions [12, 19]. When using a formula from another paper, check the following before inserting it into this monograph:

1. Does the paper use $e^{-i\omega t}$ or $e^{+i\omega t}$?
2. Is the gyrotropic tensor written with $+ig$ or $-ig$ in the xy component?
3. Is the Berry connection defined as $+i\langle u|\nabla u\rangle$ or $-i\langle u|\nabla u\rangle$?
4. Is the inner product the bare L^2 product or the electromagnetic energy product with $\mathbf{M}$ or $\mathbf{M}_g$?
5. Is material dispersion ignored, approximated, or included through auxiliary oscillator variables?
6. Are the quoted Chern numbers for individual bands, composite band groups, or gaps?

Only after these convention checks should formulas be compared or cited.

D.12 Kramers–Kronig relations and passivity

[33, 43]

D.12.1 The dispersion relations

Any causal linear response function $\chi(\omega)$ satisfies the Kramers–Kronig relations, which connect the real and imaginary parts:

$$\operatorname{Re} \chi(\omega) = \frac{1}{\pi} \text{P.V.} \int_{-\infty}^{\infty} \frac{\operatorname{Im} \chi(\omega')}{\omega' - \omega} d\omega', \quad (\text{D.26})$$

$$\operatorname{Im} \chi(\omega) = -\frac{1}{\pi} \text{P.V.} \int_{-\infty}^{\infty} \frac{\operatorname{Re} \chi(\omega')}{\omega' - \omega} d\omega'. \quad (\text{D.27})$$

Here P.V. denotes the Cauchy principal value. For the dielectric permittivity $\varepsilon(\omega) = 1 + \chi_e(\omega)$, the Kramers–Kronig relations guarantee that $\operatorname{Im} \varepsilon(\omega) > 0$ for $\omega > 0$ in a passive medium, and that the dispersive energy metric $\mathbf{M}_g(\omega) = \partial_\omega[\omega \mathbf{M}(\omega)]$ is positive definite at all real frequencies outside the absorption bands [19, 20].

D.12.2 Passivity and the positive-definite metric

A medium is passive if it does not provide energy to the field. Mathematically, this requires that the power dissipated by any field configuration is non-negative: $\mathbf{f}^\dagger \operatorname{Im}[\omega \mathbf{M}(\omega)] \mathbf{f} \geq 0$ for all $\omega > 0$ and all $\mathbf{f}$. Combined with the Kramers–Kronig relations, passivity implies [19]

$$\mathbf{M}_g(\omega) = \frac{\partial[\omega \mathbf{M}(\omega)]}{\partial \omega} > 0 \quad (\text{positive definite}) \quad (\text{D.28})$$

at all real frequencies in the transparency windows (where $\operatorname{Im} \mathbf{M} \approx 0$). This result, proved rigorously by Silveirinha [20], is the foundation for the Hermitian structure of the photonic eigenproblem and the well-definedness of the Chern number.

D.13 Worked examples of sign-convention translation

D.13.1 Example 1: Haldane–Raghu to this book

Haldane and Raghu [3] use the convention $e^{+i\omega t}$ for the time dependence. In their notation, the gyrotropic permittivity tensor has $+ig$ in the off-diagonal position. Translating to this book’s convention ($e^{-i\omega t}$), the gyrotropic coupling changes sign: $+ig \rightarrow -ig$. The Chern numbers are unaffected because the sign change affects both the Hamiltonian and the inner product symmetrically.

D.13.2 Example 2: Gangaraj–Silveirinha–Hanson to this book

Gangaraj et al. [19] use $e^{-i\omega t}$ (same as this book) and define the Berry connection as $\mathcal{A}_n = i\langle \mathbf{f}_n | \mathbf{M} | \nabla_{\mathbf{k}} \mathbf{f}_n \rangle$, consistent with our convention. Their material matrix uses $\mathbf{M} = \operatorname{diag}(\varepsilon_0 \varepsilon, \mu_0 \mu)$, which matches our definition. Formulas from this source can be used directly without sign changes.

D.14 Summary of convention choices in this book

For reference, the complete set of convention choices used throughout this monograph is collected here [12]:

Convention	Choice
Time dependence	$e^{-i\omega t}$
Gyrotropic off-diagonal	$-i\kappa_\mu$ (upper off-diagonal)
Berry connection	$\mathcal{A}_n = i\langle u_n, \nabla_{\mathbf{k}} u_n \rangle_{\mathbf{M}_g}$
Inner product	$\langle f_1, f_2 \rangle_{\mathbf{M}_g} = \int f_1^\dagger \mathbf{M} f_2 dV$
Dispersive metric	$\mathbf{M}_g = \partial_\omega [\omega \mathbf{M}(\omega)]$
Chern number	$\mathcal{C}_n = \frac{1}{2\pi} \int_{\text{BZ}} \mathcal{F}_n d^2k$
Bloch convention	$\mathbf{f}_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} \mathbf{u}_{\mathbf{k}}(\mathbf{r})$
Peierls phase	rightward hopping carries $+\phi$

D.15 Magneto-optical tensors for specific materials

D.15.1 Yttrium iron garnet (YIG)

YIG ($\text{Y}_3\text{Fe}_5\text{O}_{12}$) is the standard material for microwave topological photonics. At a bias field $B_0 = 0.2$ T and frequency $f = 4.5$ GHz, the Polder parameters are [4, 12]:

Parameter	Value
Saturation magnetisation $4\pi M_s$	1780 G
Gyromagnetic ratio $\gamma / (2\pi)$	28.0 GHz/T
Resonance frequency $\omega_0 / (2\pi)$	5.6 GHz
Perpendicular permeability $\mu_\perp$	3.47
Gyrotropic coupling κ_μ	1.98
Dielectric constant ϵ_r	15.0
Gyrotropic ratio $\kappa_\mu / \mu_\perp$	0.57

The large gyrotropic ratio makes YIG the ideal material for demonstrating Chern-insulator physics with wide topological gaps.

D.15.2 Indium antimonide (InSb)

InSb is a narrow-gap semiconductor with strong magneto-plasma effects at terahertz frequencies. Under a magnetic field $B_0 = 0.3$ T at $f = 1$ THz, the relevant parameters are [13]:

Parameter	Value
Plasma frequency $\omega_p/(2\pi)$	2.0 THz
Cyclotron frequency $\omega_c/(2\pi)$	0.5 THz
Perpendicular permittivity $\varepsilon_\perp$	-3.0
Gyrotropic coupling κ_ε	0.75
Gyrotropic ratio $ \kappa_\varepsilon/\varepsilon_\perp $	0.25

The negative perpendicular permittivity indicates operation above the plasma frequency, where the material supports propagating modes with strong gyrotropic splitting.

D.15.3 Bismuth-substituted YIG (Bi:YIG)

Bi:YIG ($\text{Bi}_x\text{Y}_{3-x}\text{Fe}_5\text{O}_{12}$) provides enhanced Faraday rotation at near-infrared wavelengths compared to undoped YIG, but the gyrotropic ratio remains small. At $\lambda = 1.55 \mu\text{m}$, the off-diagonal permittivity is $\kappa_\varepsilon \sim 10^{-2}$, giving a Faraday rotation of about $0.5 \text{ deg}/\mu\text{m}$. This is too weak for practical topological gaps but sufficient for proof-of-concept magneto-optical devices [12].

D.16 Experimental material parameters for topological photonics

This section collects the material parameters used in the principal experimental demonstrations discussed in this book, providing a single reference for numerical computations and design.

D.16.1 Yttrium iron garnet (YIG) at microwave frequencies

YIG is the standard material for microwave Chern insulators (Chapter 9). At saturation magnetisation $M_s = 1780 \text{ G}$ and applied field $B_0 = 0.20 \text{ T}$:

Parameter	Value
ε_r	15
$\omega_0/(2\pi)$	5.6 GHz
$\omega_m/(2\pi)$	4.9 GHz
$\mu_\perp$	3.47
κ_μ	1.98
$\mu_\pm = \mu_\perp \mp \kappa_\mu$	1.49, 5.45
FMR linewidth ΔH	$\approx 0.3 \text{ Oe}$
$\tan \delta_m$	$\approx 3 \times 10^{-4}$

These parameters correspond to the Polder permeability tensor (D.9) with saturation magnetisation along $\hat{z}$ [4, 53].

D.16.2 Silicon for valley and pseudospin designs

Silicon is the dominant material for reciprocal topological photonic designs at telecommunications wavelengths:

Parameter	Value
ε_r at $\lambda = 1.55 \mu\text{m}$	11.7
Refractive index n	3.42
Absorption α	$< 0.1 \text{ dB/cm}$
Thermo-optic dn/dT	$1.86 \times 10^{-4} \text{ K}^{-1}$
Kerr coefficient n_2	$4.5 \times 10^{-18} \text{ m}^2/\text{W}$

The negligible absorption and high refractive index make silicon ideal for valley photonic crystals (Chapter 12) and Wu–Hu pseudospin designs (Chapter 11) [12, 17].

D.16.3 Fused silica for Floquet waveguide arrays

Fused silica is the substrate for laser-written waveguide arrays used in Floquet topological experiments (Chapter 13):

Parameter	Value
n_0 at $\lambda = 633 \text{ nm}$	1.457
Waveguide Δn	$\approx 6 \times 10^{-4}$
Propagation loss	$\approx 0.5 \text{ dB/cm}$
Mode diameter	$\approx 10 \mu\text{m}$

The low index contrast produces weakly guided modes with large evanescent tails, enabling controllable inter-waveguide coupling at separations of $d \approx 20\text{--}30 \mu\text{m}$ [16].

D.17 Sign-convention translation guide

Different communities use different time-harmonic conventions, and translating results between conventions is a frequent source of error in topological photonics. This section provides explicit translation rules.

D.17.1 Mapping between the two time-harmonic conventions

Let quantities with a tilde denote the $e^{+i\omega t}$ convention. The mapping is:

$$\begin{aligned}
 \tilde{\varepsilon}_{ij} &= \varepsilon_{ij}^*, & \tilde{\mu}_{ij} &= \mu_{ij}^*, \\
 \tilde{\zeta}_{ij} &= -\zeta_{ij}^*, & \tilde{\zeta}_{ij} &= -\zeta_{ij}^*, \\
 \tilde{\kappa}_\mu &= -\kappa_\mu^*, & \tilde{\kappa}_\varepsilon &= -\kappa_\varepsilon^*.
 \end{aligned} \tag{D.29}$$

For lossless media where all tensor elements are real, the mapping reduces to sign flips of the off-diagonal gyrotropic elements: $\tilde{\kappa}_\mu = -\kappa_\mu$ and $\tilde{\kappa}_\varepsilon = -\kappa_\varepsilon$.

D.17.2 Practical examples

YIG at 4.5 GHz. In this book ($e^{-i\omega t}$): $\mu_{\perp} = 3.47$, $\kappa_{\mu} = +1.98$. In the condensed-matter convention ($e^{+i\omega t}$): $\tilde{\mu}_{\perp} = 3.47$, $\tilde{\kappa}_{\mu} = -1.98$. The Chern number is convention-independent: $\mathcal{C} = +1$ in both cases, because both the Berry connection and the integration orientation transform consistently.

Haldane–Raghu perturbation. Haldane and Raghu (2008) use $e^{+i\omega t}$, so their gyrotropic parameter η has the opposite sign relative to ours. When importing their Dirac-mass formula $m \propto \eta \langle u_1 | \delta \varepsilon | u_2 \rangle$, the sign of η must be flipped to match the $e^{-i\omega t}$ inner product used in this book [3, 19].

Bender–Boettcher PT symmetry. The original PT-symmetry literature uses $e^{+i\omega t}$, where a PT-symmetric potential satisfies $V^*(-x) = V(x)$. In our convention, the same condition reads $V(-x) = V^*(x)$ —i.e. the real part is even and the imaginary part is odd, with gain ($\text{Im} \varepsilon < 0$) and loss ($\text{Im} \varepsilon > 0$) interchanged relative to the $e^{+i\omega t}$ literature [22, 23].

D.18 Passivity, causality, and the Kramers–Kronig connection

The sign convention for absorption is not arbitrary: it is fixed by the requirement that the material response be causal and passive.

For a causal, passive material with the $e^{-i\omega t}$ convention, the permittivity $\varepsilon(\omega)$ is analytic in the lower half of the complex ω -plane, and the Kramers–Kronig relations connect the real and imaginary parts:

$$\begin{aligned} \text{Re } \varepsilon(\omega) &= 1 + \frac{2}{\pi} \mathcal{P} \int_0^{\infty} \frac{\omega' \text{Im } \varepsilon(\omega')}{\omega'^2 - \omega^2} d\omega', \\ \text{Im } \varepsilon(\omega) &= -\frac{2\omega}{\pi} \mathcal{P} \int_0^{\infty} \frac{\text{Re } \varepsilon(\omega') - 1}{\omega'^2 - \omega^2} d\omega'. \end{aligned} \quad (\text{D.30})$$

Passivity requires that the power dissipated by any field distribution be non-negative: $\frac{1}{2} \omega \text{Im } \varepsilon(\omega) \geq 0$ for all $\omega > 0$. Since $\omega > 0$, this implies $\text{Im } \varepsilon(\omega) \geq 0$ for passive media, confirming the sign convention used throughout this book [12, 20].

For active (gain) media, $\text{Im } \varepsilon(\omega) < 0$ in the amplifying frequency range, and the Kramers–Kronig relations remain valid but the passivity condition is violated. This is the physical basis for the gain/loss classification used in Chapter 17.

D.19 Kramers–Kronig relations and causality constraints

The frequency dependence of the material tensors is not arbitrary: it is constrained by causality through the Kramers–Kronig relations. These relations connect the real and imaginary parts of any causal response function and provide important consistency checks for material models used in topological photonics simulations.

D.19.1 Derivation from causality

A linear, time-invariant, causal response function $\chi(\omega)$ (e.g. the electric susceptibility $\varepsilon(\omega) - 1$) is analytic in the upper half of the complex frequency plane. By applying Cauchy's integral theorem to a contour that closes in the upper half-plane, one obtains the Kramers–Kronig relations [33, 43]:

$$\operatorname{Re} \chi(\omega) = \frac{1}{\pi} \text{P.V.} \int_{-\infty}^{\infty} \frac{\operatorname{Im} \chi(\omega')}{\omega' - \omega} d\omega', \quad (\text{D.31})$$

$$\operatorname{Im} \chi(\omega) = -\frac{1}{\pi} \text{P.V.} \int_{-\infty}^{\infty} \frac{\operatorname{Re} \chi(\omega')}{\omega' - \omega} d\omega', \quad (\text{D.32})$$

where P.V. denotes the Cauchy principal value.

D.19.2 Sum rules for the gyrotropic response

For the gyrotropic permeability $\kappa_\mu(\omega)$ of a ferrite (Section D.8), the Kramers–Kronig relations imply a sum rule:

$$\int_0^\infty \omega \operatorname{Im} \kappa_\mu(\omega) d\omega = \frac{\pi}{2} \omega_M \omega_0, \quad (\text{D.33})$$

where $\omega_M = \gamma \mu_0 M_s$ and $\omega_0 = \gamma \mu_0 H_0$ are the magnetisation and precession frequencies. This sum rule provides a useful check on the Polder model: any deviation indicates that the model is being used outside its range of validity [4, 19].

D.19.3 Implications for passivity and topological gaps

The Kramers–Kronig relations also constrain the size of the topological gap. Because the gyrotropic coupling κ_μ must satisfy the sum rule, increasing κ_μ at one frequency necessarily decreases it at other frequencies. This means that the topological gap width is bounded by the total oscillator strength available in the gyrotropic response.

For ferrites, the oscillator strength is concentrated near the ferromagnetic resonance, so the largest topological gaps occur at frequencies just below ω_0 . Moving to higher frequencies (e.g. optical) reduces the available gyrotropic coupling and narrows the gap, as discussed in Chapter 7 [13, 44].

D.20 Material parameter tables for topological photonics

D.20.1 Common microwave materials

Table D.1 lists the key material parameters for the most commonly used microwave-frequency topological photonic materials.

D.20.2 Common optical and infrared materials

Table D.2 lists the parameters for optical-frequency topological photonic crystals, where the gyrotropic response is replaced by geometric or Floquet engineering.

Table D.1: Material parameters for microwave topological photonics. All values at room temperature; YIG parameters at $B_0 = 0.20$ T, $f = 4.5$ GHz.

Material	ϵ_r	$\mu_\perp$	$\kappa_\mu / \mu_\perp$	Topological gap
YIG (single crystal)	15.0	3.47	0.57	$\sim 3\%$
BaFe ₁₂ O ₁₉	18.0	1.8	0.35	$\sim 1.5\%$
Li ferrite	14.5	2.5	0.45	$\sim 2\%$

Table D.2: Material parameters for optical/infrared topological photonics.

Material	ϵ_r	Platform	Mechanism	Topological gap
Silicon ($\lambda = 1550$ nm)	12.1	Valley PhC	Inversion breaking	$\sim 5\%$
SiN ($\lambda = 780$ nm)	4.0	Ring resonator	Synthetic gauge	$\sim 1\%$
Fused silica (fs-written)	1.45	Floquet waveguide	Helix modulation	$\sim 2\%$
Ce:YIG ($\lambda = 1550$ nm)	5.4	Magneto-optical	Faraday effect	$\sim 0.01\%$

These tables illustrate the fundamental trade-off in topological photonics: microwave materials provide the largest topological gaps through genuine time-reversal breaking, while optical materials offer compatibility with integrated photonic platforms at the cost of either weaker topological protection (valley and Floquet mechanisms) or extremely narrow gaps (magneto-optical route) [12, 13, 28, 67].

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